

Chapter 1 Basics of signals & systems

Signal:

A signal is defined as a function of one or more independent variables that conveys information. (on the nature of physical phenomenon)

A signal is a physical quantity that varies with space, time or any independent variable.

If the signal depends upon only one variable, then it is known as one dimensional signal.

Ex:- A speech signal whose amplitude varies with time.

If the signal depends upon two independent variables, then it is known as two dimensional signal.

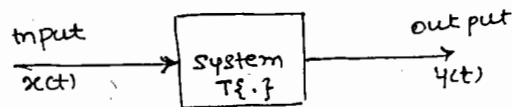
Ex:- A picture or image depends upon both horizontal and vertical co-ordinates represents a two dimensional signal

Signal modelling is known as representation of signals mathematically as function of one or more independent variables.

System:-

A system is a physical device that generates a response or output signal for a given input signal. The response of the system depends on the transfer function of the system.

The system basically transforms the input signal to o/p signal.

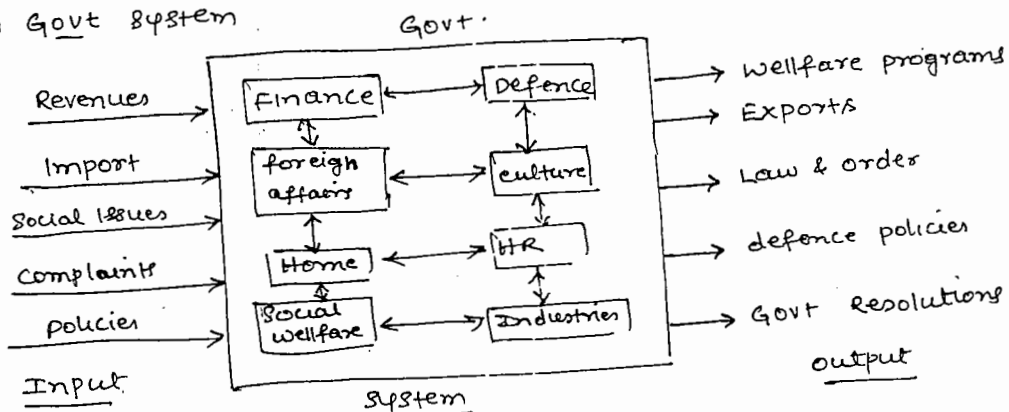


The relationship b/n input and output can be given as

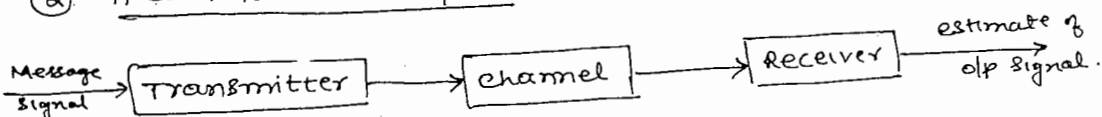
$$y(t) = T[x(t)]$$

Ex:- Communication system; Remote sensing system, etc.

(i) A Govt system



(2) A communication system



The Transmitter changes the message signal into a form suitable for transmission over a channel. The receiver processes the channel o/p to produce an estimate of message signal.

Chapter 1

Signals and systems are mathematically analyzed so that it is useful to predict the output of the system for a given input signal.

Signals and systems are analyzed with the help of Fourier Series, Fourier Transforms, DTFS, DTFT, Z-transforms etc.

a. Compare analog signal processing and Digital signal processing?

	Analog signal processing	Digital signal processing.
1	Uses analog circuit elements like resistors, capacitors, diode amplifiers etc.	Uses digital computer elements like adders, multipliers, and memory
2.	Uses differential equations without any approximations and the solutions obtained are in real time	Digital approach relies on numerical computations for its operations. The time required for these operations/ computations determines the real nature
3.	Simple approach and simple circuitry to implement	Requires a complex circuitry to implement.
4	Lack of flexibility and Repeatability	Has the advantage of flexibility & repeatability.

Classification of signals

Based on the properties and features, the signals are classified into (a) Continuous Time (CT) & Discrete Time (DT) signals.

(b) Even and odd signals (c) periodic & non periodic signals

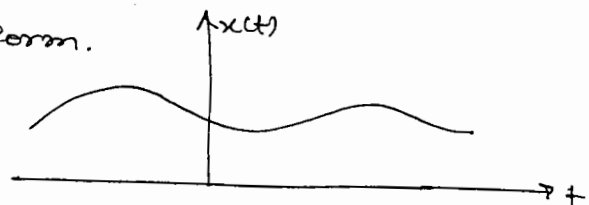
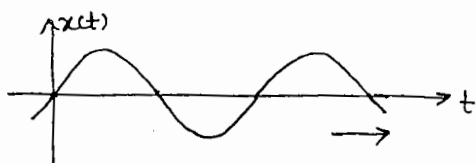
(d) Energy and power signals, (e) Deterministic and Random signals.

Continuous time and discrete time signals.

A signal $x(t)$, is said to be continuous if it is defined for all values of time. A CT signal is an analog signal.

A CT signal arises naturally when a physical phenomenon (heart beat; acoustic pressure) is converted into an electrical signal using an appropriate transducer.

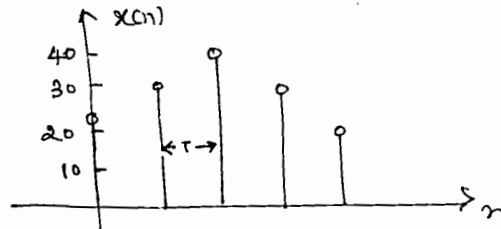
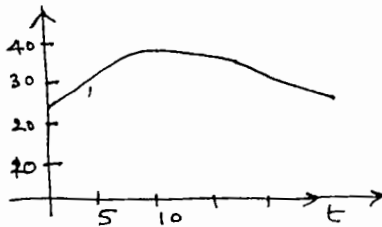
Ex:- AC power; ECG waveform.



Discrete time signals are defined at discrete instants of time and is represented by $x(n)$, where 'n' is the index.

Discrete time signals are obtained by sampling a CT signal at regular intervals.

Ex:- Share market index, temp^r variation over a day recorded per hour interval



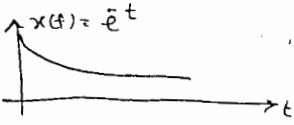
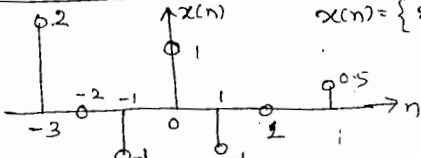
Hence sampling of CT signal at a time $nT = t$ yields a sample with a value $x(nT)$.

$$\therefore x(n) = x(nT) = x(t) \big|_{t=nT} \quad \text{where } n = \pm 0, \pm 1, \pm 2, \pm 3, \dots$$

where 'T' is known as sampling period.

A signal discretized in time and quantized in amplitude is known as a digital signal.

Distinguish between CT signals and DT signals

CT Signal	DT Signal
1. A CT signal is a continuous f ⁿ of time. Ex: $x(t) = \sin t$; for all t	A DT signal is defined only at specific interval or regular intervals only Ex: $x(n) = \{1, 4, 5\}$; $0 \leq n < 3$
2. Analog circuits like opamp, analog amplifiers, filters etc are used to process the CT signal	Digital CKTs like microcontrollers, fip's, flipflops, Registers, Counters etc are used to process DT signals.
A CT signal can be represented using Fourier series or Fourier Transform depending upon whether it is periodic or aperiodic respectively	A DT signal is represented using DTFS if periodic or DTFT if non-periodic
 $x(t) = e^{-t}$ $x(t) = e^{-t}$; for all t	 $x(n) = \{2, 0, -1, 1, -1, 0, 0.5\}$

Periodic and non periodic signals

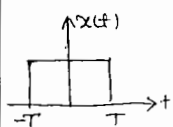
A CT signal $x(t)$ is said to be periodic if it satisfies the condition

$$x(t+T) = x(t) \quad \text{for all 't'}$$

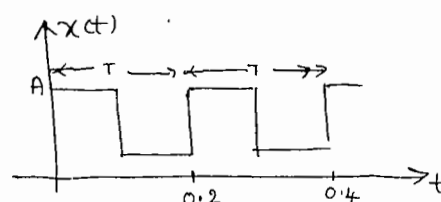
where 'T' is the fundamental period. (The smallest value of 't' for which the above condition is satisfied).

The fundamental frequency $f = \frac{1}{T}$ Hz.

we have $\omega = 2\pi f = \frac{2\pi}{T}$ rad/sec



$$T = \frac{2\pi}{\omega} \text{ sec}$$



$$T = 0.2 \text{ sec}$$

$$f = 5 \text{ Hz}$$

A signal is said to be non periodic or aperiodic if $x(t) \neq x(t+T)$ for all t .

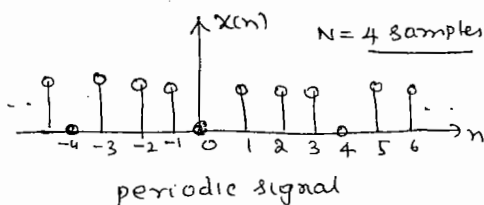
Similarly a DT signal is periodic if $x(n) = x(n+N)$ for all n .

where 'N' is the fundamental period.

The fundamental angular frequency $\Omega = \frac{2\pi}{N}$ radians

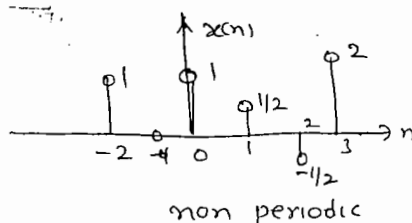
$$\text{or } N = \frac{2\pi}{\Omega} \text{ samples}$$

A DT signal is non periodic if $x(n) \neq x(n+N)$ for all n



periodic signal

EX: $(-1)^n$ for all n .



non periodic

Corollary

If $x_1(t)$ and $x_2(t)$ are periodic with periods T_1 and T_2 respectively, under what condition the sum $x(t) = x_1(t) + x_2(t)$ is periodic and what is the fundamental period if periodic.

$$x_1(t) = x_1(t+T_1) = x_1(t+mT_1) \quad \text{where } m \text{ \& } k \text{ are integers.}$$

$$x_2(t) = x_2(t+T_2) = x_2(t+kT_2)$$

$$x(t) = x_1(t) + x_2(t) = x_1(t+mT_1) + x_2(t+kT_2) \quad \text{--- (1)}$$

$$x(t+T) = x_1(t+T) + x_2(t+T) \quad \text{--- (2)}$$

In order for $x(t)$ to be periodic with period T ,

$$T = mT_1 = kT_2 \Rightarrow \frac{T_1}{T_2} = \frac{k}{m} = \text{rational number}$$

Hence the sum of two periodic signals is periodic iff the ratio of their respective periods can be expressed as a rational number.

The fundamental period is Lcm of T_1 & T_2 .

Similarly for a DT signal with $x_1(n) = x_1(n+mN_1)$ & $x_2(n) = x_2(n+kN_2)$

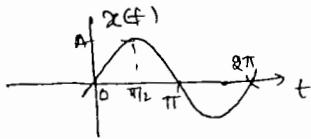
$x(n) = x_1(n) + x_2(n)$ is periodic with period N iff

$$\frac{N_1}{N_2} = \frac{k}{m} = \text{rational number}$$

Then $N = \text{Lcm of } N_1 \text{ \& } N_2$

Deterministic and Random Signals

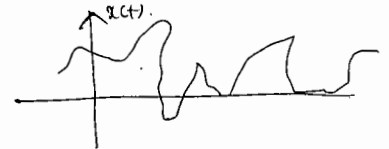
A signal which exhibits no uncertainty of values at any given instant of time is known as Deterministic signal. The instantaneous value can be accurately predicted by the mathematical equation.



$$x(t) = A \sin \omega t$$

A random signal is characterized by uncertainty before its actual occurrence. The signal takes different values at each time for which the signal value is defined.

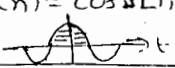
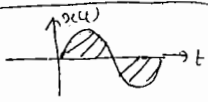
Ex: noise, Radiosignal comm. systems, EMI.



Distinguish b/w (i) even and odd signal

(ii) periodic and non periodic signals

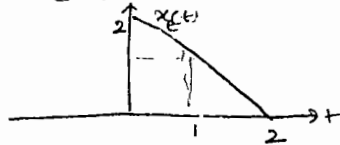
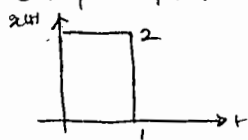
(iii) Energy and power signal.

	Even signal	odd signal.
1	A signal is said to be even if it satisfies the condition: $x(t) = x(-t)$ $x(n) = x(-n)$	A signal is said to be odd if it satisfies the condition $x(t) = -x(-t)$ $x(-n) = -x(n)$
2	Even signals are symmetrical about vertical axis (time axis)	odd signals are anti symmetric about vertical time axis.
3	Even signals have a definite value at $t=0$ or $n=0$	However odd signals have a zero value at $n=0$ & $t=0$
4	The product of even signals is an even signal	The product of 2 odd signal is even while product of odd & even signal is odd
5	Ex: $x(t) = \cos t$; $x(n) = \cos 2n$ 	$x(t) = \sin t$ $x(n) = \sin 2n$ 

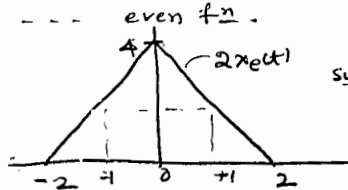
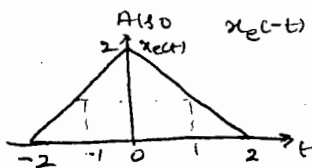
	periodic signal	non periodic signal
1	A CT or DT signal which satisfies the condition $x(t) = x(t+T)$ $x(n) = x(n+N)$ is said to be periodic. where $T(N)$ is the fundamental period	A signal which satisfies the condition $x(t) \neq x(t+T)$ $x(n) \neq x(n+N)$ is called non periodic signal

	periodic signal.	non periodic signal.
2	Generally periodic signals are power signals and their power is $< \infty$	non periodic signals are basically energy signals such that $0 < E < \infty$. where 'E' is the total energy
3	periodic signals are represented using Fourier series or DTFS	FT and DTFT are used to represent non periodic signals
4	The spectrum of periodic signal is discrete in nature	The spectrum of non periodic signal is continuous in nature.
5	Ex:- $x(t) = \sin \omega t$; for all t 	 $x(t) = \begin{cases} A & ; t < 1 \\ 0 & ; t > 1 \end{cases}$

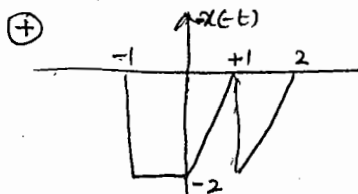
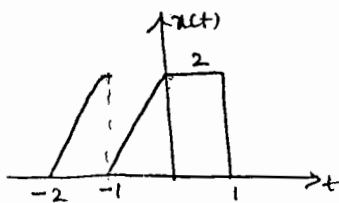
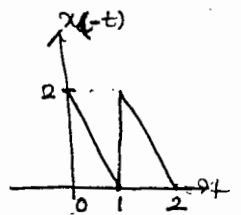
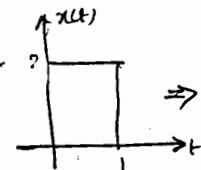
problem Fig shows a part of the signal $x(t)$ & its even part $x_e(t)$ only. plot $x(t)$, $x_e(t)$ & $x_o(t)$.



$$x_e(t) = \frac{x(t) + x(-t)}{2} \Rightarrow x(t) = 2x_e(t) - x(-t)$$

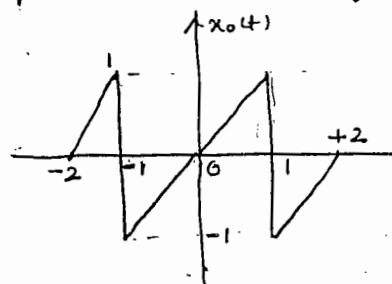


Subtraction



⇒

Also $x_o(t) = \frac{x(t) - x(-t)}{2}$



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$$x(n) = \delta(n-8) + 2(-1)^{n-1}u(n-7) - (-1)^{n-1}u(n-1)$$

Even and odd signals (Symmetric and anti symmetric signals)

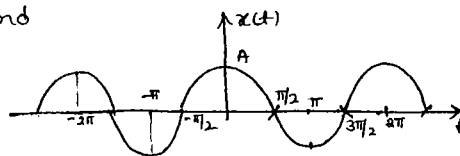
Even signal

A signal is said to be even or symmetric if it satisfies the condition

$$x(t) = x(-t) \text{ for all values of } t \text{ and}$$

$$x(n) = x(-n) \text{ for all values of } n.$$

Ex: $x(t) = A \cos t$



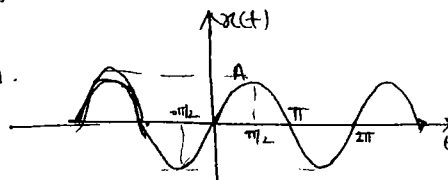
odd signal

A signal is said to be odd if it satisfies

the condition $x(-t) = -x(t)$; for all t

$$x(-n) = -x(n) ; \text{ for all } n.$$

Ex: $x(t) = A \sin t$



- ① Show that the product of two even signals or two odd signals is an even signal and the product of even and odd signal is an odd signal.

Let the two signals be $x_1(t)$ and $x_2(t)$. such that

$$x(t) = x_1(t) \cdot x_2(t)$$

(i) when both the signals are even

$$x(-t) = x_1(-t) \cdot x_2(-t) = x_1(t) \cdot x_2(t) = x(t)$$

$$\therefore \boxed{x(-t) = x(t)}$$

(ii) when both the signals are odd

$$x(-t) = x_1(-t) \cdot x_2(-t) = [-x_1(t)] [-x_2(t)] = x_1(t) \cdot x_2(t) = x(t).$$

$$\therefore \boxed{x(-t) = x_1(t) \cdot x_2(t) = x(t)}$$

(iii) when $x_1(t)$ is even and $x_2(t)$ is odd

$$x(-t) = x_1(-t) \cdot x_2(-t) = x_1(t) [-x_2(t)] = -x_1(t) \cdot x_2(t) = -x(t)$$

$$\therefore \boxed{x(-t) = -x_1(t)}$$

||| ^{by} we can show the same for discrete signals.

$$x(-n) = x_1(-n) \cdot x_2(-n) = x_1(n) \cdot x_2(n) = x(n) \quad \text{— when signals are even}$$

$$x(-n) = x_1(-n) \cdot x_2(-n) = [-x_1(n)] [-x_2(n)] = x(n) \quad \text{— when the signals are odd}$$

$$x(-n) = x_1(-n) \cdot x_2(-n) = x_1(n) [-x_2(n)] = -x(n) \rightarrow \begin{matrix} x_1(n) \text{ — even} \\ x_2(n) \text{ — odd} \end{matrix}$$

Decomposition of a signal

A continuous time signal $x(t)$ or $x(n)$ can be decomposed in to sum of two signals namely $x_e(t)$ and $x_o(t)$. where $x_e(t)$ is the even part of the signal and $x_o(t)$ is the odd part of the signal.

$$x(t) = x_e(t) + x_o(t) \text{ --- ①}$$

put $t = -t$ in eqn ①

$$x(-t) = x_e(-t) + x_o(-t) = x_e(t) - x_o(t) \text{ --- ②}$$

Solving ① & ② we get

$$x_e(t) = \frac{x(t) + x(-t)}{2}$$

$$x_o(t) = \frac{x(t) - x(-t)}{2}$$

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$$x_e(n) = \frac{x(n) + x(-n)}{2} \quad x_o(n) = \frac{x(n) - x(-n)}{2}$$

② Show that (a) If $x(t)$ and $x(n)$ are even then

$$\int_{-a}^a x(t) dt = 2 \int_0^a x(t) dt \quad \text{and} \quad \sum_{n=-k}^k x(n) = x(0) + 2 \sum_{n=1}^k x(n)$$

proof $\int_{-a}^a x(t) dt = \int_{-a}^0 x(t) dt + \int_0^a x(t) dt \text{ --- ③}$

put $t = -\lambda$ in the first term of RHS. changing the limits

$$\begin{aligned} \int_{-a}^0 x(t) dt &= \int_{\lambda=-a}^0 x(-\lambda) d(-\lambda) = - \int_{-a}^0 x(-\lambda) d\lambda = - \int_{-a}^0 x(\lambda) d\lambda \quad \text{Since } x(t) \text{ or } x(\lambda) \text{ is even} \\ &= \int_0^a x(\lambda) d\lambda = \int_0^a x(t) dt \quad [\text{by changing the variable to } t] \end{aligned}$$

$$\therefore \int_{-a}^a x(t) dt = \int_0^a x(t) dt + \int_0^a x(t) dt = 2 \int_0^a x(t) dt$$

$$\sum_{n=-k}^k x(n) = \sum_{n=-k}^{-1} x(n) + x(0) + \sum_{n=1}^k x(n)$$

putting $n = -m$ in the first term of RHS.

$$\sum_{n=-k}^{-1} x(n) = \sum_{m=k}^1 x(-m) = \sum_{m=1}^k x(-m) = \sum_{m=1}^k x(m) \quad \text{Since } x(m) \text{ is even}$$

$$\therefore \sum_{n=-k}^k x(n) = \sum_{m=1}^k x(n) + x(0) + \sum_{n=1}^k x(n)$$

$$\sum_{n=-k}^k x(n) = x(0) + 2 \sum_{n=1}^k x(n)$$

If $x(n)$ is an odd signal ; show that $\sum_{n=-\infty}^{\infty} x(n) = 0$ given $x(0) = 0$.

$$\begin{aligned}
 \sum_{n=-\infty}^{\infty} x(n) &= \sum_{n=-\infty}^{-1} x(n) + x(0) + \sum_{n=1}^{\infty} x(n) \\
 &= \sum_{n=-\infty}^{-1} x(-n) + x(0) + \sum_{n=1}^{\infty} x(n) \\
 &= \sum_{n=1}^{\infty} x(-n) + x(0) + \sum_{n=1}^{\infty} x(n) \\
 &= - \sum_{n=1}^{\infty} x(n) + x(0) + \sum_{n=1}^{\infty} x(n) \quad \text{Since } x(n) \text{ is an odd signal.}
 \end{aligned}$$

given $x(0) = 0$

$$\therefore \sum_{n=-\infty}^{\infty} x(n) = 0$$

For any arbitrary signal $x(n)$; show that $\sum_{n=-\infty}^{\infty} x^2(n) = \sum_{n=-\infty}^{\infty} [x_o^2(n) + x_e^2(n)]$

Any arbitrary signal $x(n)$ can be given as

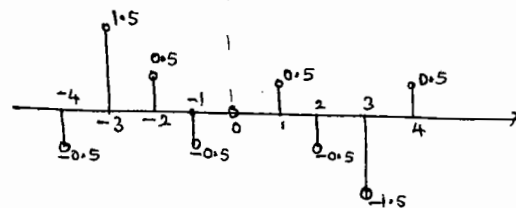
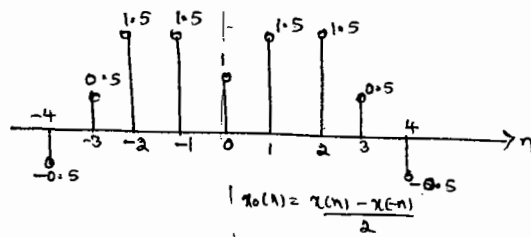
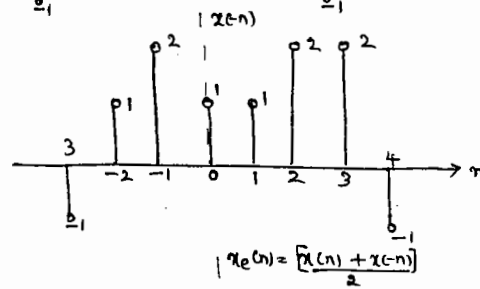
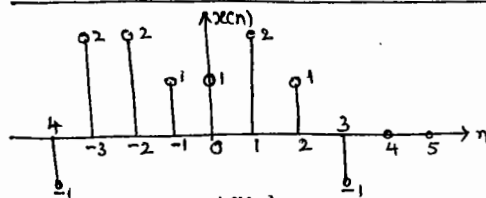
$$x(n) = x_o(n) + x_e(n)$$

$$\sum_{n=-\infty}^{\infty} [x(n)]^2 = \sum_{n=-\infty}^{\infty} [x_e(n) + x_o(n)]^2 = \sum_{n=-\infty}^{\infty} [x_e^2(n) + x_o^2(n) + 2x_e(n) \cdot x_o(n)]$$

Since the product of odd and even signal is an odd signal and whose value is zero.

$$\therefore \sum_{n=-\infty}^{\infty} [x(n)]^2 = \sum_{n=-\infty}^{\infty} [x_e^2(n) + x_o^2(n)]$$

Find the even and odd part of the given signal



Find the even and odd components of

$$x(t) = 1 + t \tan t + t^2 \tan^2 t$$

$$x(-t) = 1 + (-t) \tan(-t) + (-t)^2 \tan^2(-t)$$

$$= 1 + t \tan t + t^2 \tan^2 t$$

$$x_e(t) = 1 + t \tan t + t^2 \tan^2 t$$

$$x_o(t) = 0$$

Hence only even component persists

$$x_e(t) = x(t) = \underline{1 + t \tan t + t^2 \tan^2 t}$$

Even and odd signals

① Find the even and odd components of each of the following signals.

(a) $x(t) = \cos t + \sin t + \sin t \cos t$

Soln:

$$x(t) = \cos t + \sin t + \sin t \cos t$$

$$x(-t) = \cos(-t) + \sin(-t) + \sin(-t) \cos(-t)$$

$$x(-t) = \cos t - \sin t - \sin t \cos t$$

$$x_e(t) = \frac{1}{2} [x(t) + x(-t)]$$

$$= \frac{1}{2} [\cancel{\cos t} + \cancel{\sin t} + \cancel{\sin t \cos t} + \cancel{\cos t} - \cancel{\sin t} - \cancel{\sin t \cos t}]$$

$$= \frac{1}{2} [2 \cos t] = \cos t$$

$$\boxed{x_e(t) = \cos t} \quad \dots \text{Even part}$$

$$x_o(t) = \frac{1}{2} [x(t) - x(-t)] = \frac{1}{2} [\cancel{\cos t} + \sin t + \sin t \cos t - \cancel{\cos t} + \sin t - \sin t \cos t]$$

$$= \frac{1}{2} [2 \sin t + 2 \sin t \cos t] = \sin t + \sin t \cos t$$

$$\boxed{x_o(t) = \sin t (1 + \cos t)} = [\text{odd part of the signal}]$$

⑤ $x(t) = 1 + 3t^2 + t + 5t^3 + 9t^4$

$$x(t) = 1 + t + 3t^2 + 5t^3 + 9t^4$$

$$x(-t) = 1 + (-t) + 3(-t)^2 + 5(-t)^3 + 9(-t)^4$$

$$= 1 - t + 3t^2 - 5t^3 + 9t^4$$

Even part

$$x_e(t) = \frac{1}{2} [x(t) + x(-t)]$$

$$x_e(t) = \frac{1}{2} [1 + 3t^2 + \cancel{t} + 5\cancel{t^3} + 9t^4 + 1 - \cancel{t} + 3t^2 - 5\cancel{t^3} + 9t^4]$$

$$\boxed{x_e(t) = 1 + 3t^2 + 9t^4}$$

$$x(t) = x_e(t) + x_o(t)$$

$$x_o(t) = x(t) - x_e(t) = 0 + t + 5t^3$$

$$\therefore \boxed{x_o(t) = t + 5t^3} \quad \dots \text{odd part}$$

$$\textcircled{c} \quad x(t) = 1 + t \cos t + t^2 \sin t + t^3 \sin t \cos t$$

$$x(t) = 1 + t \cos t + t^2 \sin t + t^3 \sin t \cos t$$

$$x(-t) = 1 + (-t) \cos(-t) + (-t)^2 \sin(-t) + (-t)^3 \sin(-t) \cos(-t)$$

$$x(-t) = 1 - t \cos t - t^2 \sin t + t^3 \sin t \cos t$$

Even part $x_e(t) = \frac{x(t) + x(-t)}{2}$

$$x_e(t) = \frac{1}{2} [1 + t \cos t + t^2 \sin t + t^3 \sin t \cos t + 1 - t \cos t - t^2 \sin t + t^3 \sin t \cos t]$$

$$x_e(t) = 1 + t^3 \sin t \cos t$$

odd part $x_o(t) = x(t) - x_e(t)$

$$x_o(t) = 1 + t \cos t + t^2 \sin t + t^3 \sin t \cos t - [1 + t^3 \sin t \cos t]$$

$$x_o(t) = t \cos t + t^2 \sin t$$

$$\textcircled{d} \quad x(t) = (1 + t^3) \cos^3(10t)$$

$$x(t) = (1 + t^3) \cos^3(10t)$$

$$x(-t) = [1 + (-t)^3] \cos^3(10(-t))$$

$$x(-t) = [1 - t^3] \cos^3(10t)$$

Even part $x_e(t) = \frac{1}{2} (x(t) + x(-t))$

$$x_e(t) = \frac{1}{2} [(1 + t^3) \cos^3(10t) + (1 - t^3) \cos^3(10t)]$$

$$x_e(t) = \cos^3 10t$$

odd part

$$x_o(t) = x(t) - x_e(t)$$

$$x_o(t) = t^3 \cos^3(10t)$$

$$\textcircled{f} \quad x(n) = \{-2, 1, 2, -1, 3\}$$

$$\text{Given } x(n) = \{-2, 1, 2, -1, 3\}$$

$$x(0) = 2, \quad x(1) = -1, \quad x(2) = 3, \quad x(-1) = 1, \quad x(-2) = -2$$

Even part $x_e(0) = \frac{x(0) + x(-0)}{2} = \frac{2+2}{2} = 2$

$$x_e(1) = \frac{x(1) + x(-1)}{2} = \frac{-1 + 1}{2} = 0$$

$$x_e(2) = \frac{x(2) + x(-2)}{2} = \frac{3 - 2}{2} = \frac{1}{2}$$

odd parts $x_o(0) = \frac{x(0) - x(-0)}{2} = \frac{2 - 2}{2} = 0$

$$x_o(1) = \frac{x(1) - x(-1)}{2} = \frac{-1 - 1}{2} = -1$$

$$x_o(2) = \frac{x(2) - x(-2)}{2} = \frac{3 + 2}{2} = 5/2$$

$$\therefore \begin{array}{|l} x_e(n) = \left\{ \frac{1}{2}, 0, \underset{\uparrow}{2}, 0, \frac{1}{2} \right\} \\ x_o(n) = \left\{ \frac{5}{2}, -1, \underset{\uparrow}{0}, -1, \frac{5}{2} \right\} \end{array}$$

(g) $x(t) = e^t u(t)$

$$x(t) = e^t u(t)$$

$$x(-t) = e^{-(-t)} u(-t) = e^t u(-t)$$

$$x_e(t) = \frac{1}{2} [x(t) + x(-t)] = \frac{e^t u(t) + e^t u(-t)}{2}$$

$$x_o(t) = \frac{1}{2} [x(t) - x(-t)] = \frac{e^t u(t) - e^t u(-t)}{2}$$

(h) $x(t) = [\sin(\pi t) + \cos(\pi t)]^2$

$$x(t) = (\sin \pi t + \cos \pi t)^2 = \sin^2 \pi t + \cos^2 \pi t + 2 \sin \pi t \cos \pi t$$

$$x(t) = 1 + \sin 2\pi t$$

$$x(-t) = 1 - \sin 2\pi t$$

$$x_e(t) = \frac{1}{2} [x(t) + x(-t)] = \underline{1}$$

$$x_o(t) = \frac{1}{2} [x(t) - x(-t)] = \underline{\sin 2\pi t}$$

(i) $x(t) = e^{-2t} \cos t$

$$x(t) = e^{-2t} \cos t; \quad x(-t) = e^{-2(-t)} \cos(-t) = e^{2t} \cos t$$

$$x_e(t) = \frac{1}{2} \{ e^{-2t} \cos t + e^{2t} \cos t \} = \underline{\cos t \cosh 2t}$$

$$x_o(t) = \frac{1}{2} \{ e^{2t} \cos t - e^{-2t} \cos t \} = \underline{-\cos t \sinh 2t}$$

$$(j) \underline{x(t) = e^{jt}}$$

$$x(t) = e^{jt} ; \quad x(-t) = e^{-jt}$$

$$\therefore x_e(t) = \frac{1}{2} (e^{jt} + e^{-jt}) = \underline{\cos t}$$

$$x_o(t) = \frac{1}{2} \{e^{jt} - e^{-jt}\} = \underline{j \sin t}$$

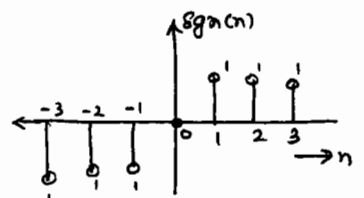
$$(k) \underline{x(n) = \alpha^n u(n)}$$

$$x(n) = \alpha^n u(n) ; \quad x(-n) = \alpha^{-n} u(-n)$$

$$x_e(n) = \frac{1}{2} \{ \alpha^n u(n) + \alpha^{-n} u(-n) \} = \frac{1}{2} \alpha^{|n|} + \frac{1}{2} \delta(n)$$

$$x_e(n) = \begin{cases} \frac{\alpha^n}{2} ; & n > 0 \\ 1 ; & n = 0 \\ \frac{\alpha^{-n}}{2} ; & n < 0 \end{cases}$$

$$\therefore \boxed{x_e(n) = \frac{1}{2} [\alpha^{|n|} + \delta(n)]}$$



$$x_o(n) = \frac{1}{2} \{ \alpha^n u(n) - \alpha^{-n} u(-n) \}$$

$$\boxed{x_o(n) = \frac{1}{2} \alpha^{|n|} \operatorname{sgn}(n)}$$

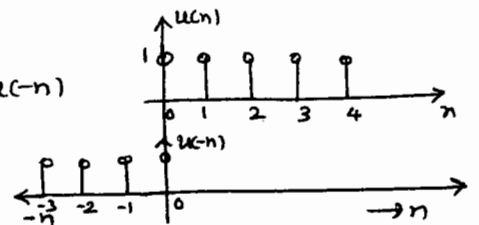
$$(l) \underline{x(n) = u(n)}$$

$$x(n) = u(n) \quad x(-n) = u(-n)$$

$$x_e(n) = \frac{x(n) + x(-n)}{2}$$

$$= \frac{u(n) + u(-n)}{2} = \begin{cases} 1 ; & n = 0 \\ 1/2 ; & n \neq 0 \end{cases}$$

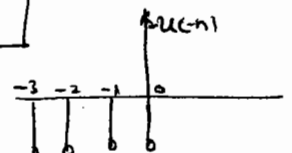
$$\therefore \boxed{x_e(n) = \frac{1}{2} + \frac{1}{2} \delta(n) = \frac{1}{2} (1 + \delta(n))}$$



$$x_o(n) = \frac{x(n) - x(-n)}{2} = \frac{u(n) - u(-n)}{2}$$

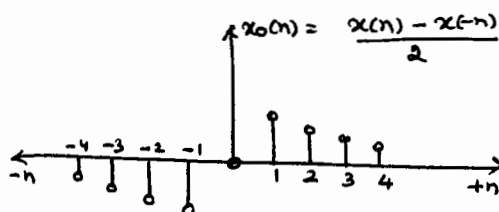
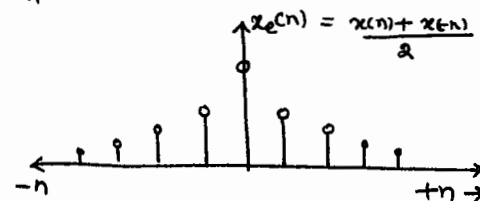
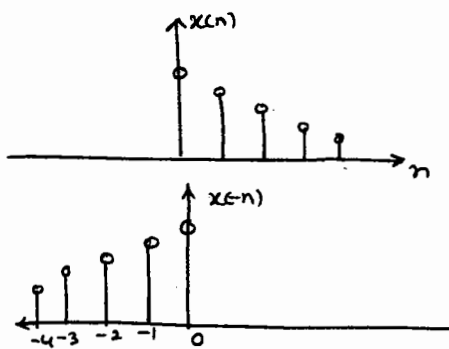
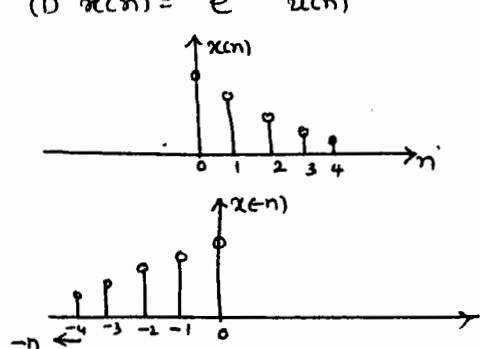
$$x_o(n) = \begin{cases} 1/2 ; & n > 0 \\ 0 ; & n = 0 \\ -1/2 ; & n < 0 \end{cases}$$

$$\text{or } \boxed{x_o(n) = \frac{1}{2} \operatorname{sgn}(n)}$$



② Find and sketch the even and odd part of the following

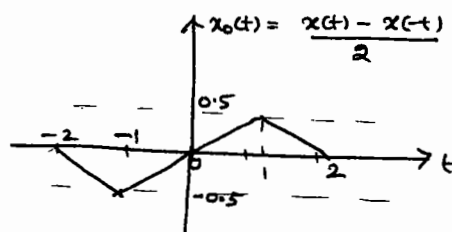
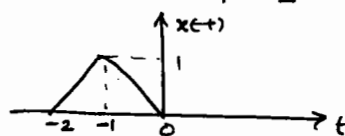
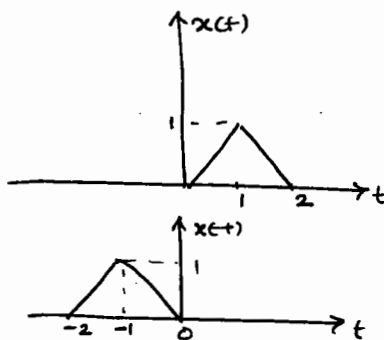
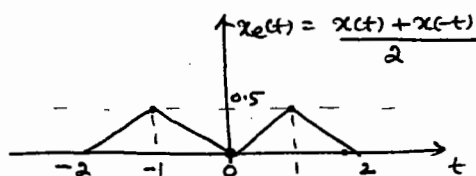
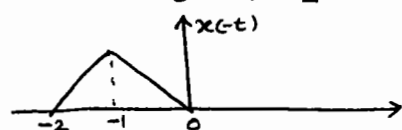
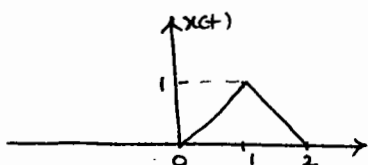
(1) $x(n) = e^{-n/4} u(n)$



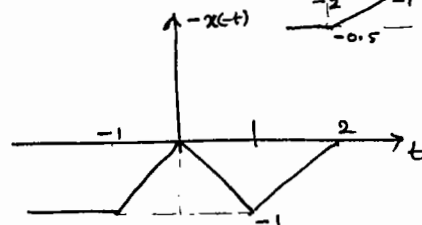
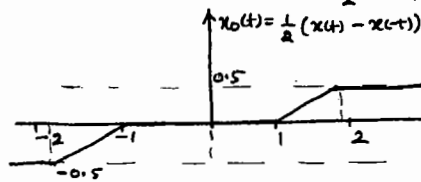
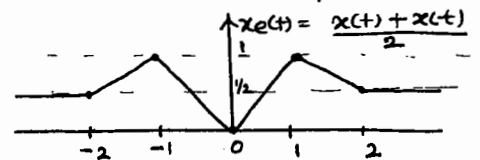
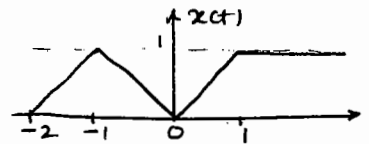
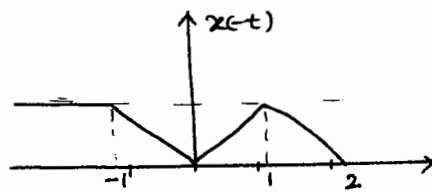
$$x_e(n) = \frac{x(n) + x(-n)}{2} = \frac{e^{-n/4} u(n) + e^{n/4} u(-n)}{2}$$

$$x_o(n) = \frac{x(n) - x(-n)}{2} = \frac{e^{-n/4} u(n) - e^{n/4} u(-n)}{2}$$

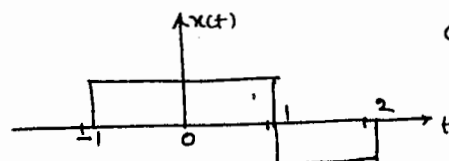
③ $x(t) = \begin{cases} t & ; 0 \leq t \leq 1 \\ 2-t & ; 1 \leq t \leq 2 \end{cases}$



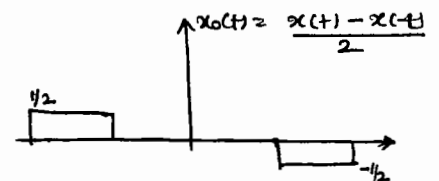
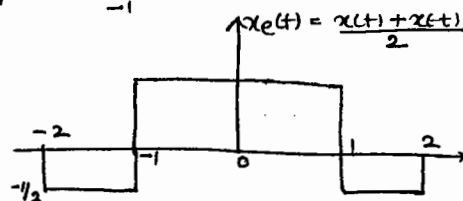
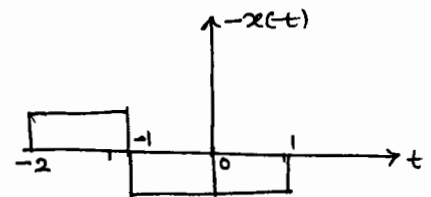
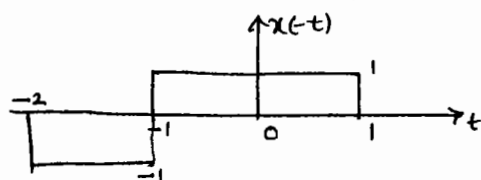
③ Given $x(t)$ as shown below



④



Given $x(t)$ as shown
determine $x_o(t)$ & $x_e(t)$



Determine the odd and even component of $x(n) = n^2 \left[\frac{1}{2} \right]^{n-2}$

$$x(-n) = (-n)^2 \left[\frac{1}{2} \right]^{-n-2}$$

$$x_e(n) = \frac{x(n) + x(-n)}{2}$$

$$= \frac{n^2}{2} \left[\left(\frac{1}{2} \right)^{n-2} + \left(\frac{1}{2} \right)^{-n-2} \right]$$

$$= n^2 \left[\left(\frac{1}{2} \right)^{n-3} + \left(\frac{1}{2} \right)^{-n-3} \right]$$

$$x_o(n) = \frac{x(n) - x(-n)}{2}$$

$$= \frac{n^2}{2} \left[\left(\frac{1}{2} \right)^{n-2} - \left(\frac{1}{2} \right)^{-n-2} \right]$$

$$= n^2 \left[\left(\frac{1}{2} \right)^{n-3} - \left(\frac{1}{2} \right)^{-n-3} \right]$$

Energy and power signals:

A signal $x(t)$ is called an energy signal if it satisfies the condition that the total energy of the signal is less than ∞ .

$$\text{i.e. } 0 < E < \infty.$$

The power of an energy signal is zero over an infinite period.

The total normalized energy is non zero and finite.

EX:- All non periodic signals defined over a finite period.

The total energy of the cont. time signal $x(t)$ } $E = \int_{-\infty}^{\infty} |x(t)|^2 dt$ Joules

|||^{ly} for a discrete time signal $x(n)$

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2 \text{ Joules}$$

Power signal.

A signal $x(t)$ is called power signal if and only if the average power of the signal satisfies the condition $0 < P < \infty$.

The average signal power is defined by (for a CT signal)

$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt \text{ Watts.}$$

for a periodic signal with fundamental period T . the expr. reduces

to

$$P = \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt \text{ Watts.}$$

|||^{ly} for a discrete time signal $x(n)$; The average power is

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N} \sum_{n=-N}^N |x(n)|^2 \text{ Watts.}$$

for a periodic signal with fundamental period N . the average

power is given by.

$$P = \frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2 \text{ Watts.}$$

Signal extending from $-\infty$ to $+\infty$

EX: All periodic signals are power signals.

A power signal has an infinite energy. i.e $E = \infty$.

Power of an Energy signal.

$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\infty}^{\infty} |x(t)|^2 dt = \lim_{T \rightarrow \infty} \frac{1}{T} E$$

$$P = \lim_{T \rightarrow \infty} \frac{E}{T} = \frac{E}{\infty} = 0 \quad \text{hence the power of an energy signal is zero over an infinite period.}$$

Energy of the power signal.

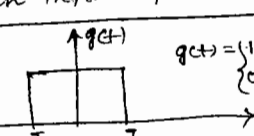
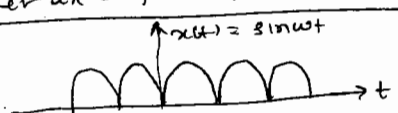
consider a power signal $x(t)$ with a finite power. The normalized energy of $x(t)$ is

$$E = \lim_{T \rightarrow \infty} \int_{-T/2}^{T/2} |x(t)|^2 dt = \lim_{T \rightarrow \infty} \left[T \cdot \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt \right]$$

$$= \lim_{T \rightarrow \infty} T \left[\lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt \right] = \lim_{T \rightarrow \infty} T [P] = \infty$$

hence the Energy of the power signal is infinite over an infinite period.

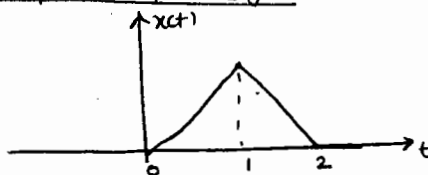
Note:- The square root of the average power P is called the RMS value of the signal $x(t)$.

	Energy signal	power signal.
1	Normalized energy is finite and non zero	Normalized power is finite non zero.
2	$E = \int_{-\infty}^{\infty} x(t) ^2 dt$ joules	$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t) ^2 dt$ Watts.
3	Energy signals are time limited	power signals can exist over infinite intervals.
4	power of an energy signal is 0 over an infinite period	Energy of the power signal is ∞ over an infinite period.
5	EX:  $g(t) = \begin{cases} 1 & t < T \\ 0 & \text{otherwise} \end{cases}$ EX: - Non periodic signals.	EX  EX: periodic signals.

Energy and power signals.

1. Check whether the following signals are energy signals or power signals, and also find the associated energy or power of the signal.

(a) $x(t) = \begin{cases} t & ; 0 \leq t \leq 1 \\ 2-t & ; 1 \leq t \leq 2 \\ 0 & ; \text{otherwise} \end{cases}$



The given signal is defined for a finite period from 0 to 2. hence it is not periodic. The given signal is an Energy signal

$$\begin{aligned} E &= \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_0^1 |x(t)|^2 dt + \int_1^2 |x(t)|^2 dt + 0 = \int_0^1 t^2 dt + \int_1^2 (2-t)^2 dt \\ &= \left. \frac{t^3}{3} \right|_0^1 + 4t \left. \left| \frac{t^3}{3} \right| \right|_1^2 - 2 \left. \frac{t^2}{2} \right|_1^2 = \frac{1}{3} + 4(2-1) + \frac{1}{3}[8-1] - [4-1] \\ &= \frac{1}{3} + 4 + \frac{7}{3} - 3 = \frac{8}{3} - 2 = \frac{8-6}{3} = \frac{2}{3} > \infty \end{aligned}$$

$\therefore$ Energy of the signal $E = \frac{2}{3}$ Joules

(b) $x(n) = \begin{cases} n & ; 0 \leq n < 5 \\ 10-n & ; 5 \leq n \leq 10 \\ 0 & ; \text{otherwise} \end{cases}$

The given signal is an Energy signal as it is defined for a finite duration and is not periodic. (discrete signal)

$$\begin{aligned} E &= \sum_{n=-\infty}^{\infty} |x(n)|^2 = \sum_{n=0}^4 |x(n)|^2 + \sum_{n=5}^{10} |x(n)|^2 = \sum_{n=0}^4 n^2 + \sum_{n=5}^{10} (10-n)^2 \\ E &= 0^2 + 1^2 + 2^2 + 3^2 + 4^2 + (10-5)^2 + (10-6)^2 + (10-7)^2 + (10-8)^2 + (10-9)^2 + (10-10)^2 \\ &= 1 + 4 + 9 + 16 + 25 + 16 + 9 + 4 + 1 = 85 < \infty \end{aligned}$$

$E = 85$ joules

(c) $x(t) = \begin{cases} 5 \cos \pi t & ; -1 \leq t \leq 1 \\ 0 & ; \text{otherwise} \end{cases}$

Given signal is an energy signal as it is finite

$$\begin{aligned} E &= \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-1}^1 [5 \cos \pi t]^2 dt = 25 \int_{-1}^1 \cos^2 \pi t dt = 25 \int_{-1}^1 \frac{1 + \cos 2\pi t}{2} dt \\ &= \frac{25}{2} \left[\int_{-1}^1 1 dt + \int_{-1}^1 \cos 2\pi t dt \right] = \frac{25}{2} \left[t \right]_{-1}^1 + \frac{\sin 2\pi t}{2\pi} \Big|_{-1}^1 = \underline{\underline{25 \text{ joules}}} \end{aligned}$$

(d) $x(t) = 5 \cos \pi t + \sin 5\pi t$; $-\infty \leq t \leq \infty$

The given signal is periodic; hence it is a power signal.

Using the concept of ^{Square of} RMS value of the signal gives the average power

$$P_{av} = P = [\text{RMS value of } x(t)]^2$$

but $x(t) = 5 \cos \pi t + \sin 5\pi t = x_1(t) + x_2(t)$

$$P = [\text{RMS value of } x_1(t)]^2 + [\text{RMS value of } x_2(t)]^2$$

$$= \left[\frac{5}{\sqrt{2}} \right]^2 + \left[\frac{1}{\sqrt{2}} \right]^2$$

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

$$= \frac{25}{2} + \frac{1}{2} = 13 \text{ watts}$$

$$P_{av} = 13 \text{ watts}$$

(e) $x(t) = t^2 u(t)$ for all t

Let $x(t)$ be an energy signal

$$E = \lim_{T \rightarrow \infty} \int_{-T}^T |x(t)|^2 dt = \lim_{T \rightarrow \infty} \int_{-T}^T t^2 u(t) dt = \lim_{T \rightarrow \infty} \int_0^T t^2 dt = \lim_{T \rightarrow \infty} \left[\frac{t^3}{3} \right]_0^T$$

$$E = \lim_{T \rightarrow \infty} \left[\frac{T^3}{3} \right] = \infty.$$

Hence the given signal is not an energy signal as $E = \infty$

Let it be a power signal

$$P_{av} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^{T/2} t^2 dt = \lim_{T \rightarrow \infty} \frac{1}{T} \left[\frac{t^3}{3} \right]_0^{T/2}$$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} \left[\frac{T^3}{24} \right] = \infty.$$

Since $P = \infty$; it is not a power signal.

Hence the given signal neither an energy signal nor a power signal.

(f) $x(t) = \begin{cases} 5 \cos \pi t & ; -0.5 \leq t \leq 0.5 \\ 0 & ; \text{otherwise} \end{cases}$

Energy signal as its period is finite & not periodic

$$E = \int_{-0.5}^{0.5} (5 \cos \pi t)^2 dt = 25 \int_{-0.5}^{0.5} \cos^2 \pi t dt = 25 \int_{-0.5}^{0.5} \frac{1 + \cos 2\pi t}{2} dt$$

$$= \frac{25}{2} \left[t + \frac{\sin 2\pi t}{2} \right]_{-0.5}^{0.5} = \underline{12.5 \text{ joules}}$$

(9) (i) $x(n) = \begin{cases} \sin n\pi & ; -4 \leq n \leq 4 \\ 0 & ; \text{otherwise} \end{cases}$ $\begin{cases} \text{since} \\ \sin n\pi = 0 \end{cases}$

By inspection; we can conclude that the signal has all samples zero. Hence it is not a periodic signal. Its energy is zero. Hence it is neither energy nor power signal. i.e. zero signal.

(ii) $x(n) = \begin{cases} \cos n\pi & ; -4 \leq n \leq 4 \\ 0 & ; \text{otherwise} \end{cases}$

The signal is defined for a finite period; hence it is not periodic. The given signal is an energy signal.

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2 = \sum_{n=-4}^4 (\cos n\pi)^2 = \sum_{n=-4}^4 (-1)^{2n} = 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 = 9 < \infty$$

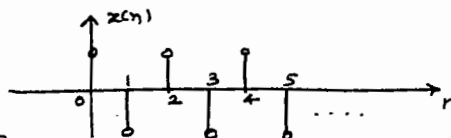
Energy $E = 9$ joules

(10) $x(n) = \begin{cases} \cos n\pi & ; n \geq 0 \\ 0 & ; \text{otherwise} \end{cases}$

The signal is defined for all $n \geq 0$; hence it is periodic; so it is a power signal.

$x(n) = \cos n\pi = (-1)^n$

$x(n)$ is periodic with $N=2$



$$P = \frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2 = \frac{1}{2} \sum_{n=0}^{1} (-1)^{2n} = \frac{1}{2} \sum_{n=0}^1 1 = \frac{1}{2} [1+1] = 1$$

$P = 1/2$ watts

For a periodic discrete signal with fundamental period N

$$P = \frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2$$

if the signal extends from $-\infty$ to $+\infty$

The average power $P = \lim_{N \rightarrow \infty} \frac{1}{2N} \sum_{n=-N}^N |x(n)|^2 \dots \dots (-\infty \leq n \leq \infty)$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N} \sum_{n=0}^{N-1} |x(n)|^2 \dots \dots (-\infty \text{ to } \infty)$$

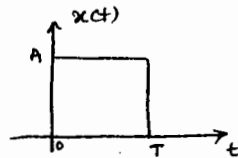
periodic signal.

In the above problem $\cos n\pi$ is extending from 0 to ∞

$\therefore$ The power will be $P = \lim_{N \rightarrow \infty} \frac{1}{2N} \sum_{n=0}^{N-1} |x(n)|^2$

Note
for any signal extending from $-a$ to a its average power is $P_{av} = \frac{1}{N} \sum_{n=-a}^a |x(n)|^2$ where $N = \text{fundamental period}$.

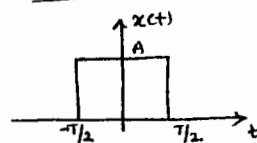
- ② Find the energy of the pulse given by $x(t) = \begin{cases} A & ; 0 \leq t \leq T \\ 0 & ; \text{elsewhere} \end{cases}$



$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_0^T |A|^2 dt = A^2 \int_0^T dt = A^2 T$$

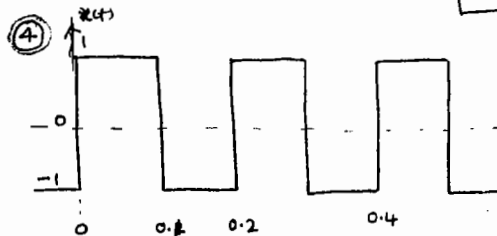
$$\therefore E = A^2 T \text{ joules}$$

- ③ Find the energy of the pulse shown below



$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-T/2}^{T/2} A^2 dt = A^2 [T/2 + T/2]$$

$$E = A^2 T \text{ joules}$$



For the periodic signal shown
compute the average power

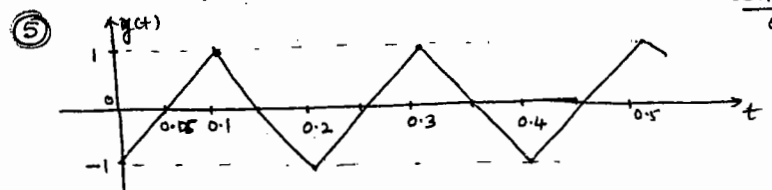
From fig $x(t) = \begin{cases} 1 & ; 0 \leq t \leq 0.1 \\ -1 & ; 0.1 \leq t \leq 0.2 \end{cases}$

from fig
 $T = 0.2$

$$P = \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt = \frac{1}{T} \int_0^T |x(t)|^2 dt$$

$$= \frac{1}{0.2} \left[\int_0^{0.1} (1)^2 dt + \int_{0.1}^{0.2} (-1)^2 dt \right] = \frac{1}{0.2} [0.1 + [0.2 - 0.1]]$$

$$P = 1 \text{ W}$$



what is the average power
of the waveform shown

For the period $0 \leq t \leq 0.1$; using the straight line equation

$$\frac{y - y_2}{y_1 - y_2} = \frac{x - x_2}{x_1 - x_2}$$

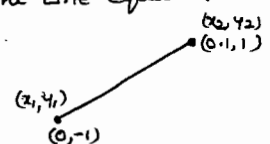
here $y = y(t)$
 $x = t$

$$\frac{y(t) - 1}{-1 - 1} = \frac{t - 0.1}{0 - 0.1}$$

$$-0.1 [y(t) - 1] = -2t + 0.2$$

$$y(t) - 1 = 20t - 2$$

$$y(t) = 20t - 1 \quad \dots 0 \leq t \leq 0.1$$



For the period $0.1 \leq t \leq 0.2$

$$\frac{y_1 - y_2}{x_1 - x_2} = \frac{x - x_2}{x_1 - x_2}$$

$$\frac{y(t) + 1}{1 + 2} = \frac{t - 0.2}{0.1 - 0.2} ; \quad y(t) + 1 = \frac{2t - 0.4}{-0.1} = -20t + 4$$

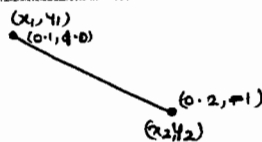
$$\therefore \boxed{y(t) = -20t + 3}$$

$$\therefore y(t) = \begin{cases} 20t - 1 ; & 0.0 \leq t \leq 0.1 \\ -20t + 3 ; & 0.1 \leq t \leq 0.2 \end{cases}$$

$$P = \frac{1}{T} \left[\int_0^T |y(t)|^2 dt \right] = \frac{1}{0.2} \left[\int_0^{0.1} (20t - 1)^2 dt + \int_{0.1}^{0.2} (-20t + 3)^2 dt \right]$$

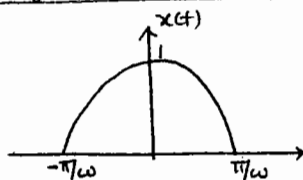
$$= \frac{1}{0.2} \left[\int_0^{0.1} (400t^2 - 40t + 1) dt + \int_{0.1}^{0.2} (400t^2 - 120t + 9) dt \right]$$

$$\boxed{P = 0.33 \text{ watts}}$$



⑥ A raised cosine pulse $x(t)$ is defined by the eqn; find the energy associated with the signal.

$$x(t) = \begin{cases} \frac{1}{2} [\cos(\omega t) + 1] ; & -\frac{\pi}{\omega} \leq t \leq \frac{\pi}{\omega} \\ 0 & ; \text{ otherwise} \end{cases}$$



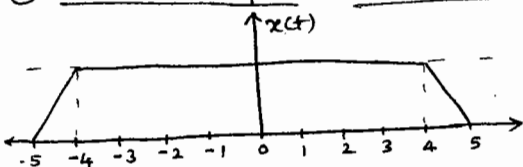
$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\pi/\omega}^{\pi/\omega} \left[\frac{1}{2} [\cos(\omega t) + 1] \right]^2 dt = \frac{1}{4} \int_{-\pi/\omega}^{\pi/\omega} (\cos^2 \omega t + 1 + 2 \cos \omega t) dt$$

$$= \frac{1}{4} \int_{-\pi/\omega}^{\pi/\omega} \left[1 + 2 \cos \omega t + \frac{1 + \cos 2\omega t}{2} \right] dt = \frac{1}{4} \left[t + \frac{2 \sin \omega t}{\omega} + \frac{1}{2} \left[t + \frac{\sin 2\omega t}{2\omega} \right] \right]_{-\pi/\omega}^{\pi/\omega}$$

$$E = \frac{1}{4} \left[\left(\frac{\pi}{\omega} + \frac{\pi}{\omega} \right) + \frac{1}{2} \left[\frac{\pi}{\omega} + \frac{\pi}{\omega} \right] \right] = \frac{1}{4} \left[\frac{2\pi}{\omega} + \frac{1}{2} \frac{2\pi}{\omega} \right] = \frac{3\pi}{4\omega}$$

$$\boxed{E = \frac{3\pi}{4\omega} \text{ joules}}$$

⑦ For the trapezoidal pulse $x(t)$ shown; calculate the total energy



$x(t)$ can be defined as

$$x(t) = \begin{cases} 5 - t ; & 4 \leq t \leq 5 \\ 1 ; & -4 \leq t \leq 4 \\ t + 5 ; & -5 \leq t \leq -4 \\ 0 ; & \text{ otherwise} \end{cases}$$

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-5}^5 |x(t)|^2 dt = \int_{-4}^4 1 dt + \int_{-5}^{-4} (t+5)^2 dt + \int_4^5 (5-t)^2 dt$$

$$E = (4+4) + \left[\frac{t^3}{3} + 10\frac{t^2}{2} + 25t \right]_{-5}^{-4} + \left[25t + \frac{t^3}{3} - 10\frac{t^2}{2} \right]_4^5$$

$$E = 8.667 \text{ joules}$$

II method $x(t)$ is symmetrical about y axis (even fⁿ).

$$E = 2 \int_0^5 |x(t)|^2 dt = 2 \left[\int_0^4 1 dt + \int_4^5 (5-t)^2 dt \right] = 8.667 \text{ Joules}$$

⑧ Determine the average power and energy of the following sequences.

(i) $x(n) = u(n)$

$$\text{Energy } E = \sum_{n=-\infty}^{\infty} |x(n)|^2 = \sum_{n=0}^{\infty} 1^2 = \infty$$



$$\text{Power } P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x(n)|^2 = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=0}^N 1$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1} [N+1] = \lim_{N \rightarrow \infty} \frac{1}{2} \left[\frac{1}{1+\frac{1}{2N}} \right] \left[1+\frac{1}{2N} \right] = \frac{1}{2}$$

$$P = \frac{1}{2} \text{ watts}$$

(ii) $x(n) = n u(n)$

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2 = \sum_{n=0}^{\infty} n^2 u(n) = \sum_{n=0}^{\infty} n^2 = \infty$$

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x(n)|^2 = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=0}^N n^2 = \infty$$

(iii) $x(n) = \left(\frac{1}{3}\right)^n u(n)$

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2 = \sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^{2n} = \sum_{n=0}^{\infty} \left(\frac{1}{9}\right)^n = \frac{1}{1-\frac{1}{9}} = \frac{9}{8} \text{ Joules.}$$

$$\text{Note } \sum_{k=0}^{\infty} a^k = \frac{1}{1-a}$$

(iv) $x(n) = e^{j(\frac{n\pi}{2} + \frac{\pi}{6})}$

$$x(n) = \cos\left(\frac{n\pi}{2} + \frac{\pi}{6}\right) + j \sin\left(\frac{n\pi}{2} + \frac{\pi}{6}\right) \text{ is of the form}$$

$$= \cos(2\pi f_1 + \theta) + j \sin(2\pi f_2 + \theta)$$

$$\therefore f_1 = f_2 = \frac{1}{4} \text{ or } N=4$$

Hence the signal is periodic.

Using the concept of square of Rms value of the signal is the average power we get

$$P_{av} = P = \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 = \underline{\underline{1 \text{ watt}}}$$

(V) $x(t) = e^{-at}$; $t > 0$

This signal is non periodic and of infinite duration. It can be an energy signal.

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_0^{\infty} e^{-2at} dt = \left[\frac{e^{-2at}}{-2a} \right]_0^{\infty} = \underline{\underline{\frac{1}{2a} \text{ joules}}}$$

(VI) $x(n) = 2e^{jan}$

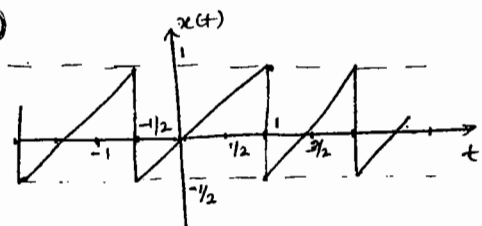
Signal is defined for all values of n . and is periodic

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x(n)|^2 = 4 \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N 1 = 4 \lim_{N \rightarrow \infty} \frac{1}{2N+1} (2N+1)$$

$P = 4 < \infty$ power is finite, hence the signal is a power signal.

$\therefore P_{av} = 4 \text{ watts}$

9



Determine the power of the following signal $x(t)$.

from fig. $T = 3/2$

$$x(t) = t ; -\frac{1}{2} \leq t \leq 1$$

$$\begin{aligned} P &= \frac{1}{T} \int_{-1/2}^1 |x(t)|^2 dt = \frac{1}{3/2} \int_{-1/2}^1 t^2 dt \\ &= \frac{2}{3} \left[\frac{t^3}{3} \right]_{-1/2}^1 = \frac{2}{3} \left[\frac{1}{3} + \frac{1}{24} \right] = \frac{2}{3} \left[\frac{9}{24} \right] = \underline{\underline{\frac{1}{4} \text{ watts}}} \end{aligned}$$

10 Determine average power of $x(t) = A \cos(\omega t + \phi)$

The given signal is periodic with fundamental period $T = \frac{2\pi}{\omega}$ or $\omega T = 2\pi$

$$P = \frac{1}{T} \int_{-T/2}^{T/2} x^2(t) dt = \frac{1}{T} \int_{-T/2}^{T/2} A^2 \cos^2(\omega t + \phi) dt = \frac{A^2}{T} \int_{-T/2}^{T/2} \frac{1 + \cos 2(\omega t + \phi)}{2} dt$$

$$P = \frac{A^2}{2T} \int_{-T/2}^{T/2} (1 + \cos 2\phi \cos 2\omega t - \sin 2\phi \sin 2\omega t) dt$$

$$= \frac{A^2}{2T} \int_{-T/2}^{T/2} (1 + \cos 2\phi \cos 2\omega t - \sin 2\phi \sin 2\omega t) dt \quad [\because \omega T = 2\pi]$$

$$= \frac{A^2}{2T} \left[t + \cos 2\phi \frac{\sin 2\omega t}{2\omega} + \sin 2\phi \frac{\cos 2\omega t}{2\omega} \right]_{-T/2}^{T/2} = \frac{A^2}{2T} \left[\frac{T}{2} + \frac{T}{2} \right]$$

$$\boxed{P = \frac{A^2}{2} \text{ watts.}}$$

- (11) The energy in a real valued signal $x(n)$ is defined as the sum of the squares of the sequence values

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

Suppose $x(n)$ has even part $x_e(n)$ equal to $x_e(n) = (\frac{1}{2})^{|n|}$

The energy in $x_e(n)$ is $E = 5$ Joules. Find the energy in the odd part $x_o(n)$ of the signal $x(n)$.

$$x(n) = x_e(n) + x_o(n)$$

$$E = \sum_{n=-\infty}^{\infty} x^2(n) = \sum_{n=-\infty}^{\infty} [x_e(n) + x_o(n)]^2 = \sum_{n=-\infty}^{\infty} [x_e^2(n) + x_o^2(n) + 2x_e(n)x_o(n)]$$

$$E = \sum_{n=-\infty}^{\infty} x_e^2(n) + \sum_{n=-\infty}^{\infty} x_o^2(n) = E_e(n) + E_o(n) = E_e + E_o$$

$$E_e = \sum_{n=-\infty}^{\infty} x_e^2(n) = \sum_{n=-\infty}^{\infty} \left(\frac{1}{2}\right)^{2|n|} = \sum_{n=-\infty}^{\infty} \left(\frac{1}{4}\right)^{|n|} = \sum_{n=-\infty}^{-1} \left(\frac{1}{4}\right)^{|n|} + \sum_{n=0}^{\infty} \left(\frac{1}{4}\right)^{|n|}$$

put $k = n+1$ in the first term
when $n = -1$, $k = 0$
when $n = \infty$, $k = \infty$

$$E_e = \sum_{n=1}^{\infty} \left(\frac{1}{4}\right)^n + \sum_{n=0}^{\infty} \left(\frac{1}{4}\right)^n$$

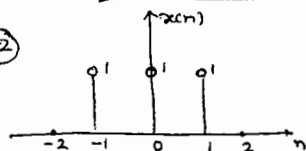
$$\therefore E_e = \sum_{k=0}^{\infty} \left(\frac{1}{4}\right)^{k+1} + \sum_{n=0}^{\infty} \left(\frac{1}{4}\right)^n = \frac{1}{4} \left[\frac{1}{1-\frac{1}{4}} \right] + \left[\frac{1}{1-\frac{1}{4}} \right] = \frac{1}{4} \left[\frac{4}{3} \right] + \frac{4}{3}$$

$$E_e = \frac{1}{3} + \frac{4}{3} = \frac{5}{3} \text{ Joules}$$

$$\therefore E = E_e + E_o \quad \therefore E_o = E - E_e = 5 - \frac{5}{3} = \frac{10}{3} \text{ Joules}$$

$$E_e = \frac{5}{3} \text{ J} ; E_o = \frac{10}{3} \text{ J}$$

(12)



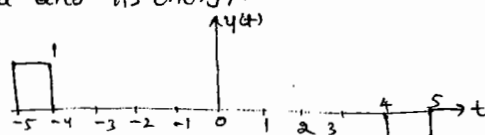
Find the energy or power associated with the given signal.

Sol: The signal is defined b/n $-1 < n < 1$ has a definite value. Hence it is an energy signal.

$$E = \sum_{n=-1}^1 |x(n)|^2 = 1^2 + 1^2 + 1^2 = 3 \text{ joules}$$

- (13) If the trapezoidal signal shown in problem (7) is passed thro' a differentiator; sketch the diff signal and its energy.

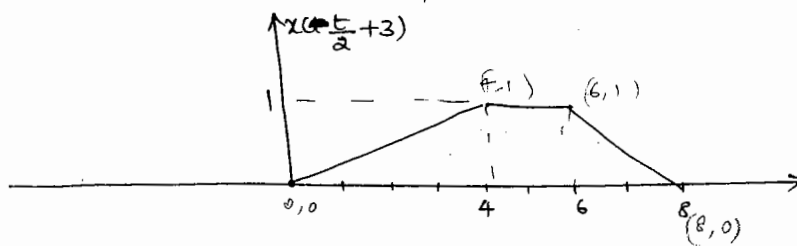
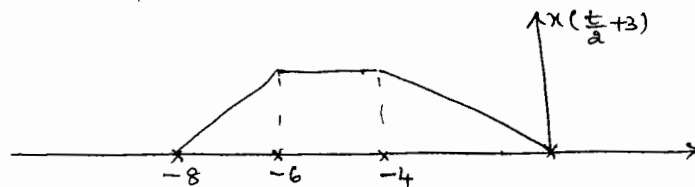
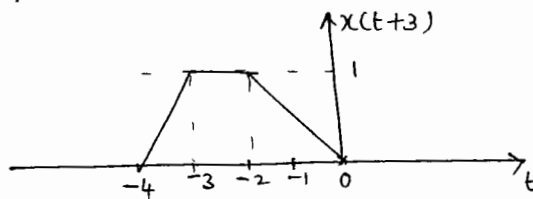
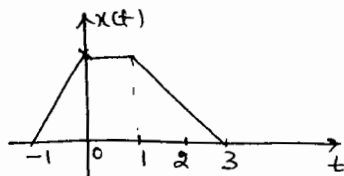
$$y(t) = \frac{d[x(t)]}{dt} = \begin{cases} -1 & ; 4 \leq t \leq 5 \\ 0 & ; -4 \leq t \leq 4 \\ 1 & ; -5 \leq t \leq -4 \\ 0 & ; \text{otherwise} \end{cases}$$



$$E = \int_{-5}^{-4} (-1)^2 dt + \int_{4}^5 (-1)^2 dt = \left[-t \right]_{-5}^{-4} + \left[-t \right]_{4}^5 = -4 + 5 + 5 - 4 = 2 \text{ Joules}$$

$$E = 2 \text{ Joules}$$

Determine the energy in the signal $x(-\frac{t}{2}+3)$ given the signal $x(t)$ as below.



$$y(t) = x\left(-\frac{t}{2}+3\right) = \begin{cases} t/4 & ; 0 \leq t \leq 4 \\ 1 & ; 4 \leq t \leq 6 \\ -\frac{t}{2}+4 & ; 6 \leq t \leq 8 \end{cases}$$

$$E = \int_0^4 |y(t)|^2 dt + \int_4^6 |y(t)|^2 + \int_6^8 |y(t)|^2 dt$$

$$= \int_0^4 \frac{t^2}{16} dt + \int_4^6 1 dt + \int_6^8 \left(\frac{t}{4} + 16 - 4t\right) dt$$

$$= \frac{1}{16} [16-0] + (6-4) + \left[\frac{1}{4} \left(\frac{8^3-6^3}{3} \right) + 16(8-6) - \frac{4}{2} (8^2-6^2) \right]$$

$$= 1 + 2 + [24.66 + 32 - 56]$$

$$= 3 + 0.66 = \underline{3.66 \text{ J only.}}$$

Let $x(t) = x_1(t) \cdot x_2(t)$

i) if $x_1(t)$ & $x_2(t)$ are both even

$$x(t) = x_1(t) \cdot x_2(t)$$

$$x(-t) = x_1(-t) \cdot x_2(-t)$$

$$= x_1(t) \cdot x_2(t)$$

$$\boxed{x(-t) = x(t)}$$

ii) if one of them is odd. Let $x_1(t)$ be odd

$$x(t) = x_1(t) \cdot x_2(t)$$

$$x(-t) = x_1(-t) \cdot x_2(-t)$$

$$x(-t) = -[x_1(t) \cdot x_2(t)]$$

$$\boxed{x(-t) = -x(t)}$$

→ Decomposition of a signal into its even and odd part.

even part $x_e(t) = \frac{x(t) + x(-t)}{2}$

$$x_e[n] = \frac{x[n] + x[-n]}{2}$$

odd part $x_o(t) = \frac{x(t) - x(-t)}{2}$

$$x_o[n] = \frac{x[n] - x[-n]}{2}$$

→ Energy and power signals

Total Energy of a signal

$$E[x(t)] = E = \int_{-\infty}^{\infty} [x(t)]^2 \cdot dt$$

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

Average Power of a signal

$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} [x(t)]^2 \cdot dt$$

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N} \sum_{n=-N}^N (x[n])^2$$

Average Power of a Periodic signal
with fundamental period = T

$$P = \frac{1}{T} \int_{-T/2}^{T/2} (x(t))^2 \cdot dt$$

Fundamental period = N

$$P = \frac{1}{N} \sum_{n=0}^{N-1} (x[n])^2$$

Square root of average power is RMS value of the signal
or. Square of the RMS value of the signal is average power of the periodic signal.

Basic signals:

Basic signals or elementary signals can be used as basic building blocks for the construction and modelling of more complicated signals. The important basic signals are as follows.

- (a) Exponential signals
- (b) Sinusoidal signals
- (c) Exponentially damped sinusoidal signals
- (d) Unit step function
- (e) Unit impulse $\delta(t)$
- (f) Unit ramp function.

(I) Exponential signals

A real exponential signal is defined as

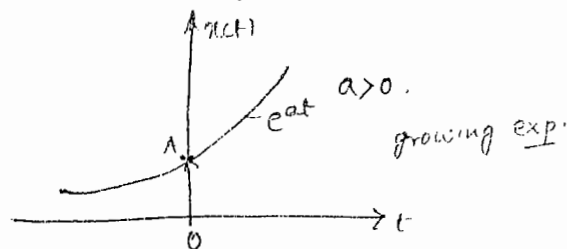
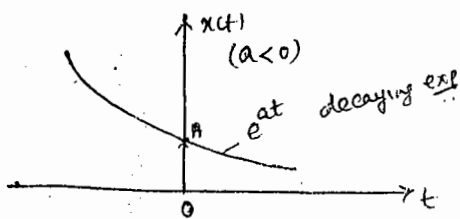
$$x(t) = A e^{at}$$

Where 'A' and 'a' are real constants.

'A' is the amplitude of the signal at $t=0$.

If $a < 0$; $x(t)$ is known as decaying exponential

If $a > 0$ $x(t)$ is known as growing exponential.



Ex:- $x(t) = 5e^{2t}$; $x(t) = 3e^{-2t}$.

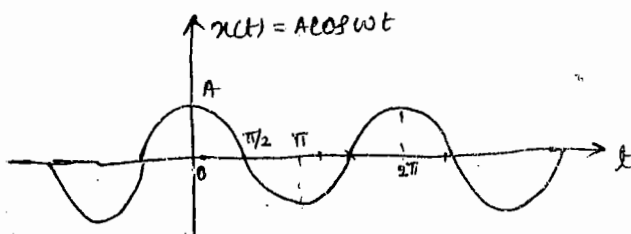
Ax - Voltage a/c the capacitor.

(II) Sinusoidal signals

A continuous time sinusoidal signal is of the form

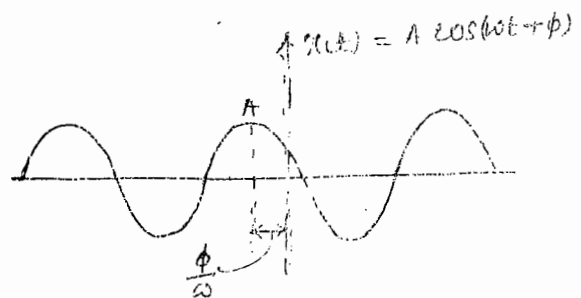
$$x(t) = A \cos(\omega_0 t + \phi)$$

Where 'A' is the amplitude at $t=0$; ' ω_0 ' is the fundamental angular frequency and ϕ is the phase shift constant.



$$T = \frac{2\pi}{\omega}$$

$$x(t) = x(t+T) = x(t + \frac{2\pi}{\omega})$$

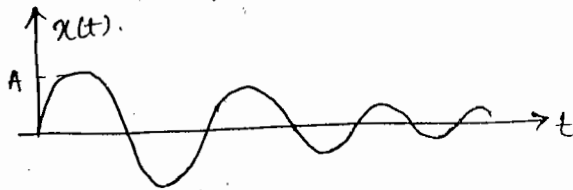


3. Exponentially damped sinusoidal signals

These signals are of the form

$$x(t) = A e^{-at} \sin \omega t.$$

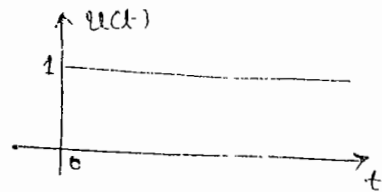
where $a > 0$; As t increases; the amplitude of the signal decreases exponentially and approaches to zero at $t = \infty$.



4. Unit step function

A CT unit step fn can be defined as

$$u(t) = \begin{cases} 1 & ; t \geq 0 \\ 0 & ; t < 0 \end{cases}$$



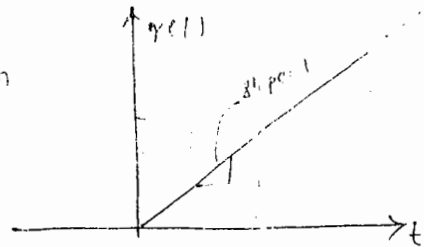
Ex: -1. Sudden application of brake

2. Turning the ignition key on; we are applying a step voltage to the starting motor.

5. Unit Ramp function

A CT unit ramp fn is of the form

$$r(t) = \begin{cases} t & ; t \geq 0 \\ 0 & ; t < 0 \end{cases}$$



Note: $r(t) = \int_0^t u(\tau) d\tau = \int_0^t 1 d\tau = \tau \Big|_0^t = t.$

Also $r(t) = t \cdot u(t)$

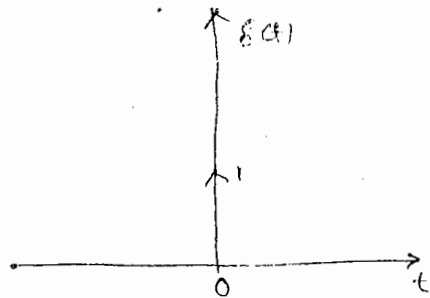
$$u(t) = \frac{d}{dt} [r(t)]$$

6. Unit impulse function

A CT unit impulse is defined as

$$\delta(t) = \begin{cases} 1 & ; t = 0 \\ 0 & ; t \neq 0 \end{cases}$$

Also $\int_{-\infty}^{\infty} \delta(t) dt = 1.$



The fn $\delta(t)$ is known Dirac delta function.

The area covered by the impulse fn is unity

(2) prove that $\int_{-\infty}^{\infty} x(t) \delta(t-t_0) dt = x(t_0)$

$$x(t) \delta(t-t_0) = x(t_0) \cdot \delta(t-t_0)$$

$$\begin{aligned} \int_{-\infty}^{\infty} x(t) \delta(t-t_0) dt &= \int_{-\infty}^{\infty} x(t_0) \delta(t-t_0) dt \\ &= x(t_0) \int_{-\infty}^{\infty} \delta(t-t_0) dt \end{aligned}$$

put $t-t_0 = \tau$. then $dt = d\tau$

$$\begin{aligned} \int_{-\infty}^{\infty} x(t) \delta(t-t_0) dt &= x(t_0) \int_{-\infty}^{\infty} \delta(\tau) d\tau \\ &= \underline{\underline{x(t_0)}} \end{aligned}$$

(3) prove that $\delta(at) = \frac{1}{|a|} \delta(t)$ & $\delta(t) = \delta(-t)$.

Let $I = \int_{-\infty}^{\infty} x(t) \delta(at) dt$ (let $a > 0$)

put $at = \tau$
 $a dt = d\tau$
 $dt = \frac{d\tau}{a}$ } The limits of integration remains same

$$\therefore I = \int_{-\infty}^{\infty} x\left(\frac{\tau}{a}\right) \delta(\tau) \frac{d\tau}{a}$$

$$I = \frac{x(\tau/a)}{a} \int_{-\infty}^{\infty} \delta(\tau) d\tau = \frac{x(\tau/a)}{a} \Big|_{\tau=0} = \frac{x(0)}{a}$$

similarly for $a < 0$; also

$$I = \int_{-\infty}^{\infty} x(t) \delta(at) = \frac{1}{a} x(0) \quad (a \neq 0)$$

hence in general $\int_{-\infty}^{\infty} x(t) \delta(at) = \frac{1}{|a|} x(0) \quad \text{--- (A)}$

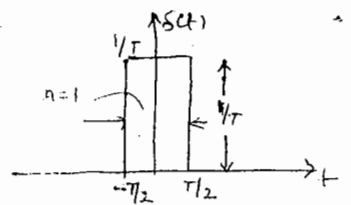
but $x(0) = \int_{-\infty}^{\infty} x(t) \cdot \delta(t) dt = \frac{1}{|a|} \int_{-\infty}^{\infty} x(t) \delta(at) dt \quad \text{--- (B)}$

~~$\int_{-\infty}^{\infty} \delta(t) dt = \int_{-\infty}^{\infty} \delta(at) dt$~~

using (B) in (A)

$$\delta(at) \int_{-\infty}^{\infty} x(t) dt = \frac{1}{|a|} \delta(t) \int_{-\infty}^{\infty} x(t) dt$$

We can visualize impulse as a rectangular pulse of unit area with the width of the pulse very small as $\tau \rightarrow 0$ and consequently its height very large. Hence unit impulse $\delta(t)$ will have infinite energy.



Impulse $\delta(t)$ exhibits even $\delta(t)$ symmetry

$$\delta(t) = \delta(-t).$$

$$\text{ii)ly } \delta(n) = \delta(-n).$$

Properties of Impulse $\delta(t)$

1. $\delta(t) = \delta(-t)$ --- Even $\delta(t)$ or Symmetry

2. If $x(t)$ is continuous at $t=0$; then

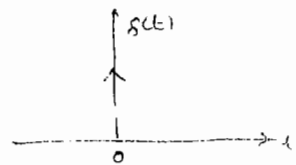
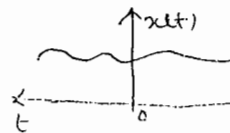
$$x(t) \cdot \delta(t) = x(0) \cdot \delta(t).$$

iii)ly If $x(t)$ is continuous at $t=t_0$; then

$$x(t) \delta(t-t_0) = x(t_0) \cdot \delta(t-t_0).$$

$$(3) \int_a^b x(t) \delta(t) dt = \begin{cases} x(0) & ; a < 0 < b \\ 0 & ; \text{otherwise} \end{cases}$$

$$(4) \int_{-\infty}^{\infty} x(t) \delta(t) dt = x(0)$$



$$5. \int_{-\infty}^{\infty} x(t) \delta(t-t_0) dt = x(t_0) \text{ --- Shifting property.}$$

$$6. \delta(at) = \frac{1}{|a|} \delta(t) \text{ --- Time scaling property.}$$

$$7. \delta(at - t_1) = \frac{1}{|a|} \delta\left(t - \frac{t_1}{a}\right)$$

$$8. \delta(t) = \frac{d[u(t)]}{dt}$$

$$9. \delta(t-t_0) = \frac{d[u(t-t_0)]}{dt}$$

1. prove that $\int_{-\infty}^{\infty} \delta(t) x(t) dt = x(0)$

We have $x(t) \delta(t) = x(0) \delta(t)$

$$\int_{-\infty}^{\infty} x(t) \delta(t) dt = \int_{-\infty}^{\infty} x(0) \delta(t) dt = x(0) \int_{-\infty}^{\infty} \delta(t) dt = x(0)$$

→ by defn of impulse $\delta(t)$

$$\therefore \boxed{\delta(at) = \frac{1}{|a|} \delta(t)} \quad \dots \text{time scaling property}$$

put $a = -1$ in the above equation

$$\boxed{\delta(-t) = \delta(t)}$$

Show that $\int_{-\infty}^{\infty} x(t) \delta(at-b) dt = \frac{1}{|a|} x\left(\frac{b}{a}\right)$

We have $\delta(at-b) = \frac{1}{|a|} \delta\left(t - \frac{b}{a}\right)$

$$\int_{-\infty}^{\infty} x(t) \cdot \delta(at-b) dt = \frac{1}{|a|} \int_{-\infty}^{\infty} x(t) \delta\left(t - \frac{b}{a}\right) dt$$

$$\boxed{\int_{-\infty}^{\infty} x(t) \delta(at-b) dt = \frac{1}{|a|} x\left(\frac{b}{a}\right)}$$

Important properties

$$\textcircled{1} \left. \begin{aligned} \int_{-\infty}^{\infty} x(t) \delta(t-t_0) dt &= x(t_0) \\ \sum_{n=-\infty}^{\infty} x(n) \delta(n-k) &= x(k) \end{aligned} \right\} \text{Time shifting property.}$$

$$\textcircled{2} \quad \begin{aligned} x(t) \cdot \delta(t) &= x(0) \delta(t) ; \quad x(t) \delta(t-t_0) = x(t_0) \delta(t-t_0) \\ x(n) \cdot \delta(n) &= x(0) \delta(n) ; \quad x(n) \delta(n-k) = x(k) \delta(n-k) \end{aligned}$$

Multiplication by an impulse.

$$\textcircled{3} \quad \delta(at) = \frac{1}{|a|} \delta(t).$$

(4) Replication property (Convolution with δ f.n)

$$\sum_{n=-\infty}^{\infty} x(n) \delta(n-k) = x(n) * \delta(n) = x(n)$$

$$\int_{-\infty}^{\infty} x(z) \delta(t-z) dz = x(t) * \delta(t) = x(t).$$

Note: Convolution of any function with impulse function leaves that function unchanged.

1. Evaluate the following

(a) $\int_{-1}^1 \left(\frac{t^2}{2} + 5\right) \delta(t) dt$ (b) $\int_{-\infty}^{\infty} (\sin \pi t + t^2 \cos \pi t) \delta(t-1) dt$

(c) $\int_{-\infty}^{\infty} e^{-t} \delta(2t-1) dt$ (d) $\int_{-3}^{-1} e^{-t} \delta(t) dt$

(a) $\int_{-1}^1 \left(\frac{t^2}{2} + 5\right) \delta(t) dt$

The integral is of the form $\int_{-1}^1 x(t) \delta(t) dt = \begin{cases} x(0); & a < 0 < b \\ 0; & \text{otherwise} \end{cases}$

here $a = -1$ $b = 1$ hence valid.

$$\therefore \int_{-1}^1 \left[\frac{t^2}{2} + 5\right] \delta(t) dt = \left. \frac{t^2}{2} + 5 \right|_{t=0} = \underline{\underline{5}}$$

~~$\int_{-1}^1 \left(\frac{t^2}{2} + 5\right) \delta(t) dt = \int_{-1}^1 \left[\frac{t^2}{2} \delta(t) + 5 \delta(t)\right] dt$~~

$$= \left. \frac{t^2}{2} \right|_{t=0} + 5 \times 1$$

(b) $\int_{-\infty}^{\infty} (\sin \pi t + t^2 \cos \pi t) \delta(t-1) dt$

The given integral is of the form

$$\int_{-\infty}^{\infty} x(t) \delta(t-t_0) dt = x(t_0) \quad \text{here } t_0 = 1$$

$$\therefore \int_{-\infty}^{\infty} (\sin \pi t + t^2 \cos \pi t) \delta(t-1) dt = (\sin \pi + 1^2 \cos \pi) \Big|_{t=1}$$

$$= (-\sin \pi + 1^2 \cos \pi) = -1$$

(c) $\int_{-\infty}^{\infty} e^{-t} \delta(2t-1) dt$

The integral is of the form $\int_{-\infty}^{\infty} x(t) \delta(at-b) dt = \frac{1}{|a|} x\left(\frac{b}{a}\right)$

Here $a = 2$, $b = 1$

$$\int_{-\infty}^{\infty} e^{-t} \delta(2t-1) dt = \frac{1}{2} e^{-1/2} = \frac{1}{2\sqrt{e}}$$

(d) $\int_{-3}^{-1} e^{-t} \delta(t) dt$

here $a = -3$, $b = -1$

$a < 0 < b$ is not satisfied

$\therefore$ Hence the integral evaluation is zero

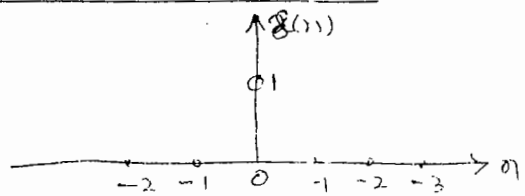
i.e. $\int_{-3}^{-1} e^{-t} \delta(t) dt = 0$

Basic discrete signals.

1. Unit sample function or Impulse function or Delta fn

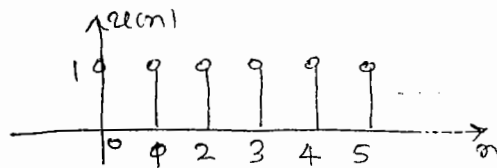
Unit impulse fn is defined as

$$\delta(n) = \begin{cases} 1 & n=0 \\ 0 & n \neq 0 \end{cases}$$



(2) Unit step function u(n)

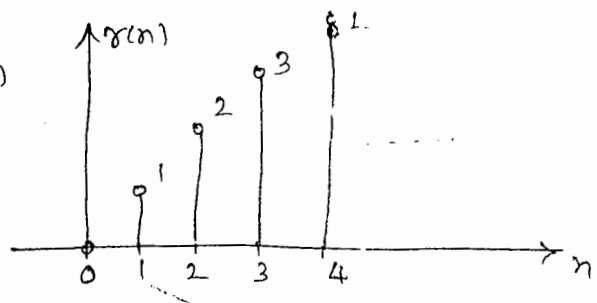
$$u(n) = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$



* When a sequence $x(n)$ is multiplied by $u(n)$; then it will become causal sequence *

(3) unit Ramp function r(n)

$$r(n) = \begin{cases} n & n \geq 0 \\ 0 & n < 0 \end{cases}$$



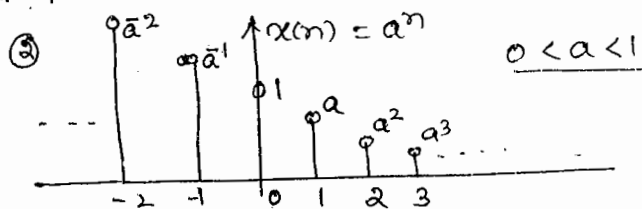
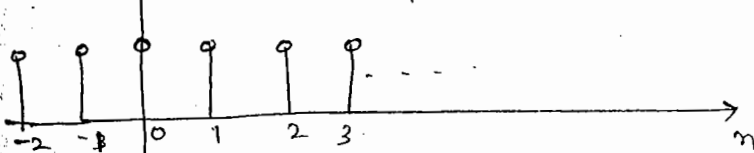
(4) Exponential sequence

The exponential sequence is of the form

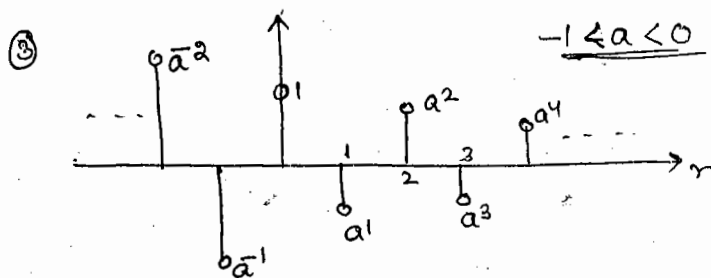
$$x(n] = a^n \text{ for all } n$$

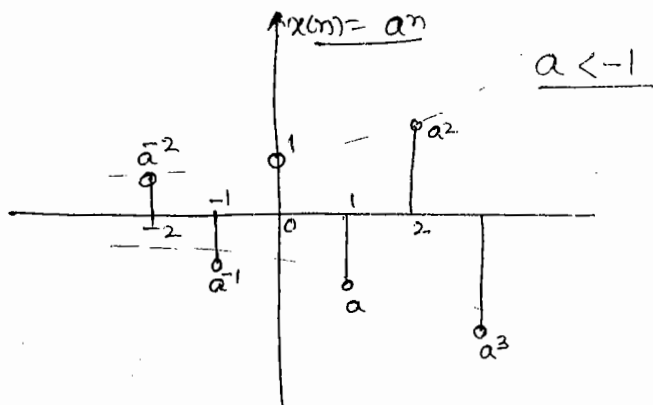
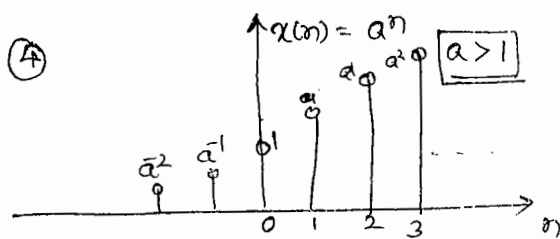
The type of sequence depends on the value of constant 'a'.

(1) $x(n] = a^n$ for $a=1$



Here $a=0.5$





Sinusoidal sequence

A discrete sinusoidal sequence is of the form

$$x[n) = A \cos[2\pi n + \phi]$$

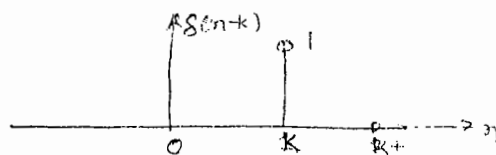
$\Omega_0 = \frac{2\pi}{N}$ 'N' is the period of the sequence

alternatively $\Omega_0 = \frac{2\pi m}{N}$ where 'm' is an integer.

Shifted functions

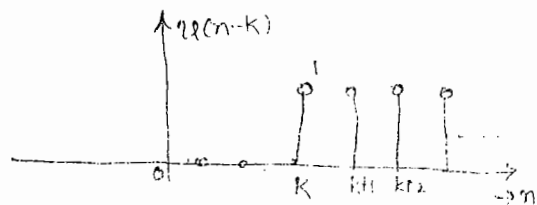
① Shifted unit sample function or shifted impulse f'n

$$\delta[n-k) = \begin{cases} 1 & ; n=k \\ 0 & ; n \neq k \end{cases}$$



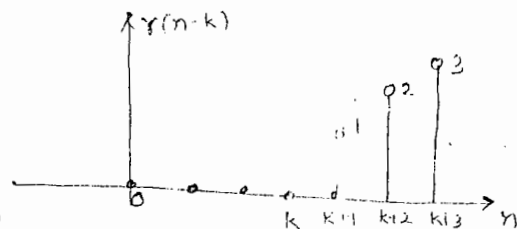
② Shifted unit step function

$$u[n-k) = \begin{cases} 1 & ; n \geq k \\ 0 & ; n < k \end{cases}$$



③ Shifted unit Ramp function

$$r[n-k) = \begin{cases} n-k & ; n \geq k \\ 0 & ; n < k \end{cases}$$



Relation ship b/n $\delta[n)$, $u[n)$ & $r[n)$

$$u[n) = \delta[n) + \delta[n-1) + \delta[n-2) + \dots$$

$$u[n) = \sum_{k=0}^{\infty} \delta[n-k)$$

$$r[n) = 0 \cdot \delta[n) + 1 \cdot \delta[n-1) + 2 \cdot \delta[n-2) + \dots$$

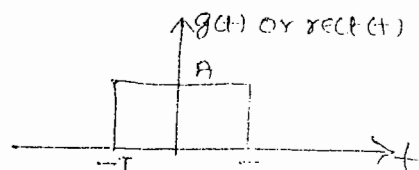
$$r[n) = \sum_{k=0}^{\infty} k \delta[n-k)$$

Also $\boxed{\delta(n) = u(n) - u(n-1)}$

Special signals

Rectangular pulse function or gate function $g(t)$ or $\text{rect}(t)$

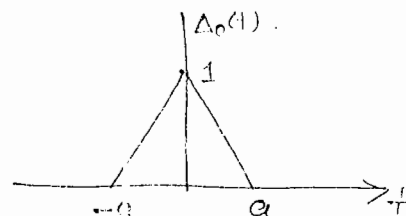
$$\text{Rect}(t) = \begin{cases} A & ; -T \leq t \leq T \\ 0 & ; \text{elsewhere} \end{cases}$$



Triangular function

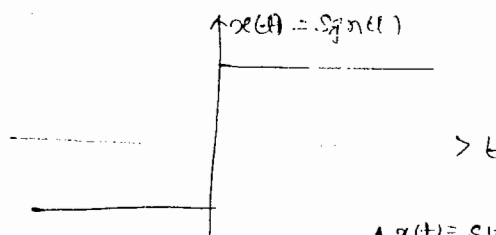
The unit Δ^r function is defined as

$$\Delta_a(t) = \begin{cases} 1 - \frac{|t|}{a} & ; |t| \leq a \\ 0 & ; |t| > a \end{cases}$$

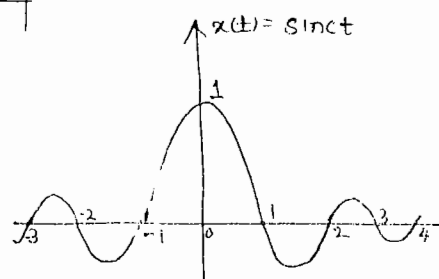


Signum function $\text{sgn}(t)$

$$\text{sgn}(t) = \begin{cases} 1 & ; t > 0 \\ 0 & ; t = 0 \\ -1 & ; t < 0 \end{cases}$$



$$\boxed{\text{sgn}(t) = -1 + 2u(t)}$$



Sinc function

$$\text{sinc}(t) = \frac{\sin \pi t}{\pi t} \quad ; \quad -\infty < t < \infty$$

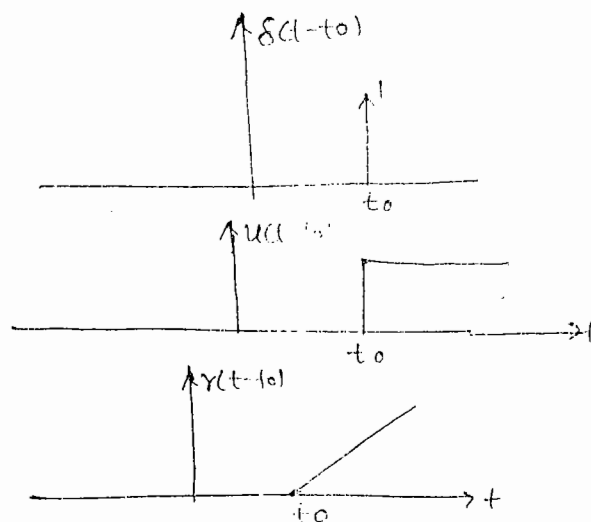
The sinc function is an oscillating function and decays with increasing $|t|$. Its values are zero at $\pm 1, \pm 2, \pm 3, \dots$. It is an even function of t .

Shifted CT signals functions

$$\delta(t-t_0) = \begin{cases} 1 & ; t = t_0 \\ 0 & ; t \neq t_0 \end{cases}$$

$$u(t-t_0) = \begin{cases} 1 & ; t \geq t_0 \\ 0 & ; t < t_0 \end{cases}$$

$$r(t-t_0) = \begin{cases} t-t_0 & ; t \geq t_0 \\ 0 & ; t < t_0 \end{cases}$$



Basic operations on signals.

(a) operations performed on dependent variable (Amplitude or value).

(i) Amplitude scaling:-

Let $x(t)$ be a continuous time signal; then the signal $y(t) = c x(t)$ is known as amplitude scaled version of $x(t)$.

Where 'c' is known as scaling factor.

IIIrd for a discrete time signal $y(n) = c x(n)$

EX:- Electronic amplifier; Voltage a/c the resistor $V(t) = R i(t)$

(ii) Addition

Let $x_1(t)$ and $x_2(t)$ are continuous time signals; then the signal $y(t) = x_1(t) + x_2(t)$ is known as addition of $x_1(t)$ & $x_2(t)$

EX:- Audio mixer which combines voice and music.

(iii) Multiplication

The product $y(t) = x_1(t) \cdot x_2(t)$ gives the multiplication of signals.

EX:- Modulation [AM signal in which $x_1(t)$ is an audio signal and $x_2(t)$ is the sinusoidal carrier].

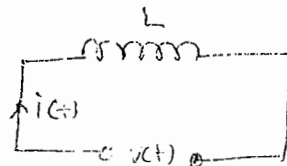
(iv) Differentiation

For a continuous time signal $x(t)$; the derivative of $x(t)$ is given by

$$y(t) = \frac{d}{dt} x(t).$$

EX:- Inductor performs the differentiation

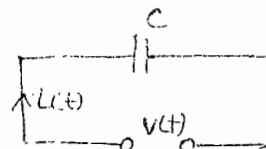
$$V(t) = L \cdot \frac{d}{dt} i(t)$$



(v) Integration for a continuous time signal $x(t)$

The integral of $x(t)$ is given by

$$y(t) = \int_{-\infty}^t x(\tau) d\tau.$$



Where τ is an independent variable.

EX: The capacitor performs the integration $V(t) = \frac{1}{C} \int_{-\infty}^t i(\tau) d\tau$

② operations performed on independent variable

(a) Time scaling:-

Let $x(t)$ be a continuous time signal; then the ^{time scaled} signal $y(t)$ is obtained by scaling the independent variable by a factor 'a' is given by

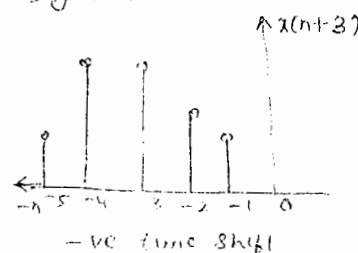
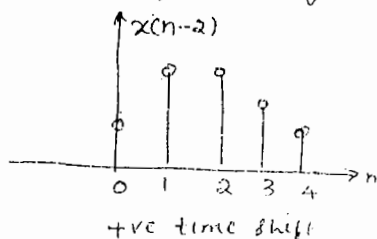
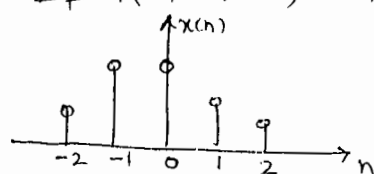
$$y(t) = x(at)$$

Similarly for DT signals

$$y(n) = x(n \pm k) \text{ where } k \text{ is the time shift operator.}$$

If k is +ve shift; shift to the right (advancing the signal)

If k is -ve; shift to the left (delay the signal).



precedence Rule for time shifting and time scaling

Let $y(t)$ be a CT signal obtained by the combination of time shifting and time scaling operations on a CT signal $x(t)$.

$$y(t) = x(at - b) \quad \text{--- (1)}$$

$$\text{Eqn (1) must satisfy } y(0) = x(-b) \quad \text{--- (2)}$$

$$y\left(\frac{b}{a}\right) = x(0) \quad \text{--- (3)}$$

once we obtain $y(t)$ from $x(t)$ by performing time shifting & time scaling; it must satisfy the eqn (2) & (3).

Hence Time shifting operation is performed first on $x(t)$ which yields $x(t - b) = v(t)$

next time scaling operation has to be performed on this time shifted signal $v(t)$ to get $y(t)$.

$$y(t) = v(at) = x(at - b).$$

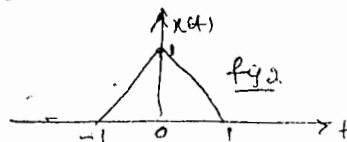
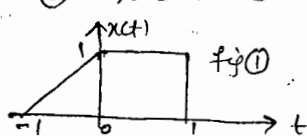
If folding is also involved; fold the signal $y(t)$ to get its reflected version.

Hence follow the Rule

① Time shift $\longrightarrow$ ② Time scale $\longrightarrow$ ③ folding

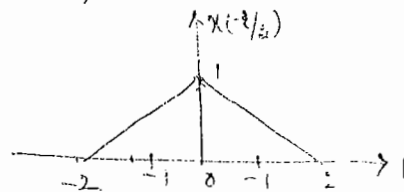
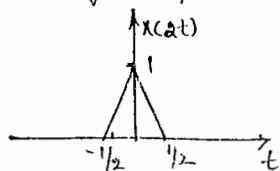
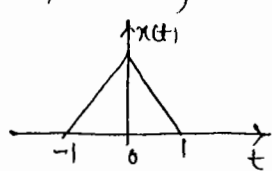
① For the signal $x(t)$ shown below. sketch the following
(a) $x(-t)$ (b) $x(t+3)$ (c) $x(2t-2)$ (d) $x\left(\frac{t}{2}-1\right)$ (e) $x(2-t)$

② for the CT signal shown in fig 2 sketch the following
(a) $x(3t)$ (b) $x(3t+2)$ (c) $x(-2t-1)$ (d) $x(2(t+2))$ (e) $x(2(t-4))$
(f) $x(3t) + x(3t+2)$.



If $a > 1$; then the signal $y(t)$ is compressed version of $x(t)$.

If $a < 1$; then the signal $y(t)$ is an expanded version of $x(t)$.



If $x(t)$ is considered as a normal recording speed;

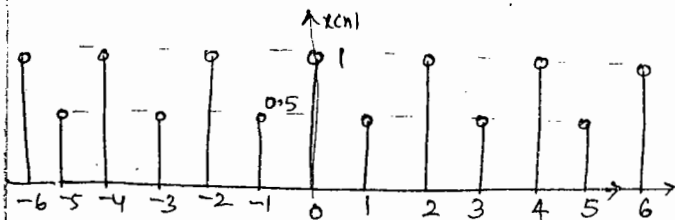
then $y(t) = x(2t)$ is the recording at double the speed (high speed dubbing).

& $y(t) = x(t/2)$ is the recording done at half the normal speed.

In the case of discrete time signals

$y(n) = x(kn)$ where 'k' is the time scale factor.

Consider the signal $x(n)$ shown below.



Let $k = 2$; $y(n) = x(2n)$

n	-1	1	2	-2	3	-3	0
y(n)	y(-1)	y(1)	y(2)	y(-2)	y(3)	y(-3)	y(0)
Value	$x(-2)$	$x(2)$	$x(4)$	$x(-4)$	$x(6)$	$x(-6)$	$x(0)$
	1	1	1	1	1	1	0

$y(n) = x(2n)$



Let $k = +0.5$; $y(n) = x(\frac{n}{2})$

0	-1	+1	2	-2	3	-3	'n'
y(n)	y(-1)	y(1)	y(2)	y(-2)	y(3)	y(-3)	y(n)
x(n)	$x(-1/2)$	$x(1/2)$	$x(1)$	$x(-1)$	$x(3/2)$	$x(-3/2)$	$x(n/2)$
Value	0	0	1	1	0	0	Value

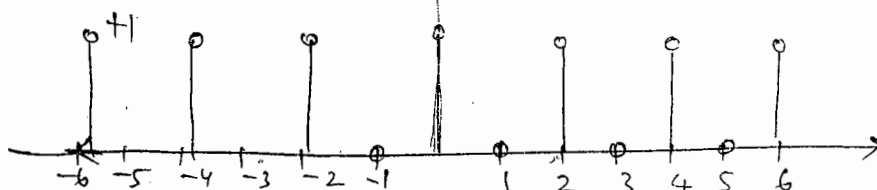
$$y(4) = x(2)$$

$$y(-4) = x(-2)$$

$$y(-6) = x(-3)$$

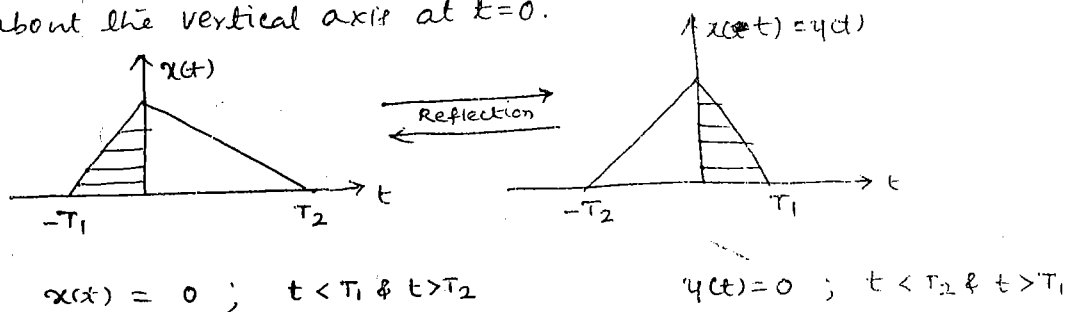
$$y(6) = x(3)$$

$y(n) = x(n/2)$



(II) Reflection or Time reversal or folding

The o/p signal obtained by replacing the time 't' with $-t$ represents the reflected or time reversal signal of $x(t)$. It is the signal obtained by folding the signal about the vertical axis at $t=0$.



*** Note ***

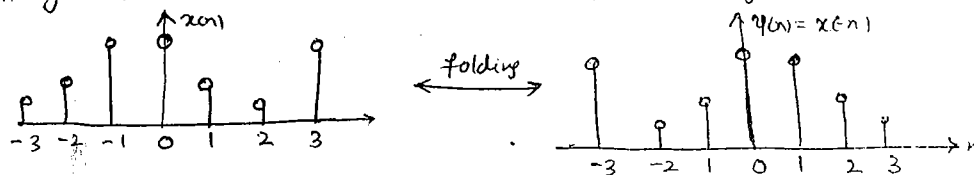
1. Even signals for which $x(t) = x(-t)$; the reflected and the original signals are same
2. In case of odd signals; the reflected output will be -ve of the i/p since $x(-t) = -x(t)$.

Ex:- If $x(t)$ represents an audio tape recording; $x(-t)$ is the same tape recorded played backwards.

In the case of discrete time signal.

$$x(n) = x(-n) = y(n)$$

The time reversal of D.T signal $x(n)$ is obtained by folding the sequence at $n=0$. i.e the o/p signal is obtained by replacing n with $-n$. will give the time reversed or folded signal.



for even signals $x(n) = x(-n)$.

for odd signals $x(-n) = -x(n)$.

3. Time Shifting

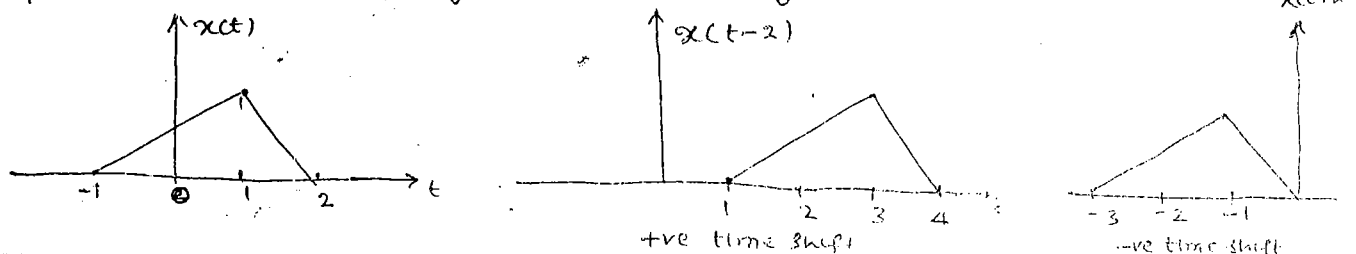
Let $x(t)$ be a cont. time signal; then the signal

$y(t) = x(t \pm t_0)$ is known as time shifted version of $x(t)$.

where ' t_0 ' is the time shift.

If t_0 is +ve; shifting delays the signal.

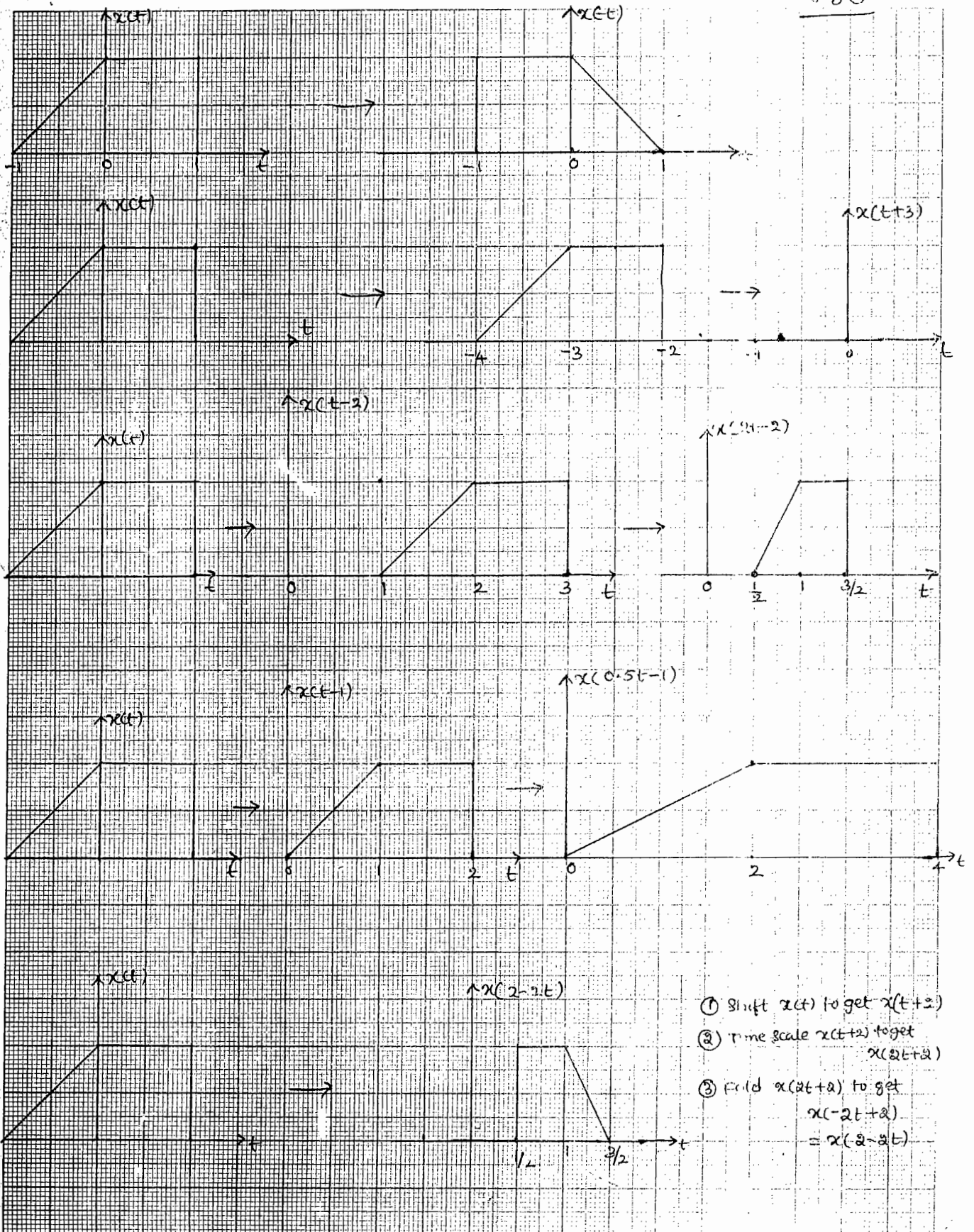
If t_0 is -ve; shifting advances the signal.



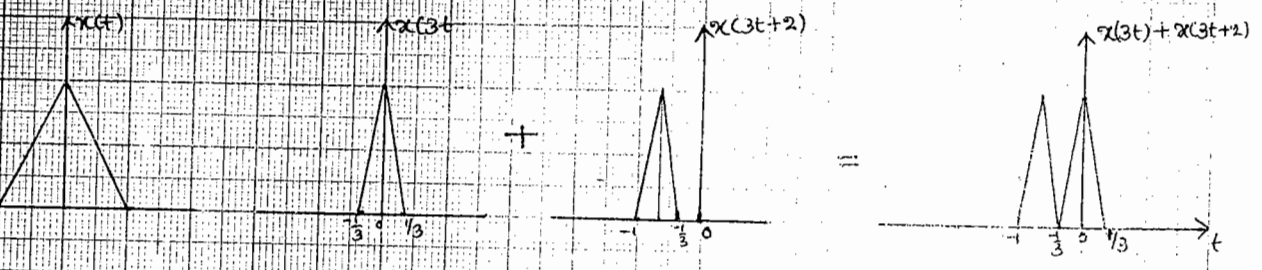
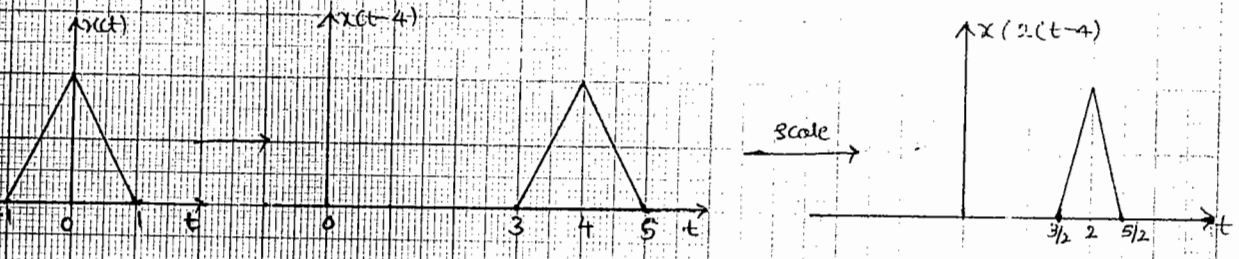
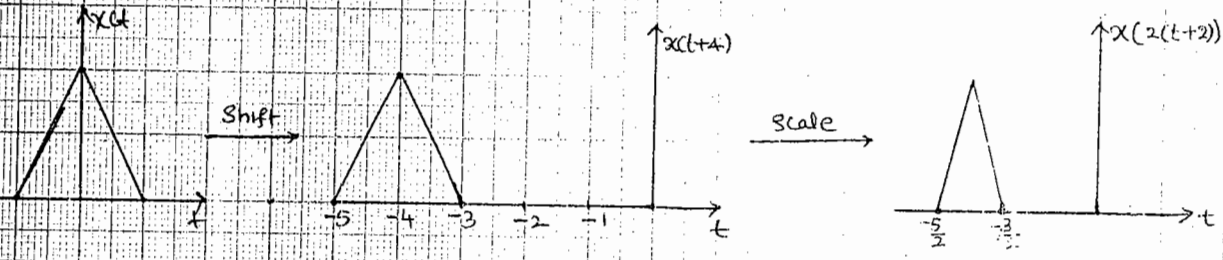
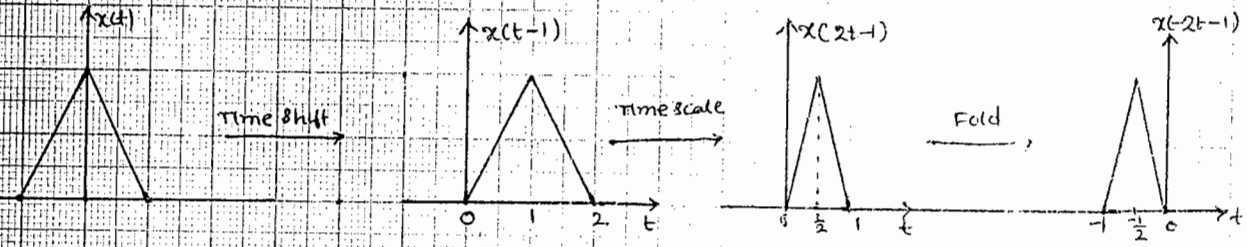
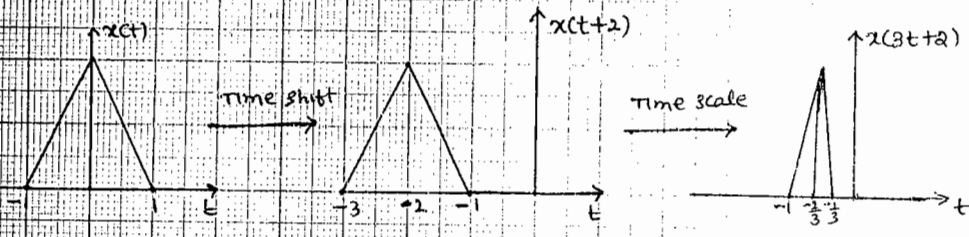
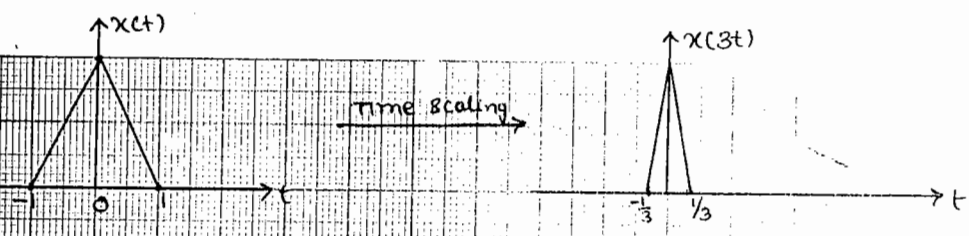
Teacher's Signature : _____

Date : _____

fig ①

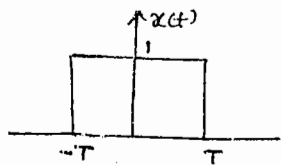


(182)

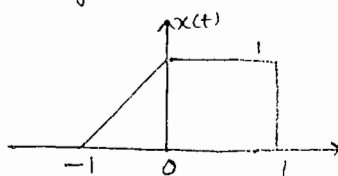


Express $x(t)$ using basic signals.

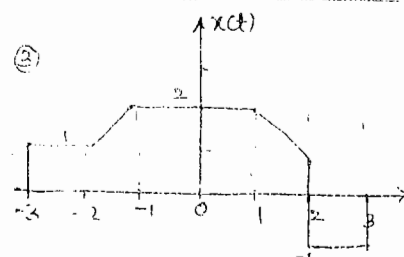
①



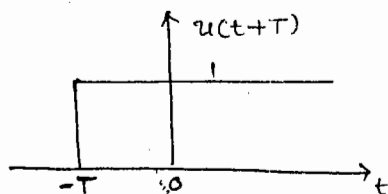
②



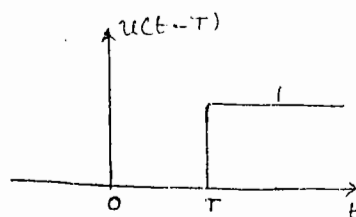
③



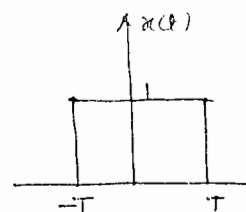
①



minus

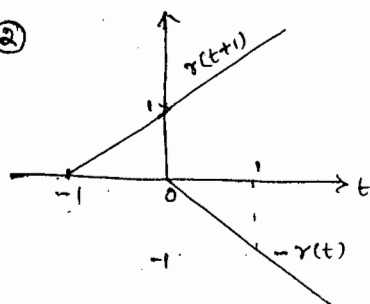


=

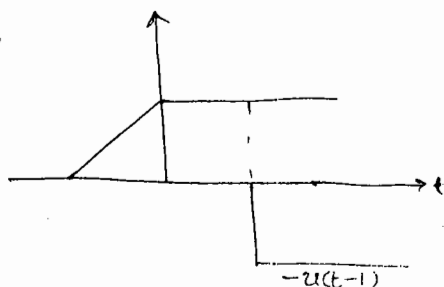


$$x(t) = u(t+T) - u(t-T)$$

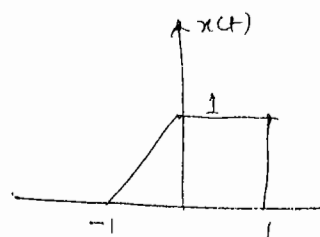
②



$\Rightarrow$

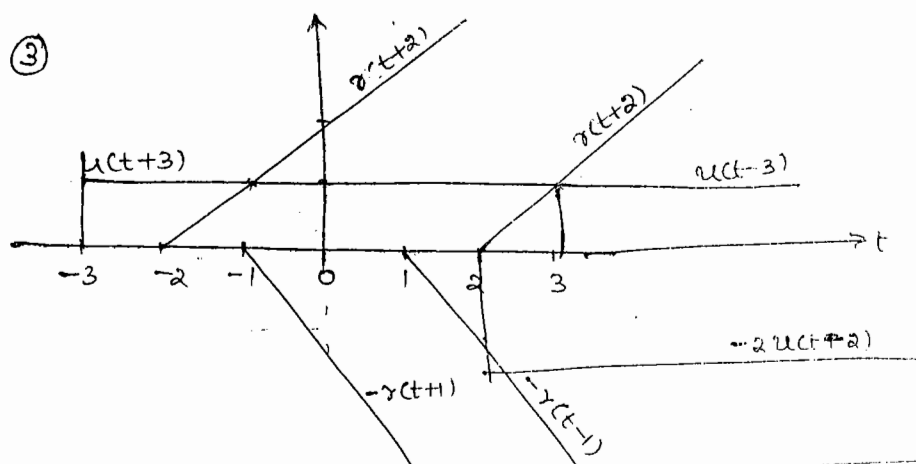


$\Rightarrow$



$$x(t) = r(t+1) - r(t) - u(t-1)$$

③

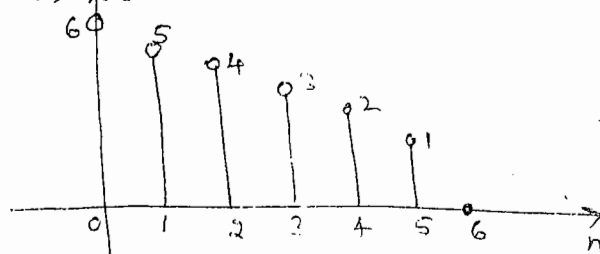
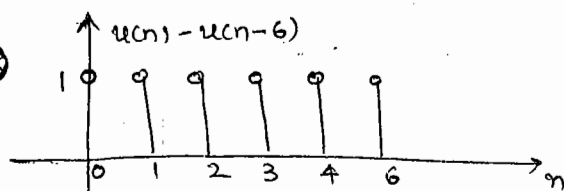


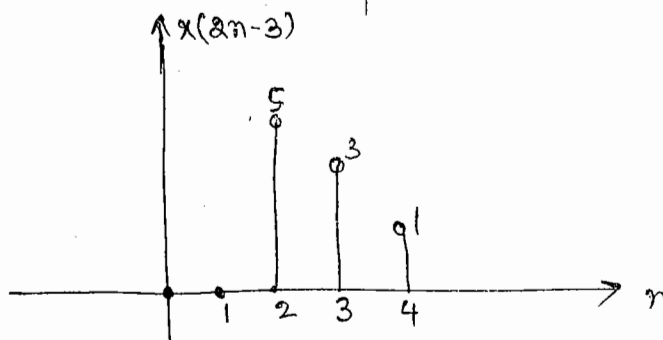
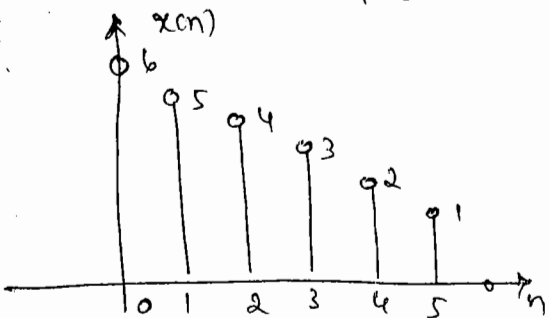
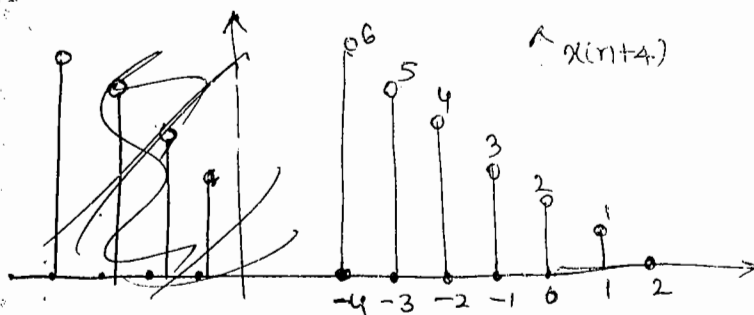
$$x(t) = u(t+3) + r(t+2) - r(t+1) - r(t-1) + r(t-2) - 2u(t-2) + u(t-3)$$

④ Given the sequence $x(n) = [6-n][u(n) - u(n-6)]$, make a sketch of

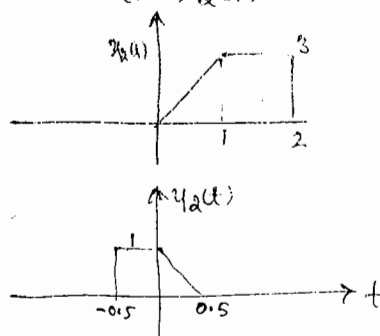
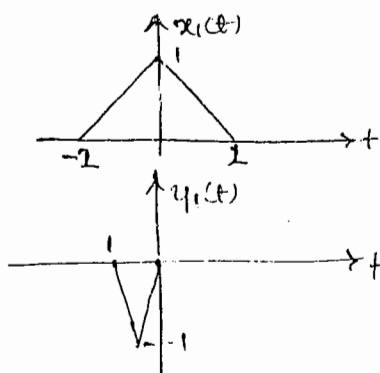
(i) $y_1(n) = x(4-n)$ (ii) $y_2(n) = x(2n-3)$ $x(n) = (6-n)[u(n) - u(n-6)]$

⑤

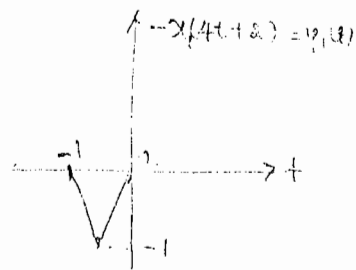
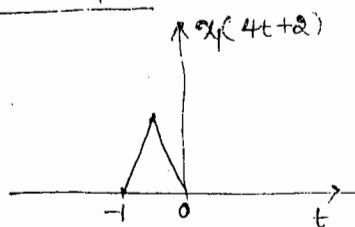
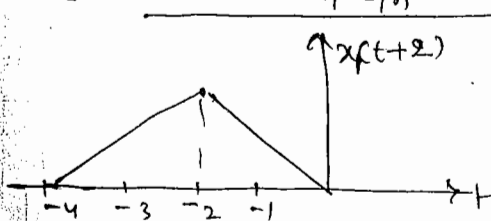




What is the relationship b/n (a) $x_1(t)$ and $y_1(t)$
(b) $x_2(t)$ and $y_2(t)$

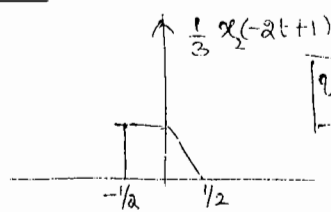
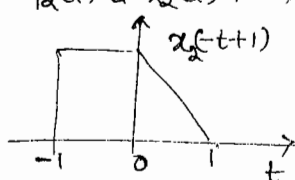
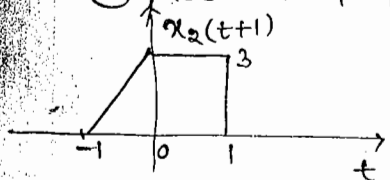


(a) Relationship b/n $x_1(t)$ and $y_1(t)$



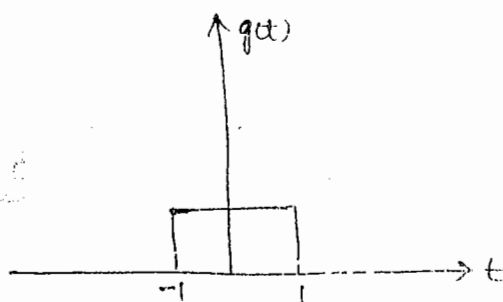
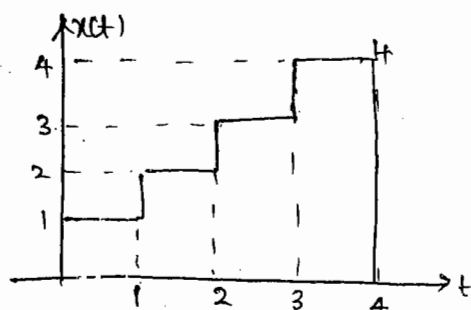
$$y_1(t) = -x_1(4t+2)$$

(b) Relationship b/n $y_2(t)$ & $x_2(t)$

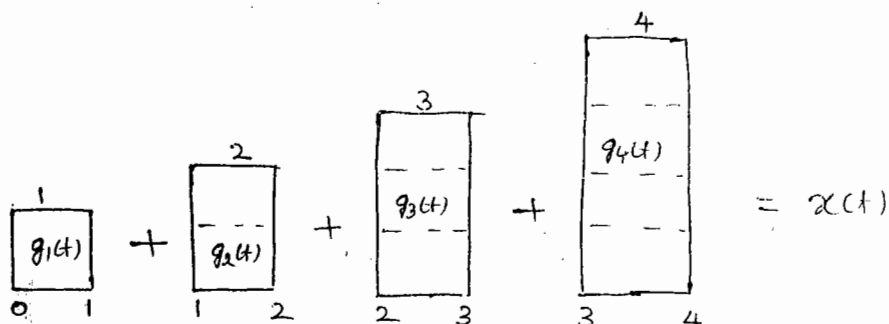


$$y_2(t) = \frac{1}{3}x_2(-2t+1)$$

Generate the staircase signal $x(t)$ using $g(t)$ shown below.



Method 1 using Amplitude scaling method



$$g_1(t) = g(2t-1)$$

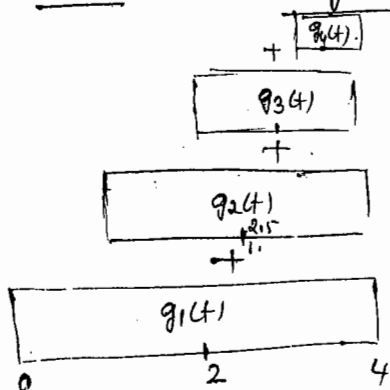
$$g_2(t) = 2g(2t-3)$$

$$g_3(t) = 3g(2t-5)$$

$$g_4(t) = 4g(2t-7)$$

$$x(t) = g(2t-1) + g(2t-3) + g(2t-5) + 2g(2t-7)$$

Method 2 using standard precedence rule method



$$x(t) = g_1(t) + g_2(t) + g_3(t) + g_4(t)$$

$$g_1(t) = g(at-b)$$

$$\text{width of } g(t) = 2 \text{ and } g_1(t) = 4$$

$$\therefore \text{scaling factor } a = \frac{2}{4} = \frac{1}{2}$$

$$\text{Mid point of } g_1(t) = 2 \text{ \& } g(t) \text{ is at } 0$$

$\therefore$ The condition to be satisfied is

$$at-b=0 \quad \text{put } t=2, a=1/2$$

$$b = at = \frac{1}{2} \cdot 2 = 1$$

$$\therefore g_1(t) = g\left(\frac{t}{2}-1\right)$$

$$\text{for } g_2(t) = g(at-b)$$

$$\text{width of } g_2(t) = 3, \text{ scaling factor } \frac{2}{3} = a$$

$$\text{Mid point of } g_2(t) = 5/2$$

condition $at-b=0$ has to be satisfied

$$\frac{2}{3} \cdot \frac{5}{2} = b = 0 \quad \therefore b = 5/3$$

$$\therefore \boxed{g_2(t) = g\left(\frac{2}{3}t - \frac{5}{3}\right)}$$

$$g_3(t) = g(at - b)$$

width required $g_3(t) = 2$ & $g(t) = 2$
scaling factor $a = 1$.

Mid point of $g_3(t) = 3$.

$$at - b = 0 \Rightarrow 1(3) - b = 0 \quad \underline{\underline{b = 3}}$$

$$g_3(t) = g(t - 3)$$

$$g_4(t) = g(at - b)$$

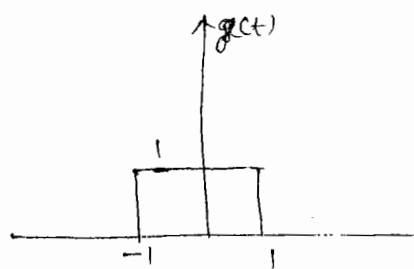
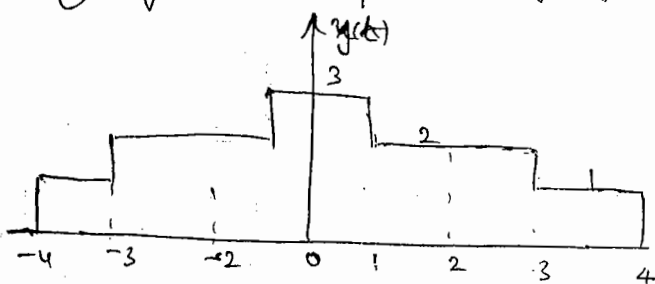
width of $g_4(t) = 1$; scaling factor = 2 (compression)
Mid point of $g_4(t) = 7/2$.

$$at - b = 0$$

$$2 \cdot \frac{7}{2} = b \Rightarrow \boxed{b = 7}$$

$$\therefore \boxed{x(t) = g\left(\frac{t}{2} - 1\right) + g\left(\frac{2}{3}t - \frac{5}{3}\right) + g(t - 3) + g(2t - 7)}$$

② generate $y(t)$ using $g(t)$

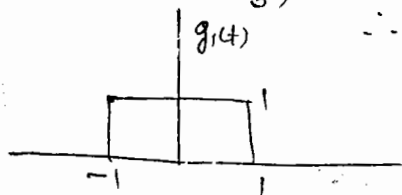


$$y(t) = g_1(t) + g_2(t) + g_3(t)$$

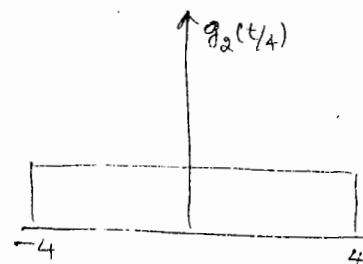
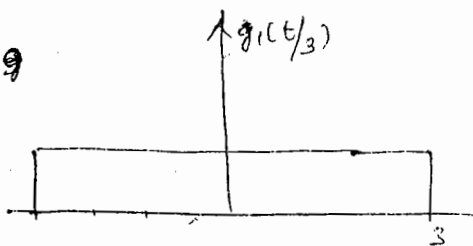
$$g_1(t) = g(t)$$

$$g_3(t) = g\left(\frac{t}{4}\right)$$

$$g_2(t) = g\left(\frac{t}{3}\right)$$

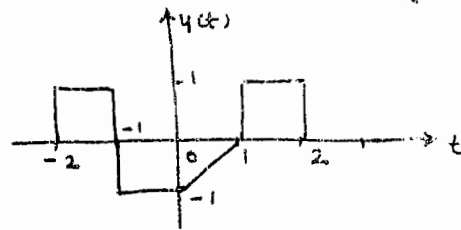
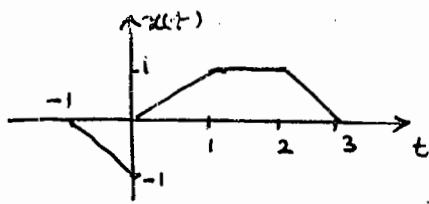


$\therefore g$



(A) Given $x(t)$ and $y(t)$ shown below. Sketch the following signals.

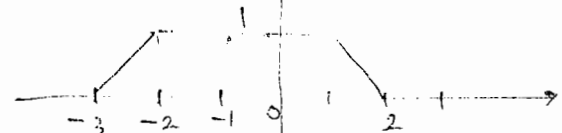
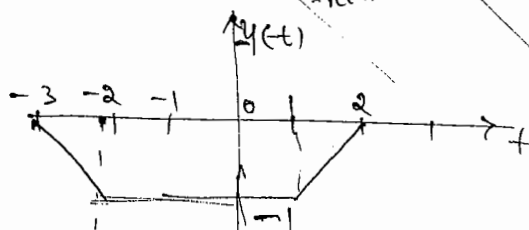
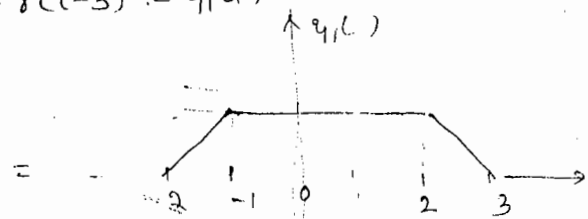
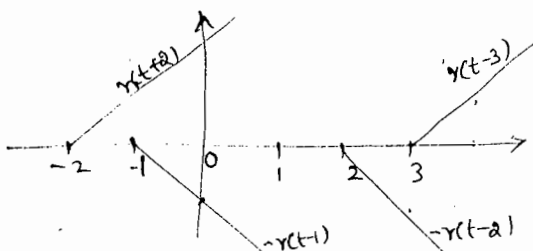
- (a) $x(t)y(t-1)$ (c) $x(t+1)y(t-2)$ (e) $x(t)y(2-t)$ (g) $x(t-y)y(t)$
 (b) $x(t-1)y(t)$ (d) $x(t)y(-1-t)$ (f) $x(2t)y(\frac{1}{2}t+1)$



(a) Sketch the following signals, and determine components.

(1) $x(t+2) - x(t+1) - x(t-2) + x(t-3) = y_1(t)$

(5) $u(n+2) - 3u(n-1) + 2u(n-5) = x(n)$

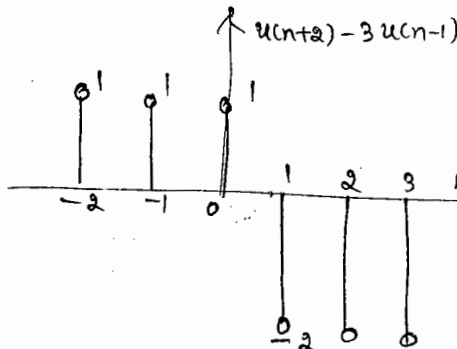
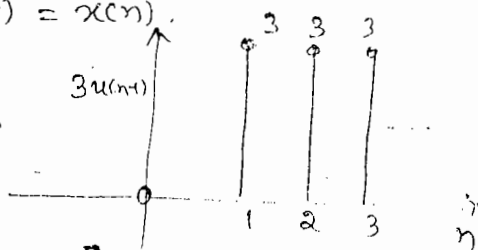
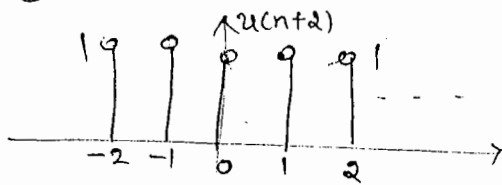


$y_0(t) = \frac{y(t) - y(-t)}{2}$

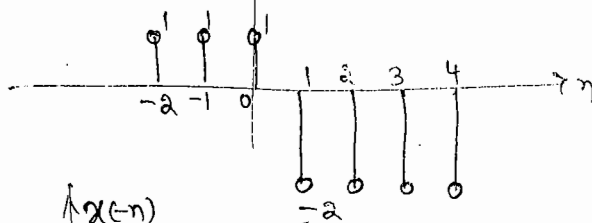


$y_e(t) = \frac{y(t) + y(-t)}{2}$

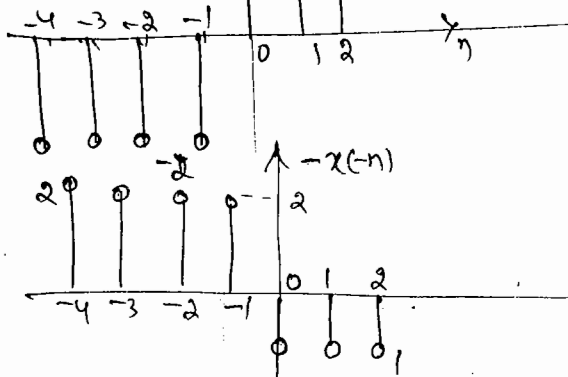
(2) $u(n+2) - 3u(n-1) + 2u(n-5) = x(n)$



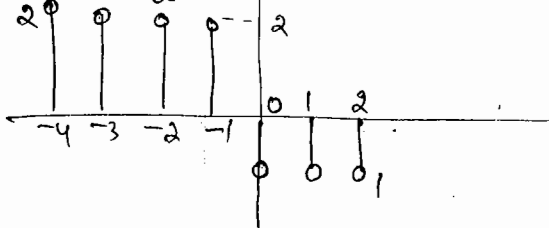
$$x(n) = u(n+2) - 3u(n-1) + 2u(n-5)$$



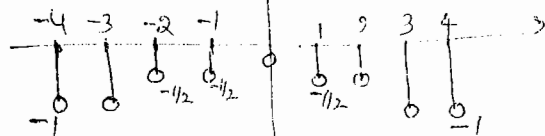
$$x(-n)$$



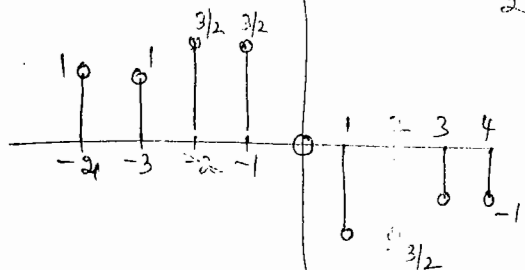
$$-x(-n)$$



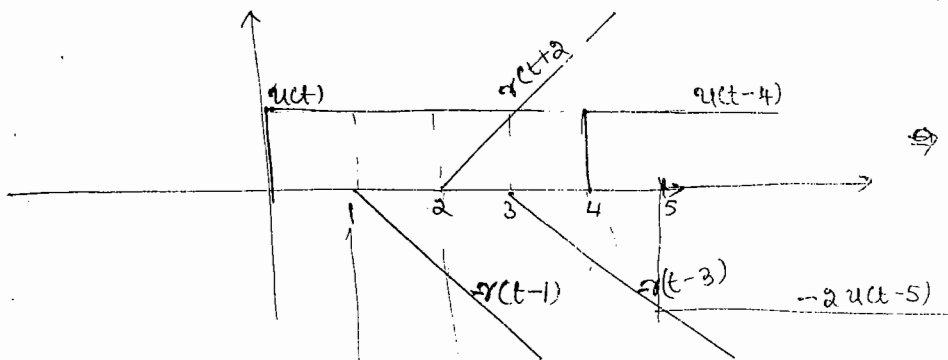
$$x_e(n) = \frac{x(n) + x(-n)}{2}$$



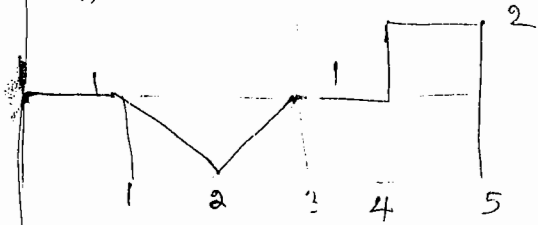
$$x_o(n) = \frac{x(n) - x(-n)}{2}$$



Sketch the following $x(t) = u(t) - \gamma(t-1) + \gamma(t+2) - \gamma(t-3) + u(t-4) - 2u(t-5)$

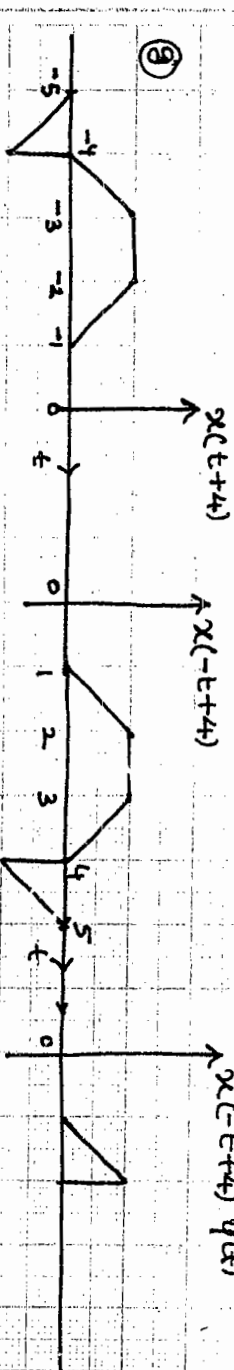
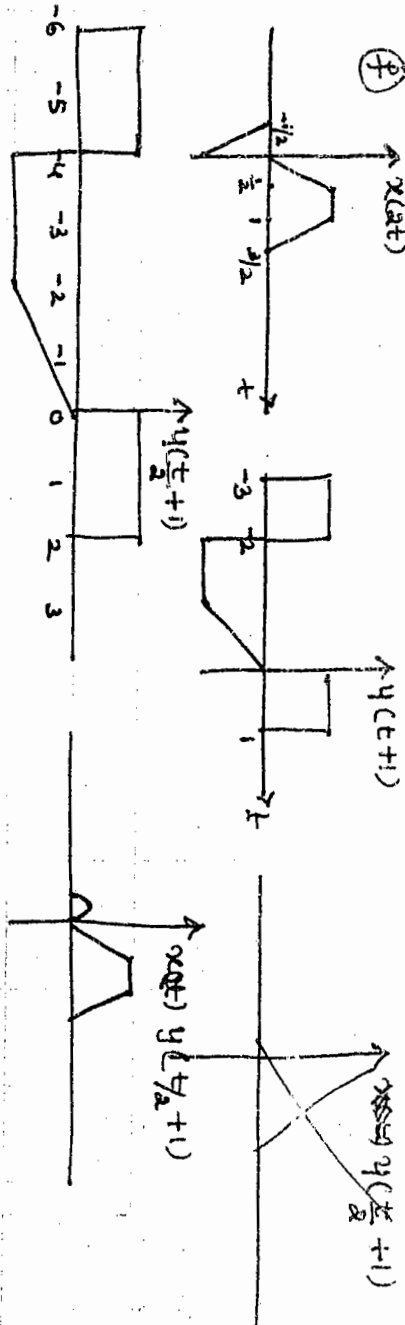
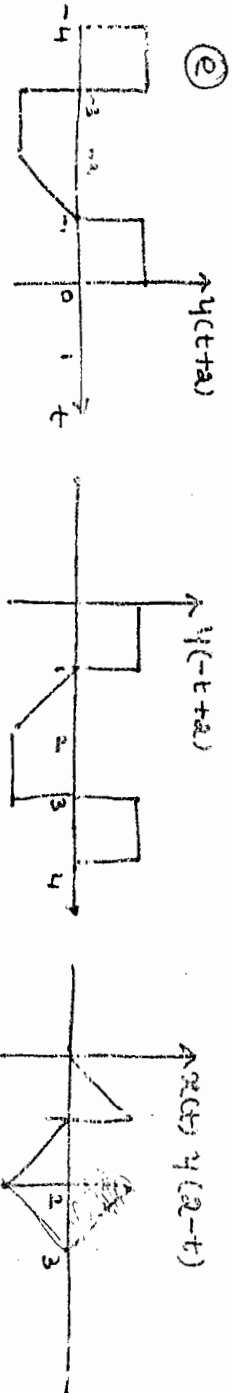


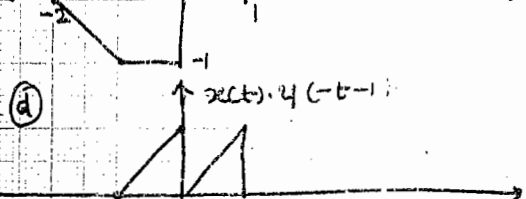
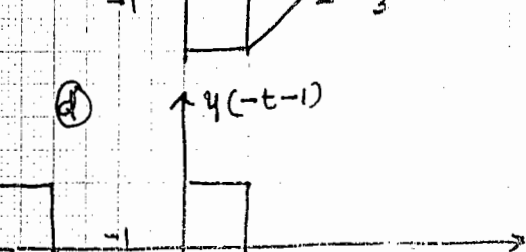
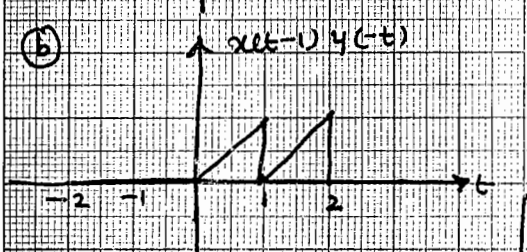
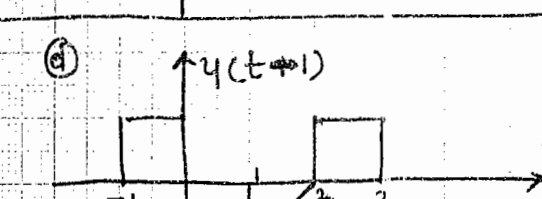
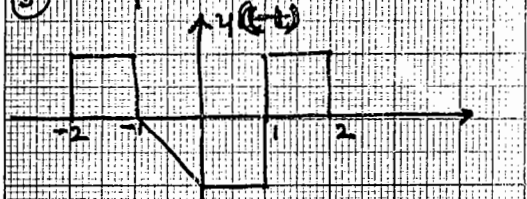
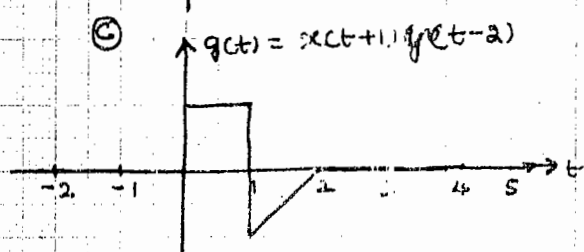
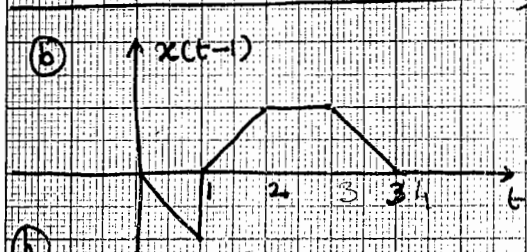
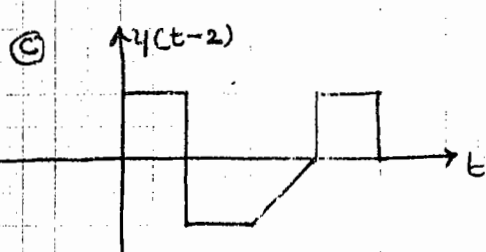
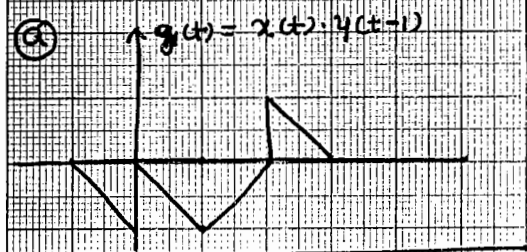
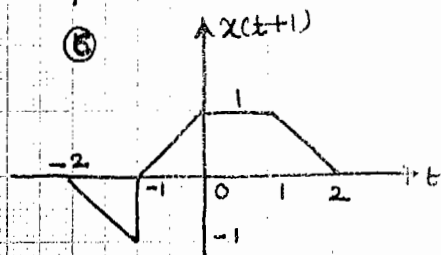
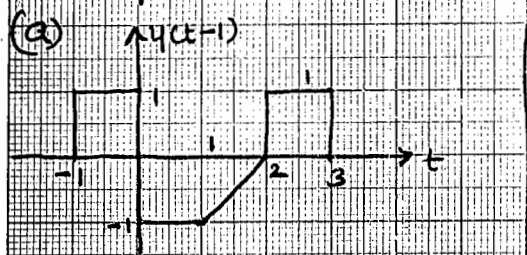
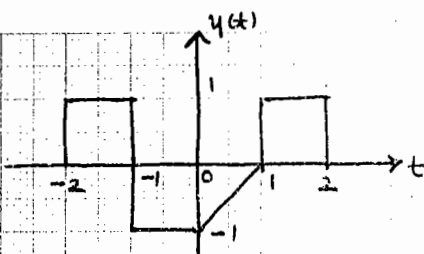
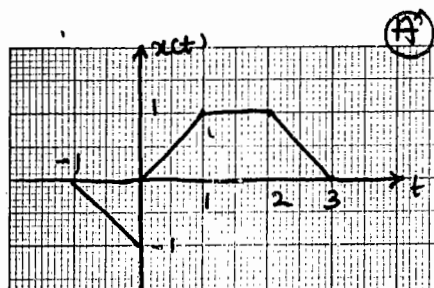
$$x(t)$$

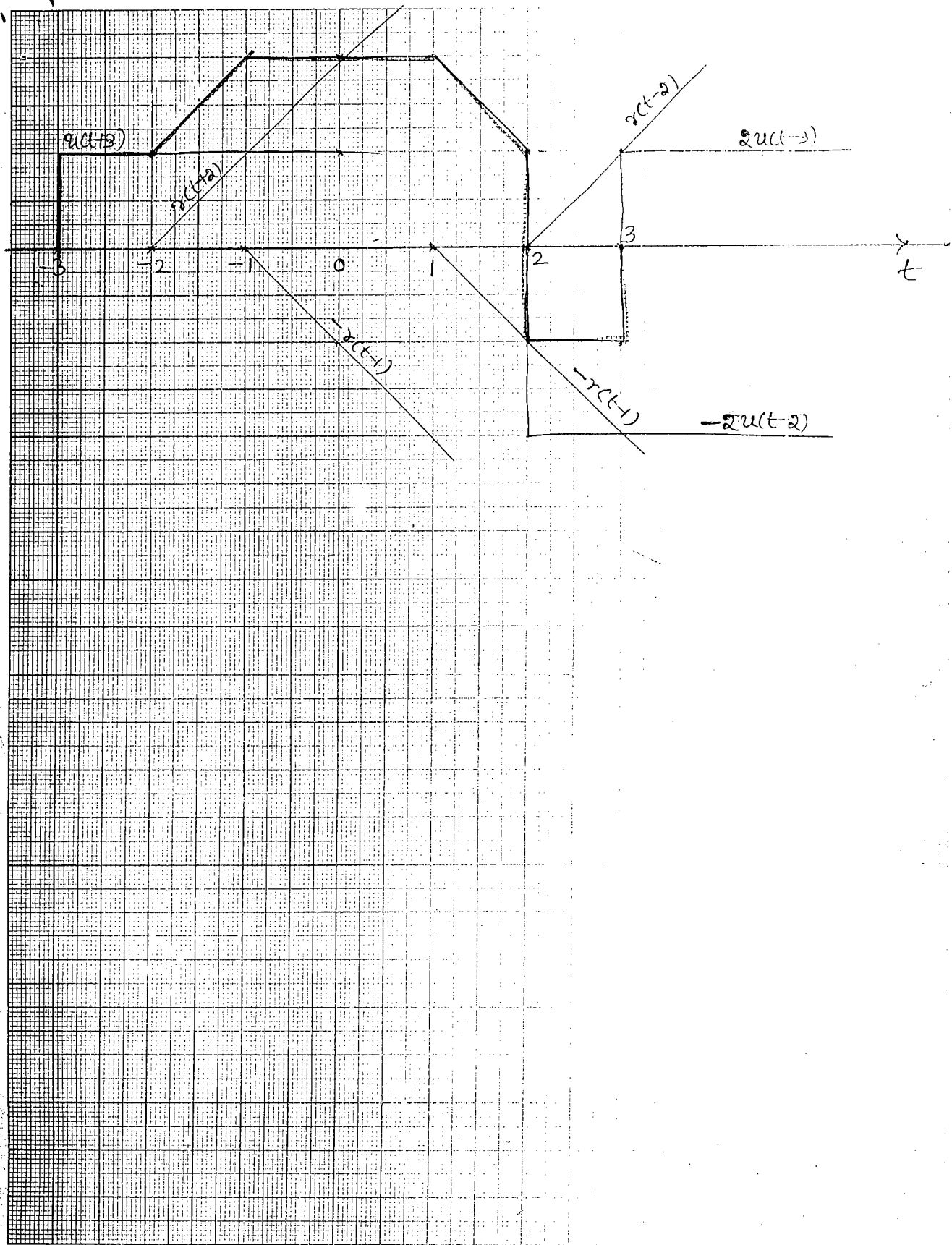


Sketch the signal.

$$x(t) = u(t+3) + \gamma(t+2) - \gamma(t+1) - \gamma(t-1) + \gamma(t-2) - 2u(t-2) + u(t-3)$$







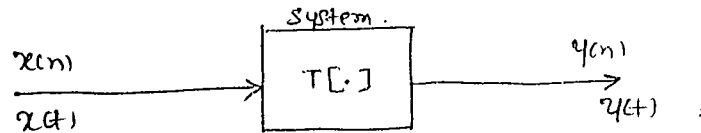
Teacher's Signature : _____

SYSTEMS

Systems viewed as Interconnections of operations

A system is an interconnection of operations that transform an input signal into an o/p signal [with properties entirely different from those of the input signal]

Consider a CT system with transformation operator $T\{\cdot\}$ or system operator $H\{\cdot\}$ as shown. with $x(t)$ as i/p & $y(t)$ as o/p.



$$y(t) = T[x(t)] \quad \text{or} \quad y(n) = T[x(n)] \quad \text{--- for a DT system}$$

Consider DT system characterized by an I/O relation

$$y(n) = \frac{1}{2} [x(n) + x(n-1)]$$

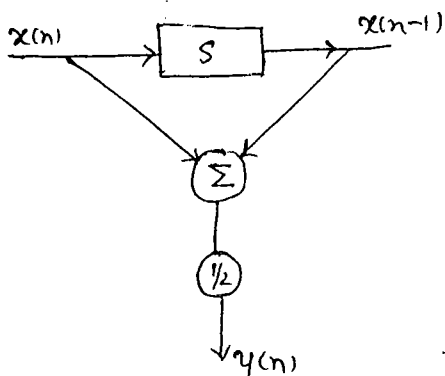
Let S^k denotes a system that shifts the input by 'k' times to produce a signal equal to $x(n-k)$.

$$\therefore \text{Here } y(n) = T[\text{Input}]$$

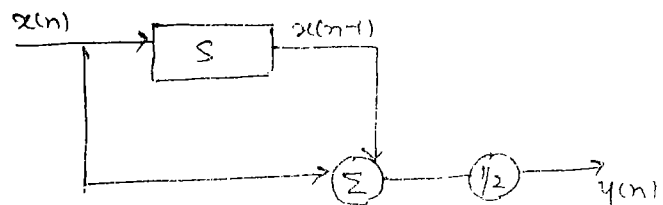
$$= \frac{1}{2} [x(n) + x(n-1)] = \frac{1}{2} [S^0 + S^1] x(n)$$

$$T[\cdot] = \frac{[1 + S]}{2}$$

$T[\cdot]$ or $H\{\cdot\}$ can be realized as follows.



cascade realization.

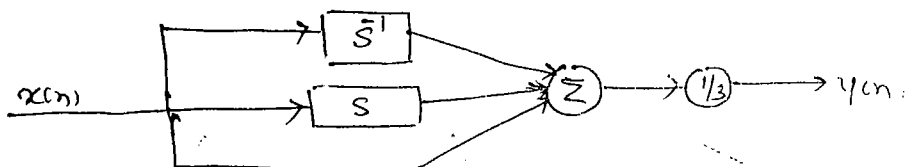


parallel realization

① Express the operator that describes the operation

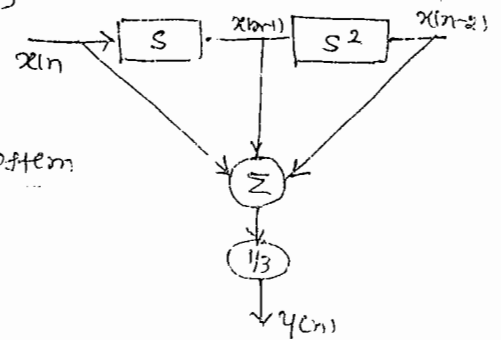
$$y(n) = \frac{1}{3} [x(n) + x(n+1) + x(n-1)]$$

$$y(n) = T[\text{Input}] = \frac{1}{3} [1 + \bar{S} + S] x(n)$$

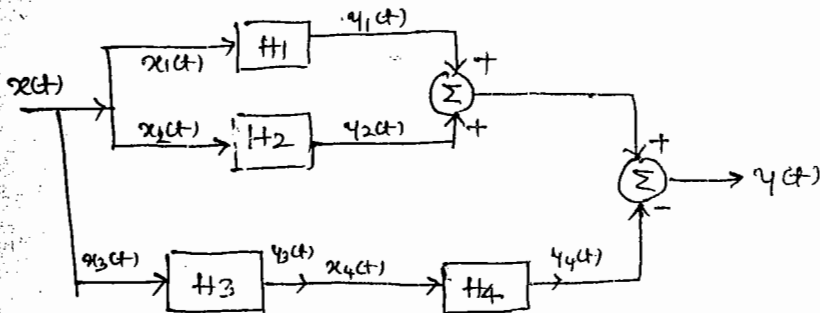


$$② \quad y(n) = \frac{1}{3} [x(n) + x(n-1) + x(n-2)]$$

$$T[z] = \frac{1}{3} [1 + s^{-1} + s^{-2}]$$



③ For the system show below find the system operator that relates the ip & o/p.



Given

$$H_1 : y_1(t) = x_1(t) x_1(t-1)$$

$$H_2 : y_2(t) = |x_2(t)|$$

$$H_3 : y_3(t) = 1 + 2x_3(t)$$

$$H_4 : y_4(t) = \cos[x_4(t)]$$

$$y(t) = y_1(t) + y_2(t) + y_4(t)$$

$$= [x_1(t) x_1(t-1) + |x_2(t)| - \cos[x_4(t)]]$$

$$= x_1(t) x_1(t-1) + |x_2(t)| - \cos[y_3(t)]$$

$$= x_1(t) x_1(t-1) + |x_2(t)| - \cos[1 + 2x_3(t)]$$

$$\text{but } x_1(t) = x_2(t) = x_3(t) = x(t)$$

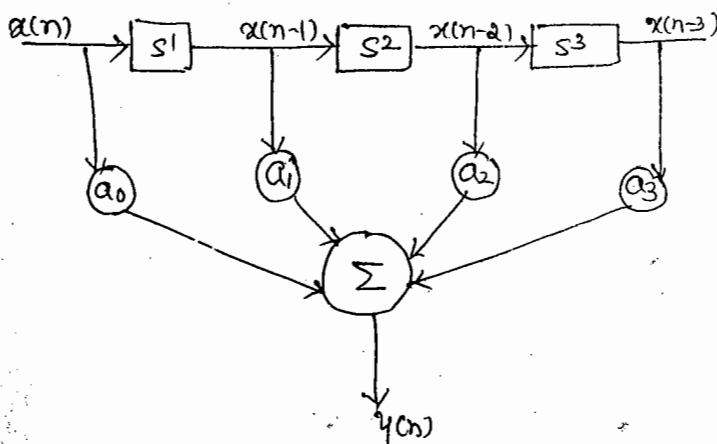
$$\therefore \boxed{y(t) = x(t) \cdot x(t-1) + |x(t)| - \cos[1 + 2x(t)]}$$

④ For the response given below; write the interconnected diagram.

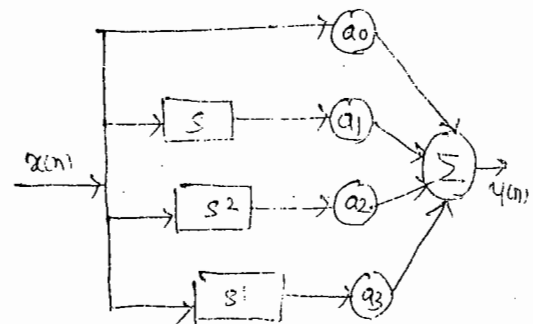
$$y(n) = a_0 x(n) + a_1 x(n-1) + a_2 x(n-2) + a_3 x(n-3)$$

$$y(n) = [a_0 + a_1 s + a_2 s^2 + a_3 s^3] x(n)$$

$$\text{where } x(n) \xrightarrow{s} x(n-1)$$



Cascaded connection



parallel connection

properties of the system (characteristics of the system)

Important properties of the system are

1. Linearity
2. Time invariance
3. Memory
4. Causality
5. Stability
6. Invertibility

Linearity

A system is said to be Linear if it satisfies the principle of superposition. If the system produces a weighted sum of outputs (responses) to corresponding weighted sum of inputs, then the system is said to be Linear.

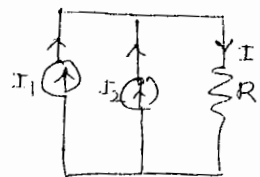
$$\text{i.e. } x_1(t) \longrightarrow y_1(t) \quad \& \quad x_2(t) \longrightarrow y_2(t)$$

$$\text{then } \boxed{a x_1(t) + b x_2(t) = a y_1(t) + b y_2(t)}$$

For a DT system

$$x_1(n) \longrightarrow y_1(n) \quad x_2(n) \longrightarrow y_2(n)$$

$$\boxed{a x_1(n) + b x_2(n) = a y_1(n) + b y_2(n)}$$



Ex:- current thro' the resistor

Time invariance

A system is said to be Time invariant if a ^{time} shift in the input signal produces an identical time shift in the output signal. The shift may be delay or advance.

$$\begin{aligned} x(t) &\longleftrightarrow y(t) & ; & & x(t-t_0) &\longleftrightarrow y(t-t_0) \\ x(n) &\longleftrightarrow y(n) & & & x(n-n_0) &\longleftrightarrow y(n-n_0) \end{aligned}$$

Ex:- 1. The response of the resistor is time invariant; whereas the response of the thermistor is time variant for a constant excitation (input).

2. The time arrival and departure of trains is time invariant provided the velocity of trains is constant; otherwise it is time variant.

3. Rainfall per month is time variant.

Causality

A system is said to be causal if its output at any time depends upon the present or past value of the ⁱⁿputs only.

$$y(k) = f[x(n); n \leq k] \quad y(t_0) = f[x(t); t \leq t_0]$$

The output of a non-causal system depends upon one or more future values of the input.

Ex:- Voltage across the capacitor depends upon past & present values of i/p.

All static systems are generally causal in nature.

Memory (Dynamic and Static systems)

A system is said to be memoryless or static if its output depends only on the present value of the input.

Ex:- current thro' the resistor. $i(t) = \frac{v(t)}{R}$

A system is said to possess memory [or dynamic] if its output depends upon the past or future value of the input.

Ex:- current thro' the inductor; voltage a/c the capacitor

$$i(t) = \frac{1}{L} \int_{-\infty}^t v(\tau) d\tau$$

$$v(t) = \frac{1}{C} \int_{-\infty}^t i(\tau) d\tau$$

$$v_L(t) = L \frac{di(t)}{dt}$$

$$v_C(t) = C \frac{dv}{dt}$$

systems with memory are also called as dynamic systems

Stability:

A system is said to be BIBO (bounded input Bounded output) stable if and only if every bounded i/p produces a bounded output. The o/p of such system does not diverge if the i/p does not diverge.

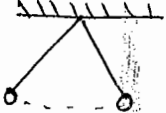
Consider an CT system with response $y(t)$ for an i/p $x(t)$ such that the system is stable if and only if

$$|y(t)| \leq M_y < \infty \text{ for all } t \text{ whenever the i/p signal satisfies the condition } |x(t)| \leq M_x < \infty \text{ for all } t$$

where M_x and M_y are certain finite values.

Similarly for a DT system

If $|x(n)| \leq M_x < \infty$; then $|y(n)| \leq M_y < \infty$; the system is stable

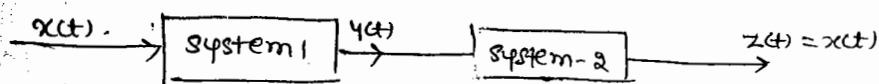
Ex:-  A stable pendulum.

Unstable systems:- chain reactions, population growth.

Invertibility

A system is said to be invertible if the input of the system can be recovered from the system output.

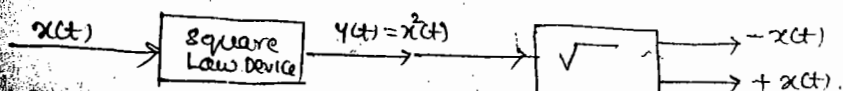
Alternatively a system is said to be invertible if distinct inputs leads to distinct outputs.



System 1 is invertible if $z(t) = x(t)$. System 2 is called inverse system

Ex:- Modulator - demodulator } are invertible systems.
 Log - Antilog operations.

A square law device is a non invertible system as square root will have both +ve and -ve value



Alternatively $-x(t)$ and $x(t)$ produces the same o/p $x^2(t)$. Hence distinct o/p produce same o/p. Hence non invertible

properties of the system : Linearity

① check whether the following systems are linear or not

(a) $y(t) = \sin[x(t)]$

Let $y_1(t)$ be the output for an input $x_1(t)$ and $y_2(t)$ be the o/p for an input $x_2(t)$

$$\therefore y_1(t) = \sin(x_1(t)) \Rightarrow ay_1(t) = a \sin x_1(t) \rightarrow \text{A}$$

$$y_2(t) = \sin(x_2(t)) \Rightarrow by_2(t) = b \sin x_2(t) \rightarrow \text{B}$$

Let $x_3(t) = ax_1(t) + bx_2(t)$ be an input which produces o/p $y_3(t)$

$$\therefore y_3(t) = \sin(x_3(t)) = \sin(ax_1(t) + bx_2(t)) \rightarrow \text{①}$$

$$A+B \Rightarrow a \sin[x_1(t)] + b \sin[x_2(t)] = ay_1(t) + by_2(t) \rightarrow \text{②}$$

Since Linear combination of input does not produce corresponding o/p

i.e. $\text{Eqn } ① \neq \text{Eqn } ②$; the given system is not linear.

(b) $y(t) = x^2(t)$

$$y_1(t) = x_1^2(t) \Rightarrow ay_1(t) = ax_1(t)$$

$$y_2(t) = x_2(t) \Rightarrow by_2(t) = bx_2(t)$$

$$\text{Let } x_3(t) = ax_1(t) + bx_2(t)$$

$$\therefore y_3(t) = x_3^2(t) = [ax_1(t) + bx_2(t)]^2 \rightarrow \text{①}$$

Since $ay_1(t) + by_2(t) = ax_1(t) + bx_2(t) \rightarrow \text{②}$

From $\text{Eqn } ① \neq \text{Eqn } ②$; the given system is not linear

(c) $y(t) = tx(t)$

$$y_1(t) = tx_1(t)$$

$$ay_1(t) = atx_1(t)$$

$$y_2(t) = tx_2(t)$$

$$by_2(t) = bt x_2(t)$$

$$ay_1(t) + by_2(t) = t[ax_1(t) + bx_2(t)] \rightarrow \text{①}$$

$$\text{Let } x_3(t) = ax_1(t) + bx_2(t)$$

$$\therefore y_3(t) = tx_3(t) = t[ax_1(t) + bx_2(t)] \rightarrow \text{②}$$

$\text{Eqn } ① = \text{Eqn } ②$; The given system is linear.

(d) $y(n) = x(n^2)$

$$y_1(n) = x_1(n^2)$$

$$; ay_1(n) = ax_1(n^2)$$

$$y_2(n) = x_2(n^2)$$

$$; by_2(n) = bx_2(n^2)$$

Let $x_3(n) = a x_1(n) + b x_2(n)$ produces a o/p $y_3(n)$

$$\therefore y_3(n) = x_3(n^2) = a x_1(n^2) + b x_2(n^2)$$

$$\text{Since } a y_1(n) + b y_2(n) = a x_1(n^2) + b x_2(n^2)$$

The given system is Linear.

(e) $y(n) = x^2(n)$

$$\text{Let } y_1(n) = x_1^2(n) \Rightarrow a y_1(n) = a x_1^2(n)$$

$$y_2(n) = x_2^2(n) \Rightarrow b y_2(n) = b x_2^2(n)$$

$$a y_1(n) + b y_2(n) = a x_1^2(n) + b x_2^2(n) \text{ --- (1)}$$

$$\text{Let } x_3(n) = a x_1(n) + b x_2(n)$$

$$\therefore y_3(n) = x_3^2(n) = [a x_1(n) + b x_2(n)]^2 \text{ --- (2)}$$

Eqⁿ (1) $\neq$ Eqⁿ (2) ; hence the system is not linear

(f) $y(n) = x^2(n) - x(n-1) + x(n+1)$

By the presence of square term $x^2(n)$, we can conclude that the given system is not linear

(g) $y(t) = x(t/2)$

$$\text{Let } y_1(t) = x_1(t/2) \Rightarrow a y_1(t) = a x_1(t/2)$$

$$y_2(t) = x_2(t/2) \Rightarrow b y_2(t) = b x_2(t/2)$$

$$\text{Let } x_3(t/2) = a x_1(t/2) + b x_2(t/2) \text{ or}$$

~~$$x_3(t) = a x_1(t) + b x_2(t)$$~~
$$\therefore y_3 \quad x_3(t) = a x_1(t) + b x_2(t)$$

$$y_3(t) = x_3(t/2) = a x_1(t/2) + b x_2(t/2) \text{ --- (1)}$$

$$a y_1(t) + b y_2(t) = a x_1(t/2) + b x_2(t/2) \text{ --- (2)}$$

from (1) & (2) ; given system is Linear

(h) $y(n) = x(n) - x(n-1)$

$$y_1(n) = x_1(n) - x_1(n-1) ; \quad a y_1(n) = a (x_1(n) - x_1(n-1))$$

$$y_2(n) = x_2(n) - x_2(n-1) ; \quad b y_2(n) = b (x_2(n) - x_2(n-1))$$

~~$$x_3(n) = a [x_1(n) - x_1(n-1)] + b [x_2(n) - x_2(n-1)] \text{ produces } y_3(n)$$~~

$$\therefore y_3(n) = x_3(n) - x_3(n-1) \quad \text{where } x_3(n) = a x_1(n) + b x_2(n)$$

$$= a x_1(n) + b x_2(n) - a x_3(n-1) - b x_3(n-1)$$

$$\text{Since } ay_1(n) + by_2(n) = y_3(n)$$

The given system is Linear

$$(i) \quad y(t) = 2 \frac{d[x(t)]}{dt}$$

$$y_1(t) = 2 \frac{d[x_1(t)]}{dt}$$

$$ay_1(t) = 2a \frac{d[x_1(t)]}{dt}$$

$$y_2(t) = 2 \frac{d[x_2(t)]}{dt}$$

$$by_2(t) = 2b \frac{d[x_2(t)]}{dt}$$

$$ay_1(t) + by_2(t) = 2 \left[a \frac{d[x_1(t)]}{dt} + b \frac{d[x_2(t)]}{dt} \right] \longrightarrow (1)$$

$$\text{Let } x_3(t) = ax_1(t) + bx_2(t)$$

$$y_3(t) = 2 \frac{d[x_3(t)]}{dt} = 2 \frac{d[ax_1(t) + bx_2(t)]}{dt}$$

$$= 2 \left[a \frac{d[x_1(t)]}{dt} + b \frac{d[x_2(t)]}{dt} \right] \longrightarrow (2)$$

From (1) & (2) we can conclude that the given system is Linear

$$(j) \quad y(t) = \text{Log}[x(t)]$$

$$y_1(t) = \text{Log}[x_1(t)]$$

$$ay_1(t) = a \text{Log}[x_1(t)]$$

$$y_2(t) = \text{Log}[x_2(t)]$$

$$by_2(t) = b \text{Log}[x_2(t)]$$

$$ay_1(t) + by_2(t) = a \text{Log}[x_1(t)] + b \text{Log}[x_2(t)] \longrightarrow (1)$$

$$\text{Let } x_3(t) = ax_1(t) + bx_2(t)$$

$$y_3(t) = \text{Log}[x_3(t)] = \text{Log}[ax_1(t) + bx_2(t)]$$

$$= \cancel{a \text{Log}[x_1(t)]} + \cancel{b \text{Log}[x_2(t)]} \longrightarrow (2)$$

from (1) & (2) the system is not Linear

$$(h) \quad y(n) = A e^{x(n)}$$

$$y_1(n) = A e^{x_1(n)}$$

$$; \quad ay_1(n) = a A e^{x_1(n)}$$

$$y_2(n) = A e^{x_2(n)}$$

$$; \quad by_2(n) = b A e^{x_2(n)}$$

$$x_3(n) = ax_1(n) + bx_2(n)$$

$$y_3(n) = A e^{x_3(n)} = A e^{[ax_1(n) + bx_2(n)]} \longrightarrow (1)$$

$$ay_1(n) + by_2(n) = A [a e^{x_1(n)} + b e^{x_2(n)}] \longrightarrow (2)$$

from (1) & (2) we can conclude that the given system is not Linear

(2) Time Invariance

check the following systems for time invariance.

(i) $y(t) = \sin[x(t)]$

Let $x_1(t)$ be an input produces an o/p $y_1(t)$

$$\therefore y_1(t) = \sin[x_1(t)]$$

Let $x_2(t)$ be the time shifted version of $x_1(t)$ which produces o/p $y_2(t)$

$$x_2(t) = x_1(t-t_0)$$

$$\therefore y_2(t) = \sin[x_2(t)] = \sin[x_1(t-t_0)] \longrightarrow \textcircled{A}$$

we have $y(t) = \sin x(t)$

time shifting both sides by t_0 we get

$$y(t-t_0) = \sin(x(t-t_0)) \longrightarrow \textcircled{B}$$

From $\textcircled{A}$ & $\textcircled{B}$ we can conclude that the given system is Time Invariant

$$\text{i.e. } y(t) = y(t-t_0)$$

(ii) $y(t) = tx(t)$

Let $y_1(t) = tx_1(t)$

Let $x_2(t) = x_1(t-t_0)$

$$y_2(t) = tx_2(t) = tx_1(t-t_0) \longrightarrow \textcircled{1}$$

$$y_1(t-t_0) = (t-t_0)x_1(t-t_0) \longrightarrow \textcircled{2}$$

from $\textcircled{1}$ & $\textcircled{2}$ we can conclude that $y(t) \neq y(t-t_0)$
the given system is Time Variant

* the presence of term 't' makes the system time variant *

(iii) $y(t) = x(t) \cos 200\pi t$

The given system is time variant as input $x(t)$ has one more time dependent term $\cos 200\pi t$.

$$y_1(t) = x_1(t) \cos 200\pi t$$

$$y_1(t-t_0) = x_1(t-t_0) \cos 200\pi(t-t_0) \longrightarrow \textcircled{1}$$

$$x_2(t) = x_1(t-t_0)$$

$$y_2(t) = x_2(t) \cos 200\pi t = x_1(t-t_0) \cos 200\pi t \longrightarrow \textcircled{2}$$

from $\textcircled{1}$ & $\textcircled{2}$ $y(t) \neq y(t-t_0)$

$\therefore$ The given system is not a time invariant system

$$(iv) \quad \underline{y(n) = x(n) - x(n-1)}$$

$$\text{Let } y_1(n) = x_1(n) - x_1(n-1)$$

$$y_1(n-n_0) = x_1(n-n_0) - x_1(n-n_0-1) \quad \text{--- (1)}$$

$$\text{Let } x_2(n) = x_1(n-n_0)$$

$$\therefore y_2(n) = x_2(n) - x_2(n-1) = x_1(n-n_0) - x_1(n-n_0-1) \quad \text{--- (2)}$$

from ① & ② $y(n-n_0) = y(n)$; hence the given system is Time invariant

$$(v) \quad \underline{y(n) = x(-n)}$$

$$y_1(n) = x_1(-n)$$

$$y_1(n-n_0) = x_1(-n+n_0) \quad \text{--- (1)}$$

$$\text{Let } x_2(n) = x_1(n-n_0)$$

$$y_2(n) = x_2(-n) = x_1(n-n_0) \quad \text{--- (2)}$$

$$\text{from ① & ② } y(n) \neq y(n-n_0)$$

The given system is a Time variant system

$$(6) \quad \underline{y(n) = n x(n)}$$

* presence of time dependent term 'n' with i/p makes the system Time variant *

$$y_1(n) = n x_1(n)$$

$$y_1(n-k) = (n-k) x_1(n-k) \quad \text{--- (1)}$$

$$x_2 = x_1(n-k)$$

$$y_2(n) = n x_2(n) = n x_1(n-k) \quad \text{--- (2)}$$

from ① & ② ; the given system is Time variant

$$(7) \quad \underline{y(n) = \cos[x(n)]}$$

$$y_1(n) = \cos[x_1(n)]$$

$$y_1(n-k) = \cos[x_1(n-k)] \quad \text{--- (1)}$$

$$x_2(n) = x_1(n-k)$$

$$\begin{aligned} y_2(n) &= \cos[x_2(n)] \\ &= \cos[x_1(n-k)] \quad \text{--- (2)} \end{aligned}$$

$$\text{from ① & ② } y(n) = y(n-k)$$

$\therefore$ The given system is Time invariant

Causal and non causal systems

check whether the given systems are causal or not.

(I) $y(t) = x(t) \cos(t+1)$

present

Here o/p depends upon the value of the i/p. Also $\cos(t+1)$ can be easily computed. Hence the given system is causal

(II) $y(t) = x(2t)$

at $t=1$; $y(1) = x(2)$

o/p depends upon the future value of i/p hence the system is non causal

(III) $y(t) = x(-t)$

$y(1) = x(-1)$ - - - - - past value of i/p

$y(0) = x(0)$ - - - - - present value

$y(-2) = x(2)$ - - - - - future value of i/p

Hence the system is not causal

(IV) $\frac{d}{dt} y(t) + 10y(t) + 5 = x(t)$

o/p depends on the present value of i/p $x(t)$, hence causal system

(V) $y(t) = \int_{-\infty}^{\infty} x(t) dt$

o/p depends on both past and future values; hence the given system is not causal.

(VI) $y(n) = x(n) + x(n-1)$

$y(0) = x(0) + x(-1)$ - - - - - past & present values of i/p

$y(-1) = x(-1) + x(-2)$ - - - - -

} Given system is causal.

(VII) $y(n) = x(n) + x(n+1)$

$y(0) = x(0) + x(1)$ - - - - - present and future values of i/p

} Given system is not causal.

(VIII) $y(t) = \int_{-\infty}^t x(\tau) d\tau$

o/p depends on present and past values of i/p. Hence the given system is causal.

(IX) $y(n) = \sum_{k=0}^{\infty} x(n-k)$

$y(n) = \underbrace{x(n)}_{\text{present value}} + \underbrace{x(n-1) + x(n-2) + x(n-3) + \dots}_{\text{past values}}$

Hence the given system is causal in nature.

$$(X) \quad y(n) = \sum_{k=-\infty}^{+\infty} x(n-k)$$

$$y(n) = \sum_{k=-\infty}^{-1} x(n-k) + \sum_{k=0}^{\infty} x(n-k)$$

Expanding the first term $\sum_{k=-\infty}^{-1} x(n-k) = x(n+1) + x(n+2) + \dots$
 $\downarrow$
 Future values.

Hence the given system is not causal.

$$(XI) \quad y(n) = \sum_{k=-\infty}^n x(k)$$

$$y(n) = \sum_{k=-\infty}^n x(k) = \underbrace{x(-1) + x(-2) + \dots}_{\text{past}} + \underbrace{x(n)}_{\text{present}}$$

The given system is causal in nature

$$(XII) \quad y(t) = \log[x(t)]$$

$$y(t) = \log[x(t)]$$

$$y(0) = \log[x(0)] \quad \dots \text{present value of } x(t)$$

$$y(1) = \log[x(1)]$$

The given system is causal

$$(XIII) \quad y(n) = x(n-1) + x(n) + x(n+1)$$

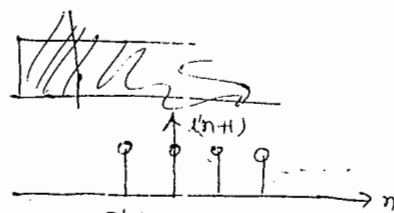
presence of $x(n+1)$ & $x(n-1)$ makes the system non causal

$$(IXV) \quad y(n) = x(n) + 3u(n+1)$$

$$y(n) = x(n) + 3u(n+1)$$

$$y(0) = x(0) + 3u(1) \quad \dots \text{present and past values of } u(t)$$

The ~~output~~ ^{value} of $3u(n+1)$ is constant.
 hence the given system is causal.



$$(XV) \quad y(n) = \max(x(n), x(n-1))$$

$$\text{Let } x(n) = \{1, 2, 3\}$$

$$x(-1)=1, \quad x(1)=3 \quad \uparrow \quad x(0)=2, \quad x(-2)=0$$

Hence the given system is causal.

$$y(0) = \max(x(0), x(-1)) = \max\{2, 1\} = 2$$

$\rightarrow$ o/p depends on present & past values

$$y(1) = \max\{3, 2\} = 3$$

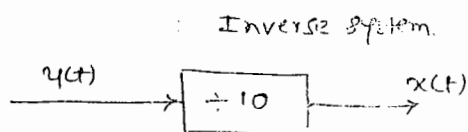
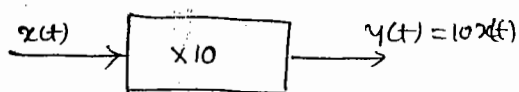
$$y(-1) = \max\{1, 0\} = 1 = x(-1)$$

o/p depends upon the present value

Invertible and inverse systems

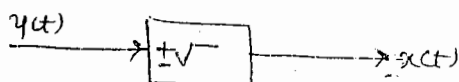
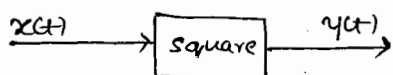
1. check whether the systems below are invertible or not

a) $y(t) = 10x(t)$



Since division operation can be easily performed (Inverse system) is available; the given system is invertible

b) $y(t) = x^2(t)$



Square root operation (inverse system) produces two distinct inputs; hence the given system is not invertible.

c) $y(n) = \sum_{k=-\infty}^n x(n-k)$

$$y(n) = \sum_{k=-\infty}^n x(n-k) = x(n) + x(n-1) + x(n-2) + \dots$$

$$y(n+1) = \sum_{k=-\infty}^{n+1} x(n+1-k) = x(n+1) + x(n) + x(n-1) + \dots$$

$$y(n+1) = x(n+1) + y(n)$$

$$\therefore y(n+2) = x(n+2) + y(n+1)$$

$$\therefore y(n) = x(n) + y(n-1)$$

$$x(n) = y(n) - y(n-1)$$

$\therefore$ 1/p is recoverable; the system is invertible in nature

d) $y(t) = \int_{-\infty}^t x(z) dz$ & $y(-\infty) = 0$

$$y(t) = \int_{-\infty}^t x(z) dz \quad \& \quad y(-\infty) = 0 \quad \text{Let } w(t) = \frac{d}{dt} y(t)$$

$$w(t) = \frac{d}{dt} \int_{-\infty}^t x(z) dz = x(t)$$

Input is recoverable; hence invertible system.

e) $y(n) = n x(n)$

Let $x(n) = \sin \omega n$.

$$y(n) = n x(n) = n \sin \omega n$$

$$y(-1) = -1 \sin(-\omega) = \sin \omega$$

$$y(1) = 1 \sin \omega = \sin \omega$$

Since $y(-1)$ & $y(1)$ are produced by same 1/p $x(n)$; the given system is not invertible

Static and dynamic systems (memory less & memory systems)

Find whether the following systems are static or dynamic

(i) $y(t) = x(t^2)$

output depends on future value of i/p.

$$\left. \begin{array}{l} y(1) = x(1) \\ y(2) = x(4) \end{array} \right\} \text{Hence the system is dynamic} \\ \text{(possess memory)}$$

(ii) $y(t) = x^2(t)$

$$\left. \begin{array}{l} y(1) = x^2(1) \\ y(2) = x^2(2) \end{array} \right\} \text{o/p depends upon the present value of} \\ \text{the input. Hence it is static.} \\ \text{(memory less system)}$$

(iii) $y(n) = x(n)$

$$\left. \begin{array}{ll} y(-1) = x(-1) & \text{--- Future value of i/p} \\ y(0) = x(0) & \text{--- present value} \\ y(1) = x(1) & \text{--- past value of i/p} \end{array} \right\} \text{The given system is dynamic.}$$

(iv) $\frac{d}{dt} y(t) + 10 y(t) = x(t)$

The given system is dynamic due to the presence of differentiation operator. (Same applicable to integral operation)

(v) $y(t) = 10 x(t) + 5$

output depends on the present value of i/p

The given system is static in nature

(vi) $y(n) = x(n+1)$

$$y(n) = x(n+1)$$

$$\left. \begin{array}{l} y(0) = x(1) \text{ --- Future} \\ y(-1) = x(0) \text{ --- Future} \end{array} \right\}$$

O/p depends upon the future value of i/p; hence the system is dynamic in nature.

(vii) $y(t) = x(t-4)$

$$y(0) = x(-4) \text{ --- past value}$$

$$y(1) = x(-3) \text{ --- " ---}$$

The given system is dynamic as o/p depends upon the past value of i/p.

Stability

check whether the following systems are stable or not

(1) $y(t) = x(t/2)$

Let $|x(t)| \leq M_x$

If $x(t)$ is finite; its scaled value is also finite

hence $|y(t)| = |x(t/2)| \leq M_y < \infty$

hence the system is stable

(2) $y(t) = t x(t)$

Let $|x(t)| \leq M_x$

If $x(t)$ is a finite signal and multiplied by a time varying parameter t .

Let $|tx(t)| = \infty$
 $t \rightarrow \infty$

hence o/p is not stable

(3) $y(t) = x(t) \sin 100\pi t$

$|x(t)| \leq M_x$ If $x(t)$ is finite and $\sin \pi$ has value b/w ± 1

hence the bounded i/p produces a bounded o/p $|y(t)| \leq M_y$

Hence the given system is stable

(4) $y(n) = x(-n)$

Let $|x(n)| \leq M_x$ be finite; then the folded signal $|x(-n)|$ is also finite. Hence $|y(n)| \leq M_y$ is also finite

The given system is stable

(5) $y(t) = \log[x(t)]$

Let $|x(t)| \leq M_x$ be finite; its $\log$ is also finite.

Hence o/p $|y(t)|$ is bounded. The given system is stable

(6) $y(n) = x(n) + 3x(n+1)$

Let $|x(n)| \leq M_x < \infty$ be a bounded i/p.

$|y(n)| = |x(n) + 3x(n+1)|$

$$= \begin{cases} M_x + 3 & ; n \geq -1 \\ M_x & ; n < -1 \end{cases}$$

hence the output is bounded

The given system is stable

1. Check whether the given system is Linear, time invariant, causal or not

① $\frac{d}{dt} y(t) + 3y(t) + 5 = x(t)$

Linearity

Due to the presence of constant term, in the equation, the system is not homogeneous. If input is zero, o/p response is not zero. Hence the system is not Linear.

Time invariance : The system is time invariant.

The system is causal as o/p depends upon present and past value of input and output.

$$y(t) = x(t) - 5 - \int_{-\infty}^t y(t) dt$$

Due to the presence of diff. operator, the system possesses memory.

② $y(n) = \text{Log}[x(n)]$

Linearity

$$y_1(n) = \text{Log}[x_1(n)] ; \quad a y_1(n) = a \text{Log}[x_1(n)] \longrightarrow \textcircled{A}$$

$$y_2(n) = \text{Log}[x_2(n)] ; \quad b y_2(n) = b \text{Log}[x_2(n)] \longrightarrow \textcircled{B}$$

$$\text{Let } x_3(n) = a x_1(n) + b x_2(n)$$

$$y_3(n) = \text{Log}[x_3(n)] = \text{Log}[a x_1(n) + b x_2(n)] \longrightarrow \textcircled{C}$$

$$\textcircled{A} + \textcircled{B} \neq \textcircled{C} \quad \text{hence the system is not linear}$$

Time invariance

$$y_1(n) = \text{Log}[x(n)]$$

$$y_1(n) = \text{Log}[x_1(n)] ; \quad y_1(n-n_0) = \text{Log}[x_1(n-n_0)] \longrightarrow \textcircled{A}$$

$$\text{Let } x_2(n) = x_1(n-n_0)$$

$$\therefore y_2(n) = \text{Log}[x_2(n)] = \text{Log}[x_1(n-n_0)] \longrightarrow \textcircled{B}$$

$$\textcircled{A} = \textcircled{B} \quad \text{hence the system is Time invariant}$$

Stability

The system is unstable as $y(n) = \infty$ as $x(n) = 0$

Causality : The system is causal as o/p depends upon the present value of the input $x(n)$.

Invertibility : As antilog operation is available ; the system is invertible in nature.

The system is memoryless.

③ $y(n) = x(n^3)$

$$y_1(n) = x_1(n^3) ; \quad a y_1(n) = a x_1(n^3) \longrightarrow \textcircled{A}$$

$$y_2(n) = x_2(n^3) ; \quad b y_2(n) = b x_2(n^3) \longrightarrow \textcircled{B}$$

$$x_3(n) = a x_1(n^3) + b x_2(n^3).$$

$$y_3(n) = x_3(n^3) = a x_1(n^3) + b x_2(n^3) \text{ --- (C)}$$

(A) + (B) = (C); The system is Linear.

Shift invariance

$$y(n) = x(n^3)$$

$$y(n-k) = \cancel{x(n-k^3)} y(n^3-k) \text{ --- (A)}$$

Let $y_1(n) = x_1(n^3)$

$x_2(n) = x_1(n-k)$ is time shifted version of $x_1(n)$

$$\therefore y_2(n) = x_2(n^3) = x_1(n-k)^3 \text{ --- (B)}$$

$\therefore$ Eqn (A) $\neq$ (B); system is Not Time invariant

Causality

$$y(n) = x(n^3)$$

$$y(0) = x(0)$$

$$y(1) = x(1)$$

$$y(2) = x(8) \rightarrow \text{Future value}$$

} present value } Hence the given system is not causal

Stability: If $|x(n)| \leq m_x < \infty$; then o/p $y(n) \leq m_y < \infty$

The system is stable

Invertibility: system is invertible as down sampling can be implemented

memory: - The system possesses memory.

$$\textcircled{3} \quad y(n) = x(n) \sum_{k=-\infty}^{\infty} \delta(n-k)$$

$$y(n) = x(n) + \delta(n \pm 2) x(n) + \delta(n \pm 4) x(n) + \dots$$

Since output depends upon only $x(n)$ i.e. present value of the input; the system is static or memoryless.

Even though $|x(n)| \leq m_x < \infty$; bounded; $y(n)$ is not bounded due to summation limits from $-\infty$ to ∞ ; Hence the system is not stable

$$y(n) = \lim_{n \rightarrow \infty} |x(n)| \left| \sum_{k=-\infty}^{\infty} \delta(n-k) \right| \rightarrow \infty$$

Since output depends upon the present value of I/P; the system is Causal.

Linearity

$$y_1(n) = x_1(n) \sum \delta(n-ak) \quad ; \quad a y_1(n) = a x_1(n) \sum \delta(n-ak) \text{ --- (A)}$$

$$y_2(n) = x_2(n) \sum \delta(n-ak) \quad ; \quad b y_2(n) = b x_2(n) \sum \delta(n-ak) \text{ --- (B)}$$

Let $x_3(n) = a x_1(n) + b x_2(n)$

$$y_3(n) = x_3(n) \sum \delta(n-ak) = [a x_1(n) + b x_2(n)] \sum \delta(n-ak) \text{ --- (C)}$$

(A) + (B) = (C); the system is Linear

Time invariance

$$y(n) = x(n) \sum \delta(n-2k)$$

$$y(n-k) = x(n-k) \sum \delta(n-k_1-2k) = x(n-k) \sum \delta(n-2k-k_1) \rightarrow (A)$$

Let $x_2(n) = x_1(n-k_1)$

$$y_2(n) = x_2(n) \sum \delta(n-2k) = x_1(n-k_1) \sum \delta(n-2k) \rightarrow (B)$$

Eqs (A) $\neq$ (B) ; the system is Time variant

$$(4) \quad y(t) = \frac{d}{dt} [\bar{e}^{-t} x(t)]$$

1. Due to the presence of diff. operator; the system is dynamic
2. $\bar{e}^{-t}$ is a decaying fn; hence the system is stable for all bounded inputs, i.e. $|x(t)| \leq m_x < \infty$.
3. Since o/p depends upon present value of I/p; system is causal.
4. $y(t) = \frac{d}{dt} [\bar{e}^{-t} x(t)]$

$$y_1(t) = \frac{d}{dt} [\bar{e}^{-t} x_1(t)] ; \quad a y_1(t) = a \frac{d}{dt} [\bar{e}^{-t} x_1(t)] \rightarrow (A)$$

$$y_2(t) = \frac{d}{dt} [\bar{e}^{-t} x_2(t)] ; \quad b y_2(t) = b \frac{d}{dt} [\bar{e}^{-t} x_2(t)] \rightarrow (B)$$

Let $x_3(t) = a x_1(t) + b x_2(t)$

$$\begin{aligned} y_3(t) &= \frac{d}{dt} [\bar{e}^{-t} x_3(t)] = \frac{d}{dt} [\bar{e}^{-t} (a x_1(t) + b x_2(t))] \\ &= a \frac{d}{dt} [\bar{e}^{-t} x_1(t)] + b \frac{d}{dt} [\bar{e}^{-t} x_2(t)] \rightarrow (C) \end{aligned}$$

Eqs (A) + (B) = (C) system is Linear

$$(5) \quad y(t) = \frac{d}{dt} [\bar{e}^{-t} x(t)]$$

due to the presence of $\bar{e}^{-t}$ in the eqn; system is not Time invariant

i.e. $y(t) = \frac{d}{dt} [\bar{e}^{-t} x(t)]$

Let $y_1(t-t_0) = \frac{d}{dt} [\bar{e}^{-(t-t_0)} x(t-t_0)] \rightarrow (1)$

Let $x_2(t) = x(t-t_0)$

$$y_2(t) = \frac{d}{dt} [\bar{e}^{-t} x_2(t)] = \frac{d}{dt} [\bar{e}^{-t} x(t-t_0)] \rightarrow (2)$$

From (1) & (2) we can conclude that $A \neq B$
The system is Time variant

⑤ $y(n) = T[x(n)] = g(n) \cdot x(n)$
 where T is a system operator.

Linearity: $y(n) = g(n) x(n)$

$y_1(n) = g(n) x_1(n)$; $ay_1(n) = a g(n) x_1(n) \rightarrow \textcircled{A}$

$y_2(n) = g(n) x_2(n)$; $by_2(n) = b g(n) x_2(n) \rightarrow \textcircled{B}$

$x_3(n) = ax_1(n) + bx_2(n)$

$y_3(n) = g(n) x_3(n) = g(n) (ax_1(n) + bx_2(n)) \rightarrow \textcircled{C}$

$\textcircled{A} + \textcircled{B} = \textcircled{C}$; the system is Linear

Time Invariance

$y(n) = g(n) x(n)$

$y(n-k) = g(n-k) x(n-k) \rightarrow \textcircled{A}$

Let $x_2(n) = x_1(n-k)$

$y_2(n) = g(n) x_2(n) = g(n) x_1(n-k) \rightarrow \textcircled{B}$

From eqn $\textcircled{A}$ & $\textcircled{B}$; Eqn $\textcircled{A} \neq \textcircled{B}$

The given system is not Time Invariant

The system is Causal as o/p depends upon present value of input. It is also memoryless.

Even though $|x(n)| \leq M_x < \infty$; bounded ;

as $\lim_{n \rightarrow \infty} g(n) |x(n)| \rightarrow y(n) \rightarrow \infty$,

the system is not stable if $g(n)$ is not bounded

⑥ $\frac{d}{dt} y(t) + t y(t) = x(t)$

$\frac{d}{dt} y_1(t) + t y_1(t) = x_1(t)$; $a \left[\frac{d}{dt} y_1(t) + t y_1(t) \right] = a x_1(t) \rightarrow \textcircled{A}$

$\frac{d}{dt} y_2(t) + t y_2(t) = x_2(t)$; $b \left[\frac{d}{dt} y_2(t) + t y_2(t) \right] = b x_2(t) \rightarrow \textcircled{B}$

Let $x_3(t) = ax_1(t) + bx_2(t)$

$\therefore \frac{d}{dt} y_3(t) + t y_3(t) = x_3(t) = ax_1(t) + bx_2(t)$

$\frac{d}{dt} a y_1(t) + a t y_1(t) + \frac{d}{dt} b y_2(t) + b t y_2(t) = a x_1(t) + b x_2(t) \rightarrow \textcircled{C}$

$\textcircled{A} + \textcircled{B} = \textcircled{C}$

Hence the system is Linear

1.e $\frac{d}{dt} y(t) + t y(t) = x(t)$

$$\frac{d}{dt} y_1(t-t_0) + (t-t_0) y_1(t-t_0) = x_1(t-t_0) \longrightarrow (A)$$

$$\therefore \frac{d}{dt} y_2(t) + t y_2(t) = x_2(t) = x_1(t - t_0) \longrightarrow (B)$$

As long as $x(t)$ is bounded; $|x(t)| \leq m_x < \infty$,

also $\lim_{t \rightarrow \infty} y(t) = \int x(t) - \int y(t) \cdot t \rightarrow \infty$

Due to the presence of diff. operator, system is dynamic

	System eqn	Memory	Stability	Causality	Linearity	Time invariant
1	$y(t) = \cos[x(t)]$	X	✓	✓	X	✓
2	$y(n) = x(n) u(n)$	X	✓	✓	✓	✓
3	$y(n) = \log_{10}(x(n))$	X	✓	✓	X	✓
4	$y(t) = \frac{d}{dt} x(t)$	✓	✓	✓	✓	✓
5	$y(n) = 2 x(2^n)$	X	✓	✓	X	✓
6	$y(t) = x(2-t)$	✓	✓	X	✓	✓
7	$y(n) = x(n) + \sum_{k=-\infty}^{n-1} x(k)$	✓	✓	✓	✓	✓
8	$y(n) = \sum_{k=-\infty}^n x(k+2)$	✓	✓	X	✓	✓
9	$y(n) = x(n) \sum_{k=-\infty}^{\infty} \delta(n-2k)$	X	✓ if $ \sum_{k=-\infty}^{\infty} \delta(n-2k) < \infty$	✓	✓	X

$\Rightarrow y(n) = \sum_{k=-\infty}^n x(k+2)$; For $n=3$; $y(3) = \sum_{k=-\infty}^3 x(k+2) = x(-6) + \dots + x(-2) + x(-1) + x(0)$

$\begin{matrix} & k=-1 & k=0 & & k=-4 & & k=-3 & & k=-2 \\ & \downarrow & & & \downarrow & & \downarrow & & \downarrow \\ & 1 & & & & & & & 2 \end{matrix}$

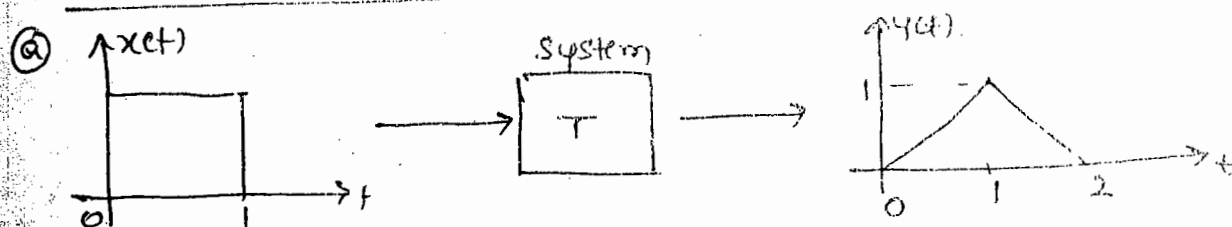
output depends upon the past values of x_p + $x(-1) + x(0) + \dots$

output depends upon the past values of y_p $+ x(n) + 2x(n-1) + \dots$
Hence the system is not memory less.

Why for $n = -1$ $y(-1) = \sum_{k=-\infty}^{-1} x(k+1) = x(-\infty) + \dots + x(-2) + x(-1) + x(0) + x(1) + \dots$

Output depends upon future values u_t ; hence not causal $K=-4, K$

① For the i/o relations shown in fig; check whether the system has memory or not, causal or not

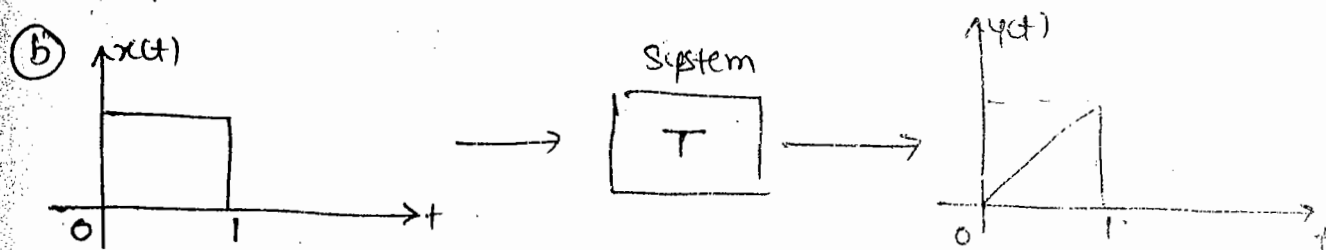


I/p appears b/n $0 \leq t \leq 1$

Output appears b/n $0 \leq t \leq 2$

; Hence the system has memory

output doesnot ~~depend~~ appear before i/p; hence the system is causal

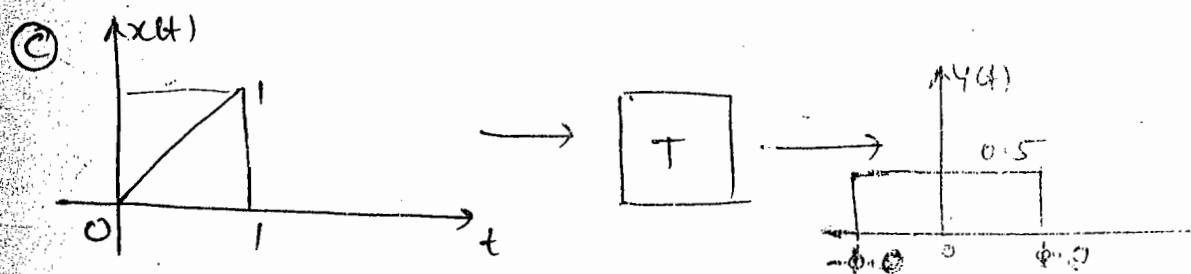
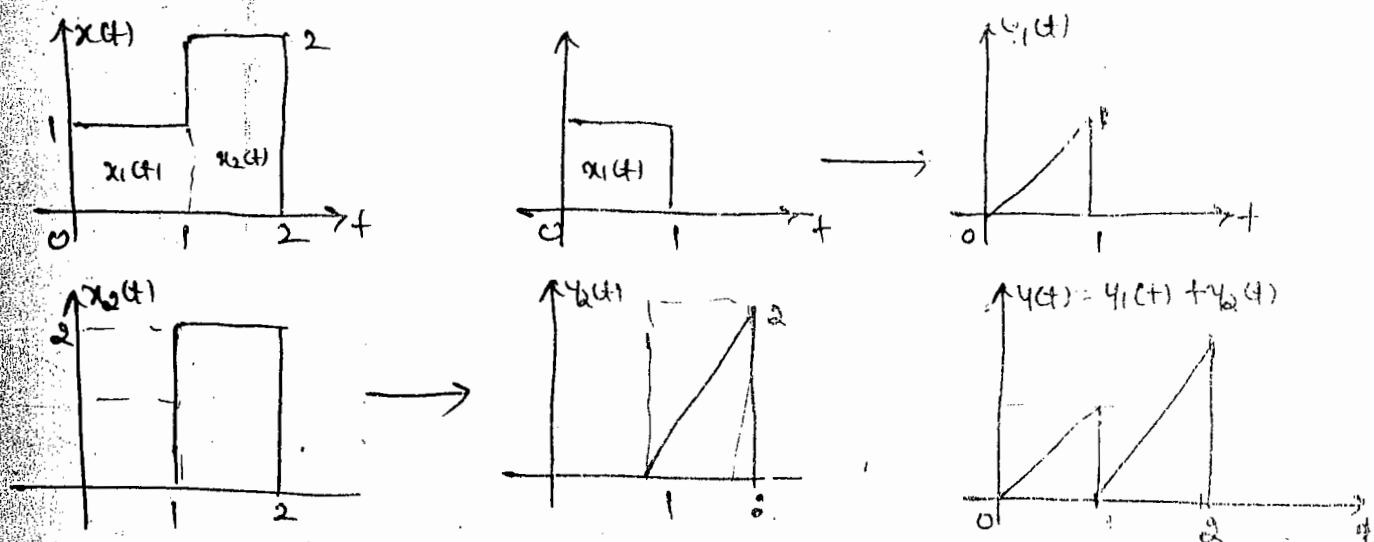


o/p is the integration of i/p.

$$y(t) = \int_0^t x(\tau) d\tau$$

hence the system has memory and o/p doesnot appear before i/p; hence it is causal.

To the above system; If $x(t)$ is given as shown below, sketch the o/p



Here o/p appears before i/p Hence the system is not causal and out doesnot depend upon the past value of the i/p; hence

LTI System

A system which follows two properties namely Linearity and Time Invariance is known as LTI System. These two properties play a vital role in the analysis of signals and systems.

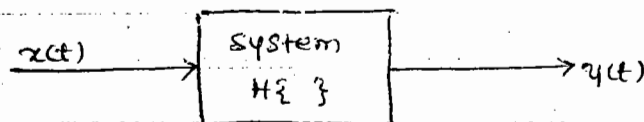
∴ LTI system can be represented by

- (i) In terms of impulse response
- (ii) In terms of differential and difference equation.
- (iii) Block diagrams.

Impulse Response

Impulse response is the output of the LTI system due to impulse input applied at various values of 't' or 'n'.

consider a LTI system shown below



The output response $y(t)$ can be related input $x(t)$ as

$$y(t) = H\{x(t)\}$$

If $x(t) = \delta(t)$; then

$$y(t) = H\{\delta(t)\} = h(t)$$

where $h(t)$ is known as impulse response of the system.

Significance:-

Since impulse response relates input and output, a complete characterization of any LTI system can be determined from its impulse response representation.

Convolution Sum:

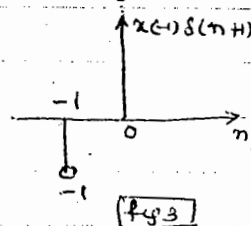
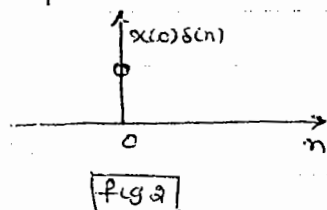
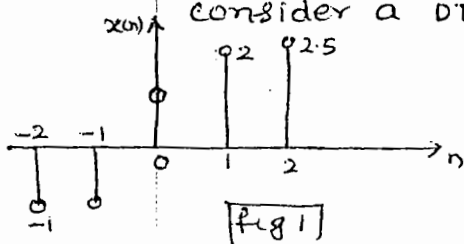
If the input to a LTI system can be expressed as weighted sum of time shifted impulses, then the output is weighted sum of time shifted responses to each of the shifted input impulses. This weighted superposition sum is known as convolution sum for a DT LTI system.

Mathematically

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k] = x[n] * h[n]$$

Expression for convolution sum:

consider a DT LTI system shown below (fig 1)

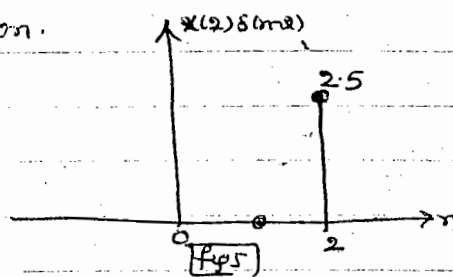
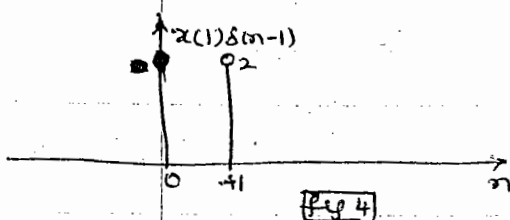


The sample at $n=0$ can be obtained by multiplying $x(0)$ with $\delta(n)$... fig 2

$$x(0) = \begin{cases} x(0)\delta(n) & ; n=0 \\ 0 & ; \text{otherwise} \end{cases}$$

Similarly we can obtain other samples as shown in fig 3

fig 4 and fig 5 and so on.



The sum of these weighted sequences is nothing but $x(n]$

$$\therefore x(n) = x(0)\delta(n) + x(-2)\delta(n+2) + x(-1)\delta(n+1) + x(1)\delta(n-1) + x(2)\delta(n-2) + \dots$$

$$\therefore x(n) = \sum_{k=-\infty}^{\infty} x(k)\delta(n-k)$$

In general

$$x(n) = \sum_{k=-\infty}^{\infty} x(k)\delta(n-k) \quad \dots \dots \dots (1)$$

Let $y(n]$ be the output response of the system with $H\{\}$ being the system operator and $x(n]$ as input.

$$\therefore y(n) = H\{x(n)\}$$

$$= H\left\{\sum_{k=-\infty}^n x(k)\delta(n-k)\right\}$$

using the Linearity property of LTI system

$$y(n) = \sum_{k=-\infty}^{\infty} x(k) H\{\delta(n-k)\}$$

$H\{\delta(n-k)\}$ is the response of the system to a time-shifted

)
 $x(n) \delta(n-k)$

impulse input $\delta(n-k)$ and denoted by $h(n-k)$
 hence $H\{\delta(n-k)\}$ corresponds to input $\delta(n-k)$ and denoted by $h(n-k)$.

$$\therefore y(n) = \sum_{k=-\infty}^{\infty} x(n-k) \cdot h(k) = x(n) * h(n) \quad \text{--- (2)}$$

Eqn (2) represents the output response of an LTI system and mathematical representation of convolution sum.
 convolution sum involves

(a) Folding (b) Time shifting (c) multiplication (d) Addition.

properties of convolution

- (i) Commutative property
- (ii) Associative property
- (iii) Distributive property.

commutative property:

consider a LTI system with impulse response $h(n)$ and input $x(n)$; we define the commutative property as

$$x(n) * h(n) = h(n) * x(n)$$

proof:

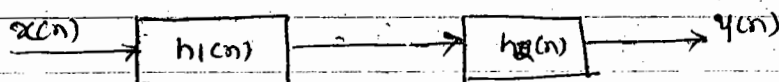
$$\text{LHS: } x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k) h(n-k)$$

$$\text{put } n-k = l \quad \therefore k = (n-l)$$

$$x(n) * h(n) = \sum_{l=-\infty}^{\infty} x(n-l) \cdot h(l) = h(n) * x(n) \quad \text{--- RHS}$$

$$\therefore x(n) * h(n) = h(n) * x(n) \quad \text{hence proved.}$$

Associative property:



$$\text{Statement: } x(n) * [h_1(n) * h_2(n)] = [x(n) * h_1(n)] * h_2(n)$$

proof:-

$$\text{Let } h(n) = h_1(n) * h_2(n) = \sum_{l=-\infty}^{\infty} h_1(l) h_2(n-l) \quad \text{--- (1)}$$

Also LHS $x(n) * [h_1(n) * h_2(n)] = x(n) * h(n)$

$$= \sum_{k=-\infty}^{\infty} x(k) h(n-k) \dots \dots \dots (2)$$

put $n=n-k$ in eqn (1)

$$h(n-k) = \sum_{l=-\infty}^{\infty} h_1(l) \cdot h_2(n-k-l) \dots \dots \dots (3)$$

using equation (3) in (2) we get.

$$x(n) * [h_1(n) * h_2(n)] = \sum_{k=-\infty}^{\infty} x(k) \sum_{l=-\infty}^{\infty} h_1(l) h_2(n-k-l)$$

put $k+l=r \therefore l=r-k$.

$$\text{LHS} = \sum_{k=-\infty}^{\infty} x(k) \sum_{r=-\infty}^{\infty} h_1(r-k) \cdot h_2(n-r)$$

using the linearity property.

$$\text{LHS} = \left[\sum_{k=-\infty}^{\infty} x(k) h_1(r-k) \right] \sum_{r=-\infty}^{\infty} h_2(n-r) = \sum_{r=-\infty}^{\infty} z(r) \cdot h_2(n-r)$$

$$= z(n) * h_2(n) \quad \text{--- (1)}$$

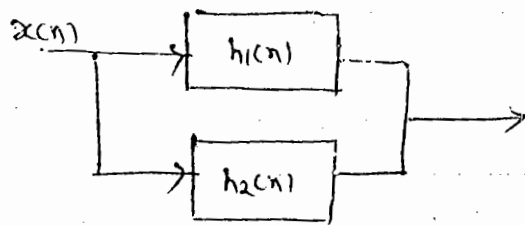
where $z(n) = x(n) * h_1(n)$

$$\therefore \text{LHS} = z(n) * h_2(n) = [x(n) * h_1(n)] * h_2(n) \dots \dots \text{RHS.}$$

$$x(n) * [h_1(n) * h_2(n)] = [x(n) * h_1(n)] * h_2(n)$$

Distributive property
statement

$$x(n) * [h_1(n) + h_2(n)] = x(n) * h_1(n) + x(n) * h_2(n)$$



proof:

$h(n)$

$$\text{Let } h(n) = h_1(n) + h_2(n)$$

$$\text{LHS: } x(n) * [h_1(n) + h_2(n)] = x(n) * h(n)$$

$$h(n) = h_1(n) + h_2(n)$$

$$h(n-k) = h_1(n-k) + h_2(n-k) \quad \dots \text{Timeshifting property}$$

$$\therefore \text{LHS} = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k) h(n-k) \quad \dots \text{--- (A)}$$

using the value of $h(n-k)$ in eqn (A) we get

$$\begin{aligned} x(n) * h(n) &= \sum_{k=-\infty}^{\infty} x(k) [h_1(n-k) + h_2(n-k)] \\ &= \sum_{k=-\infty}^{\infty} x(k) h_1(n-k) + \sum_{k=-\infty}^{\infty} x(k) h_2(n-k) \\ &= x(n) * h_1(n) + x(n) * h_2(n) \quad \dots \text{--- RHS} \end{aligned}$$

$$\boxed{x(n) * [h_1(n) + h_2(n)] = x(n) * h_1(n) + x(n) * h_2(n)}$$

$x(n)$

$h_2(n)$

- ① prove that (a) $x(n) * \delta(n) = x(n)$
(b) $x(n) * \delta(n-m) = x(n-m)$

RHS.

$x(n)$

$$(a) \quad x(n) * \delta(n) = \sum_{k=-\infty}^{\infty} x(k) \delta(n-k)$$

$$\text{but } \delta(n-k) = \begin{cases} 1 & ; k=n \\ 0 & ; k \neq n \end{cases}$$

$$\therefore x(n) * \delta(n-k) = x(n)$$

convolution of any arbitrary signal and impulse signal results in the same signal.

$x(n) * h_2(n)$

$$(b) \quad x(n) * \delta(n-m) = \sum_{k=-\infty}^{\infty} x(k) \delta(n-k-m)$$

$$\delta(n-m-k) = \begin{cases} 1 & ; n-m=k \\ 0 & ; n-m \neq k \end{cases}$$

$$\therefore x(n) * \delta(n-m) = x(n-m)$$

(4)

Hence the convolution of any arbitrary sequence & a delayed impulse results in delayed version of input signal defined by delay of impulse signal.

(2) prove that $x(n) * u(n-m) = \sum_{k=-\infty}^{\infty} |x(k)|$

$$x(n) * u(n-m) = \sum_{k=-\infty}^{\infty} x(k) u(n-m-k)$$

put $p = n-m$

$$\therefore u(n-m-k) = u(p-k) = \begin{cases} 1; & k \leq p \\ 0; & k > p \end{cases}$$

$$\therefore x(n) * u(n-m) = \sum_{k=-\infty}^{\infty} x(k) u(p-k) = \sum_{k=-\infty}^{\infty} |x(k)| \quad \text{--- RHS}$$

(3) Express the following sequences as weighted sum of time shifted impulses

(a) $x(n) = \{ 3, 2, 0, 4, 0, 0, 6 \}$

$$x(n) = \sum_{k=-3}^3 x(k) \delta(n-k)$$

$$\begin{cases} x(-3) = 3, & x(3) = 6 \\ x(-2) = 2 & x(2) = 0 \\ x(-1) = 0 & x(1) = 0 \\ & x(0) = 4 \end{cases}$$

$x(n)$ can be expressed as

$$x(n) = x(-3) \delta(n+3) + x(-2) \delta(n+2) + x(-1) \delta(n+1) + x(0) \delta(n) + x(1) \delta(n-1) + x(2) \delta(n-2) + x(3) \delta(n-3)$$

Ans

$$x(n) = 3 \delta(n+3) + 2 \delta(n+2) + 4 \delta(n) + 6 \delta(n-3)$$

(5) $x(n) = \begin{cases} 3; & \text{for } n = \pm 1, \pm 3 \\ 2; & \text{for } n = \pm 2 \\ 0; & \text{otherwise} \end{cases}$

Ans: $x(n) = 3 \delta(n+3) + 2 \delta(n+2) + 3 \delta(n+1) + 3 \delta(n-1) + 2 \delta(n-2) + 3 \delta(n-3)$

④ Determine the convolution of the following sequences

(a) $x(n) = \{1, 2, 3, 4\}$ $h(n) = \{1, 1, 3, 2\}$

Let $M =$ No of elements in $x(n) = 4$

Let $N =$ No of elements in $h(n) = 4$

Let $y(n) = x(n) * h(n)$

Let $Z =$ No of elements in $y(n)$ after convolution

$= M + N - 1 = 4 + 4 - 1 = 7$

-ve index of $x(n) = 1$

-ve index of $h(n) = 0$

$\therefore$ Total -ve index after convolution $= 1 + 0 = 1$

$x(n) \rightarrow$	1	2	3	4
$h(n) \downarrow$	1	2	3	4
	1	2	3	4
	1	2	3	4
	3	6	9	12
	2	4	6	8

$y(-1) = 1$

$y(0) = 1 + 2 = 3$

$y(1) = 3 + 2 + 3 = 8$

$y(2) = 2 + 6 + 3 + 4 = 15$

$y(3) = 4 + 9 + 4 = 17$

$y(4) = 6 + 12 = 18$

$y(5) = 8$

$\therefore y(n) = \{1, 3, 8, 15, 17, 18, 8\}$

⑤ $h(n) = \{1, 2, 1, 2\}$ $x(n) = \{3, 2, 1, 2\}$

Ans: $y(n) = \{3, 8, 8, 12, 9, 4, 4\}$

$y(3) = 6$

$y(2) = 0$

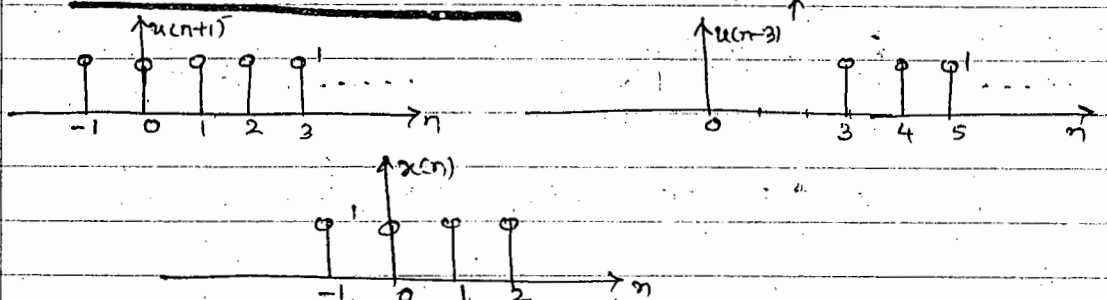
$y(1) = 0$

$y(4) = 4$

$y(0) = 8(n)$

$y(n-3)$

⑥ $x(n) = u(n+1) - u(n-3)$ $h(n) = \{1, 3, 2, -1, 1\}$



$x(n) = \{1, 1, 1, 1\}$

$y(1)$

$y(n-3)$

$$x(n) = \{1, 1, 1, 1\}$$

$$h(n) = \{1, 3, 2, -1, 1\}$$

$$M = 4$$

$$N = 5$$

$$Z = M + N - 1 = 08$$

$\therefore y(n) = x(n) * h(n)$ will have 8 elements

$$\text{-ve index of } y(n) = 1 + 1 = 02$$

hence the sequence of $y(n)$ starts from $y(-2)$

$x(n) \rightarrow$	1	1	1	1	
	1	3	2	-1	1
$\downarrow h(n)$	2	2	2	2	2
	-1	-1	-1	-1	-1
	1	1	1	1	1

$$y(n) = \{1, 4, 6, 5, 5, 2, 0, 1\}$$

(d) $x(n) = n+2 ; 0 \leq n \leq 3$ $h(n) = a^n u(n)$

$$\text{Let } y(n) = x(n) * h(n)$$

$$x(n) = n+2 ; 0 \leq n \leq 3$$

$$= \{2, 3, 4, 5\}$$

$$= 2\delta(n) + 3\delta(n-1) + 4\delta(n-2) + 5\delta(n-3)$$

$$y(n) = x(n) * h(n)$$

$$= [2\delta(n) + 3\delta(n-1) + 4\delta(n-2) + 5\delta(n-3)] * a^n u(n)$$

$$= 2a^n u(n) + 3a^{n-1} u(n-1) + 4a^{n-2} u(n-2) + 5a^{n-3} u(n-3)$$

$$y(n) = a^n \left[2u(n) + \frac{3}{a} u(n-1) + \frac{4}{a^2} u(n-2) + \frac{5}{a^3} u(n-3) \right]$$

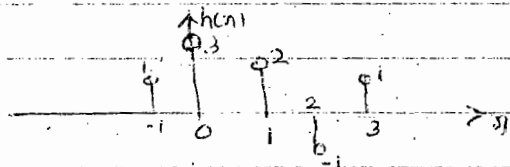
(e) A discrete LTI system has impulse response shown

below. Determine the

system output for

(a) $x(n] = 3\delta(n) - 2\delta(n-1)$

(b) $x(n] = x(n+1) - x(n-1)$



$$(a) \quad x(n) = 3\delta(n) - 2\delta(n-1)$$

$$= \{3, -2\}$$

$$h(n) = \{1, 3, 2, -1, 1\}$$

$$M=2, \quad N=5, \quad Z=M+N-1=6$$

$$\text{-ve index of } y(n) = x(n) * h(n) = 1$$

2) The o/p sequence starts from $y(-1)$

$\rightarrow h(n)$	1	3	2	-1	1	
$\downarrow x(n)$	3	3	9	6	-3	3
	-2	-2	-6	-4	2	2

$$y(n) = \{3, 7, 0, -7, 5, 2\}$$

⑤ Refer problem 4(c)

⑥ convolve $x(n) = \{1, 2, 3\}$ $h(n) = \{2, 1, 4\}$ using graphical method.

Step 1: represent the signals on time scale n .

Step 2: Represent the signals $x(n)$ & $h(n)$ as a fⁿ of independent variable k as $x(k)$ and $h(k)$

Step 3: fold the sequence $h(k)$ about $k=0$ to get $h(-k)$

Step 4: shift the reflected sequence $h(-k)$ to get $h(n-k)$

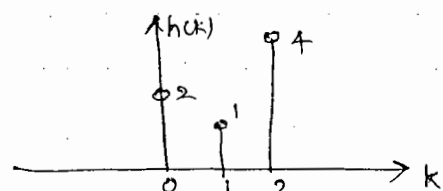
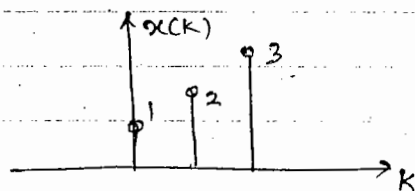
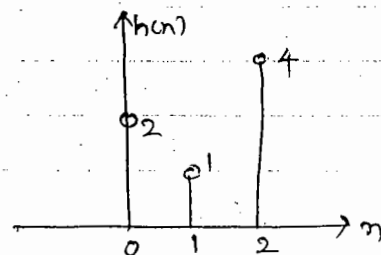
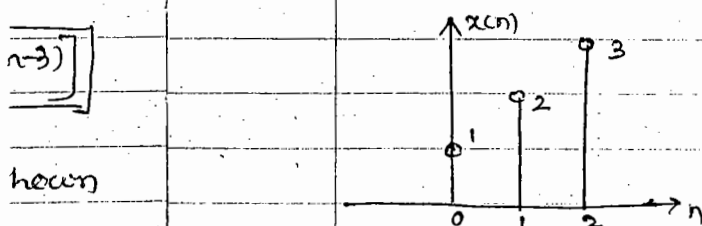
If n is +ve $\rightarrow$ shift to right

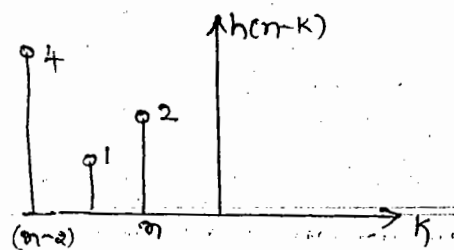
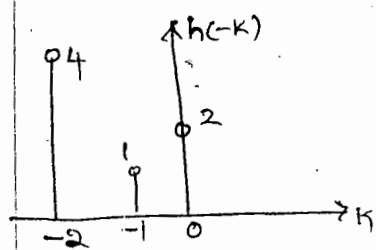
If n is -ve $\rightarrow$ shift to left

Step 5: multiply the sequences $x(k)$ and $h(n-k)$

Step 6: Add all the values of the product $x(k)h(n-k)$ to get $y(n)$ for one value of n .

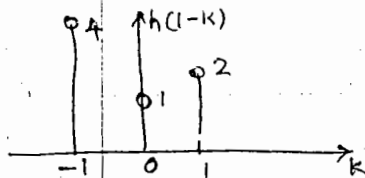
Step 6: Repeat steps 4, 5, 6 for all possible values of ' n '.





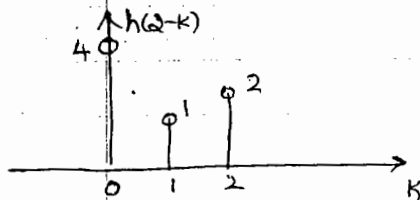
(7)

$$y(0) = 2 \cdot (1) = \underline{2} = x(k) h(n-k) \rightarrow \boxed{n=0}$$



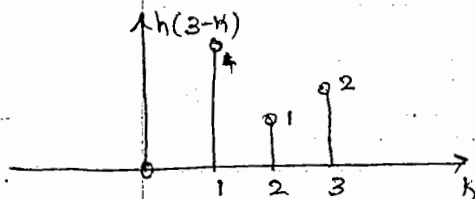
$$y(1) = x(k) h(n-k) = 1 \times 1 + 2 \times 2 + 4 \times 0 = 5$$

$$\rightarrow \boxed{n=1}$$



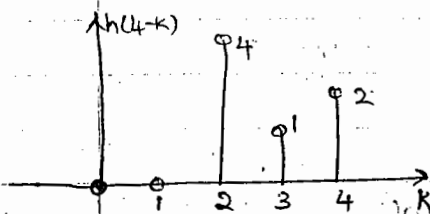
$$y(2) = 4 \times 1 + 1 \times 2 + 2 \times 3 = 12$$

$$\rightarrow \boxed{n=2}$$



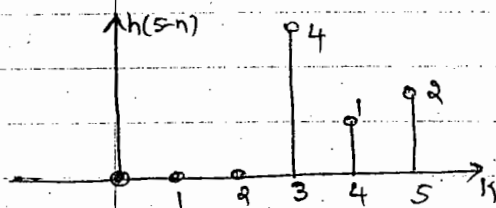
$$y(3) = 0 \times 1 + 4 \times 2 + 1 \times 3 + 3 \times 0 = 11$$

$$\rightarrow \boxed{n=3}$$



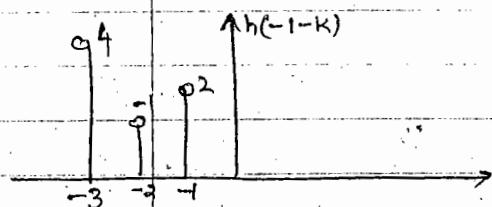
$$y(4) = 0 \times 1 + 0 \times 2 + 4 \times 3 + (1 \times 0 + 2 \times 0) = \underline{12}$$

$$\rightarrow \boxed{n=4}$$



$$y(5) = 0 \times 1 + 0 \times 2 + 0 \times 0 = \underline{0}$$

$$\rightarrow \boxed{n=5}$$



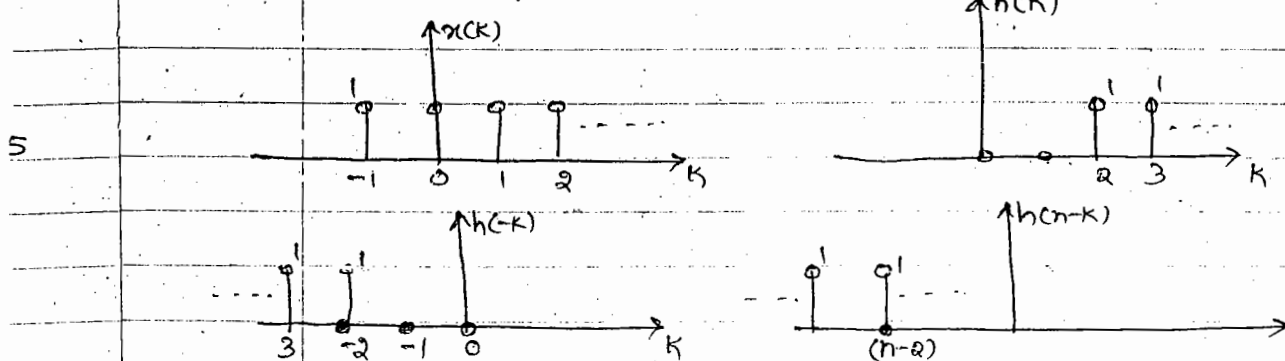
$$y(6) = \underline{0}$$

Hence convolution starts at $n=0$ & ends at $n=5$

$$\therefore y(n) = \{ \underset{\uparrow}{2}, 5, 12, 11, 12 \}$$

⑦ convolve $x(n) = u(n+1)$; $h(n) = u(n-2)$

Let $y(n) = x(n) * h(n)$



for $n < -1$; $y(n) = 0$ since there is no overlap.

for $n \geq -1$; convolution starts at $k = -1$ and ends at $(n-2)$

$$\therefore y(n) = \sum_{k=-1}^{n-2} x(k) h(n-k) = \sum_{k=-1}^{n-2} 1 \cdot 1$$

$$= (n-2) - (-1) + 1 \quad \text{put } l = k+1$$

$$\text{when } k = -1, l = 0$$

$$\text{when } k = n-2; l = n-2+1 = n-1$$

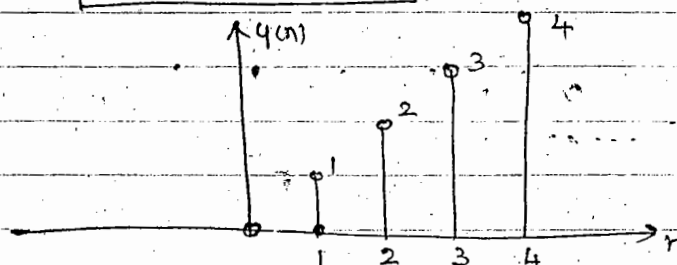
$$\therefore y(n) = \sum_{l=0}^{n-1} 1 = (n-1) + 1 = \underline{n}$$

$$\therefore y(n) = \begin{cases} n & ; -1 \leq n \leq n-2 \\ 0 & ; \text{otherwise} \end{cases}$$

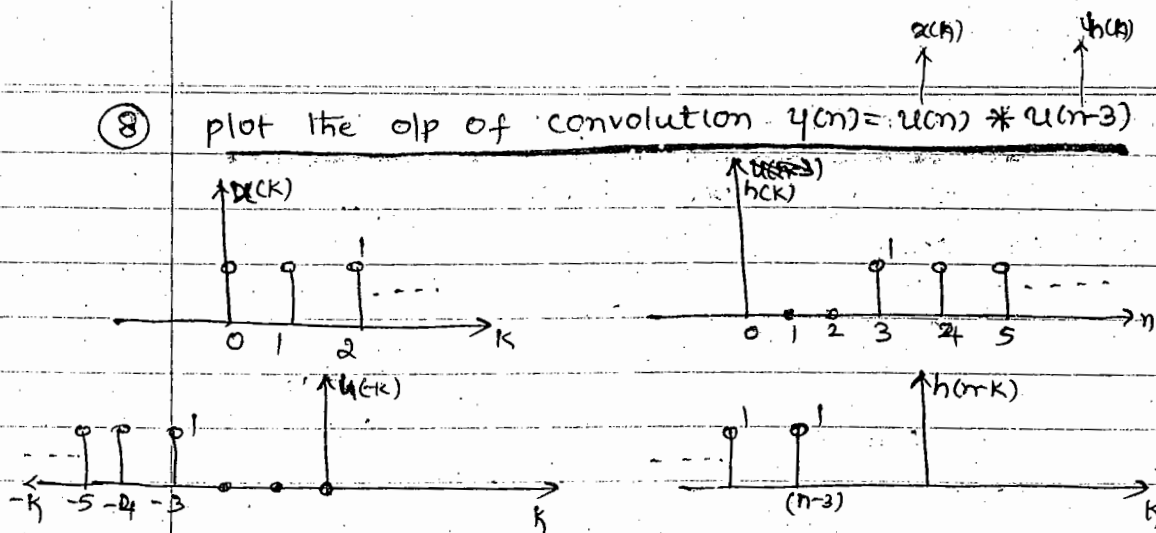
$$\text{hence } y(n) = \begin{cases} n & ; n \geq -1 \\ 0 & ; n < -1 \end{cases}$$

$$\boxed{y(n) = n \cdot u(n-1)}$$

at $n = 5$



⑧ plot the o/p of convolution $y(n) = u(n) * u(n-3)$



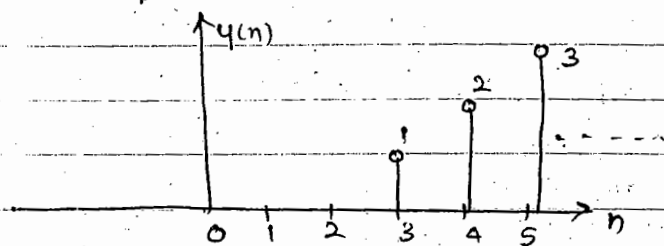
for $n < 3$, $y(n) = 0$ as there is no overlap.

for $n \geq 3$, convolution starts at $k=0$ and ends at $n-3$

$$y(n) = \sum_{k=0}^{n-3} x(k) h(n-k) = \sum_{k=0}^{n-3} 1 = n-3+1 = n-2$$

$$y(n) = n-2 \quad \text{for } n \geq 3$$

$$y(n) = (n-2) u(n-3)$$



⑨ $y(n) = x_1(n) * x_2(n)$ given $x_1(n) = u(n) - u(n-4)$
 $x_2(n) = n; 0 \leq n \leq 4$

$$x_1(n) = \{1, 1, 1, 1\} \quad x_2(n) = \{0, 1, 2, 3, 4\}$$

		$\rightarrow x_2(n)$				
		0	1	2	3	4
$\downarrow x_1(n)$	1	0	1	2	3	4
	1	0	1	2	3	4
	1	0	1	2	3	4
	1	0	1	2	3	4

$$y(n) = \{0, 1, 3, 6, 10, 9, 7, 4\}$$

⑩

⑪

4

$$= \alpha^n \left[\frac{1 - \alpha^{-(n+1)}}{1 - \alpha^{-1}} \right] =$$

$$\alpha^n \left[\frac{\alpha^{n+1} - 1}{\alpha^n (\alpha - 1)} \right]$$

$$= \alpha^n \left[\frac{\alpha^{n+1} - 1}{\alpha - 1} \right] \frac{\alpha}{\alpha^{n+1}} = \frac{\alpha^{n+1}}{\alpha^{n+1}} \left[\frac{\alpha^{n+1} - 1}{\alpha - 1} \right]$$

$$y(n) = \frac{\alpha^{n+1} - 1}{\alpha - 1} ; 0 \leq n \leq 4$$

for $4 \leq n \leq 6$ ($0 \leq k \leq 4$)

convolution starts at $k=0$ and ends at 4.

The sequence $h(n-k)$ is moved completely to the right.

The last 4 impulses of $h(n-k)$ overlap with $x(k)$ in this region resulting in max. convolution of p.

$$y(n) = \sum_{k=0}^4 x(k) h(n-k) = \sum_{k=0}^4 1 \cdot \alpha^{(n-k)}$$

$$= \alpha^n \sum_{k=0}^4 \left(\frac{1}{\alpha}\right)^k = \alpha^n \left[\frac{1 - \frac{1}{\alpha^5}}{1 - \frac{1}{\alpha}} \right]$$

$$= \alpha^n \left[\frac{\alpha^5 - 1}{\alpha - 1} \right] \frac{\alpha}{\alpha^5} = \alpha^{n-4} \left[\frac{\alpha^5 - 1}{\alpha - 1} \right]$$

$$y(n) = \frac{\alpha^{n+1} - \alpha^{n-4}}{\alpha - 1} ; 4 \leq n \leq 6$$

for $6 \leq n \leq 10$

In this range $h(n-k)$ moves further right, overlapping of samples decreases. Convolution starts at $n-6$ and ends at $k=4$.

$$y(n) = \sum_{k=n-6}^4 x(k) h(n-k) = \sum_{k=n-6}^4 1 \cdot \alpha^{(n-k)} = \alpha^n \sum_{k=n-6}^4 \left(\frac{1}{\alpha}\right)^k$$

put $r = k - n + 6$ $\therefore k = r + n - 6$

$\therefore n - k = 6 - r$

$$r = k - n + 6$$

when $k = n - 6$; $r = n - 6 - n + 6 = 0$

when $k = 4$; $r = 4 - n + 6 = 10 - n$

$$\therefore y(n) = \sum_{r=0}^{10-n} \left(\frac{1}{\alpha}\right)^{r+n-6} \cdot \alpha^n = \alpha^6 \sum_{r=0}^{10-n} \left(\frac{1}{\alpha}\right)^r$$

$$= \alpha^6 \left[\frac{1 - \left(\frac{1}{\alpha}\right)^{10-n+1}}{1 - \frac{1}{\alpha}} \right]$$

$$= \alpha^6 \left[\frac{1 - \alpha^{-(10-n+1)}}{1 - \alpha^{-1}} \right]$$

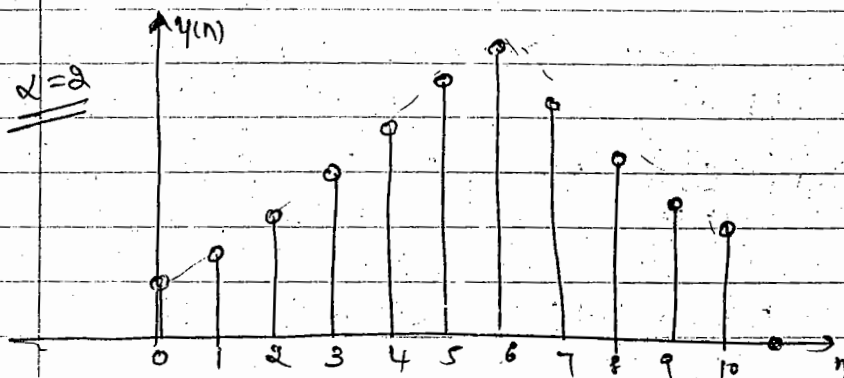
$$= \alpha^6 \left[\frac{1 - \alpha^{n-11}}{\alpha - 1} \right] \cdot \alpha = \alpha^7 \left[\frac{\alpha^{n-11} - 1}{1 - \alpha} \right]$$

$$y(n) = \frac{\alpha^{n-4} - \alpha^7}{1 - \alpha} ; 6 \leq n \leq 10$$

for $n > 10$; there is no overlap; hence $y(n) = 0$

$$y(n) = \begin{cases} 0; & n < 0 \text{ \& } n > 10 \\ \frac{1 - \alpha^{n+1}}{1 - \alpha}; & 0 \leq n \leq 4 \\ \frac{\alpha^{n-4} - \alpha^{n+1}}{1 - \alpha}; & 4 \leq n \leq 6 \\ \frac{\alpha^{n-4} - \alpha^7}{1 - \alpha}; & 6 \leq n \leq 10 \end{cases}$$

o/p for $\alpha = 2$

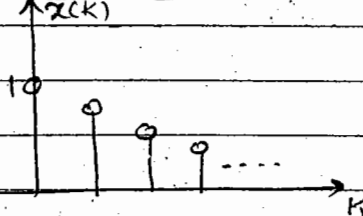


⑩

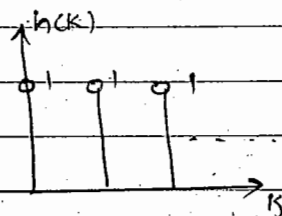
$$x(n) = \alpha^n u(n) \quad h(n) = u(n) \quad \& \quad 0 < \alpha < 1$$

$$\text{find } y(n) = x(n) * h(n)$$

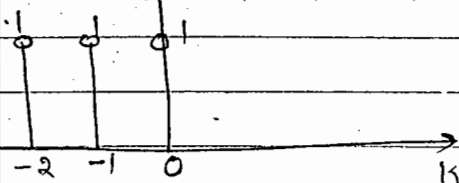
$x(k)$



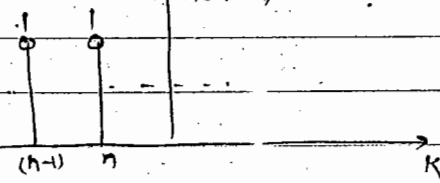
$h(k)$



$h(-k)$



$h(n-k)$



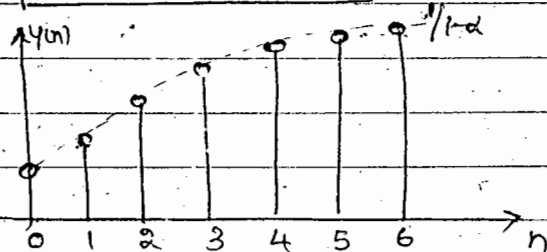
for $n < 0$; $y(n) = 0$ as there is no overlap.

for $n \geq 0$; convolution starts at $k=0$ & ends at $k=n$

$$\therefore y(n) = \sum_{k=0}^n x(k) h(n-k) = \sum_{k=0}^n \alpha^k u(k) \cdot u(n-k)$$

$$= \sum_{k=0}^n \alpha^k \cdot 1 = \frac{1 - \alpha^{n+1}}{1 - \alpha}; \quad n \geq 0$$

$$y(n) = \frac{1 - \alpha^{n+1}}{1 - \alpha} \cdot u(n)$$



⑪

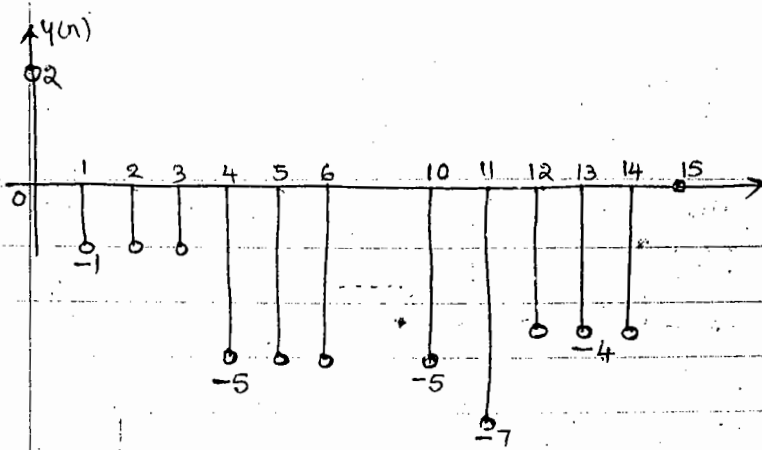
A discrete LTI system has a response $h(n) = u(n) - u(n-11)$
determine the system o/p if $x(n) = 2\delta(n) - 3\delta(n-1) - 4\delta(n-4)$

$$y(n) = x(n) * h(n)$$

$$= [2\delta(n) - 3\delta(n-1) - 4\delta(n-4)] * h(n)$$

$$= 2h(n) - 3h(n-1) - 4h(n-4) \quad \text{by distributive Law.}$$

$$y(n) = 2[u(n) - u(n-11)] - 3[u(n-1) - u(n-12)] - 4[u(n-4) - u(n-15)]$$

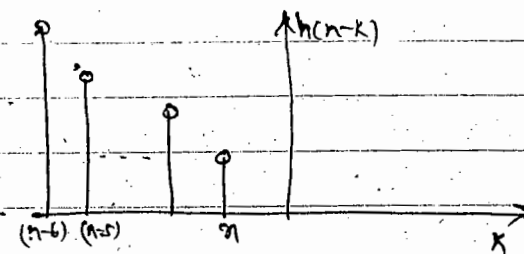
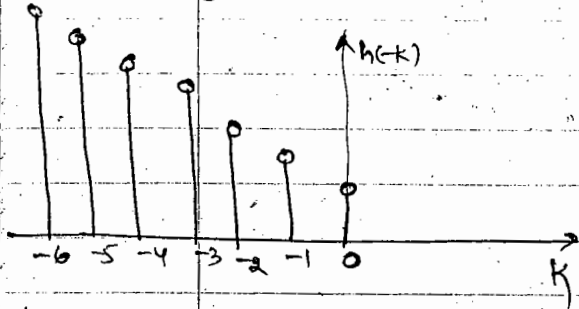
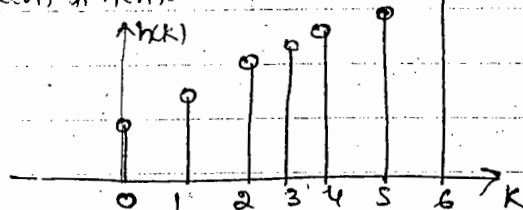
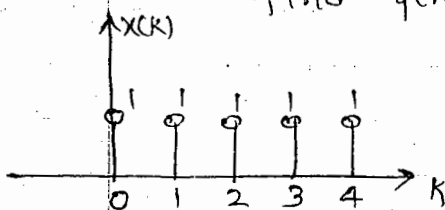


$$y(n) = \begin{cases} 2; & n=0 \\ -1; & 1 \leq n \leq 3 \\ -5; & 4 \leq n \leq 10 \\ -7; & n=11 \\ -4; & 12 \leq n \leq 14 \\ 0; & \text{otherwise} \end{cases}$$

12) $x(n) = \begin{cases} 1; & 0 \leq n \leq 4 \\ 0; & \text{otherwise} \end{cases}$

$$h(n) = \begin{cases} \alpha^n; & 0 \leq n \leq 6 \\ 0; & \text{otherwise} \end{cases}$$

find $y(n) = x(n) * h(n)$.



For $n < 0$; $y(n) = 0$ (no overlap)

for $0 \leq n \leq 4$

Convolution starts at $k=0$ and ends at n

$$y(n) = \sum_{k=0}^n x(k) h(n-k) = \sum_{k=0}^n 1 \cdot \alpha^{(n-k)} = \alpha^n \sum_{k=0}^n \alpha^{-k}$$

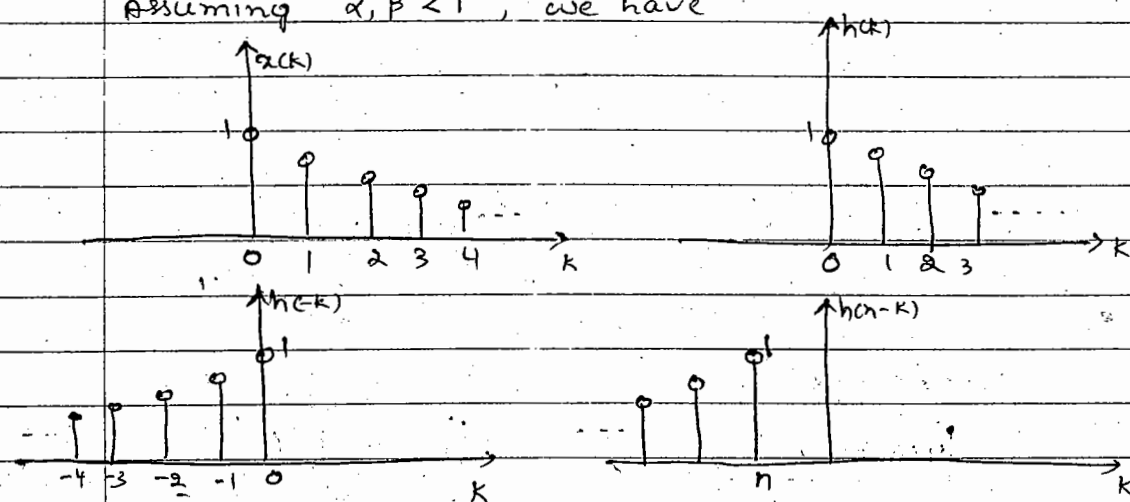
⑬ Convolve $x(n) = \alpha^n u(n)$ $h(n) = \beta^n u(n)$

Let $y(n) = x(n) * h(n)$

$$x(k) = \alpha^k u(k)$$

$$h(k) = \beta^k u(k)$$

Assuming $\alpha, \beta < 1$; we have



For $n < 0$, $y(n) = 0$ as there is no overlap of $x(k)$ & $h(n-k)$

For $0 \leq k \leq n$ or For $n \geq 0$, we have overlap.

convolution starts at $k=0$ & ends at $k=n$.

$$\begin{aligned} y(n) &= \sum_{k=0}^n x(k)h(n-k) = \sum_{k=0}^n \alpha^k u(k) \cdot \beta^{n-k} u(n-k) \\ &= \sum_{k=0}^n \left(\frac{\alpha}{\beta}\right)^k \cdot \beta^n u(k)u(n-k) = \beta^n \sum_{k=0}^n \left(\frac{\alpha}{\beta}\right)^k \cdot 1 \rightarrow \textcircled{A} \\ &= \beta^n \left[\frac{1 - \left(\frac{\alpha}{\beta}\right)^{n+1}}{1 - \frac{\alpha}{\beta}} \right] = \beta^n \left[\frac{\beta^{n+1} - \alpha^{n+1}}{\beta - \alpha} \right] \cdot \frac{\beta}{\beta^{n+1}} \end{aligned}$$

$$y(n) = \frac{\alpha^{n+1} - \beta^{n+1}}{\alpha - \beta}; \quad n \geq 0 \text{ \& \; } \alpha \neq \beta$$

When $\alpha = \beta$ put $\alpha = \beta$ in eqⁿ $\textcircled{A}$

$$y(n) = \beta^n \sum_{k=0}^n 1^k = \beta^n (n+1)$$

$$y(n) = \beta^n (1+n); \quad n \geq 0 \text{ with } \alpha = \beta$$

Method 2

$$x(n) = \alpha^n u(n) \quad h(n) = \beta^n u(n)$$

$$y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k) h(n-k)$$

$$= \sum_{k=-\infty}^{\infty} \alpha^k u(k) \beta^{n-k} u(n-k)$$

but $u(k) \cdot u(n-k) = \begin{cases} 1; & 0 \leq k \leq n \\ 0; & \text{otherwise} \end{cases}$

$$\therefore y(n) = \sum_{k=0}^n \alpha^k \beta^{n-k} \cdot 1$$

$$= \beta^n \sum_{k=0}^n \left(\frac{\alpha}{\beta}\right)^k$$

for $\alpha \neq \beta$

$$y(n) = \frac{\alpha^{n+1} - \beta^{n+1}}{\alpha - \beta}$$

for $\alpha = \beta$

$$y(n) = \beta^n (n+1)$$

(14) convolve $x(n) = \alpha^n u(n)$ $h(n) = u(n)$.

$$\text{Let } y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k) h(n-k)$$

$$= \sum_{k=-\infty}^{\infty} \alpha^k u(k) \cdot u(n-k)$$

$$u(k) u(n-k) = \begin{cases} 1; & 0 \leq k \leq n \\ 0; & \text{otherwise} \end{cases}$$

$$\therefore y(n) = \sum_{k=0}^n \alpha^k \cdot 1 = \frac{1 - \alpha^{n+1}}{1 - \alpha}; \quad n \geq 0$$

$$y(n) = \frac{\alpha^{n+1} - 1}{\alpha - 1}; \quad n \geq 0$$

(15) convolve $x(n) = \begin{cases} n+1; & n=0,1,2 \\ 0; & \text{otherwise} \end{cases}$ $h(n) = 2^n u(n)$

$$x(n) = \{1, 2, 3\} = \delta(n) + 2\delta(n-1) + 3\delta(n-2)$$

$$y(n) = x(n) * h(n)$$

$$= \delta(n) + 2\delta(n-1) + 3\delta(n-2) * 2^n u(n)$$



$$y(n) = 2^n u(n) + 2 \cdot 2^{n-1} u(n-1) + 3 \cdot 2^{n-2} u(n-2)$$

$$y(n) = 2^n \left[u(n) + u(n-1) + \frac{3}{4} u(n-2) \right]$$

(16) Given $x(n) = n+2$; $0 \leq n \leq 3$

$h(n) = a^n u(n)$; for all n

find $y(n) = x(n) * h(n)$

$$x(n) = \{2, 3, 4, 5\} = 2\delta(n) + 3\delta(n-1) + 4\delta(n-2) + 5\delta(n-3)$$

$$h(n) = a^n u(n)$$

→ Refer previous problem

Ans: $y(n) = a^n [2u(n) + 3u(n-1) + 4u(n-2) + 5u(n-3)]$

(17) $y(n) = x(n) * h(n)$ given $x(n) = \beta^n u(n)$ $h(n) = u(n-3)$

Assuming $\beta < 1$;

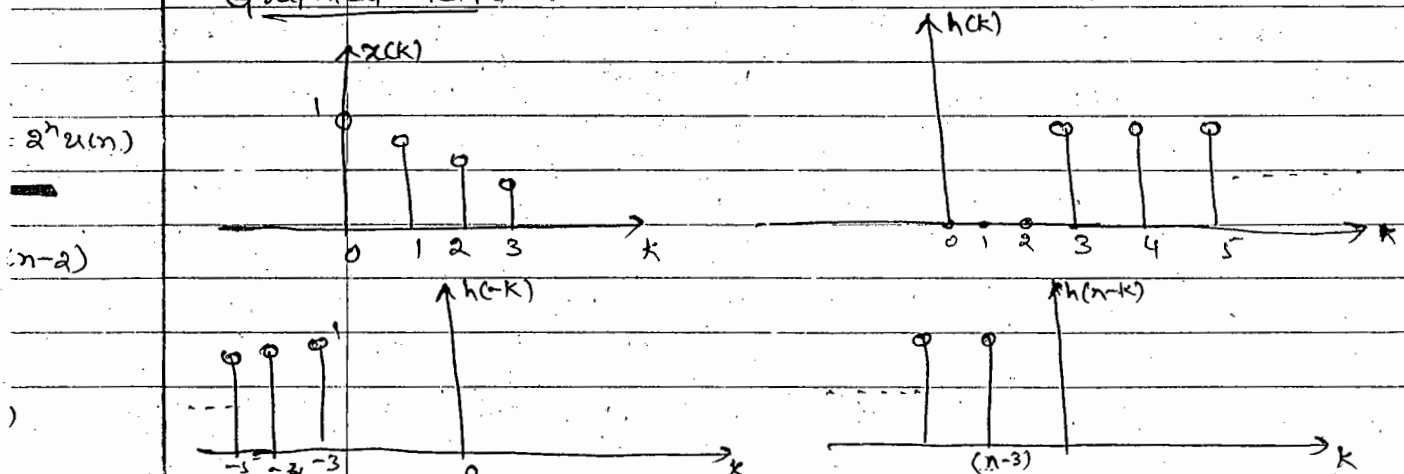
$$y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k) h(n-k) = \sum_{k=-\infty}^{\infty} \beta^k u(k) u(n-k-3)$$

$$u(k) \cdot u(n-k-3) = \begin{cases} 1 & ; 0 \leq k \leq n-3 \\ 0 & ; \text{otherwise} \end{cases}$$

$$y(n) = \sum_{k=0}^{n-3} \beta^k \cdot 1 = \frac{1 - \beta^{n-2}}{1 - \beta}$$

$$y(n) = \frac{1 - \beta^{n-2}}{1 - \beta} ; n \geq 3$$

Graphical method



for $n \leq 3$; $y(n) = 0$ as there is no overlap

for $n \geq 3$; convolution starts at $k=0$ & ends at $n-3$

$$\therefore y(n) = \sum_{k=0}^{n-3} x(k) \cdot h(n-k)$$

$$y(n) = \left[\frac{1 - \beta^{n-2}}{1 - \beta} \right] u(n-2)$$

(18)

Obtain the convolution sum of $x(n) = \alpha^n u(n)$ $h(n) = \alpha^{-n} u(n)$

(19)

$$\begin{aligned} y(n) &= x(n) * h(n) = \sum_{k=-\infty}^{\infty} \alpha^k u(k) \cdot \alpha^{-(n-k)} u(-(n-k)) \\ &= \sum_{k=-\infty}^{\infty} \alpha^{2k} \alpha^{-n} u(k) u(-n+k) \end{aligned}$$

we have $u(k) = \begin{cases} 0 & ; k < 0 \\ 1 & ; k \geq 0 \end{cases}$

$$u(-n+k) = \begin{cases} 1 & ; k \geq n \\ 0 & ; k < n \end{cases}$$

(20)

For $n \leq 0$

$$u(k) \cdot u(-n+k) = \begin{cases} 1 & ; k \geq 0 \\ 0 & ; \text{otherwise} \end{cases}$$

$$\therefore y(n) = \sum_{k=0}^{\infty} \alpha^{2k} \alpha^{-n} \cdot 1 = \alpha^{-n} \sum_{k=0}^{\infty} (\alpha^2)^k = \alpha^{-n} \left[\frac{1}{1 - \alpha^2} \right]$$

$$y(n) = \frac{\alpha^{-n}}{1 - \alpha^2}$$

For $n \geq 0$

$$u(k) \cdot u(-n+k) = \begin{cases} 1 & ; k \geq n \\ 0 & ; \text{otherwise} \end{cases}$$

$$y(n) = \sum_{k=n}^{\infty} \alpha^{2k} \alpha^{-n} (1)$$

put $l = k - n$; when $k = n$; $l = 0$
when $k = \infty$; $l = \infty$

$$\therefore y(n) = \alpha^{-n} \sum_{l=0}^{\infty} \alpha^{2(l+n)} = \alpha^{-n} \sum_{l=0}^{\infty} \alpha^{2l} \alpha^{2n}$$

$$= x^n \sum_{k=0}^{\infty} x^{2k} = \frac{x^n}{1-x^2}$$

at $n=3$

$$y(n) = \frac{x^n}{1-x^2} \text{ for } n \geq 0$$

$$y(n) = \frac{x^{|n|}}{1-x^2} \text{ for all } n$$

$y(n) = x^n u(n)$

(19) compute $x_1(n) = u(n)$ $x_2(n) = u(n)$.

$$y(n) = x_1(n) * x_2(n)$$

$$y(n) = \sum_{k=-\infty}^{\infty} x(k) h(n-k) = \sum_{k=-\infty}^{\infty} u(k) \cdot u(n-k)$$

$$= \sum_{k=0}^n 1 = (n+1)$$

$$y(n) = (n+1)u(n)$$

$$u(k) u(n-k) = \begin{cases} 1 & ; n \geq 0 \\ 0 & ; \text{otherwise} \end{cases}$$

(20) Given $x_1(n) = (\frac{1}{2})^n u(n)$ $x_2(n) = \{1, 1, 1\}$ compute the value of $y(n) = x_1(n) * x_2(n)$ for the range $0 \leq n \leq 5$

$$y(n) = x_1(n) * x_2(n)$$

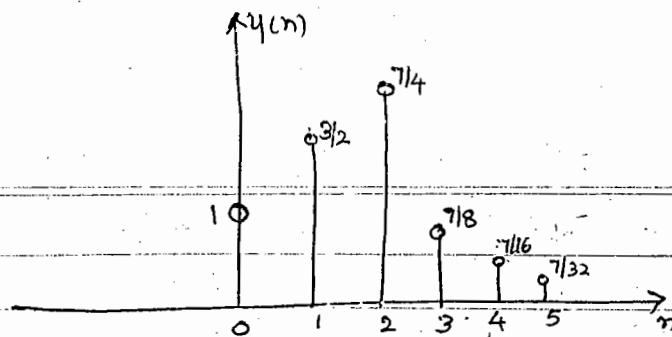
$$= (\frac{1}{2})^n u(n) * \{1, 1, 1\}$$

$$= (\frac{1}{2})^n u(n) * [\delta(n) + \delta(n-1) + \delta(n-2)]$$

$$y(n) = (\frac{1}{2})^n u(n) + (\frac{1}{2})^{n-1} u(n-1) + (\frac{1}{2})^{n-2} u(n-2)$$

n	$(\frac{1}{2})^n u(n)$	$(\frac{1}{2})^{n-1} u(n-1)$	$(\frac{1}{2})^{n-2} u(n-2)$	$y(n)$
0	1	0	0	1
1	1/2	1	0	1.5
2	1/4	1/2	1	1.75
3	1/8	1/4	1/2	7/8
4	1/16	1/8	1/4	7/16
5	1/32	1/16	1/8	7/32

$$y[n] = \left[\frac{1}{1-x^2} \right]$$



21) compute $y(n) = \beta^n u(n) * \alpha^n u(n-10)$; $|\alpha|, |\beta| < 1$

$$y(n) = \beta^n u(n) * \alpha^n u(n-10)$$

for $n < 10$; $y(n) = 0$

for $10 \leq n \leq (n-10)$

$$y(n) = \sum_{k=0}^{n-10} \beta^k u(k) \alpha^{n-k} u(n-k-10)$$

$$y(n) = \alpha^n \sum_{k=0}^{n-10} \left(\frac{\beta}{\alpha}\right)^k \cdot 1 \quad \text{--- (A)}$$

$$y(n) = \alpha^n \left[\frac{1 - \left(\frac{\beta}{\alpha}\right)^{n-10+1}}{1 - \beta/\alpha} \right]$$

$$y(n) = \alpha^n \left[\frac{1 - \left(\frac{\beta}{\alpha}\right)^{n-9}}{1 - \beta/\alpha} \right]; \quad \alpha \neq \beta$$

for $\alpha = \beta$; use this value in eqn (A)

$$y(n) = \alpha^n \sum_{k=0}^{n-10} 1 = \alpha^n (n-10+1)$$

$$y(n) = \alpha^n (n-9)$$

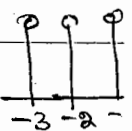
22) $y(n) = \left(\frac{1}{4}\right)^n u(n) * u(n+2)$

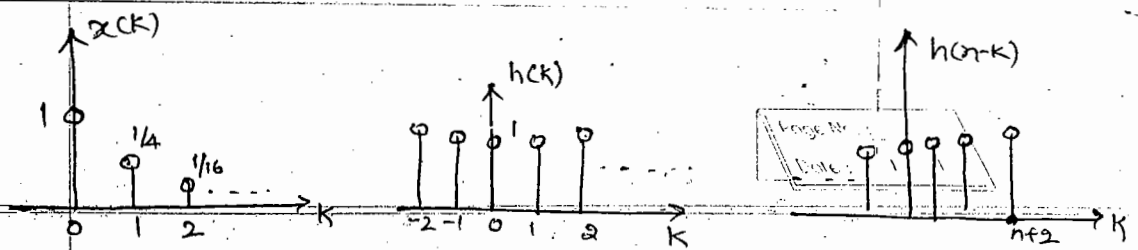
for $(n+2) < 0$

$$y(n) = 0$$

for $(n+2) \geq 0$; convolution starts at $k=0$ & ends at $k=n+2$.

$$y(n) = \sum_{k=-\infty}^{\infty} \left(\frac{1}{4}\right)^k u(k) u(n-k+2)$$





$$y(n) = \sum_{k=0}^{n+2} \left(\frac{1}{4}\right)^k \cdot 1 = \frac{1 - \left(\frac{1}{4}\right)^{n+2+1}}{1 - \frac{1}{4}}$$

$$y(n) = \frac{4}{3} \left[1 - \left(\frac{1}{4}\right)^{n+3} \right] ; n \geq -2$$

(23)

$x(n) = u(n+3)$ $h(n) = u(n+3)$ compute the value of $y(n) = x(n) * h(n)$

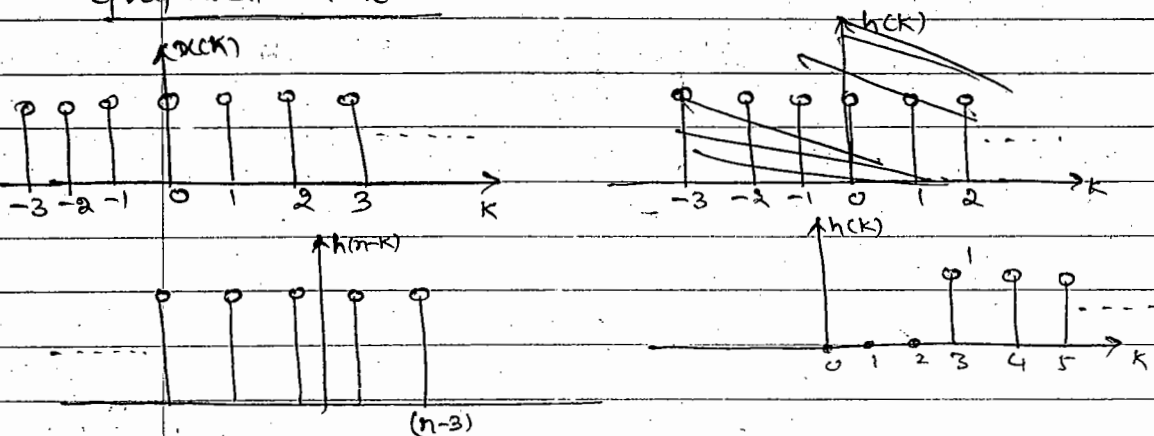
$$y(n) = \sum_{k=-\infty}^{\infty} x(k) h(n-k) = \sum_{k=-\infty}^{\infty} u(k+3) u(n-k+3)$$

$$u(k+3) u(n-k+3) = \begin{cases} 1 & -3 \leq k \leq n-3 \\ 0 & \text{otherwise} \end{cases}$$

$$y(n) = \sum_{k=-3}^{n-3} 1 = (n-3) + 1 + 3 = n+1$$

$$y(n) = (n+1) u(n) \quad n \geq 0$$

Graphical method



for $n < -3$, $y(n) = 0$.

for $n \geq -3$, convolution starts at $k=-3$ & ends $n-3$.

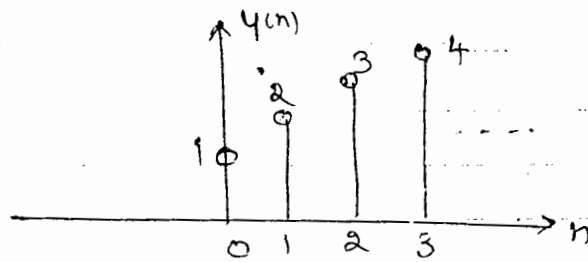
= 0 & ends

$$y(n) = \sum_{k=-3}^{n-3} 1 \cdot 1 = n+1$$

$$y(n) = \sum_{l=0}^n 1 = \underline{\underline{n+1}}$$

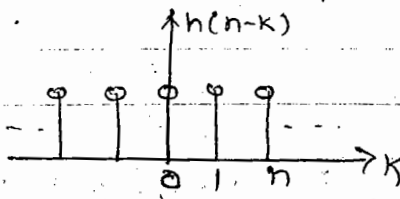
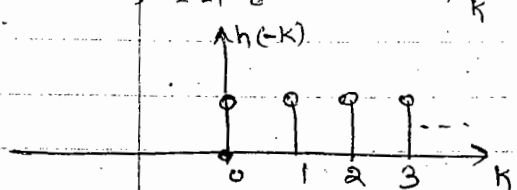
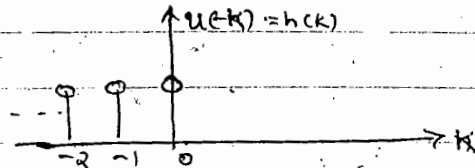
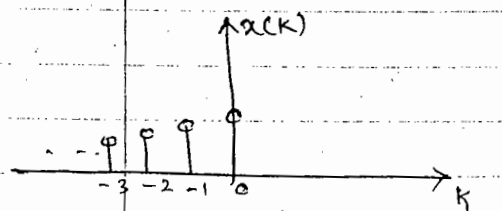
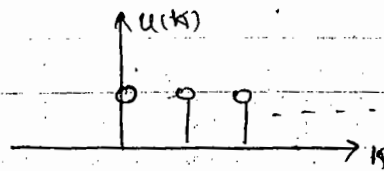
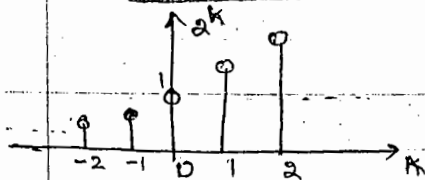
put $l = k+3$
when $k=-3$, $l=0$
when $k=n-3$,
 $l = n-3+3 = n$

$$\therefore y(n) = (n+1) \cdot u(n)$$



Q4

$x(n) = 2^n u(-n)$ $h(n) = u(n)$ compute $y(n) = x(n) * h(n)$



For $n < 0$

Convolution starts at $-\infty$ and ends at $k=n$

$$\begin{aligned} y(n) &= \sum_{k=-\infty}^{\infty} 2^k u(-k) u(n-k) = \sum_{k=-\infty}^{\infty} 2^k \cdot 1 \\ &= \sum_{k=-n}^{\infty} 2^k \end{aligned}$$

put $m = k+n$ or $k = m-n$

when $k = -n$, $m = 0$
when $k = \infty$, $m = \infty$

$$y(n) = \sum_{m=0}^{\infty} 2^{-(m-n)} = 2^n \sum_{m=0}^{\infty} 2^{-m} = 2^n \sum_{m=0}^{\infty} \left(\frac{1}{2}\right)^m$$

$$y(n) = 2^n \cdot \left[\frac{1}{1 - \frac{1}{2}} \right] = 2^{n+1}$$

$$y(n) = 2^{n+1} ; n < 0$$

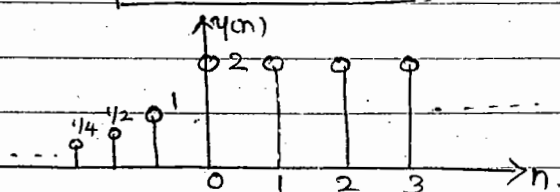
For $n \geq 0$

Convolution starts from $k = -\infty$ & ends at $k = 0$

$$y(n) = \sum_{k=-\infty}^0 2^k = \sum_{k=0}^{\infty} 2^{-k} = \frac{1}{1 - \frac{1}{2}} = \underline{\underline{2}}$$

$$y(n) = 2 ; n \geq 0$$

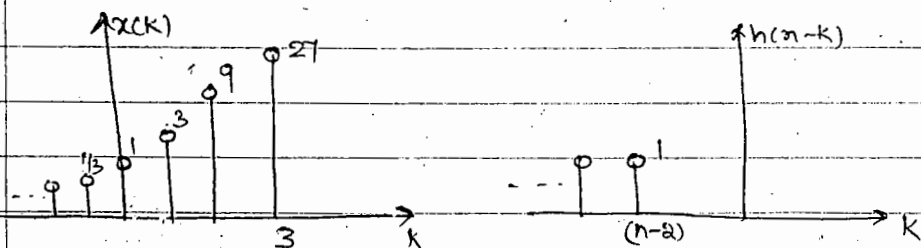
$$\text{or } y(n) = 2 u(n)$$



25)

$$y(n) = 3^n u(-n+3) * u(n-2)$$

$$\text{Let } x(n) = 3^n u(-n+3) \quad h(n) = u(n-2)$$



Convolution sum has a value from $-\infty$ to $n-2$.

$$y(n) = \sum_{k=-\infty}^{n-2} x(k) h(n-k) = \sum_{k=-\infty}^{n-2} 3^k u(-k+3) \cdot u(n-k-2)$$

$$y(n) = \sum_{k=-\infty}^{n-2} 3^k \cdot 1 = \sum_{k=-n+2}^{\infty} 3^k$$

$$\text{put } m = k + n - 2 \quad \therefore k = m - n + 2$$

Limits of m are 0 to ∞

$$3^n \sum_{m=0}^{\infty} \left(\frac{1}{3}\right)^m$$

$$y(n) = \sum_{m=0}^{\infty} \left(\frac{1}{3}\right)^{m-n+2} = \left(\frac{1}{3}\right)^{-n+2} \sum_{m=0}^{\infty} \left(\frac{1}{3}\right)^m = \left(\frac{1}{3}\right)^{-n+2} \left[\frac{1}{1 - \frac{1}{3}} \right]$$

$$y(n) = \frac{3^n}{9} \cdot \frac{3}{2} = \frac{3^n}{6} ; (n-2) \leq 3$$

$$y(n) = \frac{3^n}{6} ; n \leq 5$$

For $(n-2) \geq 4$ or $n \geq 6$

convolution starts from $k=-\infty$ & ends at $k=3$

$$y(n) = \sum_{k=-\infty}^3 3^k \cdot 1 = \sum_{k=-3}^{\infty} 3^k$$

put $l = k+3$ limits of l are 0 to ∞
or $k = l-3$.

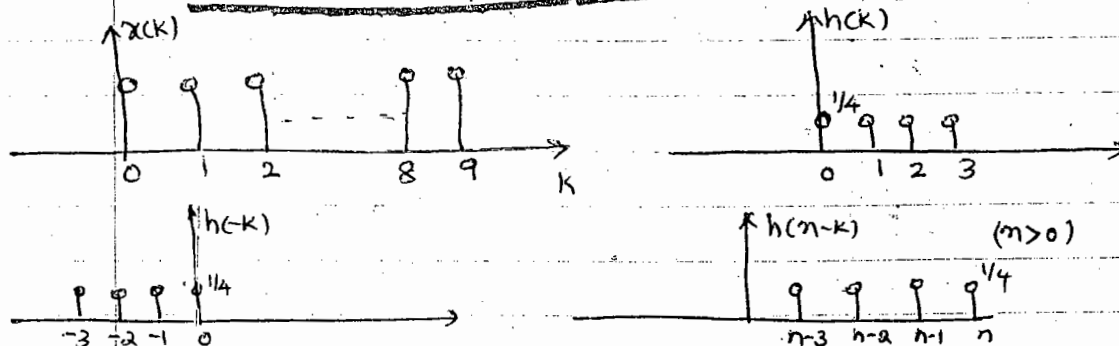
$$y(n) = \sum_{l=0}^{\infty} (3)^{-(l-3)} = 3^3 \sum_{l=0}^{\infty} \left(\frac{1}{3}\right)^l = 81 \left[\frac{1}{1-\frac{1}{3}} \right]$$

$$y(n) = \frac{81}{2} ; n \geq 6$$

(26)

Given $x(n) = u(n) - u(n-10)$

$h(n) = \frac{1}{4} [u(n) - u(n-4)]$ find $y(n) = x(n) * h(n)$



For $n < 0$; there is no overlap hence $y(n) = 0$

$$y(n) = 0 ; n < 0$$

For $0 \leq n \leq 3$; overlapping starts at $k=0$ & ends at n

$$\begin{aligned} y(n) &= \sum_{k=0}^n x(k) \cdot h(n-k) = \sum_{k=0}^n [u(k) - u(k-10)] \cdot \frac{1}{4} [u(n-k) - u(n-k-4)] \\ &= \sum_{k=0}^n [u(k) - u(k-10)] \cdot \frac{1}{4} [u(n-k) - u(n-k-4)] \\ &= \sum_{k=0}^n \frac{1}{4} (1) = \frac{1}{4} (n+1) ; 0 \leq n \leq 3. \end{aligned}$$

$$y(n) = \frac{n+1}{4} ; 0 \leq n \leq 3$$

for $n > 3$; all impulses of $h(n-k)$ crossed $k=0$ and maximum convolution takes place
convolution starts from $(n-3)$ till $k=n$. in
the range $3 \leq n \leq 9$

$$y(n) = \sum_{k=n-3}^n \left(\frac{1}{4}\right) \cdot 1$$

put $k-n+3 = m$

Lts of m are 0 & 3

$$y(n) = \sum_{m=0}^3 \left(\frac{1}{4}\right) \cdot 1 = \frac{1}{4} \sum_{m=0}^3 1$$

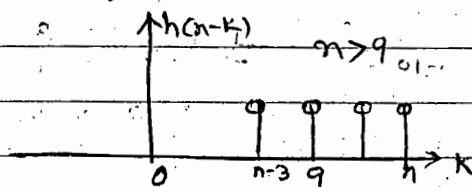
$$= \frac{1}{4} [1+1+1+1] = 1$$

$$y(n) = 1 ; 3 \leq n \leq 9$$

n)

For $n > 9$; the overlapping of sequences reduces
for $9 \leq n \leq 12$

$$y(n) = \sum_{k=n-3}^9 \left(\frac{1}{4}\right) \cdot 1$$



$$y(n) = \frac{9 - (n-3) + 1}{4} = \frac{13-n}{4} ; 9 \leq n \leq 12$$

for $n > 12$; there is no overlapping $y(n) = 0$.

III method

$$x(n) = \{1, 1, 1, 1, 1, 1, 1, 1, 1, 1\} \quad n=10$$

$$h(n) = \left\{\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right\} \quad N=4$$

h at n

~~h at n~~

≤ 3

		$x(n)$									
		1	1	1	1	1	1	1	1	1	1
$\downarrow$ $h(n)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

$$y(n) = \left\{\frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1, 1, 1, 1, 1, 1, \frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right\}$$

compute $y(n) = x(n) * h(n)$ with

$$x(n) = 0.25 [u(n) - u(n-4)] \quad h(n) = \eta; 0 \leq n \leq 4$$

(28)

→ $h(n)$

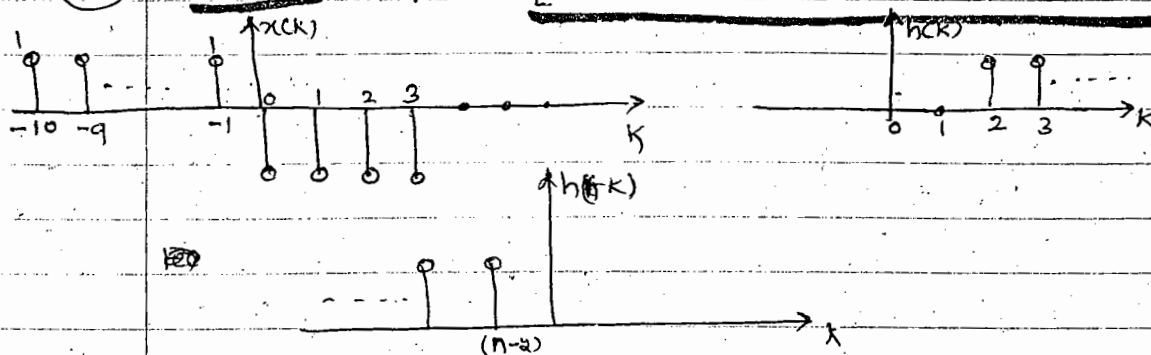
	0	1	2	3	4
$\downarrow x(n)$	1/4	0	1/4	1/2	3/4
	1/4	0	1/4	1/2	3/4
	1/4	0	1/4	1/2	3/4
	1/4	0	1/4	1/2	3/4

$$h(n) = \{0, 1, 2, 3, 4\}$$

$$y(n) = \left\{ 0, \frac{1}{4}, \frac{3}{4}, \frac{3}{2}, \frac{5}{2}, \frac{9}{4}, \frac{7}{4}, 1 \right\}$$

(27)

compute $y(n) = [u(n+10) - 2u(n) + u(n-4)] * u(n-2)$



Coroll

for $(n-2) < -8$; $y(n) = 0$

for $n-2 > 0$; convolution has a definite value.

For $-8 < n < 2$

$$y(n) = \sum_{k=-10}^{n-2} 1 = n-2+10+1$$

$$y(n) = n+9$$

for $2 \leq n \leq 5$;

$$y(n) = \sum_{k=-10}^{n-2} x(k) \cdot h(n-k) = \sum_{k=-10}^{-1} 1 + \sum_{k=0}^{n-2} (-1)$$

$$= (9+1) - (n-2+1) = 11-n$$

$$y(n) = 11-n$$

for $n \geq 6$;

$$y(n) = \sum_{k=-10}^{-1} 1 + \sum_{k=0}^3 (-1) = (9+1) - (3+1) = \underline{\underline{6}}$$

$$y(n) = \begin{cases} n+9 & ; -8 \leq n \leq 2 \\ 11-n & ; 2 \leq n \leq 5 \\ 6 & ; n > 6 \end{cases}$$

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(28)

show that if input $x(n)$ is periodic with period 'N' then the o/p of the LTI system $y(n)$ is also periodic

$n \leq 4$

43

$$\begin{aligned} \text{Let } y(n) &= x(n) * h(n) = h(n) * x(n) \\ &= \sum_{k=-\infty}^{\infty} h(k) x(n-k) \end{aligned}$$

Let $n = P + N$ where 'N' is the period

$$y(P+N) = \sum_{k=-\infty}^{\infty} h(k) x(P+N-k)$$

Since $x(n)$ is periodic with period N
i.e. $x(n+N) = x(n)$

(n-2)

$$x(P-k+N) = x(P-k)$$

$$\therefore y(P+N) = \sum_{k=-\infty}^{\infty} h(k) \cdot x(P-k)$$

$$y(P+N) = h(k) * x(k) \quad \text{--- R.H.S.}$$

$$\therefore y(n) = y(n+N)$$

Corollary:- convolution of finite seq with infinite sequence ~~can~~ ^{can} be finite. or Infinite length.

(a) Let $x(n) = \delta(n)$; $h(n) = \alpha^n u(n)$

$$\begin{aligned} x(n) * h(n) &= \delta(n) * \alpha^n u(n) \\ &= \alpha^n u(n) \quad \text{--- Infinite length} \end{aligned}$$

(b) Let $x(n) = \delta(n) - \delta(n-1)$ $h(n) = \alpha^n u(n)$

$$x(n) * h(n) = [\delta(n) - \delta(n-1)] * \alpha^n u(n)$$

$$= \alpha^n u(n) - \alpha^{n-1} u(n-1)$$

$$= \alpha^n \left[u(n) - \frac{1}{\alpha} u(n-1) \right]$$

→ finite sequence of finite length.

6

Convolution of CT signals.

Convolution integral.

The weighted superposition of time shifted impulse responses for a corresponding weight shifted inputs is known as convolution integral for a CT LTI system

proof:

Mathematically
$$y(t) = \int_{-\infty}^{\infty} x(\tau) \cdot h(t-\tau) d\tau = x(t) * h(t)$$

Expression for convolution Integral

A CT signal $x(t)$ can be expressed as the weighted superposition of shifted impulses

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \cdot \delta(t-\tau) d\tau$$

where ' τ ' is the integral variable.

Let $H\{\cdot\}$ be the system operator of a system to which the $x(t)$ is applied as input. The o/p $y(t)$ can be given as

proof:

$$y(t) = H\{x(t)\} = H\left\{\int_{-\infty}^{\infty} x(\tau) \delta(t-\tau) d\tau\right\}$$

using the linearity property

$$y(t) = \int_{-\infty}^{\infty} x(\tau) H\{\delta(t-\tau)\} d\tau$$

The impulse response can be written as

$$h(t) = H\{\delta(t)\}$$

$$\Rightarrow h(t-\tau) = H\{\delta(t-\tau)\}$$

using these values we get

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau = x(t) * h(t)$$

The above eqⁿ is mathematical representation of the convolution integral for a CT LTI system.

LHS =

properties of convolution Integral

- (a) commutative property
- (b) Associative property
- (c) Distributive property.

commutative property

inputs

TI system

*h(t)

weighted

$$x(t) * h(t) = h(t) * x(t)$$

proof: we have $x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$

put $t-\tau = k$ or $\tau = t-k$

$$x(t) * h(t) = \int_{-\infty}^{\infty} x(t-k) \cdot h(k) dk = h(t) * x(t)$$

$$\boxed{x(t) * h(t) = h(t) * x(t)}$$

Associative property

$$x(t) * [h_1(t) * h_2(t)] = [x(t) * h_1(t)] * h_2(t)$$

proof: Let $h(t) = h_1(t) * h_2(t) = \int_{-\infty}^{\infty} h_1(\lambda) h_2(t-\lambda) d\lambda$

$$h(t-\tau) = \int_{-\infty}^{\infty} h_1(\lambda) h_2(t-\tau-\lambda) d\lambda \quad \text{--- (A)}$$

$$x(t) * [h_1(t) * h_2(t)] = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau \quad \text{--- (B)}$$

using eqn (A) in (B)

$$x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) \left[\int_{-\infty}^{\infty} h_1(\lambda) h_2(t-\tau-\lambda) d\lambda \right] d\tau$$

put $\tau + \lambda = \theta$ $\lambda = \theta - \tau$

$d\lambda = d\theta$ & using linearity property;

$$\text{LHS} = x(t) * h(t) = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x(\tau) \cdot h_1(\theta-\tau) d\tau \right] h_2(t-\theta) d\theta$$

$$= \int_{-\infty}^{\infty} z(\theta) \cdot h_2(t-\theta) d\theta$$

where $z(t) = x(t) * h_1(t)$

$$= z(t) * h_2(t) = [x(t) * h_1(t)] * h_2(t)$$

→ RHS

Distributive property

②

$$x(t) * [h_1(t) + h_2(t)] = x(t) * h_1(t) + x(t) * h_2(t)$$

proof:

$$\text{Let } h(t) = h_1(t) + h_2(t)$$

$$h(t-\tau) = h_1(t-\tau) + h_2(t-\tau)$$

$$\text{LHS: } x(t) * h_1(t) + h_2(t) = x(t) * h(t)$$

$$= \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

$$= \int_{-\infty}^{\infty} x(\tau) h_1(t-\tau) d\tau + \int_{-\infty}^{\infty} x(\tau) h_2(t-\tau) d\tau$$

③

$$= x(t) * h_1(t) + x(t) * h_2(t) \longrightarrow \text{RHS.}$$

pro. lems

①

$$\text{P.T } (a) x(t) * \delta(t) = x(t)$$

$$(b) x(t) * \delta(t-t_0) = x(t-t_0)$$

$$(a) x(t) * \delta(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t-\tau) d\tau$$

$$\delta(t-\tau) = \begin{cases} 1; & t=\tau \\ 0; & t \neq \tau \end{cases}$$

④

$$\therefore x(t) * \delta(t) = x(\tau) \Big|_{\tau=t} = x(t) \longrightarrow \text{RHS}$$

$$(b) x(t) * \delta(t-t_0) = \int_{-\infty}^{\infty} x(\tau) \delta(t-\tau-t_0) d\tau$$

$$\delta(t-(\tau+t_0)) = \begin{cases} 1; & \tau = t-t_0 \\ 0; & \tau \neq t-t_0 \end{cases}$$

①

$$= x(\tau) \Big|_{\tau=t-t_0} = x(t-t_0)$$

$$\boxed{x(t) * \delta(t-t_0) = x(t-t_0)}$$

7) ② If $y(t) = x(t) * h(t)$; show that

$x(t) * h(t-t_0) = y(t-t_0)$

we have $y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$

||| ④ $x(t) * h(t-t_0) = \int_{-\infty}^{\infty} x(\tau) h(t-t_0-\tau) d\tau$
 $= y(t-t_0) \rightarrow \text{RHS}$
 (by comparison)

③ S.T. $\int_{-\infty}^{t-t_0} x(\tau) d\tau = x(t) * u(t-t_0)$

RHS $x(t) * u(t-t_0) = \int_{-\infty}^{\infty} x(\tau) \cdot u(t-t_0-\tau) d\tau$

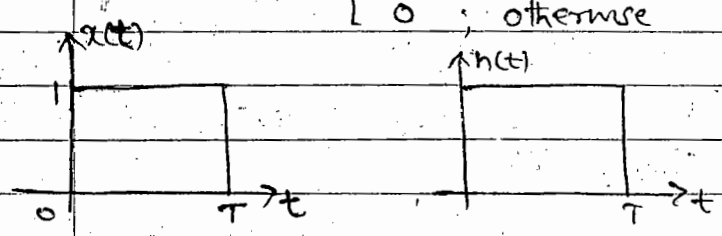
$u(t-\tau-t_0) = \begin{cases} 1; \tau < t-t_0 \\ 0; \tau > t-t_0 \end{cases}$

$\therefore x(t) * u(t-t_0) = \int_{-\infty}^{t-t_0} x(\tau) \cdot 1 d\tau \rightarrow \text{LHS}$

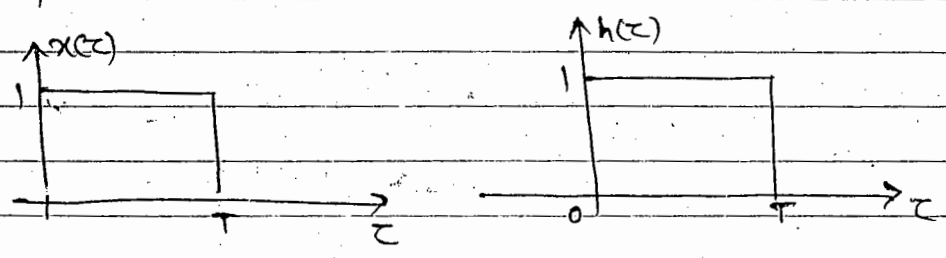
④ Find the convolution of $x(t)$ and $h(t)$

$x(t) = \begin{cases} 1; 0 \leq t \leq T \\ 0; \text{otherwise} \end{cases}$ $h(t) = \begin{cases} 1; 0 \leq t \leq T \\ 0; \text{otherwise} \end{cases}$

RHS

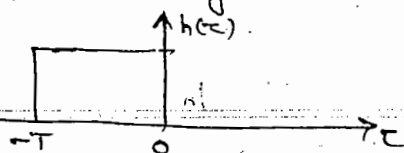


① Represent $x(t)$ and $h(t)$ as a fⁿ independent variable τ



Step 2

Fold the signal $h(\tau)$ to get $h(-\tau)$



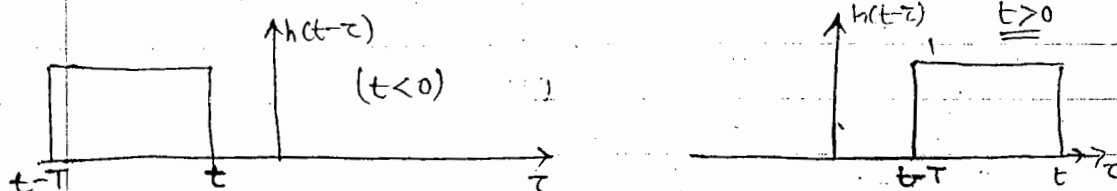
Step 3:- shift the signal $h(-\tau)$ on variable t on ' τ ' base

i.e represent $h(t-\tau)$

where ' t ' is the shifting parameter

for $t > 0$; $h(t-\tau)$ will be shifted to right

$t < 0$; $h(t-\tau)$ will be shifted to left.



(5)

Step 4 check for overlap.

for $t < 0$; there is no common area b/n $x(\tau)$ & $h(t-\tau)$

So/2:

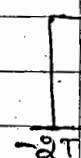
$$\therefore y(t) = 0$$

for $0 \leq t \leq T$

there exists a common area b/n $x(\tau)$ & $h(t-\tau)$

convolution starts at $\tau=0$ & ends at $\tau=t$.

$$y(t) = \int_{\tau=-\infty}^{\infty} x(\tau) h(t-\tau) d\tau = \int_{\tau=0}^t 1 \cdot 1 d\tau = \underline{t}$$



for $T \leq t \leq 2T$

convolution starts at $\tau = t-T$ & ends at $\tau = T$

$$y(t) = \int_{\tau=t-T}^T x(\tau) h(t-\tau) d\tau = \tau \Big|_{t-T}^T = T - t + T$$

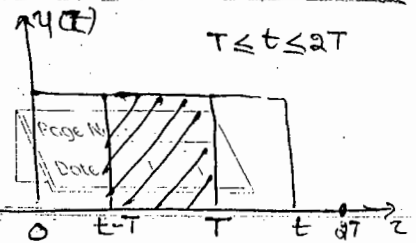
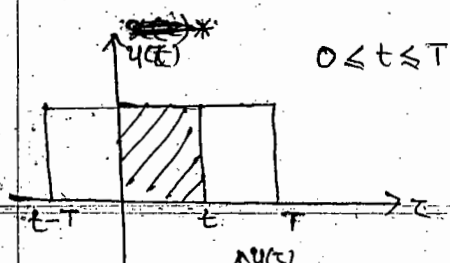
for

$$y(t) = 2T - t$$

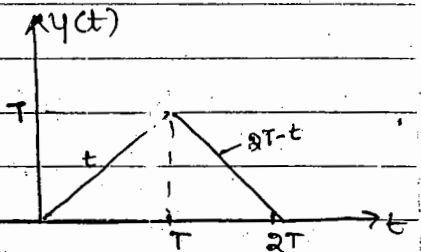
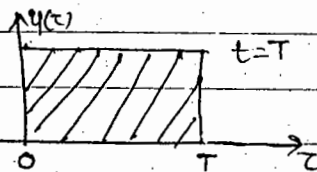
for $t \geq 2T$;

$$y(t) = 0$$

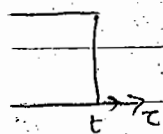
$$\therefore y(t) = \begin{cases} t; & 0 \leq t \leq T \\ 2T - t; & T < t \leq 2T \\ 0; & \text{otherwise} \end{cases}$$



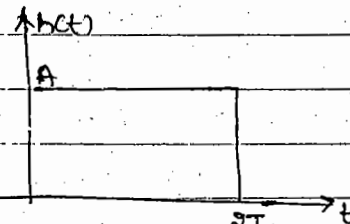
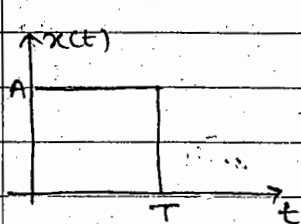
base



$t > 0$



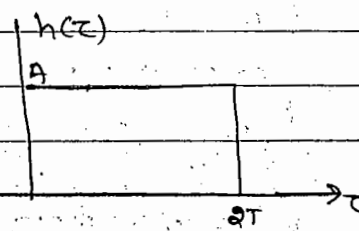
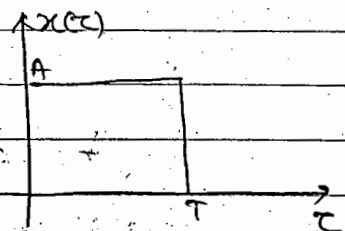
⑤



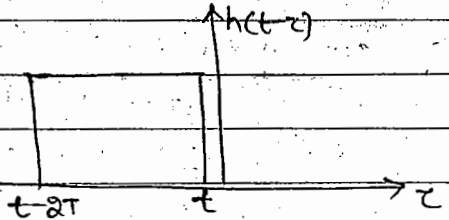
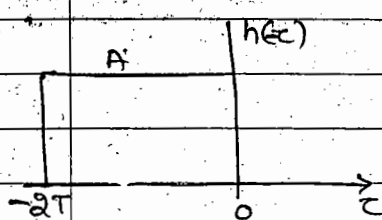
find $y(t) = x(t) * h(t)$

$h(t-\tau)$

So/2:



$h(t-\tau)$



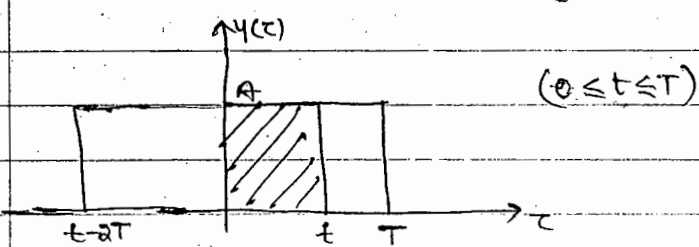
For $t < 0$; There is no common area b/w $x(\tau)$ & $h(t-\tau)$
hence $y(t) = 0$

$\tau = T$

FT

for $0 \leq t \leq T$ control starts at $\tau = 0$ & ends at $\tau = t$

$$y(t) = \int_{\tau=0}^t x(\tau) h(t-\tau) d\tau = \int_{\tau=0}^t A \cdot A \cdot d\tau = \underline{A^2 t}$$

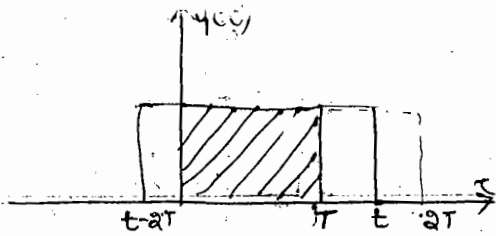


$(0 \leq t \leq T)$

for $T \leq t \leq 2T$

$$y(t) = \int_{\tau=0}^T x(\tau) h(t-\tau) d\tau$$

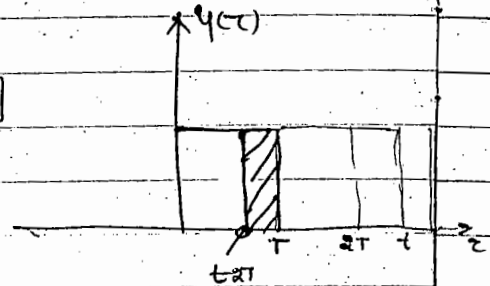
$$= A^2 T$$



for $2T \leq t \leq 3T$;

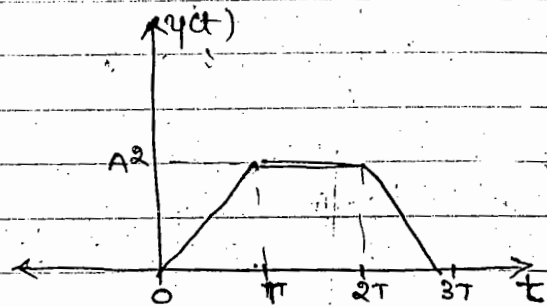
$$y(t) = \int_{\tau=t-2T}^T A^2 d\tau = A^2 [T - t + 2T]$$

$$y(t) = (3T - t) A^2$$



for $t > 3T$; $y(t) = 0$

$$y(t) = \begin{cases} A^2 t & ; 0 \leq t \leq T \\ A^2 T & ; T \leq t \leq 2T \\ A^2 (3T - t) & ; 2T \leq t \leq 3T \\ 0 & ; \text{otherwise} \end{cases}$$



⑥ Find $x(t) * y(t)$ for signals given below. also sketch the convolved signal.

$$x(t) = \delta(t) - 2\delta(t-1) + 8\delta(t-2)$$

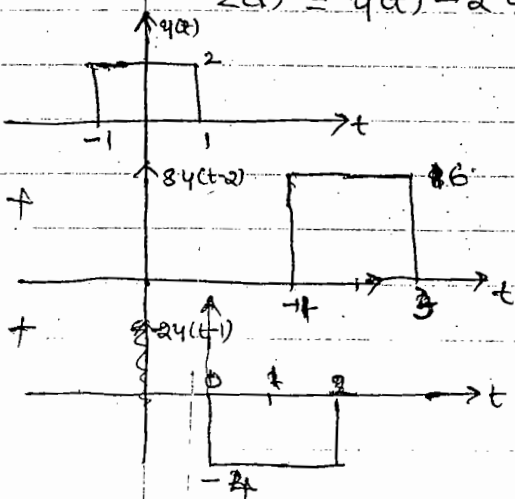
$$y(t) = 2; -1 \leq t \leq 1$$

Solⁿ:

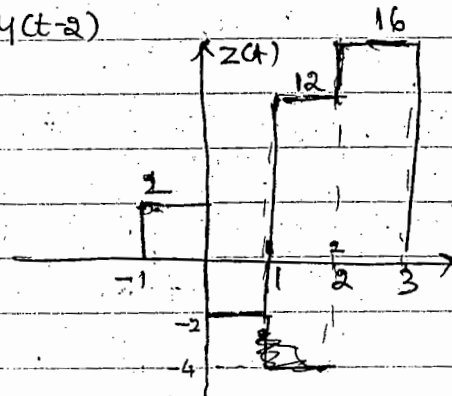
$$\text{Let } z(t) = x(t) * y(t)$$

$$= [\delta(t) - 2\delta(t-1) + 8\delta(t-2)] * y(t)$$

$$z(t) = y(t) - 2y(t-1) + 8y(t-2)$$



$\Rightarrow$



for

7

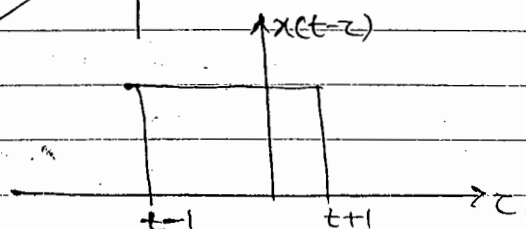
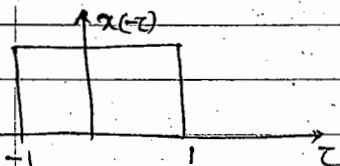
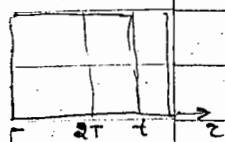
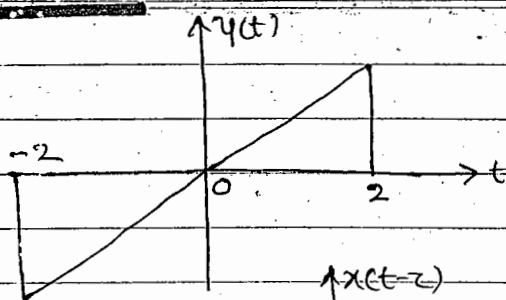
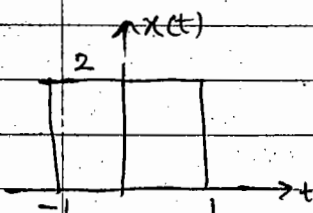
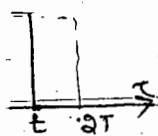
$$x(t) = 2; \quad -1 \leq t \leq 1$$

$$y(t) = t; \quad -2 \leq t \leq 2$$

find $z(t) = x(t) * y(t)$

Page No.:

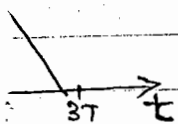
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for $t < -3$, there is no common area b/w $x(\tau)$ & $y(\tau)$

hence $z(t) = x(t) * y(t) = 0$

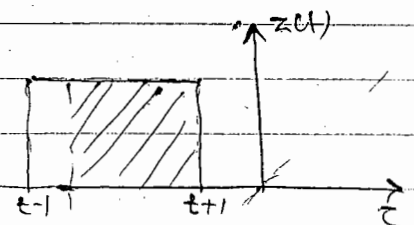
for $-3 \leq t \leq -1$



Sketch the

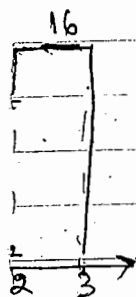
$$z(t) = \int_{\tau=-2}^{t+1} x(\tau) \cdot y(t-\tau) d\tau = \int_{\tau=-2}^{t+1} 2 \cdot \tau \cdot d\tau$$

$$= 2 \left[\frac{\tau^2}{2} \right]_{\tau=-2}^{t+1} = t^2 + 2t + 3$$



$z(t) = t^2 + 2t + 3$

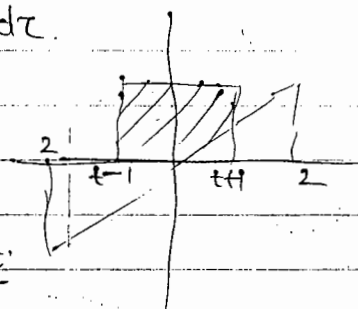
for $-1 \leq t \leq 1$



$$z(t) = \int_{\tau=t-1}^{t+1} x(\tau) y(t-\tau) d\tau = \int_{\tau=t-1}^{t+1} 2 \cdot \tau \cdot d\tau$$

$$= 2 \left[\frac{\tau^2}{2} \right]_{\tau=t-1}^{t+1} = 2 \left[\frac{(t+1)^2}{2} - \frac{(t-1)^2}{2} \right]$$

$$= t^2 + 1 + 2t - t^2 - 1 + 2t = 4t$$

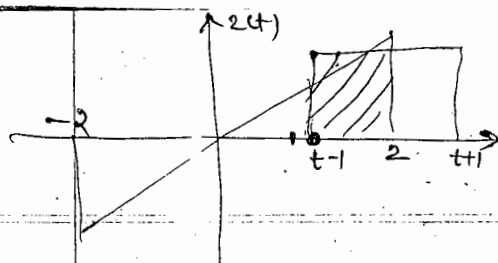


$z(t) = 4t$

for $1 \leq t \leq 3$

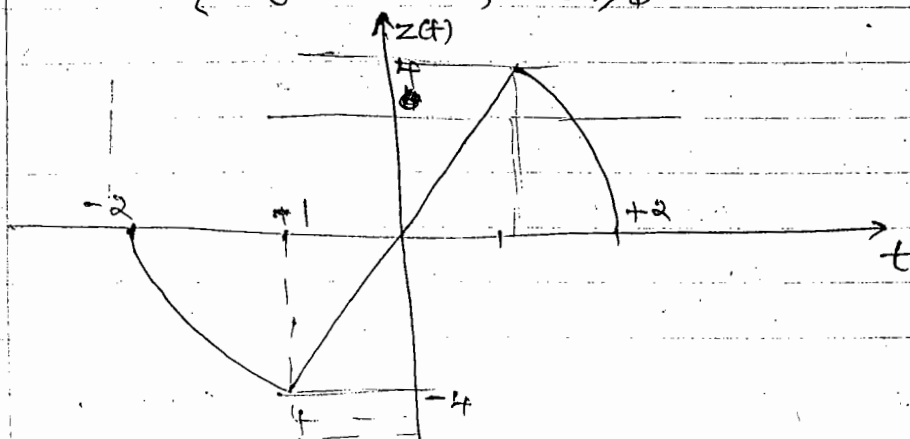
$$z(t) = \int_{\tau=t-1}^2 x(\tau) y(t-\tau) d\tau = \int_{\tau=t-1}^2 2 \cdot \tau \cdot d\tau = 2 \left[\frac{\tau^2}{2} \right]_{\tau=t-1}^2 = 4 - t^2 + 1 + 2t$$

$z(t) = -t^2 + 2t + 3$



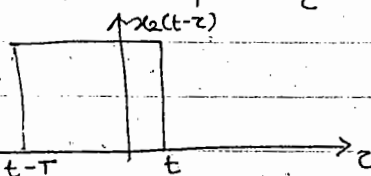
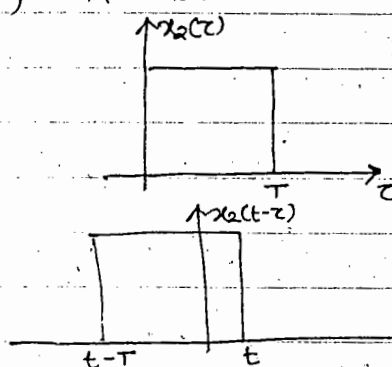
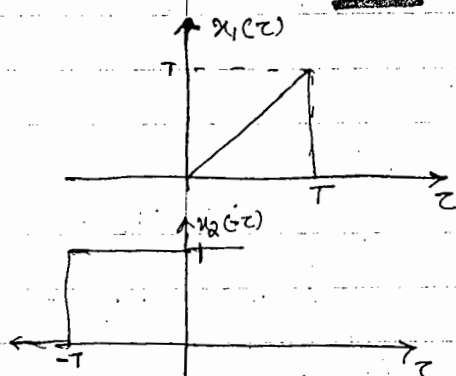
for $t \geq 3$; $\boxed{z(t) = 0}$

$$\therefore z(t) = \begin{cases} t^2 + 2t - 3 & ; -3 \leq t \leq -1 \\ 4t & ; -1 \leq t \leq 1 \\ -t^2 + 2t + 3 & ; 1 \leq t \leq 3 \\ 0 & ; t \geq 3 \end{cases}$$



⑧ Find the convolution of $x_1(t) = \begin{cases} t; & 0 \leq t \leq T \\ 0; & \text{otherwise} \end{cases}$

$$x_2(t) = \begin{cases} 1; & 0 \leq t \leq T \\ 0; & \text{otherwise} \end{cases}$$



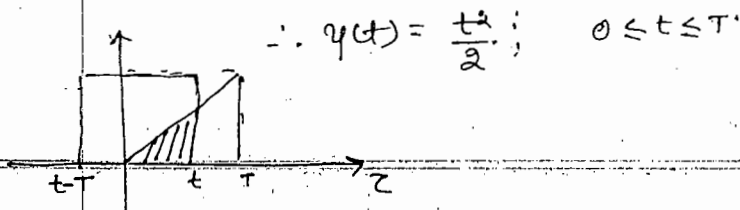
for $t < 0$; There is no common area b/w $x_1(z) + x_2(t-z)$

hence the convolution result $y(t) = x_1(t) * x_2(t) = 0$

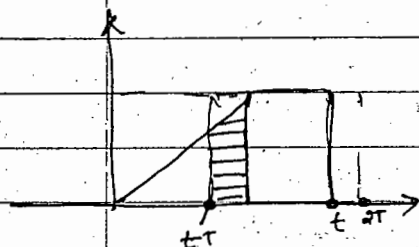
$$\boxed{\therefore y(t) = 0} ; t < 0$$

for $0 \leq t \leq T$; convolution starts at $\tau = 0$ and ends at $\tau = T$.

$$\therefore y(t) = \int_{\tau=0}^t x_1(\tau) x_2(t-\tau) d\tau = \int_{\tau=0}^t \tau \cdot 1 d\tau = \frac{t^2}{2}$$



for $T \leq t \leq 2T$, convolution starts at $\tau = t-T$ and ends at $\tau = T$.



$$y(t) = \int_{\tau=t-T}^T x_1(\tau) x_2(t-\tau) d\tau$$

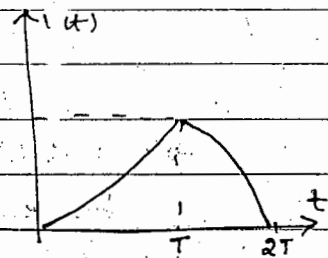
$$= \int_{\tau=t-T}^T \tau \cdot 1 d\tau = \frac{\tau^2}{2} \Big|_{t-T}^T$$

$$= \frac{1}{2} [T^2 - t^2 + 2Tt]$$

$$y(t) = t \left[T - \frac{t}{2} \right]$$

for $t \geq 2T$; There is no common area b/w $x_1(\tau)$ & $x_2(t-\tau)$ hence output $y(t) = 0$

$$y(t) = \begin{cases} \frac{t^2}{2} & ; 0 \leq t \leq T \\ t(T - \frac{t}{2}) & ; T \leq t \leq 2T \\ 0 & ; t < 0 \text{ or } t > 2T \end{cases}$$



Q) convolve $x(t) = e^{-at} u(t)$; $h(t) = e^{-bt} u(t)$, & $a \neq b$

Sol: Let $y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$

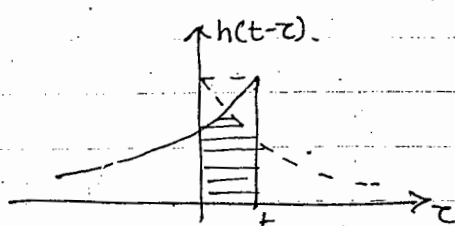
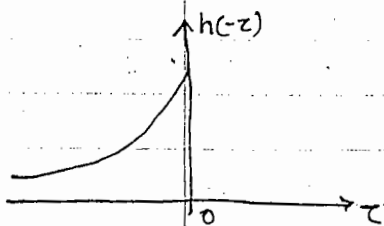
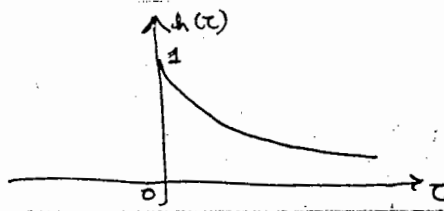
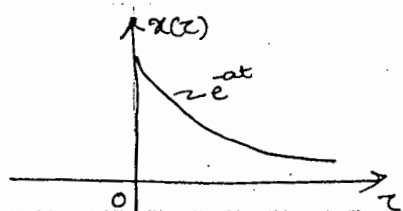
$$= \int_{-\infty}^{\infty} e^{-a\tau} u(\tau) e^{-b(t-\tau)} u(t-\tau) d\tau = \int_{-\infty}^{\infty} e^{-bt} e^{-\tau(a-b)} u(\tau) u(t-\tau) d\tau$$

but $u(\tau) u(t-\tau) = \begin{cases} 1; & 0 \leq \tau \leq t \\ 0; & \text{otherwise} \end{cases}$

$$\therefore y(t) = \int_{\tau=0}^t e^{-bt} e^{-\tau(a-b)} \cdot 1 \cdot d\tau = e^{-bt} \cdot \frac{e^{-\tau(a-b)}}{-(a-b)} \Big|_0^t$$

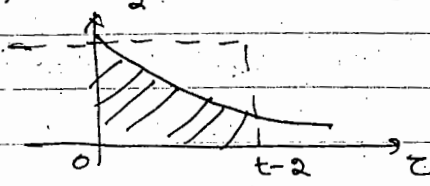
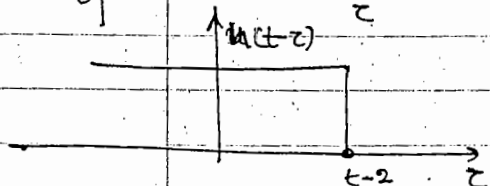
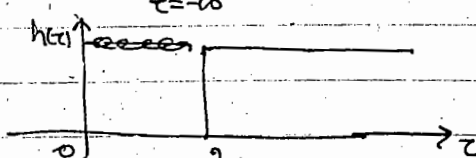
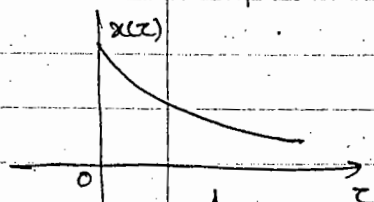
$$y(t) = e^{-bt} \left[\frac{e^{-(b-a)t} - 1}{b-a} \right]; \quad t \geq 0$$

$$= \frac{t^2}{2}$$



⑩ Convolve $x(t) = e^{-3t} u(t)$ $h(t) = u(t-2)$

Let $y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$



for $t < 2$; There is no common area of convolution Integral
hence $y(t) = 0 ; t \leq 2$

for $t \geq 2$; there exists a common area b/w $x(z)$ & $h(t-z)$, hence convolution starts from $\tau=0$ and ends at $\tau=t-2$

$$y(t) = \int_{\tau=0}^{t-2} x(\tau) \cdot h(t-\tau) d\tau = \int_{\tau=0}^{t-2} e^{-3\tau} \cdot \underbrace{u(\tau) \cdot u(t-\tau-2)}_1 d\tau$$

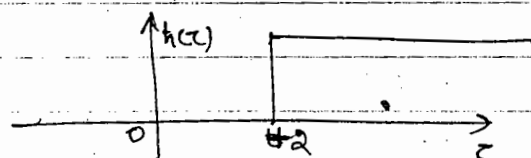
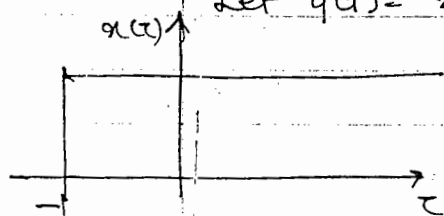
$$= \int_{\tau=0}^{t-2} e^{-3\tau} d\tau = \left. \frac{-e^{-3\tau}}{-3} \right|_0^{t-2}$$

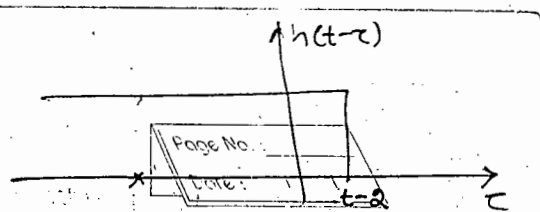
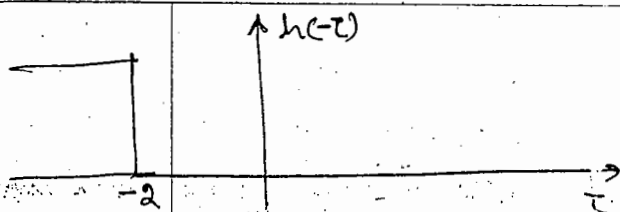
$$y(t) = \left[\frac{1 - e^{-3t}}{3} \right] u(t-2)$$

⑪ Find the output $y(t) = u(t+1) * u(t-2)$

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Let $y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$

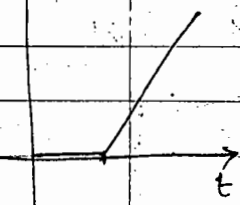




for $t < -1$; $y(t) = 0$

for $t \geq -1$; there exists a common area b/n $h(t-\tau)$ & $x(\tau)$
convolution starts from $\tau = -1$ and ends at $\tau = t-2$

$$y(t) = \int_{\tau=-1}^{t-2} x(\tau) h(t-\tau) d\tau = \int_{-1}^{t-2} 1 d\tau = t-2+1$$

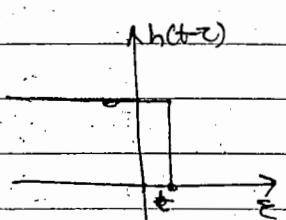
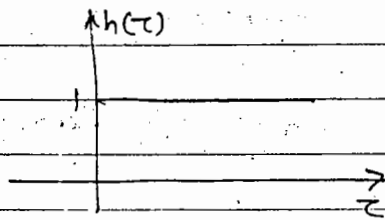
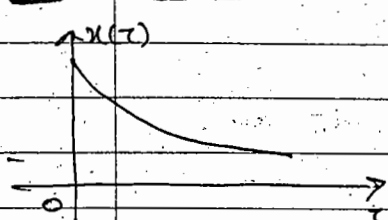


$$y(t) = t-1; t \geq -1$$

or

$$y(t) = (t-1)u(t+1)$$

(12) find $y(t) = x(t) * h(t)$ with $x(t) = e^{-at}u(t)$, $h(t) = u(t)$



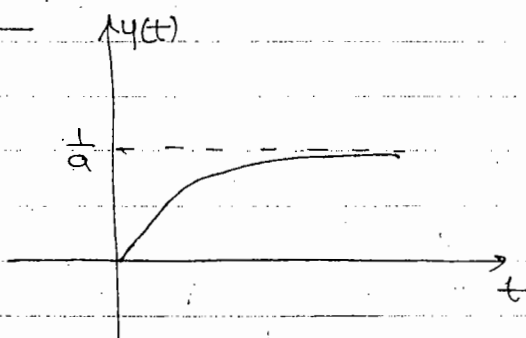
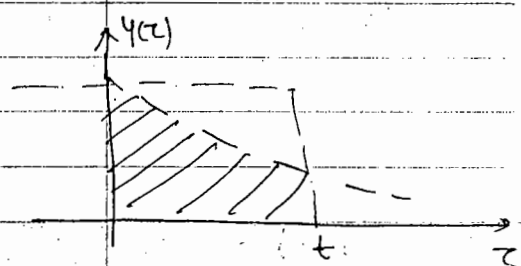
for $t < 0$; no convolution hence $y(t) = 0$

for $t \geq 0$; convolution starts from $\tau = 0$ & ends at $\tau = t$

$$y(t) = \int_{\tau=0}^t x(\tau) h(t-\tau) d\tau = \int_{\tau=0}^t e^{-a\tau} u(\tau) u(t-\tau) d\tau$$

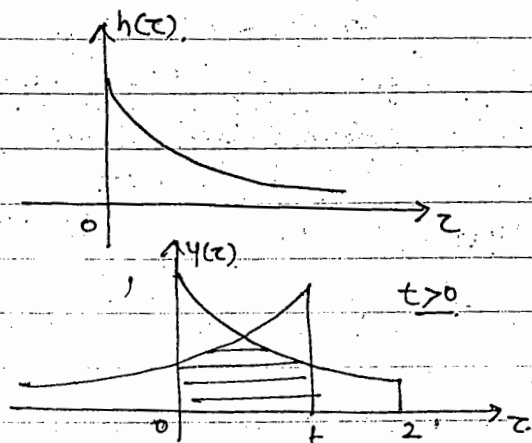
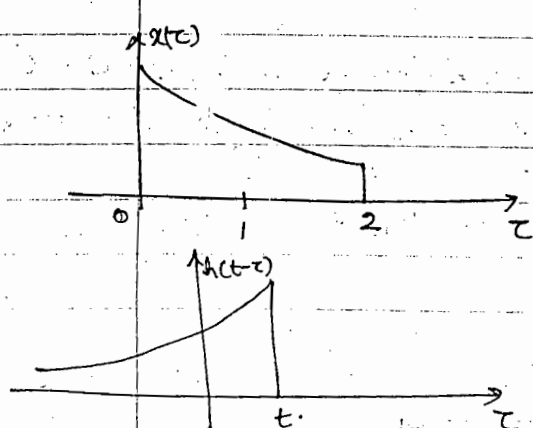
$$= \left. \frac{e^{-a\tau}}{-a} \right|_0^t = \frac{1 - e^{-at}}{a}$$

$$y(t) = \left[\frac{1 - e^{-at}}{a} \right] u(t)$$



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(13) $x(t) = e^{-3t} [u(t) - u(t-2)]$ $h(t) = e^{-t} u(t)$ find $y(t) = x(t) * h(t)$



$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau.$$

for $t < 0$; $y(t) = 0$

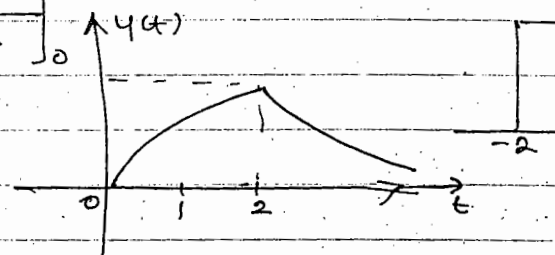
for $t > 0$; convolution has a definite value.

for $0 \leq t \leq 2$;

$$y(t) = \int_{\tau=0}^t x(\tau) h(t-\tau) d\tau = \int_{\tau=0}^t e^{-3\tau} (u(\tau) - u(\tau-2)) e^{-(t-\tau)} u(t-\tau) d\tau$$

$$= e^{-t} \int_{\tau=0}^t e^{-3\tau} e^{\tau} d\tau = e^{-t} \left[\frac{e^{-2\tau}}{-2} \right]_0^t$$

$$y(t) = e^{-t} \left[\frac{1 - e^{-2t}}{2} \right] \quad \text{N/A}$$



for $t \geq 2$; convolution starts from $\tau=0$ & ends at $\tau=2$

$$y(t) = \int_{\tau=0}^2 e^{-t} e^{-2\tau} d\tau = e^{-t} \left[\frac{e^{-2\tau}}{-2} \right]_0^2 = e^{-t} \left[\frac{-e^{-4} + 1}{2} \right]$$

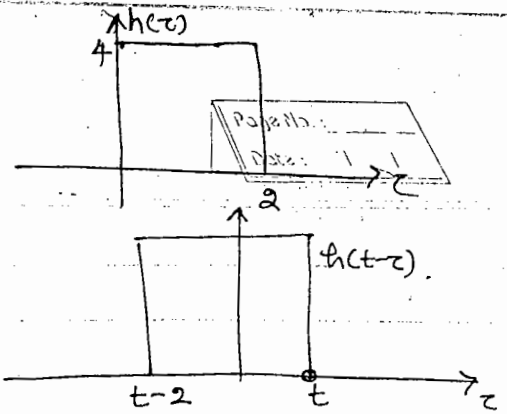
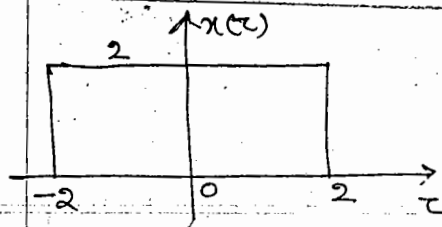
$$y(t) = e^{-t} \left[\frac{1 - e^{-4}}{2} \right]$$

$$y(t) =$$

(14) Given $x(t) = \begin{cases} 2 & -2 \leq t \leq 2 \\ 0 & \text{otherwise} \end{cases}$ $h(t) = \begin{cases} 4 & 0 \leq t \leq 2 \\ 0 & \text{otherwise} \end{cases}$

find $y(t) = x(t) * h(t)$.

$$y(t) = x(t) * h(t)$$

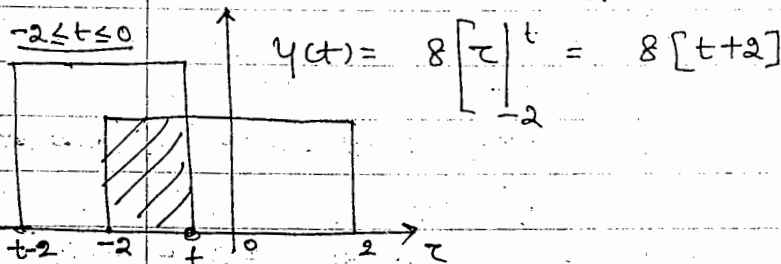


$t > 0$

for $t < -2$; $y(t) = 0$.

for $-2 \leq t \leq 0$; convolution starts from $\tau = -2$ and ends at $\tau = t$

$$y(t) = \int_{\tau=-2}^t x(\tau) h(t-\tau) d\tau = \int_{\tau=-2}^t 2 \cdot 4 \cdot d\tau$$

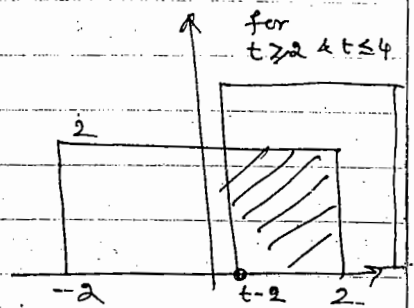
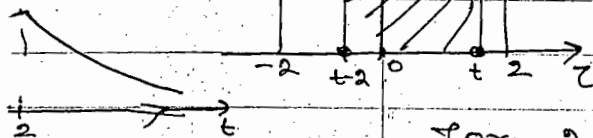


$$y(t) = \int_{\tau=-2}^t x(\tau) h(t-\tau) d\tau$$

for $0 \leq t \leq 2$

$$y(t) = \int_{\tau=t-2}^t 2 \cdot 4 \cdot d\tau = 8[t-t+2]$$

$$y(t) = 16$$

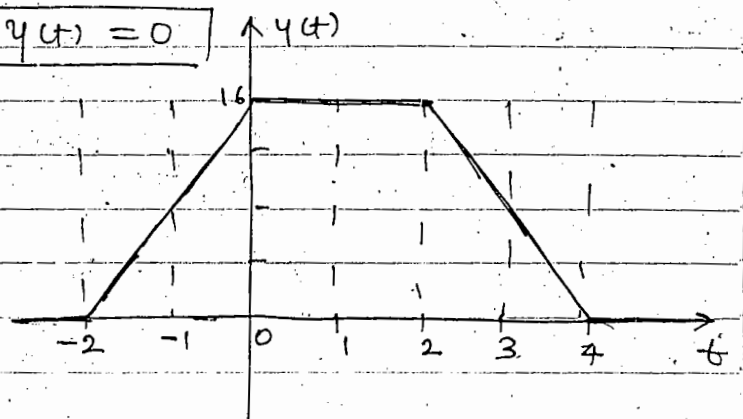


for $2 \leq t \leq 4$

$$y(t) = \int_{\tau=t-2}^2 2 \cdot 4 \cdot d\tau = 8[2-t+2] = 8[4-t]$$

for $t > 4$; $y(t) = 0$

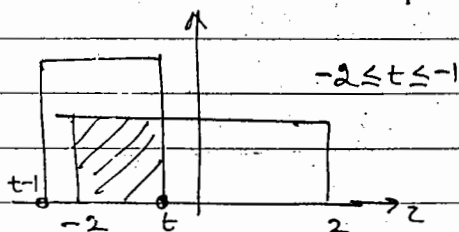
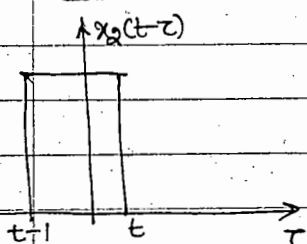
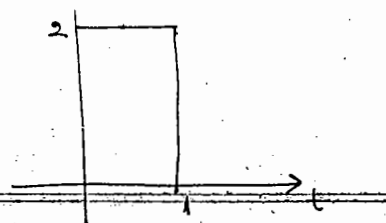
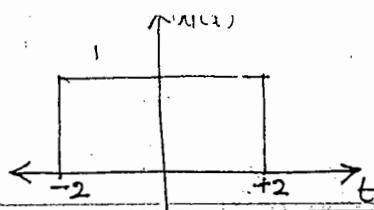
$$y(t) = \begin{cases} 0; & t < -2 \text{ \& } t > 4 \\ 8(t+2); & -2 \leq t \leq 0 \\ 16; & 0 \leq t \leq 2 \\ 8(4-t); & 2 \leq t \leq 4 \end{cases}$$



$0 \leq t \leq 2$
otherwise

⑮

convolve

① for $t < -2$; $y(t) = 0$

$$\textcircled{2} \text{ for } -2 \leq t \leq -1; \quad y(t) = \int_{\tau=-2}^t x(\tau) h(t-\tau) d\tau = \int_{\tau=-2}^t 1 \cdot 2 d\tau$$

$$y(t) = 2[t+2]$$

③ for $-1 \leq t \leq 2$; Convolution starts from $\tau = t-1$ & ends at t

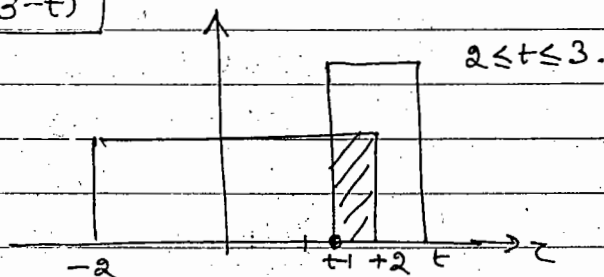
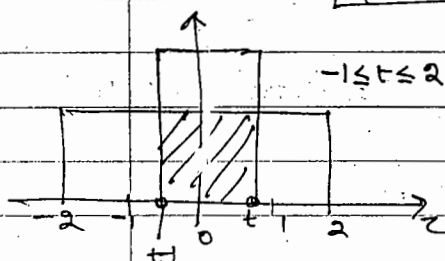
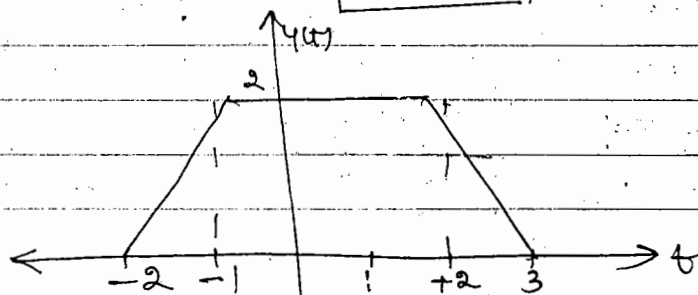
$$y(t) = \int_{\tau=t-1}^t x(\tau) h(t-\tau) d\tau = \int_{\tau=t-1}^t 1 \cdot 2 d\tau = 2[t - (t-1)]$$

$$y(t) = 2$$

④ for $2 \leq t \leq 3$

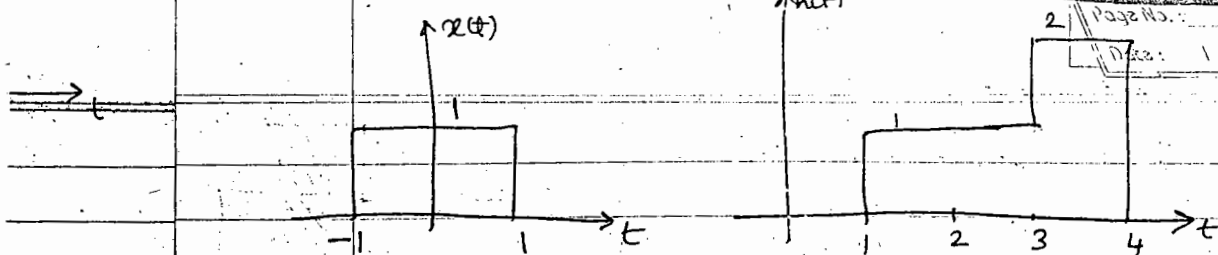
$$y(t) = \int_{\tau=t-1}^2 2 d\tau = 2[2 - (t-1)] = 2(3-t)$$

$$y(t) = 2(3-t)$$

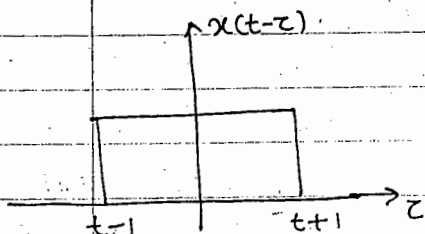
for $t \geq 3$ $y(t) = 0$ as there is no common area

16

Find the convolution of $x(t)$ & $h(t)$ shown below



Let $y(t) = x(t) * h(t)$.



for $t < 0$; $y(t) = 0$

For $0 < t < 1$

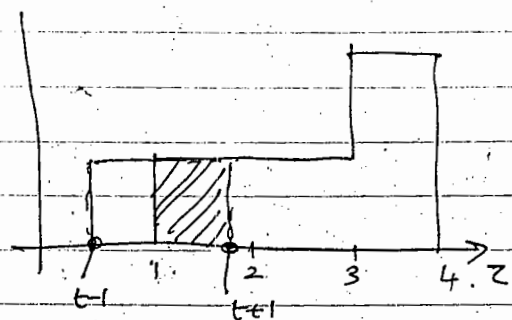
$$y(t) = \int_{\tau=1}^{t+1} x(\tau) h(t-\tau) d\tau$$

& ends at $t+1$

$-t+1$

$$= \int_{\tau=1}^{t+1} 1 \cdot 1 \cdot d\tau = \tau \Big|_1^{t+1} = t+1-1 = t$$

$y(t) = t$



For $1 \leq t \leq 2$

$$y(t) = \int_{\tau=1}^{t+1} x(\tau) h(t-\tau) d\tau = \int_{\tau=1}^{t+1} d\tau$$

b)

$\leq t \leq 3$

$$= \tau \Big|_1^{t+1} = t+1-1 = t$$

$\Rightarrow z$

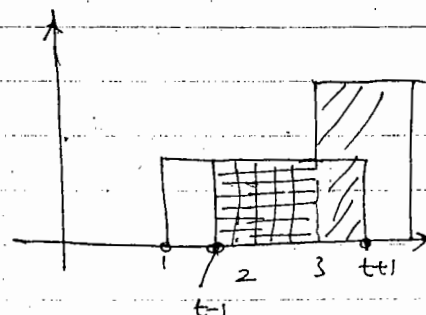
Common area

$y(t) = t$

for $2 \leq t \leq 3$;

$$y(t) = \int_{\tau=t-1}^{t+1} x(\tau) h(t-\tau) d\tau$$

$$= \int_{\tau=t-1}^3 x(\tau) h(t-\tau) d\tau + \int_{\tau=3}^{t+1} x(\tau) h(t-\tau) d\tau = \int_{\tau=t-1}^3 1 d\tau + \int_{\tau=3}^{t+1} 2 d\tau = 4-t + 2t+2-6 = t$$



$$\therefore y(t) = t$$

hence for $0 \leq t \leq 3$; $y(t) = 2$

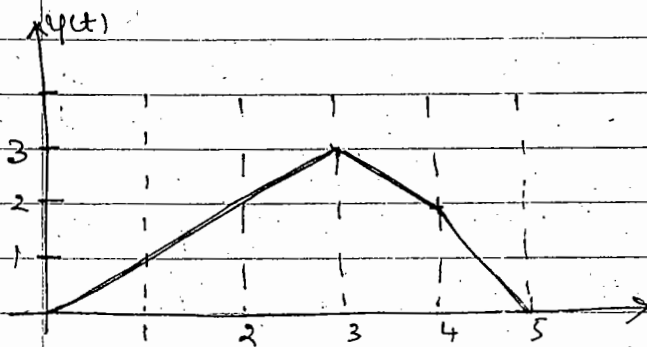
for $4 \leq t \leq 5$

$$y(t) = \int_{t-1}^4 1 \cdot 2 \cdot d\tau = 8 - 2t + 2$$

$$= 10 - 2t$$

$$y(t) = 10 - 2t$$

for $t \geq 5$; $y(t) = 0$



$$y(t) = \begin{cases} t; & 0 \leq t \leq 3 \\ 6-t; & 3 \leq t \leq 4 \\ 10-2t; & 4 \leq t \leq 5 \\ 0; & t \leq 0 \text{ and } t \geq 5. \end{cases}$$

for $3 \leq t \leq 4$

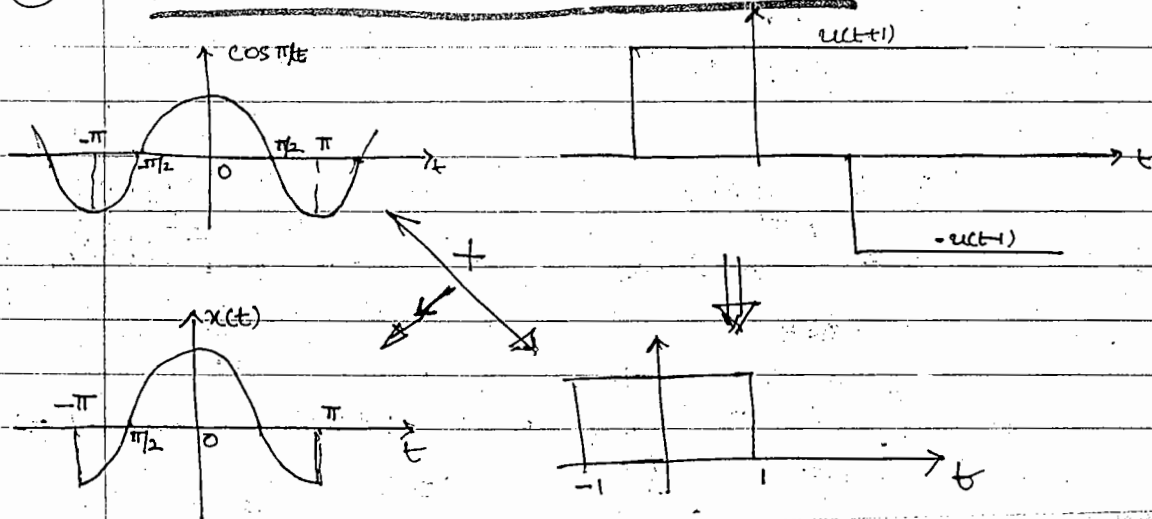
$$y(t) = \int_{t-1}^3 x(\tau)h(t-\tau)d\tau + \int_3^4 2(\tau)h(t-\tau)d\tau$$

$$= \int_{t-1}^3 1 d\tau + \int_3^4 2 d\tau = 3 - t + 1 + 8 - 6$$

$$= 6 - t$$

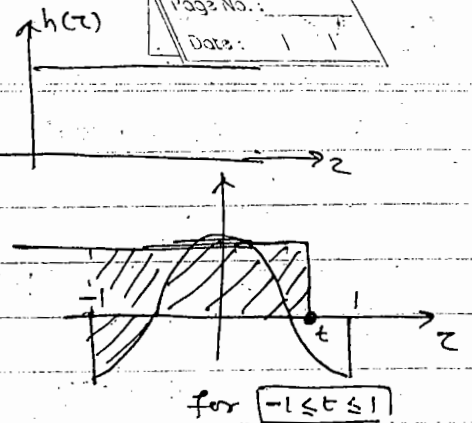
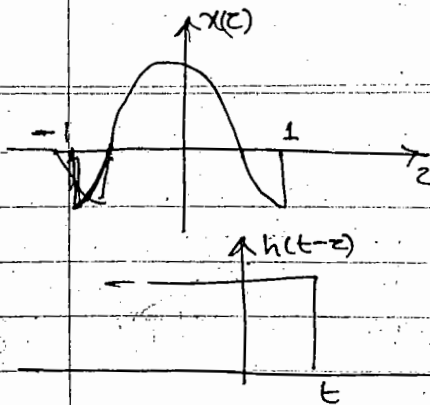
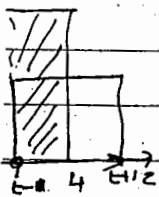
$$y(t) = 6 - t; \quad 3 \leq t \leq 4$$

(17) $y(t) = \cos \pi t [u(t+1) - u(t-1)] * u(t)$



(18)

Let $x(t) = \cos \pi t (u(t+1) - u(t-1))$
 $h(t) = u(t)$.



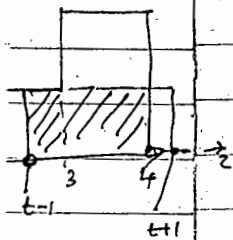
for $t < -1$; $y(t) = 0$

for $-1 \leq t \leq 1$;

$$y(t) = \int_{-1}^t x(\tau) h(t-\tau) d\tau = \int_{-1}^t \cos \pi \tau \cdot 1 d\tau = \left. \frac{\sin \pi \tau}{\pi} \right|_{-1}^t$$

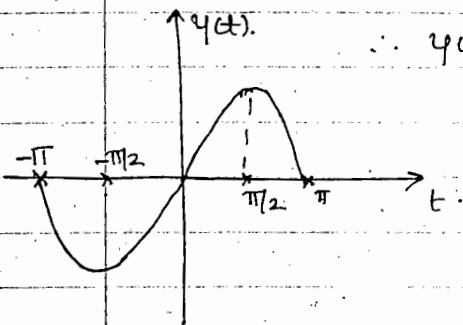
$$= \frac{\sin \pi t}{\pi} - 0 = \frac{\sin \pi t}{\pi}$$

$\therefore y(t) = \frac{\sin \pi t}{\pi}$ for $-1 \leq t \leq 1$



for $t > 1$; $y(t) = \int_{-1}^1 \cos \pi \tau d\tau \cdot 1 = \left. \frac{\sin \pi \tau}{\pi} \right|_{-1}^1 = 0$

$\therefore y(t) = 0$; $t > 1$



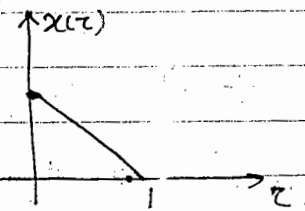
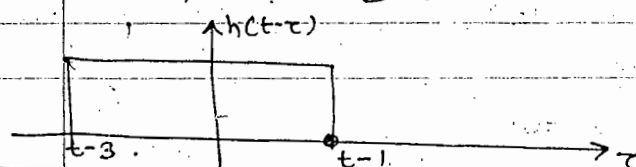
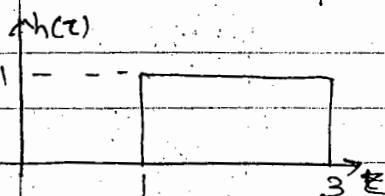
$$y(t) = \begin{cases} \frac{\sin \pi t}{\pi} & ; -1 \leq t \leq 1 \\ 0 & ; t < -1 \text{ \& } t > 1 \end{cases}$$

(8)

Given $h(t) = 1$; $1 \leq t \leq 3$
 0 ; otherwise

$x(t) = 1-t$; $0 \leq t \leq 1$
 $= 0$; otherwise

find $y(t) = x(t) * h(t)$



for $t < 1$; $y(t) = 0$

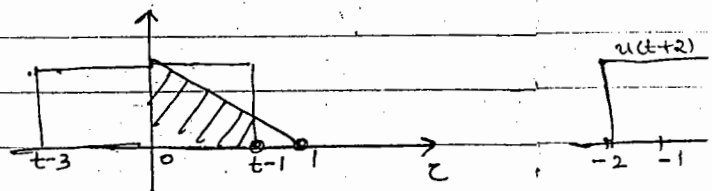
for $1 \leq t < 2$

$$y(t) = \int_{\tau=0}^{t-1} x(\tau) h(t-\tau) d\tau = \int_{\tau=0}^{t-1} \cancel{2p(2-\tau)} (1-\tau) d\tau$$

$$= \left[\tau - \frac{\tau^2}{2} \right]_0^{t-1} = (t-1) - \frac{(t-1)^2}{2}$$

$$= t-1 - \frac{t^2-1+2t}{2}$$

$$= \frac{2t-2-t^2-1+2t}{2} = \frac{-t^2+4t-3}{2}$$

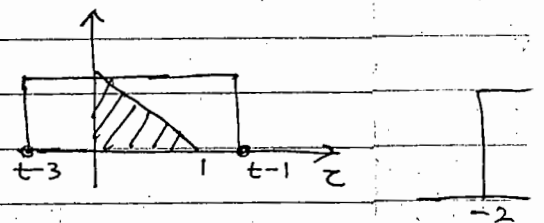


for $2 \leq t \leq 3$

$$y(t) = \int_{\tau=0}^1 (1-\tau) d\tau = \tau - \frac{\tau^2}{2} \Big|_0^1$$

$$= 1 - \frac{1}{2} = \frac{1}{2}$$

$$y(t) = \frac{1}{2}$$



for $3 \leq t \leq 4$

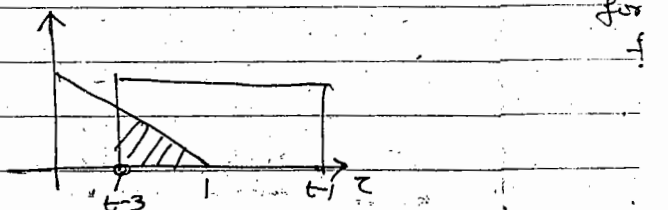
$$y(t) = \int_{\tau=t-3}^1 (1-\tau) d\tau = \tau - \frac{\tau^2}{2} \Big|_{t-3}^1$$

$$= 1 - \frac{1}{2} - t + 3 + \frac{(t-3)^2}{2}$$

$$= \frac{1}{2} - t + 3 + \frac{t^2 - 6t + 9}{2}$$

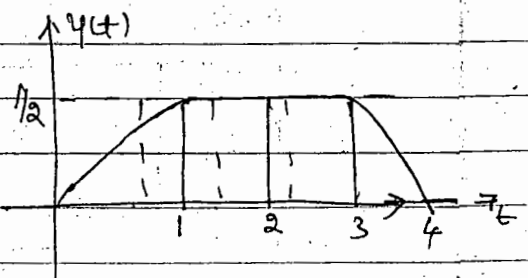
$$= \frac{1 - 2t + 6 + t^2 - 6t + 9}{2}$$

$$y(t) = \frac{t^2 - 8t + 16}{2}$$



for $t \geq 4$, $y(t) = 0$

$$y(t) = \begin{cases} 0, & t < 1 \text{ \& } t > 4 \\ \frac{-t^2+4t-3}{2}, & 1 \leq t \leq 2 \\ \frac{1}{2}, & 2 \leq t \leq 3 \\ \frac{t^2-8t+16}{2}, & 3 \leq t \leq 4 \end{cases}$$



⑤ Evaluate $y(t) = [u(t+2) - u(t-1)] * u(-t+2)$

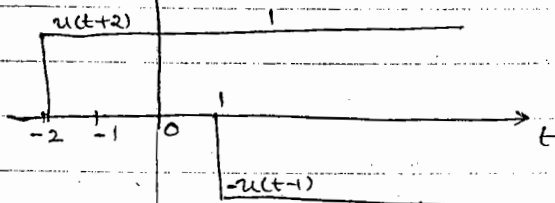
Let $x(t) = u(t+2) - u(t-1)$ and

$h(t) = u(-t+2)$

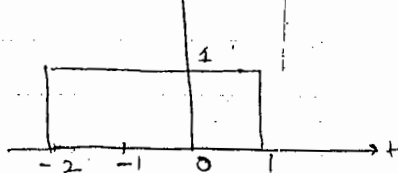
$$y(t) = x(t) * h(t)$$

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Date: ____/____/____

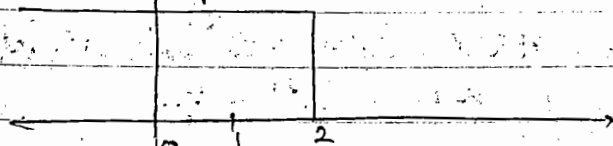
$x(t)$



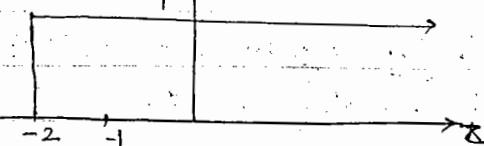
$x(t) = u(t+2) - u(t-1)$



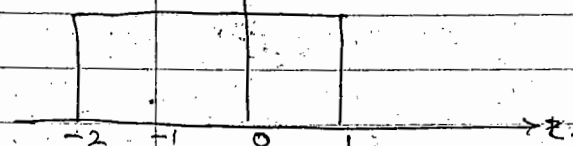
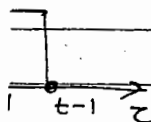
$h(-t+2) = h(t)$



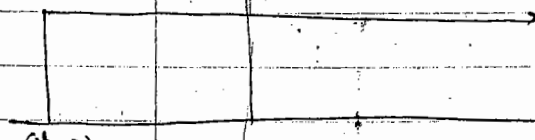
$h(-t)$



$x(t)$

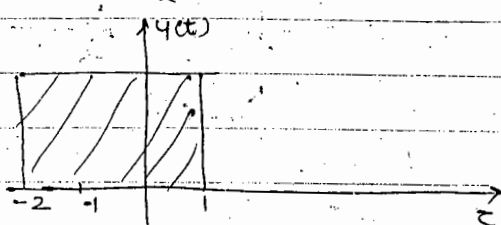


$h(t-t_0)$



For $t < 0$,

convolution starts at $\tau = -2$ and ends at $\tau = 1$



$$y(t) = \int_{\tau=-2}^1 x(\tau) h(t-\tau) d\tau = \int_{\tau=-2}^1 1 \cdot 1 d\tau = \tau \Big|_{-2}^1 = 3$$

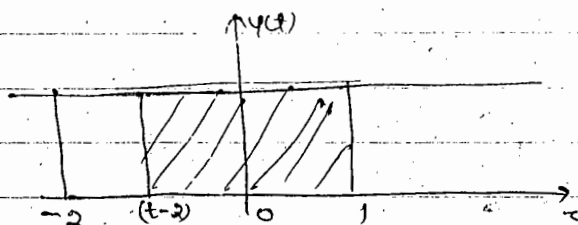
for $0 \leq t < 3$

$$y(t) = 3 ; t < 0$$

for $t > 0$; convolution starts at $\tau = t-2$ and ends at $\tau = 1$

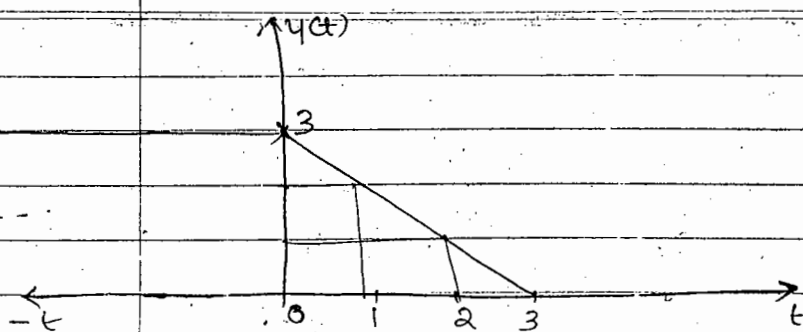
$$y(t) = \int_{\tau=t-2}^1 x(\tau) h(t-\tau) d\tau = \int_{\tau=t-2}^1 1 \cdot 1 d\tau = \tau \Big|_{t-2}^1 = 1 - t + 2$$

$$y(t) = 3 - t ; t > 0$$



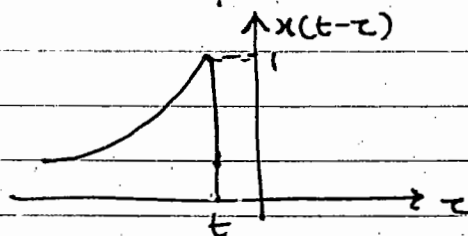
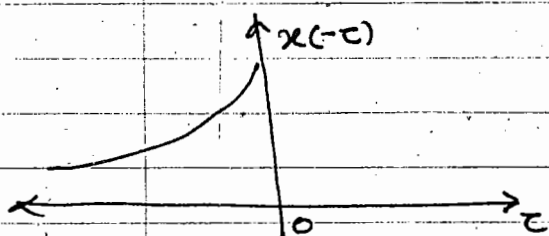
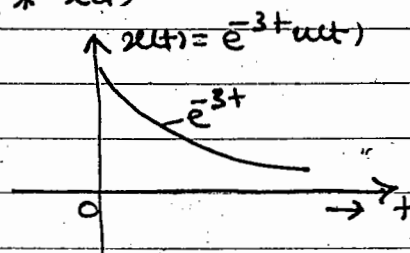
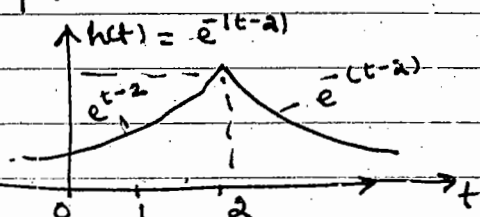
for $t > 3$; $y(t) = 0$ as there is no overlap b/w $x(t)$ & $h(t)$.

$$y(t) = \begin{cases} 3 & ; t < 0 \\ 3-t & ; 0 < t < 3 \\ 0 & ; t \geq 3 \end{cases}$$



A LTI system has impulse response $h(t) = e^{-|t-2|}$. Find the output of the system if $x(t) = e^{3t}u(t)$.

$$y(t) = x(t) * h(t) = h(t) * x(t)$$



$$\text{for } t \leq 2; \quad y(t) = \int_{-\infty}^t e^{-(t-2)} \cdot e^{-3(t-\tau)} d\tau$$

$$\begin{aligned} y(t) &= \int_{-\infty}^t e^{+2} \cdot e^{-2} e^{-3t} \cdot e^{3\tau} d\tau = e^{-3t-2} \int_{-\infty}^t e^{4\tau} d\tau \\ &= e^{-3t-2} \left[\frac{e^{4\tau}}{4} \right]_{-\infty}^t = e^{-3t-2} \frac{e^{4t}}{4} \end{aligned}$$

$$\boxed{y(t) = \frac{e^{t-2}}{4}}$$

For $t \geq 2$

$$\begin{aligned} y(t) &= \int_{-\infty}^2 e^{-3(t-\tau)} e^{-(\tau-2)} d\tau + \int_2^t e^{-3(t-\tau)-(\tau-2)} d\tau \\ &= \int_{-\infty}^2 e^{-3t+2} e^{4\tau} d\tau + \int_2^t e^{-3t+2} e^{4\tau} d\tau \end{aligned}$$

$$= \frac{e^{-3t-2}}{4} \left[\frac{e^{4t}}{4} \right]_{-\infty}^2 + e^{-3t+2} \left[\frac{e^{2t}}{2} \right]_t^2$$

$$= \frac{e^{-3t-2}}{4} \cdot \frac{e^8}{4} + e^{-3t+2} \left[\frac{e^{2t}}{2} - \frac{e^4}{2} \right]$$

$$= \frac{1}{4} \left[\frac{e^{-3t+6}}{4} + e^{-t-2} \frac{e^{2t}}{2} - e^{-3t+6} \right]$$

$$= \frac{e^{-3t+6}}{4} \left[1 + e^{-t+2} \right]$$

$$= \frac{e^{-3t+6}}{4} + \left[\frac{e^{-t+2}}{2} - \frac{e^{-3t+6}}{2} \right]$$

$$= \cancel{\frac{e^{-3t+6}}{4}} + \frac{e^{-3t+6}}{4} + \frac{e^{-t+2}}{2} - \frac{e^{-3t+6}}{2}$$

$$y(t) = \frac{e^{-t+2}}{2} + \frac{e^{-3t+6}}{4}$$

Find

5)

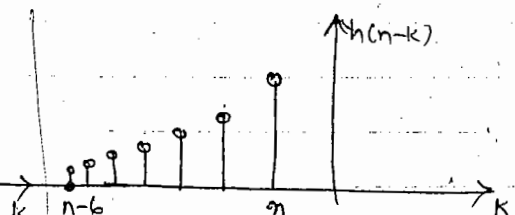
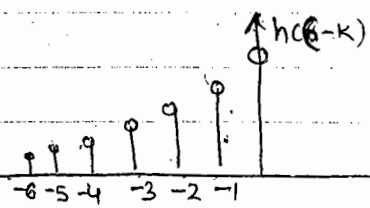
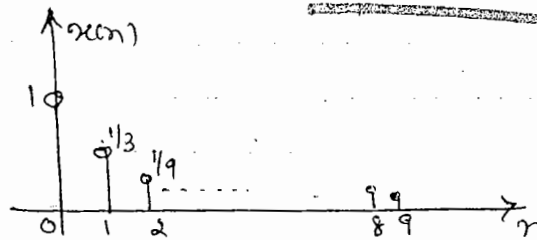
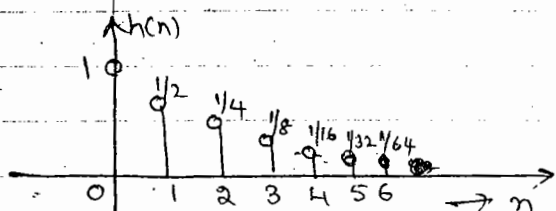
→ +

A discrete LTI system has impulse response ~~h(n)~~

$$h(n) = \left(\frac{1}{2}\right)^n [u(n) - u(n-7)] \quad \& \quad \text{input } x(n) = \left(\frac{1}{3}\right)^n; 0 \leq n \leq 9.$$

$$= 0; \text{ otherwise}$$

$$\text{find } y(n) = x(n) * h(n).$$



$$\text{for } n < 0; \quad y(n) = 0.$$

$$\text{for } 0 \leq n \leq 6;$$

$$y(n) = \sum_{k=0}^n \left(\frac{1}{3}\right)^k \left(\frac{1}{2}\right)^{n-k} u(k) [u(n-k) - u(n-k-7)] = \sum_{k=0}^n \left(\frac{1}{2}\right)^n \left(\frac{2}{3}\right)^k$$

$$= \left(\frac{1}{2}\right)^n \left[\frac{1 - \left(\frac{2}{3}\right)^{n+1}}{1 - \frac{2}{3}} \right] = \left(\frac{1}{2}\right)^n \left[\frac{1 - \left(\frac{2}{3}\right)^{n+1}}{1/3} \right]$$

for $6 \leq n \leq 9$

$$y(n) = \sum_{k=n-6}^n \left(\frac{1}{2}\right)^k \left(\frac{2}{3}\right)^{n-k} = \left(\frac{1}{2}\right)^n \sum_{k=n-6}^n \left(\frac{2}{3}\right)^k$$

put $l = k - n + 6$

when $k = n-6$, $l = 0$

when $k = n$, $l = 6$

$$y(n) = \left(\frac{1}{2}\right)^n \sum_{l=0}^6 \left(\frac{2}{3}\right)^{l+n-6} = \left(\frac{1}{2}\right)^n \left(\frac{2}{3}\right)^{n-6} \left[\sum_{l=0}^6 \left(\frac{2}{3}\right)^l \right]$$

$$= \left(\frac{1}{2}\right)^n \left(\frac{2}{3}\right)^{n-6} \left[\frac{1 - \left(\frac{2}{3}\right)^{6+1}}{1 - \frac{2}{3}} \right] = \left(\frac{1}{2}\right)^n \left[\frac{\left(\frac{2}{3}\right)^{n-6} - \left(\frac{2}{3}\right)^{n-6+7}}{1/3} \right]$$

$$y(n) = \left(\frac{1}{2}\right)^n \left[\frac{\left(\frac{2}{3}\right)^{n-6} - \left(\frac{2}{3}\right)^{n+1}}{1/3} \right]$$

for $9 \leq n \leq 15$

$$y(n) = \sum_{k=n-6}^9 \left(\frac{1}{2}\right)^k \left(\frac{2}{3}\right)^{n-k}$$

put $l = k - n + 6$

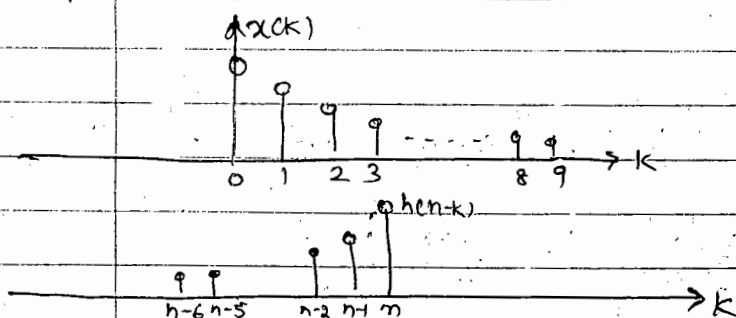
when $k = n-6$, $l = 0$; & $k = 9$, $l = -n+15$

$$y(n) = \sum_{k=0}^{-n+15} \left(\frac{1}{2}\right)^n \left(\frac{2}{3}\right)^{l+n-6} = \left(\frac{1}{2}\right)^n \left(\frac{2}{3}\right)^{n-6} \left[\frac{1 - \left(\frac{2}{3}\right)^{-n+15+1}}{1 - \frac{2}{3}} \right]$$

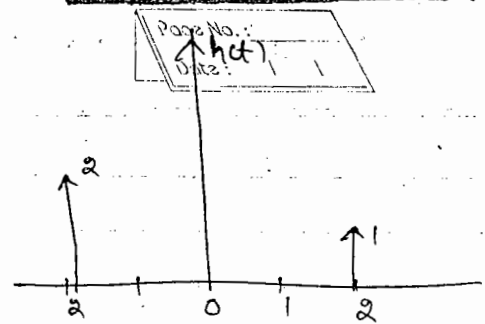
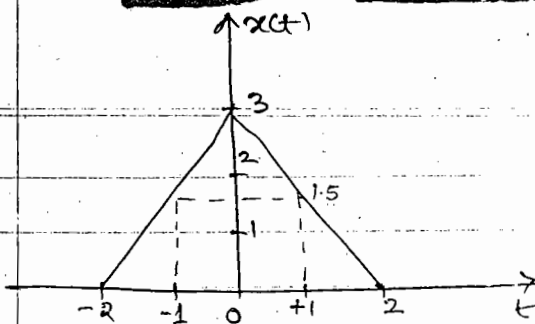
$$= \left(\frac{1}{2}\right)^n \left[\frac{\left(\frac{2}{3}\right)^{n-6} - \left(\frac{2}{3}\right)^{-n+16+n-6}}{1/3} \right]$$

$$y(n) = \left(\frac{1}{2}\right)^n \left[\frac{\left(\frac{2}{3}\right)^{n-6} - \left(\frac{2}{3}\right)^{10}}{1/3} \right]$$

for $n > 15$; $y(n) = 0$



Sketch the convolution of $x(t)$ & $h(t)$ shown below



We can define $h(t)$ as $h(t) = \delta(t-2) + 2\delta(t+2)$

Let $y(t) = x(t) * h(t) = x(t) * [\delta(t-2) + 2\delta(t+2)]$

$$y(t) = x(t-2) + 2x(t+2)$$

$$y(0) = x(-2) + 2x(2) = 0$$

$$y(1) = x(-3) + 2x(1) = 0 + 3 = 3$$

$$y(-2) = x(-4) + 2x(0) = 0 + 6 = 6$$

$$y(-3) = x(-5) + 2x(-1) = 0 + 3 = 3$$

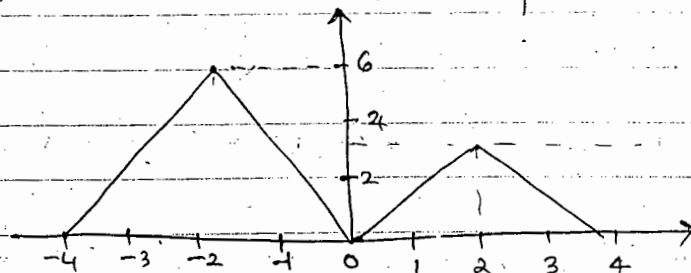
$$y(-4) = x(-6) + 2x(-2) = 0$$

$$y(1) = x(-1) + 2x(3) = 1.5$$

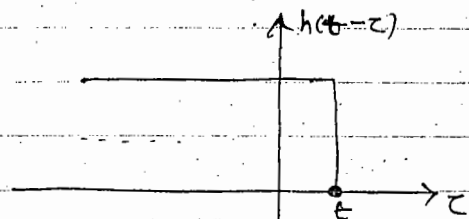
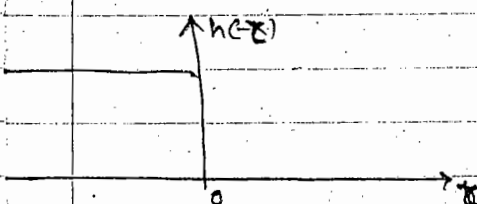
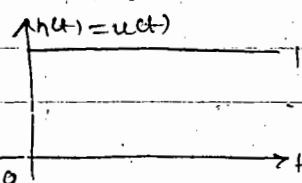
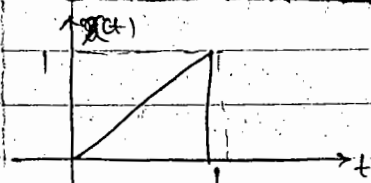
$$y(2) = x(0) + 2x(4) = 0$$

$$y(3) = x(1) + 2x(5) = 1.5$$

$$y(4) = x(2) + 2x(6) = 0$$



Find the convolution of $x(t)$ & $h(t)$ shown below



for $t < 0$, $y(t) = 0$

$$\text{for } t > 0 \text{ and } t \leq 1, \quad y(t) = \int_{\tau=0}^t x(\tau) \cdot h(t-\tau) d\tau =$$

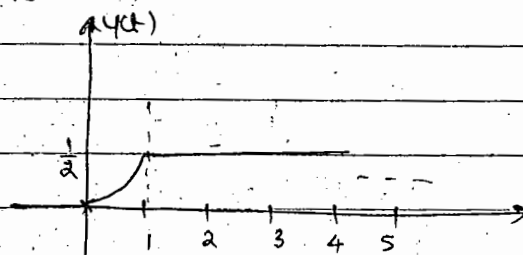
$x(t)$ can be defined as $x(t) = \begin{cases} t, & 0 < t < 1 \\ 0, & \text{otherwise} \end{cases}$

$$y(t) = \int_{\tau=0}^t t \, d\tau = \frac{t^2}{2}$$

for $1/2 \leq t$ for $t > 1$;

$$y(t) = \int_{\tau=0}^1 t \, d\tau = \left. \frac{t^2}{2} \right|_0^1 = \frac{1}{2}$$

$$y(t) = \begin{cases} 0 & ; t < 0 \\ \frac{t^2}{2} & ; 0 < t \leq 1 \\ \frac{1}{2} & ; t > 1 \end{cases}$$

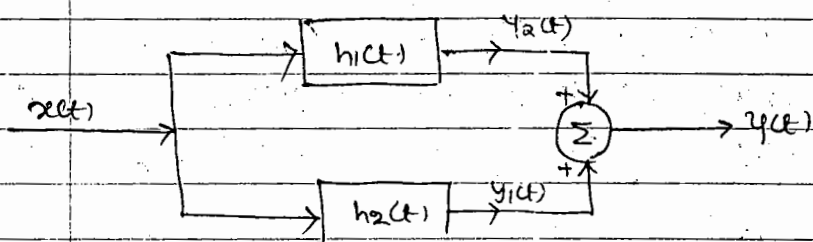


and

Interconnection of systems

parallel connection of systems

Consider two LTI systems with impulse responses $h_1(t)$ and $h_2(t)$ connected in parallel as shown.



$$y(t) = y_1(t) + y_2(t) = [x(t) * h_1(t)] + [x(t) * h_2(t)]$$

$$\text{Let } h(t) = h_1(t) + h_2(t)$$

where

$$h(t-\tau) = h_1(t-\tau) + h_2(t-\tau) \quad \text{--- (LTI systems)}$$

$$y(t) = \int_{\tau=-\infty}^{\infty} x(\tau) h_1(t-\tau) d\tau + \int_{\tau=-\infty}^{\infty} x(\tau) h_2(t-\tau) d\tau$$

$$= \int_{\tau=-\infty}^{\infty} x(\tau) [h_1(t-\tau) + h_2(t-\tau)] d\tau = \int_{\tau=-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

$$= x(t) * h(t) = x(t) * (h_1(t) + h_2(t))$$

using commutative property

$$y(t) = [h_1(t) + h_2(t)] * x(t)$$

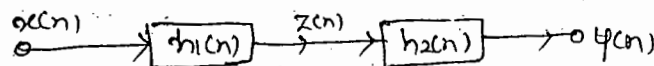
$$= x(t) * h_1(t) + x(t) * h_2(t)$$

conclusion:- The impulse response of the overall system represented by LTI systems connected in parallel is the sum of their individual responses. Hence convolution is distributive

also for a DT LTI system.

$$y(n) = x(n) * [h_1(n) + h_2(n)] = x(n) * h_1(n) + x(n) * h_2(n)$$

Cascade connection of LTI systems



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consider two DT LTI systems with impulse responses $h_1(n)$ and $h_2(n)$ connected in series (cascade) as shown in figure.

$$z(n) = x(n) * h_1(n)$$

$$y(n) = z(n) * h_2(n) = \sum_{k=-\infty}^{\infty} z(k) h_2(n-k)$$

$$= \sum_{k=-\infty}^{\infty} h_2(n-k) [x(k) * h_1(k)] = \sum_{k=-\infty}^{\infty} h_2(n-k) \sum_{l=-\infty}^{\infty} x(l) h_1(k-l)$$

put $n-k=m$; $k=n-m$.

$$y(n) = \sum_{m=-\infty}^{\infty} h_2(m) \sum_{k=-\infty}^{\infty} x(l) h_1(-l+m+n) =$$

$$= \sum_{m=-\infty}^{\infty} x(l) \sum_{l=-\infty}^{\infty} h_2(m) h_1(m-l+n)$$

$$= \sum_{m=-\infty}^{\infty} x(l) \sum_{l=-\infty}^{\infty} h_2(m) h_1(n-(m+l))$$

$$= \sum_{m=-\infty}^{\infty} x(l) h(n-m) = x(n) * h(n)$$

$$y(n) = x(n) * (h_1(n) * h_2(n))$$

where $h(n) = h_1(n) * h_2(n) = \sum_{l=-\infty}^{\infty} h_1(l) h_2(n-l)$

$$h(n-m) = \sum_{l=-\infty}^{\infty} h_1(l) h_2(n-m-l)$$

Hence the Impulse response of the overall system represented by LTI systems connected in cascade is the convolution of their individual responses.

Similarly for a CT LTI system

$$y(t) = x(t) * (h_1(t) * h_2(t)) = [x(t) * h_1(t)] * h_2(t)$$

Hence convolution is associative in nature.

Also since convolution is commutative, the ordering of the systems does not affect the output of the system.

$$h(t) = h_1(t) * h_2(t) = h_2(t) * h_1(t)$$

$$x(t) \xrightarrow{\quad} [h_1(t)] \xrightarrow{\quad} [h_2(t)] \xrightarrow{\quad} y(t)$$

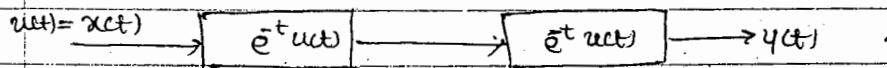
$$x(t) \xrightarrow{\quad} [h_2(t)] \xrightarrow{\quad} [h_1(t)] \xrightarrow{\quad} y(t)$$

} identical

$(x(n) * h_2(n))$

*: Show that the overall system response does not depend on the order of the system interconnection when the systems are connected in series or parallel *

① Find the response $y(t)$ of the cascaded system shown



Let $h_1(t) = e^{-t}u(t)$, $h_2(t) = e^{-t}u(t)$

$h(t) = h_1(t) * h_2(t)$

$= \int_{-\infty}^{\infty} h_1(\tau) h_2(t-\tau) d\tau$

for $t \geq 0$

$h(t) = \int_{\tau=0}^t e^{-\tau} e^{-(t-\tau)} d\tau = e^{-t}(t) = t e^{-t}$

$y(t) = x(t) * h(t) = \int_{\tau=-\infty}^{\infty} x(\tau) h(t-\tau) d\tau = \int_{-\infty}^{\infty} h(\tau) x(t-\tau) d\tau$

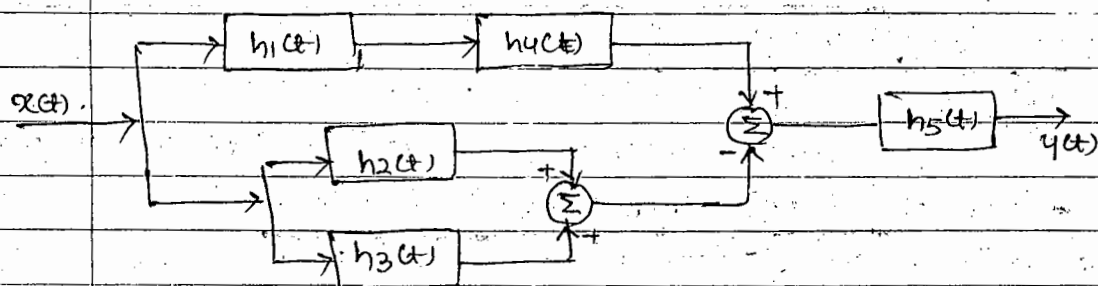
for $t \geq 0$

$y(t) = \int_{\tau=0}^t \tau e^{-\tau} d\tau = \tau \frac{e^{-\tau}}{-1} - \int_{\tau=0}^t \frac{e^{-\tau}}{-1} d\tau$
 $= -\tau e^{-\tau} + e^{-\tau} \Big|_0^t$
 $= -t e^{-t} - e^{-t} + 1$

$y(t) = 1 - e^{-t} - t e^{-t}$

or $y(t) = 1 - e^{-t} - t e^{-t} u(t)$, for $t \geq 0$

② Find the overall response of the system shown below



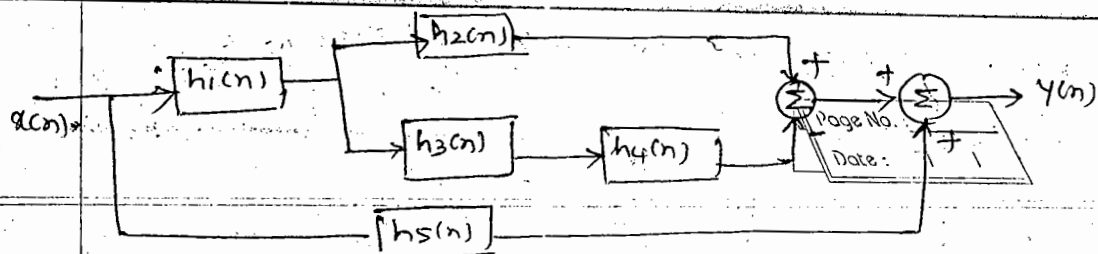
$y(t) = x(t) * [(h_1(t) * h_4(t)) + (h_2(t) + h_3(t))] * h_5(t)$

③ Find the overall response of the system shown below.

at when

*

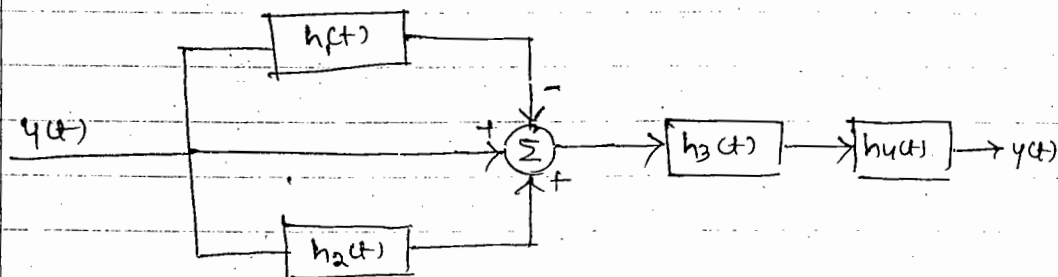
n



$$y(n) = x(n) * [h_1(n) * [h_2(n) + h_3(n) * h_4(n)] + h_5(n)]$$

$$\text{or } h_1(n) = h_5(n) + [h_1(n) * [h_2(n) + [h_3(n) * h_4(n)]]$$

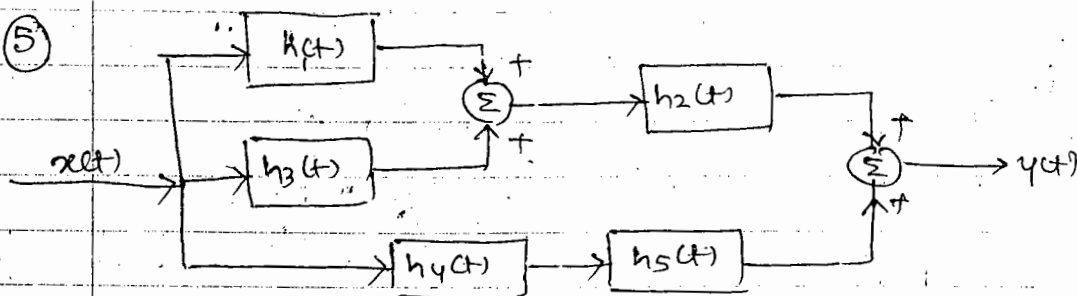
④



-c).

$$y(t) = [h_2(t) + \delta(t) - h_1(t)] * h_3(t) * h_4(t) * x(t)$$

⑤

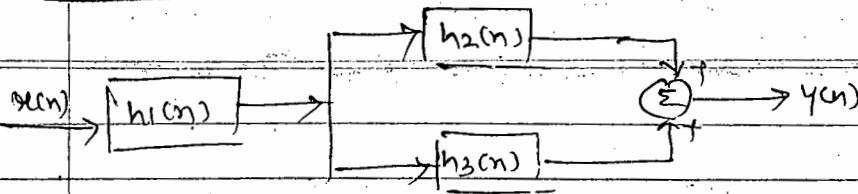


$$y(t) = x(t) * [[h_1(t) + h_2(t)] * h_3(t) + [h_4(t) * h_5(t)]]$$

y(t)

Find the overall response of the system. given

$h_1(n) = (\frac{1}{2})^n u(n+2)$; $h_2(n) = u(n-1)$, $h_3(n) = \delta(n)$



⑦ If

$$h(n) = h_1(n) * [h_2(n) + h_3(n)]$$

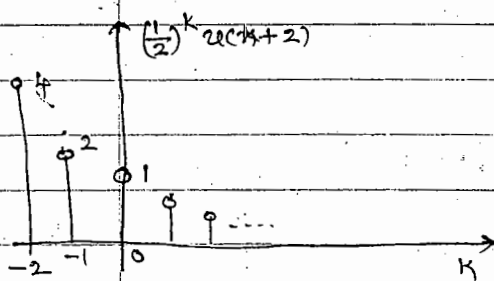
$$= (\frac{1}{2})^n u(n+2) * [\delta(n) + u(n-1)]$$

$$= (\frac{1}{2})^n u(n+2) * \delta(n) + (\frac{1}{2})^n u(n+2) * u(n-1)$$

$$= (\frac{1}{2})^n u(n+2) + z(n).$$

20

$$z(n) = (\frac{1}{2})^n u(n+2) * u(n-1) = \sum_{k=-\infty}^{\infty} (\frac{1}{2})^k u(k+2) u(n-k+1)$$



For $n < -1$ $z(n) = 0$

For n
put $k = n - 1$
or $k = n - 1$
 $z(n) = \sum_{r=0}^{\infty} (\frac{1}{2})^{n-1+r} \cdot 1$

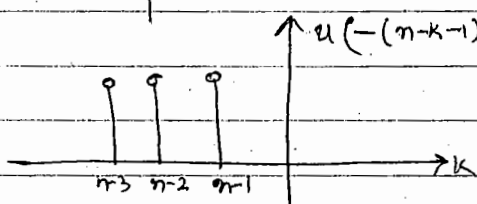
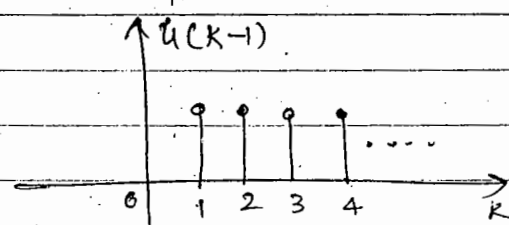
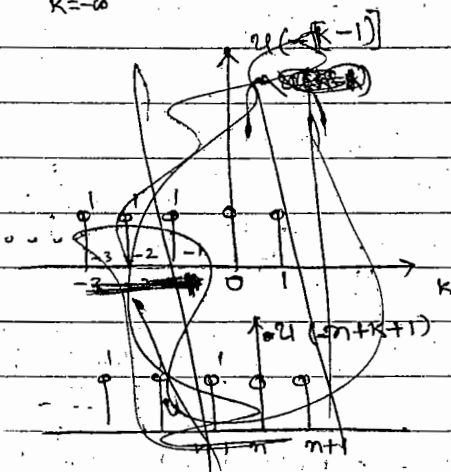
for $n \geq -1$;

$$z(n) = \sum_{k=-2}^{n-1} (\frac{1}{2})^k \cdot 1$$

put $k = k+2$

$$z(n) = \sum_{r=0}^{n+1} (\frac{1}{2})^{r-2} \cdot 1 = (\frac{1}{2})^{-2} \left[\frac{1 - (\frac{1}{2})^{n+1+1}}{1 - \frac{1}{2}} \right]$$

$$z(n) = 8 \left[1 - (\frac{1}{2})^{n+2} \right]$$



M5
K=0

Fin

Wh

$$y(n) = \left(\frac{1}{2}\right)^n \cdot u(n+2) + 8 \left[1 - \left(\frac{1}{2}\right)^{n+2} \right]$$

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⑦ If $h_1(n) = u(n-2)$ and $h_3(n) = 8(n-2)$
 $h_2(n) = n u(n)$ find $h(n)$.

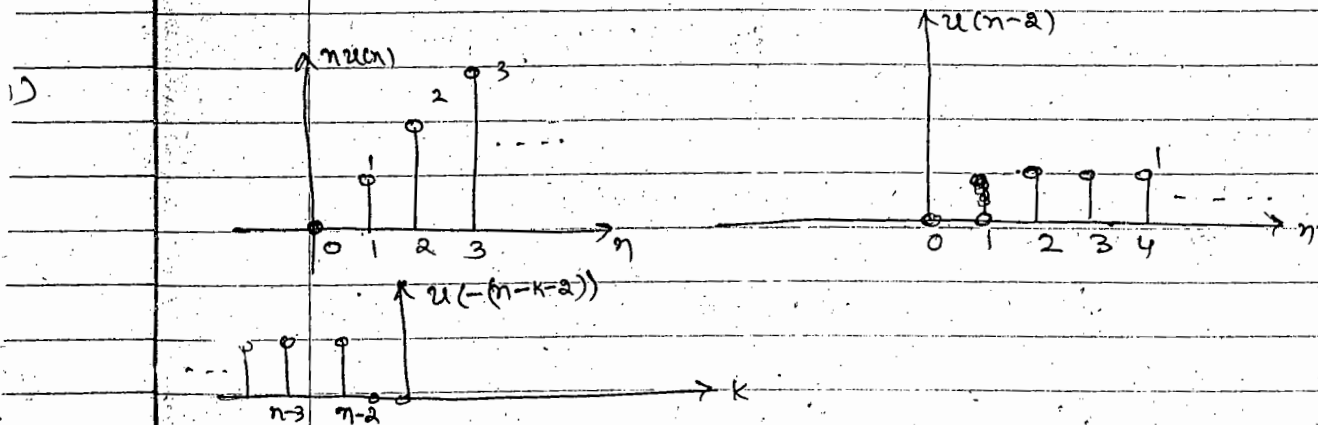
$$h(n) = h_1(n) * [h_2(n) + h_3(n)]$$

$$= u(n-2) * [8(n-2) + n u(n)]$$

$$= u(n-4) + n u(n) * u(n-2)$$

$$= u(n-4) + z(n)$$

$$z(n) = n u(n) * u(n-2)$$



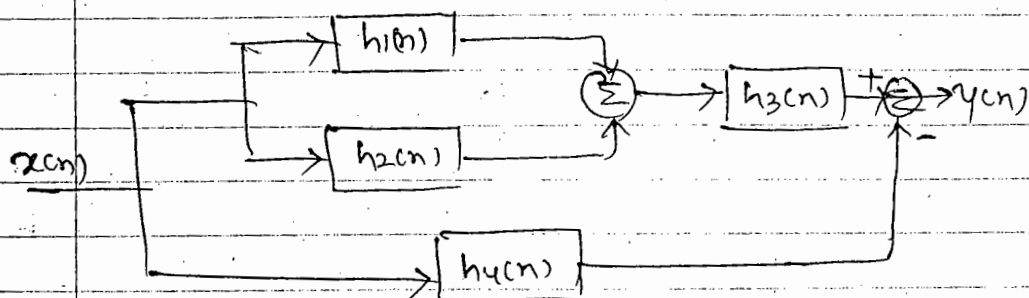
$$z(n) = \sum_{k=0}^{n-2} k \cdot u(k) \cdot u(-(n-k-2)) = \sum_{k=0}^{n-2} k = \frac{(n-2)(n-2+1)}{2}$$

$$\sum_{k=0}^n k = \frac{n(n+1)}{2}$$

$$= \frac{n^2 - 2n - 1n + 2}{2} = \frac{n^2 - 3n + 2}{2}$$

$$h(n) = u(n-4) + \frac{n^2 - 3n + 2}{2}$$

Find the impulse response of the system shown below



Where $h_1(n) = u(n)$; $h_2(n) = u(n+2) - u(n)$; $h_3(n) = 8(n-2)$,
 $h_4(n) = \alpha^n u(n)$

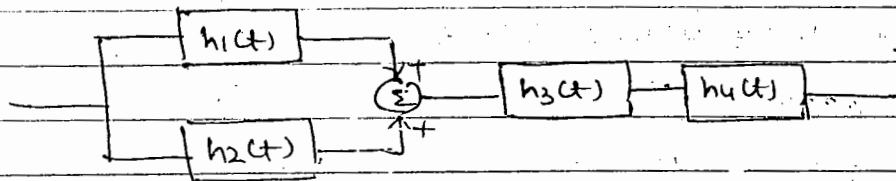
$$\begin{aligned}
 h(n) &= [h_1(n) + h_2(n)] * h_3(n) + h_4(n) \\
 &= [u(n) + u(n+2) - u(n)] * \delta(n-2) + \alpha^n u(n) \\
 &= [u(n+2) * \delta(n-2)] + \alpha^n u(n) \\
 &\quad u(n) - \alpha^n u(n) \\
 \boxed{h(n) = [1 - \alpha^n] u(n)}
 \end{aligned}$$

Mem

Draw the interconnections of systems to obtain the given response equations

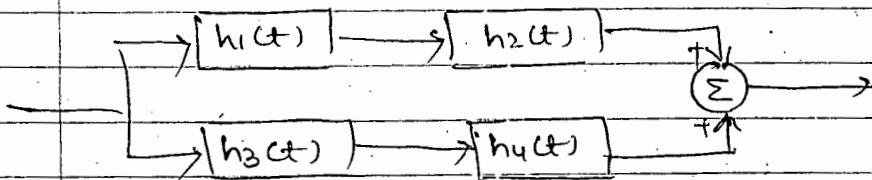
For
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(a) $h(t) = [h_1(t) + h_2(t)] * h_3(t) * h_4(t)$



En

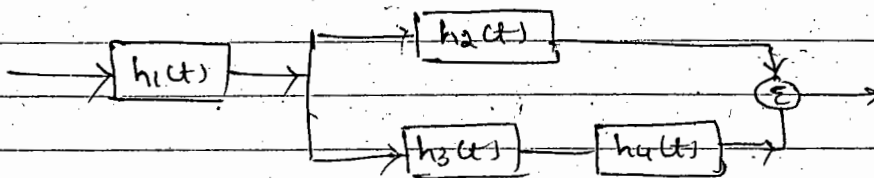
(b) $h(t) = h_1(t) * h_2(t) + h_3(t) * h_4(t)$



mp
em

(c) $h(t) = h_1(t) * [h_2(t) + h_3(t) * h_4(t)]$

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Properties of LTI System using Impulse Response

Memory-Less System

$$\text{Let } y(n) = x(n) * h(n) = h(n) * x(n)$$

$$= \sum_{k=-\infty}^{\infty} h(k) x(n-k)$$

$$= \underbrace{+ h(-2) x(n+2)}_{n < 0} + \underbrace{h(-1) x(n+1)}_{n=0} + \underbrace{x(0) h(0)}_{n > 0} + h(1) x(n-1) + \dots + h(2) x(n-2) + \dots$$

Given

For a system to be memoryless, $y(n)$ should depend only on the present value of $x(n)$ and must be independent of other input terms. Hence the required condition is

$$h(-1) = h(-2) = h(1) = h(2) = 0$$

$$\text{i.e. } \boxed{h(n) = 0; n \neq 0}$$

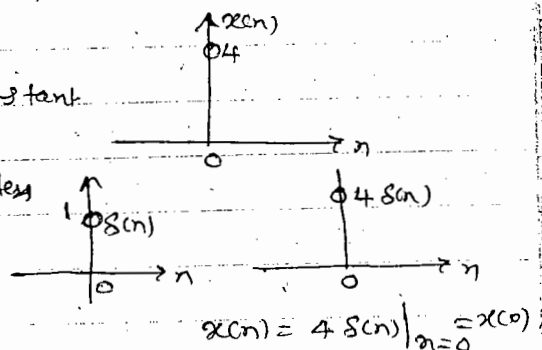
In other words a system is memoryless if and only if

$$\boxed{h(n) = c \delta(n); n=0}$$

where 'c' is any arbitrary constant.

ii) For a CT LTI system is memoryless under the following conditions

$$\boxed{\begin{aligned} h(t) &= 0; t \neq 0 \\ h(t) &= c \delta(t); t=0 \end{aligned}}$$



Causality

$$\text{we have } y(n) = h(n) * x(n)$$

$$y(n) = \underbrace{+ h(-2) x(n+2)}_{n < 0} + \underbrace{h(-1) x(n+1)}_{n=0} + \underbrace{h(0) x(0)}_{n > 0} + h(1) x(n-1) + \dots$$

In causal systems, output depends on the present and past values of the input. Hence the condition required is that the future values of the x_p associated with the index $n < 0$ must be zero.

$$\text{i.e. } \boxed{h(n) = 0; n < 0} \text{ is the condition for causality}$$

iii) A CT LTI system is causal if & only if

$$\boxed{h(t) = 0; t < 0}$$

Stability

A system is BIBO stable if & only if a bounded input produces a bounded output.

$$i.e. |x(n)| < m_x < \infty$$

$$\text{then } y(n) < m_y < \infty$$

proper

1. Memory

$$\text{Let } y(n) = \sum_{k=-\infty}^{\infty} h(k) x(n-k) = h(n) * x(n)$$

2. Causal

$$\text{If } |x(n)| < m_x < \infty, \text{ then } |x(n-k)| < m_x < \infty.$$

$$y(n) = \sum_{k=-\infty}^{\infty} |h(k)| |x(n-k)|$$

3. Stable

4. Invertible

$$\therefore y(n) \leq m_x \sum_{k=-\infty}^{\infty} |h(k)|$$

1. Ch.

$$\text{output } y(n) \text{ is bounded if } \sum_{k=-\infty}^{\infty} |h(k)| < \infty.$$

imp

(a) 1

causal

If impulse response is absolutely summable, then for a bounded input, output is also bounded and system will be stable. Hence stability condition is

Th

$$\sum_{k=-\infty}^{\infty} |h(k)| < \infty$$

i.e.

Sys.

Memory

Rule for a CT LT system

$$\int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty$$

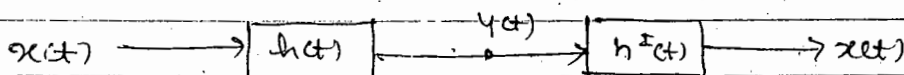
is the required condition

The

then

Stable

Invertible System



$$y(t) = x(t) * h(t)$$

$$x(t) = h^{-1}(t) * y(t) = [h^{-1}(t) * h(t)] * x(t)$$

$$= g(t) * x(t) = x(t)$$

where $g(t) = h^{-1}(t) * h(t)$ is the condition of invertibility.

Also

Rule for a DT LT system

(b)

$$h(n) * h^{-1}(n) = g(n)$$

is the required condition for invertibility.

in

sm.

property	CT LTI system	DT LTI system
1. Memoryless	$h(t) = c \delta(t); t=0$ $h(t) = 0; t \neq 0$	$h(n) = c \delta(n); n=0$ $h(n) = 0; n \neq 0$
2. Causality	$h(t) = 0; t < 0$	$h(n) = 0; n < 0$
3. Stability	$\int_{-\infty}^{\infty} h(\tau) d\tau < \infty$	$\sum_{k=-\infty}^{\infty} h(k) < \infty$
4. Invertible	$h^{\pm}(t) * h(t) = \delta(t)$	$h^{\pm}(n) * h(n) = \delta(n)$

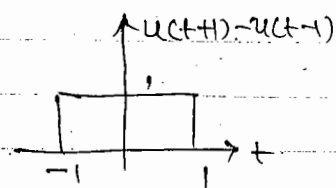
1. Check whether the following systems defined by the impulse responses are causal, stable & memoryless.

(a) $h(t) = u(t+1) - u(t-1)$

for a
system will

causality:

$$h(-1) = u(0) - u(-2) = 1 - 0 = 1.$$



The condition $h(t) = 0; t < 0$ is not satisfied, i.e. output depends upon the future values. Hence the system is not causal.

memoryless or

Memoryless:

$$h(0) = u(1) - u(-1) = u(0) = 1.$$

The condition $h(t) = c \delta(t); t=0$ is not satisfied. Hence the system is not memoryless. (dynamic)

Stability

$$\begin{aligned} \int_{-\infty}^{\infty} |h(\tau)| d\tau &= \int_{-\infty}^{\infty} |u(\tau+1) - u(\tau-1)| d\tau = \int_{-1}^1 1 d\tau \\ &= \int_{-1}^1 1 d\tau < \infty. \end{aligned}$$

(4).

hence $|h(t)| < \infty$; the system is stable

stability.

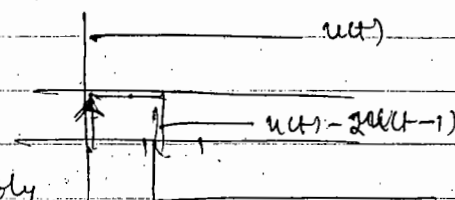
Also $h(t)$ is a energy signal.

(b) $h(t) = u(t) - 2u(t-1)$

$$h(0) = u(0) - 2u(-1) = 1 - 0 = 1$$

Since $h(t)$ relates to input directly,

the system is memoryless.



Causality: Since $h(t) = 0$; $t < 0$;
the system is causal in nature

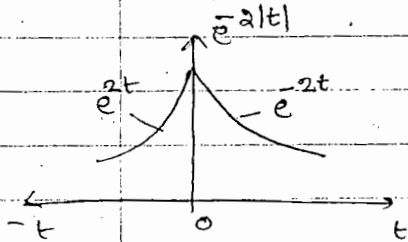
Stability: $\int_{-\infty}^{\infty} h(t) dt = \int_{t=0}^{\infty} h(t) dt \rightarrow \infty$.

Hence the system is not stable.

(f)

Memory:

(c) System $h(t) = e^{-2|t|}$



Memory:

$h(0) = 1$; $t \neq 0$

hence $h(t) = 0$; $t = 0$

The system is memoryless.

Stable

Causality: $h(t) = e^{-2|t|} = \frac{1}{e^2}$ i.e. $h(t) \neq 0$; $t < 0$

The system is not causal.

(8)

Stability: $\int_{-\infty}^{\infty} h(t) dt = \int_{-\infty}^0 e^{2t} dt + \int_0^{\infty} e^{-2t} dt = \frac{1}{2} + \frac{1}{2} < \infty$

Hence the system is stable.

(d) $h(t) = 3\delta(t)$

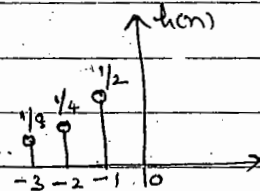
$h(t) = 3$; $t = 0$; hence $h(t) = 3\delta(t)$ at $t = 0$

The system is memoryless

also $h(t) = 0$; $t < 0$; hence it is causal

$\int_{-\infty}^{\infty} h(t) dt = \int_{-\infty}^{\infty} 3\delta(t) dt = 3 < \infty$, system is stable

(e) $h(n) = \begin{cases} 2^n & \text{for } n < 0 \\ 0 & \text{for } n > 0 \end{cases}$



Causality

$h(n) \neq 0$; $n < 0$; the system is not causal

Memory

since $h(n) \neq 0$; for $n \neq 0$; the system has memory

Stability

$\sum_{k=-1}^{\infty} h(k) = \sum_{k=-1}^{-1} 2^k = \sum_{k=1}^{\infty} 2^{-k} = \sum_{k=0}^{\infty} (2^{-1})^{k+1}$

$= 2 \left[\frac{1}{1 - \frac{1}{2}} \right] = 4 < \infty$;

The given system is stable

(9)

$$(f) \quad h(n) = u(n-1) \cdot e^{2n}$$

Memory: $h(n) \neq 0$; $n \neq 0$; hence the system is not memoryless

$$h(2) = u(2-1) \cdot e^4 = u(1) \cdot e^4 = e^4 \neq 0$$

causality: since $u(n-1)$ is defined for $n \geq 1$; the condition $h(n) = 0$; $n < 0$ is satisfied.

Hence the system is causal.

Stability: $\sum_{k=-\infty}^{\infty} |h(k)| = \sum_{k=1}^{\infty} e^{2k} = \sum_{k=1}^{\infty} e^2 \cdot e^{2(k-1)} = e^2 \left[\frac{1}{1-e^2} \right] \rightarrow \infty$

Since $e^2 > 1$; the system is not stable

$$(g) \quad h(n) = \left(\frac{1}{2}\right)^n u(n)$$

$h(0) \neq 0$; $n \neq 0$; the system has memory

Since $h(2) = \left(\frac{1}{2}\right)^2 \cdot u(2) = 0.25$

Since $u(n)$ is defined for $n \geq 0$; the system is causal

Since $\left(\frac{1}{2}\right)^n u(n)$ is a decaying exponential seq. system is stable in nature

$$\sum_{k=-\infty}^{\infty} |h(k)| = \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k = \frac{1}{1 - \frac{1}{2}} = 2 < \infty$$

$$(h) \quad h(n) = \delta(n) + \sin(\pi n)$$

$$h(1) = \delta(1) + \sin(\pi) = 0$$

$$h(-1) = \delta(-1) + \sin(-\pi) = 0$$

$$h(0) = \delta(0) + \sin 0 = 1$$

hence $h(n) = c\delta(n)$ or $h(n) = 0$; $n \neq 0$ condition is satisfied; the system is memoryless.

also $h(n) = 0$; for $n < 0$; the system is causal.

Since $\sin(\pi n)$ value = 0 for all n & $\delta(n) = 1$ at $n = 0$

the system is stable

$$(i) \quad h(t) = \cos \pi t$$

$h(0) \neq 0$; $t \neq 0$; the system has memory.

$h(0) \neq 0$; $t < 0$; the system is not causal.

$h(t)$ has value between ~~and~~

System is not stable

$h(t) / h(n)$	Memoryless	Causality	Stability
$h(t) = e^{-2t} u(t-1)$	X	✓	✓
$h(t) = u(t+1)$	X	X	X
$h(t) = 3\delta(t)$	✓	✓	✓
$h(t) = \cos \pi t \cdot u(t)$	X	✓	X
$h(n) = (-1)^n u(n)$	X	X	X
$h(n) = \left(\frac{1}{2}\right)^{ n }$	X	X	✓
$h(n) = \cos \frac{\pi n}{4} (u(n) - u(n-10))$	X	✓	✓
$h(n) = 2u(n) - 2u(n-5)$	X	✓	✓
$h(n) = \sin \frac{\pi n}{2}$	X	X	X
$h(n) = \sum_{p=-1}^{\infty} \delta(n-2p)$	X	X	X

Unit Step Response of an LTI system

Step signals are often ~~can~~ used to characterize the ~~response~~ response of an LTI system to sudden changes in the input. The step response is defined as the o/p due to a unit step signal.

Let $h(n)$ be the impulse response of a DT LTI system and let $u(n)$ be the input, the step response $s(n)$ is defined as

$$s(n) = h(n) * u(n)$$

$$s(n) = \sum_{k=-\infty}^{\infty} h(k) u(n-k)$$

$$\text{Since } u(n-k) = \begin{cases} 1 & ; -\infty \leq k \leq n \\ 0 & ; k > n \end{cases} \quad \text{or } u(n-k) = \begin{cases} 1 & ; k \leq n \\ 0 & ; k > n \end{cases}$$

$$s(n) = \sum_{k=-\infty}^n h(k)$$

Hence the step response of a DT LTI system is the running sum of impulse response.

Relationship b/w $s(n)$ & $h(n)$

$$s(n) = \sum_{k=-\infty}^n h(k) = h(n) + h(n-1) + h(n-2) + \dots$$

$$s(n-1) = h(n-1) + h(n-2) + \dots$$

$$s(n) = h(n) + s(n-1)$$

$$\text{or } h(n) = s(n) - s(n-1)$$

Similarly for a CT LTI system

$$s(t) = \int_{-\infty}^t h(\tau) d\tau$$

$$h(t) = \frac{d}{dt} s(t)$$

Step response of a CT LTI system is the running integral of impulse response.

1. Evaluate the step response of LTI systems whose input-like responses are given below.

(a) $h(n) = \left(\frac{1}{2}\right)^n u(n)$

$$s(n) = h(n) * u(n) = \left(\frac{1}{2}\right)^n u(n) * u(n)$$

$$= \sum_{k=-\infty}^n \left(\frac{1}{2}\right)^k u(k) \cdot u(n-k) = \sum_{k=0}^n \left(\frac{1}{2}\right)^k$$

$$\text{Since } u(k) \cdot u(n-k) = \begin{cases} 1 & ; 0 \leq k \leq n \\ 0 & ; k \leq 0 \end{cases}$$

$$s(n) = \frac{1 - \left(\frac{1}{2}\right)^{n+1}}{1 - 1/2} = 2 \left[1 - \left(\frac{1}{2}\right)^{n+1} \right]$$

(b) $h(n) = \left(-\frac{3}{4}\right)^n u(n)$

$$s(n) = h(n) * u(n) = \sum_{k=-\infty}^n \left(-\frac{3}{4}\right)^k u(k) \cdot u(n-k)$$

$$= \sum_{k=0}^n \left(-\frac{3}{4}\right)^k \cdot 1 = \frac{1 - \left(-\frac{3}{4}\right)^{n+1}}{1 - (-3/4)} = \frac{4}{7} \left[1 - \left(-\frac{3}{4}\right)^{n+1} \right]$$

(c) $h(t) = \frac{1}{4} [u(t) - u(t-4)]$

$$s(t) = h(t) * u(t) = \int_{-\infty}^t \frac{1}{4} (u(\tau) - u(\tau-4)) u(t-\tau) d\tau$$

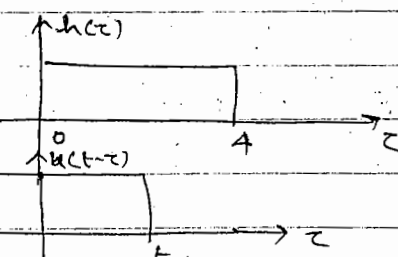
for $t < 0$, $s(t) = 0$

for $t \geq 0$ and $t \leq 4$

$$s(t) = \int_{\tau=0}^t \left(\frac{1}{4}\right) \cdot 1 \cdot d\tau$$

$$s(t) = \frac{t}{4} \cdot u(t)$$

for $t > 4$
 $s(t) = \int_{\tau=0}^4 \frac{1}{4} d\tau = \frac{1}{4} \tau \Big|_0^4 = 1$



(14) $h(t) = e^{-|t|}$

$$S(t) = \int_{-\infty}^t e^{-|\tau|} u(t-\tau) d\tau$$

For $t < 0$

$$= \int_{-\infty}^t e^{\tau} d\tau = e^{\tau} \Big|_{-\infty}^t = e^t$$

$$S(t) = e^t$$

For $t > 0$

$$S(t) = \int_{-\infty}^0 e^{+\tau} d\tau + \int_0^t e^{-\tau} d\tau = 1 + e^{-t}$$

$$S(t) = 2 + e^{-t}$$

$$\therefore S(t) = \begin{cases} e^t & ; t < 0 \\ 2 + e^{-t} & ; t > 0 \end{cases}$$

(5) $h(n) = 2\delta(n) + 3u(n-1)$

$$S(n) = h(n) * u(n)$$

$$S(n) = \sum_{k=-\infty}^n h(k) u(n-k) = \sum_{k=0}^n [2\delta(k) + 3u(k-1)] u(n-k)$$

$$= 2 + \sum_{k=1}^n 3 = 2 + 3[n+1-1] = 2 + 3n$$

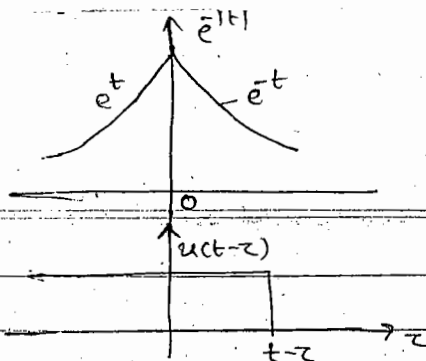
$$S(n) = 2u(n) ;$$

$$= 2u(n) + 3 \cdot n ; n > 1$$

$$= \sum_{k=0}^n 2\delta(k) + \sum_{k=1}^n 3$$

$$= 2[\delta(n) + \delta(n-1) + \delta(n-2) + \dots] + 3[(n+1)-1] \cdot u(n-1)$$

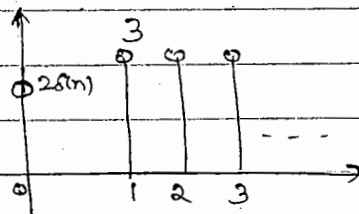
$$S(n) = 2u(n) + 3n \cdot u(n-1)$$



(7) $h(t)$

11/4

(8)



(9)

pn

check

(6)

$$h(t) = u(t)$$

$$S(t) = h(t) * u(t) = \int_{-\infty}^t u(\tau) \cdot u(t-\tau) d\tau$$

$$= \int_{\tau=0}^t 1 \cdot d\tau = \underline{\underline{t}}$$

$$\textcircled{7} \quad h(t) = s(t) - s(t-1) \quad \& \quad h(n) = s(n) - s(n-1)$$

$$s(t) = h(t) * u(t)$$

$$= [s(t) - s(t-1)] * u(t)$$

$$s(t) = u(t) - u(t-1)$$

11/4

$$s(n) = u(n) - u(n-1)$$

\textcircled{8}

$$h(n) = nu(n)$$

$$s(n) = h(n) * u(n) = \sum_{k=-\infty}^n k \cdot u(k) \cdot u(n-k)$$

$$= \sum_{k=0}^n k = \frac{n \cdot (n+1)}{2}$$

$$s(n) = \frac{n(n+1)}{2} u(n)$$

 $n-k$

\textcircled{9}

$$h(n) = (-1)^n [u(n+2) - u(n-3)]$$

$$s(n) = \sum_{k=-\infty}^n (-1)^k [u(k+2) - u(k-3)] u(n-k) = \sum_{k=-\infty}^n (-1)^k \cdot u(k-3) u(n-k)$$

$$= \sum_{k=2}^{n+3} (-1)^k - \sum_{k=-3}^{n-1} (-1)^k \cdot 1 = p(n) + q(n)$$

Put

$$k+2=m$$

$$k+3=m$$

$$s(n) = \sum_{m=0}^{n+2} (-1)^{m-2} - \sum_{m=0}^{n+3} (-1)^{m-3}$$

 $(n-1)$

$$s(n) = \frac{1 - (-1)^{n+3}}{1 - (-1)} + \frac{1 - (-1)^{n+4}}{1 - (-1)}$$

$$= \frac{1}{2} [1 - (-1)^{n+3} + 1 - (-1)^{n+4}]$$

$$= 1$$

Check

Note for n even
 $-(-1)^{n+3} = +1$
 $-(-1)^{n+4} = -1$

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③ I

④

Block diagram representation of LTI Systems

A block diagram is an interconnection of elementary operations that act on the input signal.

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A block diagram is a more detailed representation of a system than the impulse response or differential or difference equations descriptions.

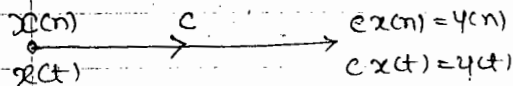
The block diagram representation describes how the system's internal computations or operations are ordered.

The impulse response and differential/difference equations represent only the I/O behavior of a system.

Block diagram representation helps in the implementation of the system using digital computer and simulation can be easily performed. It is simple, efficient and flexible way of implementing the system.

Determine the Elementary operational blocks

1. Scalar multiplication

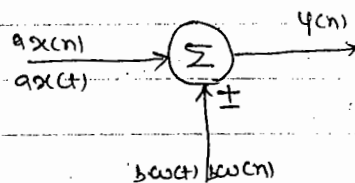


$$y(t) = c x(t)$$

$$y(n) = c x(n)$$

where 'c' is a scalar which alters the magnitude of the input signal $x(n)$ or $x(t)$.

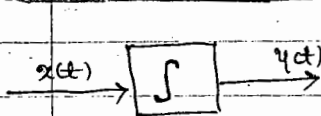
2. Summer block:



$$y(n) = a x(n) + b w(n)$$

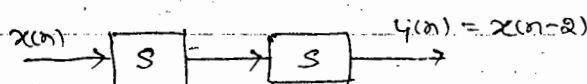
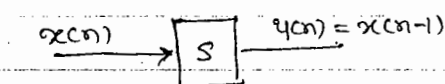
$$y(t) = a x(t) + b w(t)$$

③ Integration block for CT-LTI systems

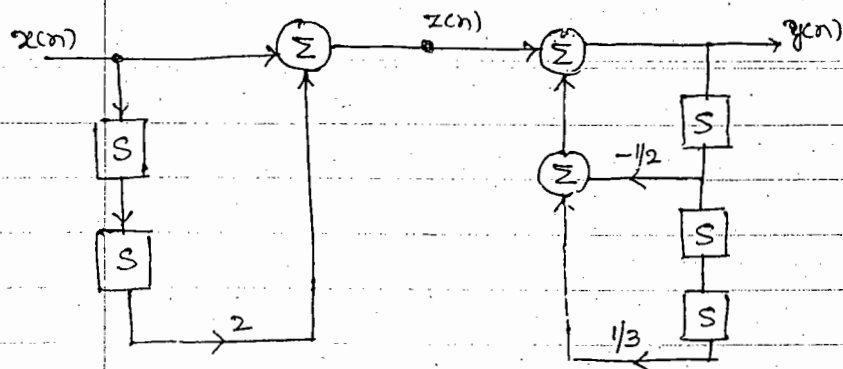


$$y(t) = \int_{-\infty}^t x(\tau) d\tau$$

④ Delay block or Time Shift block for DT-LTI system



1. Determine the difference eqⁿ corresponding to the block diagram shown below.



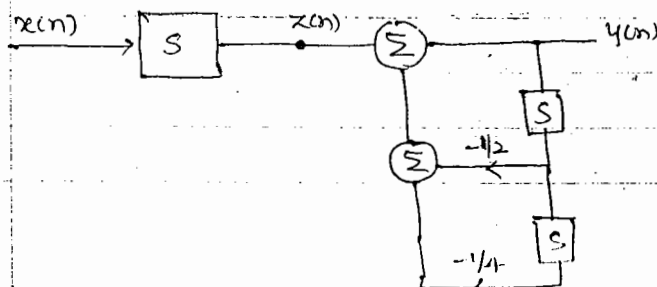
$$z(n) = x(n) + 2x(n-2)$$

$$y(n) = z(n) - \frac{1}{2}y(n-1) + \frac{1}{3}y(n-3)$$

$$y(n) = x(n) + 2x(n-2) - \frac{1}{2}y(n-1) + \frac{1}{3}y(n-3)$$

$$\therefore y(n) + \frac{1}{2}y(n-1) - \frac{1}{3}y(n-3) = x(n) + 2x(n-2)$$

2



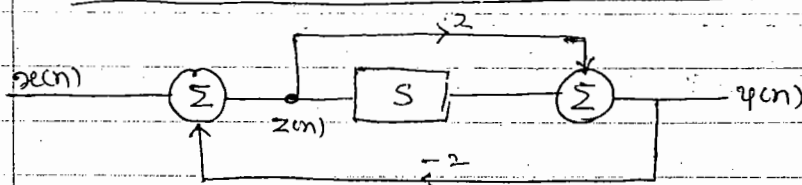
$$z(n) = x(n-1)$$

$$y(n) = z(n) - \frac{1}{4}y(n-2) - \frac{1}{2}y(n-1)$$

$$= x(n-1) - \frac{1}{4}y(n-2) - \frac{1}{2}y(n-1)$$

$$\therefore y(n) + \frac{1}{2}y(n-1) + \frac{1}{4}y(n-2) = x(n-1)$$

3



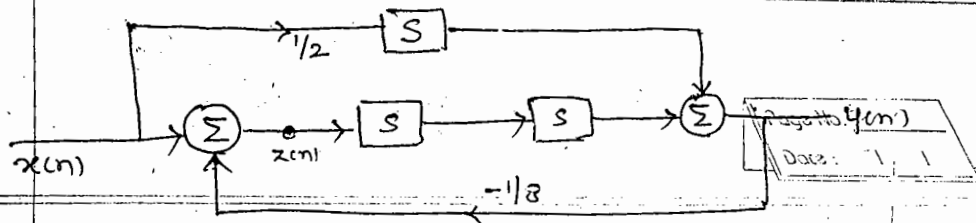
$$z(n) = x(n) - 2y(n)$$

$$y(n) = 2z(n) + z(n-2)$$

$$= 2x(n) - 4y(n) + x(n-2) - 2y(n-2)$$

$$5y(n) + 2y(n-2) = 2x(n) + x(n-2)$$

block



$$z[n] = x[n] - \frac{1}{8} y[n]$$

$$y[n] = \frac{1}{2} x[n-1] + z[n-2]$$

$$y[n] = \frac{1}{2} x[n-1] + x[n-2] - \frac{1}{8} y[n-2]$$

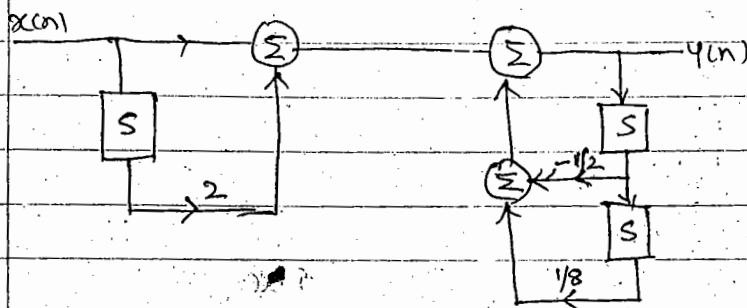
$$y[n] - \frac{1}{8} y[n-2] = \frac{1}{2} x[n-1] + x[n-2]$$

change in
* Since the order of interconnected systems does not affect the I/O relation of the overall system, there two forms of block diagram representations as Direct form I & Direct form II

1. Draw the Direct form I & II representation for the following difference equations.

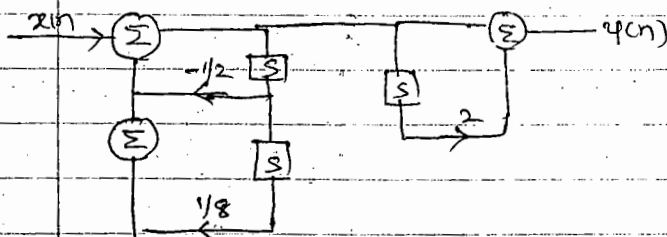
(a) $y[n] + \frac{1}{2} y[n-1] - \frac{1}{8} y[n-2] = x[n] + 2x[n-1]$

$$y[n] = x[n] + 2x[n-1] - \frac{1}{2} y[n-1] + \frac{1}{8} y[n-2]$$

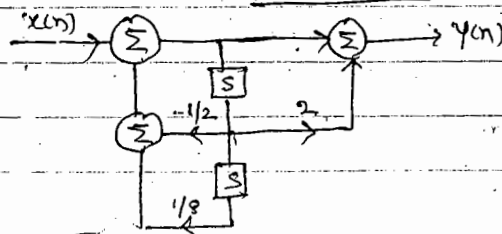


Direct Form I

Intermediate form

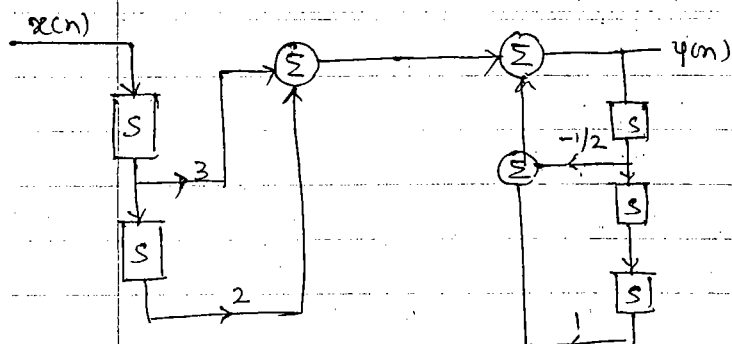


Direct Form II

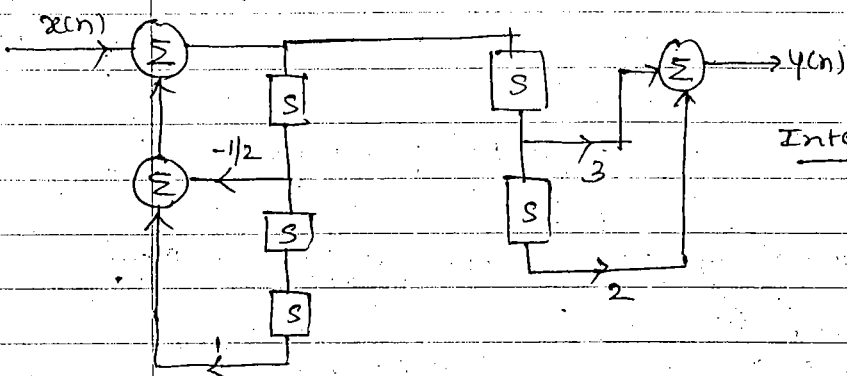


(b) $y(n) + \frac{1}{2} y(n-1) - y(n-3) = 3x(n-1) + 2x(n-2)$

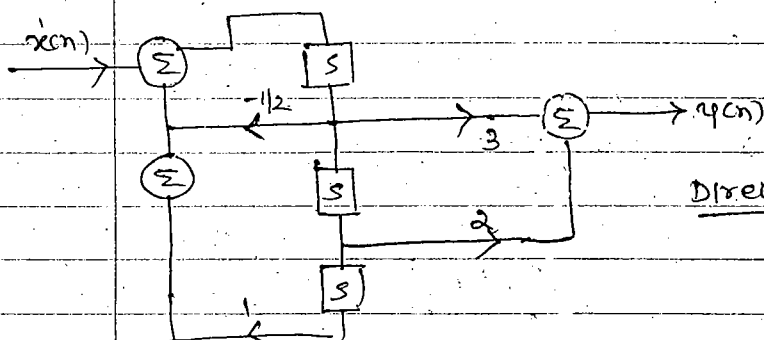
$y(n) = 3x(n-1) + 2x(n-2) - \frac{1}{2} y(n-1) + y(n-3)$



Direct Form I



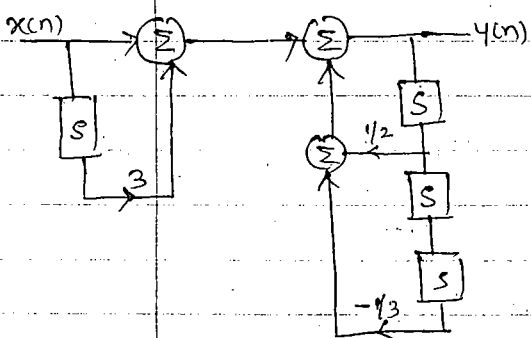
Intermediate form



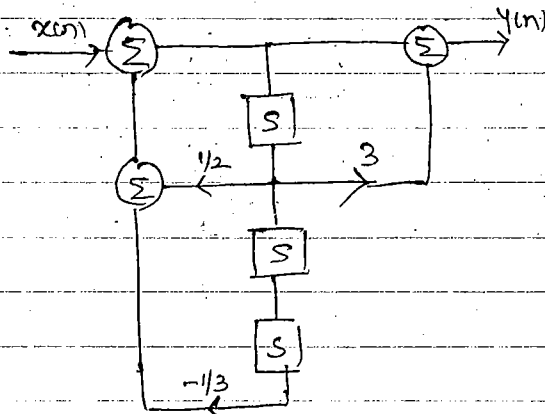
Direct Form 2

(c) $y(n) + \frac{1}{2} y(n-1) + \frac{1}{3} y(n-3) = x(n) + 3x(n-1)$

$y(n) = x(n) + 3x(n-1) + \frac{1}{2} y(n-1) - \frac{1}{3} y(n-3)$



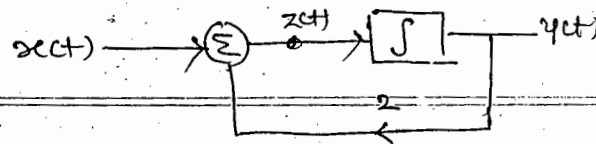
Direct Form I



Direct Form II

Find the Differential eqⁿ description of the systems shown below.

①



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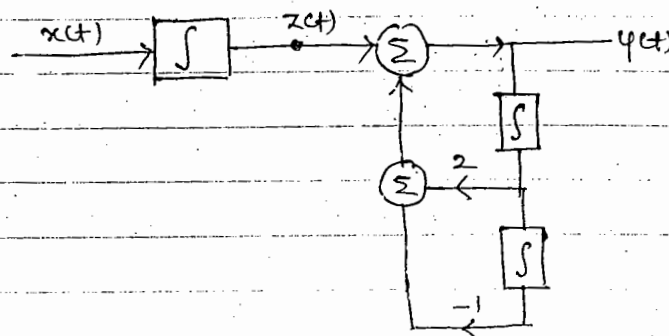
$$z(t) = x(t) + 2y(t)$$

$$y(t) = \int z(t) dt = \int (x(t) + 2y(t)) dt$$

$$\frac{dy(t)}{dt} = x(t) + 2y(t)$$

$$\therefore \frac{dy(t)}{dt} - 2y(t) = x(t)$$

②



$$z(t) = \int x(t) dt$$

$$y(t) = z(t) + 2 \int y(t) dt - \int \int y(t) dt$$

$$= - \int x(t) dt + 2 \int y(t) dt - \int \int y(t) dt$$

$$\frac{d^2 y(t)}{dt^2} = \frac{dx(t)}{dt} + 2 \frac{dy(t)}{dt} - y(t)$$

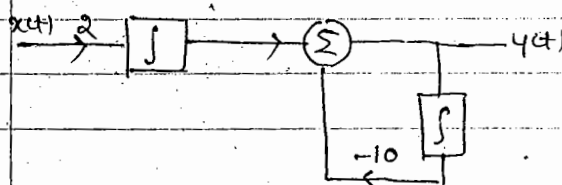
$$\therefore \left[\frac{d^2 y(t)}{dt^2} - 2 \frac{dy(t)}{dt} + y(t) \right] = \frac{dx(t)}{dt}$$

③ Draw the direct form I & 2 representation of the following systems described by the diff. eqⁿ

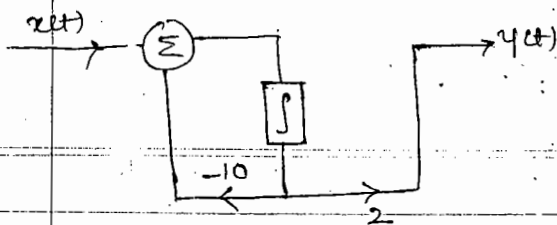
(a) $\frac{d}{dt} y(t) + 10y(t) = 2x(t)$

$$\frac{d}{dt} y(t) + 10y(t) = 2x(t)$$

$$y(t) + \int 10y(t) dt = \int 2x(t) dt$$



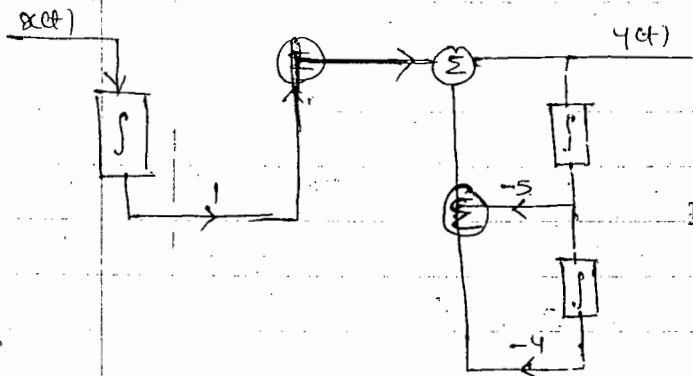
Direct form I



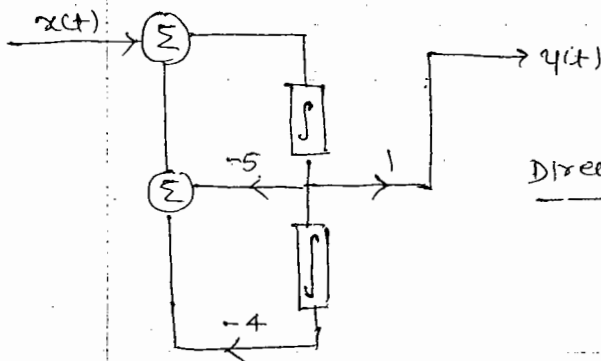
Direct form II

⑤ $\frac{d^2 y(t)}{dt^2} + 5 \frac{dy(t)}{dt} + 4y(t) = \frac{d}{dt} x(t)$

$y(t) + 5 \int y(t) + 4 \iint y(t) = \int x(t)$



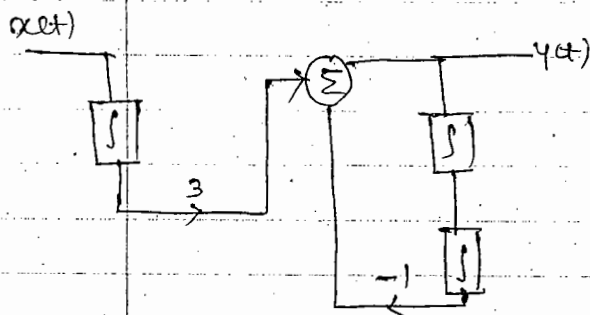
Direct form I



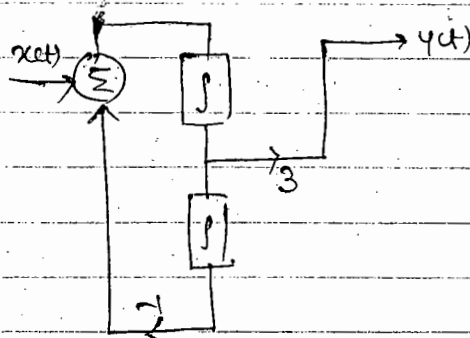
Direct form II

⑥ $\frac{d^2 y(t)}{dt^2} + y(t) = 3 \frac{d}{dt} x(t)$

$y(t) = 3 \int x(t) - \iint y(t)$



Direct form I

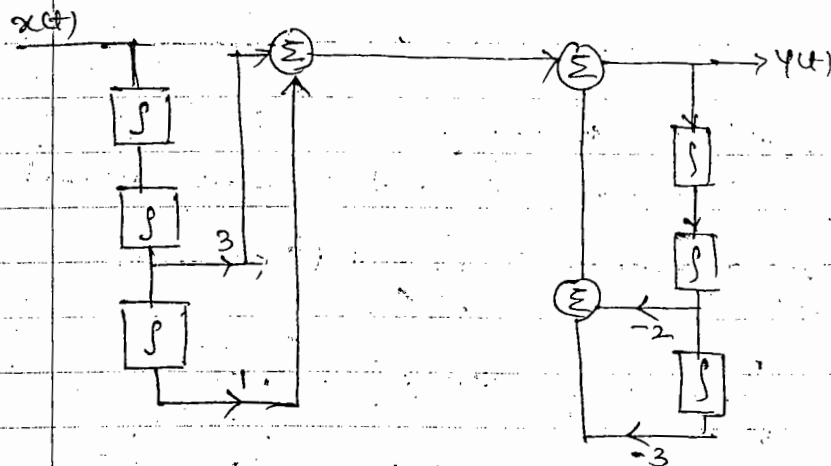


Direct form II

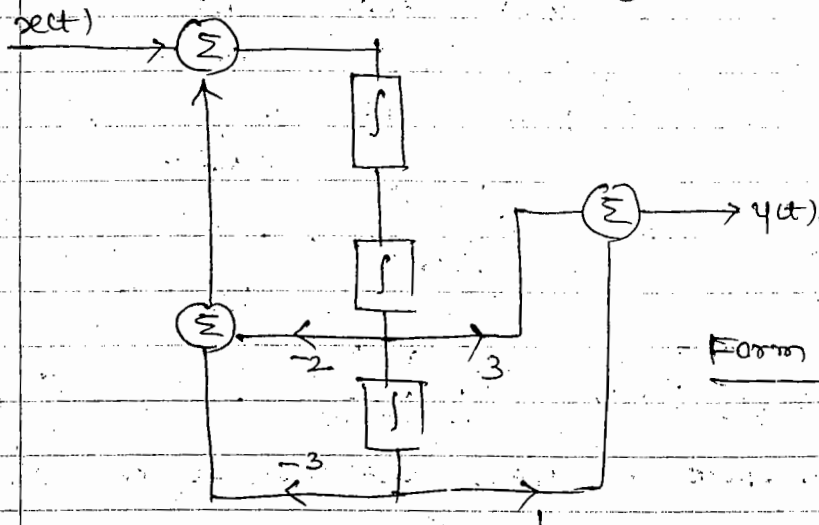
(d) ~~$y''(t) + 2y'(t) + 3y(t) = x(t)$~~

$\frac{d^3 y(t)}{dt^3} + 2 \frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} = x(t) + 3 \frac{d^2 x(t)}{dt^2}$

$y(t) = \iiint x(t) + 3 \iint x(t) - 2 \iint y(t) - 3 \iiint y(t)$

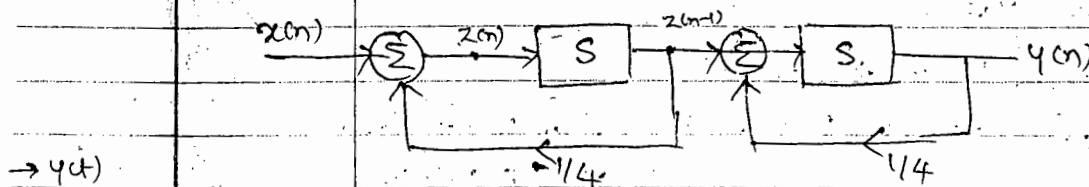


Form I



Form 2

Find the diff. eqn for the block diagram shown below.



$z(n) = x(n) - \frac{1}{4} z(n-1) \rightarrow \text{A}$

$y(n) = z(n-1) + \frac{1}{4} y(n-1) \rightarrow \text{B}$

from A $z(n) = x(n) - \frac{1}{4} z(n-1)$

$z(n-1) = x(n-1) - \frac{1}{4} z(n-2)$

$z(n-2) = x(n-2) - \frac{1}{4} z(n-3)$

from B $z(n-1) = y(n) - \frac{1}{4} y(n-1)$

$z(n-2) = y(n-1) - \frac{1}{4} y(n-2)$

$$\begin{aligned}
 y(n) &= 2(n-2) + \frac{1}{4} y(n-1) \\
 \therefore y(n) &= x(n-2) - \frac{1}{4} 2(n-3) + \frac{1}{4} y(n-1) \\
 &= x(n-2) - \frac{1}{4} \left[y(n-1) - \frac{1}{4} y(n-2) \right] + \frac{1}{4} y(n-1)
 \end{aligned}$$

$$y(n) - \frac{1}{16} y(n-2) = x(n-2)$$

convolve $u(n+3) * u(n-3)$

$$\text{Let } y(n) = x_1(n) * x_2(n) = u(n+3) * u(n-3)$$

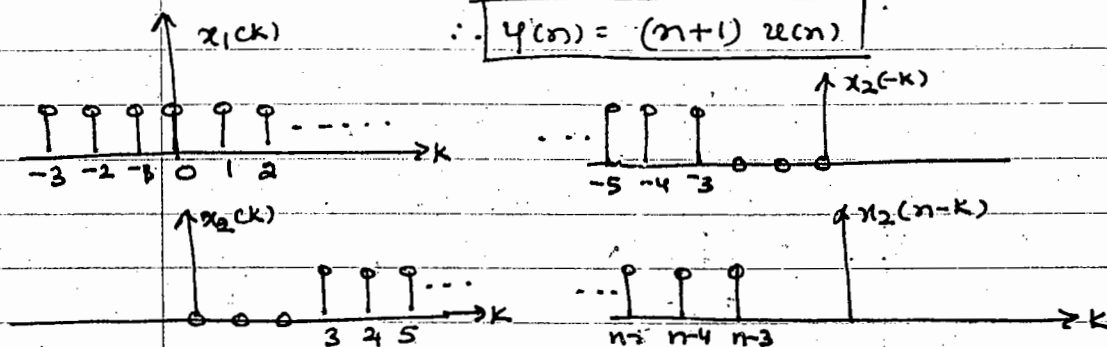
$$= \sum_{k=-\infty}^{\infty} x_1(k) x_2(n-k) = \sum_{k=-\infty}^{\infty} u(k+3) \cdot u(n-k)$$

$$= \sum_{k=-3}^{n-3} 1 = (n-3+1) + 3 = n+1$$

$$\therefore y(n) = n+1 ; n \geq 0$$

$$= 0 ; n < 0$$

$$\therefore y(n) = (n+1) u(n)$$



convolution starts at $k = -3$ and ends at $k = n-3$.

convolve $[\delta(n) - \delta(n-2)] * u(n) = y(n)$

and determine $y(0)$.

$$\text{Let } y(n) = x_1(n) * x_2(n)$$

$$\begin{aligned}
 x_1(n) &= [\delta(n) - \delta(n-2)] * u(n) = u(n) - u(n-2) \\
 &= \delta(n) + \delta(n-1)
 \end{aligned}$$

$$x_2(n) = u(n)$$

$$\therefore y(n) = [\delta(n) + \delta(n-1)] * u(n)$$

$$= \delta(n) * u(n) + \delta(n-1) * u(n)$$

$$y(n) = u(n) + u(n-1)$$

$$\therefore y(0) = y(0) + u(-1) =$$

$$\therefore y(0) = \delta(n)$$

Sinusoidal steady state response

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(n-1)

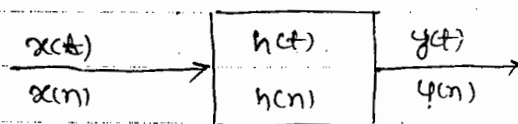
Frequency Response:

The response of an LTI system to a sinusoidal input leads to a characterization of system behavior and is termed as a frequency response of the system.

This characterization is obtained in terms of impulse response and complex sinusoidal input signal.

3)

consider an LTI system shown below



$$x(n) = e^{j\Omega n}$$

$$x(t) = e^{j\omega t}$$

$$y(n) = x(n) * h(n) = h(n) * x(n) = \sum_{k=-\infty}^{\infty} h(k) \cdot x(n-k)$$

$$= \sum_{k=-\infty}^{\infty} h(k) \cdot e^{-j\Omega k} \cdot e^{j\Omega n} =$$

$$y(n) = H(e^{j\Omega}) \cdot e^{j\Omega n}$$

($H(e^{j\Omega})$ = Freq. Response)

where $H(e^{j\Omega}) = H(\Omega) = \sum_{k=-\infty}^{\infty} h(k) \cdot e^{-j\Omega k}$

$H(\Omega)$ is not a function of time 'n' but only a fⁿ of frequency Ω and is termed as frequency response of the system (DTLTI).

Similarly for a CT LTI system

= y(t)

$$y(t) = \int_{-\infty}^{\infty} h(\tau) \cdot x(t-\tau) d\tau = \int_{-\infty}^{\infty} h(\tau) \cdot e^{-j\omega\tau} \cdot e^{j\omega t} d\tau$$

$$= e^{j\omega t} \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau$$

$$y(t) = H(j\omega) \cdot e^{j\omega t}$$

where $H(j\omega) = H(\omega) = \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau$ is known as the frequency response of the CT LTI system.

$$H(\omega) = |H(\omega)| e^{j\arg(H(\omega))}$$

$|H(\omega)| \rightarrow$ Magnitude response

$\angle H(\omega) \rightarrow$ phase response

$$y(t) = |H(\omega)| \cdot e^{j[\omega t + \angle H(\omega)]}$$

Evaluate the freq. response of the LTI systems whose impulse responses are given below

(a) $h(t) = -\delta(t+1) + \delta(t) - \delta(t-1)$

$$\begin{aligned} H(\omega) &= \int_{-\infty}^{\infty} h(t) \cdot e^{-j\omega t} dt = \int_{-\infty}^{\infty} [-\delta(t+1) + \delta(t) - \delta(t-1)] e^{-j\omega t} dt \\ &= -\left[e^{-j\omega t} \right]_{t=-1} + \left[e^{-j\omega t} \right]_{t=0} - \left[e^{-j\omega t} \right]_{t=1} \\ &= -e^{j\omega} + 1 - e^{-j\omega} = 1 - [e^{j\omega} + e^{-j\omega}] \\ \boxed{H(\omega) = 1 - 2\cos\omega} \end{aligned}$$

(b) $h(t) = e^{at} u(-t)$

$$\begin{aligned} H(\omega) &= \int_{-\infty}^0 e^{at} u(-t) \cdot e^{-j\omega t} dt = \int_{-\infty}^0 e^{at} e^{-j\omega t} dt = \int_{-\infty}^0 e^{(a-j\omega)t} dt \\ &= \left[\frac{e^{(a-j\omega)t}}{a-j\omega} \right]_{-\infty}^0 = \frac{1}{a-j\omega} \end{aligned}$$

$$\boxed{H(\omega) = \frac{1}{a-j\omega}}$$

(c) $u(n+2) - u(n-3)$

(c) $h(n) = (-1)^n [u(n+2) - u(n-3)]$

$$u(n+2) - u(n-3) = \delta(n+2) + \delta(n+1) + \delta(n) + \delta(n-1) + \delta(n-2)$$

$$\begin{aligned} H(\Omega) &= \sum_{n=-\infty}^{\infty} h(n) \cdot e^{-j\Omega n} = \sum_{n=-\infty}^{\infty} (-1)^n [u(n+2) - u(n-3)] e^{-j\Omega n} \\ &= \sum_{n=-\infty}^{\infty} [(-1)^n \delta(n+2) + (-1)^n \delta(n+1) + \delta(n) (-1)^n + (-1)^n \delta(n-1) + (-1)^n \delta(n-2)] e^{-j\Omega n} \\ &= (-1)^2 e^{-j\Omega 2} + (-1)^1 e^{-j\Omega 1} + (-1)^0 e^{-j\Omega 0} + (-1)^{-1} e^{-j\Omega (-1)} + (-1)^{-2} e^{-j\Omega (-2)} \\ &= e^{+2j\Omega} - e^{j\Omega} + 1 - e^{-j\Omega} + e^{-j2\Omega} \end{aligned}$$

$$\boxed{H(\Omega) = 1 + 2\cos 2\Omega - 2\cos \Omega}$$

LTI System representation using Differential and difference eqns

Differential and difference equations are used for the time domain representation of LTI systems. They relate the input and output of the LTI systems.

Differential equations are used for the representation of CT-LTI systems and Difference equations are used for DT-LTI systems.

DE method can not be used to find the impulse response of the system directly. However if step response is known; then the impulse response can be determined by using the relation b/w step response & impulse response i.e. $h(n) = S(n) - S(n-1)$

However the step response assumes that the system is at rest. Hence step response is the response of the system for zero initial conditions. Hence there is no provision for initial conditions when one is using impulse response.

Hence impulse response applies to systems that are initially at rest or when the input is known for all time.

Differential and Difference equation descriptions are more flexible since they apply to systems that are at rest or with non zero initial conditions.

Differential equation representation of LTI system

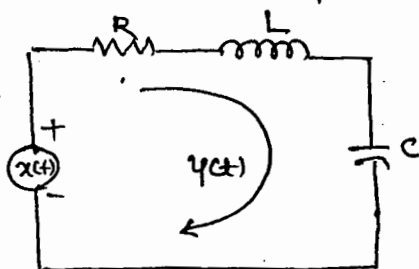
The general form of Linear constant coefficient DE is of the form

$$\sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k}$$

Where a_k, b_k are constant coefficients of the system, $x(t)$ is the input applied to the system and $y(t)$ is the output.

The order of the D.E is N and $(N > M)$.

The order of the diff. eqn represents the number of energy storage devices in the system.



Applying KVL

$$x(t) = R y(t) + L \frac{d}{dt} y(t) + \frac{1}{C} \int y(t) dt$$

$$L \frac{d^2 y(t)}{dt^2} + R \frac{d}{dt} y(t) + \frac{1}{C} y(t) = \frac{d}{dt} x(t)$$

Here L, R, C are constant coefficients

The order of the DE = $N = 2$. Hence it has two storage elements namely inductor and capacitor.

Initial conditions (IC)

Initial conditions summarize all the information about the system's past that is needed to determine the future o/p's. In general no. of IC's required to determine the output is equal to the maximum memory of the system. These are directly related to the initial values of energy storage devices of the system such as initial voltages on capacitors or initial currents thro' inductors.

Also number of initial conditions required is equal to the order of the D.E.

If the order of DE = n ; then it is required a minimum of n initial conditions to solve the equation.

Solving the Differential and Difference equations

The output response can be expressed as two parts

$$\text{output} = \text{Homogeneous solution} + \text{Forced response}$$

or. $\boxed{\text{output} = \text{Natural response} + \text{forced response}}$

Natural response (Zero input response or Homogeneous solution)

Natural response of the system is the system output for zero input and describes the behavior of the system for the initial conditions present. Natural response describes the transient behavior of the system. It describes the transition of the system from initial conditions to an equilibrium state defined by the I/p.

Forced Response (Zero State response)

The forced response ~~is~~^{is} the system output due to the input signal only assuming zero initial conditions. Hence forced response is also known as Zero State response. Forced response is of the same form as the complete solution of DE. The forced response describes the system behavior that is 'forced' by the system I/p when the system is at rest.

Forced response has two parts

$$\text{forced Response} = \text{homogeneous sol}^n + \text{particular solution}$$

Particular Solution!

particular solution represents any solution of DE for the given input. and is not unique. A particular solⁿ is obtained by assuming an o/p of the same general form as I/p.

Note: when the I/p is of the form one of the components of

the homogeneous solution; assume a particular solution that is independent of all terms in the homogeneous solution.

The complete solution or total response

Complete solution = homogeneous soln + particular solution.

$$\boxed{\text{Total response} = \text{Natural response} + \text{forced response}}$$

the homogeneous solution constants should be found out using initial conditions.

Importance of Natural response & characteristic equation page 17

Find the total response of the system

$$x(t) = e^{2t} u(t)$$

$$y''(t) + 5y'(t) + 4y(t) = x'(t) \quad \text{with } y(0) = 0, \quad y'(0) \Big|_{t=0} = 1$$

$$\text{Char. eqn is } y^2 + 5y + 4 = 0$$

$$\boxed{r_1 = -4, r_2 = -1}$$

$$\boxed{y_h(t) = c_1 e^{-4t} + c_2 e^{-t}} \quad \text{--- (A)}$$

$$\text{put } t=0 \text{ in (A)} \quad c_1 + c_2 = y_h(0) = y(0) = 0$$

$$\text{put } t=0 \text{ in diff eqn (A)} \quad y'_h(t) = -4c_1 e^{-4t} - c_2 e^{-t}$$

$$\text{put } t=0 \quad y'_h(0) = y'(0) = 1 = -4c_1 - c_2$$

$$\boxed{c_1 = -1/3}$$

$$\boxed{c_2 = 1/3}$$

$$\boxed{y_h(t) = -\frac{1}{3} e^{-4t} + \frac{1}{3} e^{-t}}$$

$$y_p(t) = K e^{2t} u(t)$$

$$K \frac{d^2}{dt^2} (e^{2t}) + K 5 \frac{d}{dt} (e^{2t}) + K 4 e^{2t} = \frac{d}{dt} [x(t)] = \frac{d}{dt} e^{2t} = 2e^{2t}$$

$$+4K e^{2t} - 10K e^{2t} + 4K e^{2t} = 2e^{2t}$$

$$\Rightarrow \boxed{K=1} \quad \therefore \boxed{y_p(t) = e^{2t} u(t)}$$

$$\therefore \boxed{y_f(t) = c_1 e^{-4t} + c_2 e^{-t} + e^{2t}} \quad \text{--- (B)}$$

$$\text{put } t=0 \text{ in eqn (B)} \quad y_f(0) = 0 = c_1 + c_2 + 1$$

& assuming zero IC.

$$\therefore \boxed{c_1 + c_2 = -1}$$

$$\text{diff eqn (B)} \quad y'_f(t) = -4c_1 e^{-4t} - c_2 e^{-t} + 2e^{2t}$$

$$\text{put } t=0; \quad y'_f(0) = 0 = -4c_1 - c_2 + 2 \Rightarrow 4c_1 + c_2 = 2$$

$$\boxed{y_f(t) = -\frac{1}{3} e^{-4t} - \frac{2}{3} e^{-t} + e^{2t}}$$

$$c_1 = -1/3, c_2 = -2/3$$

$$\boxed{y(t) = y_h(t) + y_f(t) = -\frac{2}{3} e^{-4t} - \frac{1}{3} e^{-t} + e^{2t}}$$

is the required total response.

Natural Response

Natural response form

1. Roots are real & distinct :

$$y_n(t) = \sum_{i=1}^n c_i e^{r_i t}$$

2. Roots are non repeated & imaginary

$$y_n(t) = \sum_{i=1}^n [c_i \cos t + c_{i+1} \sin t]$$

3. Roots are non repeated and complex

$$y_n(t) = e^{-t} \left[\sum_{i=1}^n [c_i \cos t + c_{i+1} \sin t] \right]$$

4. Roots are repeated :

$$y_n(t) = \sum_{i=1}^n [c_i e^{r_i t} + c_{i+1} t e^{r_i t}]$$

(1) Find the natural response of the system described by the following differential equations.

(a) $5 \frac{d}{dt} y(t) + 10 y(t) = 2x(t)$ with $y(0) \Big|_{t=0} = 3$.

$$5 \frac{d}{dt} y(t) + 10 y(t) = 2x(t)$$

Step 1 : put $x(t) = 0$ to get homogenous equation.

$$5 \frac{d}{dt} y(t) + 10 y(t) = 0$$

Step 2 : write the characteristic equation by replacing $\frac{d^k y(t)}{dt^k}$ by r^k

$$5r + 10 = 0$$

$$5r + 10 = 0$$

$$r = -\frac{10}{5} = -2$$

The root of the char. eqn $r = -2$

The natural response is of the form

$$y_n(t) = y^h(t) = \sum_{i=1}^n c_i e^{r_i t}$$

$$\boxed{y_n(t) = c_1 e^{-2t}} \quad \text{--- (A)}$$

using the initial condition $y(0) \Big|_{t=0}$ by putting $t=0$ in eqn (A)

$$y_n(0) = y(0) = 3 = c_1 e^0 \Rightarrow \boxed{c_1 = 3}$$

$$\therefore \boxed{y_n(t) = 3 e^{-2t}} \text{ is the required natural response.}$$

⑥ $\frac{d^2 y(t)}{dt^2} + 6 \frac{dy(t)}{dt} + 8y(t) = \frac{d}{dt} x(t)$ with $y(0) = 0, y'(0) \Big|_{t=0} = 1$

Homogeneous equation is

$$\frac{d^2 y(t)}{dt^2} + 6 \frac{dy(t)}{dt} + 8y(t) = 0$$

Characteristic equation is

$$r^2 + 6r + 8 = 0 \Rightarrow (r+2)(r+4) = 0$$

The roots are $r_1 = -2, r_2 = -4$ } $r = -2 \neq r = -4$
Roots are real & distinct

$$y_n(t) = c_1 e^{-2t} + c_2 e^{-4t} \text{ is the natural response}$$

→ (A)

put $t=0$ in eqn (A)

$$y(0) = 0 = c_1 e^0 + c_2 e^0 \quad \text{or} \quad \boxed{c_1 + c_2 = 0} \quad \text{--- (1)}$$

Differentiating eqn (A) & putting $t=0$; we get

$$\frac{dy_n(t)}{dt} = -2c_1 e^{-2t} + c_2 (-4e^{-4t})$$

put $t=0$; $\frac{d^2 y(0)}{dt^2} = \frac{d}{dt} y(0) = 1 = -2c_1 e^0 - 4c_2 e^0$

$$\therefore \boxed{c_1 + 2c_2 = -0.5} \quad \text{--- (2)}$$

solving (1) & (2) we get $\boxed{c_2 = -\frac{1}{2}} \quad \therefore \boxed{c_1 = \frac{1}{2}}$

$$\therefore \boxed{y_n(t) = \frac{1}{2} e^{-2t} - \frac{1}{4} e^{-4t}} \text{ is the required natural response}$$

⑦ $\frac{d^2 y(t)}{dt^2} + 4y(t) = 3 \frac{d}{dt} x(t)$ with $y(0) = -1$ & $y'(0) = 1$

homogeneous eqn is $\frac{d^2 y(t)}{dt^2} + 4y(t) = 0$

Characteristic eqn is $r^2 + 4 = 0$

$$\therefore r = \pm j2$$

→ [Roots are distinct & imaginary]

Let $r_1 = j2, r_2 = -j2$ be the roots of the char eqn

The natural response is

$$\boxed{y_n(t) = c_1 e^{j2t} + c_2 e^{-j2t}}$$

The other form is

$$\boxed{y(t) = c_1 \cos 2t + c_2 \sin 2t} \text{ is the natural response} \quad \text{--- (A)}$$

put $t=0$; in eqn (A)

$$y(0) = y(0) = -1 = c_1 \cos(0) + c_2 \sin(0) = c_1$$

$$\therefore \boxed{c_1 = -1}$$

differentiating eqn (A)

$$\frac{d^2 y(t)}{dt^2} = -2c_1 \sin 2t + 2c_2 \cos 2t$$

$$\text{put } t=0; \frac{d^2 y(0)}{dt^2} = y''(0) = 1 = -2c_1 \sin 0 + 2c_2 \cos 0$$

$$\therefore 2c_2 = 1 \quad \boxed{c_2 = 1/2}$$

$$\therefore \boxed{y(t) = -\cos 2t + \frac{1}{2} \sin 2t} \text{ is the required natural response.}$$

$$\textcircled{d} \quad \underline{\underline{\frac{d^2 y(t)}{dt^2} + 2 \frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt} \text{ with } y(0)=1, y'(0)=0}}}$$

Homogeneous eqn is

$$\frac{d^2 y(t)}{dt^2} + 2 \frac{dy(t)}{dt} + 2y(t) = 0$$

Characteristic eqn is $r^2 + 2r + 2 = 0$

$$r = \frac{-2 \pm \sqrt{2^2 - 4(2)(1)}}{2} = -1 \pm j1 \quad \left[\begin{array}{l} \text{Roots are distinct +} \\ \text{Complex} \end{array} \right]$$

Let $r_1 = -1 + j1$ and $r_2 = -1 - j1$ be the roots of Char. Eqn

The natural response is of the form

$$\boxed{y_{pn}(t) = e^{-t} [c_1 \cos t + c_2 \sin t]} \text{ --- (A)}$$

$$\text{diff. eqn (A) } \frac{d^2 y(t)}{dt^2} = e^{-t} [-c_1 \sin t + c_2 \cos t] - [c_1 \cos t + c_2 \sin t] e^{-t} \text{ --- (B)}$$

put $t=0$ in eqn (A)

$$y(0) = y(0) = 1 = e^0 [-c_1 \sin 0 + c_2 \cos 0]$$

$$\boxed{c_2 = 1}$$

$$\text{put } t=0 \text{ in eqn (B); } y'(0) = e^0 [c_1 \sin 0 + c_2 \cos 0] - [c_1 \cos 0 + c_2 \sin 0] e^0$$

$$0 = c_2 - c_1 \Rightarrow \boxed{c_1 = +c_2 = +1}$$

$\therefore y_n(t) = e^{-t} [\cos t + \sin t]$ is the required natural response

(c) $\frac{d^2 y(t)}{dt^2} + 2 \frac{dy(t)}{dt} + y(t) = \frac{d}{dt} x(t)$ with $y(0) = 1, y'(0) = +1$

Homogeneous eqn is $\frac{d^2 y(t)}{dt^2} + 2 \frac{dy(t)}{dt} + y(t) = 0$

Characteristic eqn is $r^2 + 2r + 1 = 0$
 $(r+1)^2 = 0 \quad r_1 = -1, r_2 = -1$

Roots are repeated and identical. Hence the natural response is of the form

$y_n(t) = c_1 e^{-t} + t c_2 e^{-t} \dots \dots \dots \textcircled{A}$

diff. eqn $\textcircled{A} \quad \frac{dy_n(t)}{dt} = -c_1 e^{-t} + c_2 e^{-t} - t c_2 e^{-t} \dots \dots \textcircled{B}$

put $t=0$; in eqn $\textcircled{A}$

$y_n(0) = y(0) = 1 = c_1 + 0 \Rightarrow \boxed{c_1 = 1}$

put $t=0$ in eqn $\textcircled{B}$

$\frac{dy_n(0)}{dt} = y'(0) = +1 = -c_1 + c_2 - 0$

$c_2 = c_1 + 1 = 1 + 1 = \underline{\underline{2}} \quad \boxed{c_2 = 2}$

$\therefore y_n(t) = e^{-t} + 2t e^{-t}$ is the required natural response.

Forced response

Forced response is due to input only with zero initial conditions

$y^f(t) = y_n(t) + y^p(t)$

The particular soln is of the form

Sln no	Input	particular soln.	Remarks.
1	$x(t) = 1$ or $k_1(t)$	$y^p(t) = K$ $y^p(t) = K u(t)$	Ex: $x(t) = u(t)$
2	$x(t) = e^{-at}$	$y^p(t) = K e^{-at}$	$x(t) = 2 e^{-3t}$
3	$x(t) = \cos(\omega t + \phi)$	$y^p(t) = k_1 \cos \omega t + k_2 \sin \omega t$	$x(t) = \cos 3t$
4	$x(t) = e^{-at} (\cos \omega t + \phi)$	$y^p(t) = e^{-at} [k_1 \cos \omega t + k_2 \sin \omega t]$	$x(t) = e^{-2t} \cos t$
5	$x(t) = t$	$y^p(t) = k_0 + k_1 t$	
6	$x(t) = t e^{-at}$	$y^p(t) = e^{-at} [k_0 + k_1 t]$	

Determine the forced response of the systems given by

$$5 \frac{d}{dt} y(t) + 10 y(t) = 2 x(t) \quad \text{with} \quad (a) x(t) = \frac{2}{e} t \quad (c) x(t) = \cos 3t.$$

$$(b) x(t) = \frac{2}{e} t$$

Natural response

Homogeneous eqn is $5 \frac{d}{dt} y(t) + 10 y(t) = 0$

characteristic eqn is $5s + 10 = 0$

$\therefore$ root $s_1 = -\frac{2}{e}$

$y_n(t) = c_1 e^{-2t}$ is the natural response.

Forced response

forced response $y_f(t) = y_n(t) + y_p(t)$.

particular solution $y_p(t)$.

(a) Given input $x(t) = 2$

Hence $y_p(t) = K$

using $y_p(t)$ in the main eqn, we get

$$5 \frac{d}{dt} y_p(t) + 10 y_p(t) = 2 x(t)$$

$$5 \frac{d}{dt} (K) + 10K = 2(2)$$

$$K = \frac{4}{10} = \frac{2}{5}$$

$$\therefore y_p(t) = \frac{1}{5}$$

$$y_f(t) = y_n(t) + y_p(t) = c_1 e^{-2t} + \frac{2}{5} \quad \text{--- (A)}$$

Assuming zero initial conditions

put $t=0$ in eqn (A) $y_f(0) = 0 = c_1 e^{-0} + \frac{2}{5}$

$$\therefore c_1 = -\frac{2}{5}$$

Hence the forced response $y_f(t) = -\frac{2}{5} e^{-2t} + \frac{2}{5}$

(b) Given input $x(t) = \frac{2}{e} t$

$\therefore y_p(t) = K e^{-t}$

using this value in the main eqn

$$5 \frac{d}{dt} (K e^{-t}) + 10 K e^{-t} = 2 e^{-t}$$

$$-5K e^{-t} + 10K e^{-t} = 2 e^{-t}$$

$$\Rightarrow K = \frac{2}{5}$$

$$\therefore y_p(t) = \frac{2}{5} e^{-t}$$

$$\therefore y_p(t) = c_1 e^{-2t} + \frac{2}{5} e^{-t} \quad \text{--- (3)}$$

Assuming zero initial conditions; put $t=0$ in eqn (3)

$$y_p(0) = 0 = c_1 e^{-0} + \frac{2}{5} e^{-0} \Rightarrow c_1 = -\frac{2}{5}$$

$$\therefore y_p(t) = -\frac{2}{5} e^{-2t} + \frac{2}{5} e^{-t} \quad \text{is the required forced response}$$

(c) Input $x(t) = \cos 3t$

$$y_p(t) = K_1 \cos 3t + K_2 \sin 3t$$

using this value in the main eqn

$$5 \frac{d}{dt} (K_1 \cos 3t + K_2 \sin 3t) + 10 [K_1 \cos 3t + K_2 \sin 3t] = 2 \cos 3t$$

$$5 [-3K_1 \sin 3t + 3K_2 \cos 3t] + 10K_1 \cos 3t + 10K_2 \sin 3t = 2 \cos 3t$$

equating the similar terms

$$[10K_1 - 15K_2] \cos 3t - (15K_1 + 10K_2) \sin 3t = 2 \cos 3t$$

$$10K_1 - 15K_2 = 2$$

$$-15K_1 + 10K_2 = 0$$

$$\text{solving } K_1 = 4/65, K_2 = 6/65$$

$$\therefore y_p(t) = \frac{4}{65} \cos 3t + \frac{6}{65} \sin 3t$$

$$\therefore y_p(t) = c_1 e^{-2t} + \frac{4}{65} \cos 3t + \frac{6}{65} \sin 3t$$

put $t=0$ in the above equation and assuming zero initial

$$\text{conditions } y_p(0) = 0 = c_1 e^{-0} + \frac{4}{65} \cos 0 + \frac{6}{65} \sin 0$$

$$\Rightarrow c_1 = -\frac{4}{65}$$

$$\therefore y_p(t) = -\frac{4}{65} e^{-2t} + \frac{4}{65} \cos 3t + \frac{6}{65} \sin 3t \quad \text{is the required forced response.}$$

(2) Find the forced response of the system defined by

$$\frac{d^2}{dt^2} y(t) + 5 \frac{d}{dt} y(t) + 6 y(t) = 2 x(t) + \frac{d}{dt} x(t) \quad \text{with input } x(t) = 2 e^{-t} u(t).$$

$$\frac{d^2 y(t)}{dt^2} + 5 \frac{d}{dt} y(t) + 6 y(t) = 2x(t) + \frac{d}{dt} x(t) \quad \text{--- (1)}$$

Natural response

Homogeneous eqn is

$$\frac{d^2 y(t)}{dt^2} + 5 \frac{d}{dt} y(t) + 6 y(t) = 0$$

characteristic eqn is

$$r^2 + 5r + 6 = 0$$

$$(r+2)(r+3) = 0$$

$$r = -2 \text{ \& } r = -3$$

Let the roots of this eqn be

$$\underline{r_1 = -2 \text{ and } r_2 = -3}$$

The natural response is of the form

$$\boxed{y_n(t) = c_1 e^{-2t} + c_2 e^{-3t}}$$

Forced response

$$\text{input } x(t) = 2 e^{-t} u(t)$$

$$\text{particular soln } y_p(t) = k e^{-t} u(t) = k e^{-t} \text{ for } t \geq 0$$

using this value in eqn (1)

$$\frac{d^2}{dt^2} (k e^{-t}) + 5 \frac{d}{dt} (k e^{-t}) + 6 k e^{-t} = 2 x(t) + \frac{d}{dt} x(t)$$

$$k \frac{d^2}{dt^2} [e^{-t}] + 5k \frac{d}{dt} e^{-t} + 6k e^{-t} = 2 \cdot 2 e^{-t} + \frac{d}{dt} 2 e^{-t}$$

$$+ k e^{-t} - 5k e^{-t} + 6k e^{-t} = 4 e^{-t} - 2 e^{-t}$$

$$\therefore \boxed{k = 1}$$

$$\therefore y_p(t) = k e^{-t} u(t) = e^{-t} u(t)$$

$$\therefore \boxed{y_f(t) = c_1 e^{-2t} + c_2 e^{-3t} + e^{-t} u(t)} \text{ is the required forced response.} \quad \text{--- (2)}$$

diff. eqn (2) we get

$$\frac{d}{dt} y_f(t) = -2c_1 e^{-2t} - 3c_2 e^{-3t} - e^{-t} \quad (\text{for } t > 0) \quad \text{--- (3)}$$

put $t=0$ in eqn (2) & (3) and assuming zero initial condition

$$y_f(0) = 0 = c_1 e^{-0} + c_2 e^{-0} + e^{-0} \Rightarrow \boxed{c_1 + c_2 = -1}$$

$$\frac{d}{dt} y_f(0) = 0 = -2c_1 e^{-0} - 3c_2 e^{-0} - 1 \Rightarrow \boxed{2c_1 + 3c_2 = -1}$$

$$\begin{aligned} c_1 + c_2 &= -1 \quad \times 2 \\ 2c_1 + 3c_2 &= -1 \end{aligned}$$

$$\Rightarrow \begin{array}{r} 2c_1 + 2c_2 = -2 \\ 2c_1 + 3c_2 = -1 \\ \hline -c_2 = -1 \end{array} \therefore \boxed{c_2 = 1}$$

$$\therefore c_1 = -1 - c_2 = -1 - 1 = -2$$

$$\therefore \boxed{c_1 = -2}$$

$$\therefore \boxed{y_p(t) = e^{-2t} - 2e^{-3t} + e^t \cos t}$$
 is the required forced response

- ③ $\frac{d^2 y(t)}{dt^2} + 5 \frac{dy(t)}{dt} + 4y(t) = \frac{d}{dt} x(t)$ with $y(0) = 0$ $y'(t)|_{t=0} = 1$ and $x(t) = 8 \sin t \cos t$.
Determine the forced response.

$$\frac{d^2 y(t)}{dt^2} + 5 \frac{dy(t)}{dt} + 4y(t) = \frac{d}{dt} x(t) \quad \text{--- (A)}$$

Natural response

Homogeneous eqn is $\frac{d^2 y(t)}{dt^2} + 5 \frac{dy(t)}{dt} + 4y(t) = 0$

characteristic eqn is $r^2 + 5r + 4 = 0$

$\therefore$ the roots are $r_1 = -4$
 $r_2 = -1$

The natural response is of the form

$$\boxed{y_n(t) = c_1 e^{-t} + c_2 e^{-4t}}$$

Forced response

Input $x(t) = 8 \sin t \cos t$

particular soln $y_p(t) = [k_2 \cos t + k_1 \sin t] \cos t$

For $t \geq 0$ $y_p(t) = k_1 \cos t + k_2 \sin t$.

using this value in eqn (A) we get

$$\begin{aligned} \frac{d^2}{dt^2} [k_2 \cos t + k_1 \sin t] + 5 \frac{d}{dt} [k_2 \cos t + k_1 \sin t] + 4[k_2 \cos t + k_1 \sin t] \\ = \frac{d}{dt} [8 \sin t] \end{aligned}$$

$$-k_1 \sin t - k_2 \cos t + 5k_2 \sin t + 5k_1 \cos t + 4k_1 \sin t + 4k_2 \cos t = 8 \cos t$$

equating the similar terms

$$(3k_1 - 5k_2) \sin t = 0$$

$$(5k_1 + 3k_2) \cos t = 8 \cos t$$

$\Rightarrow$

$$3k_1 - 5k_2 = 0$$

$$5k_1 + 3k_2 = 8$$

$$\Rightarrow k_1 = 5/34, \quad k_2 = 3/34$$

$$i. y_p(t) = \frac{5}{34} \sin t + \frac{3}{34} \cos t$$

$$\therefore y_f(t) = c_1 e^{-t} + c_2 e^{-4t} + \frac{5}{34} \sin t + \frac{3}{34} \cos t \quad \text{--- (B)}$$

diff. eqn (B) we get

$$y_f'(t) = -c_1 e^{-t} - 4c_2 e^{-4t} + \frac{5}{34} \cos t - \frac{3}{34} \sin t \quad \text{--- (C)}$$

put $t=0$ in eqn (B) & (C) and assuming $y_f(0)=0, y_f'(0)=0$
we get

$$0 = c_1 e^0 + c_2 e^0 + \frac{5}{34} \sin 0 + \frac{3}{34} \cos 0$$

$$\Rightarrow c_1 + c_2 = -3/34$$

consider eqn (C)

$$0 = -c_1 e^0 - 4c_2 e^0 + \frac{5}{34} \cos 0 - \frac{3}{34} \sin 0$$

$$\Rightarrow c_1 + 4c_2 = 5/34$$

Solving for c_1 & c_2 ; we get

$$c_2 = \frac{4}{51}$$

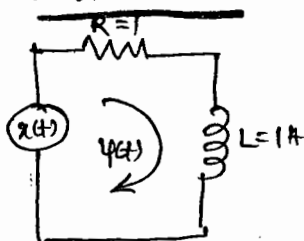
$$c_1 = -\frac{1}{6}$$

$$\therefore y_f(t) = -\frac{1}{6} e^{-t} + \frac{4}{51} e^{-4t} + \frac{5}{34} \sin t + \frac{3}{34} \cos t$$

is the required forced response of the system.

Find the forced response of the system shown below with

$$x(t) = \cos t$$



applying KVL

$$x(t) = R y(t) + L \frac{d}{dt} y(t)$$

$$\text{put } x(t) = \cos t \\ R=L=1$$

$$\left[\frac{d}{dt} y(t) + y(t) = \cos t \right]$$

$$y_h(t) = c_1 e^{-t}$$

$$y_p(t) = k_1 \cos t + k_2 \sin t$$

$$y_p(t) = \frac{1}{2} \cos t + \frac{1}{2} \sin t$$

$$y_f(t) = c_1 e^{-t} + \frac{1}{2} \cos t + \frac{1}{2} \sin t$$

using zero IC & solving for c_1 we get $c_1 = -1/2$

$$\therefore y_f(t) = -\frac{1}{2} e^{-t} + \frac{1}{2} \cos t + \frac{1}{2} \sin t \quad \text{is the required forced Response}$$

Total Response (o/p of the system)

Total response = Natural response + Forced response.

1. Calculate the natural response and determine the constants using given initial conditions.
- (2) Determine the forced response with zero initial conditions.
- (3) Add natural response to forced response to get the complete solution or total response of the system.

① Find the complete response of the system described by the DE

$$\frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt} \quad \text{with } x(t) = 2e^{-t}u(t) \\ y(0) = 1, \quad y'(0) = 0 \quad t=0$$

$$\boxed{\frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt}} \quad \text{----- ①}$$

Natural response

Homogeneous eqn is $\frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = 0$

Characteristic eqn is $r^2 + 3r + 2 = 0$

The roots are $r_1 = -2, r_2 = -1$

$\therefore y_n(t) = c_1 e^{-t} + c_2 e^{-2t}$ is the required natural response

using diff using $y(0) = 1$ at $t=0$ in eqn (A)

$$y_n(0) = y(0) = 1 = c_1 e^{-0} + c_2 e^{-2 \cdot 0} \Rightarrow \boxed{c_1 + c_2 = 1}$$

Diff. eqn (A) & using $y'(0) = -1$ at $t=0$ we get

$$\frac{d}{dt} y_n(t) = -c_1 e^{-t} - c_2 \cdot 2e^{-2t}$$

$$\text{at } t=0 \quad y'_n(0) = -1 = -c_1 e^{-0} - 2c_2 e^{-2 \cdot 0} \Rightarrow \boxed{c_1 + 2c_2 = 0}$$

solving for c_1 and c_2 we get $\boxed{c_2 = -1} \quad \boxed{c_1 = 2}$

$\therefore y_n(t) = 2e^{-t} - e^{-2t}$ is the natural response of the system.

Forced response

Input $x(t) = 2e^{-t}u(t)$.

particular solution

Since e^{-t} is already in the natural response eqn

$y_p(t)$ takes the form $y_p(t) = kt e^{-t}$

using this value in eqn (A)

$$\frac{d^2}{dt^2} [kt e^{-t}] + 3 \frac{d}{dt} [kt e^{-t}] + 2kt e^{-t} = \frac{d}{dt} (2e^{-t})$$

$$* k \left[\frac{d}{dt} (-t e^{-t} + e^{-t}) \right] + 3k [-t e^{-t} + e^{-t}] + 2kt e^{-t} = -2e^{-t}$$

$$k[-(-t e^{-t} + e^{-t}) - e^{-t}] + 3kt e^{-t} + 3k e^{-t} + 2kt e^{-t} = -2e^{-t}$$

$$kt e^{-t} - k e^{-t} - k e^{-t} + 3kt e^{-t} + 3k e^{-t} + 2kt e^{-t} = -2e^{-t}$$

$$+k(2+3)e^{-t} = -2e^{-t}$$

$$k(1+3+2)t e^{-t} = 0$$

$$\Rightarrow \boxed{K = -2}$$

$$\therefore \boxed{y_p(t) = -2t e^{-t}}$$

$$\boxed{y_f(t) = c_1 e^{-t} + c_2 e^{-2t} - 2t e^{-t}} \text{ is the forced response.}$$

$$\text{put } t=0; \quad y_f(0) = 0 = c_1 e^{-0} + c_2 e^{-0} - 2(0) e^{-0}$$

$$\Rightarrow \boxed{c_1 + c_2 = 0}$$

diff. the forced response eqn

$$* y_f'(t) = -c_1 e^{-t} - 2c_2 e^{-2t} - 2e^{-t} + 2t e^{-t}$$

put $t=0$; and assuming zero initial conditions

$$y_f'(0) = 0 = -c_1 - 2c_2 - 2 + 0$$

$$\Rightarrow \boxed{c_1 + 2c_2 = -2}$$

$$\text{solving for } c_1 \text{ \& } c_2 \text{ we get } \boxed{c_2 = -2} \quad \boxed{c_1 = +2}$$

$$\boxed{y_f(t) = 2e^{-t} - 2e^{-2t} - 2t e^{-t}} \text{ is the required forced response}$$

Total response = natural response + forced response

$$= 2e^{-t} - e^{-2t} + 2e^{-t} - 2e^{-2t} - 2t e^{-t}$$

$$\therefore \boxed{y(t) = 4e^{-t} - 3e^{-2t} - 2t e^{-t}} \text{ is the complete soln}$$

II method

The output response can also be calculated by substituting the initial conditions in the forced response eqn.

$y(t) = y_f(t)$ with initial conditions

we have $y_f(t) = c_1 e^{-t} + c_2 e^{-2t} - 2t e^{-t}$ ----- (A)

put $t=0$ in the above eqn

$$y_f(0) = y(0) = 1 = c_1 e^{-0} + c_2 e^{-0} - 2(0) e^{-0}$$

$$\Rightarrow \boxed{c_1 + c_2 = 1}$$

diff. eqn (A)

$$y_f'(t) = -c_1 e^{-t} - 2c_2 e^{-2t} + 2t e^{-t} - 2e^{-t}$$

put $t=0$; $y_f'(0) = y'(0) = 0 = -c_1 e^{-0} - 2c_2 e^{-0} + 2(0) e^{-0} - 2e^{-0}$

$$\Rightarrow \boxed{c_1 + 2c_2 = -2}$$

solving for c_1 & c_2 ; we get $c_1 = 4$, $c_2 = -3$

$\therefore y(t) = 4e^{-t} - 3e^{-2t} - 2te^{-t}$ is the required soln

$$\frac{d^2 y(t)}{dt^2} + 2 \frac{dy(t)}{dt} + y(t) = \frac{dx(t)}{dt} \text{ with } x(t) = 2e^{-t} u(t)$$

and $y(0) = 1$ & $y'(0) = 0$

Determine Total response of the system.

$$\frac{d^2 y(t)}{dt^2} + 2 \frac{dy(t)}{dt} + y(t) = \frac{dx(t)}{dt} \text{ ----- (1)}$$

natural response

homogeneous eqn is $\frac{d^2 y(t)}{dt^2} + 2 \frac{dy(t)}{dt} + y(t) = 0$

char. eqn is $r^2 + 2r + 1 = 0$

$$\therefore r_1 = -1, r_2 = -1$$

The roots are repeated & identical. Hence natural response is of the form $y_n(t) = c_1 e^{-t} + c_2 t e^{-t}$ ----- (A)

put $t=0$ in eqn (A) $y_n(0) = y(0) = c_1 e^{-0} + c_2(0) e^{-0}$

Given $y(0) = 1$; $\therefore \boxed{c_1 = 1}$

diff. eqn (A)

$$y_1'(t) = -c_1 e^{-t} + c_2 e^{-t} - t c_2 e^{-t}$$

put $t=0$ and use the IC $y_1'(0)=0$

$$0 = -c_1 e^{-0} + c_2 e^{-0} - 0 e^{-0} c_2$$

$$\Rightarrow -c_1 + c_2 = 0$$

$$\therefore \boxed{c_1 = c_2}$$

$$\therefore \boxed{c_2 = 1}$$

$$\therefore \boxed{y_1(t) = e^{-t} + t e^{-t}}$$

Forced response

$$\text{Input } x(t) = 2 e^{-t} u(t).$$

particular solution since there exists a term e^{-t} and $t e^{-t}$ are already present in the natural response eqn; the $y_p(t)$ takes the form $y_p(t) = K t^2 e^{-t} u(t)$

$$y_p(t) = K t^2 e^{-t} ; \text{ for } t \geq 0$$

using this value in the main eqn (A) we get

$$\frac{d^2}{dt^2} [K t^2 e^{-t}] + 2 \frac{d}{dt} [K t^2 e^{-t}] + K t^2 e^{-t} = \frac{d}{dt} [2 e^{-t}]$$

Equating the similar terms we get $\boxed{K = -1}$

$$\boxed{y_p(t) = -t^2 e^{-t}}$$

$$\boxed{y_f(t) = c_1 e^{-t} + t c_2 e^{-t} - t^2 e^{-t}} \quad \xrightarrow{\text{2}} \text{ is the required forced response.}$$

put $t=0$ in the above eqn & assuming zero IC's,

$$y_f(0) = 0 = c_1 e^{-0} + 0 c_2 e^{-0} - t^2 e^{-0} \Rightarrow \boxed{c_1 = 0}$$

diff. eqn (2) and putting $t=0$; we get

$$y_f'(t) = c_1 (-e^{-t}) + c_2 e^{-t} - t e^{-t} c_2 + t^2 e^{-t} - 2 t e^{-t}$$

put $t=0$ and assuming $y_f'(0)=0$

$$y_f'(0) = 0 = -c_1 e^{-0} + c_2 e^{-0} - 0 + 0 - 0 \Rightarrow \boxed{c_1 = c_2}$$

$$\therefore \boxed{y_f(t) = y_p(t)} = -t^2 e^{-t}$$

$$\text{Total response } y(t) = y_1(t) + y_f(t) = \underline{\underline{e^{-t} + t e^{-t} - t^2 e^{-t}}}$$

Importance of Natural response & characteristic equation.

(7)

The forced response depends upon both the input and roots of the char. eqn. The basic form of natural response is dependent entirely on the roots of char. eqn.

The imp. response of LTI system also depends upon roots of char. eqn since it contains some terms of natural response.

Hence roots of char eqn afford a considerable amount of information about the LTI system behavior.

The stability characteristics are directly related to the roots of the char. eqn.

A DT-LTI system is stable iff all the roots of the char. eqn have magnitude less than unity.

A CT LTI system is stable iff the real parts of the roots are -ve.

The IC represent the energy present in the system. with zero i/p; the stored energy eventually dissipates and the o/p approaches to zero.

Hence the Nat. response content describes the transient behavior of the system. It describes the transition of the system from its initial conditions to an equilibrium state defined by the input.

Hence in a DT-LTI system; the response of the system to a transient is directly proportional to the roots of the char. eqn. with largest magnitude. while in a CT LTI system, it is determined by the root with a largest real component.

for a CT LTI system to have fast response time; all the roots of the char. eqn must have large real parts.

Investigate the stability of the following systems.

(a) $y(n) - y(n-1) = x(n)$

homogeneous eqn

$$y(n) - y(n-1) = 0$$

char. eqn is $r - 1 = 0$

$$\therefore \boxed{r=1}$$

The magnitude of the root is

$|r| < 1$ Hence the system is unstable

(b) $y(n) - \frac{1}{6}y(n-1) - \frac{1}{6}y(n-2) = x(n)$

$$y(n) - \frac{1}{6}y(n-1) - \frac{1}{6}y(n-2) = 0$$

$$1 - \frac{1}{6}r - \frac{1}{6}r^2 = 0$$

$$6r^2 - r - 1 = 0$$

$$r_1 = -1/3 \quad \& \quad r_2 = +1/2$$

$$|r_1| < 1 \quad \& \quad |r_2| < 1$$

Hence the system is stable

Difference equation representation for DT-LTI system

The general form of a constant coeff. Linear Diff. eqn is

$$\sum_{k=0}^N a_k y(n-k) = \sum_{k=0}^M b_k x(n-k).$$

where 'N' is the order of diff eqn & $N > M$.

$x(n)$ is the input to the system and $y(n)$ is the output of the system.

The complete soln or total response $y(n)$ of the system is

Total response = Natural response + Forced response

$$y(n) = y_n(n) + y_f(n)$$

Natural Response

- ① Find the natural response of the system described by the difference eqn $y(n) - \frac{9}{16} y(n-2) = x(n-1)$ with $y(-1)=0$ and $y(-2)=1$

$$y(n) - \frac{9}{16} y(n-2) = x(n-1) \quad \text{--- (A)}$$

Obtain the homogeneous eqn by letting input to zero

$$y(n) - \frac{9}{16} y(n-2) = 0 \quad \text{--- (B)}$$

Obtain the characteristic eqn by replacing $y(n-k)$ by r^{-k} .

$$r^0 - \frac{9}{16} r^{-2} = 0$$

$$16r^2 - 9 = 0 \quad \therefore r = \pm \frac{3}{4}$$

Let $r_1 = \frac{3}{4}$ and $r_2 = -\frac{3}{4}$ be the roots of char eqn.

The roots are real and non repeated.

The natural response is of the form

$$y_n(n) = \sum_{i=1}^n c_i (r_i)^n.$$

$$y_n(n) = c_1 \left(\frac{3}{4}\right)^n + c_2 \left(-\frac{3}{4}\right)^n \quad \text{--- (C)}$$

Consider eqn (A) and put $n=0$

$$y(0) - \frac{9}{16} y(-2) = 0 \quad \Rightarrow y(0) = \frac{9}{16} (1) = \frac{9}{16}$$

$$\text{put } n=1; \quad y(1) - \frac{9}{16} y(-1) = 0.$$

Use these values in eqn (C)

put $n=1$ in eqn ③

$$y(1) = 0 = c_1 \left(\frac{3}{4}\right) - c_2 \left(\frac{3}{4}\right)$$

$$\therefore \boxed{c_1 - c_2 = 0}$$

solving $\boxed{c_1 = \frac{9}{32}} \quad \boxed{c_2 = +\frac{9}{32}}$

$$\therefore \boxed{y_n(n) = \frac{9}{32} \left(\frac{3}{4}\right)^n + \frac{9}{32} \left(-\frac{3}{4}\right)^n}$$
 is the required natural response.

② Find the natural response of the system;

$$y(n) - \frac{1}{4}y(n-1) - \frac{1}{8}y(n-2) = x(n) + x(n-1) \quad \text{with } y(-1)=0, y(-2)=1$$

Homogeneous eqn is

$$y(n) - \frac{1}{4}y(n-1) - \frac{1}{8}y(n-2) = 0 \quad \text{--- ①}$$

characteristic eqn is

$$r^0 + \frac{1}{4}r^{-1} - \frac{1}{8}r^{-2} = 0$$

$$\boxed{8r^2 - 2r + 1 = 0}$$

$$8r^2 - 4r + 2r - 1 = 0$$

$$4r(2r-1) + 1(2r-1) = 0$$

$$(4r+1)(2r-1) = 0$$

$$\therefore \text{roots are } r_1 = -1/4, r_2 = +1/2$$

$$\text{roots } r_1 = -1/4$$

$$r_2 = 1/2$$

natural response $\boxed{y_n(n) = \left(-\frac{1}{4}\right)^n c_1 + \left(\frac{1}{2}\right)^n c_2}$ --- ②

put $n=0$ in eqn ①

$$y(0) - \frac{1}{4}y(-1) - \frac{1}{8}y(-2) = 0 \Rightarrow y(0) = \frac{1}{4}(0) + \frac{1}{8}(+1)$$

$$\therefore \boxed{y(0) = \frac{1}{8}}$$

put $n=1$; in eqn ①

$$y(1) = \frac{1}{4}y(0) + \frac{1}{8}y(-1) = \frac{1}{4}\left(\frac{1}{8}\right) + 0 = \frac{1}{32}$$

put $n=0$; in eqn ②

$$y_n(0) = y(0) = \frac{1}{8} = \left(-\frac{1}{4}\right)^0 c_1 + \left(\frac{1}{2}\right)^0 c_2 \Rightarrow \boxed{c_1 + c_2 = 1/8}$$

put $n=1$; $y_n(1) = y(1) = \frac{1}{32} = -\frac{c_1}{4} + \frac{c_2}{2} \Rightarrow \boxed{-c_1 + 2c_2 = \frac{1}{8}}$

solving $c_1 = 1/24 \quad c_2 = 1/12$

$y_n(n) = \frac{1}{24} \left(-\frac{1}{4}\right)^n + \frac{1}{12} \left(\frac{1}{2}\right)^n$ is the required natural Response of the system.

③ $y(n) + \frac{9}{16} y(n-2) = x(n-1)$ with $y(-1)=1, y(-2)=-1$

Homogeneous eqⁿ is $y(n) + \frac{9}{16} y(n-2) = 0$ — (1)

characteristic eqⁿ is $16r^2 + 9 = 0$

roots are $r_1 = +j\frac{3}{4}$ $r_2 = -j\frac{3}{4}$

the natural response $y_n(n) = c_1 \left(j\frac{3}{4}\right)^n + c_2 \left(-j\frac{3}{4}\right)^n$ — (2)

put $n=0$ in eqⁿ (1)

$$y(0) = -\frac{9}{16} y(-2) = -\frac{9}{16} y(-2) = \frac{9}{16}$$

$$y(1) = -\frac{9}{16} y(-1) = -\frac{9}{16}$$

put $n=0$; in eqⁿ (2)

$$y_n(0) = y(0) = \frac{9}{16} = c_1 + c_2$$
 — (3)

put $n=1$; in eqⁿ (2)

$$y_n(1) = y(1) = -\frac{9}{16} = j\frac{3}{4} c_1 - j\frac{3}{4} c_2$$
 — (4)

$$c_1 + c_2 = \frac{9}{16}$$

$$j\frac{3}{4} c_1 - j\frac{3}{4} c_2 = -\frac{9}{16}$$

Solving $c_1 = \frac{9}{32} + j\frac{3}{8}$

$$c_2 = \frac{9}{32} - j\frac{3}{8}$$

∴ The natural response is

$$y_n(n) = \left[\frac{9}{32} + j\frac{3}{8}\right] \left(j\frac{3}{4}\right)^n + \left[\frac{9}{32} - j\frac{3}{8}\right] \left[-j\frac{3}{4}\right]^n$$

Forced response (with zero IC).

Forced response has two parts, ~~one~~

(a) a term resembling natural response

(b) A particular solution $y_p(n)$.

$$y_f(n) = y_n(n) + y_p(n).$$

① Find the forced response of the system with input $x(n) = \left(\frac{1}{8}\right)^n u(n)$
 $y(n) - \frac{1}{4}y(n-1) - \frac{1}{8}y(n-2) = x(n) + x(n-1)$.

	Input	particular sol ⁿ
1	$x(n) = \delta$	$y_p(n) = K$
2	$x(n) = u(n)$	$y_p(n) = K u(n)$
3	$x(n) = a^n u(n)$	$y_p(n) = K a^n u(n)$
4	$x(n) = n$	$y_p(n) = K_1 + K_2 n$
5	$x(n) = A \cos(\Omega n + \phi)$	$y_p(n) = K_1 \cos \Omega n + K_2 \sin \Omega n$

Homogeneous eqⁿ is

$$y(n) - \frac{1}{4}y(n-1) - \frac{1}{8}y(n-2) = 0$$

char. eqⁿ is $8r^2 - 2r - 1 = 0$

$$\text{roots are } r_1 = -\frac{1}{4} \quad r_2 = \frac{1}{2}$$

The natural response is

$$y_h(n) = c_1 \left(-\frac{1}{4}\right)^n + \left(\frac{1}{2}\right)^n c_2$$

Forced response

$$\text{Input } x(n) = \left(\frac{1}{8}\right)^n u(n)$$

$$\text{particular solⁿ is } y_p(n) = K \left(\frac{1}{8}\right)^n u(n).$$

Using this in the main eqⁿ

$$y(n) - \frac{1}{4}y(n-1) - \frac{1}{8}y(n-2) = x(n) + x(n-1) \quad \text{we get } \textcircled{1}$$

$$K \left(\frac{1}{8}\right)^n u(n) - \frac{1}{4} K \left(\frac{1}{8}\right)^{n-1} u(n-1) - \frac{1}{8} K \left(\frac{1}{8}\right)^{n-2} u(n-2) = \left(\frac{1}{8}\right)^n u(n) + \left(\frac{1}{8}\right)^{n-1} u(n-1)$$

For $n \geq 2$;

$$K \left(\frac{1}{8}\right)^n - \frac{K}{4} \left(\frac{1}{8}\right)^{n-1} - \frac{1}{8} K \left(\frac{1}{8}\right)^{n-2} = \left(\frac{1}{8}\right)^n + \left(\frac{1}{8}\right)^{n-1}$$

$$K - \frac{K}{4}(8) - \frac{1}{8}K(8)^2 = 1 + 8$$

$$K - 2K - 8K = 9 \quad \Rightarrow \quad \boxed{K = -1}$$

$$\therefore \quad \boxed{y_p(n) = -\left[\frac{1}{8}\right]^n u(n)}$$

$$y_p(n) = c_1 \left(-\frac{1}{4}\right)^n + c_2 \left(\frac{1}{2}\right)^n - \left(\frac{1}{8}\right)^n u(n) \quad \text{is the forced response. } \textcircled{2}$$

put $n=0$ in eqn ① & assuming zero IC's

$$y(0) - \frac{1}{4} y(-1)^0 - \frac{1}{8} y(-2)^0 = x(0) + x(-1) = u(0) + u(-1) = 1$$

$$\boxed{y(0) = u(0) = 1}$$

put $n=1$ in eqn ① & assuming zero IC's.

$$y(1) - \frac{1}{4} y(0) - \frac{1}{8} y(-1)^0 = x(1) + x(0) = u(1) + u(0)$$

$$y(1) - \frac{1}{4} (1) = \frac{1+1}{8} \Rightarrow \boxed{y(1) = \frac{3}{4}} \quad \frac{1+1}{8} + \frac{1}{4} = \frac{11}{8}$$

put $n=0$ in eqn ②

$$y_f(0) = y(0) = 1 = c_1 + c_2 - \left(\frac{1}{8}\right)^0 \Rightarrow \boxed{c_1 + c_2 = 1}$$

put $n=1$ in eqn ②

$$y_f(1) = y(1) = \frac{3}{4} = c_1 \left(-\frac{1}{4}\right) + c_2 \left(\frac{1}{2}\right) - \frac{1}{8}$$

$$-\frac{c_1}{4} + \frac{c_2}{2} = \frac{1}{8} + \frac{1}{8} = \frac{1}{4} \Rightarrow \frac{12}{8} = \frac{3}{2}$$

$$\boxed{-c_1 + 2c_2 = 3}$$

$$\boxed{-c_1 + 2c_2 = 3}$$

Solving for c_1 & c_2 we get

$$\boxed{c_2 = \frac{3}{2}} \quad \boxed{c_1 = -\frac{5}{2}}$$

$$\therefore \boxed{y_f(n) = \frac{5}{12} \left(-\frac{1}{4}\right)^n + \frac{19}{12} \left(\frac{1}{2}\right)^n - \frac{1}{8}}$$

Solving $c_1 = -\frac{2}{3}$; $c_2 = \frac{8}{3}$

$$\boxed{y_f(n) = -\frac{2}{3} \left(-\frac{1}{4}\right)^n + \frac{8}{3} \left(\frac{1}{2}\right)^n - \frac{1}{8}}$$

$$\rightarrow y(0) - \frac{1}{4} y(-1)^0 - \frac{1}{8} y(-2)^0 = \left(\frac{1}{8}\right)^0 u(0) + \left(\frac{1}{8}\right)^{-1} u(-1)$$

$$\boxed{y(0) = 1}$$

when $n=1$

$$y(1) - \frac{1}{4} (y(0)) - \frac{1}{8} y(-1)^0 = \left(\frac{1}{8}\right)^1 u(1) + \left(\frac{1}{8}\right)^0 u(0)$$

$$y(1) - \frac{1}{4} (1) = \frac{1}{8} + 1$$

$$\Rightarrow y(1) = \frac{1}{8} + 1 + \frac{1}{4} = \frac{1+2+1}{2} = \frac{4}{2} = 2$$

③ Find the forced response of the system.

$$y(n) + \frac{1}{4}y(n-1) - \frac{1}{8}y(n-2) = x(n) + x(n-1)$$

with input $x(n] = (-1)^n u[n]$.

Homogeneous eqⁿ is

$$y(n) + \frac{1}{4}y(n-1) - \frac{1}{8}y(n-2) = 0$$

char. eqⁿ is $8r^2 + 2r - 1 = 0$

$$\therefore r_1 = -\frac{1}{4} \text{ \& } r_2 = \frac{1}{2}$$

$$\therefore \boxed{y_h(n) = c_1 \left(-\frac{1}{4}\right)^n + c_2 \left(\frac{1}{2}\right)^n}$$

$$y_p(n) = K(-1)^n u[n]$$

using this in the main eqⁿ

$$K(-1)^n u[n] + \frac{1}{4}K(-1)^{n-1} u[n-1] - \frac{1}{8}K(-1)^{n-2} u[n-2] =$$

$$(-1)^n u[n] + (-1)^{n-1} u[n-1]$$

For $n \geq 2$

$$K(-1)^n (1) + \frac{1}{4}K(-1)^{n-1} (1) - \frac{1}{8}K(-1)^{n-2} (1) = (-1)^n + (-1)^{n-1}$$

$$K + \frac{1}{4}(-1)K - \frac{1}{8}K(1) = 1 - 1 = 0$$

$$\therefore \boxed{K=0}$$

$$\therefore \boxed{y_p(n) = 0}$$

$$\therefore y_f(n) = c_1 \left(-\frac{1}{4}\right)^n + c_2 \left(\frac{1}{2}\right)^n \longrightarrow \textcircled{A}$$

using zero initial conditions in main eqⁿ with $n=0$

$$y(0) + \frac{1}{4}y(-1) - \frac{1}{8}y(-2) = (-1)^0 u(0) + (-1)^{-1} u(-1)$$

$$\therefore \boxed{y(0)=1}$$

with $n=1$ $y(1) + \frac{1}{4}y(0) - \frac{1}{8}y(-1) = (-1)^1 u(1) + (-1)^0 u(0)$

$$y(1) + \frac{1}{4}(1) = -1 + 1 = 0$$

$$\therefore \boxed{y(1) = -1/4}$$

put $n=0$ in eqⁿ $\textcircled{A}$

$$y_f(0) = y(0) = 1 = c_1 + c_2$$

put $n=1$

$$y_f(1) = y(1) = -\frac{1}{4} = -\frac{c_1}{4} + \frac{c_2}{2}$$

$$\boxed{-c_1 + 2c_2 = 1}$$

$$c_1 = -\frac{1}{3} \quad c_2 = \frac{1}{3}$$

$$\therefore \boxed{y_f(n) = -\frac{1}{3} \left(-\frac{1}{4}\right)^n + \frac{1}{3} \left(\frac{1}{2}\right)^n}$$

Forced Response + natural response = Total Response

① Determine the complete response of the system

(a) $4y(n) + 4y(n+1) + y(n+2) = x(n)$ with input $x(n) = 4^n u(n)$ with $y(-1) = 0$; $y(-2) = 1$

$$4y(n) + 4y(n+1) + \frac{1}{4}y(n+2) = \frac{1}{4}x(n)$$

Since the system is an LTI system, time shift by $(n-2)$

$$4y(n-2) + 4y(n-1) + \frac{1}{4}y(n) = \frac{1}{4}x(n-2) \quad \text{--- ①}$$

Homogeneous eqn is

$$4y(n) + 4y(n-1) + \frac{1}{4}y(n-2) = 0 \quad \text{--- ②}$$

Char eqn is

$$\frac{1}{4} + 4r^{-1} + 4r^{-2} = 0$$

$$r^2 + 4r + 4 = 0$$

$$r^2 + 2r + 2r + 4 = 0$$

$$2r(r+1) + (2r+1) = 0$$

$$(r+2)(r+2) = 0$$

$$\therefore r_1 = -2, r_2 = -2$$

Repeated roots.

$$\boxed{y_h(n) = c_0 (-2)^n + c_1 n (-2)^n} \quad \text{--- ③}$$

put $n=0$ in eqn ②

$$y(0) + 4y(-1) + 4y(-2) = 0$$

$$\boxed{y(0) = -4} \quad \checkmark$$

put $n=1$ in eqn ②; $y(1) + 4y(0) + 4y(-1) = 0$
 $\boxed{y(1) = +46}$

$$(2r+1)^2 = 0$$

$$\therefore r = -\frac{1}{2}$$

put $n=0$ in eqⁿ ③

$$y_n(0) = y(0) = -4 = c_0 + c_1(0) \Rightarrow \boxed{c_0 = -4}$$

put $n=1$; $y_n(1) = y(1) = 16 = (-2)[-4 + c_1(1)]$

$$\boxed{c_1 = -8}$$

$\therefore y_n(n) = (-2)^n [-4 + 8n]$ is the natural response.

forced Response

Input $x(n) = 4^n u(n)$

Particular solⁿ $y_p(n) = K 4^n u(n)$.

using this value in eqⁿ ① and for $n \geq 2$;

$$4^n K u(n) + 4K 4^{n-1} u(n-1) + 4K 4^{n-2} u(n-2) = 4^{n-2} u(n-2)$$

for $n \geq 2$

$$4^n \left[K + 4K \left(\frac{1}{4} \right) + 4K \left(\frac{1}{16} \right) \right] = \frac{4^n}{16}$$

$$\Rightarrow K \left[1 + 1 + \frac{1}{4} \right] = \frac{1}{4}$$

$$K \left[\frac{4+4+1}{4} \right] = \frac{1}{16} = \frac{1}{16}$$

$$\therefore \boxed{K = \frac{1}{36}}$$

$$\therefore \boxed{y_p(n) = \frac{4^n}{36} u(n)}$$

$$\therefore \boxed{y_f(n) = (-2)^n [c_0 + c_1(n)] + \frac{4^n}{36} u(n)} \quad \text{--- ④}$$

put $n=0$ in eqⁿ ① & assuming zero IC

$$y(0) + 4y(-1) + 4y(-2) = 4^{-2} u(-2)$$

$$\boxed{y(0) = 0}$$

put $n=1$; in eqⁿ ① & assuming zero IC

$$y(1) + 4y(0) + 4y(-1) = 4^{-1} u(-1)$$

$$y(1) + 0 + 0 = 0 \Rightarrow \boxed{y(1) = 0}$$

(26)

Using these values in eqn (4)

put $n=0$ in eqn (4)

$$y_f(0) = y(0) = 0 = (-2)^0 [c_0 + c_1(0)] + \frac{4^0}{36}$$

$$\Rightarrow \boxed{c_0 = -\frac{1}{36}}$$

put $n=1$ in eqn (4)

$$y_f(1) = y(1) = 0 = (-2)^1 [c_0 + c_1(1)] + \frac{4}{36}$$

$$0 = \frac{2}{36} - 2c_1 + \frac{1}{9}$$

$$\therefore 2c_1 = \frac{2+4}{36} = \frac{6}{36} = \frac{1}{6}$$

$$\boxed{c_1 = \frac{1}{12}}$$

$$\therefore \boxed{y_f(n) = (-2)^n \left[-\frac{1}{36} + \frac{n}{12} \right] + \frac{4^n}{36}}$$

Total response = Natural response + forced response

$$y(n) = (-2)^n \left[\left(-4 - \frac{1}{36} \right) + n \left[-8 + \frac{1}{12} \right] \right] + \frac{4^n}{36}$$

$$\boxed{y(n) = (-2)^n \left[\frac{-145}{36} - \frac{95}{12} n \right] + \frac{4^n}{36}}$$

(2) Find the total response of the system given

$$y(n) + \frac{1}{4} y(n-1) - \frac{1}{8} y(n-2) = x(n) + x(n-1)$$

$$\text{with } y(-1) = 4, \quad y(-2) = 2, \quad x(n) = 2^n u(n)$$

Natural response

$$\text{Homogeneous eqn } y(n) + \frac{1}{4} y(n-1) - \frac{1}{8} y(n-2) = 0$$

$$\text{Char. eqn is } r^2 + \frac{1}{4}r - \frac{1}{8} = 0$$

$$8r^2 + 2r - 1 = 0$$

$$r_1 = \frac{1}{4}, \quad r_2 = -\frac{1}{2}$$

$$\therefore \boxed{y_n(n) = c_1 \left(\frac{1}{4} \right)^n + c_2 \left(\frac{1}{2} \right)^n}$$

Forced response

Input $x(n) = 2^n u(n)$

particular solⁿ $y_p(n) = K 2^n u(n)$

using this value in the main eqⁿ.

$$K \cdot 2^n u(n) + \frac{1}{4} K 2^{n-1} u(n-1) - \frac{1}{8} K 2^{n-2} u(n-2) = 2^n u(n) + 2^{n-1} u(n-1)$$

for $n \gg 2$

$$2^n \left[K + \frac{K}{4} \cdot \frac{1}{2} - \frac{K}{8} \cdot \frac{1}{4} \right] = 2^n \left[1 + \frac{1}{2} \right]$$

$$K \left[1 + \frac{1}{8} - \frac{1}{32} \right] = \frac{3}{2} \quad \Rightarrow \quad \boxed{K = \frac{48}{35}}$$

$$\boxed{y_p(n) = \left(\frac{48}{35} \right) 2^n u(n)}$$

$$\therefore \boxed{y_f(n) = c_1 \left(+\frac{1}{4} \right)^n + c_2 \left(-\frac{1}{2} \right)^n + \left(\frac{48}{35} \right) 2^n u(n)} \quad \text{--- (B)}$$

If we use the given initial conditions in the forced response eqⁿ; the response become total response.

put $n=0$ in the main eqⁿ

$$y(0) + \frac{1}{4} y(-1) - \frac{1}{8} y(-2) = 2^0 u(0) + \cancel{2^{-1} u(-1)}^0$$

$$\therefore y(0) = -\left[\frac{1}{4} (2) - \frac{1}{8} (-1) \right] + 1 = \left(\frac{3}{8} \right)$$

put $n=1$ in the main eqⁿ.

$$y(1) + \frac{1}{4} y(0) - \frac{1}{8} y(-1) = 2^1 u(1) + 2^0 u(0)$$

$$y(1) + \frac{1}{4} \left(\frac{3}{8} \right) - \frac{1}{8} (2) = 2 + 1 = 3$$

$$\boxed{y(1) = \frac{71}{32}}$$

use these values in eqⁿ (B) to get c_1 & c_2

put $n=0$; $y_f(0) = y(0) = \frac{3}{8} = c_1 + c_2 + \frac{48}{35}$

put $n=1$; $y_f(1) = y(1) = \frac{71}{32} = \frac{c_1}{4} - \frac{c_2}{2} + \frac{96}{35}$

solving for c_1 & c_2 ; we get

'Z' Transform

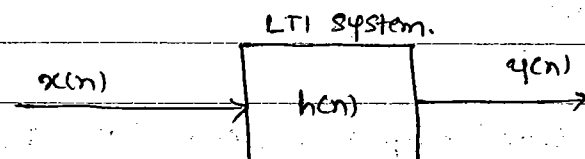
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Let $z = r e^{j\Omega}$ be a complex quantity with $|z| = r$ & $\angle(z) = \Omega$

Let $x[n] = z^n$ be a complex exponential signal applied to an LTI system with impulse response $h[n]$.



$$x[n] = r^n e^{j\Omega n} = r^n \cos(\Omega n) + j r^n \sin(\Omega n)$$

$$y[n] = T[x[n]] = h[n] * x[n]$$

$$= \sum_{k=-\infty}^{\infty} h[k] \cdot x[n-k] = \sum_{k=-\infty}^{\infty} h[k] \cdot r^{n-k} e^{j\Omega(n-k)}$$

$$= \sum_{k=-\infty}^{\infty} h[k] \cdot z^{n-k} = z^n \sum_{k=-\infty}^{\infty} h[k] \cdot z^{-k} = z^n H(z)$$

We define $H(z) = \sum_{k=-\infty}^{\infty} h[k] z^{-k}$ as the transfer function and $H(z)$ is called 'z' transform signal $h[n]$.

In general the z-transform of any arbitrary signal $x[n]$ is defined as

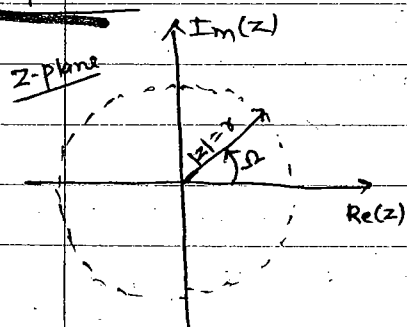
$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} \quad x[n] \xrightarrow{z} X(z) \quad \text{--- (A)}$$

Eqn (A) is an Infinite summation. For the existence of $X(z)$ this summation must converge. Hence the required condition

$$\text{is } \sum_{n=-\infty}^{\infty} x[n] z^{-n} < \infty ; \quad \sum_{n=-\infty}^{\infty} [x[n] r^{-n}] e^{-j\Omega n} < \infty$$

The range of 'r' for which the above condition is satisfied is known as "Region of Convergence" [ROC]

z-plane:



The z-plane is a complex plane. The point $r e^{j\Omega}$ is at a distance of 'r' from the origin and at an angle Ω from the +ve real axis.

If $x(n)$ is absolutely summable, then the DTFT is obtained from Z-T by setting $r=1$ or by substituting $z=e^{j\Omega}$ in eqn (A)

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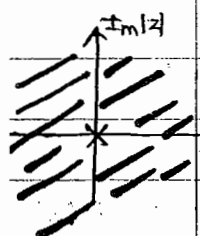
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$$X(e^{j\Omega}) = X(z) \Big|_{z=e^{j\Omega}}$$

The equation $z=e^{j\Omega}$ describes a circle of unit radius centered at origin in the z -plane.

problems

1. Find Z.T of right sided sequence $x(n)=\{1, 2, 2, 1\}$



$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} = \sum_{n=0}^3 x(n) z^{-n}$$

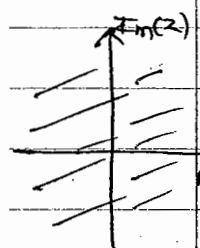
$$= x(0)z^0 + x(1)z^{-1} + x(2)z^{-2} + x(3)z^{-3}$$

$$X(z) = 1 + \frac{2}{z} + \frac{2}{z^2} + \frac{1}{z^3}$$

ROC is entire z -plane except at $z=0$

The summation exists for all values of z except at $z=0$
Hence ROC is entire z -plane except $z=0$

2. Find the Z.T of left sided sequence $x(n)=\{1, 1, 2, 2\}$



$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} = \sum_{n=-3}^0 x(n) z^{-n}$$

$$= x(-3)z^3 + x(-2)z^2 + x(-1)z^1 + x(0)z^0$$

$$= z^3 + z^2 + 2z + 2$$

The summation exists for all values of z except at $z=\infty$

3. Find the Z.T of the double sided seq, $x(n)=\{2, 1, 1, 2\}$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} = \sum_{n=-2}^2 x(n) z^{-n}$$

$$= x(-2)z^2 + x(-1)z + x(0)z^0 + x(1)z^{-1} + x(2)z^{-2}$$

$$= 2z^2 + z + 1 + z^{-1} + 2z^{-2}$$

The summation exists at all values of z in the z -plane

except at $z=0$ & $z=\infty$

Hence ROC is $0 < |z| < \infty$

ZT of basic signals

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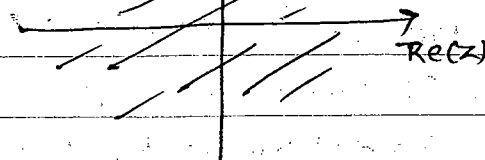
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(a) $x(n) = \delta(n)$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) \bar{z}^n = \sum_{n=-\infty}^{\infty} \delta(n) \bar{z}^n$$

$$= \bar{z}^n \Big|_{n=0} = 1$$

Roc is entire z-plane.

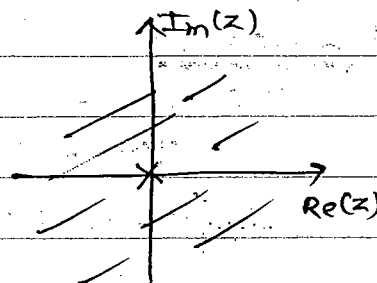


(b) $x(n) = \delta(n-k)$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) \cdot \bar{z}^n = \sum_{n=-\infty}^{\infty} \delta(n-k) \cdot \bar{z}^n$$

$$= \bar{z}^n \Big|_{n=k} = \bar{z}^k = \frac{1}{z^k}$$

Roc is entire z-plane except at $z=0$. There exists 'k' multiple poles at $z=0$.



(c) $x(n) = u(n)$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) \cdot \bar{z}^n = \sum_{n=-\infty}^{\infty} u(n) \bar{z}^n$$

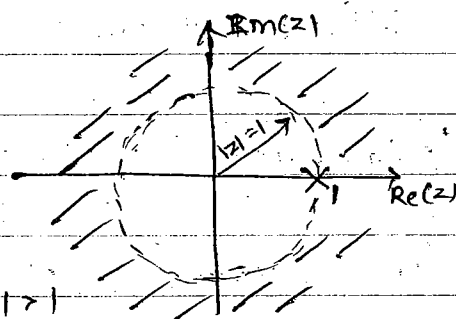
$$= \sum_{n=0}^{\infty} 1 \cdot \bar{z}^n = \frac{1}{1-\bar{z}}$$

$$X(z) = \frac{1}{1-\bar{z}} = \frac{z}{z-1}, \quad |z| > 1$$

The summation converges at all values of 'z' except at $z=1$

Hence Roc is entire z-plane ^{outside} $|z| > 1$

Hence the Roc lies outside the unit circle with $|z| > 1$



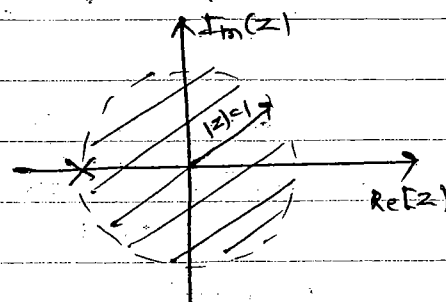
(d) $x(n) = -u(-n-1)$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) \cdot \bar{z}^n = \sum_{n=-\infty}^{\infty} -u(-n-1) \bar{z}^n$$

$$= -\sum_{n=-\infty}^{-1} 1 \cdot \bar{z}^n = \sum_{n=1}^{\infty} \bar{z}^n$$

put $l = n+1$; $X(z) = \sum_{l=0}^{\infty} \bar{z}^l \cdot z$
 $n = l+1$

$$X(z) = -z \cdot \frac{1}{1-\bar{z}} = \frac{-z}{1-\bar{z}}; \quad |z| < 1$$



$$XC(z) = \frac{z}{z-1} = \frac{1}{1-\frac{1}{z}} \quad |z| < 1$$

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*** ZT of $x(n)$ & $x(n-1)$ are same, however the ROC's are different. This ambiguity is the general feature of one sided sequence. Hence we must know the ~~correct~~ ^{ROC} to determine the correct inverse Z-transform ***

properties of Z-transform.

Scaling

I:

Proof:-

1. Linearity:

yc

If $x(n) \xrightarrow{Z} X(z)$ with ROC R_x &
 $y(n) \xrightarrow{Z} Y(z)$ with ROC R_y

then $\boxed{ax(n) + by(n) \xrightarrow{Z} aX(z) + bY(z)}$ with ROC $R_x \cap R_y$

Proof: for a signal $x(n)$,

114 $Z[e^n]$

$$Z[x(n)] = X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

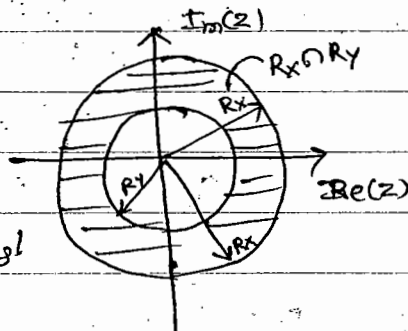
$$Z[ax(n) + by(n)] = \sum_{n=-\infty}^{\infty} [ax(n) + by(n)] z^{-n} = a \sum_{n=-\infty}^{\infty} x(n) z^{-n} + b \sum_{n=-\infty}^{\infty} y(n) z^{-n}$$

4. Time

I

$$\boxed{Z[ax(n) + by(n)] = aX(z) + bY(z)}$$

The Z transform of sum of signals is just the sum of the individual transforms!



Proof:-

2. Time Shifting

If $x(n) \xrightarrow{Z} X(z)$ with ROC R_x .

$x(n-n_0) \xrightarrow{Z} z^{-n_0} X(z)$ with ROC R_x except possibly at $z=0$ or $|z|=\infty$

proof:-

Let $y(n) = x(n-n_0)$

$$Y(z) = \sum_{n=-\infty}^{\infty} y(n) z^{-n} = \sum_{n=-\infty}^{\infty} x(n-n_0) z^{-n}$$

put $(n-n_0) = l$
 or $n = l + n_0$

$$Y(z) = \sum_{l=-\infty}^{\infty} x(l) z^{-(l+n_0)}$$

$$= z^{-n_0} \sum_{l=-\infty}^{\infty} x(l) z^{-l} = \underline{\underline{z^{-n_0} X(z)}}$$

proof:-

$Z[$

5. conv

If

then

$$\therefore \boxed{Z[x(n-n_0)] \longleftrightarrow z^{-n_0} X(z)}$$

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however the

feature
ROC

Scaling in z-domain or multiplication by an exp. sequence

If $x(n) \xrightarrow{Z} X(z)$ with ROC R then

$$a^n x(n) \xrightarrow{Z} X\left(\frac{z}{a}\right) \text{ with ROC } = |a| \cdot R$$

Proof:-

$$\text{Let } y(n) = a^n x(n)$$

$$Y(z) = Z[a^n x(n)] = \sum_{n=-\infty}^{\infty} y(n) z^{-n} = \sum_{n=-\infty}^{\infty} a^n x(n) z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} x(n) \left(\frac{z}{a}\right)^{-n} = X\left(\frac{z}{a}\right) \text{ (RHS) } \checkmark$$

with ROC

$$R_x \cap R_y$$

$$\therefore \boxed{Z[a^n x(n)] \xrightarrow{Z} X\left(\frac{z}{a}\right)}$$

$$\text{Ex } Z[e^{j\omega n} x(n)] \xrightarrow{Z} X\left(\frac{z}{e^{j\omega}}\right)$$

4. Time Reversal.

If $x(n) \xrightarrow{Z} X(z)$ with ROC R , then

$$x(-n) \xrightarrow{Z} X\left(\frac{1}{z}\right) \text{ with ROC } = 1/R.$$

Proof:-

$$\text{Let } y(n) = x(-n).$$

$$Y(z) = Z[x(-n)] = \sum_{n=-\infty}^{\infty} y(n) z^{-n} = \sum_{n=-\infty}^{\infty} x(-n) z^{-n}$$

$$\text{put } l = -n$$

$$Y(z) = \sum_{n=-\infty}^{\infty} x(l) z^{-n} = \sum_{l=-\infty}^{\infty} x(l) \left(\frac{1}{z}\right)^{-l}$$

$$\boxed{Z[x(-n)] = X\left(\frac{1}{z}\right)} \text{ with ROC } = \frac{1}{R}$$

R_x except

0 or $|z| = \infty$

5. convolution (in time domain)

If $x(n) \xrightarrow{Z} X(z)$ with ROC R_x

$y(n) \xrightarrow{Z} Y(z)$ with ROC R_y

then $x(n) * y(n) \xrightarrow{Z} X(z) \cdot Y(z)$ with ROC $= R_x \cap R_y$

proof:-

$$\text{Let } h(n) = x(n) * y(n)$$

$$Z[h(n)] = H(z) = \sum_{n=-\infty}^{\infty} h(n) z^{-n}$$

$$\xrightarrow{Z} X(z)$$

$$\mathcal{Z}[h(n)] = \sum_{n=-\infty}^{\infty} [x(n) * y(n)] z^{-n}$$

X*

$$= \sum_{n=-\infty}^{\infty} \left[\sum_{k=-\infty}^{\infty} x(k) \cdot y(n-k) \right] z^{-n}$$

$$= \sum_{k=-\infty}^{\infty} x(k) \sum_{n=-\infty}^{\infty} y(n-k) \cdot z^{-n} \cdot [z^k \cdot z^{-k}]$$

$$= \sum_{k=-\infty}^{\infty} x(k) \cdot z^{-k} \sum_{n=-\infty}^{\infty} y(n-k) \cdot z^{-(n-k)} = X(z) \cdot Y(z)$$

$$\therefore \boxed{\mathcal{Z}[x(n) * y(n)] \xleftrightarrow{\mathcal{Z}} X(z) \cdot Y(z)}$$

problems:

Determini

$$\boxed{X(z) = \sum_{n=-\infty}^{\infty} x(n) e^{j\omega n}}$$

Hence convolution in time domain is equivalent to multiplication of Z-transforms in z-domain.

Determin

(a) $x(n) =$

∴ Differentiation (in z-domain) (Multiplication by a ramp)

If $x(n) \xleftrightarrow{\mathcal{Z}} X(z)$ with ROC = R, then

$$n x(n) \xleftrightarrow{\mathcal{Z}} -z \frac{d}{dz} [X(z)] \text{ with ROC} = R.$$

Proof:

$$\text{We have } \mathcal{Z}[x(n)] = \sum_{n=-\infty}^{\infty} x(n) z^{-n} = X(z)$$

differentiating both sides w.r.t 'z'

$$\frac{d}{dz} (X(z)) = \sum_{n=-\infty}^{\infty} x(n) \left[\frac{d}{dz} z^{-n} \right] = \sum_{n=-\infty}^{\infty} x(n) [-n z^{-n-1}]$$

$$= - \sum_{n=-\infty}^{\infty} x(n) n z^{-n} z^{-1} = -\frac{1}{z} \sum_{n=-\infty}^{\infty} n x(n) z^{-n}$$

$$\boxed{\sum_{n=-\infty}^{\infty} n x(n) z^{-n} = -z \frac{d}{dz} [X(z)]}$$

The above sequence

(b) $x(n) =$

X(z)

7. Conjugate of a complex sequence.

If $x(n) \xleftrightarrow{\mathcal{Z}} X(z)$

$$x^*(n) \xleftrightarrow{\mathcal{Z}} X^*(z^*)$$

Proof:-

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

Taking complex conjugate on both sides.

$$X^*(z) = \sum_{n=-\infty}^{\infty} x^*(n) (z^*)^{-n}$$

ROC is

Replace z by z* on both sides

$$X^*(z^*) = \sum_{n=-\infty}^{\infty} x^*(n) z^{-n} = Z[x^*(n)]$$

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$$Z[x^*(n)] \xleftrightarrow{z} X^*(z^*)$$

problems:

Determine the ZT of $x(n) =$

$$\begin{cases} 1; & n=-1 \\ 2; & n=0 \\ -1; & n=+1 \\ 1; & n=2 \\ 0; & \text{otherwise} \end{cases}$$

$Y(z)$

$$X(z) = z + 2 - \bar{z} + \bar{z}^2$$

$$X(e^{j\Omega}) = X(-\Omega) = e^{j\Omega} + 2 - \frac{1}{e^{j\Omega}} + \frac{1}{e^{j2\Omega}} \quad \text{is the DTFT}$$

multiplication

Determine the ZT of the following sequences.

(a) $x(n) = \alpha^n u(n)$.

a ramp)

$z\eta$

$= R$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} = \sum_{n=0}^{\infty} \alpha^n u(n) z^{-n}$$

$$X(z) = \sum_{n=0}^{\infty} \alpha^n \bar{z}^{-n} = \sum_{n=0}^{\infty} \left(\frac{z}{\alpha}\right)^{-n} = \frac{1}{1 - \left(\frac{z}{\alpha}\right)^{-1}}$$

$$X(z) = \frac{1}{1 - \alpha \bar{z}^{-1}} = \frac{z}{z - \alpha}; \quad |z| > \alpha$$

The above sequence is a right-sided sequence or causal sequence whose ROC lies outside $|z| = \alpha$ on the z -plane.

$\bar{z}^{-n-1}$

$n \bar{z}^n$

(b) $x(n) = \alpha^n u(-n)$.

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

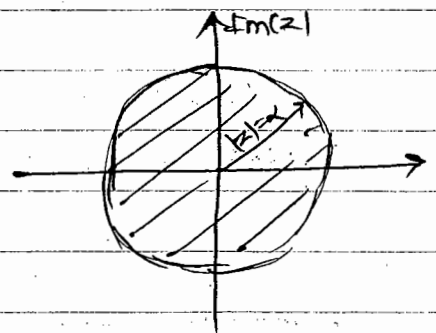
$$= \sum_{n=-\infty}^{\infty} \alpha^n u(-n) z^{-n}$$

$$= \sum_{n=0}^{\infty} \alpha^n \bar{z}^{-n} = \sum_{n=0}^{\infty} \left(\frac{z}{\alpha}\right)^n = \alpha \frac{1}{1 - \frac{z}{\alpha}}; \quad |z| < \alpha$$

$$X(z) = \frac{\alpha}{\alpha - z}; \quad |z| < \alpha$$

conjugate on

ROC lies inside the radius of circle $|z| = \alpha$



© $x(n) = -\alpha^n u(n-1)$

$$X(z) = -\left[\frac{z}{\alpha - z}\right]; |z| < \alpha$$

$$\boxed{X(z) = \frac{-z}{z - \alpha} = \frac{1}{1 - \alpha \bar{z}^{-1}}}; |z| < \alpha$$

④ $x(n) = -u(n-1) + \left(\frac{1}{2}\right)^n u(n)$

$$X(z) = \sum_{n=-\infty}^{\infty} \left[-u(n-1) + \left(\frac{1}{2}\right)^n u(n) \right] \bar{z}^n$$

$$= \sum_{n=-\infty}^{\infty} -u(n-1) \bar{z}^n + \sum_{n=-\infty}^{\infty} \left(\frac{1}{2}\right)^n u(n) \bar{z}^n$$

$$= -\sum_{n=-\infty}^{-1} \bar{z}^n + \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n \bar{z}^n$$

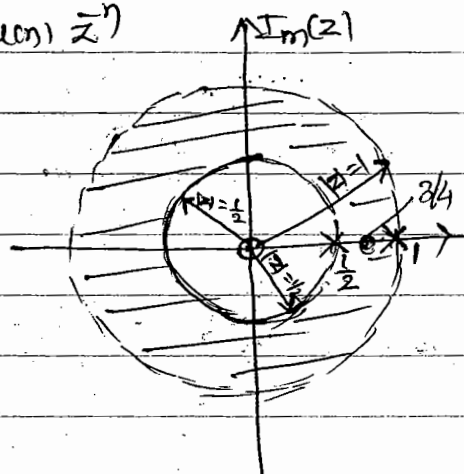
$$\frac{z}{z-1} + \frac{z}{z-\frac{1}{2}} \quad \begin{array}{l} |z| < 1 \\ |z| > \frac{1}{2} \end{array}$$

$$\boxed{\text{ROC: } \frac{1}{2} < |z| < 1}$$

$$= \frac{z(z-\frac{1}{2}) + z(z-1)}{(z-1)(z-\frac{1}{2})} = \frac{z(2z-3/2)}{(z-1)(z-\frac{1}{2})}$$

$X(z)$ has poles at $z=1$ & $z=1/2$

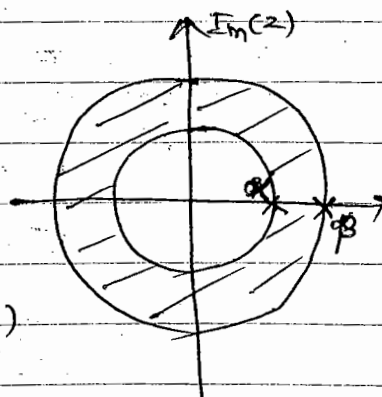
Zeros at $z=0$ & $z=3/4$.



© $x(n) = \alpha^n u(n) + \beta^n u(n-1)$

$$X(z) = \sum_{n=-\infty}^{\infty} \alpha^n u(n) \bar{z}^n + \sum_{n=-\infty}^{\infty} \beta^n u(n-1) \bar{z}^n$$

$$= \frac{z}{z-\alpha} + \frac{z}{z-\beta} \quad \begin{array}{l} |z| > \alpha \\ |z| < \beta \end{array}$$



$$(\alpha < \beta)$$

When $\alpha < \beta$, Intersection of ROC's exist, hence the common ROC is $\alpha < |z| < \beta$.

⑤ $x(n) =$

Refer p

$X(z)$

⑥ $x(n) =$

$X(z) =$

⑦ $x(n) =$

$$X(z) = \sum_{n=-\infty}^{\infty}$$

Hence

$X(z)$

⑧ $x(n) =$

$X(z) =$

When $\alpha > \beta$; no common Roc exists.

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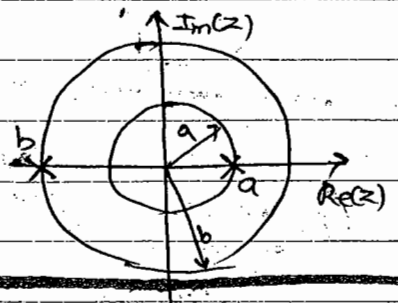
$\alpha < \beta$

⑧ $x(n) = a^n u(n) + b^n u(-n-1)$

Refer previous problem.

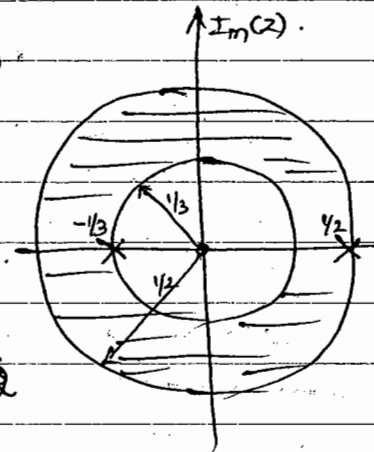
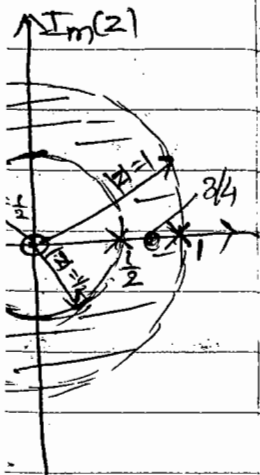
$$X(z) = \frac{z}{z-a} \Big|_{|z|>a} + \frac{z}{b-z} \Big|_{|z|<b}$$

$|a| < |b|$



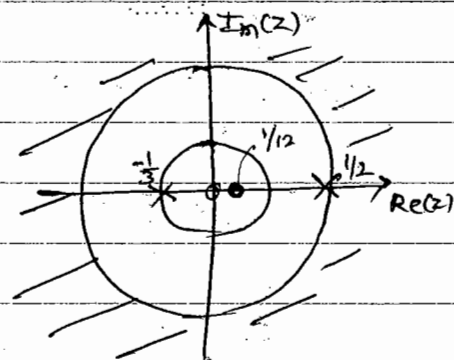
⑨ $x(n) = (\frac{1}{2})^n u(n) + (-\frac{1}{3})^n u(-n-1)$

$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{\infty} (\frac{1}{2})^n u(n) z^{-n} + \sum_{n=-\infty}^{\infty} (-\frac{1}{3})^n u(-n-1) z^{-n} \\ &= \sum_{n=0}^{\infty} (\frac{1}{2})^n z^{-n} + \sum_{n=-\infty}^{-1} (-\frac{1}{3})^n z^{-n} \\ &= \frac{z}{z-\frac{1}{2}} \Big|_{|z|>\frac{1}{2}} + \frac{z}{z+\frac{1}{3}} \Big|_{|z|<\frac{1}{3}} \end{aligned}$$



⑩ $x(n) = (\frac{1}{2})^n u(n) + (-\frac{1}{3})^n u(n)$

$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{\infty} (\frac{1}{2})^n u(n) z^{-n} + \sum_{n=-\infty}^{\infty} (-\frac{1}{3})^n u(n) z^{-n} \\ &= \frac{z}{z-\frac{1}{2}} \Big|_{|z|>\frac{1}{2}} + \frac{z}{z+\frac{1}{3}} \Big|_{|z|>\frac{1}{3}} \end{aligned}$$



Hence the Roc lies outside $|z| > 1/2$.

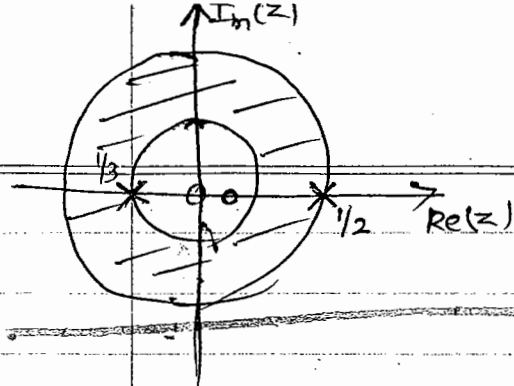
$$X(z) = \frac{z(z+\frac{1}{3}) + z(z-\frac{1}{2})}{(z-\frac{1}{2})(z+\frac{1}{3})} = \frac{z(2z-\frac{1}{6})}{(z-\frac{1}{2})(z+\frac{1}{3})} \Big|_{|z|>\frac{1}{2}}$$

$\alpha < \beta$

⑪ $x(n) = -(\frac{1}{2})^n u(-n-1) - (\frac{1}{3})^n u(-n-1)$

$$\begin{aligned} X(z) &= -\sum_{n=-\infty}^{\infty} (\frac{1}{2})^n u(-n-1) z^{-n} - \sum_{n=-\infty}^{\infty} (\frac{1}{3})^n u(-n-1) z^{-n} \\ &= -\sum_{n=-\infty}^{-1} (\frac{1}{2})^n z^{-n} - \sum_{n=-\infty}^{-1} (\frac{1}{3})^n z^{-n} = \frac{z}{z-\frac{1}{2}} \Big|_{|z|<\frac{1}{2}} + \frac{z}{z+\frac{1}{3}} \Big|_{|z|<\frac{1}{3}} \end{aligned}$$

since the



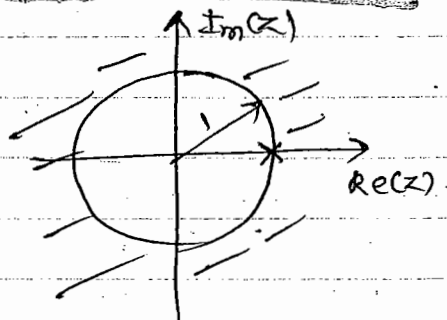
$$X(z) = \frac{z(2z - 1/6)}{(z - 1/2)(z + 1/3)}$$

$$\text{Roc is } \boxed{\frac{1}{3} < |z| < \frac{1}{2}}$$

(j) $x(n) = e^{j\Omega_0 n} u(n)$

$$X(z) = \frac{z}{z - e^{j\Omega_0}} \quad |z| > e^{j\Omega_0}$$

$$\boxed{\text{Roc is } |z| > 1}$$



$$e^{j\Omega_0} = \cos \Omega_0 + j \sin \Omega_0$$

$$|e^{j\Omega_0}| = [\cos^2 \Omega_0 + \sin^2 \Omega_0]^{1/2} = 1$$

Hence Roc lies outside the unit circle

(k) $x(n) = \alpha^{|n|}$ with $\alpha < 1$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

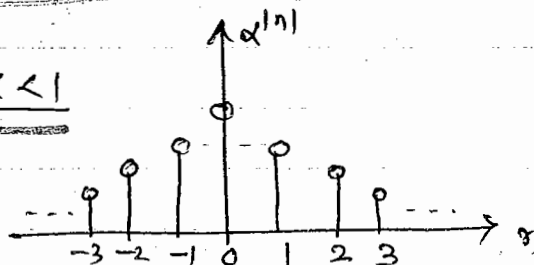
$$= \sum_{n=-\infty}^{-1} \alpha^{-n} z^{-n} + \sum_{n=0}^{\infty} \alpha^n z^{-n}$$

$$= \sum_{n=-\infty}^{-1} (\alpha z)^{-n} + \sum_{n=0}^{\infty} \left(\frac{\alpha}{z}\right)^n$$

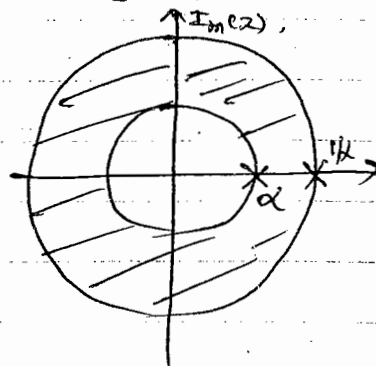
$$= \frac{1}{1 - \frac{1}{\alpha z}} \quad |z| < \frac{1}{\alpha} + \frac{z}{z - \alpha} \quad |z| > \alpha$$

$$\frac{z}{z - \alpha} - \frac{z}{z - \frac{1}{\alpha}}$$

The Roc is $\boxed{\alpha < |z| < \frac{1}{\alpha}}$



$$x(n) = \begin{cases} \alpha^{-n} & n < 0 \\ \alpha^n & n \geq 0 \end{cases}$$



There ex
 $z = \alpha$. A
 pole zero c
 in the exp
 does not
 Hence R

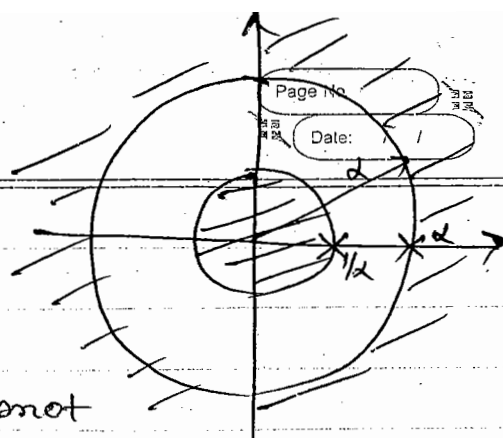
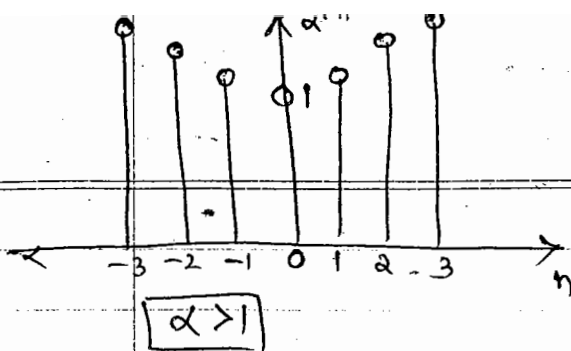
(m) $x(n)$

$$x(n) =$$

$$x(n) =$$

$$=$$

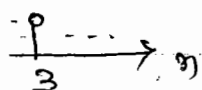
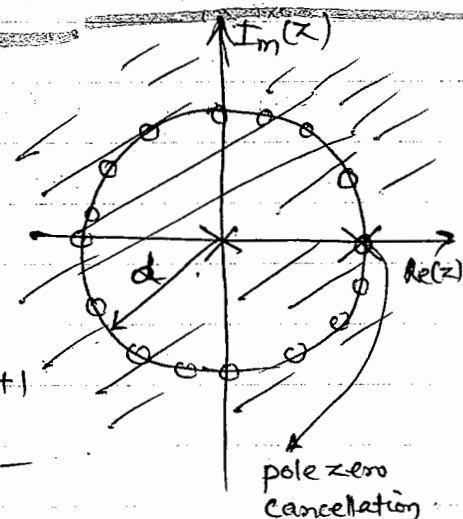
$$1 < \frac{1}{\alpha}$$



For $\alpha > 1$, the ROC's does not overlap, hence there is no convergence. ROC is an empty set.

$$\textcircled{1} \quad x(n) = \begin{cases} \alpha^n & 0 \leq n \leq N-1 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{\infty} x(n) z^{-n} = \sum_{n=0}^{N-1} \alpha^n z^{-n} \\ &= \sum_{n=0}^{N-1} \left(\frac{\alpha}{z}\right)^n = \frac{1 - \left(\frac{\alpha}{z}\right)^{N-1+1}}{1 - \frac{\alpha}{z}} \\ &= \frac{(z^N - \alpha^N) \cdot z}{(z - \alpha) z^N} \end{aligned}$$



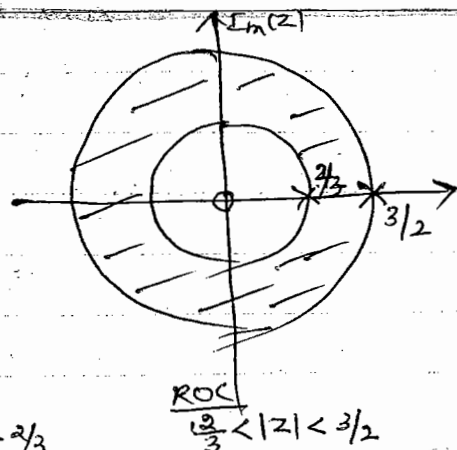
There exists multiple poles at $z=0$, and a single pole at $z=\alpha$. Also multiple zeros at $z=\alpha$ and a single ^{zero} pole at $z=0$. pole zero cancellation takes place at $z=0$ and $z=\alpha$, resulting in the expansion of ROC from $z > \alpha$ to $z > 0$. The ROC does not include the point $z=0$.

Hence ROC is the entire z -plane except at $z=0$.

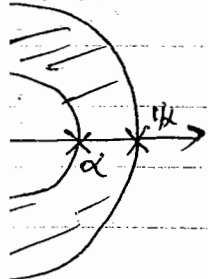
$$\textcircled{2} \quad x(n) = \left(\frac{2}{3}\right)^{|n|}$$

$$x(n) = \begin{cases} \left(\frac{2}{3}\right)^{-n} & n < 0 \\ \left(\frac{2}{3}\right)^n & n \geq 0 \end{cases}$$

$$\begin{aligned} x(n) &= \left(\frac{2}{3}\right)^n u(n-1) + \left(\frac{2}{3}\right)^n u(n) \\ &= \left(\frac{-z}{z - \frac{3}{2}}\right) \Big|_{|z| < \frac{3}{2}} + \frac{z}{z - \frac{2}{3}} \Big|_{|z| > \frac{2}{3}} \end{aligned}$$



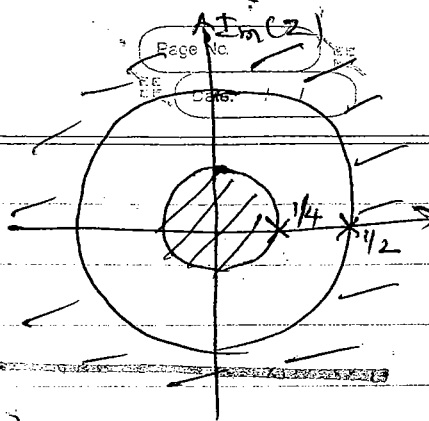
1) $n < 0$
2) $n \geq 0$
Im(z),



(n) $x(n) = \left(\frac{1}{2}\right)^n u(n) + \left(\frac{1}{4}\right)^n u(n-1)$

$$X(z) = \frac{z}{z - \frac{1}{2}} \Big|_{|z| > \frac{1}{2}} - \frac{z}{z - \frac{1}{4}} \Big|_{|z| < \frac{1}{4}}$$

There is no common ROC available
hence ROC = 0



(m) $x(n) = \left(-\frac{1}{2}\right)^n u(-n) + 2\left(\frac{1}{4}\right)^n u(n)$

$$X(z) = X_1(z) + X_2(z)$$

Let $x_1(n) = \left(-\frac{1}{2}\right)^n u(-n)$

we have $\mathcal{Z}\left[\left(-\frac{1}{2}\right)^n u(-n)\right] \xleftrightarrow{z} \frac{z}{z - \frac{1}{2}}$

using time reverse property

$$\mathcal{Z}\left[\left(-\frac{1}{2}\right)^n u(-n)\right] \longleftrightarrow \frac{\bar{z}^{-1}}{\bar{z}^{-1} + \frac{1}{2}}$$

$$\therefore X_1(z) = \frac{2}{z+2} \Big|_{|z| < 2}; \quad X_2(z) = \frac{2z}{z - \frac{1}{4}} \Big|_{|z| > \frac{1}{4}}$$

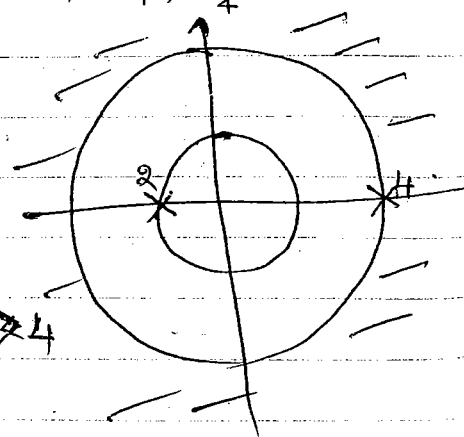
$$X(z) = \frac{2}{z+2} + \frac{2z}{z - \frac{1}{4}}; \quad \boxed{\frac{1}{4} < |z| < 2}$$

(o) $x(n) = \left(-\frac{1}{2}\right)^n u(-n) + 2\left(\frac{1}{4}\right)^{-n} u(-n)$

$$X(z) = \frac{\bar{z}^{-1}}{\bar{z}^{-1} + \frac{1}{2}} \Big|_{|z| < \frac{1}{2}} + \frac{2\bar{z}^{-1}}{\bar{z}^{-1} - \frac{1}{4}} \Big|_{|z| < \frac{1}{4}}$$

$$= \left(\frac{2}{\bar{z} + 2}\right) \Big|_{|z| < \frac{1}{2}}$$

$$= \frac{z}{z(1 + \frac{z}{2})} \Big|_{|z| > 2} + \left(\frac{2}{1 - \frac{z}{4}}\right) \Big|_{|z| > 4}$$



Hence No Common ROC

P) $x(n) = n \left(-\frac{1}{2}\right)^n u(n) * \left(\frac{1}{4}\right)^n u(n)$
 $= x_1(n) * x_2(n)$

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$x_1(n) = n \left(-\frac{1}{2}\right)^n u(n)$

Let $y(n) = \left(-\frac{1}{2}\right)^n u(n)$

$Y(z) = \leftarrow z \rightarrow \left[\frac{z}{z + \frac{1}{2}} \right] |z| > \frac{1}{2}$

Let $x_1(n) = n y(n) \leftarrow z \rightarrow -z \frac{d}{dz} [Y(z)] = X_1(z)$

$= -z \left[\frac{d}{dz} \left[\frac{z}{z + \frac{1}{2}} \right] \right] = -z \left[\frac{(z + \frac{1}{2}) - z(1)}{(z + \frac{1}{2})^2} \right]$

$X_1(z) = \left[\frac{-z/2}{(z + \frac{1}{2})^2} \right] |z| > \frac{1}{2}$

$x_2(n) = \left(-\frac{1}{4}\right)^n u(n)$

~~we have let~~

Let $m(n) = \left(\frac{1}{4}\right)^n u(n)$

$M(z) = \left[\frac{z}{z - \frac{1}{4}} \right] |z| > \frac{1}{4}$

Let $x_2(n) = m(n)$

$X_2(z) = M\left(\frac{1}{z}\right) = \left[\frac{\frac{1}{z}}{\frac{1}{z} - \frac{1}{4}} \right] \left| \frac{1}{|z|} \right| > \frac{1}{4}$

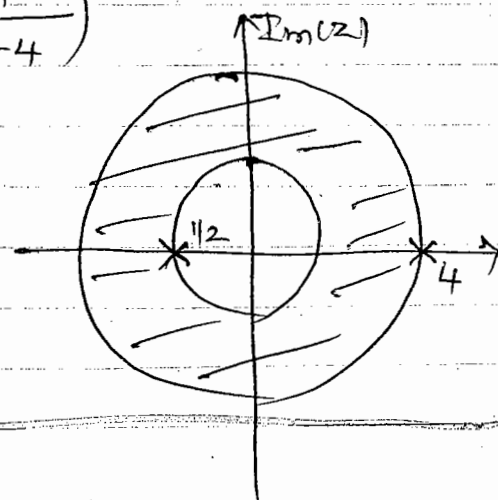
$= \frac{1}{z} \left[\frac{1}{\frac{1}{z} - \frac{1}{4}} \right] = \frac{4}{4 - z} = \frac{-4}{z - 4}$

$X(z) = X_1(z) \cdot X_2(z)$

$= \frac{-z/2}{(z + \frac{1}{2})^2} \cdot \left(\frac{-4}{z - 4} \right)$

$= \frac{2z}{(z + \frac{1}{2})^2 (z - 4)}$

Roc is $\left[\frac{1}{2} < |z| < 4 \right]$



⑥ $x(n) = n \left(\frac{1}{2} \right)^n u(n) * \left(\frac{1}{4} \right)^n u(n-2)$

Let $a(n) = \left(\frac{1}{2} \right)^n u(n) \xleftrightarrow{Z} A(z) = \frac{z}{z - \frac{1}{2}} \quad |z| > \frac{1}{2}$

Let $b(n) = \left(\frac{1}{4} \right)^n u(n) \xleftrightarrow{Z} B(z) = \frac{z}{z - \frac{1}{4}} \quad |z| > \frac{1}{4}$

Let $c(n) = \left(\frac{1}{4} \right)^{n-2} u(n-2) \cdot \left(\frac{1}{4} \right)^2 = \frac{1}{16} \left(\frac{1}{4} \right)^{n-2} u(n-2)$

$C(z) = \left[\frac{z^{-2} \cdot z}{z - \frac{1}{4}} \right] \frac{1}{16} = \frac{z/16}{z - 1/4}$

$X(z) = Z \left[n [a(n) * c(n)] \right]$

$= -z \frac{d}{dz} [A(z) \cdot C(z)]$

$= -z \left[\frac{d}{dz} \left[\frac{z}{z - \frac{1}{2}} \cdot \frac{z/16}{z - 1/4} \right] \right] = \frac{-z}{16} \frac{d}{dz} \left[\frac{1}{(z - \frac{1}{2})(z - \frac{1}{4})} \right]$

$= \frac{-z}{16} \frac{d}{dz} \left[\frac{1}{z^2 + \frac{1}{8} - \frac{3z}{4}} \right] = \frac{-z}{16} \left[\frac{-[2z - \frac{3}{4}]}{(z^2 - \frac{3}{4}z + \frac{1}{8})^2} \right]$

$X(z) = \frac{\frac{z^2}{8} - \frac{3z}{64}}{\left[z^2 - \frac{3}{4}z + \frac{1}{8} \right]^2} \quad |z| > \frac{1}{2}$

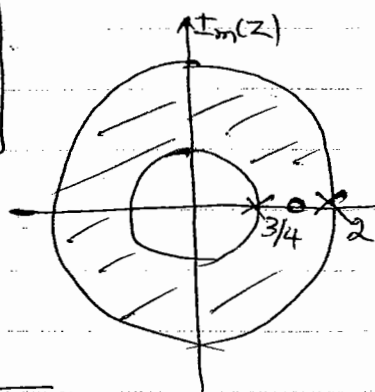
⑦ $x(n) = \left(\frac{3}{4} \right)^n u(n) + (2)^n u(-n-1)$

$X(z) = \left. \frac{z}{z - \frac{3}{4}} \right|_{|z| > \frac{3}{4}} + \left[\frac{-z}{z - 2} \right]_{|z| < 2}$

$= \frac{z(z-2) - z(z-3/4)}{(z-3/4)(z-2)}$

$= \frac{z^2 - 2z - z^2 + 3/4z}{(z-3/4)(z-2)} = \frac{-5/4z}{(z-3/4)(z-2)}$

Roc: $\frac{3}{4} < |z| < 2$



⑧ $x(n) = (n-3)^2 \left(\frac{1}{3}\right)^{n-3} u(n-3)$

Here Time shifting & diff. properties are used

Let $x_1(n) = \left(\frac{1}{3}\right)^n u(n)$

$X_1(z) = \frac{z}{z - 1/3}$

$x_2(n) = n \left(\frac{1}{3}\right)^n u(n) \xrightarrow{z} X_2(z) = -z \frac{d}{dz} [X_1(z)]$
 $= -z \left[\frac{d}{dz} \left[\frac{z}{z - 1/3} \right] \right] = -z \left[\frac{(z - 1/3) - z}{(z - 1/3)^2} \right]$

$X_2(z) = \frac{z/3}{(z - 1/3)^2}$

Let $x_3(n) = n^2 x_2(n) = n^2 \left(\frac{1}{3}\right)^n u(n) \xrightarrow{z} X_3(z)$

$X_3(z) = -z \frac{d}{dz} [X_2(z)] = -z \frac{d}{dz} \left[\frac{z/3}{(z - 1/3)^2} \right]$

$= -z \left[\frac{(z - 1/3)^2 \cdot \frac{1}{3} - \frac{z}{3} \cdot 2(z - 1/3)}{(z - 1/3)^4} \right]$

$= -z \left[\frac{(z - 1/3) \cdot \frac{1}{3} - \frac{2z}{3}}{(z - 1/3)^3} \right] = -z \left[\frac{\frac{z}{3} - \frac{1}{9} - \frac{2z}{3}}{(z - 1/3)^3} \right]$

$X_3(z) = \frac{\frac{z^2}{3} + \frac{z}{9}}{(z - 1/3)^3}$

$x_3(n) = n^2 \left(\frac{1}{3}\right)^n u(n) \xrightarrow{z} X_3(z) = \frac{\frac{z^2}{3} + \frac{z}{9}}{(z - 1/3)^3}$

Time shift by $n = n-3$.

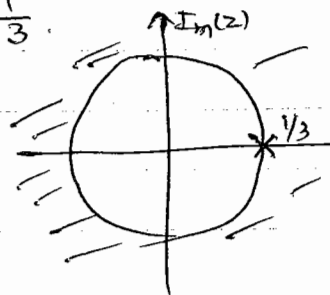
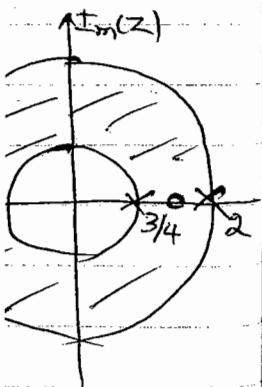
$(n-3)^2 \left(\frac{1}{3}\right)^{n-3} u(n-3) \xrightarrow{z} \bar{z}^3 \left[\frac{\frac{z^2}{3} + \frac{z}{9}}{(z - 1/3)^3} \right]$

$X(z) = \frac{\frac{\bar{z}^3}{3} + \frac{\bar{z}^2}{9}}{(z - 1/3)^3}$

$|z| > \frac{1}{3}$

Multiple poles at $z = 1/3$

$X(z) = \left[\frac{\bar{z}^3}{3} + \frac{\bar{z}^2}{9} \right] / (z - 1/3)^3 \Big|_{|z| > 1/3}$



Find the ZT of a) $x(n) = \cos \Omega_0 n u(n)$
 b) $x(n) = \sin \Omega_0 n u(n)$.

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we have

Let $y(n) = e^{j\Omega_0 n} u(n)$

$$Y(z) = \frac{z}{z - e^{j\Omega_0}} \quad |z| > 1$$

We know that $e^{j\Omega_0} = \cos \Omega_0 + j \sin \Omega_0$.

$$Y(z) = \frac{z}{z - \cos \Omega_0 - j \sin \Omega_0}$$

Rationalizing the denominator

$$Y(z) = \frac{z[(z - \cos \Omega_0) + j \sin \Omega_0]}{(z - \cos \Omega_0)^2 + (\sin \Omega_0)^2}$$

$$= \frac{z^2 - z \cos \Omega_0 + j z \sin \Omega_0}{z^2 - 2z \cos \Omega_0 + \cos^2 \Omega_0 + \sin^2 \Omega_0}$$

$$= \frac{z^2 - z \cos \Omega_0}{z^2 - 2z \cos \Omega_0 + 1} + j \frac{z \sin \Omega_0}{z^2 - 2z \cos \Omega_0 + 1}$$

$$\therefore e^{j\Omega_0 n} u(n) = [\cos \Omega_0 n + j \sin \Omega_0 n] u(n)$$

$$\therefore Y[e^{j\Omega_0 n} u(n)] = Y[(\cos \Omega_0 n + j \sin \Omega_0 n) \cdot u(n)]$$

Equating the real and imaginary parts

$$\underline{x(n) = \cos \Omega_0 n u(n)} \xrightarrow{Z} \boxed{X(z) = \frac{z^2 - z \cos \Omega_0}{z^2 - 2z \cos \Omega_0 + 1}} \quad |z| > 1$$

$$\underline{x(n) = \sin \Omega_0 n u(n)} \xrightarrow{Z} \boxed{X(z) = \frac{z \sin \Omega_0}{z^2 - 2z \cos \Omega_0 + 1}} \quad |z| > 1$$

$$Z[\cos \Omega_0 n u(n)] = \frac{1 - z^{-1} \cos \Omega_0}{1 - 2z^{-1} \cos \Omega_0 + z^{-2}} \quad |z| > 1$$

$$Z[\sin \Omega_0 n u(n)] = \frac{z^{-1} \sin \Omega_0}{1 - 2z^{-1} \cos \Omega_0 + z^{-2}} \quad |z| > 1$$

Let

Find the ZT of $x(n) = a^n \cos \Omega_0 n u(n)$

We have $x(n) \xleftrightarrow{Z} X(z)$

$$a^n x(n) \xleftrightarrow{Z} X(z/a)$$

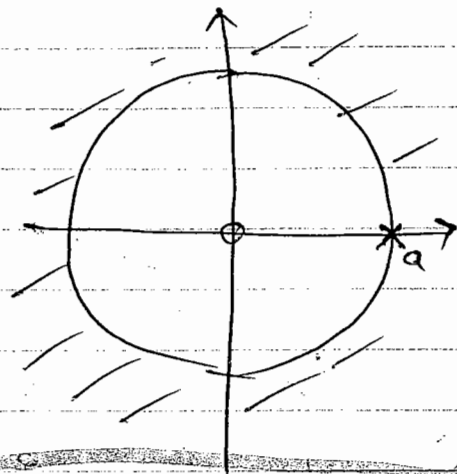
$$\therefore Z[a^n \cos \Omega_0 n u(n)] = \frac{1 - a z^{-1} \cos \Omega_0}{1 - 2a z^{-1} \cos \Omega_0 + (a z^{-1})^2}$$

put $\Omega_0 = 2\pi$

$$X(z) = \frac{1 - a z^{-1} \cos 2\pi}{1 - 2a z^{-1} \cos 2\pi + (a z^{-1})^2}$$

$$= \frac{1 - a z^{-1}}{(1 - a z^{-1})^2} = \frac{1}{1 - a z^{-1}}$$

$$X(z) = \frac{z}{z - a} \quad |z| > a$$



Find the ZT of $x(n) = a^n \sin \Omega_0 n u(n)$

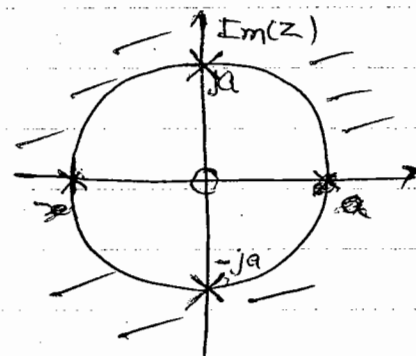
$$X(z) = \frac{a z^{-1} \sin \Omega_0}{1 - 2a z^{-1} \cos \Omega_0 + (a z^{-1})^2}$$

put $\Omega_0 = \pi/2$

$$X(z) = \frac{a z^{-1} \sin \pi/2}{1 - 2a z^{-1} \cos \pi/2 + (a z^{-1})^2}$$

$$= \frac{a z^{-1}}{1 + (a z^{-1})^2} = \frac{za}{z^2 + a^2}$$

$$= \frac{za}{(z^2 + a^2)} \quad |z| > a$$



$$z^2 + a^2 = 0$$

$$z = \pm ja$$

$$x(n) = n \left(\frac{1}{3}\right)^{n+3} u(n+3)$$

$$x(n) = n \cdot \left(\frac{1}{3}\right)^n \cdot \left(\frac{1}{3}\right)^3 u(n+3)$$

$$= \left(\frac{1}{3}\right)^3 \cdot n \left(\frac{1}{3}\right)^n u(n+3)$$

$$\text{Let } y(n) = \left(\frac{1}{3}\right)^3 \left(\frac{1}{3}\right)^n u(n+3)$$

$$\frac{0.3 \Omega_0}{\Omega_0 + 1} \quad |z| > 1$$

$$\frac{1}{z^2 + a^2} \quad |z| > 1$$

$$Y(z) = \sum_{n=-\infty}^{\infty} \left(\frac{1}{3}\right)^3 \cdot \left(\frac{1}{3}\right)^n u(n+3) \cdot \bar{z}^n$$

$$= \left(\frac{1}{3}\right)^3 \sum_{n=-3}^{\infty} \left(\frac{1}{3}\right)^n \bar{z}^n$$

$$= \left(\frac{1}{3}\right)^3 \sum_{n=-3}^{\infty} \left(\frac{1}{3z}\right)^n$$

put: $l = n+3$;
when $n = -3$, $l = 0$
when $n = \infty$, $l = \infty$

$$Y(z) = \sum_{l=0}^{\infty} \left(\frac{1}{3}\right)^3 \cdot \left(\frac{1}{3z}\right)^{l-3}$$

$$= \left(\frac{1}{3}\right)^3 \left(\frac{1}{3z}\right)^{-3} \sum_{l=0}^{\infty} \left(\frac{1}{3z}\right)^l$$

$$= \left(\frac{1}{z}\right)^{-3} \left[\frac{1}{1 - \frac{1}{3z}} \right]$$

$$Y(z) = \frac{3z^4}{3z-1}$$

contd

$$x(n) = n y(n) = n \left(\frac{1}{3}\right)^{n+3} u(n+3)$$

$$X(z) = -z \frac{d}{dz} [Y(z)] = -z \left[\frac{d}{dz} \left[\frac{3z^4}{3z-1} \right] \right]$$

$$= -z \left[\frac{(3z-1)12z^3 - 3z^4(3)}{(3z-1)^2} \right]$$

$$= -z \left[\frac{36z^4 - 12z^3 - 9z^4}{(3z-1)^2} \right] = \frac{-27z^5 + 12z^4}{(3z-1)^2}$$

$$= \frac{3z(4z^3 - 9z^4)}{(3z-1)^2} \quad |z| > \frac{1}{3}$$

$$x(n) = n \left(\frac{1}{2}\right)^n u(n) * \left[\delta(n) + \frac{1}{2} \delta(n-1) \right]$$

$$x(n) = x_1(n) * \delta(n)$$

$$x(n) = n \left(\frac{1}{2}\right)^n u(n) * \delta(n) + n \left(\frac{1}{2}\right)^n u(n) * \frac{1}{2} \delta(n-1)$$

$$= n \left(\frac{1}{2}\right)^n u(n) + \frac{1}{2} (n-1) \left(\frac{1}{2}\right)^{n-1} u(n-1)$$

$$= x_1(n) + x_2(n)$$

$$\text{Let } x_1(n) = n \left(\frac{1}{2}\right)^n u(n)$$

$$\text{Let } y(n) = \left(\frac{1}{2}\right)^n u(n)$$

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$$Y(z) = \frac{z}{z-1/2} \quad |z| > 1/2$$

$$X_1(z) = -z \frac{d}{dz} [Y(z)] = -z \left[\frac{d}{dz} \left[\frac{z}{z-1/2} \right] \right]$$

$$= -z \left[\frac{(z-1/2) - z(1)}{(z-1/2)^2} \right] = \frac{+z/2}{(z-1/2)^2}$$

$$x_2(n) = \frac{1}{2} (n-1) \left(\frac{1}{2}\right)^{n-1} u(n-1)$$

$x_2(n)$ is the time shifted version $x_1(n)$ by $(n-1)$

$$\therefore X_2(z) = \frac{1}{2} \frac{z^{-1} \cdot (+z/2)}{(z-1/2)^2} = \frac{+1/2}{(z-1/2)^2}$$

$$\therefore X(z) = X_1(z) + \frac{1}{2} X_2(z)$$

$$= \frac{-z/2}{(z-1/2)^2} + \frac{1}{2} \left[\frac{+1/2}{(z-1/2)^2} \right]$$

$$= \frac{\left(\frac{1}{4} + z/2\right)}{(z-1/2)^2} \quad |z| > 1/2$$

$$\therefore X(z) = \frac{\frac{1}{4} + z/2}{(z-1/2)^2} = \frac{\frac{z^{-1}}{2} \left[1 + \frac{z^{-1}}{2} \right]}{\left(1 - \frac{z^{-1}}{2}\right)^2} \quad |z| > 1/2$$

$$x(n) = n^2 u(n)$$

$$\text{Let } x_1(n) = u(n)$$

$$X_1(z) = \frac{z}{z-1} \quad |z| > 1$$

$$\text{Let } x_2(n) = n x_1(n) \longleftrightarrow X_2(z) = -z \frac{d}{dz} (X_1(z))$$

$$= -z \left[\frac{(z-1) - z(1)}{(z-1)^2} \right] = \frac{z}{(z-1)^2}$$

$$X(z) = \sum_{n=-3}^{\infty} \left(\frac{1}{3}\right)^n$$

$$\text{Let } x_3(n) = n x_2(n) = n^2 x_1(n)$$

$$\begin{aligned} X_3(z) &= -z \frac{d}{dz} [X_2(z)] = -z \frac{d}{dz} \left[\frac{z}{(z-1)^2} \right] \\ &= -z \left[\frac{(z-1)^2 \cdot 1 - z \cdot 2(z-1)}{(z-1)^4} \right] = -z \left[\frac{z-1-2z}{(z-1)^3} \right] \\ &= -z \left[\frac{-1-z}{(z-1)^3} \right] = \frac{z^2+z}{(z-1)^3} \end{aligned}$$

$$X(z) = X_3(z) = \frac{z(z+1)}{(z-1)^3} \quad |z| > 1$$

Properties of ROC

1. The ROC of a finite duration sequence includes entire z-plane except $z=0$ and/or $z=\infty$

$$\text{Let } x(n) = \{1, 2, 3\}$$

$$X(z) = z + 2 + \frac{3}{z}$$

for $z=0$, & $z=\infty$, $X(z) = \infty$

2. ROC does not contain any poles. (If it includes a pole the system become unstable).

$$\text{Let } x(n) = a^n u(n)$$

$$X(z) = \frac{z}{z-a} \quad |z| > a$$

$X(z)$ has a pole at $z=a$

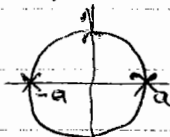
If $X(z)$ includes $z=a$ then the system become

unstable as $X(z) \rightarrow \infty$ at $z=a$.

3. ROC is a ^{circular} ring in the z-plane centered at origin.

$$\text{we } a^n u(n) \longleftrightarrow \frac{z}{z-a} \quad |z| > a$$

$$a^n u(-n-1) \longleftrightarrow \frac{z}{a-z} \quad |z| < a$$



4. The ROC of a right handed sequence is of the form $|z| > r$ where 'r' is the location of outermost pole. ROC lies outside the circle of radius 'r'.

5. ROC of a left sided sequence is of the form $|z| < r$ where 'r' is the ^{location} of innermost pole. ROC lies

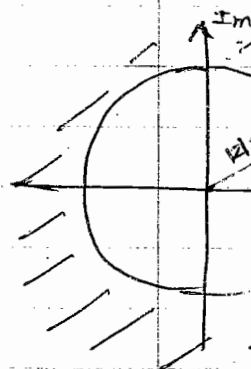
ms.

6. ROC of in the z-plane has a pole, b. ROC is 0. location of most pole.

7. For a: z-plane. Let $x(n)$ $X(z) =$

8. The ROC in the z-plane

9. The ROC sequence of points. He



$x(n) = a^n u(n)$ Right handed

inside the circle of radius r .

- ⑥ ROC of a two sided sequence is a concentric ring in the z -plane bounded on the interior and exterior by a pole, but does not include any poles.

ROC is of the form $r_1 < |z| < r_2$ where r_1 is the location of innermost pole & r_2 is the location of outermost pole.

- ⑦ For a finite causal sequence, the ROC is the entire z -plane except at $z=0$.

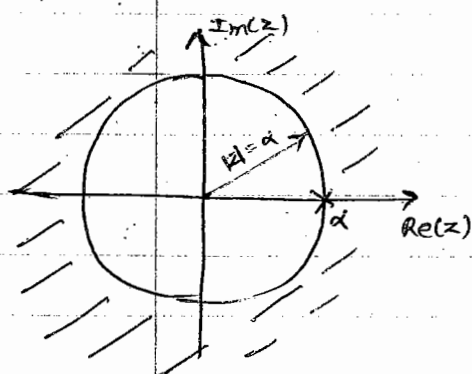
Let $x(n) = \{1, 2, 3\}$ be a finite causal sequence.

$$X(z) = 1 + \frac{2}{z} + \frac{3}{z^2}$$

The sequence converges on the entire z -plane except at $z=0$.

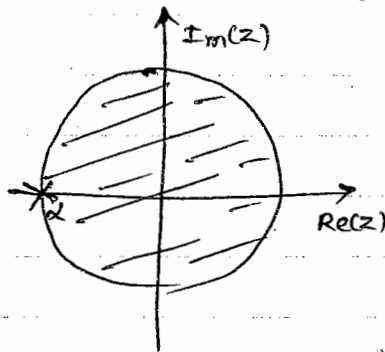
- ⑧ The ROC of a stable LTI system is a unit circle in the z -plane

- ⑨ The ROC is a connected region. The convergence of sequence exists over a certain area rather than discrete points. Hence ROC is a connected region.



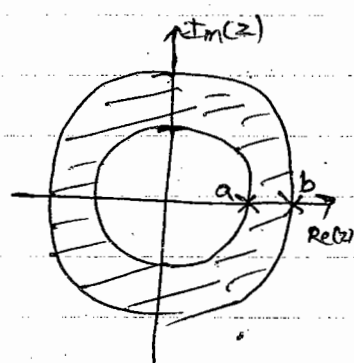
$$x(n) = a^n u(n)$$

Right handed sequence



$$x(n) = a^n u(-n-1)$$

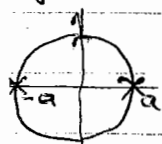
Left sided sequence



$$x(n) = -b^n u(-n-1) + a^n u(n)$$

Two sided sequence

if in.



of the form
most pole.

from $|z| < r$

ROC lies

Find the ZT of

$$x(n) = \sin\left(\frac{\pi}{8}n - \frac{\pi}{4}\right) u(n-2)$$

here $\Omega_0 = \pi/8$

$$Z[\sin(\Omega_0 n - \pi/4) u(n)] = \frac{z^{-1} \sin \pi/8}{1 - 2z^{-1} \cos \frac{\pi}{8} + z^{-2}}$$

~~Time shifting the sequence~~

Find the ZT of the following

(a) $x(n] = \left(\frac{1}{2}\right)^n u(n) * 2^n u(-n-1)$

$$X(z) = \left[\frac{z}{z - \frac{1}{2}} \right]_{|z| > \frac{1}{2}} \cdot \left[\frac{z}{2 - z} \right]_{|z| < 2}$$

$$= \frac{z^2}{(z - \frac{1}{2})(2 - z)} \quad \frac{1}{2} < |z| < 2$$

(b) $x(n] = n \left[\left(\frac{1}{2}\right)^n u(n) * \left(\frac{1}{4}\right)^n u(n-2) \right]$

$$a(n] = \left(\frac{1}{2}\right)^n u(n) \xleftrightarrow{Z} A(z) = \frac{1}{1 - z^{-1}/2} \quad |z| > \frac{1}{2}$$

$$b(n] = \left(\frac{1}{4}\right)^n u(n) \xleftrightarrow{Z} B(z) = \frac{1}{1 - z^{-1}/4} \quad |z| > \frac{1}{4}$$

$$c(n] = b(n-2) \xleftrightarrow{Z} C(z) = \frac{z^{-2}}{(1 - z^{-1}/4)} \quad |z| > \frac{1}{4}$$

$$x(n] = n[a(n] * b(n-2)] \xleftrightarrow{Z} -z \frac{d}{dz} \left[\left(\frac{1}{1 - z^{-1}/2} \right) \left(\frac{z^{-2}}{1 - z^{-1}/4} \right) \right]$$

$$X(z) = \frac{2z^{-2} - \frac{3}{4}z^{-3}}{\left[1 - \frac{3}{4}z^{-1} + \frac{z^{-2}}{8} \right]^2} \quad |z| > \frac{1}{2}$$

(b) $x(n] = n \sin\left(\frac{\pi}{2}n\right) u(n]$

we have $Z[\sin \frac{\pi}{2}n] u(n] \xleftrightarrow{Z} \frac{z}{z^2 + 1} \quad |z| > 1$

$$Z\left[-\sin\left(-\frac{\pi}{2}n\right)\right] u(n] \xleftrightarrow{Z} \frac{-z^{-1}}{1 + z^{-2}} \quad |z| > 1$$

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(c) $x(n]$

$$a(n] = \left(\frac{n}{9}\right) u(n]$$

$$b(n] = \cos\left(\frac{\pi n}{6}\right)$$

B

$$X(z) = A(z)$$

Given the properties

(a) $y(n] =$

$$y(n] = n$$

(b) $y(n] = \left(\frac{1}{2}\right)^n$

$$y(n] = \left(\frac{1}{2}\right)^n$$

(c)

(c) $y(n] =$

$$y(n] =$$

$$-n \left[\sin\left(\frac{\pi}{2}\right)n \right] u(-n) = + \left[-2 \frac{d}{dz} \left[\frac{-z^{-1}}{1+z^2} \right] \right]$$

$$= \frac{\cancel{1}}{\cancel{1+z^2}} + \frac{\cancel{2z}}{\cancel{(1+z^2)^2}} = \frac{z^3 - z}{(1+z^2)^2} \quad |z| > 1$$

(c) $x(n) = 3^{n-2} u(n) * \cos\left(\frac{\pi}{6}n + \frac{\pi}{3}\right) u(n).$

$$= \cancel{\left(\frac{1}{9}\right)} \cancel{3^n} u(n) * \left[\cancel{\cos\left(\frac{\pi}{6}\right)} \cancel{\cos\frac{\pi}{3}} - \cancel{\sin\left(\frac{\pi}{6}\right)} \cancel{\sin\frac{\pi}{3}} \right] u(n)$$

$$= \left(\frac{1}{9}\right) \cdot 3^n u(n) * \cos\left(\frac{\pi}{6}n + \frac{\pi}{3}\right) u(n)$$

$$a(n) = \left(\frac{1}{9}\right) 3^n u(n) \xrightarrow{z} A(z) = \left[\frac{1/9}{1-3z^{-1}} \right] \quad |z| > 3.$$

$$b(n) = \cos\left[\frac{\pi}{6}n + \frac{\pi}{3}\right] u(n) \xrightarrow{z} B(z) = \frac{1 - z^{-1} \cos \pi/6}{1 - 2z^{-1} \cos \frac{\pi}{6} + z^{-2}} \quad |z| > 1$$

$$B(z) = \left[\frac{2 - \sqrt{3} z^{-1}}{2 - 2\sqrt{3} z^{-1} + \sqrt{3} z^{-2}} \right]$$

$$X(z) = A(z) \cdot B(z) = \left[\frac{1/9}{1-3z^{-1}} \right] \left[\frac{2 - \sqrt{3} z^{-1}}{2 - 2\sqrt{3} z^{-1} + \sqrt{3} z^{-2}} \right] \quad |z| > 3.$$

Given the z-T pair $x(n) \xleftrightarrow{z} \frac{z^2}{z^2-16} \quad |z| < 4$; use the z-T properties to determine the z-T of the following signals.

(a) $y(n) = x(n-2)$

$$y(n) = x(n-2) \xleftrightarrow{z} Y(z) = z^{-2} \left[\frac{z^2}{z^2-16} \right] = \frac{1}{z^2-16} \quad |z| < 4$$

(Time shifting property)

(b) $y(n) = \left(\frac{1}{2}\right)^n x(n).$

$$y(n) = \left(\frac{1}{2}\right)^n x(n) \xleftrightarrow{z} Y(z) = \frac{(z/1/2)}{(z/1/2)^2 - 16} = \frac{2z^2}{4z^2 - 16} = \frac{z^2}{z^2 - 4}$$

(Multiplication by an exponential property)

(c) $y(n) = x(-n) * x(n)$

$$y(n) = x(-n) * x(n) \xleftrightarrow{z} \left[\frac{1/z^2}{\frac{1}{z^2} - 16} \right] * \left[\frac{z^2}{z^2 - 16} \right]$$

(Time reverse & convolution)

$$Y(z) = \frac{z^2}{z^2 - 16z + 257z^2 - 16} \quad |z| < 4$$

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(d) $y(n) = n \cdot x(n)$

$$\begin{aligned} y(n) = n x(n) &\leftrightarrow -z \left[\frac{d}{dz} X(z) \right] = -z \left[\frac{d}{dz} \left[\frac{z^2}{z^2 - 16} \right] \right] \\ &= -z \left[\frac{(z^2 - 16)(2z) - z^2[2z]}{(z^2 - 16)^2} \right] = -z \left[\frac{2z^3 - 32z - 2z^3}{(z^2 - 16)^2} \right] \\ &= \frac{-32z}{(z^2 - 16)^2} \quad |z| < 4 \quad (\text{differentiation}) \end{aligned}$$

(e) $y(n) = x(n+1) + x(n-1)$

$$\begin{aligned} y(n) = x(n+1) + x(n-1) &\leftrightarrow Y(z) = \frac{z \cdot z^2}{z^2 - 16} + \frac{z^{-1} \cdot z^2}{z^2 - 16} \\ &(\text{Time shifting and linearity}) \\ &= \frac{z^3 + z}{z^2 - 16} \quad |z| < 4 \end{aligned}$$

(f) $y(n) = x(n) * x(n-3)$

$$\begin{aligned} y(n) = x(n) * x(n-3) &\leftrightarrow Y(z) = \frac{z^2}{z^2 - 16} \cdot \frac{z^{-3} \cdot z^2}{z^2 - 16} \\ &(\text{convolution \& time shifting}) \\ Y(z) &= \frac{z}{(z^2 - 16)^2} \quad |z| < 4 \end{aligned}$$

we have

The inverse

Inverse

$x(n)$ from

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$X(z)$

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of $\frac{B(z)}{A(z)}$

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Inverse Z-Transform

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We have

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

is the ZT of any arbitrary seq. $x(n)$.

The inverse ZT is defined as

$$x(n) = \oint \frac{X(z) z^{n-1} dz}{2\pi j} = \frac{1}{2\pi j} \oint X(z) z^{n-1} dz$$

Inverse ZT are used to recover the time domain signal $x(n)$ from the ZT $X(z)$ of the signal. Direct evaluation of inversion integral requires the knowledge of complex variables theory. Hence alternative simple methods like partial fraction expansion method; power series method etc, are employed.

Partial Fraction Expansion method (PFE).

This method uses the knowledge of several basic ZT pairs and ZT properties to invert the large class of ZT's.

The approach relies mainly on the properties of ROC. Right sided sequence has ROC lies outside the circle of outermost pole, and for all left handed sequence ROC lies inside the circle of radius equal to innermost pole.

$$X(z) = \frac{\tilde{B}(z)}{A(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots + b_m z^{-m}}{a_0 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_n z^{-n}}$$

The numerator polynomial $\tilde{B}(z)$ has the order one less than that of denominator polynomial $A(z)$.

PFE can be applied to determine the inverse ZT of $\frac{\tilde{B}(z)}{A(z)}$.

Note: The Nr polynomial power must be always at least one less than that of Dr polynomial power.

Input (Time domain Signal)	Z.T	ROC
1. $x(n) = \delta(n)$	$X(z) = 1$	Entire z plane
2. $x(n) = \delta(n-k)$	$X(z) = z^{-k}$	Entire z plane except at $z=0$
3. $x(n) = u(n)$	$X(z) = \frac{1}{1-z^{-1}} = \frac{z}{z-1}$	$ z > 1$
(4) $x(n) = a^n u(n)$	$X(z) = \frac{1}{1-a z^{-1}} = \frac{z}{z-a}$	$ z > a$
(5) $x(n) = n u(n)$	$X(z) = \frac{z^{-1}}{(1-a z^{-1})^2} = \frac{z}{(z-1)^2}$	$ z > a$
(6) $x(n) = a^n n u(n)$	$X(z) = \frac{a z^{-1}}{(1-a z^{-1})^2} = \frac{za}{(z-a)^2}$	$ z > a$
(7) $x(n) = \cos \Omega_0 n u(n)$	$X(z) = \frac{1 - z^{-1} \cos \Omega_0}{1 - 2 z^{-1} \cos \Omega_0 + z^{-2}}$	$ z > 1$
(8) $x(n) = \sin \Omega_0 n u(n)$	$X(z) = \frac{z^{-1} \sin \Omega_0}{1 - 2 z^{-1} \cos \Omega_0 + z^{-2}}$	$ z > 1$
(9) $x(n) = a^n \cos \Omega_0 n u(n)$	$X(z) = \frac{1 - a z^{-1} \cos \Omega_0}{1 - 2 a z^{-1} \cos \Omega_0 + a^2 z^{-2}}$	$ z > a$
(10) $x(n) = a^n \sin \Omega_0 n u(n)$	$X(z) = \frac{a z^{-1} \sin \Omega_0}{1 - 2 a z^{-1} \cos \Omega_0 + a^2 z^{-2}}$	$ z > a$
(11) $x(n) = -u(-n-1)$	$X(z) = \frac{1}{1-z^{-1}} = \frac{z}{z-1}$	$ z < 1$
(12) $x(n) = -a^n u(-n-1)$	$X(z) = \frac{1}{1-a z^{-1}} = \frac{z}{z-a}$	$ z < a$
(13) $x(n) = -n u(-n-1)$	$X(z) = \frac{z}{(z-1)^2} = \frac{+z^{-1}}{(1-z^{-1})^2}$	$ z < 1$
(14) $x(n) = -a^n n u(-n-1)$	$X(z) = \frac{a z^{-1}}{(1-a z^{-1})^2} = \frac{za}{(z-a)^2}$	$ z < a$
(15) $x(n-k)$	$= z^{-k} X(z)$	
(16) $x(n) \oplus y(n)$	$= X(z) \cdot Y(z)$	
(17) $n x(n)$	$= -z \frac{d}{dz} (X(z))$	

(a) $X(z)$

Express

 $X(z) =$

$$\frac{z^{-1}}{4} =$$

put z^{-1} put $z^{-1} =$ $\therefore X(z) =$

Taking

$$x(n) =$$

(b) $X(z)$ $X(z)$

The p.

the num

 $-z^{-2} + 2z^{-1}$

S

where

Find the Inverse ZT of the following

ROC

(a) $X(z) = \frac{z^{-1}/4}{(1 - z^{-1}/2)(1 - z^{-1}/4)}$; $\frac{1}{4} < |z| < \frac{1}{2}$

Entire z plane

Entire z plane except at $z=0$

$|z| > 1$

$|z| > a$

$|z| > a$

$|z| > a$

$|z| > 1$

$|z| > 1$

$|z| > a$

$|z| > a$

$|z| < 1$

$|z| < a$

$|z| < 1$

$|z| < a$

Express $X(z)$ as a PFE

$$X(z) = \frac{A}{(1 - z^{-1}/2)} + \frac{B}{1 - z^{-1}/4}$$

$$\frac{z^{-1}}{4} = A(1 - z^{-1}/4) + B(1 - z^{-1}/2)$$

put $z^{-1} = 4$; $\frac{4}{4} = 1 = A(0) + B(1 - \frac{4}{2})$

$$B = -1$$

put $z^{-1} = 2$; $\frac{2}{4} = A(1 - \frac{2}{4}) + 0 \Rightarrow A = +1$

$$\therefore X(z) = \frac{1}{1 - z^{-1}/2} + \frac{1}{1 - z^{-1}/4}$$

Taking IZT on both sides

$$x(n) = -\left(\frac{1}{2}\right)^n u(-n-1) - \left(\frac{1}{4}\right)^n u(n)$$

Since ROC is

$$\left(\frac{1}{4}\right) < |z| < \frac{1}{2}$$

⑤ $X(z) = \frac{16z^2 - 2z + 1}{8z^2 + 2z - 1}$ with $|z| > \frac{1}{2}$

$$X(z) = \frac{z^2(16 - 2z^{-1} + z^{-2})}{z^2(8 + 2z^{-1} - z^{-2})} = \frac{16 - 2z^{-1} + z^{-2}}{8 + 2z^{-1} - z^{-2}}$$

The powers of N_r & D_r are same, Hence normalize the numerator

$$\begin{array}{r} -z^2 + 2z^{-1} + 8 \quad \cancel{z^2} - 2\cancel{z^{-1}} + 16 \quad (-1) \\ \underline{-z^2 + 2z^{-1} + 8} \\ 24 \end{array}$$

Sub

24

$$\therefore X(z) = -1 + \frac{24}{8 + 2z^{-1} - z^{-2}} = -1 + p(z)$$

where $p(z) = \frac{24}{8 + 2z^{-1} - z^{-2}} = \frac{24}{(4 + z^{-1})(2 + z^{-1})}$

$$P(z) = \frac{A}{(4-z^{-1})} + \frac{B}{(2+z^{-1})}$$

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$$\therefore 24 = A(2+z^{-1}) + B(4-z^{-1})$$

$$\text{put } z^{-1} = -2; \quad 24 = A(0) + B(4+2)$$

$$\Rightarrow \boxed{B=4}$$

$$\text{put } z^{-1} = 4; \quad 24 = A(2+4) \neq 0$$

$$\Rightarrow \boxed{A=4}$$

$$\therefore P(z) = \frac{4}{4-z^{-1}} + \frac{4}{2+z^{-1}}$$

$$= \frac{1}{1-z^{-1}/4} + \frac{2}{1+z^{-1}/2}$$

$$\therefore X(z) = -1 + \frac{1}{1-z^{-1}/4} + \frac{2}{1+z^{-1}/2}$$

Taking IZT on both sides.

$$\boxed{x(n) = -\delta(n) + \left(\frac{1}{4}\right)^n u(n) + 2\left(\frac{-1}{2}\right)^n u(n)}$$

$$\textcircled{3} \quad X(z) = \frac{2z^3 + 2z^2 + 3z + 1}{2z^4 + 3z^3 + z^2} \quad \text{with } |z| > 1$$

$$X(z) = \frac{z^3(2 + 2z^{-1} + 3z^{-2} + z^{-3})}{z^4(2 + 3z^{-1} + z^{-2})} = z^{-1} \left[\frac{2 + 2z^{-1} + 3z^{-2} + z^{-3}}{2 + 3z^{-1} + z^{-2}} \right]$$

Normalizing the denominator numerator

$$\frac{z^{-2} + 3z^{-1} + 2}{z^3 + 3z^2 + 2z} \cdot \frac{z^3}{z^3} = \frac{z^{-2} + 3z^{-1} + 2}{z^3 + 3z^2 + 2z}$$

$$X(z) = z^{-1} \left[z^{-1} + \frac{2}{z^2 + 3z^{-1} + 2} \right] = z^{-1} \left[z^{-1} + P(z) \right]$$

$$\text{KAT } P(z) = \frac{2}{z^2 + 3z^{-1} + 2} = \frac{2}{(z^{-1} + 1)(2 + z^{-1})}$$

$\therefore 2:$

put z^{-1}

put z^{-1}

$\therefore X(z) =$

Taking

$x(n) =$

$$\boxed{x(n) =}$$

$\textcircled{4} \quad X(z) =$

$X(z) =$

powers e

$-6z$

$\therefore X(z) =$

where
 $P(z) =$

$$P(z) = \frac{A}{1+z^{-1}} + \frac{B}{2+z^{-1}}$$

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$$\therefore 2 = A(2+z^{-1}) + B(1+z^{-1})$$

$$\text{put } z^{-1} = -2; \quad 2 = B(-1) \Rightarrow \boxed{B = -2}$$

$$\text{put } z^{-1} = -1; \quad 2 = A(2-1) \Rightarrow \boxed{A = 2}$$

$$\therefore P(z) = \frac{2}{1+z^{-1}} - \frac{2}{2+z^{-1}} = \frac{2}{1+z^{-1}} - \frac{1}{1+\frac{z^{-1}}{2}}$$

$$\therefore X(z) = z^{-1} \left[z^{-1} + \frac{2}{1+z^{-1}} - \frac{1}{1+\frac{z^{-1}}{2}} \right]$$

Taking IZT;

$$x(n) = \delta(n-1) \left[\delta(n-1) + 2(-1)^n u(n) - \left(\frac{1}{2}\right)^n u(n) \right]$$

$$\boxed{x(n) = \delta(n-2) + 2(-1)^{n-1} u(n-1) - \left(\frac{1}{2}\right)^{n-1} u(n-1)}$$

$$\textcircled{4} \quad X(z) = \frac{4 - 3z^{-1} + 3z^{-2}}{(1+2z^{-1})(1-3z^{-1})} \quad \text{with ROC } |z| > 3$$

$$X(z) = \frac{4 - 3z^{-1} + 3z^{-2}}{1 - z^{-1} - 6z^{-2}}$$

Powers of Nr & Dr are same; normalize Nr.

$$\frac{2+2z^{-1}+3z^{-2}+z^{-3}}{2+3z^{-1}+z^{-2}}$$

$$\begin{array}{r} -6z^{-2} - z^{-1} + 1 \quad \left| \begin{array}{l} 3z^{-2} - 3z^{-1} + 4\left(-\frac{1}{2}\right) \\ 3z^{-1} + \frac{z^{-1}}{2} + \frac{1}{2} \\ (-)\frac{7}{2}z^{-1} + \frac{9}{2} \end{array} \right. \\ \hline \end{array}$$

$$\therefore X(z) = -\frac{1}{2} - \left[\frac{\frac{7}{2}z^{-1} - 9/2}{1 - z^{-1} - 6z^{-2}} \right] = -\frac{1}{2} + P(z)$$

where

$$P(z) = \frac{-(7/2)z^{-1} + 9/2}{1 - z^{-1} - 6z^{-2}} = \frac{-(7/2)z^{-1} + 9/2}{(1+2z^{-1})(1-3z^{-1})}$$

$$z^{-1} + P(z)$$

$$1+z^{-1}$$

$$p(z) = \frac{A}{1+2z^{-1}} + \frac{B}{1-3z^{-1}}$$

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$$x(z) =$$

$$A = \frac{(-7/2)z^{-1} + 9/2}{1-3z^{-1}} \Big|_{z^{-1} = -\frac{1}{2}} = \frac{\frac{7}{4} + \frac{9}{2}}{1 + 3/2} = \underline{\underline{5/2}}$$

$$B = \frac{-\frac{7}{2}z^{-1} + 9/2}{1+2z^{-1}} \Big|_{z^{-1} = \frac{1}{3}} = \frac{-\frac{7}{2}(\frac{1}{3}) + \frac{9}{2}}{1 + 2(\frac{1}{3})} = \underline{\underline{2}}$$

$$\therefore p(z) = \frac{5/2}{1+2z^{-1}} + \frac{2}{1-3z^{-1}}$$

$$x(z) = -\frac{1}{2} - \frac{5/2}{1+2z^{-1}} - \frac{2}{1-3z^{-1}}$$

Taking IZT

$$x(n) = -\frac{1}{2}\delta(n) + \frac{5}{2}\left(-\frac{1}{2}\right)^n u(n) + 2\left(\frac{1}{3}\right)^n u(n)$$

$$\textcircled{5} \quad x(z) = \frac{1 - z^{-1} + z^{-2}}{(1 - \frac{z^{-1}}{2})(1 - 2z^{-1})(1 - z^{-1})}$$

$$\text{ROC } 1 < |z| < 2$$

$$x(z) = \frac{A}{1 - z^{-1}/2} + \frac{B}{1 - 2z^{-1}} + \frac{C}{1 - z^{-1}}$$

$$1 - z^{-1} + z^{-2} = A(1 - 2z^{-1})(1 - z^{-1}) + B(1 - \frac{z^{-1}}{2})(1 - z^{-1}) + C(1 - \frac{z^{-1}}{2})(1 - 2z^{-1})$$

$$\text{put } z^{-1} = 1;$$

$$1 - 1 + 1^2 = A(0) + B(0) + C[1 - \frac{1}{2}][1 - 2]$$

$$\Rightarrow \boxed{C = -2}$$

$$\text{put } z^{-1} = 2;$$

$$1 - 2 + 4 = A(1 - 4)(1 - 2) + B(0) + C(0)$$

$$\boxed{A = 1}$$

$$\text{put } z^{-1} = \frac{1}{2}; \quad 1 - \frac{1}{2} + \frac{1}{4} = A(0) + B(1 - \frac{1}{4})(1 - \frac{1}{2}) + C(0)$$

$$\Rightarrow \boxed{B = 2}$$

Taking

$$x(n) =$$

$$x(n) =$$

poles

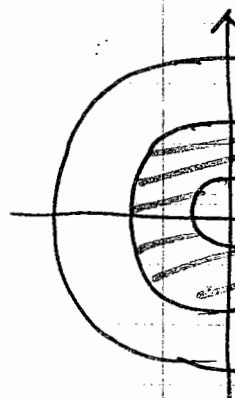
⑥ find

$$x(z) =$$

$$x(n) =$$

⑦ find

$$x(n) =$$



ROC

$$X(z) = \frac{1}{1-\frac{z^{-1}}{2}} + \frac{2}{1-2z^{-1}} - \frac{2}{1-z^{-1}}$$

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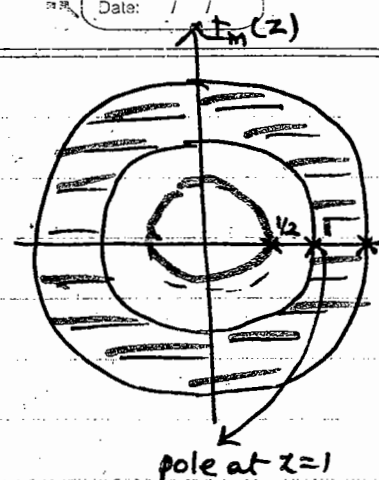
Taking IZT

$$x(n) = \left(\frac{1}{2}\right)^n u(n) + 2(2)^n u(-n-1) - 2u(n)$$

$$x(n) = \left(\frac{1}{2}\right)^n u(n) - (2)^{n+1} u(-n-1) - 2u(n)$$

poles are at $z = 1/2, 1$ and 2

$$\therefore \text{Roc is } \left[\frac{1}{2} < |z| < 2 \right]$$



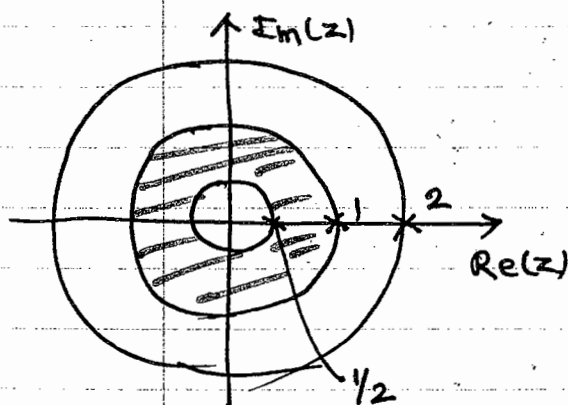
⑥ find $x(n)$ if Roc is $\frac{1}{2} < |z| < 1$.

$$X(z) = \frac{1}{1-\frac{z^{-1}}{2}} + \frac{2}{1-2z^{-1}} - \frac{2}{1-z^{-1}}$$

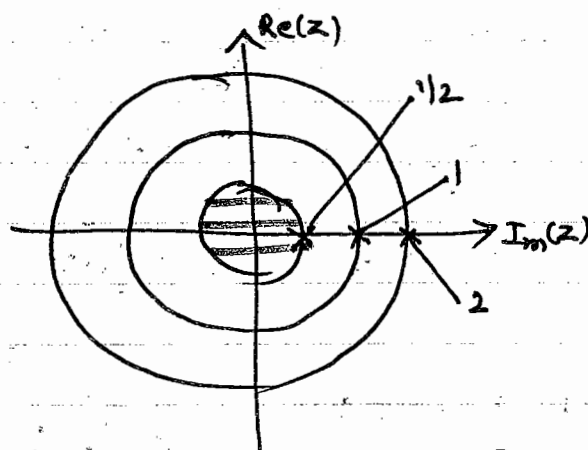
$$x(n) = \left(\frac{1}{2}\right)^n u(n) - (2)^{n+1} u(-n-1) + 2u(-n-1)$$

⑦ find Roc if Roc is $|z| < 1/2$.

$$x(n) = -\left(\frac{1}{2}\right)^n u(-n-1) - (2)^{n+1} u(-n-1) + 2(u(-n-1))$$



Roc of problem ⑥



Roc of problem ⑦

⑧ Find $x(n)$ if $X(z) = \frac{1}{1 - z^{-1}/2} + \frac{2}{1 - 2z^{-1}}$

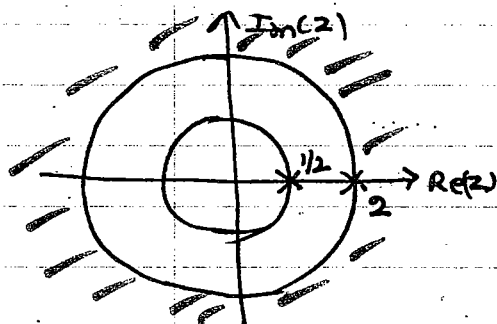
(a) Assuming the signal is causal.

$$x(n) = \left(\frac{1}{2}\right)^n u(n) + (2)^{n+1} u(n)$$

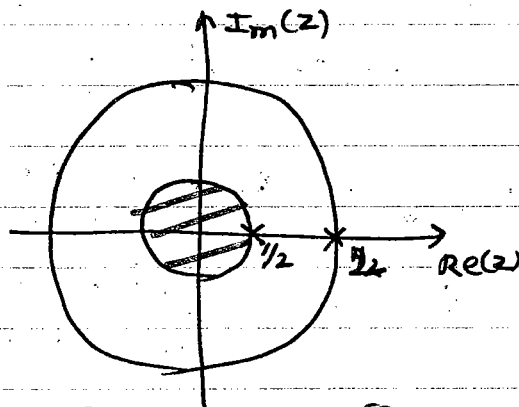
(b) Assuming the signal has DTFT

Since the signal has DTFT; it must be left handed

$$\text{Hence } x(n) = -\left(\frac{1}{2}\right)^n u(-n-1) - (2)^{n+1} u(-n-1)$$



ROC for part (a)
Causal sequence



ROC for part (b)
Signal has DTFT

⑨ $X(z) = \frac{1 + \frac{7}{6}z^{-1}}{\left(1 - \frac{z^{-1}}{2}\right)\left(1 + \frac{z^{-1}}{3}\right)}$ with $|z| > 1/2$

$$X(z) = \frac{A}{1 - z^{-1}/2} + \frac{B}{1 + z^{-1}/3}$$

$$1 + \frac{7}{6}z^{-1} = A\left(1 + \frac{z^{-1}}{3}\right) + B\left(1 - \frac{z^{-1}}{2}\right)$$

put $z^{-1} = -3$; $1 + \frac{7}{6}(-3) = 0 + B\left(1 + \frac{-3}{2}\right)$

$$B = -1$$

put $z^{-1} = 2$; $1 + \frac{7}{6}(2) = A\left(1 + \frac{2}{3}\right) + B(0)$

$$A = 2$$

$$X(z) = \frac{2}{1 - \frac{z^{-1}}{2}} + \frac{-1}{1 + \frac{z^{-1}}{3}}$$

⑩ Find

$$x(n)$$

⑪ Find

$$x(n)$$

$$x(n)$$

⑫ $X(z)$

$$X(z) =$$

$$X(z) =$$

$$1 - 3z^{-1} =$$

$$\text{put } z^{-1} =$$

$$\text{put } z^{-1} =$$

$$X(z)$$

Taking

$$x(n) =$$

$$x(n) = 2\left(\frac{1}{2}\right)^n u(n) - \left(\frac{1}{3}\right)^n u(n)$$

(10) Find $x(n]$ if ROC is $|z| < 1/3$

$$x(n) = -2\left(\frac{1}{2}\right)^n u(-n-1) + \left(\frac{1}{3}\right)^n u(-n-1)$$

e left handed

(11) Find $x(n]$ if ROC is $\frac{1}{3} < |z| < \frac{1}{2}$

$$x(n) = -2\left(\frac{1}{2}\right)^n u(n+1) - \left(\frac{1}{3}\right)^n u(n)$$

$$x(n) = -2\left(\frac{1}{2}\right)^n u(-n-1) - \left(\frac{1}{3}\right)^n u(n)$$

(12) $X(z) = \frac{z^2 - 3z}{z^2 + 3z - 1}$; $\frac{1}{2} < |z| < 2$

$$X(z) = \frac{z^2(1 - 3z^{-1})}{z^2(1 + 3\frac{z^{-1}}{2} - z^{-2})} = \frac{1 - 3z^{-1}}{1 + 3\frac{z^{-1}}{2} - z^{-2}}$$

$$X(z) = \frac{1 - 3z^{-1}}{(1 - \frac{z^{-1}}{2})(1 + 2z^{-1})} = \frac{A}{(1 - \frac{z^{-1}}{2})} + \frac{B}{(1 + 2z^{-1})}$$

$$1 - 3z^{-1} = A(1 + 2z^{-1}) + B(1 - \frac{z^{-1}}{2})$$

put $z^{-1} = 2$; $1 - 3(2) = A(1 + 4) + B(0)$

$$\Rightarrow \boxed{A = -1}$$

put $z^{-1} = -\frac{1}{2}$; $1 + \frac{3}{2} = A(0) + B(1 + \frac{1}{4})$

$$\boxed{B = 2}$$

$$X(z) = \frac{-1}{1 - z^{-1}/2} + \frac{2}{1 + 2z^{-1}}$$

Taking IZT

$$x(n) = -\left(\frac{1}{2}\right)^n u(n) - 2(-2)^n u(-n-1)$$

(13)

$$X(z) = \frac{3z^2 - \frac{z}{2}}{z^2 - 16}; |z| > 4$$

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$$X(z) = \frac{3 - \frac{z^{-1}}{2}}{1 - 16z^{-2}} = \frac{A}{(1 - 4z^{-1})} + \frac{B}{1 + 4z^{-1}}$$

$$A = \frac{1}{2}$$

$$A = \left. \frac{3 - \frac{z^{-1}}{2}}{1 + 4z^{-1}} \right|_{z^{-1} = \frac{1}{4}} = \frac{3 - \frac{1}{8}}{1 + 4(\frac{1}{4})} = \frac{23}{16}$$

$$B = \frac{1}{2}$$

$$B = \left. \frac{3 - \frac{z^{-1}}{2}}{1 - 4z^{-1}} \right|_{z^{-1} = -\frac{1}{4}} = \frac{3 + \frac{1}{8}}{1 + 4/4} = \frac{25}{16}$$

$$X(z) = \frac{23/16}{1 - 4z^{-1}} + \frac{25/16}{1 + 4z^{-1}}$$

Taking IZT

$$x(n) = \frac{23}{16} (4)^n u(n) + (-4)^n \frac{25}{16} u(n)$$

$$x(n) = \left[23 (4)^{n-2} + (-4)^{n-2} \frac{25}{16} \right] u(n)$$

$$(14) X(z) = \frac{2z^3 + z^2 + \frac{3}{2}z + \frac{1}{2}}{z^3 + \frac{3}{2}z^2 + \frac{z}{2}}; |z| < 1/2$$

$$X(z) = \frac{z^3 \left[1 + z^{-1} + \frac{3}{2}z^{-2} + \frac{1}{2}z^{-3} \right]}{z^3 \left(1 + \frac{3}{2}z^{-1} + \frac{z^{-2}}{2} \right)}$$

Normalize the numerator.

$$\left(\frac{z^{-3}}{2} + \frac{3}{2}z^{-1} + 1 \right) \frac{\frac{1}{2}z^{-3} + \frac{3}{2}z^{-2} + \frac{1}{2}z^{-1} + 1}{\frac{1}{2}z^{-3} + \frac{3}{2}z^{-1} + \frac{z^{-2}}{2}} \left(z^{-1} \right)$$

(sub)

1

$$\therefore X(z) = z^{-1} + \frac{1}{z^2 + 3z^{-1} + 2} = z^{-1} + 2P(z)$$

$$(15) X(z)$$

$$X(z) =$$

$$X(z) =$$

$$P(z) =$$

$$W(z) =$$

$$A =$$

$$W(z) =$$

$$X(z) =$$

$$= 2$$

$$P(z) = \frac{1}{z^2 + 3z + 2} = \frac{A}{(z+1)} + \frac{B}{(z+2)}$$

$$\frac{B}{1+4z^{-1}}$$

$$A = \frac{1}{z+2} \Big|_{z=-1} = 1$$

$$B = \frac{1}{z+1} \Big|_{z=-2} = -1$$

$$X(z) = z^{-1} + \frac{2}{z^{-1}+1} - \frac{2}{2+2z^{-1}}$$

Taking IZT

$$x(n) = \delta(n-1) + 2(-1)^n u(n) - 2(-\frac{1}{2})^n u(n)$$

$$x(n) = \delta(n-1) + 2(-1)^n u(n-1) + (-\frac{1}{2})^n u(n-1)$$

$$x(n) = \delta(n-1) - 2(-1)^n u(n-1) + (-\frac{1}{2})^n u(n-1)$$

$$(15) \quad X(z) = \frac{2z^4 - 2z^3 - 2z^2}{z^2 - 1}$$

with $|z| > 1$

$$X(z) = \frac{2z^4(1 - z^{-1} - z^{-2})}{z^2(1 - z^{-2})} = 2z^2 \left[\frac{1 - z^{-1} - z^{-2}}{1 - z^{-2}} \right]$$

$|z| < 1/2$

$$X(z) = 2z^2 \cdot P(z)$$

$$P(z) = \frac{1 - z^{-1} - z^{-2}}{1 - z^{-2}} = 1 - \frac{z^{-1}}{1 - z^{-2}} = 1 - W(z)$$

$$W(z) = \frac{z^{-1}}{1 - z^{-2}} = \frac{A}{1 - z^{-1}} + \frac{B}{1 + z^{-1}}$$

$$A = \frac{z^{-1}}{1 + z^{-1}} \Big|_{z^{-1}=1} = \frac{1}{2}; \quad B = \frac{z^{-1}}{1 - z^{-1}} \Big|_{z^{-1}=-1} = -\frac{1}{2}$$

$$W(z) = \frac{1/2}{1 - z^{-1}} - \frac{1/2}{1 + z^{-1}}$$

$$X(z) = 2z^2 \left[1 - \frac{1/2}{1 - z^{-1}} + \frac{1/2}{1 + z^{-1}} \right] \quad \text{Taking IZT}$$

$$z^{-1} + 2P(z)$$

$$= 2\delta(n+2) * \left[\delta(n) - \frac{1}{2}u(n) + (-1)^n \frac{1}{2}u(n) \right]$$

$$x(n) = 2\delta(n+2) - u(n+2) + (-1)^{n+2}u(n+2)$$

(16) $X(z) = \frac{z^3 - 10z^2 - 4z + 4}{2z^2 - 2z - 4}$ with $|z| < 1$

$$X(z) = \frac{z^3}{2z^2} \left[\frac{1 - 10z^{-1} - 4z^{-2} + 4z^{-3}}{1 - z^{-1} - 2z^{-2}} \right]$$

$$= \frac{z}{2} [P(z)]$$

$$P(z) = \frac{4z^3 - 4z^2 - 10z + 1}{-2z^2 - z + 1}$$

Normalize $P(z)$ to apply PFE

$$\begin{array}{r} -2z^2 - z + 1 \overline{) 4z^3 - 4z^2 - 10z + 1} \\ \underline{4z^3 + 2z^2 + 2z} \\ -6z^2 - 8z + 1 \\ \underline{-6z^2 - 3z + 3} \\ -5z + 2 \end{array}$$

$$P(z) = 2z + 3 + \left[\frac{5z + 2}{-2z - z + 1} \right] = -2z + 3 - W(z)$$

$$W(z) = \frac{5z + 2}{-2z - z + 1} = \frac{A}{(1+z)} + \frac{B}{(1-2z)}$$

$$A = \left. \frac{5z + 2}{1 - 2z} \right|_{z=-1} = -1$$

$$B = \left. \frac{5z + 2}{1 - 2z} \right|_{z=\frac{1}{2}} = 3$$

$$X(z) = \frac{z}{2} \left[-2z + 3 + \frac{1}{1+z} - \frac{3}{1-2z} \right]$$

Taking IZT.

$$x(n) = \frac{\delta(n+1)}{2} * \left[-2\delta(n-1) + 3\delta(n) - (-1)^n u(-n-1) + 3(2)^n u(-n-1) \right]$$

$$x(n) = -\delta(n)$$

PFE

(1) $X(z) =$

$$X(z) =$$

$$4 - 3z^{-1} + 3$$

put $z^{-1} = -\frac{1}{2}$

put $z^{-1} = \frac{1}{3}$

put $z^{-1} =$

$$X(z) = \frac{1}{1}$$

Taking :

$$x(n)$$

(2) $X(z) =$

$$X(z) =$$

$$3 - 2z^{-1}$$

$$n+2)$$

$$= -\delta(n) + \frac{3}{2} \delta(n+1) - \frac{(-1)^{n+1}}{2} u(-n-2) + \frac{3}{2} (2)^{n+1} u(-n-2)$$

$$K |z| < 1$$

$$x(n) = -\delta(n) + \frac{3}{2} \delta(n+1) - \frac{1}{2} (-1)^n u(-n-2) + 3(2)^n u(-n-2)$$

PFE with multiple repeated poles

$$(1) X(z) = \frac{4 - 3z^{-1} + 3z^{-2}}{(1+2z^{-1})(1-3z^{-1})^2} \quad \text{with ROC } |z| > 3.$$

$$X(z) = \frac{A}{1+2z^{-1}} + \frac{B}{1-3z^{-1}} + \frac{C}{(1-3z^{-1})^2}$$

$$4 - 3z^{-1} + 3z^{-2} = A(1-3z^{-1})^2 + B(1-3z^{-1})(1+2z^{-1}) + C(1+2z^{-1})$$

$$\text{put } z^{-1} = -\frac{1}{2}; \quad 4 + \frac{3}{2} + \frac{3}{4} = \left(1 + \frac{3}{2}\right)^2 A + B(0) + C(0)$$

$$\Rightarrow \boxed{A = 1}$$

$$\text{put } z^{-1} = \frac{1}{3}; \quad 4 + 1 + \frac{3}{9} = A(0) + B(0) + C\left(1 + \frac{2}{3}\right)$$

$$\boxed{C = 2}$$

$$+3 - W(z)$$

$$\text{put } z^{-1} = 0; \quad 4 = A + B + C \Rightarrow \boxed{B = 1}$$

$$X(z) = \frac{1}{1+2z^{-1}} + \frac{1}{1-3z^{-1}} + \frac{2}{(1-3z^{-1})^2} = \frac{1}{1+2z^{-1}} + \frac{1}{1-3z^{-1}} + \frac{2z \cdot 3z^{-1}}{(1-3z^{-1})^2}$$

$$\text{Taking IZT } x(n) = (-2)^n u(n) + (3)^n u(n) + 2\delta(n+1) * n(3)^n u(n)$$

$$x(n) = (-2)^n u(n) + (3)^n u(n) + \frac{2}{3} (n+1) (3)^{n+1} u(n+1)$$

$$(2) X(z) = \frac{3 - 2z^{-1} + z^{-2}}{(1-z^{-1})(1-\frac{1}{3}z^{-1})^2} = \quad |z| > 1$$

$$X(z) = \frac{A}{1-z^{-1}} + \frac{B}{1-\frac{z^{-1}}{3}} + \frac{C}{(1-\frac{z^{-1}}{3})^2}$$

$$3 - 2z^{-1} + z^{-2} = A(1-\frac{z^{-1}}{3})^2 + B(1-z^{-1})(1-\frac{z^{-1}}{3}) + C(1+z^{-1})$$

$$(-n-1) + u(-n-1]$$

put $\bar{z} = 1$; $A = 9/2$

put $\bar{z} = 3$; $C = -3$

put $\bar{z} = 0$; $B = 3/2$

FO

$$X(z) = \frac{9/2}{1-\bar{z}} + \frac{3/2}{1-\bar{z}/3} - \frac{3}{(1-\bar{z}/3)^2} = \frac{9/2}{1-\bar{z}} + \frac{3/2}{1-\bar{z}/3} - \frac{3\bar{z}/3 \cdot \bar{z}/3}{(1-\bar{z}/3)^2}$$

$x(n) = (-2)^n$

(4) Fmd

Taking IZT on both sides

$$x(n) = \frac{9}{2} u(n) + \frac{3}{2} \left(\frac{1}{3}\right)^n - \left[n \left(\frac{1}{3}\right)^n * \frac{1}{9} \delta(n+1)\right]$$

$$x(n) = \frac{9}{2} u(n) + \frac{3}{2} \left(\frac{1}{3}\right)^n - \frac{1}{9} (n+1) \left(\frac{1}{3}\right)^{n+1} u(n+1)$$

(3) $X(z) = \frac{4-3\bar{z}^1+3\bar{z}^2}{(z+2)(z-3)^2}$

$$X(z) = \bar{z}^3 \left[\frac{4-3\bar{z}^1+3\bar{z}^2}{(1+2\bar{z}^1)(1-3\bar{z}^1)^2} \right] = \bar{z}^3 P(z)$$

$$P(z) = \frac{4-3\bar{z}^1+3\bar{z}^2}{(1+2\bar{z}^1)(1-3\bar{z}^1)^2} = \frac{A}{1+2\bar{z}^1} + \frac{B}{1-3\bar{z}^1} + \frac{C}{(1-3\bar{z}^1)^2}$$

$$4-3\bar{z}^1+3\bar{z}^2 = A(1-3\bar{z}^1)^2 + B(1+2\bar{z}^1)(1-3\bar{z}^1) + C(1+2\bar{z}^1)$$

put $\bar{z}^1 = -\frac{1}{2}$; $A = 1$; put $\bar{z}^1 = \frac{1}{3}$; $C = 2$; put $\bar{z}^1 = 0$; $B = 1$

$$X(z) = \bar{z}^3 \left[\frac{1}{1+2\bar{z}^1} + \frac{1}{1-3\bar{z}^1} + \frac{2}{(1-3\bar{z}^1)^2} \right]$$

$$= \bar{z}^3 \left[\frac{1}{1+2\bar{z}^1} + \frac{1}{1-3\bar{z}^1} + \frac{2}{3} \bar{z} \cdot \frac{3\bar{z}^1}{(1-3\bar{z}^1)^2} \right]$$

taking IZT for ROC $|z| > 3$

$$x(n) = \delta(n-3) * \left[(-2)^n u(n) + (3)^n u(n) + \left[\frac{2}{3} \delta(n+1) * n(3)^n u(n) \right] \right]$$

$$x(n) = (-2)^{n-3} u(n-3) + (3)^{n-3} u(n-3) + \frac{2}{3} (n-2) (3)^{n-2} u(n-2)$$

For $|z| < 2$

$$x(n) = -(-2)^{n-3} u(-n+2) - (3)^{n-3} u(-n+2) - \frac{2}{3} (n-2) 3^{n-2} u(-n+1)$$

$\frac{X(z)}{z}$

$z^2 - 4z + 1$

put $z = 1$

put $z =$

$\frac{X(z)}{z}$

$X(z) =$

For ROC

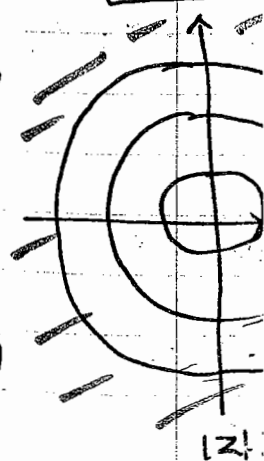
$x(n) =$

For ROC

$x(n) =$

For ROC

$x(n) =$



For ROC $2 < |z| < 3$

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$$x(n) = (-2)^{n-3} u(n-3) - (3)^{n-3} u(-n+2) - \frac{2}{3} (n-2) 3^{n-2} u(-n+1)$$

$$- 3 \frac{2}{3} \frac{z(3)}{(1-\frac{2}{3})^2}$$

(4) Find $x(n)$ for $x(z) = \frac{z(z^2 - 4z + 5)}{(z-3)(z-2)(z-1)}$ for $|z| > 3$
 $|z| < 1$
 $2 < |z| < 3$

$$\frac{x(z)}{z} = \frac{z^2 - 4z + 5}{(z-3)(z-2)(z-1)} = \frac{A}{z-3} + \frac{B}{z-2} + \frac{C}{z-1}$$

$$z^2 - 4z + 5 = A(z-2)(z-1) + B(z-3)(z-1) + C(z-3)(z-2)$$

put $z=1$; $1 - 4 + 5 = 0 + 0 + C[-2][-1] \Rightarrow \boxed{C=1}$

put $z=3$; $\boxed{A=1}$ put $z=2$; $\boxed{B=-1}$

$$\frac{x(z)}{z} = \frac{1}{z-3} - \frac{1}{z-2} + \frac{1}{z-1}$$

$$x(z) = \frac{z}{z-3} - \frac{z}{z-2} + \frac{z}{z-1}$$

For ROC $|z| > 3$

$$x(n) = \left[\left(\frac{1}{3}\right)^n - \left(\frac{1}{2}\right)^n + 1 \right] u(n)$$

For ROC $|z| < 1$;

$$x(n) = -\left(\frac{1}{3}\right)^n u(-n-1) + \left(\frac{1}{2}\right)^n u(-n-1) + u(-n-1)$$

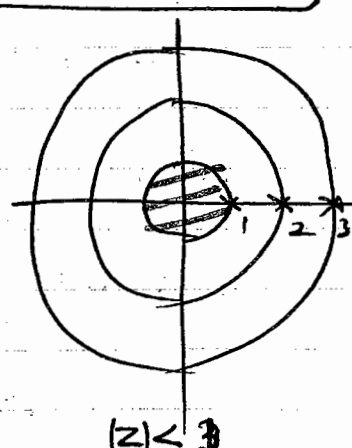
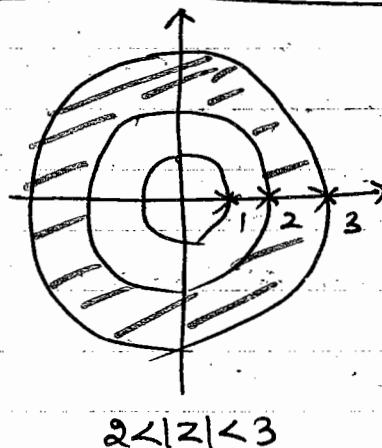
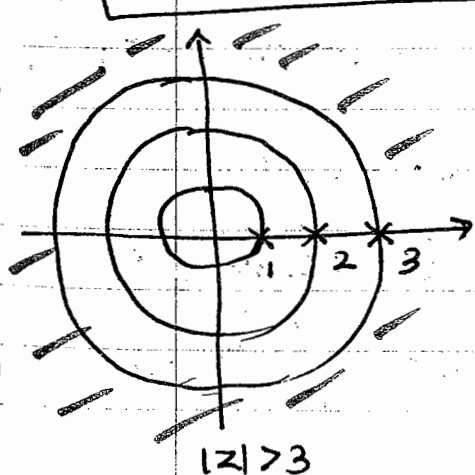
For ROC $2 < |z| < 3$

$$x(n) = -\left(\frac{1}{3}\right)^n - (3)^n u(-n-1) + (2)^n u(n) - u(n)$$

$$(n+1) * n \left(\frac{1}{3}\right)^n u(n)$$

$$(n-2)$$

$$(n-2) 3^{n-2} u(n+1)$$



⑤ Find $x(n)$ using convolution property

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$$X(z) = \frac{1}{(1 - \frac{z^{-1}}{2})(1 + \frac{z^{-1}}{4})}$$

Let $X(z) = X_1(z) \cdot X_2(z)$

$$x(n) = x_1(n) * x_2(n)$$

$$= (\frac{1}{2})^n u(n) * (-\frac{1}{4})^n u(n)$$

$$= \sum_{k=-\infty}^{\infty} (\frac{1}{2})^k u(k) (-\frac{1}{4})^{n-k} u(n-k)$$

$$= \sum_{k=0}^n (\frac{1}{2})^k (-\frac{1}{4})^{n-k}$$

since $u(n)u(n-k) = \begin{cases} 1 & 0 \leq k \leq n \\ 0 & \text{otherwise} \end{cases}$

put $(-\frac{1}{4}) = a$

$$x(n) = \sum_{k=0}^n (\frac{1}{2})^k (a)^{n-k} = a^n \sum_{k=0}^n (\frac{1}{2a})^k = a^n \left[\frac{1 - (\frac{1}{2a})^{n+1}}{1 - \frac{1}{2a}} \right]$$

$$= a^n \left[\frac{1 - (\frac{1}{2})^{n+1} (\frac{1}{a})^{n+1}}{1 - \frac{1}{2a}} \right] = \frac{a^n - (\frac{1}{2})^{n+1} \cdot a^n \cdot a^{-n-1}}{1 - \frac{1}{2(-\frac{1}{4})}}$$

$$= \frac{(-\frac{1}{4})^n - (\frac{1}{2})^{n+1} \left[\frac{1}{-1/4} \right]}{1 + 2} = \frac{(-\frac{1}{4})^n + (\frac{1}{2})^{n-1}}{3}$$

$$\therefore x(n) = \frac{(\frac{1}{2})^{n-1} + (-\frac{1}{4})^n}{3}$$

⑥ obtain the sequence $x(n)$ from its transform

$$X(z) = \frac{z^2}{(z-2)(z-3)} \text{ by using convolution property}$$

$$X(z) = \frac{z}{z-2} \cdot \frac{z}{z-3} = X_1(z) \cdot X_2(z)$$

$$x(n) = x_1(n) * x_2(n) = (2)^n u(n) * (3)^n u(n)$$

$$= \sum_{k=-\infty}^{\infty} 2^k u(k) \cdot (3)^{n-k} u(n-k) = \sum_{k=0}^n (\frac{2}{3})^k \cdot (3)^n$$

Ex:- with

only

o. / /

$$= 3^n \left[\frac{1 - \left(\frac{2}{3}\right)^{n+1}}{1 - \frac{2}{3}} \right]$$

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$$= \frac{3^n}{3^{n+1}} \left[\frac{3^{n+1} - 2^{n+1}}{1} \right] \times 3$$

$$x(n) = 3^{n+1} - 2^{n+1}$$

$$(7) \quad X(z) = \frac{1}{1-z^{-2}}; \quad |z| > 1$$

$$= \begin{cases} 1 & ; 0 \leq k \leq n \\ 0 & ; \text{otherwise} \end{cases}$$

$$X(z) = \frac{1}{(1-z^{-1})(1+z^{-1})} = \frac{A}{1-z^{-1}} + \frac{B}{1+z^{-1}}$$

$$A = \frac{1}{1+z^{-1}} \Big|_{z^{-1}=-1} = \frac{1}{2}$$

$$B = \frac{1}{1-z^{-1}} \Big|_{z^{-1}=1} = \frac{1}{2}$$

$$= a^n \left[\frac{1 - \left(\frac{1}{2a}\right)^{n+1}}{1 - \frac{1}{2a}} \right]$$

$$n: a^{-n-1}$$

$$x(n) = \frac{1}{2} u(n) + \frac{1}{2} \left(-\frac{1}{2}\right)^n u(n)$$

$$x(n) = \left[\frac{1}{2} u(n) + \frac{1}{2} (-1)^n \right] u(n)$$

$$+ \left(\frac{1}{2}\right)^{n-1}$$

$$x(n) = \begin{cases} u(n) & ; \text{for } n \text{ even} \\ 0 & ; n \text{ odd} \end{cases}$$

II method

$$X(z) = \frac{1}{1-z^{-2}} \quad |z| > 1$$

its transform
on property

$$\text{we have } \sum_{k=0}^{\infty} a^k = \frac{1}{1-a}$$

$$\text{put } z^{-2} = a \quad \therefore X(z) = \sum_{k=0}^{\infty} z^{2k} = \delta(n-2k)$$

$$\neq (3)^n u(n)$$

$$x(n) = \delta(n-2k)$$

$$\left(\frac{2}{3}\right)^k \cdot (3)^n$$

$$\delta(n-2k) = \begin{cases} 1 \\ 0 \end{cases}$$

when 'n' is
integer multiple of 2
 $k \geq 0$

$$\text{Ex:- when } k=0; \quad x(n) = \delta(n) = 1;$$

When $k=1$, & $n=2$,

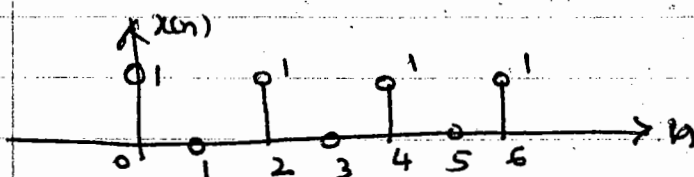
$$x(n) = \delta(n) = 1$$

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When $k=2$, $n=4$;

$$x(n) = \delta(n) = 1$$



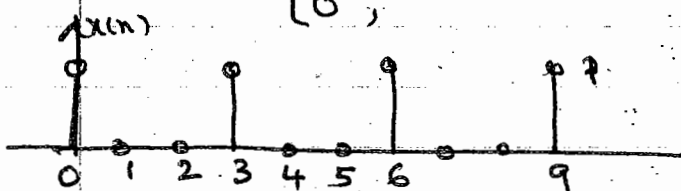
$$(8) \quad X(z) = \frac{1}{1 - z^{-3}}; |z| > 1$$

$$\sum_{k=0}^{\infty} a^k = \frac{1}{1-a}; \text{ put } z^{-3} = a$$

$$\therefore X(z) = \sum_{k=0}^{\infty} (z^{-3})^k =$$

$$x(n) = \delta(n-3k)$$

$$x(n) = \delta(n-3k) = \begin{cases} 1; & \text{when 'n' is integer multiples of 3} \\ 0; & \text{and } n \geq 0 \end{cases}$$



$$x(n) = \delta(n-3k); n \geq 0$$

$$(9) \quad X(z) = \frac{1}{1 - \frac{z^{-2}}{4}}; |z| > \frac{1}{4}$$

$$X(z) = \frac{1}{1 - \left(\frac{z^{-1}}{2}\right)^2} = \sum_{k=0}^{\infty} \left(\frac{z^{-1}}{2}\right)^{2k}$$

$$x(n) = \left(\frac{1}{4}\right)^k \delta(n-2k) = \left(\frac{1}{4}\right)^k \delta(n-2k)$$

$$x(n) = \left(\frac{1}{4}\right)^k \delta(n-2k);$$

for 'n' being the integer multiples of 2 & $n \geq 0$

PO
Z' Trans
Series
 $X(z) = \sum_{n=0}^{\infty}$

By compo
can deter
coeff. of
This inver
sided end
we expres
sided ser
Z' to

(1) Determ

(a) $x(z)$

$$1 - \frac{z^{-1}}{2}$$

$$X(z) =$$

Taking

$$x(n) =$$

$$x(n) =$$

power series expansion method

'Z' Transform of a signal be expressed as a power series

$$X(z) = \sum_{n=-\infty}^{+\infty} x(n) \bar{z}^n = x(-\infty) \bar{z}^{\infty} + \dots + x(-2) \bar{z}^2 + x(-1) \bar{z}^1 + x(0) \bar{z}^0 + x(1) \bar{z}^1 + x(2) \bar{z}^2 + \dots + x(\infty) \bar{z}^{\infty}$$

By comparing with the ZT of the standard signals; we can determine the discrete time sequence $x(n)$ because the coeff. of $\bar{z}^n$ is in the sequence values of $x(n)$.

This inversion method is limited signals that are one sided only. i.e. ROC is of the form $|z| < a$ or $|z| > a$. We express $x(z)$ as a power series in terms of 'z' to get a Left sided sequence & and we express $x(z)$ in terms of $\bar{z}^1$ to get right sided sequence.

① Determine $x(n)$ using power series expn method

Multiplies of 3
& $n \geq 0$

(a) $x(z) = \frac{2 + \bar{z}^1}{1 - \bar{z}^1/2}$ with ROC $|z| > 1/2$

$$1 - \frac{\bar{z}^1}{2} \Bigg) \begin{array}{r} 2 + 2\bar{z}^1 + \bar{z}^2 + 2\bar{z}^3 + \dots \\ \underline{2 + \bar{z}^1} \\ 2\bar{z}^1 \\ \underline{2\bar{z}^1 - \bar{z}^2} \\ \bar{z}^2 \\ \underline{\bar{z}^2 - \frac{\bar{z}^3}{2}} \\ \frac{\bar{z}^3}{2} \\ \underline{\frac{\bar{z}^3}{2} - \frac{\bar{z}^4}{4}} \\ \frac{\bar{z}^4}{4} \\ \dots \end{array}$$

1-2K)

$$X(z) = 2 + 2\bar{z}^1 + \bar{z}^2 + \frac{\bar{z}^3}{2} + \frac{\bar{z}^4}{4} + \dots$$

the integer
of 2 & $n \geq 0$

Taking IZT

$$x(n) = 2\delta(n) + 2\delta(n-1) + \delta(n-2) + \frac{1}{2}\delta(n-3) + \frac{1}{4}\delta(n-4)$$

$$x(n) = \left\{ 2, 2, 1, \frac{1}{2}, \frac{1}{4}, \dots \right\}$$

⑤ Repeat part (a) for $|z| < 1/2$

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③ R

$0.5z^2 - 1$

$$\begin{array}{r} -2 - 8z - 16z^2 - 32z^3 - 64z^4 \\ \underline{-1z + 1} \\ 2 \end{array}$$

$$\begin{array}{r} 4 - 8z \\ \underline{+ 8z} \\ 8z - 16z^2 \end{array}$$

$$\begin{array}{r} 16z^2 \\ \underline{16z^2 - 32z^3} \\ 32z^3 \end{array}$$

$$\begin{array}{r} 32z^3 \\ \underline{32z^3 - 64z^4} \end{array}$$

$\therefore X(z) = -2 - 8z - 16z^2 - 32z^3 - 64z^4$
Taking IZT

$x(n) = -2\delta(n) - 8\delta(n+1) - 16\delta(n+2) - 32\delta(n+3) - \dots$
 $= \{-2, -8, -16, -32, \dots\}$

$X(z) =$

Taking

$x(n)$

$x(n) =$

④ $X(z)$

② $X(z) = \frac{1}{1 - 1.5z^{-1} + 0.5z^{-2}}$ with $|z| > 1$

$1 + 3z^0$

Determine $x(n)$ using Long division method.

$$\begin{array}{r} 1 - 1.5z^{-1} + 0.5z^{-2} \overline{) 1 + 1.5z^{-1} + 1.75z^{-2} + 1.875z^{-3} + \dots} \\ \underline{1 - 1.5z^{-1} + 0.5z^{-2}} \\ 1.5z^{-1} - 0.5z^{-2} \\ \underline{1.5z^{-1} - 2.25z^{-2} + 0.75z^{-3}} \\ 1.75z^{-2} - 0.75z^{-3} \\ \underline{1.75z^{-2} - 2.625z^{-3} + 0.875z^{-4}} \\ 1.875z^{-3} - 0.875z^{-4} \end{array}$$

$\therefore X(z) = 1 + 1.5z^{-1} + 1.75z^{-2} + 1.875z^{-3} + \dots$

$X(z) =$

Taking IZT

$x(n)$

$x(n) = \delta(n) + 1.5\delta(n-1) + 1.75\delta(n-2) + 1.875\delta(n-3) + \dots$

$= \{1, 1.5, 1.75, 1.875, \dots\}$

$x(n) = \delta(n)$

③ Repeat the above problem for $|z| < 1$

$$\begin{array}{r} 2z^2 + 6z^3 + 14z^4 + 30z^5 \\ 0.5z^2 - 1.5z^1 + 1 \overline{) } \\ \underline{-3z + 2z^2} \\ 3z - 2z^2 \\ \underline{3z - 9z^2 + 6z^3} \\ 7z^2 - 6z^3 \\ \underline{7z^2 - 21z^3 + 14z^4} \\ 15z^3 - 14z^4 \\ \underline{15z^3 - 45z^4 + 30z^5} \end{array}$$

$$X(z) = 2z^2 + 6z^3 + 14z^4 + 30z^5 + \dots$$

Taking IZT

$$x(n) = 2\delta(n+2) + 6\delta(n+3) + 14\delta(n+4) + 30\delta(n+5) + \dots$$

$$x(n) = \{2, 6, 14, 30, \dots\}$$

$\delta(n+3) + \dots$

④ $X(z) = \frac{z}{2z^2 - 3z + 1}$ ROC $|z| < 1/2$

$$z + 3z^2 + 7z^3 + 15z^4 + 31z^5 + \dots$$

$$\begin{array}{r} 1 + 3z^2 + 3z^2 \overline{) } z \\ \underline{z + 3z^2 + 3z^3} \\ 3z^2 - 2z^3 \\ \underline{3z^2 - 9z^3 + 6z^4} \\ + 7z^3 - 6z^4 \\ \underline{7z^3 - 21z^4 + 14z^5} \\ 15z^4 - 14z^5 \\ \underline{15z^4 - 45z^5 + 30z^6} \\ 31z^5 - 30z^6 \\ \underline{31z^5 - 93z^6 + 62z^7} \end{array}$$

$$X(z) = z + 3z^2 + 7z^3 + 15z^4 + 31z^5 + \dots$$

$$x(n) = \{1, 3, 7, 15, 31, \dots\}$$

$$x(n) = \delta(n+1) + 3\delta(n+2) + 7\delta(n+3) + 15\delta(n+4) + 31\delta(n+5) + \dots$$

⑤ Find $x(n)$ if $X(z) = \sin z$

from Taylor's series expansion

$$\sin z = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+1)!}$$

$$= z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \frac{z^9}{9!} - \frac{z^{11}}{11!} + \dots \rightarrow \textcircled{A}$$

but $X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} = \dots + z^{-7} x(-7) + z^{-6} x(-6) + z^{-5} x(-5) +$
 $z^{-4} x(-4) + z^{-3} x(-3) + z^{-2} x(-2) + z^{-1} x(-1) +$
 $z(0) x(0) + \dots \rightarrow \textcircled{B}$

Comparing eqs ① & ② equating similar terms

$$x(n) = \left\{ -\frac{1}{11!}, 0, \frac{1}{9!}, 0, -\frac{1}{7!}, 0, \frac{1}{5!}, 0, -\frac{1}{3!}, 0, 1 \right\}$$

⑥ $X(z) = \cos(2z)$ with $|z| < \infty$ find $x(n)$

$$\cos \theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \frac{\theta^8}{8!} - \frac{\theta^{10}}{10!} + \dots$$

$$= \sum_{k=0}^{\infty} \frac{(-1)^k \theta^{2k}}{2k!}$$

Here $\theta = 2z$; $X(z) = \sum_{k=0}^{\infty} \frac{(-1)^k (2z)^{2k}}{2k!} = \sum_{k=0}^{\infty} \frac{(-1)^k 4^k z^{2k}}{2k!}$

$$= \sum_{k=0}^{\infty} \frac{(-4)^k z^{2k}}{2k!} \quad \text{Taking IZT}$$

$$x(n) = \sum_{k=0}^{\infty} \frac{(-4)^k}{2k!} \delta(n+2k)$$

⑦ $X(z) = \cos(\bar{z}^2)$; $|z| > 0$; find $x(n)$

put $\theta = \bar{z}^2$; $X(z) = \sum_{k=0}^{\infty} \frac{(-1)^k (\bar{z}^2)^{4k}}{2k!}$

$$x(n) = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k!} \delta(n-4k)$$

for $\cos(\bar{z}^3)$; $x(n) = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k!} \delta(n-6k)$

(8) $X(z) = \ln(1+z^{-1})$; $|z| > 0$ find $x(n)$

We have $\ln(1+\alpha) = \alpha - \frac{\alpha^2}{2} + \frac{\alpha^3}{3} - \frac{\alpha^4}{4} + \frac{\alpha^5}{5} - \dots$

$$\ln(1+\alpha) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} \alpha^k$$

$$X(z) = \ln(1+z^{-1}) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} z^{-k}$$

$$x(n) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} \cdot \delta(n-k) = \underline{\underline{\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} \delta(n-k)}}$$

(9) $X(z) = \ln(1-2z)$.

We have $\ln(1-\alpha) = -\alpha - \frac{\alpha^2}{2} - \frac{\alpha^3}{3} - \frac{\alpha^4}{4} - \dots$

$$= - \sum_{k=1}^{\infty} \frac{\alpha^k}{k}$$

$$\therefore X(z) = \ln(1-2z) = - \sum_{k=1}^{\infty} \frac{(2z)^k}{k}$$

$$x(n) = - \sum_{k=1}^{\infty} \frac{2^k}{k} \delta(n+k)$$

||| for $X(z) = \ln(1-az^{-1})$ will have

$$x(n) = - \sum_{k=1}^{\infty} \frac{a^k}{k} \delta(n-k)$$

(10) $X(z) = \ln \left[\frac{a}{a-z^{-1}} \right]$ $|z| > a^{-1}$

$$X(z) = \ln \left[\frac{1}{1-(az)^{-1}} \right] = \ln(1) - \ln(1-(az)^{-1})$$

$$X(z) = -\ln(1-(az)^{-1}) = \sum_{k=1}^{\infty} \frac{(az)^{-k}}{k} = \sum_{k=1}^{\infty} \frac{a^{-k}}{k} z^{-k}$$

Taking IZT

$$x(n) = \sum_{k=1}^{\infty} \frac{1}{a^k} \cdot \frac{1}{k} \delta(n-k)$$

Transform Analysis of LTI system

Relationship between transfer fn & diff eq²

Consider a DT LTI system described by the diff eqⁿ

$$\sum_{k=0}^N a_k y(n-k) = \sum_{k=0}^M b_k x(n-k) \quad \text{where } (M < N)$$

Taking ZT on both sides; $\sum_{k=0}^N a_k Y(z) z^{-k} = \sum_{k=0}^M b_k X(z) z^{-k}$

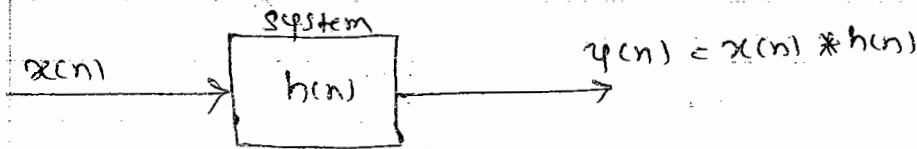
$$\frac{Y(z)}{X(z)} = H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}}$$

where $H(z)$ is called system function or Rational transfer function.

The transfer fn of a LTI system described by a difference eqⁿ is the ratio of polynomials in z^{-k} .

Transfer function and impulse Response

consider an LTI system with impulse response $h(n)$ with $x(n)$ as input and $y(n)$ as output response



$$y(n) = x(n) * h(n)$$

$$\boxed{Y(z) = X(z) \cdot H(z)} \Rightarrow H(z) = \frac{Y(z)}{X(z)}$$

Hence transfer function is the ratio of Z-transforms of output to input. This defⁿ holds good for all values of z in the ROC's $X(z)$ and $Y(z)$ for which $X(z)$ is non zero.

Impulse response is the inverse ZT of the transfer function for which $X(z)$ is non zero. In order to define the impulse response uniquely, we know the correct ROC. Otherwise stability or causality of the system must be known.

$$h(n) = \mathcal{Z}^{-1}[H(z)] = \mathcal{Z}^{-1}\left[\frac{Y(z)}{X(z)}\right]$$

① Let $y(n)$

Taking

$Y(z)$

$$\frac{Y(z)}{X(z)}$$

$$H(z) =$$

$h(n)$

$h(n)$

② $y(n) =$

Taking

$$H(z) =$$

$$h(n) =$$

$H(z)$

Taking

$h(n) =$

$$h(n) =$$

Design a and $y(n) =$

$$X(z) =$$

$$Y(z) =$$

$$H(z) =$$

① Determine the transfer fn & impulse response of the system

$$y(n) - \frac{1}{2} y(n-1) = 2x(n-1)$$

Taking ZT on both sides

$$Y(z) - \frac{1}{2} Y(z) z^{-1} = 2 z^{-1} X(z)$$

$$\frac{Y(z)}{X(z)} = H(z) = \left[\frac{2 z^{-1}}{1 - z^{-1}/2} \right] \quad \text{Transfer fn.}$$

$$H(z) = 2 z^{-1} \cdot \frac{1}{1 - z^{-1}/2} \quad \text{Taking IZT}$$

$$h(n) = 2 \delta(n-1) * \left(\frac{1}{2}\right)^n u(n) = 2 \left(\frac{1}{2}\right)^{n-1} u(n-1)$$

$$h(n) = \left(\frac{1}{2}\right)^{n-2} u(n-1) \quad \text{--- required imp Response}$$

② $y(n) = \frac{8}{5} y(n-1) - \frac{16}{25} y(n-2) = 2x(n) - x(n-1)$

Taking ZT: $Y(z) \left[1 - \frac{8}{5} z^{-1} - \frac{16}{25} z^{-2} \right] = [2 - z^{-1}] X(z)$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{2 - z^{-1}}{1 - \frac{8}{5} z^{-1} - \frac{16}{25} z^{-2}} = \frac{2 - z^{-1}}{\left(1 - \frac{4}{5} z^{-1}\right)^2}$$

$h(n) = \checkmark$ $H(z) = \frac{2 - z^{-1}}{\left(1 - \frac{4}{5} z^{-1}\right)^2}$ is the transfer fn

$$H(z) = \frac{2 \cdot \frac{5}{4} z^{-1} \times \frac{5}{4}}{\left(1 - \frac{4}{5} z^{-1}\right)^2} - \frac{\frac{5}{4} z^{-1} \times \frac{5}{4}}{\left(1 - \frac{4}{5} z^{-1}\right)^2}$$

Taking IZT

$$h(n) = 5 \delta(n+1) * n \left(\frac{4}{5}\right)^n u(n) - \frac{5}{4} n \left(\frac{4}{5}\right)^n u(n)$$

$$h(n) = 5(n+1) \left(\frac{4}{5}\right)^{n+1} u(n+1) - \frac{5}{4} n \left(\frac{4}{5}\right)^n u(n)$$

Design a causal LTI system with $x(n) = \left(\frac{1}{2}\right)^n u(n) - \frac{1}{2} \left(\frac{1}{2}\right)^{n-1} u(n-1)$ and $y(n) = \left(\frac{1}{3}\right)^n u(n)$. Determine $H(z)$ & $h(n)$.

$$X(z) = \frac{1}{1 - z^{-1}/2} - \frac{z^{-1}/4}{1 - z^{-1}/2} = \frac{1 - z^{-1}/4}{1 - z^{-1}/2}$$

$$Y(z) = \frac{1}{1 - z^{-1}/3}$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 - z^{-1}/2}{(1 - z^{-1}/3)(1 - z^{-1}/4)} = \frac{A}{1 - \frac{z^{-1}}{3}} + \frac{B}{1 - \frac{z^{-1}}{4}}$$

↓
transfer fn

$$H(z) = \frac{-2}{1-z^{-1}/3} + \frac{3}{1-z^{-1}/4}$$

taking IZT

$$h(n) = -2 \left(\frac{1}{3}\right)^n u(n) + 3 \left(\frac{1}{4}\right)^n u(n)$$

A DT LTI system is given by $y(n] - \frac{3}{4} y(n-1) + \frac{1}{8} y(n-2) = x(n]$

(a) determine the system function (b) impulse response of the system (c) step response of the system (d) frequency response.

$$y(n] - \frac{3}{4} y(n-1) + \frac{1}{8} y(n-2) = x(n]$$

$$Y(z) \left[1 - \frac{3}{4} z^{-1} + \frac{1}{8} z^{-2} \right] = X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{1 - \frac{3}{4} z^{-1} + \frac{1}{8} z^{-2}} \quad \text{[Tr. f.]} \quad \text{system f.}$$

$$H(z) = \frac{1}{\left(1 - \frac{z^{-1}}{2}\right) \left(1 - \frac{z^{-1}}{4}\right)} = \frac{A}{1 - \frac{z^{-1}}{2}} + \frac{B}{1 - \frac{z^{-1}}{4}}$$

$$H(z) = \frac{2}{1 - \frac{z^{-1}}{2}} - \frac{1}{1 - \frac{z^{-1}}{4}} \quad \text{Taking IZT}$$

$$h(n) = 2 \left(\frac{1}{2}\right)^n u(n) - \left(\frac{1}{4}\right)^n u(n) \quad \text{impulse response}$$

Step response

$$S(z) = H(z) \cdot X(z)$$

$$x(n) = u(n] \\ X(z) = \frac{1}{1-z^{-1}}$$

$$S(z) = \frac{1}{\left(1 - \frac{z^{-1}}{2}\right) \left(1 - \frac{z^{-1}}{4}\right)} \times \frac{1}{1-z^{-1}}$$

$$= \frac{A}{1 - \frac{z^{-1}}{2}} + \frac{B}{1 - \frac{z^{-1}}{4}} + \frac{C}{1 - z^{-1}}$$

$$S(z) = \frac{-2}{1 - \frac{z^{-1}}{2}} + \frac{1/3}{1 - \frac{z^{-1}}{4}} + \frac{8/3}{1 - z^{-1}}$$

$$s(n) = -2 \left(\frac{1}{2}\right)^n u(n) + \frac{1}{3} \left(\frac{1}{4}\right)^n u(n) + \frac{8}{3} u(n]$$

Freq. response replace z^{-1} by $e^{j\Omega}$ in $H(z)$ eqn

$$H(e^{j\Omega}) = H(z) = \frac{1}{1 - \frac{3}{4} e^{j\Omega} + \frac{1}{8} e^{j2\Omega}}$$

where

$$H(z) =$$

$$Y(z) =$$

$$\frac{-z^{-1}}{2} =$$

$$\text{put } z^{-1} =$$

$$\text{put } z^{-1} = -2$$

$$\text{put } z^{-1} = 0$$

$$Y(z) =$$

$$X_3(z) =$$

$$x_3(n) =$$

$$y(n) =$$

(a) A stable

impulse

(b) If $x(n]$

(a) $y(n) -$

$$Y(z) [1$$

$$H(z) =$$

Determine the output of the LTI system
where $h(n) = (\frac{1}{2})^n u(n)$

$$x(n) = n(-\frac{1}{2})^n u(n).$$

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$$H(z) = \frac{1}{1 - z^{-1}/2}; \quad X(z) = \frac{-\frac{z^{-1}}{2}}{(1 + \frac{z^{-1}}{2})^2}$$

$$\frac{1}{8} y(n-2) = x(n)$$

response of the
by response.

$$Y(z) = X(z) \cdot H(z) = \frac{-z^{-1}/2}{(1 - z^{-1}/2)(1 + z^{-1}/2)^2}$$

$$= \frac{A}{1 - z^{-1}/2} + \frac{B}{1 + z^{-1}/2} + \frac{C}{(1 + z^{-1}/2)^2}$$

$$-\frac{z^{-1}}{2} = A(1 + \frac{z^{-1}}{2})^2 + B(1 + \frac{z^{-1}}{2})(1 - \frac{z^{-1}}{2}) + C(1 - \frac{z^{-1}}{2})$$

[tr. for]
system for

$$\text{put } z^{-1} = 2; \quad -1 = A(2)^2 \Rightarrow \boxed{A = -1/4}$$

$$\text{put } z^{-1} = -2; \quad 1 = C[2] \Rightarrow \boxed{C = 1/2}$$

$$\text{put } z^{-1} = 0; \quad 0 = A + B + C \Rightarrow \boxed{B = -1/4}$$

$$Y(z) = \frac{-1/4}{1 - z^{-1}/2} - \frac{1/4}{1 + z^{-1}/2} + \frac{1/2}{(1 + z^{-1}/2)^2} = X_1(z) + X_2(z) + X_3(z)$$

$$X_3(z) = \frac{1/2}{(1 + z^{-1}/2)^2} = \frac{-[-\frac{z^{-1}}{2}] \cdot z}{(1 + \frac{z^{-1}}{2})^2};$$

$$x_3(n) = -\delta(n+1) * n(-\frac{1}{2})^n u(n) = -(n+1)(-\frac{1}{2})^{n+1} u(n+1)$$

$$\therefore y(n) = -\frac{1}{4}(\frac{1}{2})^n u(n) - \frac{1}{4}(-\frac{1}{2})^n u(n) - (n+1)(-\frac{1}{2})^{n+1} u(n+1)$$

(a) A stable system is described by the diff. eqⁿ Determine the impulse response. $y(n) - y(n-1) + \frac{1}{4}y(n-2) = x(n) + \frac{1}{4}x(n-1) - \frac{1}{8}x(n-2)$

(b) If $x(n) = [\frac{1}{4})^n + (-\frac{1}{2})^n] u(n)$; determine the o/p.

$$(a) \quad y(n) - y(n-1) + \frac{1}{4}y(n-2) = x(n) + \frac{1}{4}x(n-1) - \frac{1}{8}x(n-2)$$

$$Y(z)[1 - z^{-1} + \frac{1}{4}z^{-2}] = X(z)[1 + \frac{z^{-1}}{4} - \frac{z^{-2}}{8}]$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 + z^{-1}/4 - z^{-2}/8}{1 - z^{-1} + \frac{1}{4}z^{-2}} \quad \dots \text{Transfer fcn}$$

Normalize the numerator

$$\frac{1}{4}z^2 - z^1 + 1 \left(-\frac{z^1}{8} + \frac{z^1}{4} + 1 \right) \left(-\frac{1}{2} \right)$$

$$\frac{-\frac{z^1}{8} + \frac{z^1}{4} - \frac{1}{2}}{-\frac{z^1}{4} + \frac{1}{2}}$$

$$H(z) = -\frac{1}{2} + \frac{\frac{1}{2} - \frac{z^1}{4}}{1 - z^1 + \frac{z^2}{4}} = -\frac{1}{2} + P(z)$$

$$P(z) = \frac{\frac{1}{2} - \frac{z^1}{4}}{1 - z^1 + \frac{z^2}{4}} = \frac{A}{(1 - \frac{z^1}{2})} + \frac{B}{(1 - \frac{z^1}{2})^2}$$

$$\frac{1}{2} - \frac{z^1}{4} = A(1 - \frac{z^1}{2}) + B$$

put $z^1 = 4$; $\frac{1}{2} - 1 = B \Rightarrow \boxed{B = -\frac{1}{2}}$

put $z^1 = 0$; $\frac{1}{2} = A + B \Rightarrow \boxed{A = 1}$

$$H(z) = -\frac{1}{2} + \frac{1}{1 - \frac{z^1}{2}} - \frac{1/2}{(1 - \frac{z^1}{2})^2}$$

$$h(n) = -\frac{1}{2} \delta(n) + \left(\frac{1}{2}\right)^n u(n) - \delta(n+1) * n \left(\frac{1}{2}\right)^n u(n)$$

$$\boxed{h(n) = -\frac{1}{2} \delta(n) + \left(\frac{1}{2}\right)^n u(n) - (n+1) \left(\frac{1}{2}\right)^{n+1} u(n+1)}$$

(b) Given $x(n) = \left[\left(\frac{1}{4}\right)^n + \left(-\frac{1}{2}\right)^n \right] u(n)$

$$X(z) = \frac{1}{1 - \frac{z^1}{4}} + \frac{1}{1 + \frac{z^1}{2}} = \frac{2 + \frac{z^1}{4}}{(1 - \frac{z^1}{4})(1 + \frac{z^1}{2})}$$

$$Y(z) = X(z) \cdot H(z) = \frac{2 + \frac{z^1}{4}}{(1 - \frac{z^1}{4})(1 + \frac{z^1}{2})} \times \left[\frac{1 + \frac{z^1}{4} - \frac{z^2}{8}}{1 - z^1 + \frac{z^2}{4}} \right]$$

$$= \frac{2 + \frac{z^1}{4}}{(1 - \frac{z^1}{4})(1 + \frac{z^1}{2})} \cdot \frac{(1 - \frac{z^1}{4})(1 + \frac{z^1}{2})}{(1 - \frac{z^1}{2})^2}$$

$2 + \frac{z^1}{4} =$
put $z^1 =$

put z

$Y(z)$

$y(n) =$

$y(n) =$

The out
when the
(i) determ
(ii) Determ

$y(n) =$
 $x(n)$

$H(z) =$

$H(z) =$

$A = \frac{2}{1 - \frac{1}{4}}$

$H(z) =$

$h(n) =$

$$Y(z) = \frac{2 + \frac{z^{-1}}{4}}{(1 - \frac{z^{-1}}{2})^2} = \frac{A}{(1 - \frac{z^{-1}}{2})} + \frac{B}{(1 - \frac{z^{-1}}{2})^2}$$

$$2 + \frac{z^{-1}}{4} = A(1 - \frac{z^{-1}}{2}) + B$$

$$\text{put } z^{-1} = 2; \quad \boxed{B = 5/2}$$

$$\text{put } z^{-1} = 0; \quad 2 = A + B \Rightarrow \boxed{A = -1/2}$$

$$Y(z) = \frac{-1/2}{1 - z^{-1}/2} + \frac{5/2}{(1 - z^{-1}/2)^2}$$

$$y(n) = -\frac{1}{2} \left(\frac{1}{2}\right)^n u(n) + 5 \delta(n+1) * \left(\frac{1}{2}\right)^n n u(n)$$

$$\boxed{y(n) = -\frac{1}{2} \left(\frac{1}{2}\right)^n u(n) + 5 \left(\frac{1}{2}\right)^{n+1} (n+1) u(n+1)}$$

The output of a discrete LTI system is $y(n) = 2 \left(\frac{1}{3}\right)^n u(n)$ when the input is $u(n)$.

(i) determine the impulse response $h(n)$ of the system.

(ii) Determine the o/p when the i/p is $\left(\frac{1}{2}\right)^n u(n)$.

$$n \left(\frac{1}{2}\right)^n u(n)$$

$$y(n) = 2 \left(\frac{1}{3}\right)^n u(n) \longleftrightarrow Y(z) = 2 \frac{1}{1 - z^{-1}/3}$$

$$x(n) = u(n) \longleftrightarrow X(z) = \frac{1}{1 - z^{-1}}$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{2(1 - z^{-1})}{1 - z^{-1}/3} = \frac{2(1 - z^{-1})}{1 - z^{-1}/3}$$

$$H(z) = \frac{A}{1 - z^{-1}/3} + \frac{B}{1 - z^{-1}}$$

$$h(n) = 2 z^{-1} \left[\frac{1}{1 - z^{-1}/3} - \frac{z^{-1}}{1 - z^{-1}/3} \right]$$

$$A = \frac{2}{1 - z^{-1}} \bigg|_{z^{-1}=3} = -1$$

$$B = \frac{2}{1 - z^{-1}} \bigg|_{z^{-1}=1} = 3$$

$$H(z) = \frac{-1}{1 - z^{-1}/3} + \frac{3}{1 - z^{-1}}$$

$$= 2 \left(\frac{1}{3}\right)^n u(n) - 2 \delta(n+1) * \left(\frac{1}{3}\right)^n u(n)$$

$$\boxed{h(n) = 3 u(n) - \left(\frac{1}{3}\right)^n u(n)}$$

$$\boxed{h(n) = 2 \left(\frac{1}{3}\right)^n - 2 \left(\frac{1}{3}\right)^{n-1} u(n-1)}$$

(b) $x(n) = \left(\frac{1}{2}\right)^n u(n) \xleftrightarrow{ZT} X(z) = \frac{1}{1-z/2}$

$Y(z) = X(z) \cdot H(z) = \frac{1}{(1-z/2)(1-z/3)}$

$Y(z) = \frac{A}{1-z/2} + \frac{B}{1-z/3}$

$A = \frac{2}{(1-\frac{1}{3})(1-\frac{1}{2})} = 3; \quad B = \frac{3}{(1-\frac{1}{2})(1-\frac{1}{3})} = 2$

$C = \frac{2}{(1-z/2)(1-z/3)} \Big|_{z=2} = -6$

$Y(z) = X(z) \cdot H(z) = \frac{1}{1-z/2} \cdot \frac{2(1-z/3)}{1-z/3}$

$Y(z) = \frac{A}{1-z/2} + \frac{B}{1-z/3}$

$2-2z = A(1-\frac{z}{3}) + B(1-\frac{z}{2})$

put $z=3; \quad -4 = B(-\frac{1}{2}) \Rightarrow B = 8$

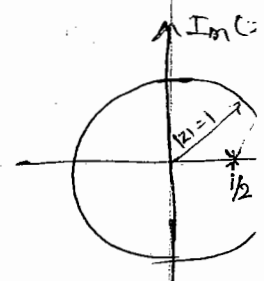
put $z=2; \quad -2 = A(1-\frac{2}{3}) = A(\frac{1}{3}) \Rightarrow A = -6$

$Y(z) = \frac{-6}{1-\frac{z}{2}} - \frac{8}{1-\frac{z}{3}}$

$y(n) = -6\left(\frac{1}{2}\right)^n u(n) + 8\left(\frac{1}{3}\right)^n u(n)$

part (a)
Answer can also be
 $h(n) = \delta(n) - 4\left(\frac{1}{3}\right)^n u(n)$

and ROC
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Stability and causality

Stability and causality can be determined from the pole zero plot

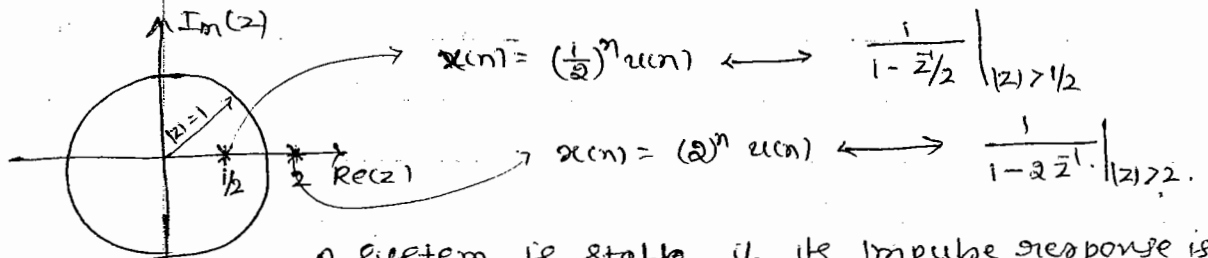
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and ROC of the transfer function.

For a system to be causal; impulse response $h(n) = 0; n < 0$
Hence for a causal system; the ROC of $H(z)$ lies outside the outermost pole.

A pole inside the unit circle contributes a decaying exp. sequence and a pole outside the unit circle corresponds to an increasing exp. sequence



A system is stable if its impulse response is absolutely summable $\sum_{n=-\infty}^{\infty} |h(n)| < \infty$.

Hence a causal system is stable; if the poles of $H(z)$ lies inside the unit circle in the z -plane.

Hence for a system to be both causal and stable; the ROC must include the unit circle; ROC must be outside the outermost pole and all the poles must lie inside the unit circle in the z -plane. Ex $x(n) = \left(\frac{1}{3}\right)^n u(n) + \left(\frac{7}{8}\right)^n u(n)$.

However for the system to be stable alone, the poles can lie outside the unit circle; but the sequence has to be a left sided decaying sequence Ex: $(3)^n u(-n-1) = x(n)$.

Inverse system and Inverse transfer function

For a system with Imp. response $h(n)$; its inverse Imp. response $h^{-1}(n)$ is such that $h(n) * h^{-1}(n) = 1$

Taking ZT on both sides $H(z) \cdot \bar{H}(z) = 1$

$$\therefore \boxed{\bar{H}(z) = \frac{1}{H(z)}}$$

The transfer function of an inverse system is the reciprocal of the transfer fn of the system.

Due to this original poles become zeros and zeros become poles. However the causality, stability properties remain same as the original system.

Fin and det HC

A system with all its poles and zeros lies inside the unit circle in the z-plane is referred to as a minimum phase system.

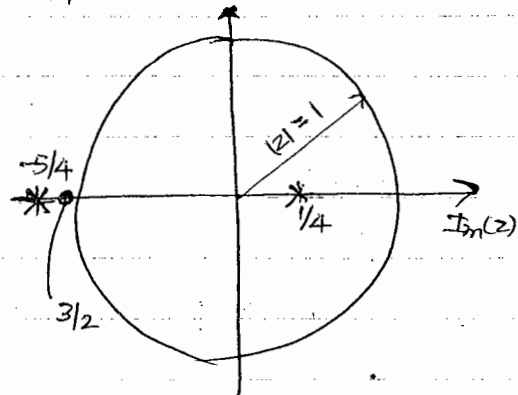
$$H^{-1}(z) =$$

① Check whether the following transfer functions/systems are stable and causal, in minimum phase or not

$$(a) H(z) = \frac{2z+3}{z^2 - z - \frac{5}{16}}$$

$$H(z) = \frac{2z+3}{(z+\frac{5}{4})(z-\frac{1}{4})}$$

poles are at $1/4, -5/4$
zeros at $z = 3/2$.



One pole lies outside the unit circle.

Since not all poles lie inside the unit circle, the system is not causal, stable and not in minimum phase.

$$(b) y(n) - y(n-1) - \frac{1}{4}y(n-2) = 3x(n) - 2x(n-1)$$

$$Y(z) \left[1 - z^{-1} - \frac{1}{4}z^{-2} \right] = [3 - 2z^{-1}] X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{3 - z^{-1}2}{1 - z^{-1} - \frac{1}{4}z^{-2}} = \frac{4(3z-2)}{4z^2 - 4z - 1}$$

$$H(z) = \frac{4(3z-2)}{4z^2 - 4z - 1}$$

$$\text{roots } r_1, r_2 \left\{ \frac{4 \pm \sqrt{16+16}}{8} = \frac{4 \pm 4\sqrt{2}}{8} = \frac{1 \pm \sqrt{2}}{2} \right.$$

Hence the poles are at $\frac{1+\sqrt{2}}{2}$ & $\frac{1-\sqrt{2}}{2}$ & zero at $\frac{2}{3}$

Not all poles lie inside the unit circle; the system is not causal and stable and also not in minimum phase.

The sy (Also in

poles causal

$$(b) y(n)$$

$$H^{-1}(z) =$$

poles are Hence 1 the pole.

(c) Give

$$H^{-1}(z)$$

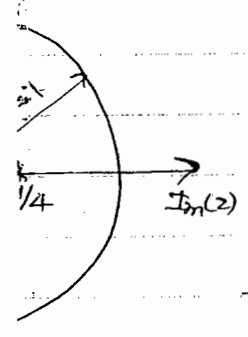
Roots

1 zero become

tem,

inside the
minimum phase
- system

systems



the system
phase.

$$\frac{z-2}{z-1}$$

$$= \frac{1 \pm j\sqrt{2}}{2}$$

$$\& \text{ zero at } \frac{2}{3}$$

a system is
minimum phase

Find the impulse response of the inverse sys.
and determine whether the system is stable and causal

(a) $H(z) = \frac{1 - 4z^{-1} + 4z^{-2}}{1 - \frac{z^{-1}}{2} + \frac{z^{-2}}{4}}$

$$H^{-1}(z) = \frac{1 - \frac{z^{-1}}{2} + \frac{z^{-2}}{4}}{1 - 4z^{-1} + 4z^{-2}} = \frac{(1 + \frac{z^{-1}}{2})^2}{(1 - 2z^{-1})^2}$$

poles are at $z=2$, Hence the system is not
causal and stable.

(b) $y(n) - \frac{1}{4}y(n-2) = 6x(n) - 7x(n-1) - 3x(n-2)$

$$H(z) = \frac{6 - 7z^{-1} - 3z^{-2}}{(1 - \frac{z^{-2}}{4})}$$

$$H^{-1}(z) = \frac{1 - \frac{z^{-2}}{4}}{6 - 7z^{-1} - 3z^{-2}} = \frac{z^2 - \frac{1}{4}}{6z^2 - 7z - 3} = \frac{z^2 - 1/4}{(3z+1)(2z-3)}$$

poles are at $z = -1/3$ and $z = 3/2$.
Hence the system is not stable and causal as one of
the pole lies outside the unit circle.

(c) Given $H(z) = \frac{6 - 7z^{-1} + 3z^{-2}}{1 - \frac{z^{-2}}{4}}$

$$H^{-1}(z) = \frac{1 - z^{-2}/4}{6 - 7z^{-1} + 3z^{-2}} = \frac{z^2 - 1/4}{6z^2 - 7z + 3}$$

Roots are at $r_1, r_2 \} = \frac{+7 \pm \sqrt{49 - 72}}{12} = \frac{7 \pm j\sqrt{23}}{12}$

$$\frac{7 \pm j\sqrt{23}}{12} < 1$$

The system is both causal and stable
(Also in minimum phase)

Unilateral z-transforms (UZT)

Unilateral z-transforms are also called one sided

z transforms defined for $n \geq 0$. For any arbitrary sequence $x(n)$; UZT can be defined as

$$X^+(z) = \sum_{n=0}^{\infty} x(n) z^{-n}$$

The bilateral and unilateral zT have same properties for a causal sequence except the time shifting property.

Applications of UZT

(i) UZT is appropriate for problems involving causal sequences and LTI system.

(ii) Since UZT is defined for $n \geq 0$; the ROC need not be specified.

(iii) These Transforms are useful in the analysis of DTI systems described by difference eq^s with Initial conditions

$$x(n) \xrightarrow{z^+} X^+(z)$$

Time shifting property (Applicable to UZT).

If $x(n) \xrightarrow{z^+} X^+(z)$ then

$$x(n-k) \xrightarrow{z^+} x(-k) + x(-k+1) z^{-1} + \dots + x(-1) z^{-k+1} + z^{-k} X^+(z)$$

Proof

$$\text{We have } X^+(z) = \sum_{n=0}^{\infty} x(n) z^{-n}$$

$$\text{Let } w(n) = x(n-1)$$

$$W^+(z) = \sum_{n=0}^{\infty} w(n) z^{-n} = \sum_{n=0}^{\infty} x(n-1) z^{-n}$$

$$= x(-1) + \sum_{n=1}^{\infty} x(n-1) z^{-n}$$

put $n-1 = m$; when $n=1$, $m=0$, & $n=\infty$, $m=\infty$

$$\therefore W^+(z) = x(-1) + \sum_{m=0}^{\infty} x(m) \cdot z^{-m} \cdot z^{-1}$$

$$W^+(z) = x(-1) + z^{-1} X^+(z)$$

$$\text{|||}^4 \quad x(n-2) \xrightarrow{z^+} x(-2) + x(-1) z^{-1} + z^{-2} X^+(z)$$

In

$$Z^+ [x(n-k)]$$

The addi-
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proof: fo

Taking

$$\text{Lt } z \rightarrow \infty$$

Final val

For any

as if

$$\text{Lt } x(n) \text{ as } N \rightarrow \infty$$

Proof we

Taking Lt_n

$$\text{Lt } z \rightarrow 1$$

As z-

$$\therefore \text{Lt } z \rightarrow 1$$

$$\text{Lt } x(n) \text{ as } N \rightarrow \infty$$

In general

$$\mathcal{Z}^+ [x(n-k)] \xleftarrow{z^+} x(-k) + x(-k+1) z^{-1} + \dots + x(-1) z^{-k+1} + z^{-k} x^+(z)$$

by arbitrary
s

The additional terms account for the value of the sequences that are shifted in to -ve portion of the signal i.e.

Initial Value Theorem

for any causal sequence $x(n)$ the initial value can be given as.

$$\lim_{n \rightarrow \infty} x(n) = x(0) = \lim_{z \rightarrow \infty} z^+ X^+(z)$$

proof: for any arbitrary seq. $x(n)$

$$X^+(z) = \sum_{n=0}^{\infty} x(n) z^{-n} = x(0) + x(1) z^{-1} + x(2) z^{-2} + \dots$$

need not be

Taking $\lim_{z \rightarrow \infty}$ on both sides

$$\lim_{z \rightarrow \infty} z^+ X^+(z) = \lim_{z \rightarrow \infty} \left[x(0) + \frac{x(1)}{z} + \frac{x(2)}{z^2} + \dots + \frac{x(n)}{z^n} \right]$$

s of DTI syst.
1 conditions

$$\boxed{x(0) = \lim_{z \rightarrow \infty} z^+ X^+(z)}$$

Final value theorem

For any causal sequence $x(n)$; the final value can be

as if $x(n) \xleftarrow{z^+} X^+(z)$ then

$$(-1)z^{-k+1} + z^{-k} X^+(z)$$

$$\lim_{N \rightarrow \infty} x(N) = x(\infty) = \lim_{z \rightarrow 1} (1 - z^{-1}) X^+(z)$$

proof

$$\text{We have } \mathcal{Z}^+ [x(n) - x(n-1)] = \sum_{n=0}^{\infty} [x(n) - x(n-1)] z^{-n}$$

$$= X^+(z) - z^{-1} X^+(z) = X^+(z) (1 - z^{-1})$$

Taking $\lim_{n \rightarrow \infty}$, $\lim_{z \rightarrow 1}$ on both sides

$$\lim_{n \rightarrow \infty} \lim_{z \rightarrow 1} \sum_{n=0}^{\infty} [x(n) - x(n-1)] z^{-n} = \lim_{n \rightarrow \infty} \lim_{z \rightarrow 1} X^+(z) [1 - z^{-1}]$$

$\infty, m=0$

$$\text{As } z \rightarrow 1; \quad z^{-1} \rightarrow 1 \quad \text{hence } \frac{1}{z} = 1$$

$$\therefore \lim_{N \rightarrow \infty} \lim_{n=0} \sum_{n=0}^{\infty} [x(n) - x(n-1)] = \lim_{z \rightarrow 1} X^+(z) [1 - z^{-1}]$$

$^+(z)$.

$$\boxed{\lim_{N \rightarrow \infty} x(N) = x(\infty) = \lim_{z \rightarrow 1} (1 - z^{-1}) X^+(z)}$$

RHS

Note $\sum_{n=0}^{\infty} x(n) - x(n-1) = \lim_{N \rightarrow \infty} x(N)$

Ex:- Let $\sum_{n=0}^5 x(n) - x(n-1) = x(0) - x(-1) + x(1) - x(0) + x(2) - x(1) + x(3) - x(2) + x(4) - x(3) + x(5) - x(4)$

$\therefore \sum_{n=0}^5 x(n) - x(n-1) = x(5)$

① If $X^+(z) = 2 + 3z^{-1} + 4z^{-2}$. Find the initial value and final value of the corresponding sequence.

Initial value $x(0) = \lim_{z \rightarrow \infty} X^+(z)$

$= \lim_{z \rightarrow \infty} [2 + 3z^{-1} + 4z^{-2}]$

$x(0) = 2$

Final value

$x(\infty) = \lim_{z \rightarrow 1} [2 + 3z^{-1} + 4z^{-2}] [1 - z^{-1}]$

$= 0$

Ex Determine the VZT of the sequence

① $x(n) = \{2, 4, 5, 7, 0, 1\}$

$X^+(z) = \sum_{n=0}^{\infty} x(n) z^{-n} = 5 + \frac{7}{z^4} + \frac{0}{z^2} + \frac{1}{z^3}$

$X^+(z) = 5 + 7z^{-4} + z^{-3}$

② $x(n) = \alpha^n u(n)$

$X^+(z) = \sum_{n=0}^{\infty} \alpha^n z^{-n} = \frac{1}{1 - \alpha z^{-1}} \quad |z| > \alpha$

③ $x(n) = \delta(n+k)$

$X^+(z) = \sum_{n=0}^{\infty} \delta(n+k) z^{-n} = z^k \Big|_{n=-k} = \underline{\underline{z^k = 0}}$

Since 'n' is not defined for $n < 0$

① Solve

taking $z^2 X^+(z)$

$z^2 X^+(z) -$

$X^+(z)$

$-z^{-1} =$

put $z^{-1} =$

put $z^{-1} =$

$\therefore X^+(z)$

Taking

$x(n)$

$x(n)$

② Find total

$y(n) - \frac{3}{2}$

taking $Y(z) -$

$\therefore Y$

$Y(z) - \frac{3}{2}$

Solving difference equations

① Solve the diff. eqn

$$x(n+2) + 18x(n) - 9x(n-1) = 0 \text{ with } x(-1)=1, x(-2)=9.$$

taking UZT

$$\bar{z}^2 x^+(z) + \bar{z}^1 x(-1) + x(-2) + 18[x^+(z)] - 9[x(-1) + \bar{z}^1 x^+(z)]$$

$$\bar{z}^2 x^+(z) + \bar{z}^1 + 9 + 18x^+(z) - 9 - 9\bar{z}^1 x^+(z) = 0$$

$$x^+(z) = \frac{-\bar{z}^1}{\bar{z}^2 - 9\bar{z}^1 + 18} = \frac{-\bar{z}^1}{(\bar{z}^1 - 3)(\bar{z}^1 - 6)}$$

$$= \frac{A}{\bar{z}^1 - 3} + \frac{B}{\bar{z}^1 - 6}$$

$$-\bar{z}^1 = A(\bar{z}^1 - 6) + B(\bar{z}^1 - 3)$$

$$\text{put } \bar{z}^1 = 6; \quad -6 = B(3) \Rightarrow B = -2$$

$$\text{put } \bar{z}^1 = 3; \quad -3 = A(3-6) \Rightarrow A = 1$$

$$\therefore x^+(z) = \frac{1}{\bar{z}^1 - 3} - \frac{2}{\bar{z}^1 - 6} = \frac{-1/3}{1 - \bar{z}^1/3} + \frac{1/3}{1 - \bar{z}^1/6}$$

Taking I-UZT we get

$$x(n) = -\frac{1}{3} \left(\frac{1}{3}\right)^n u(n) + \left(\frac{1}{6}\right)^n \cdot \frac{1}{3} u(n)$$

$$x(n) = -\left(\frac{1}{3}\right)^{n+1} u(n) + \frac{1}{3} \left(\frac{1}{6}\right)^n u(n)$$

② Find the natural response, forced response and total response of the system described by Diff. eqn

$$y(n) - \frac{3}{2}y(n-1) + \frac{1}{2}y(n-2) = x(n) \text{ with } y(-2)=10, y(-1)=4$$

$$x(n) = \left(\frac{1}{4}\right)^n u(n).$$

taking UZT;

$$Y(z) - \frac{3}{2}[y(-1) + \bar{z}^1 Y(z)] + \frac{1}{2}[y(-2) + \bar{z}^1 y(-1) + \bar{z}^2 Y(z)] = X(z)$$

$$\therefore Y(z) = \frac{3}{2}(4 + 10Y(z))$$

$$Y(z) - \frac{3}{2}(\bar{z}^1 Y(z) + 4) + \frac{1}{2}[10 + \bar{z}^1(4) + \bar{z}^2 Y(z)] = X(z)$$

$$Y(z) \left[1 - \frac{3}{2}\bar{z}^1 + \frac{\bar{z}^2}{2} \right] - 6 + 5 + 2\bar{z}^1 = X(z)$$

$$Y(z) \left[1 - \frac{3}{2}z^{-1} + \frac{z^{-2}}{2} \right] = \frac{1 - 2z^{-1}}{2} + X(z)$$

Given $x(n) = \left(\frac{1}{4}\right)^n u(n)$

$$X(z) = \frac{1}{1 - \frac{1}{4}z^{-1}}$$

$$Y(z) = \frac{1 - 2z^{-1}}{1 - \frac{3}{2}z^{-1} + \frac{z^{-2}}{2}} + \frac{1}{(1 - \frac{1}{4}z^{-1})(1 - \frac{3}{2}z^{-1} + \frac{z^{-2}}{2})}$$

↓
Natural response

↓
Forced response

Natural response: $Y_n(z) = \frac{1 - 2z^{-1}}{1 - \frac{3}{2}z^{-1} + \frac{z^{-2}}{2}} = \frac{1 - z^{-1}/2}{(1 - \frac{z^{-1}}{2})(1 - z^{-1})}$

$$Y_n(z) = \frac{A}{1 - z^{-1}/2} + \frac{B}{(1 - z^{-1})}$$

$$1 - \frac{z^{-1}}{2} = A(1 - z^{-1}) + B(1 - z^{-1}/2)$$

put $z^{-1} = 1$; $-\frac{1}{2} = B \cdot \frac{1}{2} \Rightarrow \boxed{B = -2}$

put $z^{-1} = 2$; $-3 = A(-1)$ $A = 3$

$$\therefore Y_n(z) = \frac{-2}{1 - z^{-1}} + \frac{3}{1 - \frac{z^{-1}}{2}} \quad \boxed{y_n(n) = -2u(n) + 3\left(\frac{1}{2}\right)^n u(n)}$$

Forced response:

$$Y_f(z) = \frac{1}{(1 - \frac{1}{4}z^{-1})(1 - \frac{z^{-1}}{2})(1 - z^{-1})} = \frac{A}{1 - \frac{1}{4}z^{-1}} + \frac{B}{1 - \frac{z^{-1}}{2}} + \frac{C}{1 - z^{-1}}$$

$$1 = A(1 - \frac{z^{-1}}{2})(1 - z^{-1}) + B(1 - \frac{1}{4}z^{-1})(1 - z^{-1}) + C(1 - \frac{1}{4}z^{-1})(1 - \frac{z^{-1}}{2})$$

put $z^{-1} = 2$; $1 = 0 + B(\frac{1}{2})(-1) + 0 \Rightarrow \boxed{B = -2}$

put $z^{-1} = 1$; $1 = 0 + 0 + C(\frac{3}{4})(\frac{1}{2}) \Rightarrow \boxed{C = 8/3}$

put $z^{-1} = 4$; $1 = A(-1)(-3) \Rightarrow \boxed{A = 1/3}$

$$Y_f(z) = \frac{1/3}{1 - \frac{z^{-1}}{4}} + \frac{-2}{1 - \frac{z^{-1}}{2}} + \frac{8/3}{1 - z^{-1}}$$

$y(n)$

Find the
difference
 $y(n) - \frac{1}{4}$

Taking LT

$$Y(z) - \frac{1}{4}$$

$$Y(z) \left[1 - \frac{z^{-1}}{2} \right]$$

$$\therefore Y(z) =$$

$$Y_n(z) =$$

$$\frac{1}{8} = A$$

$$Y_n(z) =$$

$$y_n(n) =$$

$$y_n(n) =$$

$$Y_f(z) =$$

$$1 + z^{-1} = A(1$$

put $z^{-1} = 2$;

$$y[n] = \frac{1}{3} \left(\frac{1}{4}\right)^n u[n] - 2 \left(\frac{1}{2}\right)^n u[n] + \frac{8}{3} u[n]$$

Computer
 $y[n] = y_h[n] + y_f[n]$

Find the total response of the system described by the difference equation

$$y[n] - \frac{1}{4} y[n-1] - \frac{1}{8} y[n-2] = x[n] + x[n-1] \text{ with } x[n] = u[n]$$

with $y[0] = 0, y[-2] = 1$

Taking ZST on both sides;

$$Y(z) - \frac{1}{4} [y[-1] + z^{-1} Y(z)] - \frac{1}{8} [y[-2] + z^{-1} y[-1] + z^{-2} Y(z)] = X(z) + z^{-1} X(z)$$

We

$$\frac{1 - z^{-1/2}}{(1 - \frac{z^{-1}}{2})(1 - z^{-1})}$$

$$Y(z) \left[1 - \frac{z^{-1}}{4} - \frac{z^{-2}}{8} \right] - \frac{1}{4} (0) - \frac{1}{8} (1) - z^{-1} (0) = X(z) (1 + z^{-1}) + 0$$

Since $u[-1] = 0$.

$$\therefore Y(z) = \underbrace{\frac{1/8}{1 - \frac{z^{-1}}{4} - \frac{z^{-2}}{8}}}_{\text{Natural response}} + \underbrace{\frac{1 + z^{-1}}{(1 - z^{-1})(1 - \frac{z^{-1}}{4} - \frac{z^{-2}}{8})}}_{\text{forced response}}$$

$$Y_h(z) = \frac{1/8}{1 - \frac{z^{-1}}{4} - \frac{z^{-2}}{8}} = \frac{A}{(1 - \frac{z^{-1}}{2})} + \frac{B}{(1 + \frac{z^{-1}}{4})}$$

$$\frac{1}{8} = A(1 + \frac{z^{-1}}{4}) + B(1 - \frac{z^{-1}}{2}) \quad \begin{matrix} A = 1/12 \\ B = 1/24 \end{matrix}$$

$$\frac{1}{3} \left(\frac{1}{2}\right)^n u[n]$$

$$\frac{B}{1 - \frac{z^{-1}}{2}} + \frac{C}{1 - z^{-1}}$$

$$Y_h(z) = \frac{1/12}{1 - z^{-1}/2} + \frac{1/24}{1 + z^{-1}/4}$$

$$y_h[n] = \frac{1}{12} \left(\frac{1}{2}\right)^n u[n] + \frac{1}{24} \left(\frac{1}{4}\right)^n u[n]$$

$$z^{-1})(1 - \frac{z^{-1}}{2})$$

$$B = -2$$

$$C = 8/3$$

$$y_h[n] = \left[\frac{1}{3} \left(\frac{1}{2}\right)^{n+2} + \frac{1}{6} \left(\frac{1}{4}\right)^{n+1} \right] u[n]$$

$$Y_F(z) = \frac{1 + z^{-1}}{(1 - z^{-1})(1 - \frac{z^{-1}}{2})(1 + \frac{z^{-1}}{4})} = \frac{A}{1 - z^{-1}} + \frac{B}{1 - \frac{z^{-1}}{2}} + \frac{C}{1 + \frac{z^{-1}}{4}}$$

$$1 + z^{-1} = A(1 - \frac{z^{-1}}{2})(1 + \frac{z^{-1}}{4}) + B(1 - z^{-1})(1 + \frac{z^{-1}}{4}) + C(1 - z^{-1})(1 - \frac{z^{-1}}{2})$$

put $z^{-1} = 2$; $3 = B(-1)(\frac{3}{2}) \Rightarrow B = -2$

put $\bar{z} = 1$; $2 = A(\frac{1}{2})(\frac{5}{4}) \Rightarrow A = \frac{16}{5}$

put $\bar{z} = -4$; $-3 = C(5)(3) \Rightarrow \boxed{C = -5}$

$$Y_f(z) = \left(\frac{16/5}{1-\bar{z}} \right) + \frac{-2}{1-\frac{\bar{z}}{2}} + \frac{-5}{1+\frac{\bar{z}}{4}}$$

$$Y_f(n) = \left[\frac{16}{5} u(n) - 2\left(\frac{1}{2}\right)^n u(n) - 5\left(-\frac{1}{4}\right)^n u(n) \right]$$

$$Y(n) = Y_h(n) + Y_f(n)$$

$$= \frac{1}{12}\left(\frac{1}{2}\right)^n u(n) + \frac{1}{24}\left(\frac{1}{4}\right)^n u(n) + \frac{16}{5} u(n) - 2\left(\frac{1}{2}\right)^n u(n) - 5\left(-\frac{1}{4}\right)^n u(n)$$

$$Y(n) = \frac{16}{5} u(n) - \frac{23}{12}\left(\frac{1}{2}\right)^n u(n) - \frac{119}{24}\left(-\frac{1}{4}\right)^n u(n)$$

Solve the following diff. eqn using UZT.

$$Y(n) - \frac{3}{9}Y(n-2) = X(n-1) \quad \text{with } X(n) = u(n)$$

$$Y(-1) = 0, Y(-2) = 1$$

$$Y(z) - \frac{1}{9}[Y(-2) + \bar{z}^2 Y(z)] = \bar{z}^{-1} X(z)$$

$$Y(z) - \frac{1}{9}[1 + \bar{z}^2 Y(z)] = 0 + \bar{z}^{-1} X(z)$$

$$Y(z) \left[1 - \frac{\bar{z}^2}{9} \right] = \frac{1}{9} + \bar{z}^{-1} X(z)$$

$$Y(z) = \frac{1/9}{1 - \frac{\bar{z}^2}{9}} + \frac{\bar{z}^{-1} X(z)}{1 - \frac{\bar{z}^2}{9}}$$

Natural response

forced response

$$Y_n(z) = \frac{1/9}{1 - \bar{z}^2/9} = \frac{1/9}{(1 - \bar{z}/3)(1 + \bar{z}/3)} = \frac{A}{1 - \bar{z}/3} + \frac{B}{1 + \bar{z}/3}$$

$$\frac{1}{9} = A(1 + \frac{\bar{z}}{3}) + B(1 - \frac{\bar{z}}{3})$$

put $\bar{z} = 3$; $\frac{1}{9} = A(2) \Rightarrow \boxed{A = 1/18}$

put $\bar{z} = -3$; $\frac{1}{9} = B(2) \Rightarrow \boxed{B = 1/18}$

$$X(n) =$$

$$Y_f(z) =$$

$$\bar{z} = A(1 - \bar{z})$$

put $\bar{z} = 1$;

put $\bar{z} = -3$

put $\bar{z} = 3$

$$Y_f(z) =$$

$$Y_f(n) =$$

$$Y(n) =$$

$$Y(n) =$$

$$Y(n) =$$

find its

Y

with no

$$Y(z) =$$

system is

$$y_n(n) = \left[\frac{1}{18} \left(\frac{1}{3}\right)^n + \frac{1}{18} \left(-\frac{1}{3}\right)^n \right] u(n)$$

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$$x(n) = u(n) \xleftrightarrow{z^T} X(z) = \frac{1}{1-z^{-1}}$$

$$Y_F(z) = \frac{z^{-1}}{(1-\frac{z^{-1}}{3})(1+\frac{z^{-1}}{3})(1-z^{-1})} = \frac{A}{1-\frac{z^{-1}}{3}} + \frac{B}{1+\frac{z^{-1}}{3}} + \frac{C}{1-z^{-1}}$$

$$z^{-1} = A(1-\frac{z^{-1}}{3})(1+\frac{z^{-1}}{3}) + B(1-\frac{z^{-1}}{3})(1-z^{-1}) + C(1-\frac{z^{-2}}{9})$$

$$\text{put } z^{-1}=1; 1 = C \left[\frac{8}{9} \right] \Rightarrow \boxed{C = 9/8}$$

$$\text{put } z^{-1}=-3; -3 = B(2)(4) \Rightarrow B = -3/8$$

$$\text{put } z^{-1}=3; 3 = A(-2)(2) \Rightarrow A = -3/4$$

$$Y_F(z) = \frac{-3/4}{1-z^{-1}/3} - \frac{3/8}{1+z^{-1}/3} + \frac{9/8}{1-z^{-1}}$$

$$y_f(n) = -\frac{3}{4} \left(\frac{1}{3}\right)^n u(n) - \frac{3}{8} \left(-\frac{1}{3}\right)^n u(n) + \frac{9}{8} u(n)$$

$$\boxed{y(n) = y_f(n) + y_n(n)}$$

$$y(n) = \left[\left[\frac{1}{18} - \frac{3}{4} \right] \left(\frac{1}{3}\right)^n + \left[-\frac{3}{8} + \frac{1}{18} \right] \left(-\frac{1}{3}\right)^n + \frac{9}{8} \right] u(n)$$

$$= \left(\frac{2-27}{36} \right) \left(\frac{1}{3}\right)^n u(n) + \left(\frac{-27+4}{72} \right) \left(-\frac{1}{3}\right)^n u(n) + \frac{9}{8} u(n)$$

$$\boxed{y(n) = \left[\frac{-25}{36} \left(\frac{1}{3}\right)^n + \frac{23}{72} \left(-\frac{1}{3}\right)^n + \frac{9}{8} \right] u(n)}$$

response

$$\frac{1}{3} + \frac{B}{1+z^{-1}/3}$$

find the forced response of the system.

$$y(n) - \frac{5}{6} y(n-1) + \frac{1}{6} y(n-2) = x(n)$$

with $x(n) = 2^n u(n)$ Assume that the system is relaxed.

$$Y(z) - \frac{5}{6} z^{-1} Y(z) + \frac{1}{6} z^{-2} Y(z) = \frac{1}{1-2z^{-1}}$$

System is relaxed Means IC are zero's

$$Y(z) = \frac{1}{(1-2z^{-1})(1-\frac{5}{6}z^{-1}+\frac{1}{6}z^{-2})}$$

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Fi

$$Y(z) = \frac{A}{1-2z^{-1}} + \frac{B}{(1-\frac{z^{-1}}{2})} + \frac{C}{(1-\frac{z^{-1}}{3})}$$

we have

$$Y(z) = \frac{8/5}{1-2z^{-1}} - \frac{1}{1-\frac{z^{-1}}{2}} + \frac{2/5}{1-\frac{z^{-1}}{3}}$$

$$y_f(n) = \frac{8}{5} (2)^n u(n) - \left(\frac{1}{2}\right)^n u(n) + \frac{2}{5} \left(\frac{1}{3}\right)^n u(n)$$

put x

$$y_f(n) = \left[\frac{1}{5} (2)^{n+3} - \left(\frac{1}{2}\right)^n + \frac{2}{5} \left(\frac{1}{3}\right)^n \right] u(n)$$

Prob 2.20

If a DT LTI system is BIBO stable; show that the ROC of its system function $H(z)$ must contain the unit circle, i.e. $|z|=1$.

Find

Proof:- A DT LTI system is BIBO stable, if its imp. response is absolutely summable

$$\sum_{n=-\infty}^{\infty} |h(n)| < \infty \quad \text{or}$$

$$H(z) = \sum_{n=-\infty}^{\infty} |h(n)| z^{-n} < \infty$$

$$\text{put } z = e^{j\Omega}; \quad H(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} |h(n)| e^{-j\Omega n} = \sum_{n=-\infty}^{\infty} |h(n)| < \infty$$

Taking IZT

Hence we can conclude that if the system is stable then $H(z)$ converges for $z = e^{j\Omega}$.

Determine

(a) $x(n)$

$(e^{j\Omega}) = 1$, constitutes a unit circle in s-plane.

$X(z) =$

Hence for a stable DT-LTI system; the ROC of $H(z)$ must contain the unit circle $|z|=1$.

Find the IZT of $x(z) = e^{-z^{-1}}$ with $|z| \neq 0$

We have
$$e^x = 1 + \frac{x^1}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

put $x = z^{-1}$;
$$x(z) = \sum_{k=0}^{\infty} \frac{z^{-k}}{k!}$$

$$x(z) = \sum_{k=0}^{\infty} \frac{\delta(n-k)}{k!} = \frac{1}{k!} u(k)$$

$$x(z) = \frac{1}{k!} u(k)$$

now that the
in the unit

Find the IZT $x(z) = e^{z^2}$

if its imp.

We have
$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

$$\therefore x(z) = (e)^{z^2} = \sum_{k=0}^{\infty} \frac{z^{2k}}{k!}$$

Taking IZT.

$$x(n) = \sum_{k=0}^{\infty} \frac{\delta(n+2k)}{k!}$$

$$= \sum_{n=-\infty}^{\infty} x(n) \delta(n)$$

is stable

s-plane.

the ROC

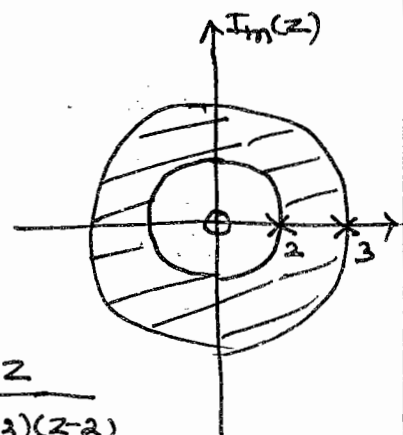
$$|z|=1$$

Determine the ZT, ROC and pole-zero plot of

(a) $x(n) = 2^n u(n) + 3^n u(-n-1)$

$$X(z) = \frac{z}{z-2} \Big|_{|z|>2} - \frac{z}{z-3} \Big|_{|z|<3}$$

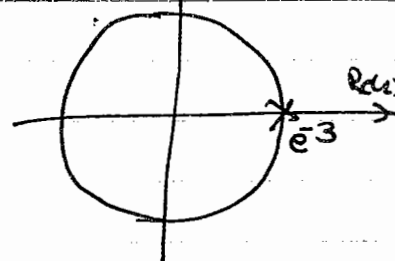
$$\begin{aligned} &= \frac{z(z-3) - z(z-2)}{(z-2)(z-3)} \\ &= \frac{z^2 - 3z - z^2 + 2z}{(z-3)(z-2)} = \frac{-z}{(z-3)(z-2)} \end{aligned}$$



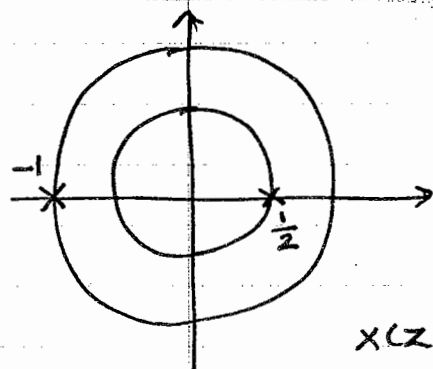
(b) $x(n) = e^{-3n} u(n-1)$
 $= \sum_{n=1}^{\infty} e^{-3n} \cdot z^{-n} = \sum_{n=1}^{\infty} \left[\frac{1}{ze^3} \right]^n$

$$= \frac{e^{-3} z^{-1}}{1 - e^{-3} z^{-1}} \quad |z| > e^{-3}$$

$$X(z) = \frac{e^{-3}}{z - e^{-3}} \quad |z| > e^{-3}$$



Find $x(n]$ given $x(z)$ has poles at $z = -1$, $z = 1/2$
 $x(1) = 1$, $x(-1) = 1$ and ROC includes a point $z = 3/4$



Since ROC includes a point $z = 3/4$

$\therefore$ ROC is of the form

$$\frac{1}{2} < |z| < 1$$

$$X(z) = \frac{A z}{z - \frac{1}{2}} \Big|_{|z| > 1/2} + \frac{B z}{z + 1} \Big|_{|z| < 1}$$

$$\therefore x(n) = A \left(\frac{1}{2} \right)^n u(n) - (-1)^n B u(-n-1)$$

given $x(1) = 1$; $= \frac{A}{2} + B(0)$

$$\therefore \boxed{A = 2}$$

$x(-1) = 1$; $= A \left(\frac{1}{2} \right)^{-1} u(-1) + B \cdot (1)$

$$\Rightarrow \boxed{B = 1}$$

$$\therefore X(z) = \frac{2}{z - 1/2} \Big|_{|z| > 1/2} + \frac{B z}{z + 1} \Big|_{|z| < 1}$$

$$\therefore x(n) = 2 \left(\frac{1}{2} \right)^n u(n) - (-1)^n u(-n-1)$$

(2) x

$\sin$

are co

$$x(0) =$$

$$x(2) =$$

$$\therefore \boxed{x}$$

(3) $x(n)$
 $x(-1) = 1$

Let $X(z)$

$$\boxed{x(n) =}$$

$$x(-1) = 1 =$$

$$x(-3) =$$

$$X(z) =$$

$$x(1) = \frac{11}{3}$$

② $x(n)$ is right sided; $X(z)$ has a single pole
 $x(0)=2$, $x(2)=1/2$

Since $x(n)$ is right sided;

Let $x(n) = k a^n u(n)$. where a & k
are constants to be determined.

$$x(0) = 2 = k a^0 u(0) \Rightarrow \boxed{k=2}$$

$$x(2) = \frac{1}{2} = k a^2 u(2) = 2 a^2 \cdot 1$$

$$a^2 = \frac{1}{4} \quad \text{or} \quad \boxed{a = \frac{1}{2}}$$

$$\therefore \boxed{x(n) = 2 \left(\frac{1}{2}\right)^n u(n)}$$

③ $x(n)$ is two sided, $X(z)$ has a pole at $z=1/4$
 $x(-1)=1$, $x(-3)=1/4$; & $x(1)=1/3$.

$$\text{Let } X(z) = \frac{A z}{z - 1/4} + \frac{B z}{z - k}$$

$$\boxed{x(n) = A \left(\frac{1}{4}\right)^n u(n) - B (k)^n u(-n-1)}$$

$$x(-1) = 1 = A \left(\frac{1}{4}\right)^{-1} u(-1) - B k^{-1} u(-2)$$

$$1 = 0 - \frac{B}{k}$$

$$\boxed{B = -k}$$

$$x(-3) = \frac{1}{4} = 0 - B k^{-3} (1) = -\frac{B}{k^3} = +\frac{k}{k^3} = \frac{1}{k^2}$$

$$\Rightarrow \boxed{k=2} \quad \& \quad \boxed{B=-2}$$

$$X(z) = \frac{A z}{z - 1/4} - \frac{2 z}{z - 2}$$

$$x(1) = \frac{1}{3} = \frac{A(1)}{1 - 1/4} - \frac{2(1)}{1 - 2} \Rightarrow A = 5/4$$

$$\therefore x(n) = \frac{5}{4} \left(\frac{1}{4}\right)^n u(n) + 2 \cdot 2^n u(-n-1)$$

$$\text{Roc: } \frac{1}{4} < |z| < 2$$

- ① The Z-transform of complex e
- ② The Z-T and system: The ZT exists DTFT exists for $x > 1$; can not If we see x^n grows summable The ability is like sign
- ③ Z-T fine ZT systems
- ④ It can initial
- ⑤ Z-T to algebra

ZT a for

DTFT X

Find the inverse ZT of $X(z) = e^{1/2}$

II method

Assume $x(n)$ is causal.

$$X(z) = e^{1/2}$$

$$\ln X(z) = \frac{1}{2} \ln e = \frac{1}{2}$$

diff. both sides; $\frac{1}{X(z)} \frac{d}{dz} X(z) = -\frac{1}{z^2}$

$$\frac{-z}{X(z)} \frac{d}{dz} X(z) = -\frac{(-z)}{z^2} = \frac{1}{z}$$

$$-z \frac{d}{dz} X(z) = \frac{1}{z} X(z) = z^{-1} x(n)$$

Taking IZT

$$n x(n) = x(n-1) \longrightarrow \textcircled{A}$$

Given the sequence as causal; using Initial value theorem

$$x(0) = \lim_{z \rightarrow \infty} z X(z) = \lim_{z \rightarrow \infty} z e^{1/2} = 1$$

$$\therefore x(0) = 1$$

when $n=1$, in eqn ①

$$x(1) = x(0) = 1$$

for $n=2$; $x(2) = 1/2$

for $n=3$; $x(3) = 1/6$

$$\therefore x(n) = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \dots$$

$$\therefore x(n) = \frac{1}{n!}; n \geq 0$$

$$\text{or } x(n) = \delta(n) + \frac{1}{n!} u(n-1)$$

practical importance of Z-Transforms

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① The Z-transform is the discrete time counterpart of Laplace transform. Here DT signals are represented in terms of complex exponential signals.

② The Z-T are used to study much larger class of signals and systems compared to a DTFT.

The ZT exists for signals that does not have a DTFT.

DTFT existence requires absolute summability of signal $x(n)$.

Ex:- Consider the DTFT of signal $x(n) = \alpha^n u(n)$.

for $\alpha > 1$; the summation leads to ∞ . hence DTFT can not exist.

If we select a value $z < \alpha$; z^{-n} decays faster than α^n grows. hence $X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$ is absolutely summable and ZT exists.

The ability to work with signals that does not have DTFT is the significant advantage offered by Z-Transform.

③ Z-T find its applications in terms of implementation of DT systems on computers (study of char & model derivation).

④ It can also be used to solve difference equations with initial conditions.

⑤ Z-T transforms the difference equations into simple algebraic expressions making the analysis simple.

ZT and DTFT are interchangeable

for any ZT substitute $z = e^{j\omega}$ to get DTFT

Ex: $x(n) = a^n u(n)$ with $|a| < 1$

$$X(z) = \frac{z}{z-a} \quad |z| > a$$

$$\text{DTFT } X(e^{j\omega}) = \frac{e^{j\omega}}{e^{j\omega} - a} = \frac{1}{1 - ae^{-j\omega}}$$

$X(e^{j\omega})$ is known as frequency
response of the system.

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Other name is frequency spectrum.

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Continuous time Fourier Transform (FT)

Date: / /

The Fourier Transform of a signal $x(t)$ is defined as

$$X(j\omega) = X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad \text{--- (A)}$$

and $x(t)$ is obtained from $X(j\omega)$ by Inverse Fourier Transform

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \quad \text{--- (B)}$$

$$x(t) \xleftrightarrow{\text{FT}} X(j\omega)$$

$X(j\omega)$ is the frequency domain representation of time domain signal $x(t)$; and is known as Spectrum of $x(t)$.

The Spectrum is non-periodic in nature.

Eqn (A) is known as Synthesis equation and eqn (B) is known as Analysis equation.

For the existence of FT of continuous non periodic signal $x(t)$; the following conditions are to be satisfied; and these conditions are known as Dirichlet conditions

(a) $x(t)$ is absolutely integrable, i.e. $\int_{-\infty}^{\infty} x(t) dt < \infty$

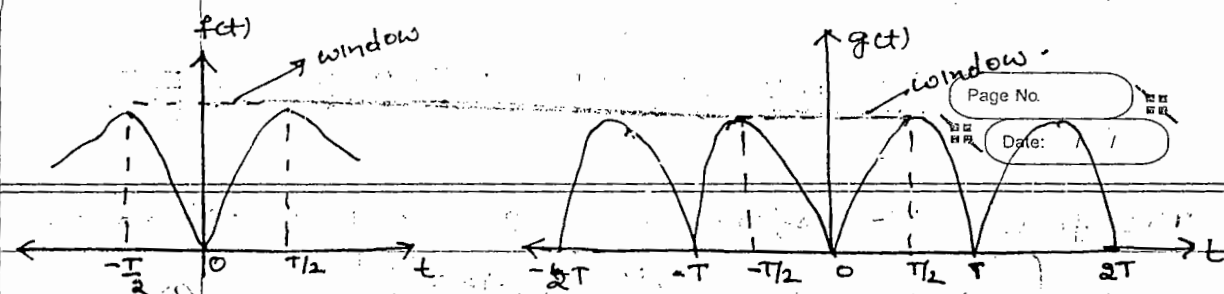
(b) $x(t)$ must have a finite number of maxima and minima within any finite interval.

(c) $x(t)$ must have finite number of discontinuities within any finite interval and the size of each discontinuity is finite.

Note:- The Fourier Transform is used to represent a non periodic CT signal as a superposition of complex sinusoids.

FT provides an effective reversible link b/w freq. domain & Time domain representation of signals.

For a non periodic signal as $T \rightarrow \infty$; $\omega_0 \rightarrow 0$. Hence the spacing b/w Spectral lines become very very small and the Spectrum appears to be continuous.

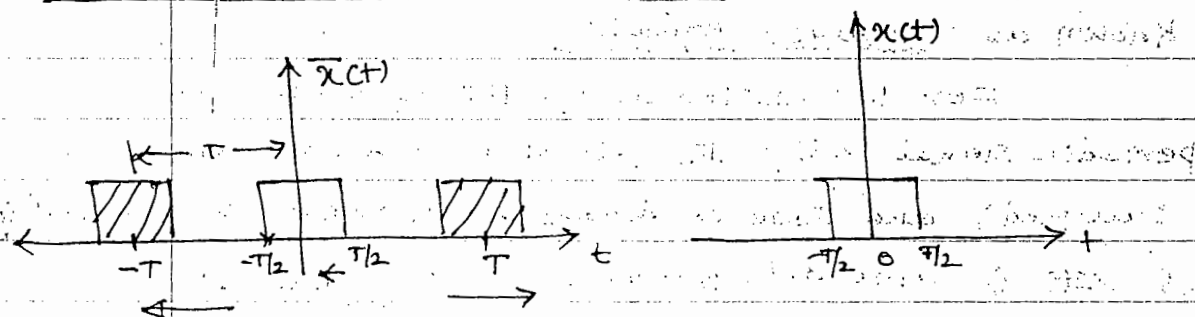


Look at the portion of non periodic signal $f(t)$ over a interval T by imagining that we are looking at a graph of $f(t)$ thro' a window of length T . Let us consider what happens as T gets larger.

In the next fig, a periodic function $g(t)$ based on the windowed portion of $f(t)$.

The graph of $f(t)$ and $g(t)$ agree on the interval $-T/2$ to $T/2$ hence the approach "fourier series expansion of signals defined over finite period" is used to define F.T.

Derivation of Fourier Transform



The F.S. Expansion for a periodic signal $\bar{x}(t)$ is

$$\bar{x}(t) = \sum_{k=-\infty}^{\infty} x(k) e^{jk\omega_0 t} \quad \text{where } x(k) = \frac{1}{T} \int_{-T/2}^{T/2} \bar{x}(t) e^{-jk\omega_0 t} dt$$

$\bar{x}(t)$ is a periodic signal with period T and $x(k)$ is its Fourier series representation

Let $x(t)$ be a finite duration non periodic signal having one period that of periodic signal $\bar{x}(t)$.

$$x(t) = \begin{cases} \bar{x}(t); & -T/2 < t < T/2 \\ 0; & |t| > T/2 \end{cases}$$

As the period T is increased, the periodic replicates of $x(t)$ in $\bar{x}(t)$ are moved farther and farther away from the origin (the shaded portions).

as $T \rightarrow \infty$, the replicates are moved to ∞ .

$$\therefore x(t) = \lim_{T \rightarrow \infty} \sum_{k=-\infty}^{\infty} \bar{x}(t)$$

as $T \rightarrow \infty$; $\omega_0 = \frac{2\pi}{T} \rightarrow 0$

Hence replace ω_0 by an infinitesimally small quantity $d\omega$ and $d\omega \rightarrow 0$. and the discrete frequency ' $k\omega_0$ ' by a continuous frequency ' ω '.

$$X(k) = \frac{1}{T} \int_{\langle T \rangle} \bar{x}(t) e^{-jk\omega_0 t} dt$$

$$T X(k) = \int_{\langle T \rangle} \bar{x}(t) e^{-jk\omega_0 t} dt$$

$$\lim_{T \rightarrow \infty} T X(k) = \lim_{T \rightarrow \infty} \int_{\langle T \rangle} \bar{x}(t) e^{-jk\omega_0 t} dt$$

$$\lim_{T \rightarrow \infty} T X(k) = \int_{-\infty}^{\infty} \bar{x}(t) e^{-j\omega t} dt = X(j\omega) \quad \text{Fourier Transform of } \bar{x}(t)$$

$X(j\omega)$ or $X(\omega)$ is known as FT of signal $x(t)$.

$$\bar{x}(t) = \sum_{k=-\infty}^{\infty} X(k) e^{jk\omega_0 t}$$

$$= \sum_{k=-\infty}^{\infty} T X(k) e^{jk\omega_0 t} \cdot \frac{1}{T}$$

$$x(t) = \lim_{T \rightarrow \infty} \bar{x}(t) = \lim_{T \rightarrow \infty} \sum_{k=-\infty}^{\infty} T X(k) e^{jk\omega_0 t} \left(\frac{\omega_0}{2\pi} \right)$$

As $T \rightarrow \infty$; $T X(k) \rightarrow X(j\omega)$

$k\omega_0 \rightarrow \omega$ $\omega_0 \rightarrow d\omega$.

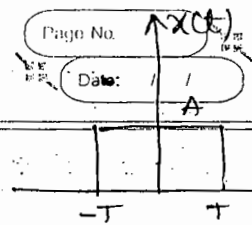
& summation becomes an integration

$$\therefore x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$$

Inverse Fourier Transform of $X(j\omega)$

① Find the Fourier Transform of $x(t)$

shown in the fig.



$$x(t) = \begin{cases} A & -T < t < T \\ 0 & \text{otherwise} \end{cases}$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_{-T}^T A e^{-j\omega t} dt = A \int_{-T}^T e^{-j\omega t} dt \quad (1)$$

$$= \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-T}^T = A \left[\frac{e^{+j\omega T} - e^{-j\omega T}}{j\omega} \right] = A \frac{2 \sin \omega T}{\omega}$$

$$X(j\omega) = 2AT \frac{\sin \omega T}{\omega T} = 2AT \text{sinc}[\omega T]$$

$$\boxed{X(j\omega) = 2AT \text{sinc}(\omega T)}$$

for $\omega=0$, we get $X(j\omega)$ as indeterminate form

hence put $\omega=0$ in eqn (1)

$$X(j\omega) = \int_{-T}^T A \cdot e^0 dt = 2AT \Big|_{\omega=0}$$

$$\therefore X(j\omega) = \begin{cases} 2AT & \omega=0 \\ 2AT \text{sinc} \omega T & \omega \neq 0 \end{cases}$$

$$|X(j\omega)| = |2A \frac{\sin \omega T}{\omega}| \quad \dots \text{Magnitude spectrum.}$$

Phase spectrum $\angle X(j\omega)$

$$\text{at } \omega=0, \quad X(j\omega) = 2AT$$

To find the zero crossings put $X(j\omega)=0$ for $\omega \neq 0$

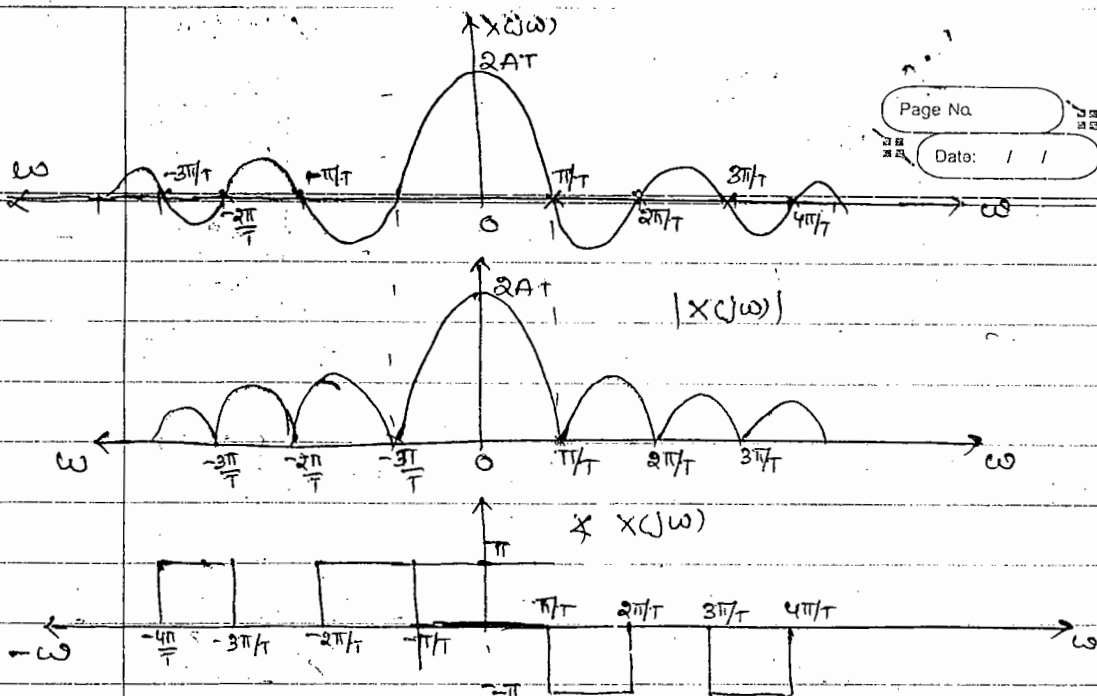
$$\therefore 2A \frac{\sin \omega T}{\omega} = 0 \Rightarrow \sin \omega T = 0$$

$$\omega T = \pi, 2\pi, 3\pi, 4\pi, \dots, n\pi$$

$$\therefore \boxed{\omega = \frac{n\pi}{T}}$$

$$\therefore \angle X(j\omega) = \begin{cases} 0 & \sin \frac{\omega T}{\omega} > 0 \\ \pi & \sin \frac{\omega T}{\omega} < 0 \text{ \& } \omega > 0 \\ -\pi & \sin \frac{\omega T}{\omega} > 0 \text{ \& } \omega < 0 \end{cases}$$

ω
0
a
2a
∞
-a
-2a
$-\infty$



② Find the f.T of $x(t) = e^{-at} u(t)$.

for $a < 0$, the transform will not converge.

hence for $a > 0$, the transform has an absolute value.

$$\begin{aligned}
 X(j\omega) &= \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} e^{-at} u(t) e^{-j\omega t} dt = \int_0^{\infty} e^{-at-j\omega t} dt \\
 &= \int_0^{\infty} e^{-(a+j\omega)t} dt = \left[\frac{-1}{(a+j\omega)} e^{-(a+j\omega)t} \right]_0^{\infty}
 \end{aligned}$$

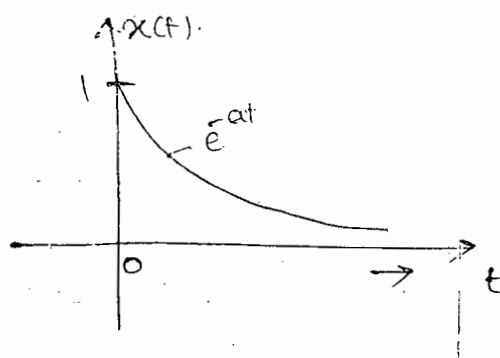
$$X(j\omega) = \frac{1}{a+j\omega}$$

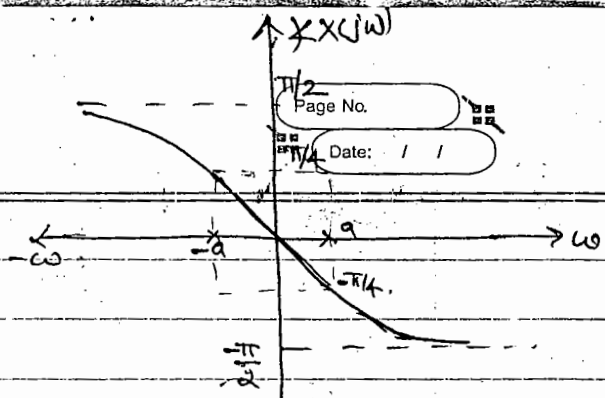
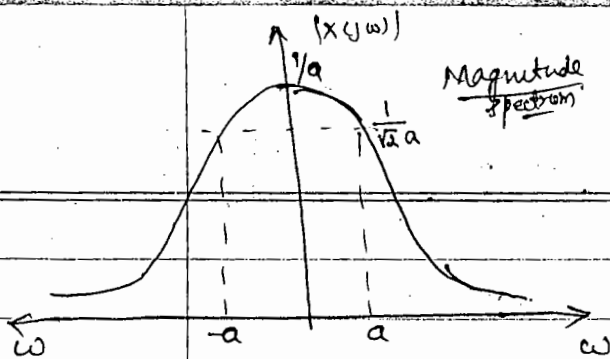
$$|X(j\omega)| = \frac{1}{\sqrt{a^2 + \omega^2}}$$

$$\begin{aligned}
 \angle X(j\omega) &= \frac{1}{a+j\omega} \cdot \frac{a-j\omega}{a-j\omega} \\
 &= \frac{a}{a^2 + \omega^2} - j \frac{\omega}{a^2 + \omega^2}
 \end{aligned}$$

$$\angle X(j\omega) = \tan^{-1}\left(\frac{b}{a}\right) = \tan^{-1}\left[\frac{-\omega/a^2 + \omega^2}{a/a^2 + \omega^2}\right] = -\tan^{-1}\left(\frac{\omega}{a}\right)$$

ω	$ X(j\omega) $	$\angle X(j\omega)$
0	$\frac{1}{a}$	0
a	$\frac{1}{\sqrt{2}a}$	$-\pi/4$
2a	$\frac{1}{\sqrt{5}} \frac{1}{a}$	-63°
∞	0	$-\pi/2$
-a	$\frac{1}{\sqrt{2}} \frac{1}{a}$	$\pi/4$
-2a	$\frac{1}{\sqrt{5}} \frac{1}{a}$	$+63^\circ$
$-\infty$	0	$\pi/2$

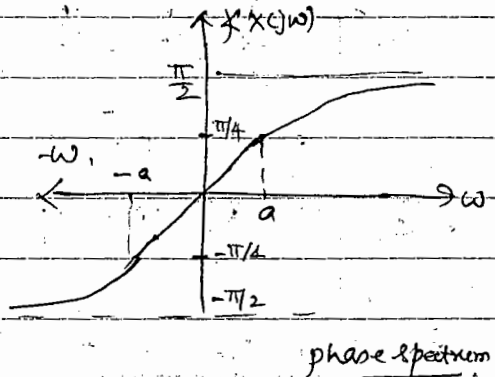




③ $x(t) = e^{at} u(-t)$ for $a > 0$

$$X(j\omega) = \int_{-\infty}^0 e^{at} e^{-j\omega t} dt = \int_{-\infty}^0 e^{(a-j\omega)t} dt$$

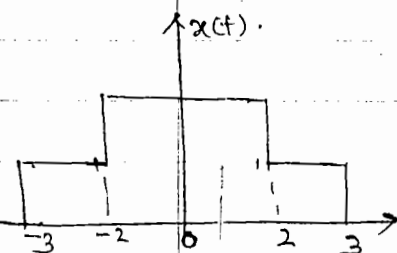
$$= \frac{e^{(a-j\omega)t}}{a-j\omega} \Big|_{-\infty}^0 = \frac{1}{a-j\omega}$$



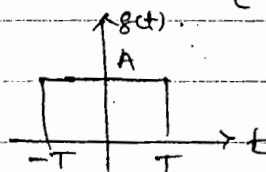
$$|X(j\omega)| = \frac{1}{(a^2 + \omega^2)^{1/2}}$$

$$\angle X(j\omega) = \tan^{-1}\left(\frac{\omega}{a}\right)$$

④ Find the Fourier transform of $x(t)$ shown in figure.



Let $g(t) = \begin{cases} A; & -T < t < T \\ 0; & \text{otherwise.} \end{cases}$



put $A=1, T=1$, in $G(\omega)$.

$$g(t) \xrightarrow{FT} 2A \frac{\sin \omega T}{\omega} = 2 \frac{\sin \omega}{\omega}$$

$$\therefore g(t/2) \xrightarrow{FT} \frac{1}{|a|} X(j\omega/a) = \frac{1}{2} \cdot 2 \frac{\sin 2\omega}{\omega/2}$$

$$g(t/3) \xrightarrow{FT} \frac{1}{|a|} X(j\omega/a) = \frac{1}{3} \cdot 2 \frac{\sin 3\omega}{\omega/3}$$

$$\therefore x(t) = g(t/2) + g(t/3) \longleftrightarrow 2 \frac{\sin 2\omega}{\omega} + 2 \frac{\sin 3\omega}{\omega}$$

$$X(j\omega) = 4 \frac{\sin 2\omega}{2\omega} + 6 \frac{\sin 3\omega}{3\omega}$$

$$X(j\omega) = 4 \frac{\sin 2\omega}{2\omega} + 6 \frac{\sin 3\omega}{3\omega}$$

Properties of Fourier Transform

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1. Linearity:

→ ω

If $x(t) \xleftrightarrow{FT} X(j\omega)$ and
 $y(t) \xleftrightarrow{FT} Y(j\omega)$; then

$$a x(t) + b y(t) \xleftrightarrow{FT} a X(j\omega) + b Y(j\omega)$$

Proof:- The FT of any arbitrary signal $x(t)$ is

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

→ ω

Let $z(t) = a x(t) + b y(t)$

$$Z(j\omega) = \int_{-\infty}^{\infty} z(t) \cdot e^{-j\omega t} dt = \int_{-\infty}^{\infty} [a x(t) + b y(t)] e^{-j\omega t} dt$$

spectrum

$$= \int_{-\infty}^{\infty} a x(t) e^{-j\omega t} dt + \int_{-\infty}^{\infty} b y(t) e^{-j\omega t} dt$$

$$Z(j\omega) = a X(j\omega) + b Y(j\omega)$$

The FT of Linear combination of signals is equal to the Linear combination of their transforms.

T

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2. Time Shifting:-

If $x(t) \xleftrightarrow{FT} X(j\omega)$ then

$$x(t-t_0) \xleftrightarrow{FT} e^{-j\omega t_0} X(j\omega)$$

Proof:-

we have $X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$

Let $y(t) = x(t-t_0)$

$$Y(j\omega) = \int_{-\infty}^{\infty} y(t) \cdot e^{-j\omega t} dt = \int_{-\infty}^{\infty} x(t-t_0) e^{-j\omega t} dt$$

put $t-t_0 = \lambda$ $\therefore t = (\lambda + t_0)$
 $dt = d\lambda$

$$\therefore Y(j\omega) = \int_{\lambda=-\infty}^{\infty} x(\lambda) e^{-j\omega(\lambda+t_0)} d\lambda = e^{-j\omega t_0} \int_{-\infty}^{\infty} x(\lambda) e^{-j\omega \lambda} d\lambda$$

is

$$Y(j\omega) = e^{-j\omega t_0} X(j\omega) \quad \text{--- RHS}$$

||| 4

$$x(t+t_0) \xleftrightarrow{FT} e^{j\omega t_0} X(j\omega)$$

Hence a shift in Time domain is equivalent to a phase-shift in frequency domain with amplitude remains same.

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(3). Frequency shift

5.

$$\text{If } x(t) \xleftrightarrow{FT} X(j\omega)$$

$$e^{j\beta t} x(t) \xleftrightarrow{FT} X(j(\omega - \beta))$$

Proof: $X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$

proof

Let $y(t) = e^{j\beta t} x(t)$

$$Y(j\omega) = \int_{-\infty}^{\infty} y(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} e^{j\beta t} x(t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} x(t) e^{-j(\omega - \beta)t} dt = X[j(\omega - \beta)] \dots \underline{\underline{RHS}}$$

4. Time Scaling:

$$\text{If } x(t) \xleftrightarrow{FT} X(j\omega)$$

then $x(at) \xleftrightarrow{FT} \frac{1}{|a|} X(j\frac{\omega}{a})$

Proof:-

we have $X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$

Let $a > 1$

Let $y(t) = x(at)$

$$Y(j\omega) = \int_{-\infty}^{\infty} y(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} x(at) e^{-j\omega t} dt$$

D4

put $at = \lambda$
 $dt = d\lambda/a$

$$Y(j\omega) = \int_{-\infty}^{\infty} x(\lambda) e^{-j\omega \lambda/a} \frac{d\lambda}{a}$$

6. F:

$$= \frac{1}{a} \int_{-\infty}^{\infty} x(\lambda) e^{-j\omega \lambda/a} d\lambda$$

$$\boxed{Y(j\omega) = \frac{1}{a} X(j\frac{\omega}{a})}$$

pre

Let $a < 1$ (a is -ve.)

put $-at = \tau \therefore dt = -\frac{d\tau}{a}$

$t = -\tau/a$

$$Y(j\omega) = \int_{-\infty}^{\infty} x(\tau) e^{-j\omega(-\tau/a)} \frac{-d\tau}{a} = -\frac{1}{a} \int_{-\infty}^{\infty} x(\tau) e^{-j(\frac{\omega}{a})\tau} d\tau$$

$$= -\frac{1}{a} [-X(j\frac{\omega}{a})] = \frac{1}{a} X(j\frac{\omega}{a})$$

hence $x(at) \xleftrightarrow{FT} \frac{1}{|a|} X(j\frac{\omega}{a})$

Compression or expansion of signal in Time domain is equivalent to expansion or compression of signal in freq. domain respectively.

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5. Time domain differentiation

If $x(t) \xleftrightarrow{FT} X(j\omega)$. then

$$\frac{d}{dt} [x(t)] \xleftrightarrow{FT} j\omega \cdot X(j\omega).$$

proof:

$$\text{we have } x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega.$$

$$\frac{d}{dt} x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) \cdot \left[\frac{d}{dt} e^{j\omega t} \right] d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) \cdot j\omega \cdot e^{j\omega t} d\omega.$$

$$= j\omega \cdot \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) \cdot e^{j\omega t} d\omega$$

$$= j\omega \cdot x(t).$$

Taking FT on both sides

$$\boxed{\mathcal{F} \left[\frac{d}{dt} x(t) \right] = j\omega \cdot X(j\omega)}$$

Differentiation in time domain corresponds to X^n by $j\omega$ in freq. domain.

6. Frequency domain differentiation

If $x(t) \xleftrightarrow{FT} X(j\omega)$. then

$$-jt \cdot x(t) \xleftrightarrow{FT} \frac{d}{d\omega} [X(j\omega)]$$

proof:

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$$

$$\frac{d}{d\omega} X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot dt \cdot \frac{d}{d\omega} (e^{-j\omega t}) = \int_{-\infty}^{\infty} x(t) \cdot dt \cdot (-jt) e^{-j\omega t}$$

$$= -jt \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt = -jt \cdot x(t)$$

Taking Inverse FT on both sides

$$\mathcal{F}^{-1} \left[\frac{d}{d\omega} X(j\omega) \right] \xleftrightarrow{FT} -jt \cdot x(t) \rightarrow \text{RHS.}$$

7. Convolution

$$\text{If } x(t) \xleftrightarrow{FT} X(j\omega) \text{ \& } y(t) \xleftrightarrow{FT} Y(j\omega) \text{ then}$$

$$x(t) * y(t) \xleftrightarrow{FT} X(j\omega) \cdot Y(j\omega)$$

proof

$$\text{Let } z(t) = x(t) * y(t) = \int_{-\infty}^{\infty} x(\tau) \cdot y(t-\tau) \cdot d\tau$$

$$Z(j\omega) = \int_{-\infty}^{\infty} z(t) \cdot e^{-j\omega t} \cdot dt = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x(\tau) \cdot y(t-\tau) \cdot d\tau \cdot e^{-j\omega t} \cdot dt$$

$$= \int_{-\infty}^{\infty} x(\tau) \cdot \int_{-\infty}^{\infty} y(t-\tau) \cdot e^{-j\omega t} \cdot dt \cdot [e^{-j\omega \tau} \cdot e^{j\omega \tau}]$$

$$= \int_{-\infty}^{\infty} x(\tau) \cdot e^{-j\omega \tau} \cdot \int_{-\infty}^{\infty} y(t-\tau) \cdot e^{-j\omega(t-\tau)} \cdot dt \cdot d\tau = \underline{X(j\omega) \cdot Y(j\omega)}$$

Convolution in time domain = $\times$ in freq. domain

$$\text{convolution} \xleftrightarrow{FT} \text{modulation}$$

8. Modulation

$$\text{If } x(t) \xleftrightarrow{FT} X(j\omega) \text{ \& } y(t) \xleftrightarrow{FT} Y(j\omega) \text{ then}$$

$$x(t) \cdot y(t) \xleftrightarrow{FT} [X(j\omega) * Y(j\omega)]$$

proof

$$\text{Let } z(t) = x(t) \cdot y(t) = x(t) \cdot y(t)$$

$$Z(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot y(t) \cdot e^{-j\omega t} \cdot dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\beta) \cdot e^{j\beta t} \cdot d\beta$$

$$Z(j\omega) = \int_{-\infty}^{\infty} \frac{1}{2\pi} \left[\int_{-\infty}^{\infty} X(j\beta) \cdot e^{j\beta t} \cdot d\beta \right] y(t) \cdot e^{-j\omega t} \cdot dt$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\beta) \cdot d\beta \int_{-\infty}^{\infty} y(t) \cdot e^{-j(\omega-\beta)t} \cdot dt$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\beta) \cdot d\beta \cdot Y(j(\omega-\beta))$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\beta) Y(j(\omega - \beta)) d\beta$$

$$= \frac{1}{2\pi} [X(j\omega) * Y(j\omega)] \quad \underline{\text{RHS}}$$

Modulation in Time domain corresponds to convolution of spectrum in frequency domain.

9. Duality

If $x(t) \xrightarrow{\text{FT}} X(j\omega)$ then

$$X(jt) \longleftrightarrow 2\pi x(-\omega)$$

proof

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) \cdot e^{j\omega t} d\omega$$

$$\int_{-\infty}^{\infty} X(j\omega) \cdot e^{j\omega t} d\omega = 2\pi x(t)$$

put $t = -t$.

$$\int_{-\infty}^{\infty} X(j\omega) \cdot e^{-j\omega t} d\omega = 2\pi x(-t)$$

Interchange t and ω .

$$\int_{-\infty}^{\infty} X(jt) \cdot e^{-j\omega t} dt = 2\pi x(-\omega)$$

$$\therefore \mathcal{F}[X(jt)] = 2\pi x(-\omega)$$

$$\therefore \boxed{X(jt) \xrightarrow{\text{FT}} 2\pi x(-\omega)}$$

This property explores the consistent symmetry existing b/n time and frequency domains. Hence we can interchange frequency and time.

10. Symmetry property

For a real signal $x(t)$;

If $x(t) \xrightarrow{\text{FT}} X(j\omega)$ then

$$x_e(t) \xrightarrow{\text{FT}} X_R(j\omega) \quad \&$$

$$x_o(t) \xrightarrow{\text{FT}} jX_I(j\omega)$$

proof If $x(t)$ is real, then $x(t) = x^*(t)$.

We have $x(t) \xleftrightarrow{FT} X_R(j\omega) + jX_I(j\omega)$

$$\therefore x(-t) \xleftrightarrow{FT} X_R(j\omega) - jX_I(j\omega)$$

$$x_e(t) = \frac{x(t) + x(-t)}{2} \xleftrightarrow{FT} \frac{1}{2} [X_R(j\omega) + jX_I(j\omega) + X_R(j\omega) - jX_I(j\omega)]$$

$$\boxed{x_e(t) \xleftrightarrow{FT} X_R(j\omega)}$$

$$\text{III}^{\text{ly}} \boxed{x_o(t) \xleftrightarrow{FT} jX_I(j\omega)}$$

Hence FT of even signal is real & odd signal is imaginary

11. Parseval's Theorem (Rayleigh's Criterion)

If $x(t) \xleftrightarrow{FT} X(j\omega)$ then

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega$$

The energy of the signal can be obtained by interchanging its energy spectrum.

proof:

$$\begin{aligned} E &= \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} x(t) \cdot x^*(t) \cdot dt \\ &= \int_{-\infty}^{\infty} x(t) \cdot \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(j\omega) e^{-j\omega t} d\omega \right] dt \end{aligned}$$

changing the order of integration.

$$\begin{aligned} E &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(j\omega) \left[\int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} \cdot dt \right] d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(j\omega) \cdot X(j\omega) \cdot d\omega \end{aligned}$$

$$\boxed{E = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega}$$

RHS

12. Time reverse property

from time scaling property

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$$x(at) \xleftrightarrow{FT} \frac{1}{|a|} X\left(\frac{j\omega}{a}\right)$$

put $a = -1$;

$$x(-t) = \frac{1}{|a|} X(-j\omega)$$

13. Integration

If $x(t) \xleftrightarrow{FT} X(j\omega)$ then

$$\int_{-\infty}^t x(\tau) \cdot d\tau \xleftrightarrow{FT} \frac{1}{j\omega} X(j\omega)$$

proof

$$\text{Let } x(t) = \frac{d}{dt} \left[\int_{-\infty}^t x(\tau) \cdot d\tau \right]$$

$$F[x(t)] = F\left[\frac{d}{dt} \left[\int_{-\infty}^t x(\tau) \cdot d\tau \right]\right]$$

$$X(j\omega) = j\omega F\left[\int_{-\infty}^t x(\tau) \cdot d\tau\right]$$

$$\therefore F\left[\int_{-\infty}^t x(\tau) \cdot d\tau\right] = \frac{1}{j\omega} X(j\omega) \quad \dots \text{RHS}$$

The above equation holds good for signals whose average value is zero. Hence for any signal in general a dc term is added to make it compatible

$$\therefore \int_{-\infty}^t x(\tau) \cdot d\tau \longleftrightarrow \frac{1}{j\omega} [X(j\omega)] + \pi X(j0) \cdot \delta(\omega)$$

$$\text{where } X(j0) = X(j\omega)|_{\omega=0}$$

Integration in time domain represents smoothening. It is like de-emphasizing of frequency components of the signal. The diff. operation destroys the dc components of $x(t)$ and $X(j\omega) = 0$ at $\omega = 0$. Hence a dc term is added to have compatibility

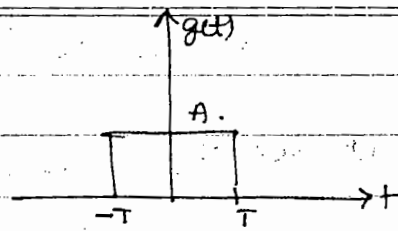
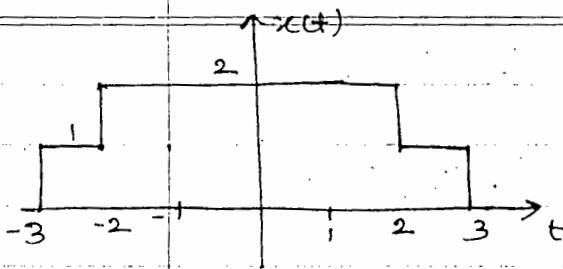
$$\begin{aligned} x_1(t) * x_2(t) &= \int_{-\infty}^{\infty} x_1(\tau) x_2(t-\tau) d\tau \\ \& \ x_1(t) = x(t) \\ x_2(t) &= u(t) \\ x(t) * u(t) &= \int_{-\infty}^t x(\tau) \cdot d\tau \\ u(t-\tau) &= \begin{cases} 1, & \tau < t \\ 0, & \tau > t \end{cases} \end{aligned}$$

problems:

Find the FT of $x(t)$ ~~using~~ ~~g(t)~~

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we have $g(t) \longleftrightarrow G(j\omega) = 2A \frac{\sin \omega T}{\omega}$

here $A=1, T=1$.

$$g(t/2) \longleftrightarrow \frac{1}{a} G(j\omega/a) = \frac{2 \cdot \sin(\omega/2)}{(1/2) \cdot (\omega/2)} = 2 \frac{\sin 2\omega}{\omega}$$

$$g(t/3) \longleftrightarrow \frac{1}{(1/3)} \left[\frac{2 \sin(\omega/3)}{(\omega/3)} \right] = \frac{6 \sin 3\omega}{3\omega} = 2 \frac{\sin 3\omega}{\omega}$$

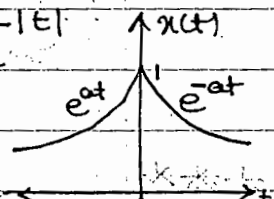
$$x(t) = g(t/2) + g(t/3)$$

$$\therefore x(t) \xrightarrow{FT} \frac{2 \sin 2\omega}{\omega} + \frac{2 \sin 3\omega}{\omega}$$

$$X(j\omega) = 4 \sin 2\omega + 6 \sin 3\omega$$

② $x(t) = e^{-at} u(t) + e^{at} u(-t) = e^{-|t|}$

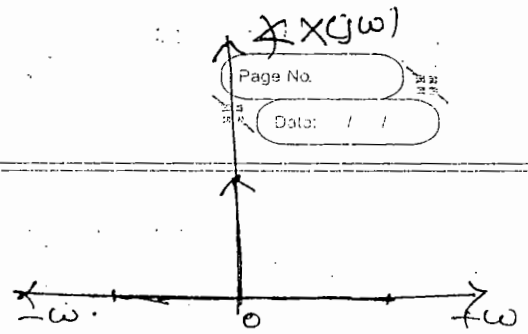
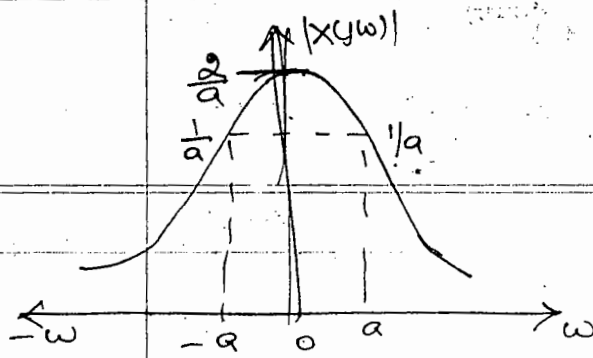
$$X(j\omega) = \int_{-\infty}^0 e^{at} e^{-j\omega t} dt + \int_0^{\infty} e^{-at} e^{-j\omega t} dt$$



$$= \frac{1}{a-j\omega} + \frac{1}{a+j\omega} = \frac{2a}{a^2 + \omega^2}$$

$$\therefore X(j\omega) = \frac{2a}{a^2 + \omega^2} = |X(j\omega)|$$

$$\angle X(j\omega) = 0$$



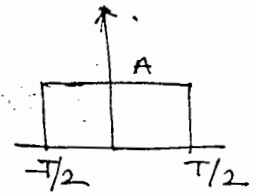
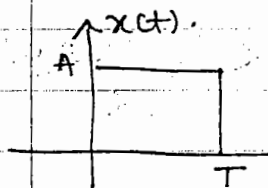
③ $x(t) = e^{-2t} u(t-1)$

$$X(j\omega) = \int_{-\infty}^{\infty} e^{-2t} u(t-1) e^{-j\omega t} dt = \int_1^{\infty} e^{-2t} e^{-j\omega t} dt$$

$$= \int_1^{\infty} e^{-(2+j\omega)t} dt = \frac{-1}{2+j\omega} \left[e^{-(2+j\omega)t} \right]_1^{\infty}$$

$$X(j\omega) = \frac{e^{-(2+j\omega)}}{2+j\omega}$$

④ Find the FT of $x(t) = \begin{cases} A; & 0 < t < T \\ 0; & \text{otherwise} \end{cases}$



We know that $g(t) \longleftrightarrow G(j\omega)$

$$G(j\omega) = 2A \frac{\sin \omega T/2}{\omega}$$

$$g(t-T/2) \longleftrightarrow e^{-j\omega T/2} G(j\omega) = e^{-j\omega T/2} \left[\frac{\sin \omega T/2}{\omega} \right] 2A$$

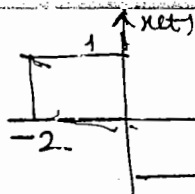
$$x(t) = g(t-T/2)$$

$$\therefore X(j\omega) = 2A e^{-j\omega T/2} \frac{\sin \omega T/2}{\omega}$$

$$= AT e^{-j\omega T/2} \left[\frac{\sin \omega T/2}{\omega T/2} \right]$$

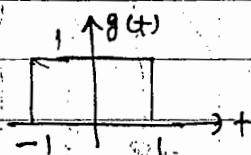
$$X(j\omega) = AT e^{-j\omega T/2} \text{sinc}(\omega T/2)$$

Find the FT of



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Let $g(t) = \begin{cases} 1; & 0 < t < 1 \\ 0; & \text{otherwise} \end{cases}$



$$x(t) = g(t+1) - g(t-1)$$

$$g(t) \longleftrightarrow G(j\omega) = 2 \frac{\sin \omega}{\omega}$$

(8)

$$g(t+1) \longleftrightarrow e^{j\omega} G(j\omega) = e^{j\omega} 2 \frac{\sin \omega}{\omega}$$

$$g(t-1) \longleftrightarrow e^{-j\omega} G(j\omega) = e^{-j\omega} 2 \frac{\sin \omega}{\omega}$$

$$\therefore x(t) = g(t+1) - g(t-1)$$

$$X(j\omega) = 2 \frac{\sin \omega}{\omega} [e^{j\omega} - e^{-j\omega}]$$

$$= 2 \frac{\sin \omega}{\omega} \cdot 2j \sin \omega$$

$$= 4j \frac{\sin^2 \omega}{\omega} = \frac{2j}{\omega} \cdot 2(1 - \cos 2\omega)$$

$$X(j\omega) = \frac{2}{j\omega} [\cos 2\omega - 1]$$

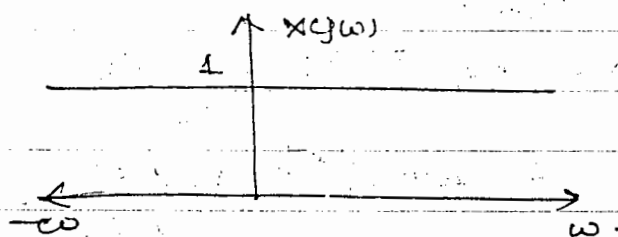
(6)

$$x(t) = \delta(t)$$

$$X(j\omega) = \int_{-\infty}^{\infty} \delta(t) \cdot e^{-j\omega t} dt = e^{-j\omega t} \Big|_{t=0} = 1$$

(9)

$$\delta(t) \xrightarrow{\text{FT}} 1$$



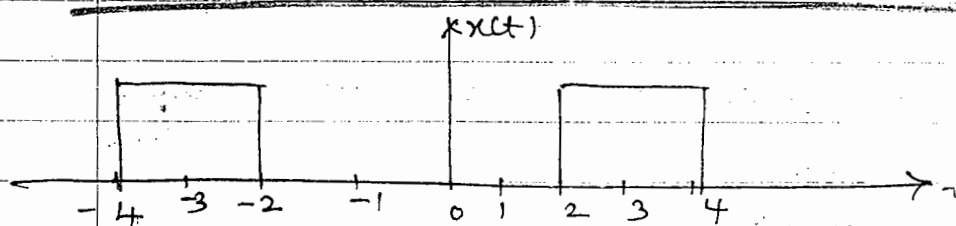
$$\textcircled{7} \quad x(t) = \delta(t - t_0)$$

$$X(j\omega) = \int_{-\infty}^{\infty} \delta(t - t_0) e^{-j\omega t} dt$$

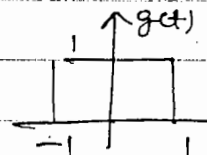
$$= \left[e^{-j\omega t} \right]_{t=t_0} = \underline{\underline{e^{-j\omega t_0}}}$$

$$\text{III} \quad \delta(t + t_0) \longleftrightarrow e^{j\omega t_0}$$

⑧ Find the FT of the wave form shown



$$\text{Let } g(t) = \begin{cases} 1 & ; 0 < t < 1 \\ 0 & ; \text{otherwise} \end{cases}$$



$$x(t) = g(t+3) + g(t-3)$$

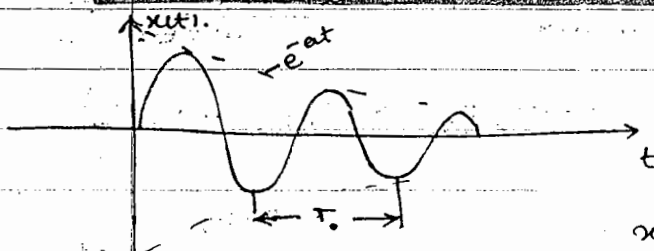
using the time shifting property

$$X(j\omega) = [e^{j\omega 3} + e^{-j\omega 3}] G(j\omega) = 2 \cos 3\omega \cdot G(j\omega)$$

$$= 2 \cos 3\omega \left[\frac{2 \sin \omega}{\omega} \right]$$

$$\boxed{X(j\omega) = \frac{4}{\omega} \cos 3\omega \cdot \sin \omega}$$

⑨ Find the FT of $x(t)$ shown below



$$\text{Let } x_1(t) = x(t) e^{-\alpha t}$$

$$\text{Let } x_1(t) = e^{\alpha t} u(t)$$

$$\therefore x(t) = x_1(t) \cdot \sin \omega_0 t$$

$$x(t) = x_1(t) \left[\frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j} \right]$$

$$= \frac{1}{2j} [e^{j\omega_0 t} x_1(t) - e^{-j\omega_0 t} x_1(t)]$$

Now using freq. shift property $e^{j\beta t} x(t) \longleftrightarrow X[j(\omega - \beta)]$

$$X_c(j\omega) = \frac{1}{2j} [X_1(j(\omega - \omega_0)) - X_1(j(\omega + \omega_0))]$$

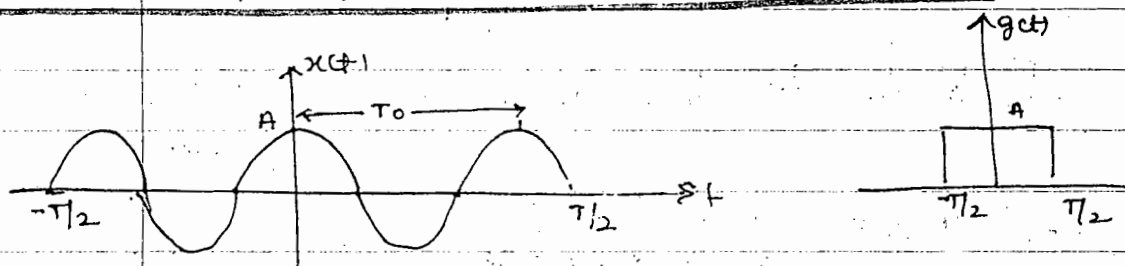
we have $x_1(t) = e^{-at} u(t)$

$$X_1(j\omega) = \frac{1}{a + aj\omega}$$

$$\therefore X_c(j\omega) = \frac{1}{2j} \left[\frac{1}{a + j(\omega - \omega_0)} - \frac{1}{a + j(\omega + \omega_0)} \right]$$

$$X_c(j\omega) = \frac{1}{2j} \left[\frac{\omega_0}{a^2 + 2aj\omega - (\omega^2 - \omega_0^2)} \right]$$

10 Find the FT of the wave $x(t)$ shown below:



$$\text{Let } g(t) = \begin{cases} A & -T/2 < t < T/2 \\ 0 & \text{otherwise} \end{cases}$$

$$G(j\omega) = \frac{2A \sin \omega T/2}{\omega}$$

$$x(t) = g(t) \cdot \cos \omega_0 t = g(t) \left[\frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} \right]$$

$$= \frac{1}{2} [e^{j\omega_0 t} g(t) + e^{-j\omega_0 t} g(t)]$$

using freq. shift property

$$X_c(j\omega) = \frac{1}{2} [G(j(\omega - \omega_0)) + G(j(\omega + \omega_0))]$$

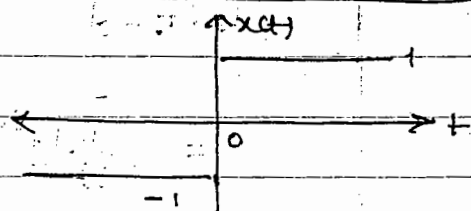
$$= \frac{1}{2} \left[\frac{2A \sin(\omega - \omega_0) T/2}{\omega - \omega_0} + \frac{2A \sin(\omega + \omega_0) T/2}{\omega + \omega_0} \right]$$

$$X_c(j\omega) = \frac{A}{\omega} [\sin(\omega - \omega_0) T/2 + \sin(\omega + \omega_0) T/2]$$

Find the FT of Signum f_n.

$$11. x(t) = \text{sgn}(t)$$

$$\text{sgn}(t) = \begin{cases} 1 & t > 0 \\ 0 & t = 0 \\ -1 & t < 0 \end{cases}$$

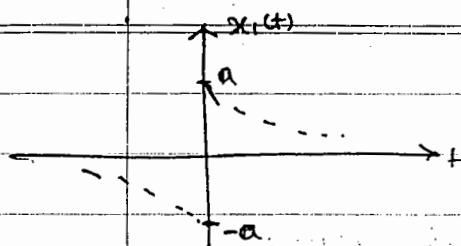


Method 1 $\text{sgn}(t)$ can be considered as

a limiting form of $x_1(t)$ shown below

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$$x_1(t) = \begin{cases} -e^{at} u(-t) & ; t < 0 \\ 0 & ; t = 0 \\ e^{at} u(t) & ; t > 0 \end{cases}$$

$$\therefore \text{sgn}(t) = \lim_{a \rightarrow 0} x_1(t)$$

$$x_1(t) = e^{at} u(t) - e^{-at} u(-t)$$

$$F[\text{sgn}(t)] = X_1(j\omega) = \lim_{a \rightarrow 0} \left[\frac{1}{a+j\omega} - \frac{1}{a-j\omega} \right]$$

$$= \lim_{a \rightarrow 0} \left[\frac{-2j\omega}{a^2 + \omega^2} \right] = -\frac{2j}{\omega} = \frac{2}{j\omega}$$

$$\therefore X(j\omega) = \frac{-2j}{\omega} = \frac{2}{j\omega}$$

Method 2

$$x(t) = \text{sgn}(t)$$

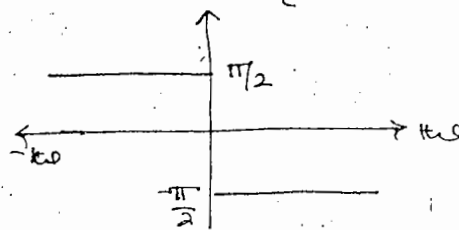
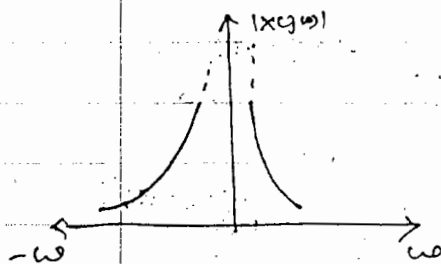
$$\frac{d}{dt} x(t) = 2\delta(t)$$

using time diff. property of FT

$$j\omega X(j\omega) = 2(1)$$

$$\therefore X(j\omega) = \frac{2}{j\omega}$$

$$|X(j\omega)| = 2/\omega \quad \& \quad \angle X(j\omega) = \begin{cases} -\pi/2 & ; \omega > 0 \\ \pi/2 & ; \omega < 0 \end{cases}$$



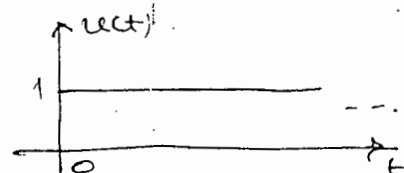
12. Find the FT of unit step f_n $u(t)$

$$\text{Let } x(t) = u(t)$$

We have the relationship

$$\text{sgn}(t) = 2u(t) - 1$$

$$\therefore u(t) = \frac{\text{sgn}(t) + 1}{2}$$



$$\therefore x(t) = \frac{\text{sgn}(t) + 1}{2}$$

$$X(j\omega) = \frac{1}{2} \left[\frac{2}{j\omega} + 2\pi\delta(\omega) \right]$$

$$1 \xleftrightarrow{FT} 2\pi\delta(\omega)$$

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$$X(j\omega) = \pi\delta(\omega) + \frac{1}{j\omega}$$

13 Find The Fourier Transform of $x(t) = 1$

15.

Here Dirichlet conditions are not satisfied. However from duality property, we have

$$x(t) \longleftrightarrow X(j\omega)$$

$$X(t) \longleftrightarrow 2\pi x(-\omega)$$

$$\therefore g(t) \longleftrightarrow 1$$

$$1 \longleftrightarrow 2\pi\delta(-\omega)$$

Since $\delta(\omega) = \delta(-\omega)$ even f?

$$1 \xleftrightarrow{FT} 2\pi\delta(\omega)$$

14 Find the FT of $x(t) = \begin{cases} 1 + \cos\pi t & |t| < 1 \\ 0 & |t| > 1 \end{cases}$

$$X(j\omega) = \int_{-\infty}^{\infty} [1 + \cos\pi t] e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} e^{-j\omega t} dt + \int_{-1}^1 e^{-j\omega t} \left[\frac{e^{j\pi t} + e^{-j\pi t}}{2} \right] dt$$

$$= \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-\infty}^{\infty} + \int_{-1}^1 \frac{e^{-j t(\omega - \pi)} + e^{-j t(\omega + \pi)}}{2} dt$$

$$= \left[\frac{e^{-j\omega} + e^{j\omega}}{j\omega} \right] + \frac{1}{2} \left[\left[\frac{e^{-j t(\omega - \pi)}}{-j(\omega - \pi)} \right]_{-1}^1 + \left[\frac{e^{-j t(\omega + \pi)}}{-j(\omega + \pi)} \right]_{-1}^1 \right]$$

$$= \frac{2}{\omega} \sin\omega + \frac{1}{2} \left[\frac{e^{j(\omega - \pi)} - e^{-j(\omega - \pi)}}{j(\omega - \pi)} + \frac{e^{j(\omega + \pi)} - e^{-j(\omega + \pi)}}{j(\omega + \pi)} \right]$$

16.

$$= \frac{2}{\omega} \sin \omega + \frac{\sin(\omega - \pi)}{\omega - \pi} + \frac{\sin(\omega + \pi)}{\omega + \pi}$$

$$X(j\omega) = \frac{2}{\omega} \sin \omega + \frac{\sin \omega}{(\pi - \omega)} - \frac{\sin \omega}{\pi + \omega}$$

15. $x(t) = e^{-3t} u(t)$

from

$$X(j\omega) = \int_{-\infty}^{\infty} e^{-3t} u(t) \cdot e^{-j\omega t} dt = \int_0^{\infty} e^{-(3+j\omega)t} dt$$

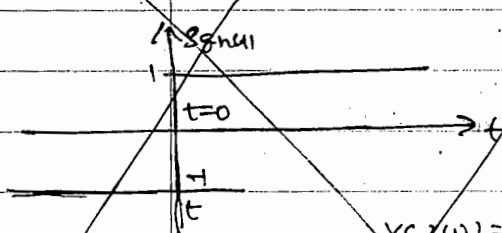
$$= \left[\frac{e^{-(3+j\omega)t}}{-(3+j\omega)} \right]_0^{\infty} = \frac{-e^{-(3+j\omega)t}}{3+j\omega} \Big|_0^{\infty} = \frac{e^{-3}}{3+j\omega}$$

$$X(j\omega) = \frac{e^{-3}}{3+j\omega}$$

$$|X(j\omega)| = \frac{e^{-3}}{\sqrt{9+\omega^2}} = \frac{1}{20\sqrt{\omega^2+9}}$$

$$e^{-3} = \frac{1}{20}$$

Find the FT of $\text{sgn}(t)$



$$\text{sgn}(t) = \begin{cases} 1 & t > 0 \\ 0 & t = 0 \\ -1 & t < 0 \end{cases}$$

$$X(j\omega) = \int_{-\infty}^{\infty} \text{sgn}(t) e^{-j\omega t} dt = \int_{-\infty}^0 -e^{-j\omega t} dt + \int_0^{\infty} e^{-j\omega t} dt$$

$$= \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-\infty}^0 + \left[\frac{e^{-j\omega t}}{-j\omega} \right]_0^{\infty}$$

$$= \frac{e^{+j\omega}}{-j\omega} + \frac{e^{-j\omega}}{j\omega} = \frac{2}{j\omega}$$

16. Find the FT of $\text{sgn}(t)$

Method 3

$$X(j\omega) = - \int_{-\infty}^0 e^{-j\omega t} dt + \int_0^{\infty} e^{-j\omega t} dt$$

$$= \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-\infty}^0 + \left[\frac{e^{-j\omega t}}{-j\omega} \right]_0^{\infty} = \frac{+1}{j\omega} + \frac{1}{j\omega} = \frac{2}{j\omega}$$

$$\left[\frac{e^{-j(\omega+\pi)t}}{-j(\omega+\pi)} \right]_{-\infty}^{\infty}$$

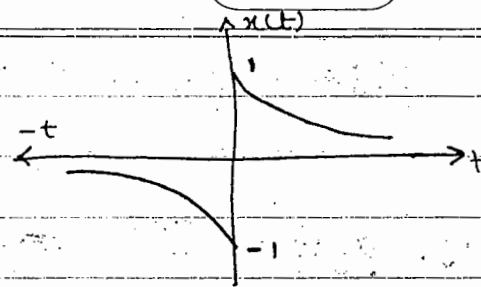
$$x(t) = e^{-a|t|} \operatorname{sgn}(t)$$

$$x(t) = \begin{cases} e^{at} & t > 0 \\ -e^{-at} & t < 0 \end{cases}$$

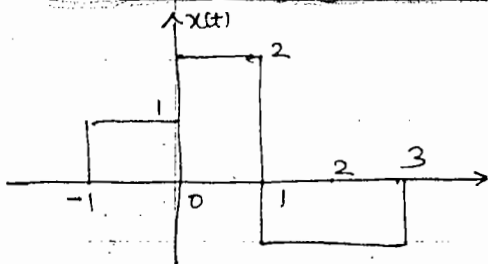
$$X(j\omega) = \int_{-\infty}^0 e^{at} e^{-j\omega t} dt + \int_0^{\infty} -e^{-at} e^{-j\omega t} dt$$

$$= \frac{1}{a+j\omega} - \frac{1}{a-j\omega} = \frac{-2j\omega}{a^2 + \omega^2}$$

$$X(j\omega) = \frac{2\omega}{j(a^2 + \omega^2)}$$



17 Compute the FT of the waveform shown below.



$$x(t) = \begin{cases} 1 & -1 < t < 0 \\ 2 & 0 < t < 1 \\ -1 & 1 < t < 3 \end{cases}$$

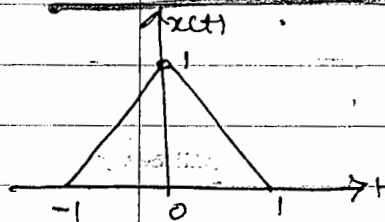
$$X(j\omega) = \int_{-1}^0 e^{-j\omega t} dt + \int_0^1 2e^{-j\omega t} dt + \int_1^3 -e^{-j\omega t} dt$$

$$= \left. \frac{e^{-j\omega t}}{-j\omega} \right|_{-1}^0 + 2 \left. \frac{e^{-j\omega t}}{-j\omega} \right|_0^1 + \left. \frac{e^{-j\omega t}}{j\omega} \right|_1^3$$

$$= \frac{1}{j\omega} [-1 + e^{j\omega} + 2e^{-j\omega} + 2 + e^{-j3\omega} - e^{-j\omega}]$$

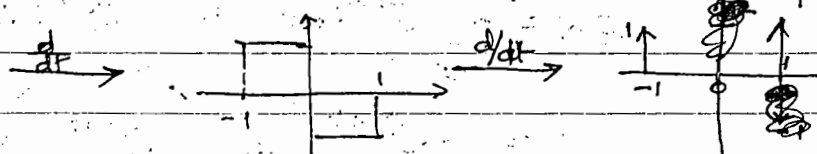
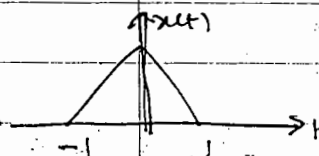
$$X(j\omega) = \frac{1}{j\omega} [1 + e^{j\omega} - 3e^{-j\omega} + e^{-j3\omega}]$$

18. Find the FT of $x(t)$



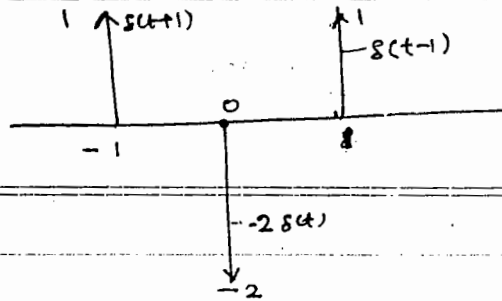
$$x(t) = \begin{cases} 1-t & 0 < t < 1 \\ 1+t & -1 < t < 0 \\ 0 & \text{otherwise} \end{cases}$$

Method 1 (using diff. property)



$$\frac{d^2}{dt^2} [x(t)] = \delta(t+1) + 2\delta(t) + \delta(t-1)$$

$$\frac{d^2}{dt^2} x(t) \Rightarrow$$



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$$\therefore \frac{d^2}{dt^2} x(t) = \delta(t+1) - 2\delta(t) + \delta(t-1)$$

$$(j\omega)^2 X(j\omega) = e^{j\omega} - 2 + e^{-j\omega}$$

$$X(j\omega) = \frac{1}{(j\omega)^2} [-2 + 2\cos\omega]$$

$$X(j\omega) = \frac{2}{\omega^2} [1 - \cos\omega]$$

Method 2

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt = \int_{-1}^0 (1+t) e^{-j\omega t} dt + \int_0^1 (1-t) e^{-j\omega t} dt$$

$$= \int_{-1}^0 t e^{-j\omega t} dt + \int_{-1}^0 e^{-j\omega t} dt + \int_0^1 e^{-j\omega t} dt - \int_0^1 t e^{-j\omega t} dt$$

Note: $\int t e^{-at} dt = -\frac{e^{-at}}{a^2} (1+at)$

$$X(j\omega) = \left[\frac{t e^{-j\omega t}}{-j\omega} + \frac{e^{-j\omega t}}{\omega^2} \right]_{-1}^0 + \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-1}^0 - \left[\frac{t e^{-j\omega t}}{-j\omega} + \frac{e^{-j\omega t}}{\omega^2} \right]_0^1 + \left[\frac{e^{-j\omega t}}{-j\omega} \right]_0^1$$

$$= -\frac{e^{j\omega}}{j\omega} + \frac{1-e^{j\omega}}{\omega^2} - \left[\frac{1-e^{j\omega}}{j\omega} \right] + \frac{e^{j\omega}}{j\omega} + \frac{1-e^{j\omega}}{j\omega} - \left[\frac{e^{j\omega}-1}{j\omega} \right]$$

$$= \frac{2}{\omega^2} - \frac{e^{j\omega}}{\omega^2} - \frac{e^{-j\omega}}{\omega^2} = \frac{2 - e^{j\omega} + e^{-j\omega}}{\omega^2}$$

$$= \frac{2 - 2\cos\omega}{\omega^2} = \frac{2}{\omega^2} [1 - \cos\omega]$$

$$\therefore X(j\omega) = \frac{2}{\omega^2} (1 - \cos\omega)$$

19. $x(t) = A \cos\omega_0 t e^{-at} u(t)$. Find $X(j\omega)$.

$$x(t) = e^{-at} u(t) \longleftrightarrow X(j\omega) = \frac{1}{a+j\omega}$$

using $x(t) = e^{-at} u(t)$

we have

$$\cos \omega_0 t = \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2}$$

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(23)

$$x(t) = \frac{A}{2} [e^{j\omega_0 t} x_1(t) + e^{-j\omega_0 t} x_1(t)]$$

Using freq. shifting property

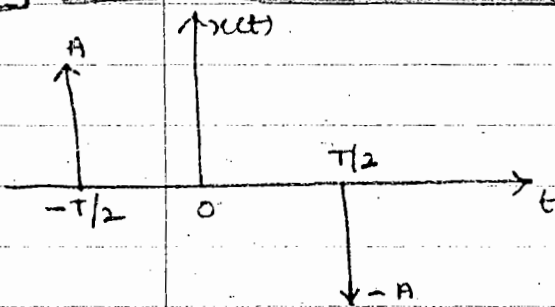
$$X(j\omega) = \frac{A}{2} [x_1(j\omega - j\omega_0) + x_1(j\omega + j\omega_0)]$$

$$X(j\omega) = \frac{A}{2} \left[\frac{1}{a + j(\omega - \omega_0)} + \frac{1}{a + j(\omega + \omega_0)} \right]$$

$$X(j\omega) = \frac{A(a + j\omega)}{a^2 + (\omega - \omega_0)^2 + 2aj\omega}$$

$$X(j\omega) = \frac{A(a + j\omega)}{a^2 + (\omega - \omega_0)^2 + 2aj\omega}$$

20 Find the FT of the signal shown below



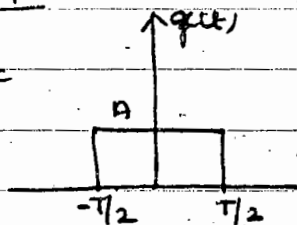
$$x(t) = A[\delta(t + T/2) - \delta(t - T/2)]$$

$$X(j\omega) = A[e^{j\omega T/2} - e^{-j\omega T/2}]$$

$$X(j\omega) = 2jA \sin(\omega T/2)$$

Method 2

Let $g(t) =$



$$G(j\omega) = \frac{2A \sin \omega T/2}{\omega}$$

$$x(t) = \frac{d}{dt} g(t)$$

$$X(j\omega) = j\omega G(j\omega) = j\omega \frac{2A \sin \omega T/2}{\omega}$$

$$\therefore X(j\omega) = 2jA \sin \omega T/2$$

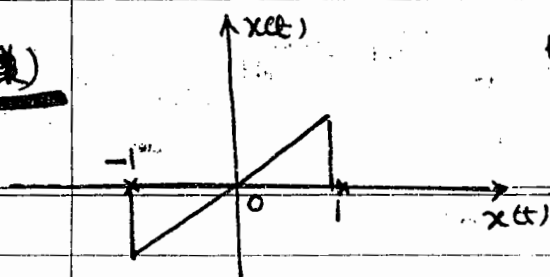
(24)

(21)

Find $X(j\omega)$

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$$x(t) = t; -1 < t < 1$$

0; otherwise.

$$X(j\omega) = \int_{-1}^1 t e^{-j\omega t} dt = \left. \frac{t e^{-j\omega t}}{-j\omega} - \frac{e^{-j\omega t}}{(-j\omega)^2} \right|_{-1}^1$$

$$= -\frac{e^{-j\omega}}{j\omega} + \frac{e^{-j\omega}}{\omega^2} - \frac{t e^{-j\omega t}}{j\omega} - \frac{e^{-j\omega t}}{\omega^2}$$

$$= -\frac{1}{j\omega} [e^{j\omega} + e^{-j\omega}] - \frac{1}{\omega^2} [e^{j\omega} - e^{-j\omega}]$$

$$= -\frac{2}{j\omega} \cos \omega - \frac{2j}{\omega^2} (\sin \omega)$$

$$= \frac{2j}{\omega} \cos \omega - \frac{2j}{\omega^2} \sin \omega$$

$$= \frac{2j}{\omega} [\cos \omega - \frac{\sin \omega}{\omega}] = \frac{2j}{\omega} [\cos \omega - \text{sinc } \omega]$$

-T/2

T/2

(22)

$$x(t) = t e^{-2t} u(t) \quad \text{Find } X(j\omega)$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} \cdot u(t) dt = \int_0^{\infty} t e^{-j\omega t} \cdot e^{-2t} dt = \int_0^{\infty} t e^{-(2+j\omega)t} dt$$

$$X(j\omega) = -e^{-(2+j\omega)t} \left[\frac{1 - (2+j\omega)t}{(2+j\omega)^2} \right]_0^{\infty} = \frac{1}{(2+j\omega)^2}$$

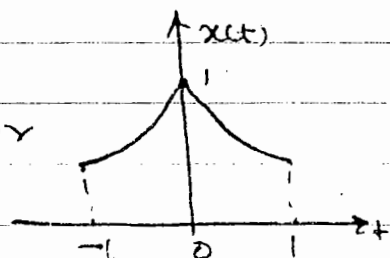
$$X(j\omega) = \frac{1}{(2+j\omega)^2}$$

T/2

24) Find the FT of $x(t) = e^{-|t|}$

or

$$x(t) = \begin{cases} e^t & -1 < t < 0 \\ e^{-t} & 0 < t < 1 \end{cases}$$



$$x(j\omega) = \int_{-1}^0 e^t e^{-j\omega t} dt + \int_0^1 e^{-t} e^{-j\omega t} dt$$

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$$= \int_{-1}^0 e^{(1-j\omega)t} dt + \int_0^1 e^{-(1+j\omega)t} dt$$

$$= \left[\frac{e^{(1-j\omega)t}}{1-j\omega} \right]_{-1}^0 - \left[\frac{e^{-(1+j\omega)t}}{1+j\omega} \right]_0^1$$

$$= \frac{1}{1-j\omega} - \frac{e^{(j\omega-1)t}}{1-j\omega} - \frac{e^{-(1+j\omega)t}}{1+j\omega} + \frac{1}{1+j\omega}$$

$$x(j\omega) = \frac{1 - e^{-(1+j\omega)t}}{1+j\omega} + \frac{1 - e^{(j\omega-1)t}}{1-j\omega}$$

Q5) $x(t) = t e^{-at} u(t)$ find $X(j\omega)$.

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_0^{\infty} t e^{-at} dt e^{-j\omega t}$$

$$X(j\omega) = \int_0^{\infty} t e^{-(a+j\omega)t} dt = \frac{1}{(a+j\omega)^2}$$

Method 2: $e^{-at} u(t) \xrightarrow{FT} \frac{1}{a+j\omega}$

using freq. domain diff. property

$$-jt x(t) \longleftrightarrow \frac{d}{d\omega} X(j\omega)$$

Let $x_1(t) = e^{-at} u(t) \longleftrightarrow x_1(j\omega) = \frac{1}{a+j\omega}$ (dk)

$x(t) = t x_1(t) \longleftrightarrow \frac{1}{j} \frac{d}{d\omega} x_1(j\omega)$

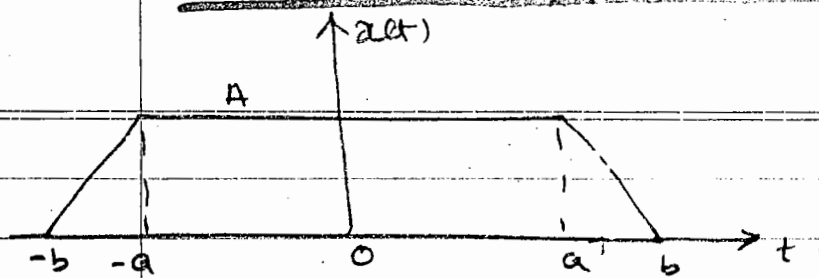
$$X(j\omega) = j \left[\frac{d}{d\omega} \left(\frac{1}{a+j\omega} \right) \right]$$

$$= j \left[\frac{0 - j}{(a+j\omega)^2} \right] = \frac{1}{(a+j\omega)^2}$$

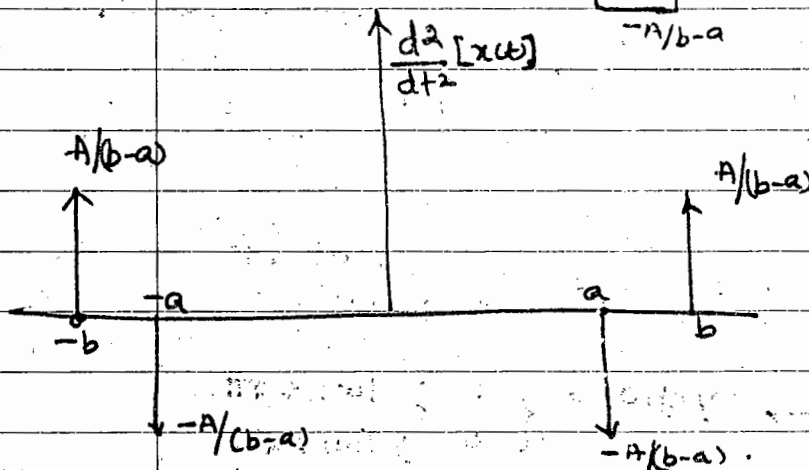
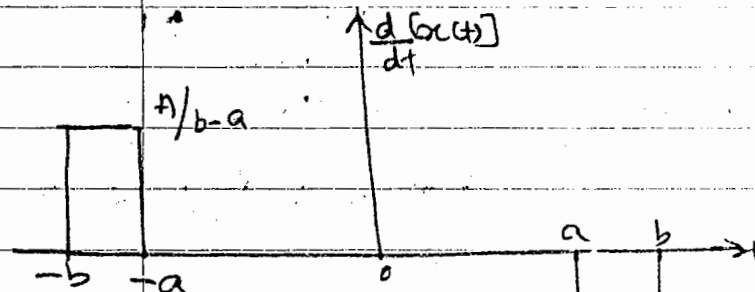
Find the FT of the waveform, $x(t)$

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$$\text{Slope} = \frac{y}{x} = \frac{A}{(b-a)}$$



$$\frac{d^2}{dt^2}[x(t)] = \frac{A}{(b-a)} [\delta(t+b) - \delta(t+a) - \delta(t-a) + \delta(t-b)]$$

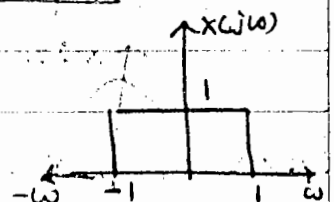
Taking FT on both sides

$$(j\omega)^2 X(j\omega) = \frac{A}{(b-a)} [e^{j\omega b} - e^{j\omega a} - e^{-j\omega a} + e^{-j\omega b}]$$

$$X(j\omega) = \frac{2A}{\omega(b-a)} [\cos\omega a - \cos\omega b]$$

Find the inverse FT of

$$X(j\omega) = \begin{cases} 1 & ; |\omega| < 1 \\ 0 & ; |\omega| > 1 \end{cases}$$



$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} x(t) e^{j\omega t} d\omega$$

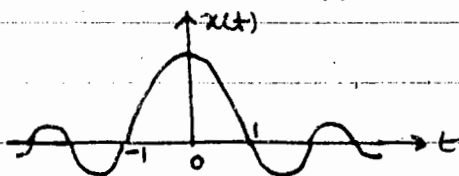
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$$= \int_{-\infty}^{\infty} \frac{1}{2\pi} e^{j\omega t} d\omega = \frac{1}{2\pi} \left[\frac{e^{j\omega t}}{jt} \right]_{-\infty}^{\infty}$$

$$= \frac{1}{2\pi jt} [e^{j\omega t} - e^{-j\omega t}] = \frac{1}{\pi t} \left[\frac{e^{j\omega t} - e^{-j\omega t}}{2j} \right]$$

$$x(t) = \frac{\sin t}{\pi t}$$



Q8) Using convolution properly find $G(j\omega)$ where $g(t) = x(t) * y(t)$. given.

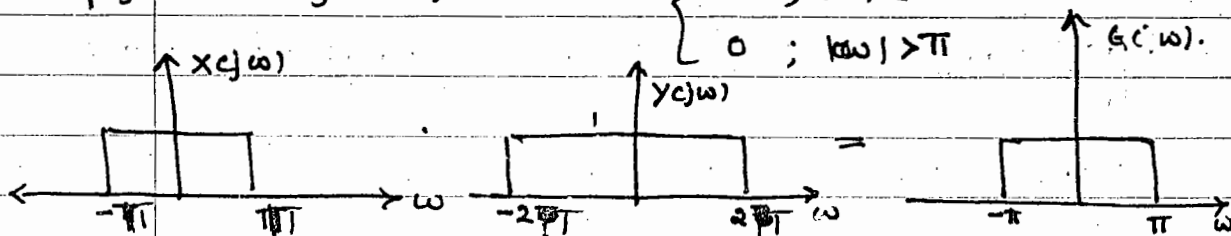
$$x(t) = \frac{1}{\pi t} \sin \pi t$$

$$y(t) = \frac{1}{\pi t} \sin 2\pi t$$

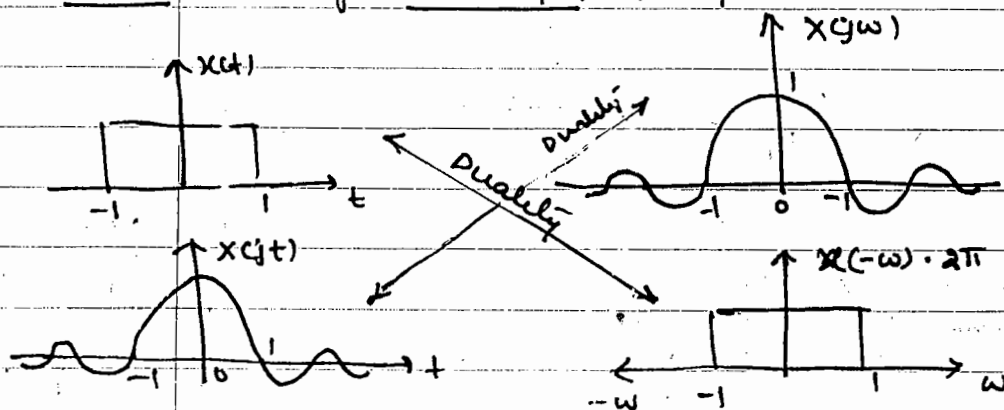
$$x(t) = \frac{\sin \pi t}{\pi t} \longleftrightarrow x(j\omega) = \begin{cases} 1 & ; |\omega| \leq \pi \\ 0 & ; |\omega| > \pi \end{cases}$$

$$y(t) = \frac{\sin 2\pi t}{\pi t} \longleftrightarrow y(j\omega) = \begin{cases} 1 & ; |\omega| \leq 2\pi \\ 0 & ; |\omega| > 2\pi \end{cases}$$

$$G(j\omega) = x(j\omega) \cdot y(j\omega) = \begin{cases} 1 & ; |\omega| \leq \pi \\ 0 & ; |\omega| > \pi \end{cases}$$



Note Using Duality properly



30

Let

29) Using Modulation Property find $x(t)$

Given $x(t) = \frac{4}{\pi^2 t^2} \sin^2 \pi t$

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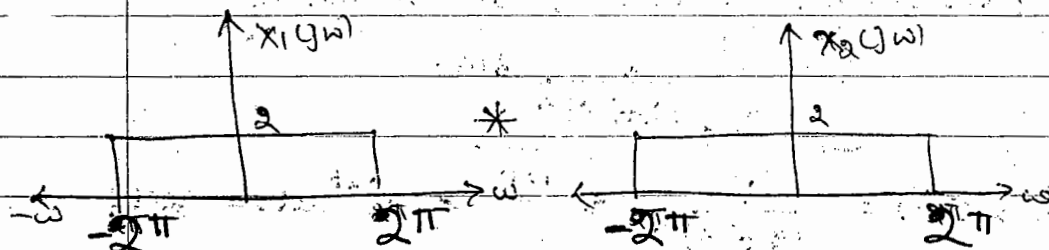
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Let $x_1(t) = \frac{2}{\pi t} \sin \pi t$ & $x_2(t) = \frac{2}{\pi t} \sin \pi t$

$\therefore x(t) = x_1(t) \cdot x_2(t)$

$\therefore X(j\omega) = \frac{1}{2\pi} X_1(j\omega) * X_2(j\omega) = \frac{1}{2\pi} X_1(j\omega) * X_1(j\omega)$

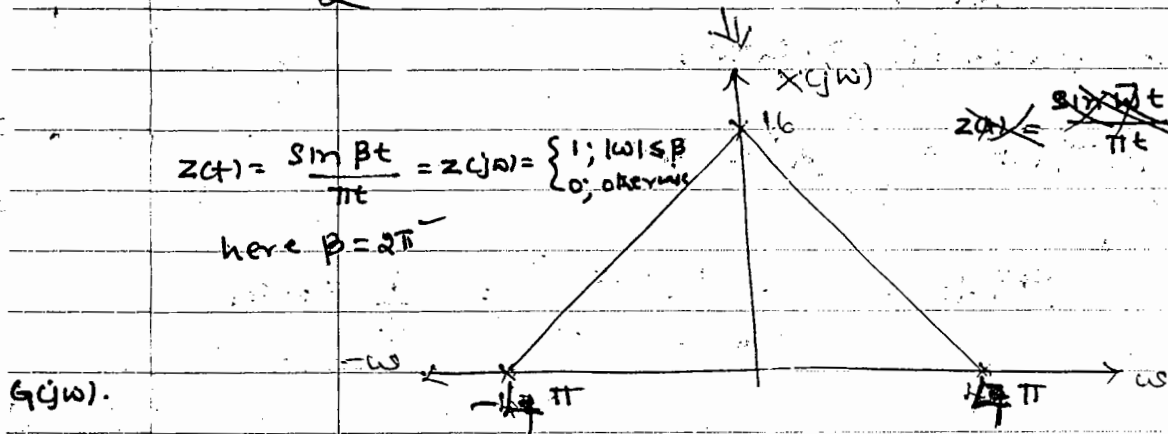
$X_1(j\omega) = X_2(j\omega) = \begin{cases} 1; & |\omega| \leq 2\pi \\ 0 & |\omega| > 2\pi \end{cases}$



$z(t) = \frac{\sin \beta t}{\pi t} = z(j\omega) = \begin{cases} 1; & |\omega| \leq \beta \\ 0; & \text{otherwise} \end{cases}$

here $\beta = 2\pi$

~~$z(t) = \frac{\sin \pi t}{\pi t}$~~ $\Rightarrow z(j\omega) = \begin{cases} 1; & |\omega| \leq 1 \\ 0 & \text{otherwise} \end{cases}$



30) Use convolution property to find $x(t)$: where

$x(t) \longleftrightarrow X(j\omega) = \frac{4}{\omega^2} \sin^2 \omega$

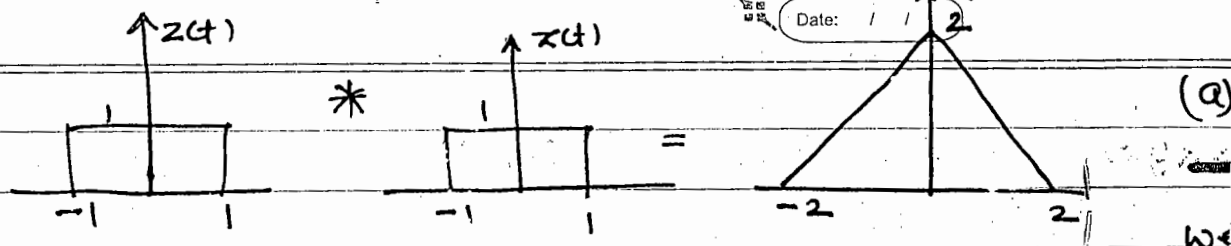
From the FT pair

Let $z(j\omega) = \frac{2 \sin \omega}{\omega} \xleftrightarrow{\text{FT}} z(t) = \begin{cases} 1; & |t| \leq 1 \\ 0; & |t| > 1 \end{cases}$

$X(j\omega) = z(j\omega) \cdot z(j\omega) = \frac{4}{\omega^2} \sin^2 \omega$

Taking inverse FT, we get the convolution of $z(t)$ & $z(t)$ in time domain.

$$\therefore x(t) = z(t) * z(t).$$



$$\therefore x(t) = \begin{cases} 2-t & ; 0 \leq t \leq 2 \\ t+2 & ; -2 \leq t \leq 0 \\ 0 & ; |t| \geq 2 \end{cases}$$

3) Use Parseval's theorem to evaluate.

$$Z = \int_{-\infty}^{\infty} \frac{2}{|2+j\omega|^2} d\omega.$$

$$\text{Let } x(j\omega) = \frac{2}{(2+j\omega)^2} \xleftrightarrow{\text{FT}} e^{-2t} u(t) = x(t)$$

using Parseval's relation

$$\begin{aligned} E &= \int_{-\infty}^{\infty} |e^{-2t} u(t)|^2 dt = \int_{-\infty}^{\infty} |x(j\omega)|^2 d\omega \\ &= \int_0^{\infty} e^{-4t} dt = \frac{1}{-4} \left[e^{-4t} \right]_0^{\infty} = \frac{1}{4} \text{ joules} \end{aligned}$$

$$\int_{-\infty}^{\infty} \frac{2}{|2+j\omega|^2} d\omega = Z$$

$$\therefore \frac{Z}{2} = \int_{-\infty}^{\infty} \frac{d\omega}{|2+j\omega|^2} = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{4}$$

$$\therefore Z = \frac{2}{4} = \frac{1}{2}$$

$$\boxed{\therefore Z = \frac{1}{2}}$$

32) Find the FT of the following signals using duality properly

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(a) $x(t) = \frac{1}{a^2 + t^2}$

we have

$$e^{-a|t|} \xleftrightarrow{FT} \frac{2a}{a^2 + \omega^2}$$

$$\frac{1}{2a} e^{-a|t|} \longleftrightarrow \frac{1}{a^2 + \omega^2}$$

using duality properly

$$\frac{1}{a^2 + t^2} \longleftrightarrow \frac{1}{2a} 2\pi e^{-a|\omega|}$$

$$\boxed{\frac{1}{a^2 + t^2} \xleftrightarrow{FT} \frac{\pi}{a} e^{-a|\omega|}}$$

(b) $x(t) = A \cos \omega_0 t$

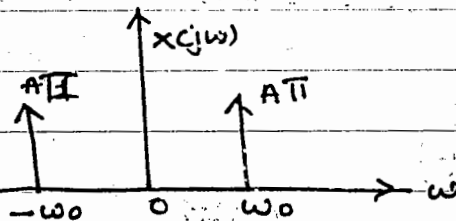
$$x(t) = \frac{A}{2} [e^{j\omega_0 t} + e^{-j\omega_0 t}]$$

we have $\delta(t) \longleftrightarrow 1$

$$1 \longleftrightarrow 2\pi \delta(\omega)$$

$$e^{j\omega_0 t} \longleftrightarrow \delta(\omega - \omega_0) 2\pi$$

$$\therefore \mathcal{F}(x(t)) \xleftrightarrow{FT} \frac{A}{2} [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] = X(j\omega)$$



(c) $x(t) = A \sin \omega_0 t$

$$x(t) = \frac{A}{2j} (e^{j\omega_0 t} - e^{-j\omega_0 t})$$

using duality properly

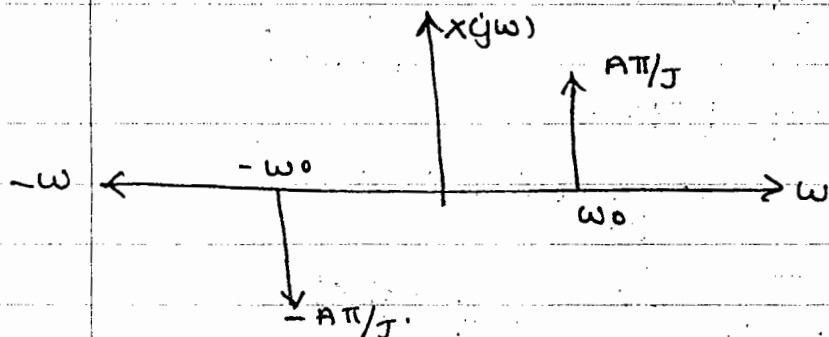
$$1. e^{\pm j\omega_0 t} \longleftrightarrow \delta(\omega \pm \omega_0) \cdot 2\pi$$

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$$x(t) \longleftrightarrow \frac{A}{2} \cdot \frac{2\pi}{j} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)] = x(j\omega)$$

$$x(j\omega) = \frac{A\pi}{j} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$$



33 Find FT of the following signals

a) $x(t) = e^{-3t} \cos \pi t u(t)$

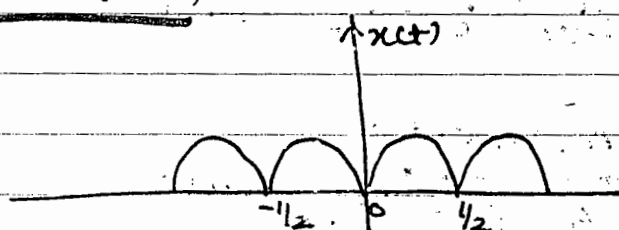
$$\text{Let } x_1(t) = e^{-3t} u(t) \longleftrightarrow \frac{1}{3+j\omega} = x_1(j\omega)$$

$$\cos \pi t = \frac{e^{+j\pi t} + e^{-j\pi t}}{2}$$

$$\therefore x(t) = \frac{x_1(t)}{2} [e^{j\pi t} + e^{-j\pi t}] \longleftrightarrow \frac{1}{2} [x_1(j(\omega - \pi)) + x_1(j(\omega + \pi))]$$

$$x(j\omega) = \frac{1}{2} \left[\frac{1}{3+j(\omega+\pi)} + \frac{1}{3+j(\omega-\pi)} \right]$$

b) $x(t) = |\sin 2\pi t|$



$$T = 1/2$$

$$\omega = 4\pi$$

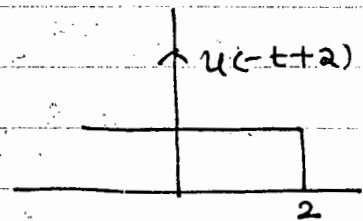
$$X(j\omega) = \frac{1}{T} \int_{\langle T \rangle} \sin 2\pi t e^{-jk\omega t} dt$$

$$X(j\omega) = 2 \int_0^{1/2} \left[\frac{e^{j2\pi t} - e^{-j2\pi t}}{2j} \right] e^{-j4\pi t} dt$$

$$= \frac{1 - e^{-j\pi(1-2k)}}{2\pi(1-2k)} + \frac{1 - e^{-j\pi(1+2k)}}{2\pi(1+2k)}$$

© $x(t) = e^{(1+t)} u(-t+2)$

$$X(j\omega) = \int_{-\infty}^2 e^{1+t} \cdot e^{-j\omega t} dt$$



$$= \int_{-\infty}^2 e \cdot e^{(1-j\omega)t} dt = e \left[\frac{e^{(1-j\omega)t}}{1-j\omega} \right]_{-\infty}^2$$

$$= e \left[\frac{e^2 - e^{-2j\omega}}{1-j\omega} \right] = \frac{e^3 - e^{-2j\omega+1}}{1-j\omega}$$

$x_1(j\omega)$

(34) Find the inverse transform of $X(j\omega) = \frac{1}{(a+j\omega)^2}$ using convolution theorem.

$x_1(j\omega)$

$$X(j\omega) = \frac{1}{(a+j\omega)} \cdot \frac{1}{(a+j\omega)} = x_1(j\omega) \cdot x_2(j\omega)$$

Taking inverse FT

$$x(t) = x_1(t) * x_2(t)$$

$$= [e^{-at} * e^{-at}]_{u(t)} = \int_{-\infty}^{\infty} e^{-a\tau} u(\tau) \cdot e^{-a(t-\tau)} u(t-\tau) d\tau$$

$$= e^{-at} u(t) * e^{-at} u(t)$$

$$= \int_{-\infty}^{\infty} e^{-a\tau} u(\tau) \cdot e^{-a(t-\tau)} u(t-\tau) d\tau$$

$$= \int_{\tau=0}^t e^{-a\tau} d\tau = e^{-a\tau} \int_{\tau=0}^t d\tau = e^{-a\tau} \tau$$

$$\therefore x(t) = te^{-at}$$

$$\text{for } t > 0, \quad x(t) = te^{-at} u(t)$$

35) Find the FT of $x(t) = \frac{2}{t^2+1}$

From duality property

$$x(t) \longleftrightarrow X(j\omega)$$

$$X(jt) \longleftrightarrow 2\pi x(-\omega)$$

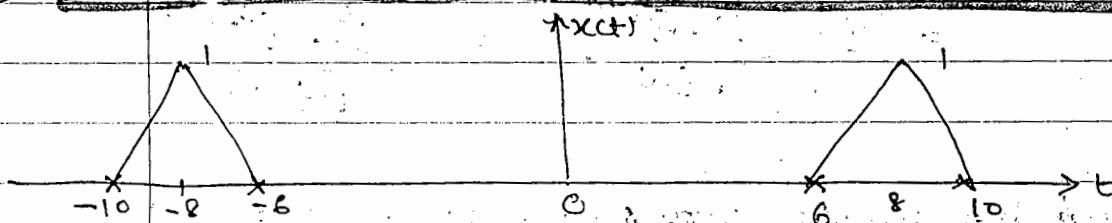
$$x(t) = e^{-|t|} \longleftrightarrow \frac{2}{1+\omega^2} = X(j\omega)$$

$$X(jt) = \frac{2}{1+t^2} \longleftrightarrow 2\pi x(-\omega) = 2\pi e^{-|\omega|}$$

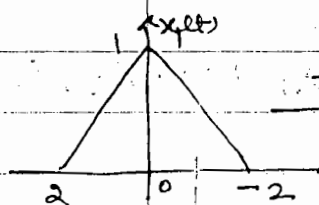
$$\therefore \frac{2}{1+t^2} \longleftrightarrow 2\pi e^{-|\omega|}$$

37) USI

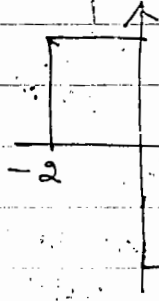
36) Find the FT of the following signal $x(t)$ shown below



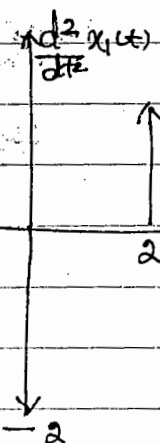
Let $x(t)$ be as shown below



$$\frac{d}{dt} x(t)$$



$$\frac{d^2}{dt^2} x(t)$$



$$\frac{d^2}{dt^2} [x(t)] = \delta(t+10) - \delta(t+6) + \delta(t-6) - \delta(t-10)$$

Taking FT on both sides

$$(j\omega)^2 X(j\omega) = e^{j\omega 10} - e^{j\omega 6} + e^{-j\omega 6} - e^{-j\omega 10}$$

$$X(j\omega) = \frac{e^{j\omega 10} - e^{j\omega 6} - e^{-j\omega 6} + e^{-j\omega 10}}{(j\omega)^2} = \frac{2}{(j\omega)^2}$$

$$-\frac{2\cos 2\omega}{\omega^2} + \frac{2}{\omega^2}$$

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$$X_1(j\omega) = \frac{2}{\omega^2} [1 - \cos 2\omega] = \frac{4 \sin^2 \omega}{\omega^2}$$

$$x(t) = x_1(t+8) + x_1(t-8)$$

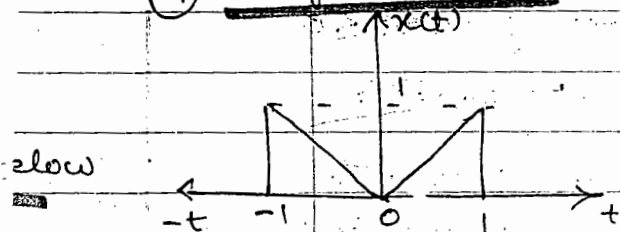
$$X(j\omega) = e^{-j\omega 8} X_1(j\omega) + e^{+j\omega 8} X_1(j\omega)$$

$$= X_1(j\omega) [e^{j\omega 8} + e^{-j\omega 8}]$$

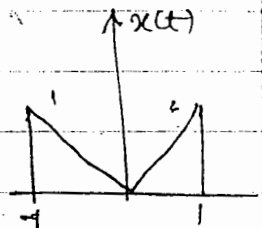
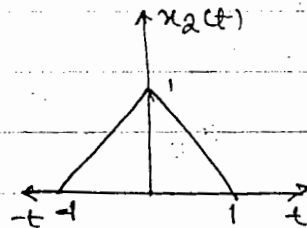
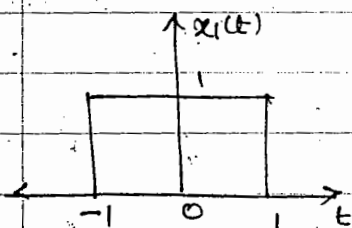
$$= \left[\frac{4 \sin^2 \omega}{\omega^2} \right] [2 \cos 8\omega]$$

$$X(j\omega) = \frac{8 \sin^2 \omega \cdot \cos 8\omega}{\omega^2}$$

(37) Using suitable FT pairs, Find FT of $x(t)$ shown.



Let



From the above figures

$$x(t) = x_1(t) - x_2(t)$$

$$X(j\omega) = X_1(j\omega) - X_2(j\omega)$$

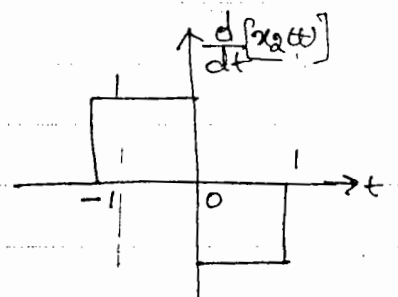
$$\frac{d[x_1(t)]}{dt} = \delta(t+1) - \delta(t-1)$$

$$X_1(j\omega) = \frac{e^{j\omega} - e^{-j\omega}}{j\omega} = \frac{2 \sin \omega}{\omega}$$

$$x_2(t) = \begin{cases} 1+t & ; -1 \leq t \leq 0 \\ -t+1 & ; 0 \leq t \leq 1 \end{cases}$$

$$\frac{d[x_2(t)]}{dt} = \begin{cases} +1 & ; -1 \leq t \leq 0 \\ -1 & ; 0 \leq t \leq 1 \end{cases}$$

$$X(j\omega) = X_1(j\omega) - X_2(j\omega)$$



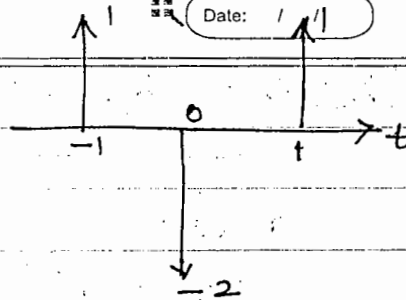
$$\frac{d^2}{dt^2} [x_2(t)] = \delta(t+1) - 2\delta(t) + \delta(t-1)$$

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$$x_2(j\omega) = \frac{e^{j\omega} + e^{-j\omega} - 2}{(j\omega)^2}$$

$$= \frac{2(1 - \cos \omega)}{\omega^2}$$



1. $x_2(t)$

$$x(j\omega) = x_1(j\omega) - x_2(j\omega)$$

$$= \frac{2}{\omega} \sin \omega - \frac{2(1 - \cos \omega)}{\omega^2}$$

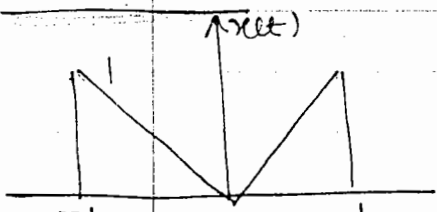
(2)

$$x(j\omega) = \frac{2}{\omega^2} [\omega \sin \omega + \cos \omega - 1]$$

(3)

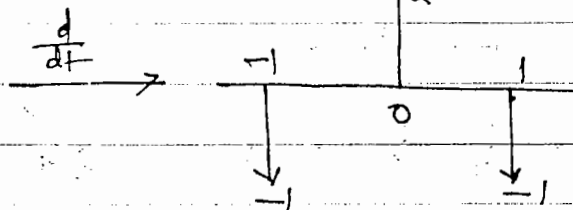
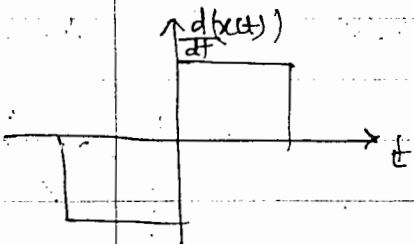
(4)

II method



$$x(t) = \begin{cases} -t & ; -1 \leq t \leq 0 \\ t & ; 0 \leq t \leq 1 \end{cases}$$

$$\frac{d}{dt} [x(t)] = \begin{cases} -1 & ; -1 \leq t \leq 0 \\ 1 & ; 0 \leq t \leq 1 \end{cases}$$



$$\frac{d^2}{dt^2} [x(t)] = -\delta(t+1) + \delta(t-1) + 2\delta(t)$$

$$x(j\omega) = \frac{1}{(j\omega)^2} [-(e^{j\omega} + e^{-j\omega}) + 2]$$

$$x(j\omega) = \frac{2\cos \omega - 2}{\omega^2}$$

(13)

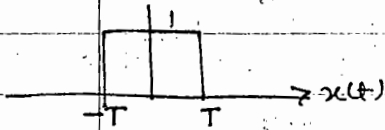
$x(t)$

FT pairs

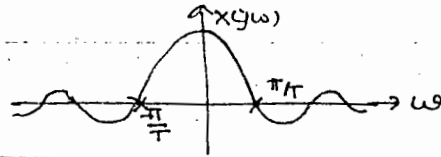
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$$1. x(t) = \begin{cases} 1 & |t| < T \\ 0 & |t| > T \end{cases}$$



$$X(j\omega) = \frac{2 \sin \omega T}{\omega}$$



$$(2) x(t) = \frac{\sin \omega_c t}{\pi t}$$

$$X(j\omega) = \begin{cases} 1 & |\omega| \leq \omega_c \\ 0 & |\omega| > \omega_c \end{cases}$$

$$(3) x(t) = \delta(t)$$

$$X(j\omega) = 1$$

$$(4) x(t) = 1$$

$$X(j\omega) = 2\pi \delta(\omega)$$

$$(5) x(t) = u(t)$$

$$X(j\omega) = \frac{1}{j\omega} + \pi \delta(\omega)$$

$$6. x(t) = e^{-at} u(t)$$

$$X(j\omega) = \frac{1}{a + j\omega}$$

$$7. x(t) = t e^{-at} u(t)$$

$$X(j\omega) = \frac{1}{(a + j\omega)^2}$$

$$(8) x(t) = e^{-|t|}$$

$$X(j\omega) = \frac{2a}{a^2 + \omega^2}$$

$$(9) x(t) = \delta(t \pm t_0)$$

$$X(j\omega) = e^{\pm j\omega t_0}$$

$$(10) x(t) = e^{j\omega_0 t}$$

$$X(j\omega) = 2\pi \delta(\omega - \omega_0)$$

$$11. x(t) = \cos \omega_0 t$$

$$\pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

$$12. x(t) = \sin \omega_0 t$$

$$X(j\omega) = -j\pi [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$$

$$(13) x(t) = \frac{1}{a^2 + t^2}$$

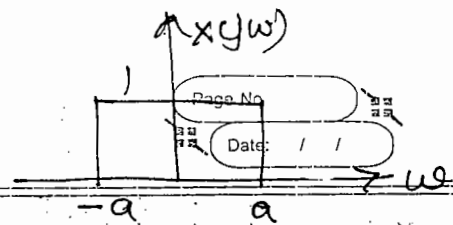
$$X(j\omega) = e^{-a|\omega|}$$

$$x(t) = \text{sgn}(t)$$

$$X(j\omega) = \frac{2}{j\omega}$$

Find the inverse FT of

$$x(\omega) = \begin{cases} 1 & -a \leq \omega \leq a \\ 0 & \text{otherwise} \end{cases}$$



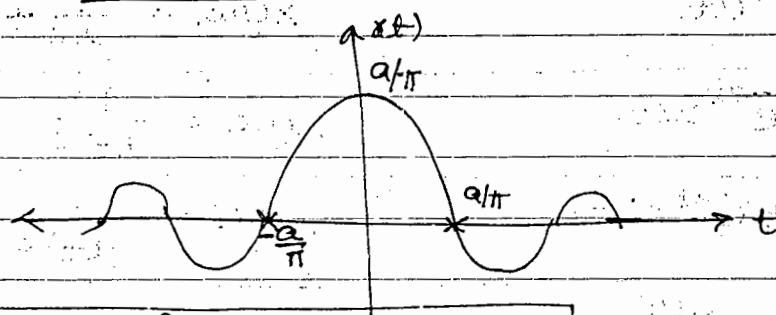
(3)

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} x(\omega) e^{j\omega t} d\omega = \frac{1}{2\pi} \int_{-a}^a 1 e^{j\omega t} d\omega$$

$$= \frac{1}{2\pi} \left[\frac{e^{j\omega t}}{jt} \right]_{-a}^a$$

$$= \frac{1}{2\pi \cdot jt} \left[e^{ja t} - e^{-ja t} \right] = \frac{\sin at}{\pi t}$$

$$\therefore x(t) = \frac{\sin at}{\pi t} = \frac{a}{\pi} \operatorname{sinc} at$$



$$x(t) = \begin{cases} \frac{\sin at}{\pi} & t \neq 0 \\ \frac{a}{\pi} & t = 0 \end{cases}$$

(2) $x(\omega) = e^{-2\omega} u(\omega)$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} x(\omega) \cdot e^{j\omega t} d\omega = \frac{1}{2\pi} \int_0^{\infty} e^{-2\omega} \cdot e^{j\omega t} d\omega$$

$$= \frac{1}{2\pi} \int_0^{\infty} e^{(jt-2)\omega} d\omega = \frac{1}{2\pi(jt-2)} \left[e^{(jt-2)\omega} \right]_0^{\infty}$$

$$= \frac{-1}{2\pi(jt-2)}$$

$$x(t) = \frac{1}{2\pi(2-jt)}$$

③

find $x(t)$

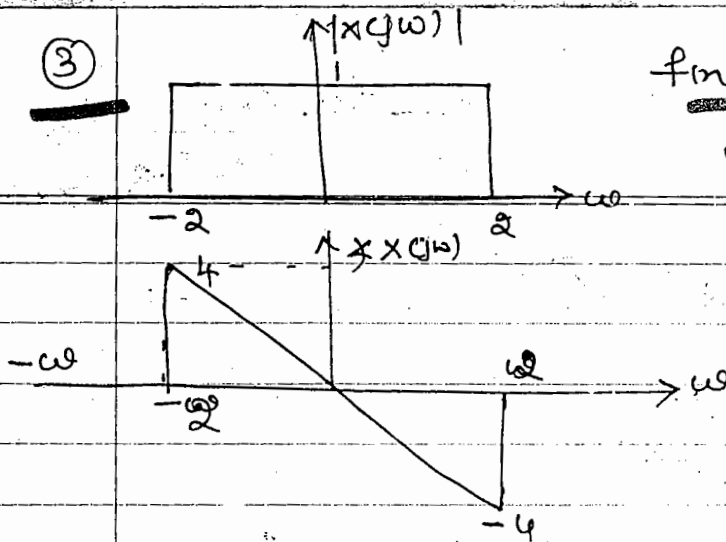
given

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$|x(j\omega)|$ and

$\angle x(j\omega)$ as shown
in figure.



$$x(j\omega) = |x(j\omega)| e^{j\angle x(j\omega)}$$

from fig(2) $\angle x(j\omega) = \frac{+4}{-2} \omega = -2\omega$

$$x(j\omega) = 1 \cdot e^{-j2\omega} = \begin{cases} e^{-j2\omega} & -2 < \omega < 2 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} x(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} x(j\omega) \cdot e^{j\omega t} d\omega = \frac{1}{2\pi} \int_{-2}^2 e^{-j2\omega} \cdot e^{j\omega t} d\omega \quad \text{--- (A)} \\ &= \frac{1}{2\pi} \int_{-2}^2 e^{-j(2-t)\omega} d\omega = \frac{+j}{2\pi(2-t)} \left[e^{-j(2-t)\omega} \right]_{-2}^2 \\ &= \frac{j}{2\pi(2-t)} \left[e^{-2j(2-t)} - e^{2j(2-t)} \right] \\ &= \frac{\sin 2(2-t)}{2\pi(2-t)} = \frac{\sin 2(t-2)}{2\pi(t-2)} \end{aligned}$$

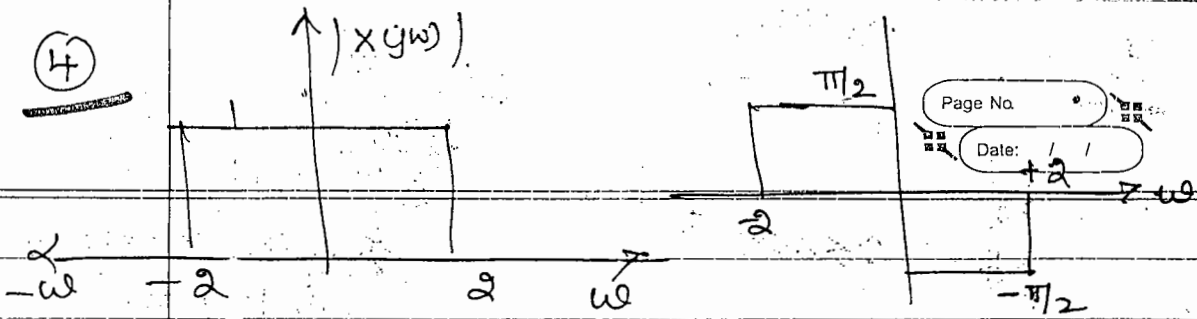
$$x(t) = \frac{\sin(2t-4)}{\pi(t-2)} \quad t \neq 2$$

also at $t=2$, $x(t)$ is indeterminate

hence $x(t) = \frac{2}{\pi}$ at $t=2$

put $t=2$ in eqn (A) to get the above expression

(4)



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find $x(t)$.

$$X(j\omega) = |X(j\omega)| e^{j\angle X(j\omega)}$$

$$X(j\omega) = \begin{cases} 1 \cdot e^{-j\pi/2} & ; 0 \leq \omega \leq 2 \\ 1 \cdot e^{j\pi/2} & ; -2 \leq \omega < 0 \end{cases}$$

$$x(t) = \frac{1}{2\pi} \left[\int_0^2 e^{-j\pi/2} e^{j\omega t} d\omega + \int_{-2}^0 e^{j\pi/2} e^{j\omega t} d\omega \right]$$

$$= \frac{1}{2\pi} \left[\int_{-2}^0 e^{j(\frac{\pi}{2} + t)\omega} d\omega + \int_0^2 e^{j(t - \frac{\pi}{2})\omega} d\omega \right] \quad \left| e^{j(t \pm \frac{\pi}{2})} = e^{jt} \right.$$

$$= \frac{1}{2\pi} \left[\int_{-2}^0 e^{j\omega t} d\omega + \int_0^2 e^{j\omega t} d\omega \right]$$

$$= \frac{1}{2\pi} \left[\frac{e^{j\omega t}}{jt} \Big|_{-2}^0 + \frac{e^{j\omega t}}{jt} \Big|_0^2 \right]$$

$$= \frac{1}{2\pi} \left(\frac{1}{jt} - \frac{e^{-j2t}}{jt} + \frac{e^{j2t}}{jt} + \frac{1}{jt} \right)$$

$$\Rightarrow \frac{-\cos 2t}{\pi t} + \frac{2}{2\pi jt}$$

$$x(t) = \frac{1}{\pi jt} [1 - \cos 2t]$$

(5) Determine the time domain signal $x(t)$ given

$$X(j\omega) = \begin{cases} \cos\left[\frac{\omega}{2}\right] + j\sin\frac{\omega}{2} & ; |\omega| < \pi \\ 0 & ; \text{otherwise} \end{cases}$$

$$\text{Note } e^{j\omega/2} = \cos \omega/2 + j\sin \omega/2$$

$$x(t) = \frac{1}{2\pi} \left[\int_{-\pi}^{\pi} \cos \frac{\omega}{2} e^{j\omega t} d\omega + \int_{-\pi}^{\pi} j \sin \frac{\omega}{2} e^{j\omega t} d\omega \right]$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j\omega/2} e^{j\omega t} d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j\omega(t+1/2)} d\omega$$

$$= \frac{1}{2\pi} \left[\frac{e^{j\omega(t+1/2)}}{j(t+1/2)} \right]_{-\pi}^{\pi} = \frac{1}{2\pi} \left[\frac{e^{j\pi(t+1/2)} - e^{-j\pi(t+1/2)}}{j(t+1/2)} \right]$$

$$= \frac{1}{\pi} \left[\frac{\sin \pi(t+1/2)}{t+1/2} \right] = \frac{1}{\pi} \left[\frac{\sin(\frac{\pi}{2} + t\pi)}{t+1/2} \right]$$

$$x(j\omega) = \frac{\cos \pi t}{\pi(t+1/2)} \quad \text{for } t \neq -1/2$$

$$\therefore x(j\omega) = \begin{cases} 1 & \text{at } t = -1/2 \\ \frac{\cos \pi t}{\pi(t+1/2)} & ; t \neq -1/2 \end{cases}$$

$$= e^{jt}$$

Find $x(t)$ given $x(j\omega) = e^{-|\omega|}$

$$x(j\omega) = e^{-|\omega|} = \begin{cases} e^{\omega} & \text{for } \omega < 0 \\ e^{-\omega} & \text{for } \omega > 0 \end{cases}$$

$$x(t) = \frac{1}{2\pi} \left[\int_{-\infty}^0 e^{\omega} e^{j\omega t} d\omega + \int_0^{\infty} e^{-\omega} e^{j\omega t} d\omega \right] = \frac{1}{2\pi} \left[\int_{-\infty}^0 e^{(1+j)t\omega} d\omega + \int_0^{\infty} e^{(-1+j)t\omega} d\omega \right]$$

$$= \frac{1}{2\pi} \left[\frac{e^{(1+j)t\omega}}{1+jt} \Big|_{-\infty}^0 + \frac{e^{(-1+j)t\omega}}{j t - 1} \Big|_0^{\infty} \right] = \frac{1}{2\pi} \left[\frac{1}{1+jt} + \frac{1}{1-jt} \right]$$

$$x(t) = \frac{2}{2\pi(t^2+1)} = \frac{1}{\pi(t^2+1)}$$

Check the above problem using duality Refer problem (35)

Find the inverse FT of step fn in freq. domain

$x(j\omega) = u(\omega)$. Use the appropriate property.

We have $x(t) = u(t) \longleftrightarrow \frac{1}{j\omega} + \pi \delta(\omega)$

~~$x(j\omega) = u(\omega)$~~

we have duality property as

$$x(t) \longleftrightarrow X(j\omega)$$

$$X(j\omega) \longleftrightarrow 2\pi x(-\omega)$$

$$2\pi x(-\omega) \longleftrightarrow X(j\omega)$$

$$2\pi u(-\omega) \longleftrightarrow \frac{1}{j\omega} + \pi \delta(\omega)$$

$$u(\omega) \longleftrightarrow \frac{1}{2\pi} \left[\frac{1}{j\omega} + \pi \delta(\omega) \right]$$

$$\delta(t) = \delta(-t) \\ \text{even f.d.}$$

$$u(\omega) \longleftrightarrow \frac{j}{2\pi\omega} + \frac{\delta(\omega)}{2}$$

Taking Inverse FT

$$u(t) = \frac{j}{2\pi t} + \frac{\delta(t)}{2}$$

Find the inverse FT of the following

(a) $X(j\omega) = \frac{j\omega + 1}{(j\omega)^2 + 5j\omega + 6}$

$$X(j\omega) = \frac{j\omega + 1}{(j\omega + 3)(j\omega + 2)} = \frac{A}{j\omega + 3} + \frac{B}{j\omega + 2}$$

$$j\omega + 1 = A(j\omega + 2) + B(j\omega + 3)$$

put $j\omega = -2$; $-1 = B$

put $j\omega = -3$; $A = 2$

$$X(j\omega) = \frac{2}{j\omega + 3} - \frac{1}{j\omega + 2}$$

Taking inverse FT on both sides

$$x(t) = 2e^{-3t}u(t) - e^{-2t}u(t)$$

$$\therefore x(t) = [2e^{-3t} - e^{-2t}]u(t)$$

(b) $X(j\omega) = \frac{-j\omega}{(j\omega)^2 + 3j\omega + 2}$

$$= \frac{A}{j\omega + 1} + \frac{B}{j\omega + 2}$$

~~So~~ $A = \frac{-j\omega}{j\omega + 2} \Big|_{j\omega = -1} = 1, \quad B = \frac{-j\omega}{j\omega + 1} \Big|_{j\omega = -2} = -1$

E =

$$X(j\omega) = \frac{1}{1+j\omega} - \frac{2}{2+j\omega}$$

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$$x(t) = [e^{-t} - 2e^{-2t}] u(t)$$

(c) $X(j\omega) = \frac{j\omega}{(2+j\omega)^2}$

$x = f(-t)$
m f n

$$X(j\omega) = \frac{A}{(2+j\omega)} + \frac{B}{(2+j\omega)^2}$$

$$j\omega = A(2+j\omega) + B$$

put $j\omega = -2$; $B = -2$

put $j\omega = 0$; $A = 1$

$$X(j\omega) = \frac{1}{2+j\omega} - \frac{2}{(2+j\omega)^2}$$

$$x(t) = e^{-2t} u(t) - t e^{-2t} u(t)$$

$$x(t) = e^{-2t} [1 - 2t] u(t)$$

Evaluate the following quantities

(a) $\int_{-\infty}^{\infty} \frac{4}{(\omega^2+1)^2} d\omega$

Let $X(j\omega) = \frac{2}{\omega^2+1}$ & $|X(j\omega)|^2 = \frac{4}{(\omega^2+1)^2}$

we have $e^{-at} \longleftrightarrow \frac{2a}{(a^2+\omega^2)^2}$

put $a=1$,

$$e^{-|t|} \longleftrightarrow \frac{2}{(1+\omega^2)^2}$$

$$x(t) = e^{-|t|} = \begin{cases} e^{-t} & ; t > 0 \\ 1 & ; = 0 \\ e^t & ; t < 0 \end{cases}$$

From parseval's theorem

$$E = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$\therefore E = 2\pi \left[\int_{-\infty}^0 e^{2t} dt + \int_0^{\infty} e^{-2t} dt \right]$$

$$= 2\pi \left[\frac{1}{2} + \frac{1}{2} \right] = \underline{2\pi} \text{ joules}$$

(b) $\int_{-\infty}^{\infty} \frac{\sin^2 \pi t}{(\pi t)^2} dt$

Let $x(t) = \frac{\sin \pi t}{\pi t}$; $|x(t)|^2 = \frac{\sin^2 \pi t}{(\pi t)^2}$

$x(t) = \frac{\sin \pi t}{\pi t} \longleftrightarrow X(j\omega) = \begin{cases} 1 & ; -\pi \leq \omega \leq \pi \\ 0 & ; |\omega| > \pi \end{cases}$

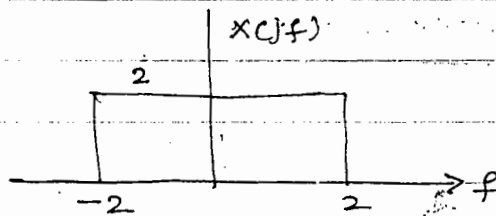
From parseval's theorem

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} 1 d\omega$$

$$= \frac{[2\pi]}{2\pi} = \underline{1 \text{ joules}} \quad (b)$$

Using parseval's theorem, determine the range of frequencies ($f_1, -f_1$) where 50% of the signal energy lies.

The spectrum of the signal is shown below



$$E = \int_{-\infty}^{\infty} |X(f)|^2 df = \int_{-2}^2 2 df = \underline{16 \text{ Joules}} \quad (c)$$

Total energy $E = 16$

50% of $E = \frac{16}{2} = 8$

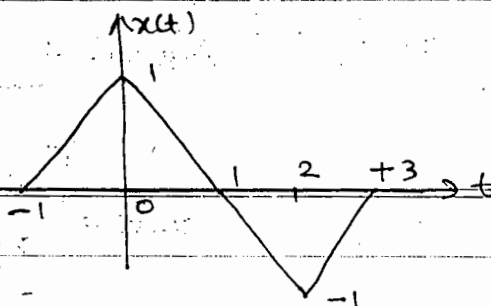
$$E = \int_{-1}^1 |X(f)|^2 df = \int_{-1}^1 2^2 df = 4 \times 2 = 8$$

50% of E lies in the freq. range $-1 \leq f \leq 1$

For the signal $x(t)$ shown below. Evaluate

① $\int_{-\infty}^{\infty} x(j\omega) d\omega$ ② $\int_{-\infty}^{\infty} |x(j\omega)|^2 d\omega$ ③ $\int_{-\infty}^{\infty} x(j\omega) e^{j2\omega} d\omega$

④ $\angle x(j\omega)$ ⑤ $x(j0)$



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we have $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} x(j\omega) d\omega \cdot e^{j\omega t}$

$$\int_{-\infty}^{\infty} x(j\omega) \cdot e^{j\omega t} d\omega = 2\pi \cdot x(t) \quad \text{--- (A)}$$

put $t=0$

(a) $\int_{-\infty}^{\infty} x(j\omega) e^{j\omega \cdot 0} d\omega = \int_{-\infty}^{\infty} x(j\omega) d\omega = 2\pi x(0) = \underline{\underline{2\pi}}$

(b) From Parseval's theorem

frequency
res.

$$E = \int_{-\infty}^{\infty} |x(j\omega)|^2 d\omega = 2\pi \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$= 2\pi \left[\int_{-1}^0 (t+1)^2 dt + \int_0^2 (1-t)^2 dt + \int_2^3 (t-3)^2 dt \right]$$

$$E = \frac{8\pi}{3} \text{ joules}$$

(c) put $t=2$ in eqn (A)

$$\int_{-\infty}^{\infty} x(j\omega) e^{j\omega \cdot 2} d\omega = 2\pi x(2) = \underline{\underline{-2\pi}}$$

(d) $x(t)$ is an odd signal $x_o(t)$ shifted by right by 1 unit

$$x(t) = x_o(t-1)$$

$$x(j\omega) = e^{-j\omega} x_o(j\omega)$$

$f \leq 1$

for an odd signal $\phi = \pi/2$.

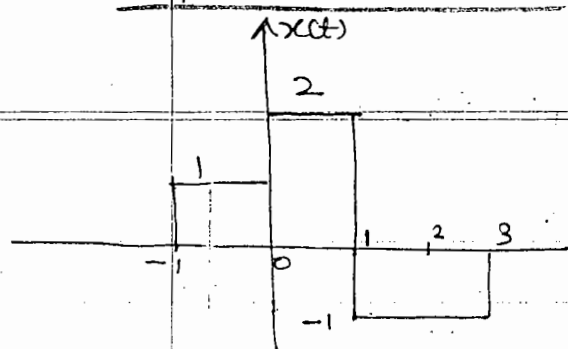
$$|x(j\omega)| = |x_o(j\omega)| e^{j\pi/2} \cdot e^{-j\omega} = |x_o(j\omega)| e^{-j(\frac{\pi}{2} - \omega)}$$

$$\therefore \angle x(j\omega) = \frac{\pi}{2} - \omega$$

(e) put $\omega=0$ in eqn (d) $x(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$

$$x(j0) = \int_{-\infty}^{\infty} x(t) dt = \underline{\underline{0}}$$

compute the FT of the signal shown below



$$x(t) = \begin{cases} -1; & -1 < t < 0 \\ 2; & 0 < t < 1 \\ -1; & 1 < t < 3 \end{cases}$$

$$X(j\omega) = \int_{-1}^0 -1 e^{-j\omega t} dt + \int_0^1 2 e^{-j\omega t} dt + \int_1^3 -1 e^{-j\omega t} dt$$

$$X(j\omega) = \frac{1}{j\omega} [1 + e^{j\omega} - 3e^{j\omega} + e^{j3\omega}]$$

Find the time domain signal x(t) given

$$X(j\omega) = \frac{\omega^2 - 4j\omega - 6}{(-\omega^2 + 3j\omega + 2)(j\omega + 4)}$$

$$= \frac{-(j\omega)^2 - 4j\omega - 6}{((j\omega)^2 + 3j\omega + 2)(j\omega + 4)}$$

$$X(j\omega) = \frac{A}{(j\omega + 1)} + \frac{B}{(j\omega + 2)} + \frac{C}{(j\omega + 4)}$$

$$-(j\omega)^2 - 4j\omega - 6 = A(j\omega + 2)(j\omega + 4) + B(j\omega + 1)(j\omega + 4) + C(j\omega + 1)(j\omega + 2)$$

$$X(j\omega) = \frac{-1}{j\omega + 1} + \frac{1}{j\omega + 2} - \frac{1}{j\omega + 4}$$

$$x(t) = [e^{-t} + e^{-2t} - e^{-4t}] u(t)$$

Find the IFT of $\frac{1 + 2j\omega}{(2 + j\omega)^2}$

$$X(j\omega) = \frac{A}{2 + j\omega} + \frac{B}{(2 + j\omega)^2}$$

$$1 + 2j\omega = A(2 + j\omega) + B$$

$$X(j\omega) = \frac{2}{2 + j\omega} - \frac{3}{(2 + j\omega)^2}$$

$$x(t) = [2e^{-2t} - 3te^{-2t}] u(t)$$

Show that ~~is~~ a real and odd CT signal which is non periodic has purely imaginary Fourier Transform.

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If $x(t)$ is real; then $x(t) = x^*(t)$.

If $x(t)$ is odd; then $x(-t) = -x(t)$.

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

put $t = -\tau$; $dt = -d\tau$.

$$X(j\omega) = - \int_{\infty}^{-\infty} x(-\tau) e^{j\omega\tau} d\tau = \int_{-\infty}^{\infty} x(\tau) e^{j\omega\tau} d\tau.$$

$$= - \int_{-\infty}^{\infty} x(\tau) e^{j\omega\tau} d\tau.$$

Since $x(\tau)$ is odd

Taking conjugate on both sides

$$X^*(j\omega) = - \int_{-\infty}^{\infty} x^*(\tau) e^{-j\omega\tau} d\tau = - \int_{-\infty}^{\infty} x(\tau) e^{-j\omega\tau} d\tau$$

$$\Rightarrow \cancel{X(j\omega)} = \cancel{X(j\omega)^*}$$

$$X^*(j\omega) = -X(j\omega)$$

$\left(\begin{array}{l} X^*(\omega) = X(\omega) \\ \because x(\tau) \text{ is real} \end{array} \right)$

Hence the FT is purely imaginary and odd.

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Discrete Time Fourier Transform

DTFT

Any discrete time signal $x(n)$ can be expressed as

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\Omega}) e^{j\Omega n} d\Omega$$

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\Omega}) e^{j\Omega n} d\Omega \quad \rightarrow \textcircled{1}$$

where

$$X(e^{j\Omega}) = X(\Omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\Omega n} \quad \rightarrow \textcircled{2}$$

$$x(n) \xleftrightarrow{\text{DTFT}} X(e^{j\Omega}) \quad \text{DTFT pair}$$

$X(e^{j\Omega})$ is known as DTFT of discrete time signal $x(n)$.

DTFT is used to represent non periodic discrete time signals as a superposition of complex sinusoids and it is represented on a continuous freq. spectrum in the interval $-\pi \leq \Omega \leq \pi$; where Ω is in radians.

The transform $X(e^{j\Omega})$ represents the signal $x(n)$ as a function of sinusoidal frequency Ω and is known as freq. domain representation of $x(n)$ (a DT signal).

Eqn ① is known as synthesis eqn and eqn ② is known as analysis equation.

conditions for existence of DTFT

1. $x(n)$ has to be absolutely summable

$$\sum_{n=-\infty}^{\infty} |x(n)| e^{-j\Omega n} < \infty \quad \text{or} \quad \sum_{n=-\infty}^{\infty} |x(n)| < \infty$$

2. If $x(n)$ is not absolutely summable; then it should have finite energy for DTFT to exist.

$$\sum_{n=-\infty}^{\infty} |x(n)|^2 < \infty$$

Note $x(n)$ is not periodic, but $X(e^{j\Omega})$ is periodic

$$X(e^{j(\Omega + 2\pi k)}) = \sum_{n=-\infty}^{\infty} x(n) [e^{-j\Omega n} e^{-j2\pi k n}] = \sum_{n=-\infty}^{\infty} x(n) e^{-j\Omega n} e^{-j2\pi k n} = X(e^{j\Omega}) \quad \text{since } e^{-j2\pi k n} = 1$$

properties of DTFT

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1. Linearity

$$\text{If } x(n) \longleftrightarrow X(\Omega); \\ y(n) \longleftrightarrow Y(\Omega)$$

$$\text{then } ax(n) + by(n) \longleftrightarrow aX(\Omega) + bY(\Omega)$$

proof

$$\text{Let } z(n) = ax(n) + by(n)$$

$$Z(\Omega) = \sum_{n=-\infty}^{\infty} z(n) e^{-j\Omega n} = \sum_{n=-\infty}^{\infty} [ax(n) + by(n)] e^{-j\Omega n} \\ = a \sum_{n=-\infty}^{\infty} x(n) e^{-j\Omega n} + b \sum_{n=-\infty}^{\infty} y(n) e^{-j\Omega n}$$

$$Z(\Omega) = aX(\Omega) + bY(\Omega) \quad \text{--- RHS}$$

(2) Time shifting property

$$\text{If } x(n) \longleftrightarrow X(\Omega) \text{ then}$$

$$x(n-k) \longleftrightarrow e^{-j\Omega k} X(\Omega)$$

proof

$$\text{Let } y(n) = x(n-k)$$

$$Y(\Omega) = \sum_{n=-\infty}^{\infty} y(n) e^{-j\Omega n} = \sum_{n=-\infty}^{\infty} x(n-k) e^{-j\Omega n}$$

$$\text{put } n-k = \lambda \quad \therefore n = k + \lambda$$

$$Y(\Omega) = \sum_{\lambda=-\infty}^{\infty} x(\lambda) e^{-j\Omega(k+\lambda)} = e^{-j\Omega k} \sum_{\lambda=-\infty}^{\infty} x(\lambda) e^{-j\Omega \lambda}$$

$$Y(\Omega) = e^{-j\Omega k} X(\Omega) \quad \text{--- RHS}$$

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$$x(n+k) \longleftrightarrow e^{j\Omega k} X(\Omega)$$

3. Frequency shift

$$\text{If } x(n) \longleftrightarrow X(\Omega) \text{ or } X(e^{j\Omega}) \text{ then}$$

$$y(n) = e^{j\beta n} x(n) \longleftrightarrow Y(\Omega) = X(e^{j(\Omega-\beta)}) = X(\Omega-\beta)$$

proof

we have

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\Omega n}$$

$$Y(\Omega) = \sum_{n=-\infty}^{\infty} y(n) e^{-j\Omega n} = \sum_{n=-\infty}^{\infty} x(n) e^{j\beta n} e^{-j\Omega n}$$

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$$= \sum_{n=-\infty}^{\infty} x(n) \cdot e^{-j(\Omega - \beta)n} = X(\Omega - \beta) = X(e^{j(\Omega - \beta)n})$$

$$e^{j\beta n} x(n) \longleftrightarrow X(\Omega - \beta) = X(e^{j(\Omega - \beta)n})$$

$$e^{-j\beta n} x(n) \longleftrightarrow X(\beta + \Omega) = X(e^{j(\beta + \Omega)n})$$

Shifting the frequency by ' β ' is equivalent to multiplying its sequence by $e^{j\beta n}$.

4. Scaling:

$$\text{If } x(n) \longleftrightarrow X(\Omega) \text{ then}$$

$$x(pn) \longleftrightarrow X\left(\frac{\Omega}{p}\right) = X(e^{j\frac{\Omega}{p}})$$

proof: Let $y(n) = x(pn)$

$$Y(\Omega) = \sum_{n=-\infty}^{\infty} x(pn) e^{-j\Omega n} \quad ; \quad \begin{matrix} \text{put } np = k \\ n = k/p \end{matrix}$$

$$Y(\Omega) = \sum_{k=-\infty}^{\infty} x(k) \cdot e^{-j\Omega \frac{k}{p}} = X\left(\frac{\Omega}{p}\right) = X\left[e^{j\frac{\Omega}{p}}\right]$$

NOTE scaling of DT signal discards informations. It is not possible to express the DTFT of the scaled signal in terms of the DTFT of the original signal.

Consider a non periodic signal $x(n)$ such that

$$x(n) = 0 \quad ; \quad \text{unless } \frac{n}{p} \text{ is an integer \& } p > 1$$

$$\therefore y(n) = x(np) \text{ is also non periodic}$$

$y(n)$ is the scaled version of $x(n)$.

Hence expanding in time domain is equivalent to compression in frequency domain.

5. Frequency domain differentiation

$$\text{If } x(n) \longleftrightarrow X(\Omega)$$

$$-jn x(n) \longleftrightarrow \frac{d}{d\Omega} [X(\Omega)]$$

proof

we have

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\Omega n} \quad \text{diff. both sides; } \textcircled{A}$$

$$\frac{d}{d\Omega} [X(\Omega)] = \frac{d}{d\Omega} \left[\sum_{n=-\infty}^{\infty} x(n) e^{-j\Omega n} \right]$$

$$= \sum_{n=-\infty}^{\infty} x(n) \cdot \frac{d}{d\Omega} (e^{-j\Omega n}) = \sum_{n=-\infty}^{\infty} [(-jn) x(n)] e^{-j\Omega n} \quad \textcircled{B}$$

comparing eqn (A) & (B)

$$(-jn) x(n) = \frac{d}{dz} X(z)$$

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$$x(n) = \frac{1}{(-jn)} \frac{d}{dz} X(z) \quad - \text{RH}$$

6. Summation.

Summation is the reverse process of differencing. The summation operation on $x(n)$ yields $y(n)$ where as difference operation on $y(n)$ yields $x(n)$.

$$\text{i.e. } x(n) = y(n) - y(n-1)$$

$$X(z) = Y(z) - e^{-j\Omega} Y(z)$$

$$Y(z) = \frac{X(z)}{1 - e^{-j\Omega}} \quad \text{--- (A)}$$

$$\text{Let } y(n) = \sum_{k=-\infty}^n x(k)$$

$$= x(n) + x(n-1) + x(n-2) + \dots$$

$$y(n-1) = x(n-1) + x(n-2) + \dots$$

$$y(n) - y(n-1) = x(n)$$

$$\therefore x(n) = y(n) - y(n-1)$$

from eqn (A) $Y(0) = Y(e^{j0})$ can be determined. Hence an impulse is added to account for non zero average value of $x(k)$. Hence eqn (A) modifies in to

$$Y(z) = \frac{X(z)}{1 - e^{-j\Omega}} + \pi X(e^{j0}) \delta(\Omega) \quad ; \quad -\pi < \Omega < \pi$$

where the first term is assumed to be zero for $\Omega=0$.

Since $Y(z)$ is periodic with period 2π ; we can write

$$Y(z) = \frac{X(z)}{1 - e^{-j\Omega}} + \pi X(0) \sum_{k=-\infty}^{\infty} \delta(\Omega - 2\pi k)$$

hence we can conclude that

$$\text{If } x(n) \longleftrightarrow X(z)$$

$$y(n) = \sum_{k=-\infty}^n x(k) \longleftrightarrow Y(z) = \frac{X(z)}{1 - e^{-j\Omega}} + \pi X(0) \sum_{k=-\infty}^{\infty} \delta(\Omega - 2\pi k)$$

7. Convolution.

If $x(n) \longleftrightarrow X(z)$, and

$y(n) \longleftrightarrow Y(z)$; then

$$g(n) = x(n) * y(n) \longleftrightarrow G(z) = X(z) \cdot Y(z)$$

proof:-

we have

$$G(z) = \sum_{n=-\infty}^{\infty} g(n) e^{-j\Omega n} = \sum_{n=-\infty}^{\infty} [x(n) * y(n)] e^{-j\Omega n}$$

$$= \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} x(k) \cdot y(n-k) \cdot e^{-j\Omega n} [e^{-j\Omega k} \cdot e^{j\Omega k}]$$

$$= \sum_{k=-\infty}^{\infty} x(k) \cdot e^{j\Omega k} \sum_{n=-\infty}^{\infty} y(n-k) \cdot e^{-j\Omega(n-k)}$$

$$Z(\Omega) = X(\Omega) \cdot Y(\Omega)$$

hence convoln $\xleftrightarrow{\text{DTFT}}$ Multiplication

The
reference

8. Modulation:

$$\text{If } x(n) \longleftrightarrow X(\Omega),$$

$$y(n) \longleftrightarrow Y(\Omega) \text{ then}$$

$$z(n) = x(n) \otimes y(n) \longleftrightarrow Z(\Omega) = \frac{1}{2\pi} [X(\Omega) * Y(\Omega)]$$

proof:-

we have

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\beta}) e^{j\beta n} d\beta \quad \text{--- (A)}$$

$$z(n) = \sum_{n=-\infty}^{\infty} [x(n) \otimes y(n)] e^{-j\Omega n} \quad \text{--- (B)}$$

using (A) in (B)

$$Z(\Omega) = \sum_{n=-\infty}^{\infty} \left[\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\beta}) e^{j\beta n} d\beta \right] \cdot y(n) e^{-j\Omega n}$$

changing the order of integration & summation

$$Z(\Omega) = \frac{1}{2\pi} \left[\int_{-\pi}^{\pi} X(e^{j\beta}) d\beta \sum_{n=-\infty}^{\infty} y(n) \cdot e^{-j(\Omega-\beta)n} \right]$$

$$= \frac{1}{2\pi} \left[\int_{-\pi}^{\pi} X(e^{j\beta}) \otimes y(e^{-j(\Omega-\beta)}) d\beta \right] \quad \text{--- periodic convolution}$$

$$= \frac{1}{2\pi} [X(e^{j\Omega}) * Y(e^{j\Omega})]$$

$$Z(\Omega) = \frac{1}{2\pi} [X(\Omega) * Y(\Omega)]$$

9. Parseval's theorem

If $x(n) \longleftrightarrow X(\Omega)$ then

$$\sum_{n=-\infty}^{\infty} |x(n)|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(\Omega)|^2 d\Omega$$

Proof:-

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2 = \sum_{n=-\infty}^{\infty} x(n) \otimes x^*(n)$$

Since $x(n) = x(n) \otimes x^*(n)$
 $x(n)$ is real

$$= \sum_{n=-\infty}^{\infty} x(n) \cdot \left[\frac{1}{2\pi} \int_{-\pi}^{\pi} x^*(e^{j\omega}) \cdot e^{-j\omega n} d\omega \right]$$

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= by changing the order of integration & summation

$$E = \frac{1}{2\pi} \int_{-\pi}^{\pi} x^*(e^{j\omega}) \sum_{n=-\infty}^{\infty} x(n) \cdot e^{-j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} x^*(e^{j\omega}) \cdot x(e^{j\omega}) d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} |x(e^{j\omega})|^2 d\omega$$

hence $\boxed{\sum_{n=-\infty}^{\infty} |x(n)|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |x(e^{j\omega})|^2 d\omega}$

Symmetry property:

If $x(n) \longleftrightarrow X(e^{j\omega})$ then

$x(n)$: real $\longleftrightarrow X^*(e^{j\omega}) = X(e^{j\omega})$

$x(n)$: real & even $\longleftrightarrow \text{Im}\{X(e^{j\omega})\} = 0$ { $x(n)$ is purely real }

$x(n)$: real & odd $\longleftrightarrow \text{Re}\{X(e^{j\omega})\} = 0$ { $x(n)$ is imaginary }

Proof:

$$x(n) = x_o(n) + x_e(n) \quad \text{--- (1)}$$

$$x(n) \longleftrightarrow X(e^{j\omega}) = X_R(e^{j\omega}) + jX_I(e^{j\omega}) \quad \text{--- (1)}$$

also $x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) \cdot e^{j\omega n} d\omega$ taking complex conjugate on both sides

$$x^*(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X^*(e^{j\omega}) \cdot e^{-j\omega n} d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} X^*(e^{j\omega}) e^{j(\omega) n} d\omega$$

$$\boxed{x^*(n) = X^*(e^{-j\omega})}$$

since $x(n) = x^*(n)$ {for real $x(n)$ }

$x(-n) = X^*(e^{-j\omega})$ taking c.c on both sides

$$\boxed{X^*(e^{j\omega}) = X(e^{-j\omega})}$$

we have $x(n) = x_o(n) + x_e(n)$

$$x(-n) = x_e(-n) + x_o(-n) = x_e(n) - x_o(n) \quad \text{--- (2)}$$

$$\cancel{x(-n)} \quad X(-\omega) = X^*(e^{j\omega}) = X_R(e^{j\omega}) - jX_I(e^{j\omega}) \quad \text{--- (2)}$$

$$2x_e(n) \longleftrightarrow 2X_R(e^{j\omega})$$

$$\boxed{x_e(n) \longleftrightarrow X_R(e^{j\omega})} \quad \& \quad \boxed{x_o(n) \longleftrightarrow X_I(e^{j\omega})}$$

The DTFT of real and odd sequence is purely imaginary.

Find the DTFT of $x(n) = \left(\frac{1}{2}\right)^n [u(n+3) - u(n-2)]$

$$x(n) = \left(\frac{1}{2}\right)^n [u(n+3) - u(n-2)]$$

$$X(\omega) = \sum_{n=-3}^1 \left(\frac{1}{2}\right)^n \cdot e^{-j\omega n} = \sum_{n=-3}^1 \left(\frac{e^{-j\omega}}{2}\right)^n$$

put $n+3=k$; $n=k-3$

$$X(\omega) = \sum_{k=0}^4 \left(\frac{e^{-j\omega}}{2}\right)^k \cdot \left(\frac{e^{-j\omega}}{2}\right)^{-3}$$

$$X(\omega) = 8 e^{-j3\omega} \left[\frac{1 - \left(\frac{e^{-j\omega}}{2}\right)^5}{1 - \frac{e^{-j\omega}}{2}} \right]$$

purely real

imaginary

$$= 8 e^{-j3\omega} \left[\frac{1 - \frac{e^{-j5\omega}}{2^5}}{1 - \frac{e^{-j\omega}}{2}} \right]$$

irregular

Given $x(n) = \{3, 0, 1, -2, -3, -4, 1, 0, -1\}$ without

$x(n)$
d n

computing $X(\omega)$; evaluate the following.

(a) $X(e^{j\pi})$

$$X(e^{j\pi}) = \sum_{n=-6}^6 x(n) \cdot e^{-j\pi n} = \sum_{n=-6}^6 x(n) \cdot (-1)^n$$

$$= \sum_{n=-3}^3 x(n) \cdot (-1)^n = -3 + 0 - 1 - 2 + 3 - 4 - 1 + 0 + 1$$

$$= 1$$

$$\therefore X(e^{j\pi}) = 1$$

(b) $\int_{-\pi}^{\pi} X(\omega) \cdot d\omega$

we have $x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) \cdot d\omega \cdot e^{j\omega n}$

put $\omega = 0$

$$\int_{-\pi}^{\pi} x(\omega) \cdot \omega d\omega = 2\pi x(0)$$

$$= 2\pi(-2) = -4\pi$$

$$\boxed{\int_{-\pi}^{\pi} x(\omega) d\omega = -4\pi}$$

(c) $\int_{-\pi}^{\pi} |x(\omega)|^2 d\omega$

From Parseval's Theorem

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} |x(\omega)|^2 d\omega = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

$$\therefore \int_{-\pi}^{\pi} |x(\omega)|^2 d\omega = 2\pi \sum_{n=-3}^5 |x(n)|^2$$

$$= 2\pi [3^2 + 0^2 + 0^2 + (-2)^2 + (-3)^2 + (-4)^2 + 0^2 + 4^2]$$

$$= 2\pi(41)$$

$$= 82\pi$$

$$\boxed{\therefore \int_{-\pi}^{\pi} |x(\omega)|^2 d\omega = 82\pi}$$

(d) $\int_{-\pi}^{\pi} \frac{d}{d\omega} (|x(\omega)|^2) d\omega$

we have $\sum_{n=-\infty}^{\infty} |x(n)|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |x(\omega)|^2 d\omega$

$$\sum_{n=-\infty}^{\infty} [-jn x(n)]^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{d}{d\omega} (|x(\omega)|^2) d\omega$$

$$\therefore \int_{-\pi}^{\pi} \frac{d}{d\omega} (|x(\omega)|^2) d\omega = 2\pi \sum_{n=-3}^5 [-jn x(n)]^2$$

$$= 2\pi [(3)^2(3)^2 + 0 + 1 + 0 + 3^2 + 64 + 9 + 0 + 25]$$

$$= 378\pi$$

$$\therefore \int_{-\pi}^{\pi} \frac{d}{d\omega} (|x(\omega)|^2) d\omega = 378\pi$$

Given $x(n) = 2 - \left(\frac{1}{2}\right)^n; |n| \leq 4$.
 $= 0$; otherwise

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Determine the DTFT of $x(n)$.

$$X(e^{j\omega}) = \sum_{n=-4}^4 \left[2 - \left(\frac{1}{2}\right)^n \right] e^{-j\omega n}$$

$$= \sum_{n=-4}^4 2 e^{-j\omega n} - \sum_{n=-4}^4 \left[\frac{e^{-j\omega n}}{2} \right]$$

$$= \sum_{m=0}^8 2 e^{-j\omega m} e^{j4\omega} - \sum_{m=0}^8 \left(\frac{e^{j\omega}}{2} \right)^4 \left[\frac{e^{-j\omega}}{2} \right]^m$$

$$= 2 e^{j4\omega} \left[\frac{1 - e^{-j9\omega}}{1 - e^{-j\omega}} \right] - \frac{e^{j4\omega}}{2^4} \left[\frac{1 - \frac{e^{-j9\omega}}{2^9}}{1 - \frac{e^{-j\omega}}{2}} \right]$$

$$= 2 e^{j4\omega} \left[\frac{1 - e^{-j9\omega}}{1 - e^{-j\omega}} \right] - \left(\frac{e^{j\omega}}{2} \right)^4 \left[\frac{1 - \frac{e^{-j9\omega}}{2^9}}{1 - \frac{e^{-j\omega}}{2}} \right]$$

$$= 2 e^{j4\omega} \left[\frac{1 - e^{-j9\omega}}{1 - e^{-j\omega}} \right] - \left[2 e^{j2\omega} \right]^4 \left[\frac{2^9 - e^{-j9\omega}}{2 - e^{-j\omega}} \right] \frac{2}{2^9}$$

$$X(e^{j\omega}) = 2 e^{j4\omega} \left[\frac{1 - e^{-j9\omega}}{1 - e^{-j\omega}} \right] - \left[\frac{e^{j4\omega}}{2} \right]^4 \left[\frac{2 - e^{-j9\omega}}{2 - e^{-j\omega}} \right]$$

DTFT contd

Find the DTFT of the following signals.

(a) $x(n) = \{2, 1, 4, 1, 2\}$

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\Omega n} = \sum_{n=-2}^2 x(n) e^{-j\Omega n}$$

$$= x(-2) e^{+j\Omega 2} + x(-1) e^{j\Omega} + x(0) + x(1) e^{-j\Omega} + x(2) e^{-2j\Omega}$$

$$= 2 e^{j\Omega 2} + 1 e^{j\Omega} + 4 + 1 e^{-j\Omega} + 2 e^{-j\Omega 2}$$

$$X(\Omega) = 4 \cos 2\Omega + 2 \cos \Omega + 4$$

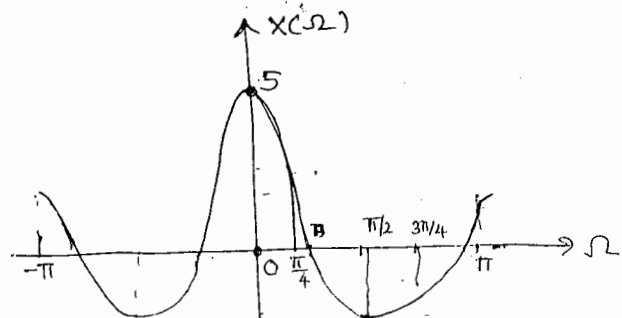
(b) $x(n) = \{1, 1, 1, 1, 1\}$. Sketch the spectrum:

$$x(n) = \delta(n+2) + \delta(n+1) + \delta(n) + \delta(n-1) + \delta(n-2)$$

$$X(\Omega) = e^{j2\Omega} + e^{j\Omega} + 1 + e^{-j\Omega} + e^{-j2\Omega}$$

$$X(\Omega) = 1 + 2 \cos \Omega + 2 \cos 2\Omega$$

Ω	$X(\Omega)$
0	5
$\pm \frac{\pi}{4}$	± 2.414
$\pm \frac{\pi}{2}$	1
$\pm \frac{3\pi}{4}$	∓ 0.414
$\pm \pi$	1



Since $X(\Omega)$ is real
phase spectrum is zero.

(c) $x(n) = 2^n [u(n) - u(n-4)]$

or $x(n) = [1, 2, 4, 8]$

or $x(n) = \begin{cases} 2^n; & 0 \leq n \leq 3 \\ 0; & |n| > 3 \end{cases}$

or $x(n) = \delta(n) + 2\delta(n-1) + 4\delta(n-2) + 8\delta(n-3)$

Find the DTFT. Sketch the spectrum.

$$X(\Omega) = 1 + 2 e^{-j\Omega} + 4 e^{-j2\Omega} + 8 e^{-j3\Omega}$$

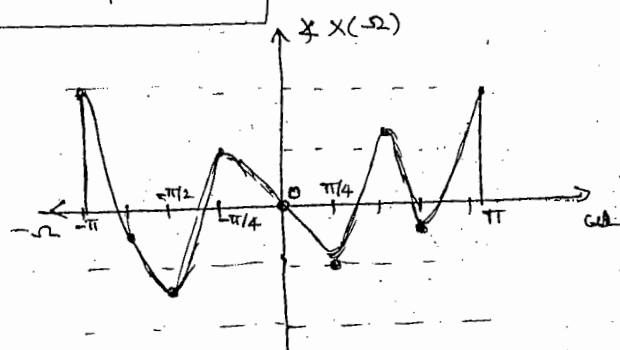
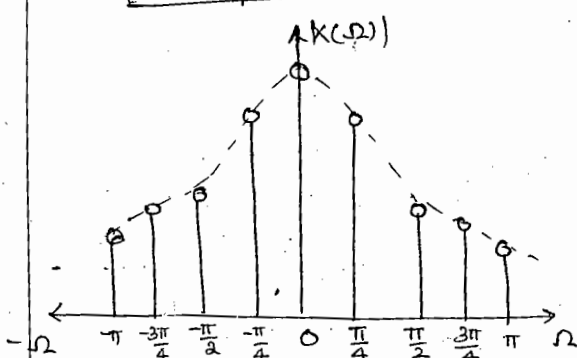
or $X(\Omega) = \sum_{n=0}^3 [2 e^{-j\Omega}]^n$

using time shifting property

$$= \frac{1 - [2 e^{-j\Omega}]^4}{1 - 2 e^{-j\Omega}}$$

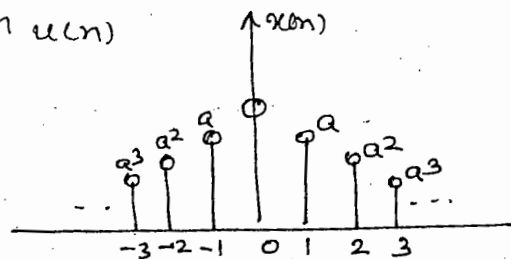
for $\Omega \neq 0, \pm 2\pi, \pm 4\pi$

Ω	$X(\Omega)$	$ X(\Omega) $	$\angle X(\Omega)$
0	15	15	0
$\pi/4$	$-3.24 - j11.07$	11.53	-1.855
$\pi/2$	$-3 + j6$	6.70	2.0344
$3\pi/4$	$5.24 + j3.07$	6.075	-0.53
π	$-5 + j0$	5	3.1416
$-\pi/4$	$-3.24 + j11.07$	11.53	1.855
$-\pi/2$	$-3 - j6$	6.70	-2.0344
$-3\pi/4$	$-5.24 + j3.07$	6.075	-0.53
$-\pi$	-5	5	3.1416



(d) $x[n] = a^{|n|}$ with $|a| < 1$ find the DTFT of $x[n]$

or $x[n] = +a^n u[n-1] + a^n u[n]$



$$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n}$$

$$= \sum_{n=-\infty}^{-1} + a^n e^{-j\Omega n} + \sum_{n=0}^{\infty} a^n e^{-j\Omega n}$$

$$= \frac{a e^{+j\Omega}}{1 - a e^{+j\Omega}} + \frac{1}{1 - a e^{-j\Omega}}$$

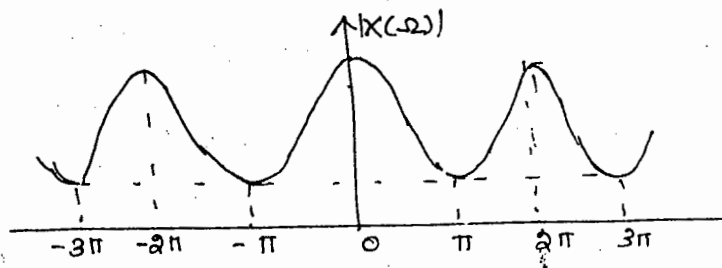
$$= \sum_{n=1}^{\infty} [a e^{j\Omega}]^n + \sum_{n=0}^{\infty} [\bar{a} e^{j\Omega}]^{-n}$$

put $n-1=m$ in first term. & changing the limits & variable

$$X(\Omega) = \sum_{m=0}^{\infty} [a e^{j\Omega}] \cdot [a e^{j\Omega}]^m + \sum_{n=0}^{\infty} a^n e^{-j\Omega n}$$

$$= \frac{a e^{j\Omega}}{1 - a e^{j\Omega}} + \frac{1}{1 - a e^{j\Omega}} = \frac{1 - a^2}{1 + a^2 - a[e^{j\Omega} + e^{-j\Omega}]}$$

$$X(\Omega) = \frac{1 - a^2}{1 - a \cos \Omega + a^2} = |X(\Omega)| \quad \& \quad \neq X(\Omega) = 0$$



(e) $x(n) = 2(3)^n u(-n)$

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\Omega n} = \sum_{n=0}^{\infty} 2(3)^n e^{-j\Omega n}$$

$$= 2 \sum_{n=0}^{\infty} \left(\frac{e^{j\Omega}}{3}\right)^n = \frac{2}{1 - \frac{e^{j\Omega}}{3}}$$

$$X(\Omega) = \frac{6}{3 - e^{j\Omega}}$$

(f) $x(n) = \alpha^n u(n)$

$$X(\Omega) = \sum_{n=0}^{\infty} \alpha^n e^{-j\Omega n} = \frac{e^{j\Omega}}{e^{j\Omega} - \alpha} \quad | \alpha | < 1$$

$$X(\Omega) = \frac{1}{1 - \alpha e^{-j\Omega}}$$

for $\alpha > 1$ the series will not converge
hence for $\alpha < 1$; the series will converge

$$X(\Omega) = \frac{1}{(1 - \alpha \cos \Omega) - j \alpha \sin \Omega}$$

$$|X(\Omega)| = \frac{1}{\sqrt{(1 - \alpha \cos \Omega)^2 + \alpha^2 \sin^2 \Omega}}$$

$$= \frac{1}{\sqrt{\alpha^2 + 1 - 2\alpha \cos \Omega}}$$

$$\angle X(\Omega) = -\tan^{-1} \left[\frac{\alpha \sin \Omega}{1 - \alpha \cos \Omega} \right]$$

$x(n) = \left(\frac{1}{2}\right)^n u(n-4)$ find the DTFT of $x(n)$.

$$x(n) = \left(\frac{1}{2}\right)^n u(n-4) = \left(\frac{1}{2}\right)^{n-4} u(n-4) \cdot \left(\frac{1}{2}\right)^4$$

using the time shifting property

$$x(n-k) \longleftrightarrow e^{-j\Omega k} X(\Omega)$$

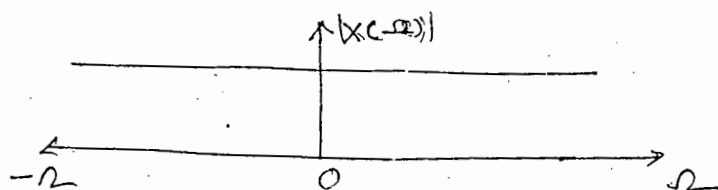
$$\text{Let } x_1(n) = \frac{1}{16} \left(\frac{1}{2}\right)^n u(n) \longleftrightarrow \frac{1/16}{1 - \frac{1}{2} e^{-j\Omega}}$$

$$x(n) = x_1(n-4) \longleftrightarrow \frac{e^{-j\Omega 4}}{16} \cdot \frac{1}{1 - \frac{1}{2} e^{-j\Omega}}$$

$$X(\Omega) = \frac{e^{-j4\Omega}}{16 - 8e^{-j\Omega}}$$

$$x(n) = \delta(n)$$

$$X(\Omega) = \sum_{n=-\infty}^{\infty} \delta(n) e^{-j\Omega n} = e^{-j\Omega n} \Big|_{n=0} = 1$$



$|X(\Omega)| = 1$ for all Ω

$\angle X(\Omega) = 0$

$$x(n) = u(n)$$

$$x(n) = \begin{cases} 1 & ; \text{ for all } n \geq 0 \\ 0 & ; n < 0 \end{cases}$$

$$X(\Omega) = \sum_{n=0}^{\infty} 1 \cdot e^{-j\Omega n} = \frac{1}{1 - e^{-j\Omega}} \quad \Omega \neq 0$$

Since $x(n)$ is not absolutely summable sequence
However $X(\Omega)$ can be evaluated for other values of Ω

$$X(\Omega) = \frac{1}{e^{j\Omega/2} \cdot e^{-j\Omega/2} - e^{-j\Omega/2} \cdot e^{-j\Omega/2}} = \frac{1}{e^{-j\Omega/2} [e^{j\Omega/2} - e^{-j\Omega/2}]}$$

$$X(\Omega) = \frac{e^{j\Omega/2}}{2j \sin \frac{\Omega}{2}} \quad \Omega \neq 0$$

II method

$$x(n) = u(n) = \sum_{k=0}^{\infty} \delta(n-k)$$

put ~~n-k~~ $n-k = m$

when $k=0$; $n=m$. & when $k=\infty$, $m=\infty$

$$\therefore x(n) = \sum_{m=n}^{-\infty} \delta(m) = \sum_{m=-\infty}^n \delta(m)$$

$$\text{or } x(n) = \sum_{k=-\infty}^n \delta(k) \quad \text{General form.}$$

using the summation property

$$\sum_{k=-\infty}^{\infty} x(k) \longleftrightarrow \frac{X(e^{j\Omega})}{1 - e^{-j\Omega}} + \pi X(e^{j\Omega}) \sum_{k=-\infty}^{\infty} \delta(\Omega - 2\pi k)$$

$$\text{comparing } x(n) = \delta(n) \longleftrightarrow X(\Omega) = 1$$

$$\sum_{k=-\infty}^n \delta(k) = u(n) \longleftrightarrow \frac{1}{1 - e^{-j\Omega}} + \pi(1) \sum_{k=-\infty}^{\infty} \delta(\Omega - 2\pi k)$$

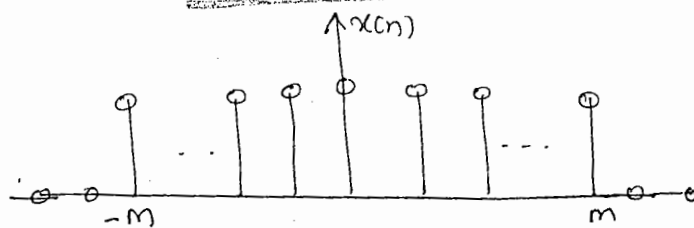
for one cycle of Ω from $-\pi$ to π ; for $k=0$;

$$u(n) \longleftrightarrow \frac{1}{1 - e^{-j\Omega}} + \pi \sum_{k=-\infty}^{\infty} \delta(\Omega)$$

$$u(n) \xleftrightarrow{\text{DTFT}} \frac{1}{1 - e^{-j\Omega}} + \pi \delta(\Omega)$$

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$$x(n) = \begin{cases} 1 & |n| \leq m \\ 0 & |n| > m \end{cases}$$



$$X(\Omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\Omega n} = \sum_{n=-m}^m 1 e^{-j\Omega n} \quad \text{--- (A)}$$

put $n+m=k$;

$$X(\Omega) = \sum_{k=0}^{2m} e^{-j\Omega k} e^{+j\Omega m} = e^{j\Omega m} \sum_{k=0}^{2m} (e^{-j\Omega})^k \rightarrow \text{(A)}$$

$$X(\Omega) = e^{j\Omega m} \left[\frac{1 - (e^{-j\Omega})^{2m+1}}{1 - e^{-j\Omega}} \right]$$

at $\Omega=0$; the $X(\Omega)$ is indeterminate

hence put $\Omega=0$ in eqn (A)

$$\boxed{X(\Omega) = 2m+1} \text{ at } \Omega=0$$

$$X(\Omega) = \begin{cases} \frac{e^{j\Omega m} [1 - (e^{-j\Omega})^{2m+1}]}{1 - e^{-j\Omega}} ; & \Omega \neq 0, \pm 2\pi, \pm 4\pi \\ = 2m+1 ; & \Omega = 0, \pm 2\pi, \pm 4\pi \end{cases}$$

$$\underline{x(n) = \delta(6-3n)}$$

$\delta(6-3n)$ exists at $6-3n=0$

$$\text{or } n=2$$

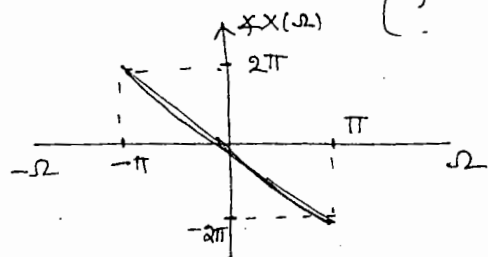
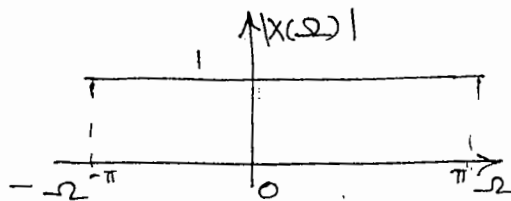
$$X(\Omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\Omega n} = \sum_{n=-\infty}^{\infty} \delta(6-3n) e^{-j\Omega n} = e^{-j\Omega n} \Big|_{n=2}$$

$$\boxed{X(\Omega) = e^{-j2\Omega}}$$

$$|X(\Omega)| = |e^{-j2\Omega}| = 1$$

$$\neq X(\Omega) = -2\Omega$$

$$\begin{cases} |e^{j0}| = 1 \\ \neq e^{j0} = 0 \end{cases}$$



→ (A)

$$\underline{x(n) = \left(\frac{1}{2}\right)^n u(n)}$$

Let $x_1(n) = \left(\frac{1}{2}\right)^n u(n)$

$$X_1(\Omega) = \sum_{n=-\infty}^{\infty} x_1(n) e^{-j\Omega n} = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n u(n) e^{-j\Omega n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n e^{-j\Omega n} = \sum_{n=0}^{\infty} \left[\frac{1}{2} e^{-j\Omega}\right]^n$$

$$= \frac{1}{1 - \frac{1}{2} e^{-j\Omega}}$$

$$\boxed{X_1(\Omega) = \frac{1}{1 - \frac{1}{2} e^{-j\Omega}}}$$

$$\frac{d}{d\Omega} [X_1(\Omega)] = \frac{-\left(\frac{j}{2}\right) e^{-j\Omega}}{\left[1 - \frac{1}{2} e^{-j\Omega}\right]^2}$$

we have

$$-jn x_1(n) = -jn \left(\frac{1}{2}\right)^n u(n) \longleftrightarrow \frac{d}{d\Omega} [X_1(\Omega)]$$

$$\therefore \text{ie } -jn x(n) \longleftrightarrow \frac{-\frac{j}{2} e^{-j\Omega}}{\left[1 - \frac{1}{2} e^{-j\Omega}\right]^2}$$

$$\therefore x(n) \longleftrightarrow \frac{\frac{1}{2} e^{-j\Omega}}{\left[1 - \frac{1}{2} e^{-j\Omega}\right]^2}$$

$j, \pm 4\pi$

$$\underline{x(n) = \left(\frac{1}{2}\right)^n u(-n-1)}$$

$$X(\Omega) = \sum_{n=-\infty}^0 x(n) e^{-j\Omega n} = \sum_{n=-1}^{-\infty} \left(\frac{1}{2}\right)^n e^{-j\Omega n}$$

$$= \sum_{n=1}^{\infty} [2e^{j\Omega}]^n \quad \text{put } n-1=m$$

$$m+1=n$$

$$X(\Omega) = \sum_{m=0}^{\infty} [2e^{j\Omega}] [2e^{j\Omega}]^m = 2e^{j\Omega} \left[\frac{1}{1-2e^{j\Omega}} \right]$$

$$\boxed{X(\Omega) = \frac{2e^{j\Omega}}{1-2e^{j\Omega}}}$$

Find the DTFT of $x(n) = \alpha^n \cos \Omega_0 n u(n)$.

$$x(n) = \alpha^n \cdot u(n) \left[\frac{e^{j\Omega_0 n} + e^{-j\Omega_0 n}}{2} \right]$$

$$x(n) = \frac{1}{2} [\alpha^n e^{j\Omega_0 n} u(n) + \alpha^n e^{-j\Omega_0 n} u(n)]$$

~~we~~ we have

$$\alpha^n u(n) \longleftrightarrow \frac{1}{1-\alpha e^{-j\Omega}}$$

from freq. shift property

$$e^{j\Omega_0 n} \alpha^n u(n) \longleftrightarrow \frac{1}{1-\alpha e^{-j(\Omega-\Omega_0)}}$$

$$\therefore x(n) \longleftrightarrow \frac{1}{2} \left[\frac{1}{1-\alpha e^{-j(\Omega-\Omega_0)}} + \frac{1}{1-\alpha e^{-j(\Omega+\Omega_0)}} \right] = X(\Omega)$$

Find the DTFT of $x(n) = \sin\left(\frac{\pi}{4}n\right) \left(\frac{1}{4}\right)^n u(n-1)$

$$\left(\frac{1}{4}\right)^n u(n) \longleftrightarrow \frac{1}{1-\frac{1}{4}e^{-j\Omega}}$$

$$\left(\frac{1}{4}\right)^{n-1} u(n-1) \longleftrightarrow \frac{\frac{1}{4} e^{-j\Omega}}{1-\frac{1}{4}e^{-j\Omega}}$$

$$\sin\left(\frac{\pi}{4}n\right) = \frac{e^{j\frac{\pi}{4}n} - e^{-j\frac{\pi}{4}n}}{2j}$$

using the frequency shift property we get

$$e^{j\Omega_0 n} y(n) \longleftrightarrow Y[e^{j(\Omega-\Omega_0)}]$$

$$X(\Omega) = \frac{1}{8j} \left[\frac{e^{-j(\Omega - \frac{\pi}{4})}}{1 - \frac{1}{4} e^{-j(\Omega - \frac{\pi}{4})}} - \frac{e^{-j(\Omega + \frac{\pi}{4})}}{1 - \frac{1}{4} e^{-j(\Omega + \frac{\pi}{4})}} \right]$$

A signal has a DTFT $X(\Omega) = \frac{1}{1 - a e^{-j\Omega}}$

Find the DTFT of $x(n) * x(n-1) = g(n)$

$$x(n-1) \xrightarrow{\text{DTFT}} \frac{e^{-j\Omega}}{1 - a e^{-j\Omega}} \quad (\text{from time shifting property})$$

$$x(n) * x(n-1) \longleftrightarrow \frac{1}{1 - a e^{-j\Omega}} \cdot \frac{e^{-j\Omega}}{1 - a e^{-j\Omega}} = G(\Omega)$$

$$G(\Omega) = \frac{e^{-j\Omega}}{[1 - a e^{-j\Omega}]^2}$$

Evaluate the following fr without computing $X(\Omega)$.
where $X(\Omega)$ is the DTFT of the sequence $x(n) = (1, 2, 3, 2, 2)$

(a) $X(0)$ (b) $\int_{-\pi}^{\pi} |X(\Omega)|^2 d\Omega$

$$(a) \quad X(0) = X(e^{j0}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j0} = \sum_{n=-\infty}^{\infty} x(n) = 1 + 2 + 3 + 2 + 2 = 10$$

$$\therefore \boxed{X(e^{j0}) = 10}$$

(b) From Parseval's theorem

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} |X(\Omega)|^2 d\Omega = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

$$\int_{-\pi}^{\pi} |X(\Omega)|^2 d\Omega = 2\pi \sum_{n=-\infty}^{\infty} |x(n)|^2 = 2\pi [1^2 + 2^2 + 3^2 + 2^2 + 2^2] = 44\pi \text{ joules}$$

Find the DTFT of $x(n) = \sin\left(\frac{\pi n}{8}\right)$

$$\begin{aligned} x(n) = \sin\frac{\pi n}{8} &= \frac{e^{j\frac{\pi n}{8}} - e^{-j\frac{\pi n}{8}}}{2j} \\ &= \frac{1}{2j} e^{j\frac{\pi n}{8}} - \frac{1}{2j} e^{-j\frac{\pi n}{8}} \end{aligned}$$

we have $A \longleftrightarrow 2\pi A \delta(\Omega)$

$$\text{here } A = \frac{1}{2j}$$

$$\frac{1}{2j} \longleftrightarrow 2\pi \cdot \frac{1}{2j} \delta(\Omega) = \frac{\pi}{j} \delta(\Omega).$$

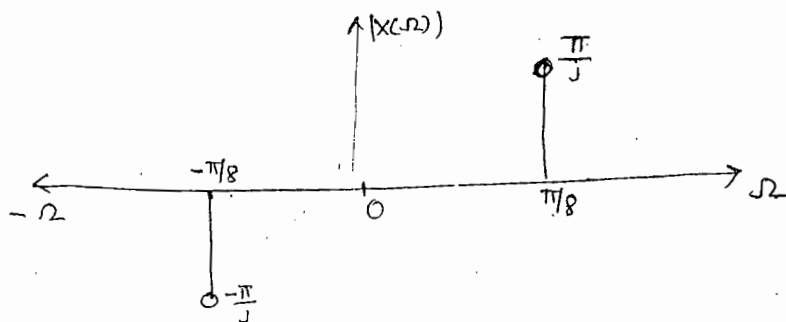
from freq. shift property

$$\frac{1}{2j} e^{j\frac{\pi}{8}n} \longleftrightarrow \frac{\pi}{j} \delta(\Omega - \frac{\pi}{8})$$

$$\therefore X(\Omega) = \frac{\pi}{j} \left[\delta(\Omega - \frac{\pi}{8}) - \delta(\Omega + \frac{\pi}{8}) \right]$$

(b) also determine the value of transform for $x(n-2)$

$$x(n-a) \longleftrightarrow \frac{e^{-j\Omega a}}{j} \left[\delta(\Omega - \frac{\pi}{8}) - \delta(\Omega + \frac{\pi}{8}) \right] = X(\Omega)$$



Find $X(\Omega)$ given $x(n) = \cos \frac{\pi}{5}n + j \sin \frac{\pi}{5}n$; $1 \leq n \leq 10$

$$x(n) = e^{j\frac{\pi}{5}n}$$

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\Omega n} = \sum_{n=-10}^{10} e^{j\frac{\pi}{5}n} e^{-j\Omega n}$$

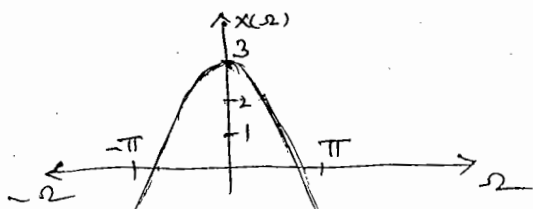
$$X(\Omega) = \sum_{m=0}^{20} e^{+j(\frac{\pi}{5}-\Omega)(m-10)} = e^{-10j(\frac{\pi}{5}-\Omega)} \sum_{m=0}^{20} e^{j(\frac{\pi}{5}-\Omega)m}$$

$$X(\Omega) = e^{-10j(\frac{\pi}{5}-\Omega)} \left[\frac{1 - e^{j21(\frac{\pi}{5}-\Omega)}}{1 - e^{j(\frac{\pi}{5}-\Omega)}} \right]$$

$x(n) = u(n+1) - u(n-2)$, find its DTFT.

$$x(n) = u(n+1) - u(n-2) = \delta(n+1) + \delta(n) + \delta(n-1)$$

$$X(\Omega) = e^{j\Omega} + 1 + e^{-j\Omega} = \underline{\underline{1 + 2\cos\Omega}}$$



For a real signal $x(n]$, $x(n) \longleftrightarrow X(\Omega)$. compute the DTFT of the following signals.

(a) $y(n) = x(-n)$.

$$Y(\Omega) = \sum_{n=-\infty}^{\infty} x(-n) e^{-j\Omega n} = X(\Omega).$$

(b) $g(n) = x(n) * x(-n)$ since $x(n)$ is real.
 $= X(\Omega) \cdot X(\Omega) = |X(\Omega)|^2$

(c) $s(n) = (-1)^n x(n) = e^{j\pi n} x(n)$
 $S(\Omega) = X(\Omega - \pi)$

(d) $z(n) = (1 + \cos n\pi) x(n) = x(n) + \cos n\pi x(n)$
 $= x(n) + e^{jn\pi} x(n)$
 $Z(\Omega) = X(\Omega) + X(\Omega - \pi)$

(e) $b(n) = (-1)^{n/2} x(n) = e^{jn\pi/2} x(n)$
 $B(\Omega) = X(\Omega - \pi/2)$

Evaluate the following using parseval's theorem.

$$\int_0^\pi \frac{4}{5 + 4 \cos \Omega} d\Omega.$$

Let $x(n) = -a^n u(-n-1)$

$$X(\Omega) = \frac{1}{1 - a e^{-j\Omega}} = \frac{1}{(1 - a \cos \Omega) - j \sin \Omega a}$$

$$|X(\Omega)|^2 = \frac{1}{1 - 2a^2 \cos \Omega + a^2}$$

put $a = -2$, $|X(\Omega)|^2 = \frac{1}{1 + 4 \cos \Omega + 4} = \frac{1}{5 + 4 \cos \Omega}$

Hence $x(n) = -(-2)^n u(-n-1)$

$$\int_0^\pi \frac{4 d\Omega}{5 + 4 \cos \Omega} = 4 \int_0^\pi \frac{d\Omega}{5 + 4 \cos \Omega} = 4 \int_0^\pi |X(\Omega)|^2 d\Omega$$

$$= 2 \int_{-\pi}^\pi |X(\Omega)|^2 d\Omega = 4\pi \sum_{n=-\infty}^{\infty} |x(n)|^2$$

$$= 4\pi \sum_{n=-\infty}^{-1} |(-2)^n|^2 = 4\pi \sum_{n=-\infty}^{-1} \left(\frac{1}{4}\right)^n$$

$$= \pi \sum_{n=0}^{\infty} \left(\frac{1}{4}\right)^n = \frac{4\pi}{3}$$

Find the inverse DTFT of

(a) $x(\Omega) = [1 + \cos \Omega] e^{-j2\Omega}$

$$x(\Omega) = 1 + \left[\frac{e^{j\Omega} + e^{-j\Omega}}{2} \right] \cdot e^{-j2\Omega}$$

$$= e^{-j2\Omega} + \frac{1}{2} e^{-j\Omega} + \frac{1}{2} e^{-j3\Omega}$$

Taking IDFT

$$x(n) = \delta(n-2) + \frac{1}{2} \delta(n-1) + \frac{1}{2} \delta(n-3)$$

$$= \left\{ 0, \frac{1}{2}, 1, \frac{1}{2} \right\}$$

(b) $x(\Omega) = \frac{1}{1 - \frac{1}{3} e^{-j\Omega}}$

we have $a^n u(n) \longleftrightarrow \frac{1}{1 - a e^{-j\Omega}}$

put $a = \frac{1}{3}$,

$$\left(\frac{1}{3}\right)^n u(n) \longleftrightarrow \frac{1}{1 - \frac{1}{3} e^{-j\Omega}}$$

$$\therefore \boxed{x(n) = \left(\frac{1}{3}\right)^n u(n)}$$

(c) $x(\Omega) = \delta(\Omega)$

$$x(n) = \frac{1}{2\pi} \int_{\langle 2\pi \rangle} x(\Omega) e^{j\Omega n} d\Omega = \frac{1}{2\pi} \int_{\langle 2\pi \rangle} \delta(\Omega) \cdot e^{j\Omega n} d\Omega$$

$$= \frac{1}{2\pi} e^{j\Omega n} \Big|_{\Omega=0} = \frac{1}{2\pi}$$

$$\therefore \boxed{x(n) = \frac{1}{2\pi}}$$

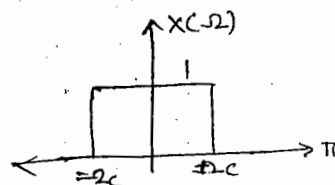
hence $\boxed{\frac{1}{2\pi} \longleftrightarrow \delta(\Omega)}$

(d) $x(\Omega) = \begin{cases} 1 & ; |\Omega| \leq \Omega_c \\ 0 & ; |\Omega| > \Omega_c \end{cases}$

$$x(n) = \frac{1}{2\pi} \int_{-\Omega_c}^{\Omega_c} 1 e^{j\Omega n} d\Omega$$

$$= \frac{1}{2\pi} \frac{e^{j\Omega n}}{jn} \Big|_{-\Omega_c}^{\Omega_c} = \frac{1}{n\pi} \left[\frac{e^{jn\Omega_c} - e^{-jn\Omega_c}}{2j} \right]$$

$$\boxed{x(n) = \frac{1}{n\pi} \sin[n\Omega_c]}$$



(e) $x(\Omega) = 2\pi \delta(\Omega - \Omega_0)$

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} x(\Omega) e^{j\Omega n} d\Omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} 2\pi \delta(\Omega - \Omega_0) e^{j\Omega n} d\Omega$$

$$= e^{j\Omega_0 n} \Big|_{\Omega=\Omega_0} = e^{j\Omega_0 n}$$

$$\therefore \boxed{x(n) = e^{j\Omega_0 n}}$$

$$e^{j\Omega_0 n} \longleftrightarrow 2\pi \delta(\Omega - \Omega_0)$$

if $\Omega_0 = 0$;

$$1 \longleftrightarrow 2\pi \delta(\Omega)$$

$$A \longleftrightarrow 2\pi A \delta(\Omega)$$

(f) $x(\Omega) = |\sin(\Omega)|$;

$$x(\Omega) = \left| \frac{e^{j\Omega} - e^{-j\Omega}}{2j} \right| = \frac{e^{j\Omega}}{2j} - \frac{e^{-j\Omega}}{2j}$$

$$x(\Omega) = \frac{+e^{j\Omega}}{2j} + 0 \cdot e^{j0} + \frac{e^{-j\Omega}}{2j} \longrightarrow \textcircled{A}$$

also $x(\Omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\Omega n}$ (B)

comparing (A) & (B)

$$x(n) = \left\{ \frac{1}{2j}, 0, \frac{-1}{2j} \right\}$$

(g) $x(\Omega) = \cos \Omega + j \sin \Omega$

$$x(\Omega) = e^{j\Omega}$$

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} x(\Omega) e^{j\Omega n} d\Omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j\Omega} e^{j\Omega n} d\Omega$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j\Omega(n+1)} d\Omega = \frac{1}{2\pi} \left[\frac{e^{j(n+1)\Omega}}{j(n+1)} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi(n+1)} \left[\frac{e^{j(n+1)\pi} - e^{-j(n+1)\pi}}{2j} \right]$$

$$\boxed{x(n) = \frac{1}{\pi(n+1)} [\sin(n+1)\pi] = \underline{\underline{\text{sinc}(n+1)}}$$

$$(g) \quad x(\Omega) = \begin{cases} 1 & ; -\pi/2 < \Omega < \pi/2 \\ 0 & ; \text{otherwise} \end{cases} \quad \& \quad \underline{\cancel{x}(\Omega) = -4\Omega}$$

$$x(\Omega) = |x(\Omega)| e^{j\angle x(\Omega)}$$

$$= 1 \cdot e^{-j4\Omega} = \underline{e^{-j4\Omega}}$$

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-j4\Omega} e^{j\Omega n} d\Omega$$

$$= \frac{1}{2\pi} \left[\frac{e^{j(n-4)\Omega}}{j(n-4)} \right]_{-\pi}^{\pi} = \frac{1}{\pi(n-4)} \sin(n-4)\pi$$

$$\boxed{x(n) = \text{sinc}(n-4)}$$

$$(h) \quad x(\Omega) = \begin{cases} 1 & ; +\pi/2 \leq \Omega < \pi \\ 0 & ; \text{otherwise} \end{cases} \quad \& \quad \underline{\cancel{x}(\Omega) = -4\Omega}$$

$$\therefore \boxed{x(\Omega) = e^{-j4\Omega}}$$

$$x(n) = \frac{1}{2\pi} \left[\frac{e^{j(n-4)\Omega}}{j(n-4)} \right]_{-\pi/2}^{\pi} + \left[\frac{e^{j(n-4)\Omega}}{j(n-4)} \right]_{\pi/2}^{\pi}$$

$$\boxed{x(n) = \frac{\sin \pi(n-4) - \sin \frac{\pi}{2}(n-4)}{\pi(n-4)}}$$

$$\text{or } \boxed{x(n) = \text{sinc}(n-4) - \frac{1}{2} \text{sinc}\left(\frac{n-4}{2}\right)}$$

$$(i) \quad x(\Omega) = \frac{6 - \frac{2}{3} e^{-j\Omega} - \frac{1}{6} e^{-j2\Omega}}{-\frac{1}{6} e^{-j2\Omega} + \frac{1}{6} e^{-j\Omega} + 1} \quad \text{find } x(n)$$

$$\text{put } e^{-j\Omega} = z$$

$$x(\Omega) = x(z) = \frac{6 - \frac{2}{3} z - \frac{1}{6} z^2}{-\frac{1}{6} z^2 + \frac{1}{6} z + 1} = \frac{36 - 4z - z^2}{-z^2 + z + 6}$$

$$x(\Omega) = \frac{z^2 + 4z - 36}{z^2 - z - 6}$$

powers are same
hence normalize the
numerator to apply
PFC

$$\begin{array}{r} z^2 - z - 6 \overline{) z^2 + 4z - 36} \quad 1 \\ \underline{z^2 - z - 6} \\ 5z - 30 \end{array}$$

$$X(z) = 1 + \frac{5z-30}{z^2-z-6} = 1 + \frac{A}{z+2} + \frac{B}{z-3}$$

$$X(z) = 1 + P(z)$$

$$P(z) = \frac{5z-30}{(z+2)(z-3)} = \frac{A}{z+2} + \frac{B}{z-3}$$

$$A = \left. \frac{5z-30}{z-3} \right|_{z=-2} = \frac{-40}{-5} = \underline{\underline{8}}$$

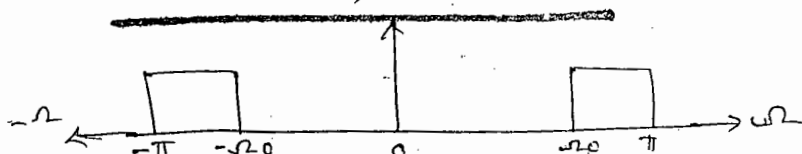
$$B = \left. \frac{5z-30}{z+2} \right|_{z=3} = -3$$

$$X(z) = 1 + \frac{8}{z+2} - \frac{3}{z-3}$$

$$X(\Omega) = 1 + \frac{4}{1 + \frac{e^{j\Omega}}{2}} + \frac{1}{1 - \frac{e^{j\Omega}}{3}}$$

$$x(n) = \delta(n) + 4\left(-\frac{1}{2}\right)^n u(n) + \left(\frac{1}{3}\right)^n u(n)$$

$$(d) \quad X(\Omega) = \begin{cases} 0 & ; 0 \leq |\Omega| \leq \Omega_0 \\ 1 & ; \Omega_0 \leq |\Omega| \leq \pi \end{cases}$$



$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\Omega) e^{j\Omega n} d\Omega = \frac{1}{2\pi} \left[\int_{-\pi}^{-\Omega_0} e^{j\Omega n} d\Omega + \int_{\Omega_0}^{\pi} e^{j\Omega n} d\Omega \right]$$

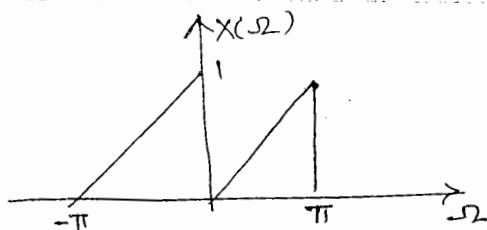
$$= \frac{1}{n\pi} \left[\sin(n\pi) - \sin(\Omega_0 n) \right]$$

$$\text{since } \sin(n\pi) = 0$$

$$x(n) = \frac{-\sin(\Omega_0 n)}{n\pi}$$

$$\frac{-z^2}{+6}$$

me
e. the
o apply
FC



$X(\Omega)$ is shown in fig

find $x(n)$

$$X(\Omega) = \begin{cases} \frac{\Omega}{\pi} + 1; & -\pi \leq \Omega \leq 0 \\ \frac{\Omega}{\pi}; & 0 \leq \Omega \leq \pi \end{cases}$$

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^0 \left[\frac{\Omega}{\pi} + 1 \right] e^{j\Omega n} d\Omega + \frac{1}{2\pi} \int_0^{\pi} \frac{\Omega}{\pi} e^{j\Omega n} d\Omega$$

Note $\int -\Omega e^{a\Omega} d\Omega = \frac{e^{a\Omega}}{a^2} (a\Omega - 1)$ & following the results

$$\frac{e^{j0} - e^{-j0}}{2j} = \sin 0 \text{ \& \; } \sin(\pi\pi) = 0$$

& using $1 - e^{jn\pi} = e^{-jn\pi/2} \left[e^{jn\pi/2} - e^{-jn\pi/2} \right] = e^{-jn\pi/2} \sin\left(\frac{n\pi}{2}\right)$

we get
$$x(n) = \frac{\sin(n\pi/2)}{n\pi} e^{-jn\pi/2}$$

Find the DTFF of $x(n) = 5(3)^n u(-n)$

$$x(n) = 5(3)^n u(-n)$$

$$X(\Omega) = \sum_{n=-\infty}^{\infty} 5(3)^n u(-n) e^{-jn\Omega} = \sum_{n=-\infty}^0 5(3)^n e^{-jn\Omega}$$

$$= \sum_{n=0}^{\infty} 5 \frac{e^{j\Omega n}}{3^n} = \frac{5}{1 - \frac{e^{j\Omega}}{3}}$$

$$5(3)^n u(-n) \xrightarrow{\text{DTFT}} \frac{5}{1 - e^{j\Omega}/3}$$

SAMPLING THEOREM

Statement :

A continuous time band limited signal can be completely represented by its samples and recovered back if the sampling frequency is twice that of the highest frequency content of the original signal.

$$\omega_s \geq 2\omega_m$$

where ω_s is the sampling frequency and ω_m is the max value of frequency content in rad/sec.

Proof : Let $x(t)$ be a CT signal and $x(n)$ or $x(nT_s)$ be the sampled signal at the rate ' T_s ' sec

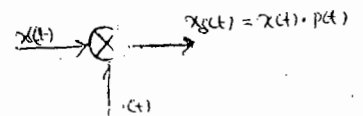
Let $x(t)$ be band limited so that whose transform has no frequency content beyond $|\omega_m|$ rad/sec.

The sampled signal $x_s(t)$ can be represented using an impulse train called Dirac comb $p(t)$.

$$x_s(t) = x(t) \cdot p(t) \quad \text{--- (A)}$$

$$\text{where } p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s) \quad \text{--- (B)}$$

$$\therefore x_s(t) = x(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$



Taking FT on both sides

$$X_s(j\omega) = \frac{1}{2\pi} \left[X(j\omega) * \sum_{n=-\infty}^{\infty} \delta(t - nT_s) \right] \quad \text{--- by mod\~{u}l property.}$$

$$= \frac{1}{2\pi} \left[X(j\omega) * \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s) \cdot \frac{2\pi}{T_s} \right]$$

and using Linearity property and rearranging the terms

$$X_s(j\omega) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X(j\omega) * \delta(\omega - k\omega_s)$$

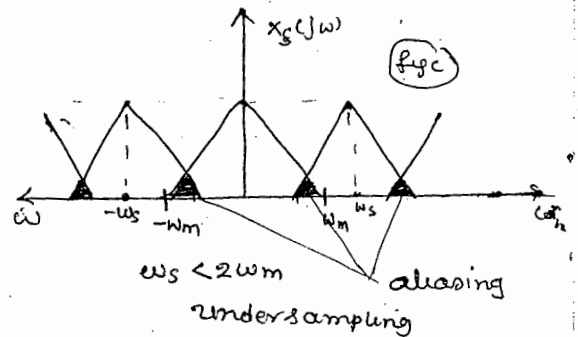
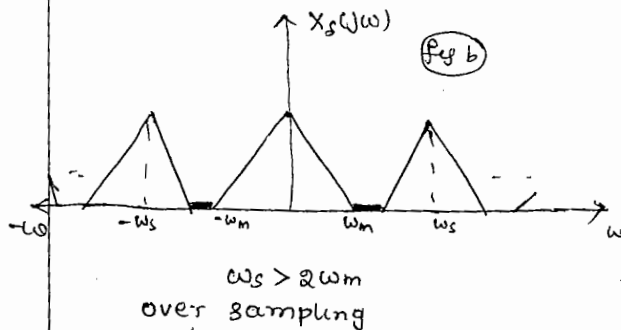
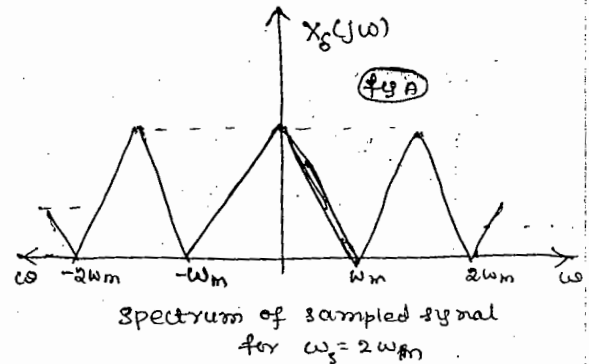
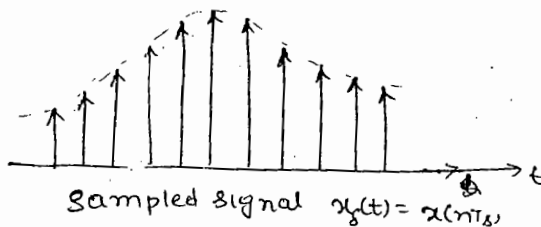
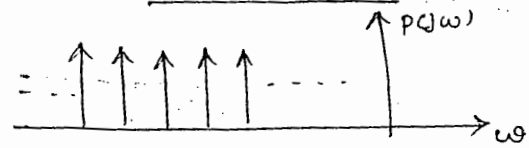
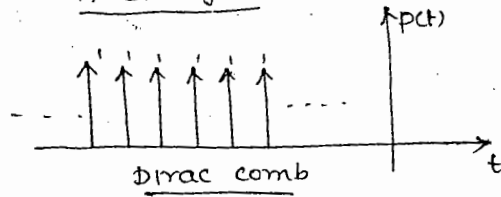
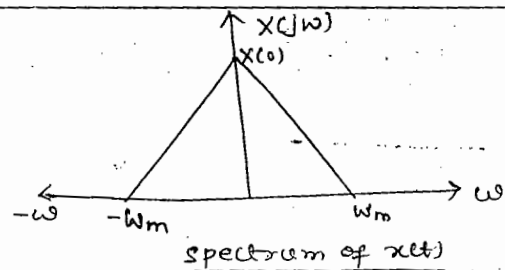
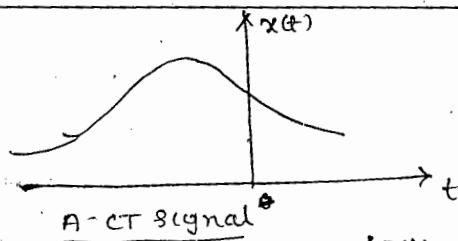
Since convolution of any signal with shifted impulse is signal itself with the time shift of impulse fn

$$\therefore X_s(j\omega) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X(j(\omega - k\omega_s)) = X(j\omega) \quad \text{Since } x(t) = \delta(t)$$

Hence the Fourier transform of sampled signal is an infinite sum of shifted versions original signals Fourier transform.

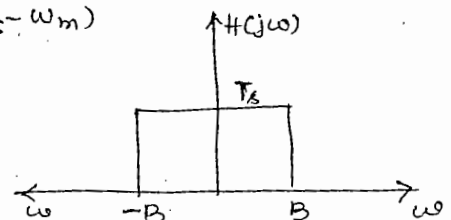
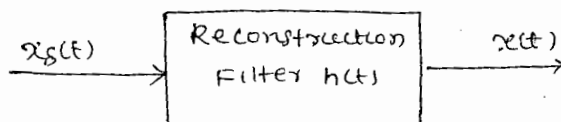
The 'shifted' versions of $X_s(j\omega)$ are offset by an integer 'k' multiples of ω_s . depending upon the value of 'k' we have the following situations

- (a) $\omega_s \geq 2\omega_m$ --- over sampling --- fig b
- (b) $\omega_s < 2\omega_m$ --- under sampling (aliasing) --- fig c
- (c) $\omega_s = 2\omega_m$ --- Nyquist sampling --- fig d



Signal Reconstruction

The message signal $x(t)$ can be recovered from the sampled signal $x_s(t)$ if $\omega_s \geq 2\omega_m$ and the bandwidth of the ideal low pass filter 'B' must lie in bin ω_m & $(\omega_s - \omega_m)$



consider a reconstruction filter with response char as shown.

$$h(t) = \int_{-\infty}^{\infty} H(j\omega) \cdot e^{j\omega t} d\omega = \int_{-B}^B T_s e^{j\omega t} d\omega$$

$$h(t) = T_s \frac{e^{jBt} - e^{-jBt}}{j2\pi} = \boxed{\frac{2T_s}{\pi} \text{sinc}(2Bt)}$$

$$\boxed{h(t) = 2T_s B \text{sinc}(2Bt)} = \boxed{2T_s B \text{sinc}(2Bt)}$$

Assuming the filter to be an LTI system

$$x(t) = h(t) * x_s(t) = x_s(t) * h(t)$$

$$= x(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t - nT_s) * h(t)$$

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \delta(t - nT_s) * 2BT_s \text{sinc} 2Bt$$

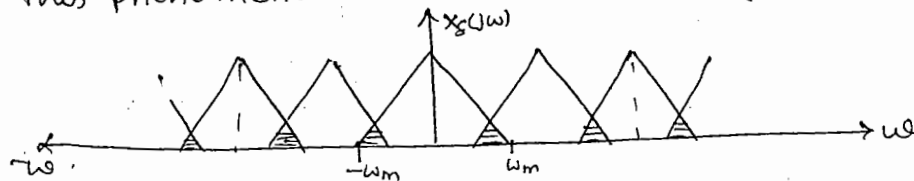
$$x(t) = 2BT_s \sum_{n=-\infty}^{\infty} x(nT_s) \delta(t - nT_s) * \text{sinc} 2Bt$$

$$x(t) = 2BT_s \sum_{n=-\infty}^{\infty} x(nT_s) \text{sinc} 2B(t - nT_s) \rightarrow \textcircled{A}$$

The above relation is called interpolation formula for signal reconstruction.

Effects of aliasing

When $\omega_s < 2\omega_m$, the high frequency components interfere with low frequency components and appear as low frequency. This phenomenon is known as Aliasing.



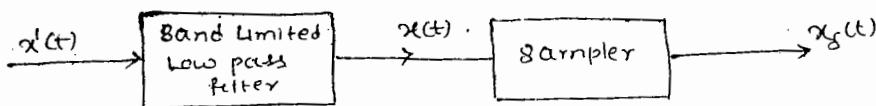
Effects: Due to the interference of high and low frequency components, distortion is generated. The information content is lost and can not be recovered.

Different ways of countering Aliasing effects

Aliasing can be avoided by two methods.

(a) Sampling rate $\omega_s > 2\omega_m$ & (b) Strictly band limit the signal to ω_m .

If we sample the signal ^{above} ~~above~~ and equal to nyquist rate, no overlapping of samples takes place. Also providing a low pass filter as shown, we can strictly avoid aliasing even at nyquist rate.



Nyquist rate when $\omega_s = 2\omega_m$; then the sampling rate for a given bandwidth ω_m is called nyquist rate and the interval T_s between the adjacent samples is known as nyquist interval.

$$\text{Nyquist rate} = 2\omega_m \text{ rad/sec}$$

$$\text{Nyquist interval} = \frac{1}{\text{Nyquist rate}} = \frac{1}{2\omega_m} \text{ sec}$$

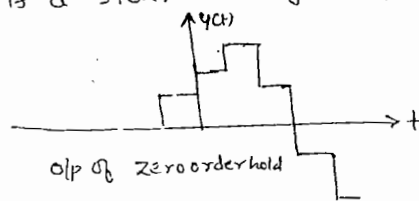
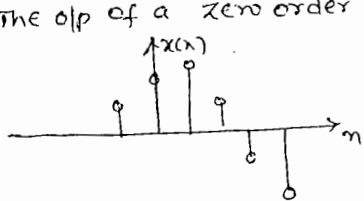
Importance of sampling theorem

1. Since CT signals can not be implemented on a digital computer, sampling theorem provides a convenient way for representation of CT signals on a computer. and it gives the rate at which the CT signals are sampled without losing data and provides adequate information about reconstruction of CT signal at other end over a bandwidth.

2. Sampling theorem indicates how fast we must sample a CT signal so that samples uniquely represent a CT signal.
3. It also gives information about smoothness or rate at which a time domain signal changes and its relationship b/w the maximum freq. content of the signal.

Practical Reconstruction or zero-order hold

A zero order hold (ZOH) is practically used for reconstruction of the signal. It simply holds the value of $x(n)$ for T sec. Here T is the sampling period. The o/p of a zero order hold is a stair-case signal that approximates $x(n)$.



Let the imp. response of zero order hold be

$$h(t) = \begin{cases} 1; & 0 \leq t < T \\ 0; & \text{elsewhere} \end{cases}$$

$$H(j\omega) = e^{-j\omega T/2} \cdot \frac{2 \sin \omega T/2}{\omega}$$

output $y(t)$ of ZOH (K) is $y(t) = x(t) * h(t) = h(t) * x(t)$

$$= h(t) * \sum_{n=-\infty}^{\infty} x(n) \delta(t - nT) = h(t) * x_s(t)$$

Taking FT on both sides; $Y(j\omega) = H(j\omega) \cdot X_s(j\omega)$

$$Y(j\omega) = 2 e^{-j\omega T/2} \frac{\sin \omega T/2}{\omega} \cdot X_s(j\omega) = 2 e^{-j\omega T/2} \frac{\sin \omega T/2}{\omega}$$

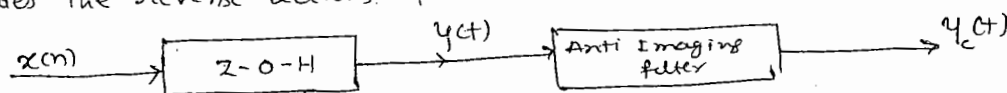
$$X_s(j\omega) = X(j\omega)$$

From the above eqn we can conclude that

① The shape of $X(j\omega)$ is modified by the main lobe $\frac{\sin \omega T/2}{\omega}$.

② There is a Linear time shift (corresponding to time delay $T/2$).

The above modifications can be reduced by increasing the sampling frequency. Also an anti-imaging filter is used for compensating the modifications. The filter provides the reverse actions of zero order hold.



① Specify the nyquist rate for each of the signals

② $x(t) = \cos 5\pi t + 0.5 \cos 10\pi t$

$$x(t) = \cos 5\pi t + 0.5 \cos 10\pi t = x_1(t) + x_2(t)$$

$$\omega_1 = 5\pi$$

$$\omega_2 = 10\pi$$

$$2\pi f_1 = 5\pi$$

$$2\pi f_2 = 10\pi$$

$$\therefore f_1 = (5/2) \text{ Hz}$$

$$f_2 = 5$$

$$f_2 > f_1$$

f_2 is the highest frequency.

$$\therefore \text{Sampling frequency} \geq 2 f_2$$

$$\therefore \text{Nyquist rate} \geq 10 \text{ Hz}$$

⑤ $x(t) = \text{sinc } 200t$

$$x(t) = \text{sinc } 200t = \frac{\sin 200\pi t}{200\pi t}$$

$$\omega = 200\pi$$

$$f = \frac{200\pi}{2\pi} = 100 \text{ Hz}$$

$$\therefore f_s \geq 2f$$

$$\therefore \text{Nyquist rate } f_s \geq 2 \times 100 = 200 \text{ Hz}$$

signal

domain
real

real
valued

(c) $x(t) = \sin^2 200t$

$$x(t) = \sin^2 200t = \left[\frac{\sin 200\pi t}{200\pi t} \right]^2 = \frac{1 - \cos 400\pi t}{2(200\pi t)^2} = \frac{1}{2(200\pi t)^2} - \frac{1}{(200\pi t)^2} \cos 400\pi t$$

$$\left. \begin{aligned} \omega &= 400\pi \\ 2\pi f &= 400\pi \\ \therefore f &= 200\text{Hz} \end{aligned} \right\}$$

Nyquist rate $f_s \geq 2f$.

$$f_s \geq 400\text{Hz}$$

(d) $m(t) = \frac{1}{2\pi} \cos[4000\pi t] \cos[1000\pi t]$

$$m(t) = \frac{1}{2\pi} [\cos 5000\pi t + \cos(3000\pi t)]$$

$$\left. \begin{aligned} \omega_1 &= 5000\pi & \omega_2 &= 3000\pi \\ f_1 &= 2500\text{Hz} & f_2 &= 1500\text{Hz} \end{aligned} \right\}$$

$$f_2 > f_1$$

$$\text{Nyquist rate } f_s \geq 2f_2 \geq 5000\text{Hz}$$

$$\text{Nyquist interval} \leq 0.2\text{msec}$$

(e) $x(n) = \cos \pi n + \frac{1 + \sin 2\pi n}{2}$

$$\omega_1 = \pi$$

$$\omega_2 = 2\pi$$

$$\omega_2 > \omega_1; \text{ hence } f_s \geq 2f_2 \geq 2 \times (2\pi) = 4\pi$$

$$\text{Nyquist rate } \omega_s \geq 4\pi$$

or ϕ

$$\text{Nyquist rate } N = \frac{4\pi}{2\pi} = 2 \text{ samples.}$$

(f) $x(t) = 25e^{j500\pi t}$

$$x(t) = 25e^{j500\pi t} = 25(\cos 500\pi t + j \sin 500\pi t)$$

$$\text{Nyquist rate } f_s \geq 500\text{Hz}$$

(g) $x(t) = 2 \sin 50t \cdot \sin 5000\pi t$

$$= 2 \frac{\sin 50\pi t}{50\pi t} \cdot \sin 5000\pi t = \frac{1}{50\pi t} [\cos 4950\pi t - \cos 5050\pi t]$$

$$\left. \begin{aligned} \omega_2 &= 5050\pi \text{ rad/sec} \\ \omega_1 &= 4950\pi \text{ rad/sec} \end{aligned} \right\}$$

$$f_s \geq 5050\text{Hz}$$

is the nyquist rate.

(h) $p(t) = \sin 20\pi t + \sin 10t + \sin^2 10t$

$$\begin{aligned} p(t) &= \sin 20\pi t + \frac{\sin 10\pi t}{10\pi t} + \frac{\sin^2 10\pi t}{(10\pi t)^2} \\ &= \sin 20\pi t + \frac{\sin 10\pi t}{10\pi t} + \frac{(1 - \cos 20\pi t)}{2(10\pi t)^2} \end{aligned}$$

$$\text{Nyquist rate } f_s \geq 20\text{Hz}$$

$$\omega_1 = 20\pi, f_1 = 10\text{Hz}$$

$$\omega_2 = 10\pi; f_2 = 5\text{Hz}$$

$$\omega_3 = 20\pi; f_3 = 10\text{Hz}$$

$$\therefore f_3 > f_2$$

$$f_s \geq 2f_3$$

(i) Determine the conditions on the sampling interval T_s so that $x(t)$ is uniquely represented by the discrete time sequence given $x(t) = \sin 10t$

$$x(t) = \sin 10t = \frac{\sin 10\pi t}{10\pi t}$$

$$\therefore f_s \geq 10\text{Hz} \text{ is the nyquist rate}$$

$$\therefore \text{Nyquist interval} = \frac{1}{10} = 0.1 \text{ sec}$$

If $T_s < 0.1 \text{ sec}$, then $x(t)$ can be uniquely discretized. This is the required condition to be satisfied.

(j) For an analog signal $m(t) = 4 \cos 100\pi t$ (a) Calculate the min sampling rate to avoid aliasing (b) If the signal is sampled at the rate of 200Hz what is the DT signal after sampling.

(a) $m(t) = 400 \cos 10\pi t$

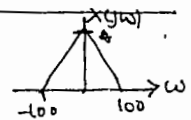
$$\therefore \text{Nyquist rate} = 100\text{Hz} \dots \text{is the min sampling rate.}$$

⑤ given $f_s = 200\text{Hz}$ $T_s = \frac{1}{200}$

we have $\frac{1}{T} = n T_s = \frac{n}{200} \text{ sec}$

$\therefore$ new sampled signal $m'(n) = 4 \cos 100\pi \left(\frac{n}{200}\right) = 4 \cos \left(\frac{n}{2}\right)\pi$

The spectrum of $x(j\omega)$ shown below; Draw the spectrum of sampled signal at (a) twice the nyquist rate (b) $\frac{1}{2}$ the nyquist rate

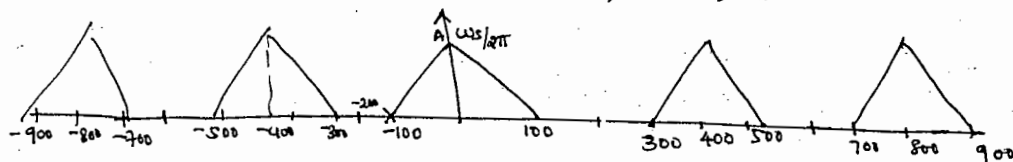


$\omega_m = 100 \text{ rad/sec}$

$\therefore$ Nyquist rate = $100 \text{ rad/sec} = \omega_s$.

(a) Twice the nyquist rate i.e. $\omega_s = 200 \text{ rad/sec}$.

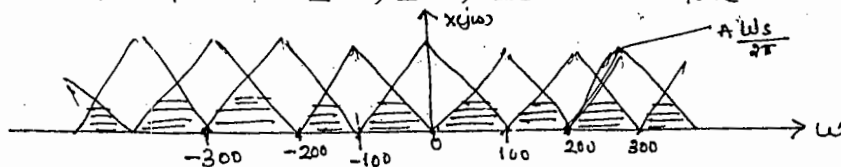
The spectrum will repeat at $\pm 400\text{Hz}$, ± 800 , ± 1200 - - -



(b) Half the nyquist rate

$\omega_s = 50 \text{ rad/sec}$.

Spectrum repeats at ± 100 , ± 200 , ± 300 - - - rad/sec



A 100Hz sinusoid $x(t)$ is sampled at 240Hz . Has aliasing occurred. How many full periods of $x(t)$ are required to obtain one period of sampled period.

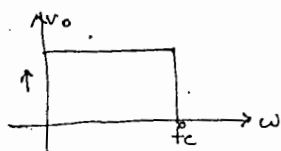
$f = 100\text{Hz}$ $f_s > 2f$; hence no aliasing takes place.

$f_s = 240\text{Hz}$

Also $T_s \leq \frac{2}{f}$ OR $T_s \leq 0.5T$

Hence number of periods of LP signal required is $1/2$

A CT signal $x(t)$ is obtained by the o/p of an ideal Lpf with $f_c = 1000\text{Hz}$. If impulse train sampling is performed on $x(t)$, which of the following periods would guarantee that $x(t)$ can be recovered from its samples version using an appropriate Low pass filter. (a) $T_s = 0.5\text{msec}$ (b) $T_s = 0.1\text{msec}$ (c) $T_s = 2\text{msec}$.



$f_c = f_m = 1000\text{Hz}$ $\therefore T_c = 1\text{msec}$

(a) $T_s = 0.5\text{msec}$ $T_s \leq \frac{1}{2}T$; signal can be recovered

(b) $T_s = 0.1\text{msec}$; $T_s \leq T$; _____

(c) $T_s = 2\text{msec}$; $T_s \neq T$; signal cannot be recovered

Applications of Fourier Representations.

Frequency response of LTI system

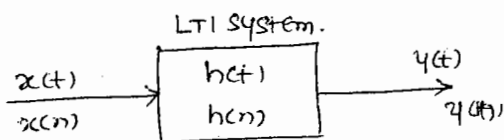
Consider a CT LTI system with $x(t)$ as input, having $h(t)$ as impulse response. The output response of the system $y(t)$ is

$$y(t) = x(t) * h(t)$$

Taking FT on both sides

$$Y(j\omega) = X(j\omega) \cdot H(j\omega)$$

$$\therefore H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} \quad \dots \text{Frequency response of CT LTI system.}$$



Similarly for a DT LTI system

$$H(e^{j\Omega}) = \frac{Y(e^{j\Omega})}{X(e^{j\Omega})} \quad \dots \text{Frequency response of DT LTI system.}$$

1. Determine the frequency response of the systems described by the following impulse responses

$$(a) h(t) = \delta(t) - 2e^{-2t}u(t)$$

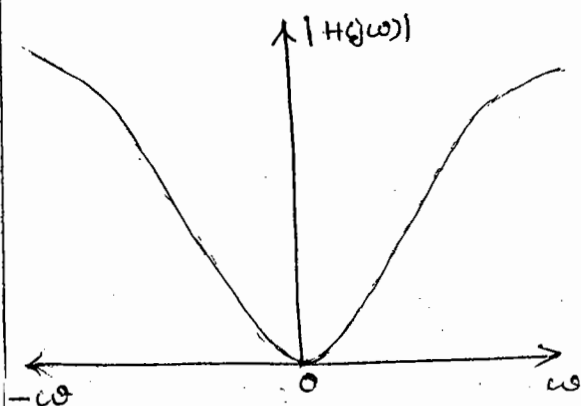
$$h(t) = \delta(t) - 2e^{-2t}u(t)$$

Taking FT on both sides

$$H(j\omega) = 1 - 2 \left[\frac{1}{2+j\omega} \right]$$

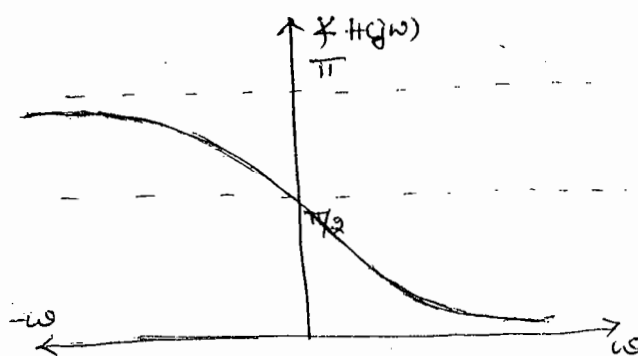
$$H(j\omega) = \frac{j\omega}{2+j\omega}$$

$$\begin{aligned} H(j\omega) &= \int_{-\infty}^{\infty} \delta(t) e^{j\omega t} dt - 2 \int_{-\infty}^{\infty} e^{-2t} u(t) e^{j\omega t} dt \\ &= 1 - 2 \int_0^{\infty} e^{-2t} e^{j\omega t} dt \\ &= 1 - 2 \int_0^{\infty} e^{-(2-j\omega)t} dt \\ &= 1 - \left[\frac{2 e^{-(2-j\omega)t}}{-(2-j\omega)} \right]_0^{\infty} \\ H(j\omega) &= 1 - \frac{2}{2-j\omega} \end{aligned}$$



$$|H(j\omega)| = \frac{\omega}{\sqrt{4+\omega^2}}$$

Magnitude spectrum.



$$\angle H(j\omega) = \pi/2 - \tan^{-1}\left(\frac{\omega}{2}\right)$$

Phase spectrum.

⑥ $h(t) = e^{-|t|}$ for all t .

$$h(t) = \begin{cases} e^t & ; t \leq 0 \\ e^{-t} & ; t \geq 0 \end{cases}$$

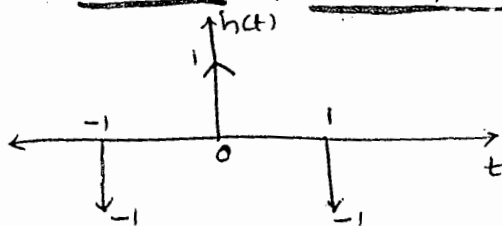
Taking Fourier transforms on both side

$$H(j\omega) = \int_{-\infty}^{\infty} h(t) e^{-j\omega t} dt = \int_{-\infty}^0 e^t e^{-j\omega t} dt + \int_0^{\infty} e^{-t} e^{-j\omega t} dt$$

$$\begin{aligned} H(j\omega) &= \int_{-\infty}^0 e^{(1-j\omega)t} dt + \int_0^{\infty} e^{-(1+j\omega)t} dt \\ &= \left. \frac{e^{(1-j\omega)t}}{(1-j\omega)} \right|_{-\infty}^0 + \left. \frac{e^{-(1+j\omega)t}}{-(1+j\omega)} \right|_0^{\infty} \\ &= \frac{1}{1-j\omega} - \frac{1}{1+j\omega} = \frac{2}{1+\omega^2} \end{aligned}$$

$$\boxed{H(j\omega) = \frac{2}{1+\omega^2} = |H(j\omega)|} \quad \dots \text{is the required freq. response.}$$

⑦ For the impulse response shown below; find $H(j\omega)$



From fig we can write

$$h(t) = -\delta(t+1) + \delta(t) - \delta(t-1)$$

Taking FT on both sides

$$\begin{aligned} H(j\omega) &= -\int_{-\infty}^{\infty} \delta(t+1) e^{-j\omega t} dt + \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt - \int_{-\infty}^{\infty} \delta(t-1) e^{-j\omega t} dt \\ &= -e^{-j\omega t} \Big|_{t=-1} + e^{-j\omega t} \Big|_{t=0} - e^{-j\omega t} \Big|_{t=1} \\ &= -e^{j\omega} + 1 - e^{-j\omega} = 1 - [e^{j\omega} + e^{-j\omega}] \end{aligned}$$

$$\boxed{H(j\omega) = 1 - 2\cos\omega} \quad \dots \text{is the required freq. response.}$$

(d) $h(t) = \frac{1}{RC} e^{-t/RC} u(t)$. Find $H(j\omega)$. Plot the spectrum.

$$h(t) = \frac{1}{RC} e^{-t/RC} u(t)$$

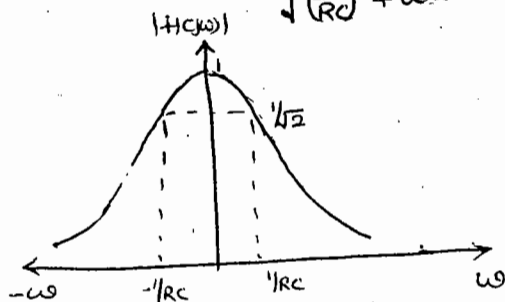
Taking FT on both sides

$$H(j\omega) = \frac{1}{RC} \int_{-\infty}^{\infty} e^{-t/RC} u(t) e^{j\omega t} dt = \frac{1}{RC} \int_0^{\infty} e^{-(\frac{1}{RC} + j\omega)t} dt$$

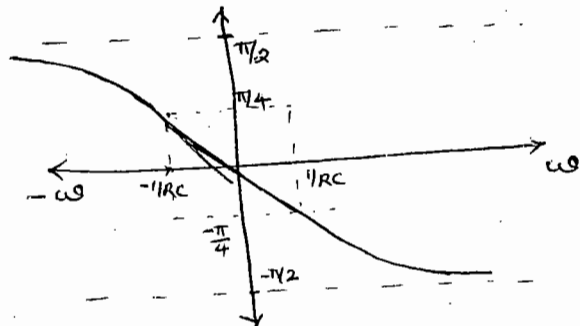
$$H(j\omega) = \frac{1}{RC} \cdot \frac{e^{-(\frac{1}{RC} + j\omega)t}}{-[\frac{1}{RC} + j\omega]} \bigg|_0^{\infty} = \frac{1/RC}{\frac{1}{RC} + j\omega}$$

$$\therefore H(j\omega) = \frac{1/RC}{\frac{1}{RC} + j\omega} \quad \dots \text{is the required forced response.}$$

$$|H(j\omega)| = \frac{1/RC}{\sqrt{(\frac{1}{RC})^2 + \omega^2}} \quad \text{and} \quad \angle H(j\omega) = -\tan^{-1}[\omega RC]$$



Amplitude spectrum



phase spectrum.

(e) Given the impulse response $h(n) = (\frac{1}{2})^n u(n)$. Find its frequency response.

$$h(n) = (\frac{1}{2})^n u(n)$$

Taking DTFT on both sides

$$\begin{aligned} H(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} h(n) e^{-j\omega n} = \sum_{n=-\infty}^{\infty} (\frac{1}{2})^n u(n) e^{-j\omega n} = \sum_{n=0}^{\infty} (\frac{1}{2})^n e^{-j\omega n} \\ &= \sum_{n=0}^{\infty} \left[\frac{1}{2} e^{-j\omega} \right]^n = \frac{1}{1 - \frac{1}{2} e^{-j\omega}} \end{aligned}$$

$$H(e^{j\omega}) = \frac{1}{1 - \frac{e^{-j\omega}}{2}} = \frac{2 e^{j\omega/2}}{2 e^{j\omega/2} - 1}$$

is the required frequency response.

(e) $h(n) = \frac{1}{8} \left(\frac{7}{8}\right)^n u(n)$. Find the freq. response.

$$h(n) = \frac{1}{8} \left(\frac{7}{8}\right)^n u(n)$$

Taking DTFT on both sides

$$\begin{aligned} H(e^{j\Omega}) &= \sum_{n=-\infty}^{\infty} h(n) e^{-j\Omega n} = \sum_{n=0}^{\infty} \frac{1}{8} \left(\frac{7}{8}\right)^n e^{-j\Omega n} \\ &= \frac{1/8}{1 - \frac{7}{8} e^{-j\Omega}} = \frac{1}{8 - 7 e^{-j\Omega}} \end{aligned}$$

$$\boxed{H(e^{j\Omega}) = \frac{1}{8 - 7 e^{-j\Omega}}}$$

2(a) Determine the frequency response and impulse response of the systems given $y(t) = e^{-2t} u(t) + e^{-3t} u(t)$ & $x(t) = e^{-t} u(t)$

$$x(t) = e^{-t} u(t)$$

$$y(t) = e^{-2t} u(t) + e^{-3t} u(t)$$

$$X(j\omega) \xrightarrow{FT} \frac{1}{1+j\omega}$$

$$Y(j\omega) = \frac{1}{2+j\omega} + \frac{1}{3+j\omega}$$

Frequency response is defined as $H(j\omega) = \frac{Y(j\omega)}{X(j\omega)}$

$$\boxed{H(j\omega) = \frac{1+j\omega}{2+j\omega} + \frac{1+j\omega}{3+j\omega}}$$

Normalize the numerators

$$H(j\omega) = 1 + \frac{1}{2+j\omega} + 1 - \frac{2}{3+j\omega} = 2 - \frac{1}{2+j\omega} - \frac{2}{3+j\omega}$$

Taking inverse FT on both sides

$$\boxed{h(t) = 2\delta(t) - e^{-2t} u(t) - 2e^{-3t} u(t)}$$

is the required impulse response

$$\left[h(t) = 2\delta(t) - \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{2+j\omega} e^{j\omega t} d\omega - \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{3+j\omega} e^{j\omega t} d\omega \right]$$

2(b) $y(n) = \frac{1}{4} \left(\frac{1}{2}\right)^n u(n) + \left(\frac{1}{4}\right)^n u(n)$

$x(n) = \left(\frac{1}{2}\right)^n u(n)$ Find the impulse response.

$$x(n) = \left(\frac{1}{2}\right)^n u(n)$$

TAKING DTFT

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} \left(\frac{1}{2}\right)^n u(n) e^{-j\omega n}$$

$$X(e^{j\omega}) = \frac{1}{1 - \frac{e^{-j\omega}}{2}}$$

$$y(n) = \frac{1}{4} \left(\frac{1}{2}\right)^n u(n) + \left(\frac{1}{4}\right)^n u(n)$$

TAKING DTFT

$$Y(e^{j\omega}) = \frac{1/4}{1 - \frac{e^{-j\omega}}{2}} + \frac{1}{1 - \frac{e^{-j\omega}}{4}}$$

we define $H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})}$

$$\begin{aligned} H(e^{j\omega}) &= \frac{1}{4} + \frac{1 - \frac{e^{-j\omega}}{2}}{1 - \frac{e^{-j\omega}}{4}} \\ &= \frac{1}{4} + \frac{1}{1 - \frac{e^{-j\omega}}{4}} - \frac{\frac{1}{2} e^{-j\omega}}{1 - \frac{e^{-j\omega}}{4}} \end{aligned}$$

TAKING INVERSE DTFT

$$h(n) = \frac{1}{4} \delta(n) + \left(\frac{1}{4}\right)^n u(n) - \frac{1}{2} \delta(n-1) * \left(\frac{1}{4}\right)^n u(n)$$

$$h(n) = \frac{1}{4} \delta(n) + \left(\frac{1}{4}\right)^n u(n) - \frac{1}{2} \left(\frac{1}{4}\right)^{n-1} u(n-1)$$

© $x(n) = \left(\frac{1}{4}\right)^n u(n)$ $y(n) = \left(\frac{1}{4}\right)^n u(n) + \left(\frac{1}{4}\right)^{n-1} u(n-1)$

Find the impulse response of the system.

$$x(n) = \left(\frac{1}{4}\right)^n u(n)$$

$$y(n) = \left(\frac{1}{4}\right)^n u(n) + \left(\frac{1}{4}\right)^{n-1} u(n-1)$$

$$X(e^{j\omega}) = \frac{1}{1 - \frac{1}{4} e^{-j\omega}}$$

$$Y(e^{j\omega}) = \frac{1}{1 - \frac{1}{4} e^{-j\omega}} + \frac{e^{-j\omega}}{1 - \frac{1}{4} e^{-j\omega}}$$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = 1 + e^{-j\omega}$$

Taking Inverse DTFT

$\boxed{h(n) = \delta(n) + \delta(n-1)}$ is the required impulse response.

Given $x(t) = e^{-2t} u(t)$; $y(t) = 2t e^{-2t} u(t)$

Find the impulse response.

$$x(t) = e^{-2t} u(t)$$

$$y(t) = 2t e^{-2t} u(t)$$

$$X(j\omega) = \frac{1}{2+j\omega}$$

$$Y(j\omega) = \frac{2}{(2+j\omega)^2}$$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{2}{2+j\omega}$$

Taking inverse FT

$\boxed{h(t) = 2 \cdot e^{-2t} u(t)}$ --- is the required impulse response

Given $x(t) = e^{-3t} u(t)$; $y(t) = e^{-3(t-2)} u(t-2)$.

Find the impulse response.

$$x(t) = e^{-3t} u(t)$$

$$y(t) = e^{-3(t-2)} u(t-2)$$

$$X(j\omega) = \frac{1}{3+j\omega}$$

$$Y(j\omega) = \frac{e^{-j\omega 2}}{3+j\omega}$$

$$H(j\omega) = e^{-j\omega 2}$$

Taking inverse FT

$\boxed{h(t) = \delta(t-2)}$

Differential and difference equations

1. Determine the frequency response and impulse response of the systems described by

(a) $\frac{d}{dt} y(t) + 3y(t) = x(t)$

$$\frac{d}{dt} y(t) + 3y(t) = x(t)$$

Taking FT on both sides

$$j\omega Y(j\omega) + 3Y(j\omega) = X(j\omega)$$

$$Y(j\omega)[3+j\omega] = X(j\omega)$$

Frequency response $H(j\omega) = \frac{Y(j\omega)}{X(j\omega)}$

$$H(j\omega) = \frac{1}{3+j\omega}$$

Taking inverse FT ; we get the impulse response $h(t)$

$$h(t) = e^{-3t} u(t)$$

(b) $\frac{d^2}{dt^2} y(t) + 5 \frac{d}{dt} y(t) + 6y(t) = -\frac{d}{dt} x(t)$

$$\frac{d^2}{dt^2} y(t) + 5 \frac{d}{dt} y(t) + 6y(t) = -\frac{d}{dt} x(t)$$

Taking FT on both sides

$$(j\omega)^2 Y(j\omega) + 5j\omega Y(j\omega) + 6Y(j\omega) = -X(j\omega)(j\omega)$$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{-j\omega}{(j\omega)^2 + 5j\omega + 6} = \frac{-j\omega}{(j\omega+3)(j\omega+2)}$$
$$= \frac{A}{j\omega+3} + \frac{B}{j\omega+2}$$

↓
freq. response

$$-j\omega = A(j\omega+2) + B(j\omega+3)$$

put $j\omega = -2$

$$2 = B$$

put $j\omega = 0$; $0 = 2A + B \cdot 3$

$$A = -\frac{2}{3}$$

$$0 = 2A + 6$$

$$\text{or } A = -3$$

$$H(j\omega) = \frac{-3}{3+j\omega} + \frac{2}{2+j\omega}$$

Taking inverse Fourier Transform

$$h(t) = -3 e^{-3t} u(t) + 2 e^{-2t} u(t) \rightarrow \text{impulse response}$$

$$\textcircled{c} \quad y(n) - \frac{1}{4} y(n-1) - \frac{1}{8} y(n-2) = 3 x(n) - \frac{3}{4} x(n-1)$$

$$y(n) - \frac{1}{4} y(n-1) - \frac{1}{8} y(n-2) = 3 x(n) - \frac{3}{4} x(n-1)$$

Taking DFT on both sides

$$Y(e^{j\Omega}) - \frac{1}{4} e^{-j\Omega} Y(e^{j\Omega}) - \frac{1}{8} e^{-j2\Omega} Y(e^{j\Omega}) = 3 X(e^{j\Omega}) - \frac{3}{4} e^{-j\Omega} X(e^{j\Omega})$$

$$H(e^{j\Omega}) = \frac{Y(e^{j\Omega})}{X(e^{j\Omega})} = \frac{3 - \frac{3}{4} e^{-j\Omega}}{1 - \frac{1}{4} e^{-j\Omega} - \frac{1}{8} e^{-j2\Omega}}$$

$$H(e^{j\Omega}) = \frac{3 - \frac{3}{4} e^{-j\Omega}}{\left[1 + \frac{1}{4} e^{-j\Omega}\right] \left[1 - \frac{1}{2} e^{-j\Omega}\right]} = \frac{A}{1 + \frac{1}{4} e^{-j\Omega}} + \frac{B}{1 - \frac{1}{2} e^{-j\Omega}}$$

$$3 - \frac{3}{4} e^{-j\Omega} = A \left(1 - \frac{1}{2} e^{-j\Omega}\right) + B \left[1 + \frac{1}{4} e^{-j\Omega}\right]$$

$$\text{put } e^{-j\Omega} = 2$$

$$3 - \frac{3}{4} (2) = \frac{3}{2} = 0 + B \left(\frac{3}{2}\right) \Rightarrow \boxed{B=1}$$

$$\text{put } e^{-j\Omega} = 0$$

$$3 = A + B = A + 1 \Rightarrow \boxed{A=2}$$

$$H(e^{j\Omega}) = \frac{2}{1 + \frac{1}{4} e^{-j\Omega}} + \frac{1}{1 - \frac{1}{2} e^{-j\Omega}}$$

Taking inverse DTFT we get

$$h(n) = 2 \left(-\frac{1}{4}\right)^n u(n) + \left(\frac{1}{2}\right)^n u(n) \quad \text{is the required impulse response.}$$

④ $y(n) + \frac{1}{2} y(n-1) = x(n) - 2x(n-1)$ find $h(n)$.

$$y(n) + \frac{1}{2} y(n-1) = x(n) - 2x(n-1)$$

Taking DTFT on both sides

$$Y(e^{j\omega}) + \frac{1}{2} e^{-j\omega} Y(e^{j\omega}) = X(e^{j\omega}) - 2e^{-j\omega} X(e^{j\omega})$$

$$\boxed{H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{1 - 2e^{-j\omega}}{1 + \frac{1}{2}e^{-j\omega}}}$$
 Frequency response.

$$H(e^{j\omega}) = \frac{1}{1 + \frac{1}{2}e^{-j\omega}} - \frac{2e^{-j\omega}}{1 + \frac{1}{2}e^{-j\omega}}$$

Taking inverse DTFT

$$h(n) = \left(\frac{-1}{2}\right)^n u(n) - 2 \cdot \delta(n-1) * \left(\frac{-1}{2}\right)^n u(n)$$

$$\boxed{h(n) = \left(\frac{-1}{2}\right)^n u(n) - 2 \left(\frac{-1}{2}\right)^{n-1} u(n-1)} \dots \text{Impulse response}$$

Determine the differential and difference eqⁿ descriptions for the systems given the impulse responses as

① $h(t) = \frac{1}{a} e^{-\frac{t}{a}} u(t)$

$$h(t) = \frac{1}{a} e^{-\frac{t}{a}} u(t)$$

Taking FT on both sides

$$H(j\omega) = \frac{1}{a} \left[\frac{1}{\frac{1}{a} + j\omega} \right] = \frac{Y(j\omega)}{X(j\omega)}$$

$$\left[\frac{1}{a} + j\omega \right] Y(j\omega) = \frac{1}{a} X(j\omega)$$

Taking inverse FT on both sides

$$\boxed{\frac{1}{a} y(t) + \frac{d}{dt} y(t) = \frac{1}{a} x(t)}$$

is the required diff. eqⁿ

$$(b) \quad h(t) = 2e^{-2t}u(t) - 2te^{-2t}u(t)$$

$$h(t) = 2e^{-2t}u(t) - 2te^{-2t}u(t)$$

$$H(j\omega) = \frac{2}{2+j\omega} - \frac{2}{(2+j\omega)^2} = \frac{2+2j\omega}{(2+j\omega)^2} = \frac{Y(j\omega)}{X(j\omega)}$$

$$Y(j\omega)(2+j\omega)^2 = (2+2j\omega)X(j\omega)$$

$$4Y(j\omega) + 4j\omega Y(j\omega) + (j\omega)^2 Y(j\omega) = 2X(j\omega) + 2X(j\omega) \cdot j\omega$$

Taking inverse FT

$$\boxed{4y(t) + 4\frac{d}{dt}y(t) + \frac{d^2}{dt^2}y(t) = 2x(t) + 2\frac{d}{dt}x(t)}$$

$$(c) \quad h(n) = \alpha^n u(n) ; |\alpha| < 1$$

$$H(e^{j\Omega}) = \frac{1}{1 - \alpha e^{-j\Omega}} = \frac{Y(e^{j\Omega})}{X(e^{j\Omega})}$$

$$Y(e^{j\Omega}) - \alpha e^{-j\Omega} Y(e^{j\Omega}) = X(e^{j\Omega})$$

Taking inverse DTFT

$$y(n) - \alpha y(n-1) = x(n)$$

$$\boxed{y(n) - \alpha y(n-1) = x(n)}$$

$$(d) \quad h(n) = \delta(n) + 2\left(\frac{1}{2}\right)^n u(n) + \left(-\frac{1}{2}\right)^n u(n)$$

$$H(e^{j\Omega}) = 1 + \frac{2}{1 - \frac{1}{2}e^{-j\Omega}} + \frac{1}{1 + \frac{1}{2}e^{-j\Omega}} = \frac{Y(e^{j\Omega})}{X(e^{j\Omega})}$$

$$= \frac{1 - \frac{1}{4}e^{-j2\Omega} + 2 + \frac{1}{2}e^{-j\Omega} + 1 - \frac{1}{2}e^{-j\Omega}}{(1 - \frac{1}{2}e^{-j\Omega})(1 + \frac{1}{2}e^{-j\Omega})} = \frac{4 + \frac{1}{2}e^{-j\Omega} - \frac{1}{4}e^{-j2\Omega}}{1 - \frac{1}{4}e^{-j2\Omega}}$$

Model paper

$$y(e^{j\omega}) \left[1 - \frac{1}{4} e^{-j2\omega} \right] = 4 x(e^{j\omega}) + \frac{1}{2} e^{-j\omega} x(e^{j\omega}) - \frac{1}{4} e^{-j2\omega} x(e^{j\omega})$$

Taking inverse DTFT

$$\boxed{y(n) - \frac{1}{4} y(n-2) = 4 x(n) + \frac{1}{2} x(n-1) - \frac{1}{4} x(n-2)}$$

① For the frequency responses given below, ^{description} determine the difference/differential eq^s of the systems

(a) $H(j\omega) = \frac{2 + 3j\omega - 3(j\omega)^2}{1 + 2j\omega}$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{2 + 3j\omega - 3(j\omega)^2}{1 + 2j\omega}$$

$$Y(j\omega) [1 + 2j\omega] = X(j\omega) [2 + 3j\omega - 3(j\omega)^2]$$

Taking inverse FT

$$\boxed{y(t) + 2 \frac{d}{dt} y(t) = 2x(t) + 3 \frac{d}{dt} x(t) - 3 \frac{d^2}{dt^2} x(t)}$$

(b) $H(j\omega) = \frac{1 - j\omega}{-\omega^2 - 4}$

$$H(j\omega) = \frac{1 - j\omega}{-\omega^2 - 4} = \frac{1 - j\omega}{(j\omega)^2 - 4} = \frac{Y(j\omega)}{X(j\omega)}$$

$$(j\omega)^2 Y(j\omega) - 4 Y(j\omega) = X(j\omega) - j\omega X(j\omega)$$

Taking inverse FT

$$\boxed{\frac{d^2}{dt^2} y(t) - 4 y(t) = x(t) - \frac{d}{dt} x(t)}$$

$$\frac{1}{4} e^{-j2\omega}$$

$$\textcircled{c} \quad H(j\omega) = \frac{1+j\omega}{(j\omega+2)(j\omega+1)}$$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{1+j\omega}{(j\omega+2)(j\omega+1)} = \frac{1+j\omega}{(j\omega)^2 + 3j\omega + 2}$$

$$(j\omega)^2 Y(j\omega) + Y(j\omega) \cdot 3j\omega + 2 Y(j\omega) = X(j\omega) + X(j\omega) j\omega$$

Taking inverse FT

$$\boxed{\frac{d^2}{dt^2} y(t) + 3 \frac{d}{dt} y(t) + 2 y(t) = x(t) + \frac{d}{dt} x(t)}$$

$$\textcircled{d} \quad H(e^{j\Omega}) = \frac{1 + e^{-j\Omega}}{e^{-j2\Omega} + 3}$$

$$H(e^{j\Omega}) = \frac{Y(e^{j\Omega})}{X(e^{j\Omega})} = \frac{1 + e^{-j\Omega}}{e^{-j2\Omega} + 3} = \frac{1 + e^{-j\Omega}}{3 + e^{-j2\Omega}}$$

$$e^{-j2\Omega} Y(e^{j\Omega}) + 3 Y(e^{j\Omega}) = X(e^{j\Omega}) + e^{-j\Omega} X(e^{j\Omega})$$

Taking inverse DTFT

$$\boxed{y(n-2) + 3y(n) = x(n) + x(n-1)}$$

$$\textcircled{e} \quad H(e^{j\Omega}) = 1 + \frac{e^{-j\Omega}}{\left[i - \frac{1}{2} e^{-j\Omega}\right] \left[1 + \frac{1}{4} e^{-j\Omega}\right]}$$

$$H(e^{j\Omega}) = \frac{Y(e^{j\Omega})}{X(e^{j\Omega})} = 1 + \frac{e^{-j\Omega}}{\left[i - \frac{1}{2} e^{-j\Omega}\right] \left[1 + \frac{1}{4} e^{-j\Omega}\right]}$$

$$= \frac{1 - \frac{1}{4} e^{-j\Omega} - \frac{1}{8} e^{-j2\Omega} + e^{-j\Omega}}{1 - \frac{1}{4} e^{-j\Omega} - \frac{1}{8} e^{-j2\Omega}} = \frac{1 - \frac{3}{4} e^{-j\Omega} - \frac{1}{8} e^{-j2\Omega}}{1 - \frac{1}{4} e^{-j\Omega} - \frac{1}{8} e^{-j2\Omega}}$$

$$Y(e^{j\Omega}) \left[1 - \frac{1}{4} e^{-j\Omega} - \frac{1}{8} e^{-j2\Omega}\right] = X(e^{j\Omega}) \left[1 - \frac{3}{4} e^{-j\Omega} - \frac{1}{8} e^{-j2\Omega}\right]$$

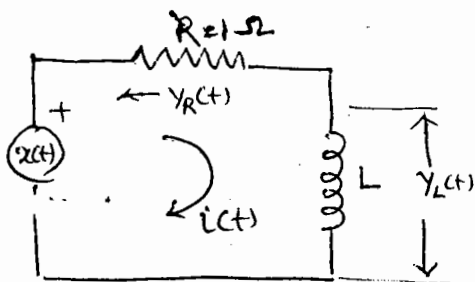
Taking inverse DTFT

$$\boxed{y(n) - \frac{1}{4} y(n-1) - \frac{1}{8} y(n-2) = x(n) + \frac{3}{4} x(n-1) - \frac{1}{8} x(n-2)}$$

② For the circuit shown in figure

(a) Let the ^{input} voltage a/c the inductor $y_L(t)$, write the differential eqⁿ description for the system

(b) Let the output be voltage a/c resistor $y_R(t)$, write the diff. eqⁿ description for the system



(a) Applying KVL

$$\begin{aligned} x(t) &= i(t)R + y_L(t) \\ &= \frac{R}{L} \int_{-\infty}^t y_L(t) dt + y_L(t) \end{aligned}$$

differentiating w.r.t 't'

$$\boxed{\frac{d}{dt} x(t) = \frac{R}{L} y_L(t) + \frac{d}{dt} y_L(t)}$$

$$L \frac{d}{dt} x(t) = R y_L(t) + L \frac{d}{dt} y_L(t)$$

Taking FT on both sides

$$L(j\omega) X(j\omega) = R Y(j\omega) + L(j\omega) Y(j\omega)$$

$$\boxed{H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{j\omega L}{R + j\omega L}} \quad \text{frequency response}$$

~~Taking inverse FT~~

$$\cancel{h(t)} =$$

⑥ Applying KVL

$$\begin{aligned} x(t) &= y_R(t) + y_L(t) \\ &= y_R(t) + L \frac{d}{dt} \frac{y_R(t)}{R} = y_R(t) + \frac{L}{R} \frac{d}{dt} y_R(t) \end{aligned}$$

$\frac{-j\omega L}{R + j\omega L}$

Taking FT on both sides

$$X(j\omega) = Y(j\omega) + \frac{L}{R} j\omega Y(j\omega)$$

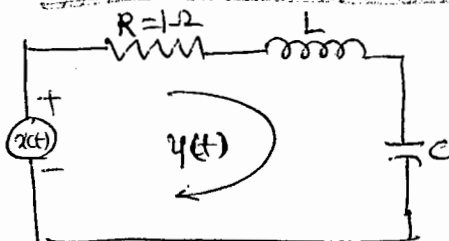
$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{1}{1 + j\omega \frac{L}{R}}$$

put $R = 1\Omega$

$$H(j\omega) = \frac{1}{1 + j\omega L} = \frac{1/L}{\frac{1}{L} + j\omega}$$

$$h(t) = \frac{1}{L} e^{-\frac{t}{L}}$$

③ For the ckt shown; determine the freq. response.



Applying KVL

$$x(t) = R y(t) + L \frac{d}{dt} y(t) + \frac{1}{C} \int_{-\infty}^{\infty} y(\omega) d\omega$$

diff. both sides w.r.t t & putting $R=1\Omega$

$$\frac{d}{dt} x(t) = \frac{d}{dt} y(t) + L \frac{d^2}{dt^2} y(t) + \frac{1}{C} y(t)$$

Taking FT on both sides

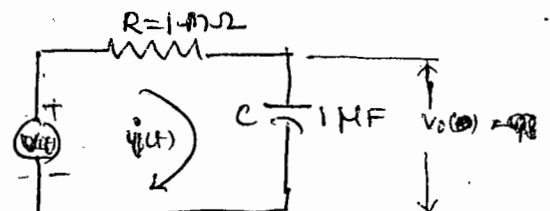
$$j\omega X(j\omega) = j\omega Y(j\omega) + L(j\omega)^2 Y(j\omega) + \frac{1}{C} Y(j\omega)$$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{j\omega}{L(j\omega)^2 + j\omega + \frac{1}{C}}$$

is the freq. response.

④ For the ckt shown below; determine the frequency at which the power frequency response falls to $\frac{1}{\sqrt{2}}$.

Find the corresponding phase at this frequency



Let i be the current thro' the ckt.

Applying KVL

$$v_i(t) = \text{---} = R i(t) + x_c i(t)$$

$$v_o(t) = \text{---} = i(t) \cdot x_c$$

$$\frac{v_o(t)}{v_i(t)} =$$

Let i be the current thro' the ckt.

$$v_i = i R + x_c i = i(R + x_c)$$

$$v_o = i \cdot x_c$$

$$\frac{v_o}{v_i} = H(j\omega) = \frac{x_c}{R + x_c} = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}}$$

$$H(j\omega) = \frac{1}{1 + j\omega RC} \quad \text{--- freq. response}$$

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}} \quad \angle H(j\omega) = -\tan^{-1}(\omega RC)$$

$$H(j\omega) = \frac{1/RC}{\frac{1}{RC} + j\omega}$$

Taking Inverse FT

$$h(t) = \frac{1}{RC} e^{-\frac{t}{RC}} u(t) \quad \text{Impulse response}$$

at

$$\text{Given } |H(j\omega)| = \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

$$1 + (\omega RC)^2 = 2$$

at

$$\text{Putting } R = 1 \text{ m}\Omega, C = 1 \text{ }\mu\text{F} ;$$

$$1 + \omega^2 = 2$$

$$\boxed{\omega = \pm 1} \text{ rad/sec}$$

The corresponding phase is given by [put $\omega = \pm 1$]

$$\angle H(j\omega) = -\tan^{-1}(\pm 1) = \underline{\underline{\mp 45^\circ}}$$

- 5) The Transfer fn of a system is $H(j\omega) = \frac{16}{4+j\omega}$. Find the Time domain response $y(t)$ for input $x(t) = u(t)$

$$x(t) = u(t)$$

$$X(j\omega) = \frac{1}{j\omega}$$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{16}{4+j\omega}$$

~~$$16 X(j\omega) = 4 Y(j\omega) + j\omega X(j\omega)$$~~

~~$$4 Y(j\omega) + j\omega Y(j\omega) = \frac{16}{j\omega} = -16j\omega$$~~

~~$$[4 Y(t) - e^{t/4} Y(t)]$$~~

$$Y(j\omega) = X(j\omega) \cdot H(j\omega)$$

$$Y(j\omega) = \frac{1}{j\omega} \left\{ \frac{16}{4+j\omega} \right\}$$

$$Y(j\omega) = \frac{A}{j\omega} + \frac{B}{4+j\omega}$$

$$A = \left. \frac{16}{4+j\omega} \right|_{j\omega=0}$$

$$A = 4,$$

$$B = \left. \frac{16}{j\omega} \right|_{j\omega=-4}$$

$$B = -4$$

$$Y(j\omega) = \frac{4}{j\omega} - \frac{4}{4+j\omega}$$

$$y(t) = 4 u(t) - 4 e^{-4t} u(t)$$

$$y(t) = 4 [1 - e^{-4t}] u(t)$$

- 5) A DT system has a unit impulse response $h(n)$ given by

$$h(n) = \frac{1}{2} \delta(n) + \delta(n-1) + \frac{1}{2} \delta(n-2)$$

Find the freq. response of the system.

$$H(e^{j\Omega}) = \frac{1}{2} + e^{-j\Omega} + \frac{1}{2} e^{-j2\Omega} = \frac{1}{2} e^{-j\Omega} \cdot e^{j\Omega} + e^{-j\Omega} + \frac{1}{2} e^{-j2\Omega}$$

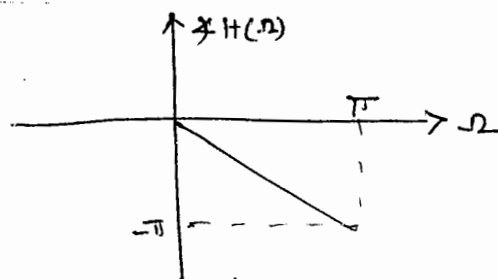
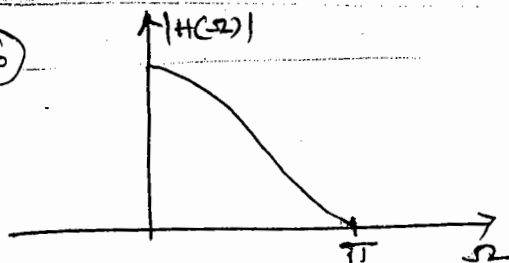
$$= e^{-j\Omega} \left\{ \frac{1}{2} \{e^{j\Omega} + e^{-j\Omega}\} + 1 \right\}$$

$$H(e^{j\Omega}) = e^{-j\Omega} \{1 + \cos \Omega\}$$

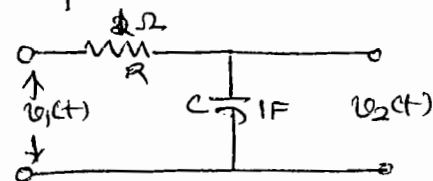
$$|H(e^{j\Omega})| = |1 + \cos \Omega|$$

$$\angle H(e^{j\Omega}) = -\Omega$$

6



A CT LTI system shown below.
Determine $h(t)$ and $S(e^{j\omega})$.
plot the responses and indicate
whether the system is stable/causal



$$v_2(t) = \frac{x_c}{R + x_c} \cdot v_1(t)$$

$$v_2(j\omega) = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} v_1(j\omega)$$

$$\frac{v_2(j\omega)}{v_1(j\omega)} = \frac{1}{1 + j\omega RC} = \frac{1}{1 + j\omega} = H(j\omega)$$

$$h(t) = \mathcal{F}^{-1}\{H(j\omega)\}$$

For unit step input $x(t) = u(t) = v_1(t)$

$$X(j\omega) = \frac{1}{j\omega} = v_1(j\omega)$$

$$v_2(j\omega) = H(j\omega) = v_1(j\omega) = \frac{1}{j\omega} \cdot \frac{1}{j\omega + 1} = \frac{A}{j\omega} + \frac{B}{j\omega + 1}$$

$$v_2(j\omega) = \frac{1}{j\omega} + \frac{\phi}{1 + j\omega}$$

$$v_2(t) = A u(t) + e^{-t} u(t) \quad \text{is the step response } s(t)$$

$$h(t) = e^{-t} u(t)$$

$h(t) = 0$ for $t < 0$; hence system is causal.

Since $\int_{-\infty}^{\infty} |h(t)| dt < \infty$; system is stable

$$\text{i.e. } \int_{-\infty}^{\infty} |h(t)| dt = \int_0^{\infty} e^{-t} dt = 1 < \infty$$

by

